

The Proto-Germanic *n*-stems

A study
in diachronic morphophonology

LEIDEN STUDIES IN INDO-EUROPEAN

18

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The Proto-Germanic *n*-stems

A study in diachronic morphophonology

Guus Kroonen



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Table of contents

PREFACE	11
LIST OF ABBREVIATIONS	13
Language abbreviations	13
Linguistic abbreviations	14
Logical symbols	15
PRELIMINARY REMARKS	17
The Germanic linguistic sources	17
Normalization and orthography	21
Presentation of the evidence	21
1 INTRODUCTION	23
2 THE INFLECTION OF THE N-STEMS	27
2.1 The Indo-European <i>n</i>-stems	28
2.1.1 The amphikinetic type	28
2.1.2 The hysterokinetic type	30
2.1.3 The proterokinetic type	31
2.2 Origins of the inflectional types	33
2.3 The Proto-Germanic <i>n</i>-stems	35
2.3.1 The masculine <i>n</i> -stems	35
2.3.2 The feminine <i>n</i> -stems	36
2.3.3 The neuter <i>n</i> -stems	38
3 THE PROTO-GERMANIC GEMINATES	41
3.1 Kluge's law	41
3.2 Shortening in over-long syllables	45
3.3 Exceptions to Kluge's law	46
3.3.1 No gemination of voiceless fricatives	46

3.3.2 No gemination of sibilants	48
3.4 Different configurations of Kluge's law	49
3.4.1 Kluge's traditional configuration	49
3.4.2 Lühr: <i>Verdoppelung</i> rather than assimilation	50
3.4.3 A glottalic approach by Kortlandt	52
 4 KLUGE'S LAW AND THE N-STEMS	 55
4.1 Gemination as <i>grammatischer Wechsel</i>	56
4.2 Reconstruction of the <i>n</i> -stem paradigm	57
4.2.1 Paradigmatic split-offs	58
4.2.2 Special cases	62
4.2.3 Evaluation	68
4.3 Paradigmatic analogy	69
4.3.1 Kluge's <i>Associationen</i>	71
4.3.2 Reception of Kluge's <i>Associationen</i>	72
4.3.3 From allomorphy to consonant gradation	78
4.3.4 Dating the rise of consonant gradation	80
4.3.5 Gemination of * <i>n</i>	83
 5 KLUGE'S LAW AND THE DIRECTIONALS	 85
5.1 The Germanic directional system	85
5.2 The Pre-Germanic situation	89
5.3 On the full grade of Go. <i>iup</i>	91
 6 CONSONANT GRADATION IN THE VERB	 93
6.1 Evidence for Osthoff's hypothesis	94
6.1.1 Internal reconstruction	94
6.1.2 The origin of the zero-grade	97
6.1.3 Direct correspondences	97
6.1.4 The iterative aspect	99
6.1.5 The "problem" of the inchoative verbs	101
6.1.6 An alternative hypothesis by Lühr	102
6.2 The iterative system	103
6.3 Evidence for de-iterativization	106
6.4 De-iterativization in Gothic	110
6.5 The rise of PGm. * <i>ū</i> as a full-grade marker	112

7 A LIFE WITHOUT KLUGE'S LAW?	119
7.1 The Expressivity Theory	119
7.2 Evaluation of the argumentation	121
7.2.1 No evidence for Kluge's law?	121
7.2.2 Expressive gemination vs analogical singulation	124
7.3 The Leiden Substrate Theory	126
7.3.1 The "Language of the Geminates"	127
7.3.2 <i>Hamartia</i> in Germanic Studies	127
 8 ROOT ABLAUT IN THE N-STEMS	 133
8.1 Kauffmann and nominal ablaut	133
8.2 The ablaut types	134
8.3 Methodology	137
8.3.1 Ablaut betrayed by Kluge's <i>Associationen</i>	138
8.3.2 Resolution of the Schwebeablaut	139
8.3.3 Over-long syllables in Upper German	140
8.4 Reconstructing the ablauting paradigm	142
 9 THE EVIDENCE	 147
9.1 The *e ~ *u type	147
*belkō, *bulk ^k az 'beam'	148
*brezdō, *burzdeni 'edge, board'	149
*dimbō, *dump ^p az 'haze'†	151
*drenō, *durraz 'drone'	152
*elm, *ulmaz 'elm (tree)'	155
*eudur, *ūdraz 'udder'	157
*finkō, *funk ^k az 'spark'†	159
*greubō, *gruppaz 'pot'	161
*helm, *hulmaz 'blade, cane, reed'?	162
*hemō, *humnaz 'heaven'	163
*hersō, *hurznaz 'brain'?	165
*hnekkō, *hnukkaz 'neck'	167
*hnellō, *hnullaz 'nape, crest'	169
*hrēhō, *hrugeni 'fish roe'?	171
*kelkō, *kulk ^k az 'jaw, throat'	172
*kernō, *kurnaz 'grain'?	174
*keudō, *kuttaz 'bag'	175
*klewō, *klunaz 'clew'	176
*klimbō, *klump ^p az 'lump, hillock'†	178

<i>*krebō, *kurp^paz</i> 'basket'	179
<i>*leuhmō, *l(a)uhmenaz</i> 'flash'	182
<i>*melhmō, *mulhnaz</i> 'cloud'?	183
<i>*melm, *mulmaz</i> 'sand'?	184
<i>*rehhō, *ruhhaz</i> 'ray'?	185
<i>*reumō, ?*rūm^maz</i> 'cream'	186
<i>*skinkō, *skunk^kaz</i> 'shank'	187
<i>*sterō, *sturraz</i> 'infertile animal'?	189
<i>*telgō, *tulk^kaz</i> 'twig'	190
<i>*timbō, *tump^paz</i> 'stub'	192
<i>*wekō, *wukkaz</i> 'wick'?	194
9.2 The *e ~ *a type	197
<i>*eulō / *aulō</i> 'hollow stalk'	197
<i>*fesō, *faznaz</i> 'fiber'†	199
<i>*hesō, *haznaz</i> 'hare'	200
<i>*kekō, *kaweni</i> 'jaw'?	202
9.3 The *a ~ *u type	207
<i>*barsō, *burznaz</i> 'perch'?	209
<i>*brahsmō, *bruhs(m)naz</i> 'bream'	210
<i>*dabō, *duppaz</i> 'puddle'?	211
<i>*galdō, *gult^taz</i> 'gelding'	212
<i>*lapō, *luttaz</i> 'shoot'	214
<i>*mapō, *muttaz</i> 'moth'	218
<i>*rapō, *ruttaz</i> 'rat'	221
<i>*swambō, *sump^paz</i> 'sponge, mushroom'?	223
<i>*tadō, *tuttaz</i> 'tuft'?	225
9.4 The *ī ~ *i type	228
<i>*bīō, *binaz</i> 'bee'	228
<i>*gīmō, *gim(e)naz</i> 'open space'?	231
<i>*hrīpō, *hrittaz</i> 'fever'	232
<i>*kībō, *kippaz</i> 'basket'	234
<i>*klīpō, *klittaz</i> 'burdock, tangle, clay'	235
<i>*rīhō, *rikkaz</i> 'stringing pole, line'	239
<i>*sīlō, *sillaz</i> 'strap, horse harness'†	243
<i>*skīō, *skinaz</i> 'shinbone'	244
<i>*skīmō, *skimenaz</i> 'shine'†	246
<i>*snībō, *snippaz</i> 'pointy nose, snipe'?	247
<i>*strīmō, *strimenaz</i> 'stripe, streak'?	250
<i>*swīmō, *swimenaz</i> 'dizziness'†	251
<i>*swīrō, *swirraz</i> 'neck, mooring-mast'	252
<i>*tīgō, *tikkaz</i> 'tick'	254
<i>*twīgō, *twikkaz</i> 'twig'?	256

<i>*wīō, *wiwaz</i> ‘kite’	258
<i>*wrihō, *wrigeni</i> ‘instep’	261
9.5 The <i>*ai ~ *ī</i> type	262
<i>*aikwernō / *īkwernō</i> ‘squirrel’	263
<i>*h(r)aignrō / *higrō</i> ‘heron’	266
9.6 The <i>*ū ~ *u</i> type	267
<i>*hrūhō, *hrukkaz</i> ‘pile’	268
<i>*hūfō, *huppaz</i> ‘heap’?	270
<i>*klūpō, *kluttaz</i> ‘clot’?	272
<i>*krūmō, *krumenaz</i> ‘crumb’?	274
<i>*kūpō, *kuttaz</i> ‘tuft’	275
<i>*mūhō, *mukkaz</i> ‘bunch’	278
<i>*mūhō, *mukkaz</i> ‘chunk’	279
<i>*pūpō, *puttaz</i> ‘pout’?	280
<i>*rūbō, *ruppaz</i> ‘caterpillar’	282
<i>*skūbō, *skuppaz</i> ‘brush’	283
<i>*stūfō, *stuppaz</i> ‘stub’†	286
<i>*pūmō, *pumenaz</i> ‘thumb’	287
<i>*pūhō, *pukkaz</i> ‘bag’?	289
<i>*pūsō, *pusnaz</i> ‘purse’?	291
<i>*snūfō, *snuppaz</i> ‘sniffing, cold’†	292
<i>*sprūtō, *spruttaz</i> ‘sprout’†	293
<i>*strūpō, *struppaz</i> ‘throat’†	294
<i>*strūtō / *prūtō, *struttaz / *pruttaz</i> ‘throat’?	295
9.7 The <i>*ū ~ *u ~ *a</i> type	297
<i>*knūbō, knuppaz</i> ‘knob’	297
<i>*knūpō, *knuttaz</i> ‘knot’	299
<i>*knūsō, *knuzzaz</i> ‘gnarl’	300
9.8 The <i>*ō ~ *a</i> type	301
<i>*gōmō, *ga(w)umnaz</i> ‘palate’?	302
<i>*hōdō, *hattaz</i> ‘hood’†	304
<i>*kōkō, *kakkaz</i> ‘cake’†	306
<i>*krōn, *kranaz</i> ‘crane’†	307
<i>*lōfō, *lappaz</i> ‘palm of the hand’	309
<i>*mōhō, *mageni</i> ‘poppy’	311
<i>*slōgō, *slakkaz</i> ‘sludge’†	314
<i>*skōgō, *skakkaz</i> ‘tip, brush’†	316
<i>*tōgō, *takkaz</i> ‘twig’	317
9.9 The <i>*ō ~ *ū</i> type	319
<i>*fōr, *funaz</i> ‘fire’	320
<i>*krōhō, *krūk^kaz</i> ‘jug’?	321
<i>*sōl, *sunnaz</i> ‘sun’	323

9.10 The *ē ~ *a type	324
<i>*dēbō, *dappaz</i> ‘paw’?	327
<i>*hēhō, *hakkaz</i> ‘hook’?	327
<i>*krēbō, *krappaz</i> ‘hook’?	329
<i>*krēgō, *krakkaz</i> ‘crook’?	331
<i>*snēgō, *snakkaz</i> ‘snake’?	333
 10 PSEUDO-ABLAUT	 335
10.1 Upper German	335
<i>*kredō, *krattaz</i> ‘basket’†	338
<i>*tebō, *tappaz</i> ‘tuft, knot, peg’†	341
<i>*skredō, *skrattaz</i> ‘demon’†	344
<i>*kredō, *kruttaz</i> ‘toad’†	346
10.2 West Norse	350
<i>*hnetō, *hnuttaz</i> ‘nut’†	350
 11 SUMMARY AND OUTLOOK	 353
11.1 Summary	353
11.2 Outlook	355
 BIBLIOGRAPHY	 357
Abbreviations	357
References	357
 INDEX OF CITED FORMS	 391

Preface

The initial goal of my Ph.D. scholarship was to investigate the influence of lost non-Indo-European languages on the Proto-Germanic lexicon. The idea was to tackle this issue with the help of a new methodology developed by Leiden linguists such as F.B.J. Kuiper, R. Beekes, P. Schrijver and D. Boutkan in the nineties of the former century. During the course of time, however, my dissertation gradually developed into a study of the Proto-Germanic *n*-stems and their typical morphology. The reason for this change of direction was that the most important formal criterion that had been used in order to isolate non-Indo-European words from the rest of the lexicon – the Proto-Germanic geminates – turned out to be significantly overrepresented in this morphological category.

The advocates of the Leiden Substrate Theory had defined the typical Germanic cross-dialectal interchange of singulate and geminate roots as the prime indicator of prehistoric language contact. For this reason, this substrate language had even been dubbed the “Language of the Geminates”. Yet, beside the fact that geminates were not at all distributed randomly across the vocabulary, as would be expected in the case of language contact, the interchanges proved to be far from erratic. In fact, they turned out to be strikingly predictable in nature. It thus became clear that the question of the Germanic substrate could not be solved without a more elaborate study of the morphology of the *n*-stems.

During my research, I have profited enormously from the knowledge and encouragements of many. I am much indebted to Harry Perridon, Arend Quak and Alexander Lubotsky for teaching and guiding me during my studies of Nordic, Germanic and Indo-European linguistics. I also wish to thank Frederik Kortlandt, Rick Derksen, Michiel de Vaan, Arjen Versloot and Patrick Stiles for their helpful comments on the manuscript. Not at least, my thanks go out to Luzius Thöny and to my *Doktorbrüder* Alwin Kloekhorst, Michaël Peyrot, Tijmen Pronk and Lucien van Beek.

List of abbreviations

Language abbreviations

Ang.	Anglian	Fri.	Frisian
Alb.	Albanian	G	German
Als.	Alsatian German	Gae.	Scottish Gaelic
App.	Appenzell Swiss	Go.	Gothic
Arm.	Armenian	Gr.	(Ancient) Greek
Av.	Avestan	Gutn.	Gutnish
Bav.	Bavarian German	Hess.	Hessian German
Bm.	Bokmål Norwegian	Hitt.	Hittite
Brab.	Brabantian Dutch	Holl.	Hollandish Dutch
Bret.	Breton	Hom.	Homer
BRu.	Byelorussian	Hsch.	Hesychius
Bulg.	Bulgarian	Icel.	Icelandic
Cimb.	Cimbrian German	Ir.	Irish
Crn.	Carinthian German	It.	Italian
Cz.	Czech	Ja.	Jaun Swiss
Da.	Danish	Lat.	Latin
Dor.	Dorian Greek	Latv.	Latvian
Du.	Dutch	LG	Low German
E	English	Limb.	Limbουργian Dutch
EDa.	Early (Modern) Danish	Lith.	Lithuanian
EDu.	Early (Modern) Dutch	Lus.	Luserna Cimbrian
EFri.	East Frisia Low German	M	Middle
EFri.	Early West Frisian	MDu.	Middle Dutch
EG	Early (Modern) German	ME	Middle English
Est.	Estonian	MHG	Middle High German
Far.	Faroese	MIr.	Middle Irish
Fi.	Finnish	MLat.	Middle Latin
Flem.	Flemish	MLG	Middle Low German
Fr.	French	Mo.	Modern
Fra.	Franconian	MRhnl.	Middle Rhinelandish

MW	Middle Welsh	Stw.	Stellingwerven Dutch
Myc.	Mycenaean Greek	Sw.	Swedish
NFri.	North Frisian	Swab.	Swabian German
Nn.	Nynorsk	Swi.	Swiss German
Nw.	Norwegian (Bm. & Nn.)	Thur.	Thuringian German
NRhnl.	North Rhinelandish	Tyr.	Tyrolean German
Nth.	Northumbrian	To.	Tocharian
O-	Old	Ukr.	Ukrainian
OCS	Old Church Slavonic	Val.	Valais/Wallis Swiss
ODa.	Old Danish	Visp.	Visperterminen Swiss
OE	Old English	W	Welsh
OFr.	Old French	Wall.	Walloon French
OFri.	Old Frisian	Wdh.	Wiedingharde Frisian
OGutn.	Old Gutnish	WFri.	West (Lauwers) Frisian
OHG	Old High German	WPhal.	Westphalian German
OIr.	Old Irish	WS	West Saxon
OLFra.	Old Low Franconian		
ON	Old Norse		
OPol.	Old Polish		
OPru.	Old Prussian		
OS	Old Saxon		
Osc.	Oscan		
OSw.	Old Swedish		
P-	Proto-		
Pal.	Palatinate German		
PBSl.	Proto-Balto-Slavic		
PCelt.	Proto-Celtic		
Pers.	Persian		
PGm.	Proto-Germanic		
PIE	Proto-Indo-European		
Rhnl.	Rhinelandish		
Rhtl.	Rheintal Swiss		
Ru.	Russian		
Sco.	Scottish		
SCr.	Serbian / Croatian		
SFri.	Saterlandic Frisian		
Skt.	Sanskrit		
Slov.	Slovene		

Linguistic abbreviations

acc	accusative
adj.	adjective
asg.	accusative singular
apl.	accusative plural
cf.	<i>confer</i>
c.	common
dat.	dative
dial.	dialectal
dsg.	dative singular
dpl.	dative plural
et al.	<i>et alii</i>
e.g.	<i>exempli gratia</i>
f.	feminine
ff.	<i>foliae</i>
fn.	footnote
gen.	genitive
gsg.	genitive singular
gpl.	genitive plural
ibid.	<i>ibidem</i>

i.e.	<i>id est</i>
inf.	infinitive
ins.	instrumental
loc.	locative
l.c.	<i>loco citato</i>
m.	masculine
n.	neuter
nom.	nominative
nsg.	nominative singular
npl.	nominative plural
obl.	oblique
obs.	obsolete
pl.	plural
poet.	poetical
pres.	present
pret.	preterit
ptc.	participle
s.v.	strong verb
s.v.	<i>sub voce</i>
sg.	singular
top.	toponym
v.	verb
viz.	<i>videlicet</i>
vs.	<i>versus</i>
w.v.	weak verb

Logical symbols

=	is (borrowed as)
<	developed from
>	developed into
⇒	served as basis for
⇐	was derived from
⇔	either ⇒ or ⇐
~	alternates with
:	contrasts with

Preliminary Remarks

The Germanic linguistic sources

Below follows an overview of the most important linguistic sources. The bulk of the material furnished in this monograph comes from the Northwest Germanic languages and dialects. The role of Gothic is more modest due to the simple fact that the evidence generally is more abundant in the Middle and Modern Germanic languages than in the oldest period.

North Germanic

For the Old Norse material, I have mainly used J. Fritzner's *Ordbog over det gamle norske sprog* (1886) at the website of Oslo University (www.edd.uio.no), and occasionally L. Heggstad's *Gamalnorsk ordbok* (1930). The English translations of the Old Norse forms are usually adopted from G.T. Zoëga's *Concise dictionary of Old Icelandic*.

The Modern Icelandic material is drawn from *Íslensk orðabók fyrir skóla og skrifstofur* (2nd ed., 1983) by Árni Böðvarsson and Ásgeir Blöndal Magnússon (eds.). For Faroese, I used M.A. Jacobsen's and Chr. Matras's *Føroysk-dönsk orðabók* (1927-1928), *Føroysk-ensk orðabók* (1985) compiled by G.V.C. Young and C.R. Clewer, and especially the new *Føroysk orðabók* (1998) by J.H.W. Poulsen (ed.).

The Old Swedish material is adopted from K.F. Söderwall's *Ordbok öfver svenska medeltids-språket* (1884), which is made available in database format by the University of Gothenburg (www.språkbanken.gu.se). For modern Swedish, I used *Svenska akademiens ordbok* (1997-2007), which has been digitalized by *Språkbanken* (spraakbanken.gu.se), and E. Hellquist's *Svensk etymologisk ordbok* (1922). All of the forms from the Swedish dialects are adopted from J.E. Rietz's *Svenskt dialektlexikon* (1872 [1962]), except for the Gutnish material, which is taken from *Ordbok över Laumålet* by M. Klintberg and H. Gustavsson (1895-1986).

The Early Danish material comes from O. Kalkar's *Ordbog over det eldre danske sprog* (1881-1907). Modern Danish forms were checked by using the online version of *Ordbog over det danske sprog* (1919-1956) at ordnet.dk/ods.

The Norwegian evidence is almost exclusively adopted from *Dokumentasjonsprosjektet* (www.dokpro.uio.no), which has published *Bokmålsordboka* (2005), *Nynorskordboka* (2006) and *Grunnmanuskriptet* (1935) on the internet. I have tried to simplify the complex formal variation in and between the two standard languages by citing as much as possible those forms that are accepted in both Bokmål and Nynorsk. These forms I have simply called *Norwegian* (Nw.). Relevant variants that exclusively occur in Nynorsk, including the material furnished by A. Torp in his *Nynorsk etymologisk ordbok* (1919), are labeled accordingly. The highly valuable dialectal material is extracted from *Grunnmanuskriptet*, which is the originally unpublished source manuscript of *Norsk Ordbok*. It contains a wealth of material that is not or no longer part of the Nynorsk standard language.

The etymological literature that has been consulted for the Nordic material primarily consists H.S. Falk & A. Torp, *Norwegisch-dänisches etymologisches Wörterbuch* (2nd ed., 1960), *Isländisches Etymologisches Wörterbuch* (1956) by A. Jóhannesson, J. de Vries's *Altnordisches etymologisches Wörterbuch* (1962) and R. Lühr's *Die Gedichte des Skalden Egill* (2000).

Anglo-Frisian

For Old English, I made use of Bosworth's and Toller's extensive *Anglo-Saxon dictionary* (1882-1972), F. Holthausen's *Altenglisches etymologisches Wörterbuch* (1934) and the *Dictionary of Old English Corpus* (1998), published at quod.lib.umich.edu/o/oec by the University of Toronto Center of Medieval Studies. For Middle English, I consulted the *Middle English Dictionary* by F. McSparran (ed.), which the same institute published online in 2001. Modern English forms as well as etymologies have been checked against the *Oxford English Dictionary* at dictionary.oed.com.

The Old Frisian material is collected from F. Holthausen's concise *Altfriesisches Wörterbuch* (1925), D. Boutkan's and S. Siebinga's *Old Frisian etymological dictionary* (2005) and the new *Altfriesisches Handwörterbuch* (2008) by D. Hofmann and A. Popkema. Modern West

Frisian forms were checked in J.W. Zantema's *Frysk Wurdboek* (1984). I have occasionally adduced evidence from Saterlandic Frisian as presented by P. Kramer in his *Düütsk-Seeltersk* glossary (1995), and from the North Frisian Wiedingharde dialect as provided by P. Jensen in his *Wörterbuch der nordfriesischen Sprache der Wiedingharde* (1927).

Low German

Only few Old Saxon (Old Low German) forms have been taken up. The material has been adopted from J.H. Gallée's *Vorstudien zu einem altniederdeutschen Wörterbuch* (1903), which despite its fallacies has proved to be a useful source and from E.H. Sehr's *Vollständiges Wörterbuch zum Heliand und zur altsächsischen Genesis* (1925). The recent publication of H. Tiefenbach's *Altsächsisches Handwörterbuch* (2010) has allowed me to check the cited forms, but I have not been able to include new material.

The Middle Low German data were subtracted from A. Lübber's & Chr. Walther's *Mittelniederdeutsches Wörterbuch* (1888 [1965]) and Schiller's and A. Lübber's *Mittelniederdeutsches Wörterbuch* (1875-1881). For Middle Dutch, I have used E. Verwijs and J. Verdam's *Middelnederlandsch handwoordenboek* (1973) as edited by C.H. Ebbinge Wubben. Cornelius Kilian's *Etymologicum Teutonicae Linguae* (1599) is the main source of the material included from Early Modern Dutch and its (neighboring) dialects.

I have made exhaustive use of the literature on Modern Dutch etymology and dialectology, so as to include material and insights that have remained unnoticed in Germanic studies. Material and etymologies were collected from *Woordenboek der Nederlandsche Taal* (1863-2001) as put online at www.wnt.inl.nl by the Instituut voor Nederlandse Lexicologie, N. van Wijk's *Franck's etymologisch woordenboek* (1912), J. Vercoullie's *Beknopt etymologisch woordenboek der Nederlandsche taal* (3rd ed., 1925), J. de Vries's and F. de Tollenaere's *Etymologisch woordenboek* (1983), and the new *Etymologisch woordenboek van het Nederlands* (2003-2009) by M. Philippa, F. de Brabandere and A. Quak (eds.), to which I have contributed myself. For the Dutch dialects, I have consulted a selection of sources, the most important of which are *Woordenboek van de Brabantse dialecten* (1967-2005), *Woordenboek van de Drentse dialecten* (1996-2000), *Woordenboek van de Limburgse dialecten* (1983-), *Woordenboek van de Vlaamse dialecten* (1979-),

Woordenboek der Zeeuwse dialecten (1964) and A.A. Weijnen's *Etymologisch dialectwoordenboek* (1996).

High German

The Old High German evidence is obtained from a variety of sources: E.G. Graff's antiquated, yet still useful *Sprachschatz oder Wörterbuch der althochdeutschen Sprache* (1834-1846), E. Siever's & E.E. Steinmeyer's *Die althochdeutschen Glossen* (1879-1923), R. Schützeichel's *Althochdeutsches Wörterbuch* (1969), T. Starck's and J.C. Well's *Althochdeutsches Glossenwörterbuch* (1972-1990), and Schützeichel's new *Althochdeutscher und Altsächsischer Glossenwortschatz* (2004). Etymological information has been retrieved from *Etymologisches Wörterbuch des Althochdeutschen* by A. Lloyd, O. Springer and R. Lühr (1988-).

For Middle High German, I have used M. Lexer's *Mittelhochdeutsches Handwörterbuch* (1872-1878) and, to a lesser extent, *Mittelhochdeutsches Wörterbuch* (1854-1866) by G.F. Benecke.

The most important source for the Modern High German material is *Deutsches Wörterbuch* (1854-1960) by J. and W. Grimm, which has been put online by the University of Trier. For etymological purposes, I have used *Etymologisches Wörterbuch* by F. Kluge and W. Mitzka (20th ed., 1967) and the most recent edition (24th ed., 2002) by E. Seebold. For the German dialects, I have primarily made use of *Bayerisches Wörterbuch* (1872-1877) by J.A. Schmeller and K. Frommann, *Pfälzisches Wörterbuch* (1965-1997) by E. Christmann et al., *Rheinisches Wörterbuch* (1923-1971) by J. Franck and J. Müller (eds.), *Schwäbisches Handwörterbuch* (1999) by H. Fischer and H. Taigel, *Wörterbuch der Elsässischen Mundarten* (1899-1907) by E. Martin and H. Lienhart, *Wörterbuch des deutsch-lothringischen Mundarten* (1909) by M.F. Follmann. The Swiss German material was taken from a selection of the *Beiträge zur Schweizerdeutschen Grammatik* (1910-), and not from *Schweizerisches Idiotikon*, because the lexicon is often difficult to analyze without the help of the descriptive grammars in question. Finally, I have incorporated some valuable material from Schmeller's and Bergmann's *Cimbrisches Wörterbuch* (1855) of the South Bavarian dialects in Italy and from *Wörterbuch der deutschen Sprachinselmundart von Zarz/Sorica und Deutschrut/Rut in Jugoslawien* (1983) by E. Kranzmayer and P. Lessiak.

Normalization and orthography

The orthographical representation of the material from the different languages has roughly been kept in accordance with the dominant conventions. This has the advantage that the legibility of the material is optimized, and the disadvantage that a certain amount of arbitrariness is imported. As a result, for instance, vowel length is marked by an acute in Old Norse, by doubling of the vowel in the Swiss dialects, and by a macron in most of the other languages, including Old English and the German dialects.

The spelling of the Old High German material is problematic, because the source dialects differ in their treatment of the Proto-Germanic stops. I have rather idiosyncratically normalized the Old High German forms according to the Low Alemannic *Lautstand*, because of the important role of this dialect group in the sources. Affricated *p*, *t* and *k* are represented as <pf>, <tz> and <ch>, the corresponding fricatives as <f>, <s> and <h> when short, and as <ff>, <zz> and <hh> when long. The continuants of PGm. **b*, *d* and *g* are represented as , <t> and <g>, <d> being reserved for the voiced stop continuing PGm. **þ*. The geminated variants are spelled <pp>, <tt> and <cc>. The outcome of long **þ* shifts from <dd> to <tt> within the Old High German period, and is indicated accordingly.

Presentation of the evidence

Throughout this monograph, the reader will encounter paradigms that are reconstructed on the basis of large clusters of different formations from a variety of North, East and West Germanic dialects, ranging from Gutnish to Flemish, from Faroese to Cimbrian. In order to present the data as clearly as possible, the material is ordered in the following way. First, the reconstructed Proto-Germanic paradigm is given in bold. Then, the different sub-reconstructions on which the main paradigm is based are given in indented lines, each different sub-reconstruction receiving a separate horizontal level. Formations that are derived from a sub-reconstruction are preceded by a $\Rightarrow$ sign. They are only indented when the derivation did not take place in the same dialect, but at an earlier stage. Language-internal derivations are given between brackets. Loanwords are presented in the same way and put directly behind the

source language. To separate the sub-reconstructions pertaining to the proto-paradigm from more indirectly related cognates, a long, horizontal bar is sometimes inserted.

The order in which the involved languages are given is determined with the help of two criteria, i.e. 1) dialectal affiliation and 2) linguistic archaicity. By the first criterion, the dialects are positioned between their closest relatives, resulting in a dialectal chain Gothic, Nordic, Anglo-Frisian, Low German, High German. In accordance with the second criterion, the more archaic dialects override the less archaic dialects. This means that, when, for instance, no Old Saxon form is attested, an Old High German attestation precedes a Middle Low German. Similarly, any Icelandic evidence always precedes an Old English attestation, because Icelandic is comparable to Old Norse when it comes to archaicity.

In the described format, the reconstruction of the Proto-Germanic paradigm of the word for 'tooth' (see Schaffner 2001: 627 ff), deriving from PIE **h₃d-ónt*, gen. **h₃d-nt-ós*, would appear as follows:

****tan, *tundaz 'tooth'***

- **tan(p)-*: ON *tǫnn*, pl. *teðr*, *tennr* f. 'id.', Icel. *tönn* f. 'id.', Far. *tonn* f. 'id.', OE *tōþ*, pl. *tēþ* m. 'id.', OFri. *tōth* m. 'id.', OHG *zan(t)* m. 'id.', MHG *zan(t)*, pl. *zende* m. 'id.', G *Zahn*, MLG, MDu. *tant* 'id.', Du. *tand* 'id.'
 ⇒ **tanþjan-*: Icel. *tenna* 'to give teeth', OE *tæðan* 'id.', MHG *zenden* 'id.'
 - **tunþu-*: Go. *aiþva-tunþus* 'thornbush'
 ⇒ **tunþska-*: OE *tux*, *tusc* m. 'tusk', OFri. *tosk*, *tusk* m. 'tooth', WFri. *tosk* 'id.'
-
- **tinda-*: ON *tindr* m. 'peak', OE *tind* 'jag, nail', MHG *zint* 'jag, merlon'

1 Introduction

The *n*-stems are one of the more fascinating inflectional categories in Proto-Germanic morphology. Whereas other nouns, such as the *a*- and *ō*-stems, show great uniformity throughout the Germanic dialect area, the *n*-stems usually exhibit a whole range of dissimilar root forms. Typically, even within the North and West Germanic continuums, neighboring dialects exhibit different roots for one and the same *n*-stem.

The most common type of variation found in the dialects consists of the root-final consonantism shifting between single and double stops. It can be observed in hundreds of both masculine and feminine *n*-stems. The following cases may exemplify this:

- Swi. Visp. *toxxa* f. 'doll' < **dukōn*-, ON *doki* m. 'shred, strip' < **dukan*- : ON *dokka* f. 'windlass', OHG *tocha* f. 'doll', Far. *dokki* m. 'wad of wool' < **dukkan*-, -*ōn*-
- Go. *fauho* f. 'vixen' < **fuhōn*- : OE *fogge* f. 'id.' < **fuggōn*-
- Icel. *hjari* m. 'hinge' < **heran*- : ON *hjarri* m. 'id.', OE *hearra* m. 'id.' < **herran*-
- NFri. *nope* 'flock of wool' < **hnupōn*- : MLG, MDu. *noppe* f. 'id.' < **hnuppōn*- : MLG *nobbe* f. 'id.' < **hnubbōn*-
- OE *piða* m. 'pith' < **piþan*- : EDu. *pitte* 'medulla arboris' < **pittan*-/**piþþan*-
- OE *pohha* m. 'bag' < **puhhan*- : ON *poki* m. 'id.' < **pukan* : OE *pocca* m. 'id.' < **pukkan*-
- MDu. *rogen* m.pl. 'supplies, rye', MHG *roge* m. 'rye' < **rugan*- : MDu., MHG *rogge* m. 'id.' < **ruggan*-
- OFri. *stera* m. 'star' < **steran*- : OE *steorra*, OHG *sterro* m. 'id.' < **sterran*-
- G *Truhe* f. 'trough' < **pruhōn*- : Swi. *trukxa* f. 'box, trunk' < **prukkōn*-

The second type of root variation is of vocalic nature. It is much less frequent, but still the number of instances amounts to dozens. Vowel

alternations typically occur in combination with the consonant alternations described above. As a result, the affected etyma may display a bewildering set of root variants:

- MHG *krebe* m. 'basket', SFri. *krääf* m. 'id.' < **kreban*- : MHG *krebbe* f. 'id.' < **krebbōn*- : MHG *kruppe* f. 'id.' < **krubbōn*- : MHG *krupfe* f. < **kruppōn*- : MHG *korb(e)* < **kurba(n)*-
- G *Zimpe(n)* m. 'tip, nozzle' < **timban*- : MLG *timpe* m. 'id.' < **timpan*- : OHG *zumpo* m. 'penis' < **tumban*- : Du. dial. *tump(e)* 'tip, corner' < **tumpan*-
- Du. dial. *tijg* 'tick' < **tīgan*- : E obs. *tyke* 'id.' < **tīkan*- : Du. *teek* 'id.', Swi. Visp. *zäxxo* m. 'id.' < **tikan*- : G *Zecke* f. 'id.' < **tikkōn*-
- G *Reihen* m. 'insep' < **wrihan*- : MDu. *rjghe* 'id.' < **wriġan*- : Du. obs. *wreeg* 'id.' < **wriġan*- : Du. dial. *wree* 'id.', Swi. Ja. *reəhə* m. 'id.' < **wrihan*-
- OHG *zuogo*, OS *tōgo* m. 'branch' < **tōgan*- : Du. dial. *toeke* 'id.' < **tōkan*- : MLG *tagge* 'id.' < **taggan*- : MLG *tack(e)*, MDu. *tac(ke)* 'id.' < **takkan*-
- Icel. *hró* 'hillock' < **hrūha*- : ON *hrúga* f. 'pile' < **hrūgōn*- : Icel. *hrúka* f. 'id.' < **hrūkōn*- : MDu. *roc* m. 'id.' < **hrukka*- : ON *hroki* m. 'id.' < **hrukan*-

The main goal of this study is to investigate the nature and origin of the vowel alternations exhibited by the *n*-stems. This cannot be done, however, without a thorough understanding of the much intertwined consonant alternations. Therefore, the first sections of this book consists of a theoretical discussion of the Germanic consonantism. After that, the focus is shifted towards the vocalism.

This monograph contains a total of eleven chapters, including this short introduction. In chapter 2, a brief outline is presented of the Proto-Indo-European and Proto-Germanic inflection of the *n*-stems. Chapters 3 and 4 treat the origin of the geminates and consonant alternations that are displayed by the *n*-stems. The chapters basically are a defense of the Assimilation Theory, which was developed by the Neogrammarians, most notably F. Kluge, in the late 19th century. I have further included a discussion on the typologically similar alternations that are found in the directional adverbs and in the iterative verbs. This is the focus of chapter

5 and 6. Alternative theories regarding the Germanic consonantism, i.e. the Expressivity Theory and the Leiden Substrate Theory, are reviewed in chapter 7.

The chapters 8, 9 and 10 treat the vowel alternations that are found in the *n*-stems. A theoretical discussion of the origin and evolution of the alternations is presented in chapter 8. In brief, it is argued that the alternations evolved out of the root ablaut that is traditionally reconstructed for the Proto-Indo-European paradigm of the *n*-stems. Chapter 9 is the most important and longest chapter of all, and consists of a discussion of the evidence for this ablaut. Chapter 10 can be regarded as an excursion: it offers a solution to the “pseudo-ablaut” that is displayed by a number of *n*-stems in the Upper German dialects.

Finally, the main conclusions of this study are summarized in chapter 11. The chapter also includes a short outlook on further research.

2 The inflection of the *n*-stems

In order to be able to investigate the consonant and vowel alternations of the Germanic *n*-stems, it is necessary to give a brief outline of the inflection of this category in the Indo-European and the Germanic proto-languages.

In § 2.1, the reader will find a concise presentation of the different *n*-stem inflections that are reconstructed for Proto-Indo-European. The *n*-stems, just as most other nouns in the Indo-European parent language consisted of a root, a suffix and an ending. (The exception are the root nouns, which only have root and an ending.) Nominal inflection was created with the help of eight cases, the nominative, vocative, accusative, genitive, dative, ablative, locative, and instrumental, all with different endings.

The cases were also characterized by a specific distribution of the ablaut and the accent. The Proto-Indo-European ablaut consisted of a systematic vowel shift of **e*, **o* and *zero* on the morpheme level; the occurrence of the ablaut grades in the root, the suffix or the ending differed from case to case, and thus formed a specific inflectional pattern. The origins of the different inflectional patterns are discussed in § 2.2.

In § 2.3, an overview is given of the inflection of the Proto-Germanic *n*-stems. The main differences between Proto-Indo-European and Proto-Germanic morphology are 1) the marginalization of the ablaut of the root, and 2) the reduction of the number of cases from eight to five, i.e. the nominative, accusative, genitive, dative and instrumental. As a result, nominal inflection chiefly became a matter of suffixes and endings in the attested Germanic languages. The form of these elements is examined from the Proto-Indo-European perspective in § 2.3.

The retention of the ablaut of the root by a relatively small number of *n*-stems is the main focus of this book, and will be discussed elaborately in the chapters 8 and 9. The present chapter primarily treats the ablaut that is found in the suffixes and the endings.

2.1 The Indo-European *n*-stems

In the earliest phase of Proto-Indo-European, the *n*-stems had paradigms in which the stressed full grade shifted between the root, the suffix and the ending. However, this primeval ablaut pattern seems to have split up into several different sub-types in late Proto-Indo-European by the leveling of either the ablaut or the accent (Beekes 1985: 154ff, 1995: 193ff). In the oldest languages, the most important sub-types thus became 1) the amphikinetic type, 2) the hysterokinetic type and 3) the proterokinetic type. The former two types were associated with the Indo-European masculines and feminines, the latter with the neuter.

2.1.1 The amphikinetic type

In Proto-Indo-European, most masculine or feminine *n*-stems had an amphikinetic inflection. The ablaut of the root is usually no longer visible in the daughter languages, but it can nevertheless be retrieved from the paradigm of the Sanskrit *mn*-stem *ātmā*, gen. *tmānas*, loc. *tmān(i)*¹, acc. *ātmānam* 'breath, soul' < **h₁eh₁t-mōn*, **h₁h₁t-mn-os*, **h₁h₁t-men(-i)*, **h₁eh₁t-mon-m*. The ablaut of the suffixes and the endings can further be reconstructed on the basis of the historical paradigms of, for instance, the Sanskrit, Lithuanian and Germanic *n*-stems:

	PIE	Skt.	Lith.	Go.
nsg.	<i>*CéC-(m)ōn</i>	<i>rājā</i> 'king'	<i>akmuō</i> 'stone'	<i>guma</i> 'man'
gsg.	<i>*CC-(m)n-és</i>	<i>rājñas</i>	<i>akmeñs</i>	<i>gumins</i>
asg.	<i>*CéC-(m)on-m</i>	<i>rājānaṃ</i>	<i>ākmenį</i>	<i>guman</i>
lsg.	<i>*CC-(m)én-i</i>	<i>rājan(i)</i>	<i>akmenyjė</i>	<i>gumin</i>
npl.	<i>*CéC-(m)on-es</i>	<i>rājānas</i>	<i>ākmenys</i>	<i>gumans</i>
gpl.	<i>*CC-(m)n-óm</i> ²	<i>rājñām</i>	<i>akmenų</i>	<i>gumane</i> ³
apl.	<i>*CC-(m)n-ís</i>	<i>rājñas</i>	<i>ākmenis</i>	<i>gumans</i>
lpl.	<i>*CC-(m)n-mis</i>	-	<i>akmenims</i>	<i>gumam</i>

¹ The genitive *tmānas*, which replaces expected ***tanás* < **h₁h₁t-mn-ós*, is based on the locative (cf. Schaffner 2001: 518).

² It was demonstrated by Kortlandt (1978; 2007) that Lith. gpl. *-ų*, OCS *-ъ* and Skt. *asmākam* 'ours' point to a PIE gpl. ending **-om* rather than **-ōm*, the latter representing **-oHom* from the *o*-stems.

³ Go. gpl. in *-e* continues the *i*-stem ending **-ejan* from **-ei-om* (Kortlandt 1978).

As can be observed, the amphikinetic type is characterized by a lengthened grade ending **-ōn* in the nominative (cf. Skt. *-ā*, Gr. *-ω(ν)*, Lat. *-ō*, Lith. *-uo*, OCS *-y*), *-n-ós* in the genitive, and **-on-m* in the accusative. The nominative ending probably lost the nasal in PIE times already (cf. Melchert 1983: 10), as is clear from the endingless nominatives in Sanskrit and Latin, and from the Greek transfer of certain *n*-stems into the *oi*-stems, e.g. ἀηδὼ(ν) f. 'nighting-gale', εἰκὼ(ν) f. 'image' etc.⁴

The amphikinetic type contains two sub-categories, viz. 1) plain *n*-stems, cf. Gr. ἄκμων m. 'anvil'; ἄξων m. 'axle'; βλήχων f. 'pennyroyal'; βραχίων m. 'lower arm'; Lat. *carō*, *carnis* m. 'meat'; Gr. κίων mf. 'pillar'; μήκων f., OSw. *val-mōghe* m. 'poppy'; Gr. πλεύμων, πνεύμων, Lat. *pulmō* 'lung'; Gr. κύων, κυνός mf. 'dog, bitch', Skt. *śnā́*, *śúnah* m. 'dog'; and 2) individualizing *n*-stems, either of deverbative or denominative origin, cf. Gr. δαίμων mf. 'demon'; εἴρων mf. 'dissembler'; γείτων mf. 'neighbor'; Lat. *Nāsō* 'the Nose'; Go. *staua* m. 'judge'; Gr. Στράβων 'the Blind one'; τέκτων, Skt. *tákṣan-* m. 'carpenter'; Lat. *virgō*, *-inis* f. 'girl'; Gr. φλέδων mf. 'chatterer' etc.

Some of the individualizing *n*-stems have been reconstructed with the allegedly possessive Hoffmann suffix **-Hon-* that was added to *o*-stems. Thus, the paradigm of e.g. Av. *mąθrā*, gen. *mąθrānō* 'prophet' (< **-oHōn*, **-oHn-os*) has been explained as a derivation from *mąθra-* 'thought' < **mn-tro-* (Hoffmann 1955). Not all individualizing *n*-stems were derived from *o*-stems, however. A clear counter-example is the word for 'man', which on the basis of Lat. *homō* (< OLat. *hemō*), Lith. *žmuō* and Go. *guma* m. 'id.' can be reconstructed as the *n*-stem **d^h(é)ǵ^hm-ōn*, gen. **d^hǵ^hm-n-ós*, acc. **d^hǵ^hm-ón-m* ('earthling'?)⁵ It was clearly derived from the old *m*-stem PIE **d^héǵ^h-m*, **d^hǵ^h-m-ós* 'land', cf. Hitt. *tēkan*, gen. *taknaš* n. 'id.' and therefore does not qualify as a Hoffmann formation.⁶ The distinction between the plain and individualizing amphikinetic *n*-stems consequently remains questionable.

⁴ Harðarson (2005: 220): "Dieser Metaplasma setzt den Zusammenfall der *oi*- und *n*-Stämme wenigstens in einer Form voraus, und das kann nur der Nominativ gewesen sein".

⁵ The full grade of the root is by no means ascertained. Lith. *žmuō* straightforwardly continues the nominative **d^hǵ^hm-ōn*. Lat. *homō* < OLat. **hemo* as well as PGm. **guman-* may have generalized the vocalized **m* from the oblique **d^hǵ^hm-n-*. There is no need to invoke Lindeman's law (**d^hǵ^hm-ōn*) in order to explain this vocalization.

⁶ Thus Nussbaum 1986: 187 fn. 6.

2.1.2 The hysterokinetic type

The hysterokinetic type is characterized by an inflection that had a nominative in **-én* (Skt. -*én*, Gr. -*ήν*, Lat. -*én*), a genitive in **-n-ós* and an accusative in **-én-m*. Some hysterokinetic *n*-stems have an *o*-grade in the root, e.g. OCS *korę* m. 'root' < **kor-én-*, Av. *karapan-* 'priest' < **kolp-én-* (or **kolp-o-Hon-?*),⁷ but the far majority appears with a zero-grade root throughout the paradigm (cf. Rix 1976: 145). This is the logical result of the fact that the stress, and therefore also the full grade, was realized in the ending.

PIE	Skt.	Gr.
nsg. <i>*CC-(m)én</i>	<i>ukṣá</i> 'bull'	πυθμήν 'bottom'
gsg. <i>*CC-(m)n-ós</i>	<i>ukṣṇás</i>	πυθμένος
asg. <i>*CC-(m)én-m</i>	<i>ukṣánam</i>	πυθμένα
lsg. <i>*CC-(m)én-i</i>	<i>ukṣan(i)</i>	πυθμένι
npl. <i>*CC-(m)én-es</i>	<i>ukṣánas</i>	πυθμένες
gpl. <i>*CC-(m)n-óm</i>	<i>ukṣnám</i>	πυθμένων
apl. <i>*CC-(m)n-ís</i>	<i>ukṣnás</i>	πυθμένας
dpl. <i>*CC-(m)n-mis</i>	-	-

As opposed to the amphikinetic *n*-stems, the hysterokinetic *n*-stems are a relatively small group. They predominantly consist of primary formations of the masculine gender, e.g. Gr. ἀδήν mf. 'gland' < **ng^w-én-*; ἀρήν m. 'lamb' < **urh₁-én-*; αὐχίν (Hsch. ἀμφήν) m. 'neck, throat; isthmus' < **h₂ng^{wh}-én-*⁸; Skt. *mūrdhán-* m. 'head', OE *molda* m. 'top of the head' < **mlHd^h-én-*; ποιμήν m. 'herd' < **ph₂-oi-mén-*⁹; πυθμήν m. 'bottom' < **b^hud^h-mén-*; σπλήν, Skt. *plīhán-*, Lat. *liēn* m. 'spleen'; Skt. *ukṣán-*, Go. *auhsa* m. 'bull' < **uks-én-*; Gr. ὑμήν m. 'film' < **siuH-mén-*; Gr. ἄρσσην, -ενος m. 'male' < **urs-én-* etc.¹⁰

⁷ For Germanic *n*-stems with *o*-grades in the root, see § 9.3.

⁸ Gr. αὐχίν developed out of ἀμφήν < **h₂ng^{wh}-én-* by dissimilation of the *m* against the ending (Pronk forthc.).

⁹ Gr. ποιμήν, Lith. *piemuō* m. 'herd' is usually interpreted as the *o*-grade form **poh₂i-men-* (thus Rix 1976: 145). It is more likely that the PIE form was **ph₂-oi-men-*, containing the stem of an *i*-present **ph₂-oi-*, cf. Hitt. 3sg. *dāi*, 3pl. *ti(i)anzi* 'to put' < **d^hh₁-ói-ei*, **d^hh₁-i-énti* (as reconstructed by Kloekhorst 2008: 806).

¹⁰ Gr. Σειρήν f. 'Siren' has no etymology and χήν mf. 'goose' is a secondary *n*-stem from **g^héh₂n-s-*.

2.1.3 The proterokinetic type

The neuter *n*-stems are rare in the Indo-European languages, and hence reveal only few traces of their original inflection. The ablaut pattern can nevertheless be safely reconstructed on the basis of the neuter *mn*-stems, which are abundant throughout the Indo-European dialects. A notorious example of an *mn*-stem with full paradigmatic ablaut is **h₃néh₃-mn*, **h₃nh₃-mén-s* ‘name’ (Beekes 1995: 186), which is nowhere attested as such, but can readily be reconstructed on the basis of the opposition of e.g. Skt. *náman-* < **h₃néh₃-mn* vs Gr. ὄνομα, OCS *imę*, OIr. *ainm*, Go. *namo* < **h₃nh₃-mén*.¹¹ The plural of the neuter proterokinetic stems was probably inflected as a collective of the type Hitt. *uātār* sg. < **uod-r* : *widār* pl. < **ud-ōr* (= Gr. ὄδωρ), in early PIE (cf. Streitberg 1896: 258). This is supported by e.g. Skt. *nāmāni*, which may consist of the ending **-ōn* plus **-h₂* (Harðarson 1987a: 96; Beekes 1995: 187). The laryngeal is also found in Lat. *nōmina* and Go. *namna*, but these forms have a different vowel grade in the suffix, i.e. **h₃n(e)h₃-mn-(e)h₂*.¹²

	PIE	Lat.	OIr.	Go.
nasg.	<i>*CéC-(m)n</i>	<i>nōmen</i>	<i>ainm</i>	<i>namo</i>
gsg.	<i>*CC-(m)én-s</i>	<i>nōminis</i>	<i>anm(a)e</i>	<i>namins</i>
napl.	<i>*CéC-(m)ōn</i> ¹³	<i>nōmina</i>	<i>anman(n)</i>	<i>namna</i>
gpl.	<i>*CC-(m)én-om</i>	<i>nōminum</i>	<i>anman(n)</i>	<i>namne</i>

The neuter *n*-stems are so rare, that their existence in the IE proto-language is generally considered to be doubtful (cf. Rasmussen 1999, 1: 147). In most cases, the neuter *n*-stems seem to have replaced earlier heteroclitc paradigms, i.e. the *r/n-* or *ur/un-*stems, cf. Av. *azan-* ‘day’ vs Skt. *áhar*, *áhnas* ‘id.’ < **(d^h)eǵ^h-r/n-* or *zruuan-* ‘life-time’ < **g^w(e)rh₂-un-* vs *zaurura-* ‘senile’ < **g^werh₂-ur-o-* (Lubotsky 1998). This may also be the

¹¹ MHG *nüemen*, MLG *nōmen*, MDu. *noemen* < **nōmjan-* is also to be derived from the full grade in the root (cf. Uhlenbeck 1896: 109), but this full grade may have been induced by the causative formation, i.e. **h₃noh₃m(n)-eie-*.

¹² Note that Go. *namna* must be an innovation anyway, because the proto-form **h₃nh₃-mn-eh₂* would have regularly yielded **numna*. The root **nam-* is either from the lsg. **h₃nh₃-mén-i*, dpl. **h₃nh₃-mṇ-mis* or from the plural **h₃neh₃-mn-éh₂* itself by pretonic shortening (cf. Petit 2004: 62).

¹³ The ending *-(m)n-eh₂* as found in Gothic is an innovation (Beekes 1995: 187).

case for many names for body parts, since heteroclisys is widespread in this semantic class, cf. Skt. *úddhar*, *úddhnas* n. 'udder' < **h₁uHd^h-r/n-* or Lat. *femur*, *feminis* n. 'thigh'. However, not all heteroclitic terms for body parts had an *r* in the strong cases, cf. Skt. *śíras*, *śírṣṇás* n. 'head' < **krh₂-os*, **krh₂-s-n-ós*; *ákṣi*, *akṣnás* n. 'eye' < **h₃ek^ws(H)*, **h₃ek^w-s-n-ós*; *ásthi*, *asthnás* n. 'bone' < **h₃estH-i*, **h₃estH-n-ós*, and this might indicate that the *n*-suffix was more or less independent. Anyhow, the *n*-stems – including the neuters – are clearly associated with body parts throughout the Indo-European languages, cf. Lat. *inguen* n. 'loin' (cf. Gr. ἄδην, -ένοϋς m. 'gland', ON *økk* m. 'tumor') < **(h₁e)ng^w-n-*, Lat. *sanguen* n. 'blood', Av. *āṇhan-* n. 'mouth' < **h₁eh₃-s-n-*. The Pan-Germanic neuter *n*-stems, such as **augōn-* 'ear', **ausōn-* 'ear', and **hertōn-* 'heart', are an especially clear illustration of this.

The *n*-suffix certainly functioned as a deverbal stem formant, as becomes clear from e.g. Lat. *unguen* n. 'fat', OIr. *imb*, gen. *imbe* n. 'butter' (OHG *ancho* m. 'id.') < **h₃ng^w-n-* to Skt. *ánakti* 'to smear' < **h₃eneg^w-*. In Anatolian, the suffix was still productive, cf. Hitt. *henkan-* n. 'death, doom, plague', *kuššan-* n. 'pay', *lagan-* n. 'inclination', *mūdan-* n. 'pig-food', *šaḥḥan-* n. 'service to the a land owner', *šāk(a)n-* n. 'oil, fat', *takšan-* n. 'centre, joint, combination'. This is further demonstrated by the inclusion of the original *m*-stem *tēkan*, *taknaš* 'earth' (PIE **d^hég^h-m*, **d^hg^h-m-ós*) into this category. Avestan, too, offers a number of instances, but the suffix is not purely deverbal in this language, cf. *masan-* n. 'greatness' < **mek^h-n-*, *mazan-* n. 'size' < **meg^h-n-*, *sāx^han-* n. 'teachings' < **keh₂-un-*, *sraiiian-* n. 'beauty' < **krei^h-n-*, *usan-* n. 'wish' < **uk^h-n-*, *vaṇhan-* 'goodness' < **h₁ues-n-*, *zauuan-* n. 'call' < **g^heuH-n-*. In these cases, the *n*-suffix is more or less isofunctional with the *ur/n-* and *mn-* suffixes, and this again means that there may be a historical link with these elements. It is still possible, however, that the *n*-suffix was derivationally independent in the parent language itself. This possibility is supported by the parallelism with Germanic, where the weak neuters ON *bjúga* n. 'sausage', *leika* n. 'doll' and Nw. *drekk* n. 'drink' seem to have been derived directly from the verbs *bjúga* 'to bend', *leika* 'to play' and *drekk* 'to drink'.

Finally, there are some remaining neuters that can represent original *n*-stems. These are Lat. *gluten* 'glue' and *pollen* 'mill dust, fine flour', Hitt. *eḫan-* 'some (evergreen) tree' < **iei-n-* (cf. ON *einir*, Lat. *iūniperus*), *eḫan-* 'some type kind of grain' < **ieu-n-* (cf. Skt. *yáva-* 'grain,

barly') and possibly also *karza(n)*- 'spool, bobbin'.¹⁴ In Germanic, ON *hnoða* n. 'clew' < **ǵnu-t-ōn-*, Go. *kaurno*, Nw. *korna* n. 'grain' < **ǵrhz₂n-ōn-* and notably Go. *barnilo* n. 'child' are in support of primary neuter *n*-stems. The category seems to have denoted profoundly non-agentive entities, as will be demonstrated in § 2.3.3.

2.2 Origins of the inflectional types

The historical relation between the ablaut of the different inflectional ablaut types has been clarified by Beekes in *The origins of the Proto-Indo-European nominal inflection* (1985). Beekes's explanation revolved around the observation that the Proto-Indo-European *e* and *o* grades are at least partially in complementary distribution: while *e* occurs under the stress more often than not, *o* is typically encountered in unstressed position, cf. Gr. πατέρα : εὐ-πάτορα. To account for this morphophonemic distribution, Beekes argued that *o* had developed out of unstressed *e* at some point in Pre-Proto-Indo-European.

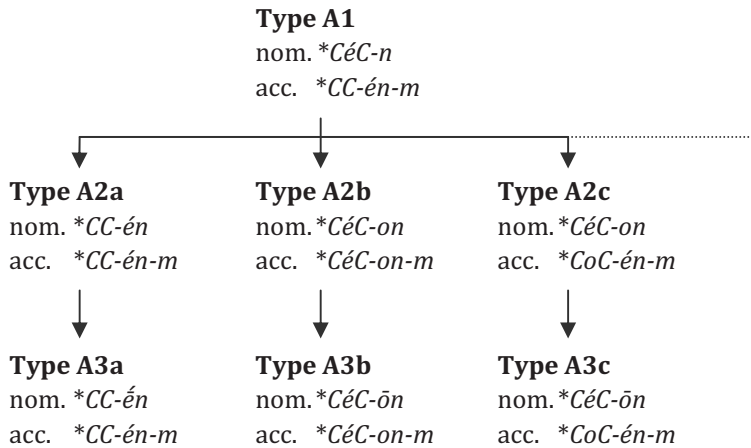
Beekes envisioned three different stages. In the oldest stage (A1), the full grade and the accent still coincided: when a syllable was stressed, it automatically received an *e*-grade. In the second stage (A2), the full grade analogically spread to unstressed syllables. Under those circumstances, it developed into *o*. In the final stage (A3), the *e*-grade again spread to unstressed syllables, but was no longer modified into *o*.

Beekes's diachronic analysis of the PIE ablaut put the correlation between the amphikinetic and hysterokinetic inflectional types in a different light. In the oldest Indo-European dialects, the strong cases of the amphikinetic type had stressed **e* in the root and unstressed **ǝ* in the suffix, cf. Skt. *rājā*, *rājānam* < **Hréh₁ǵ-ōn*, **Hréh₁ǵ-on-m*. The hysterokinetic type had a zero grade in the root and stressed **ě* in the suffix, cf. *ukṣā* : *ukṣānam* < **uks-én*, **uks-én-m*.¹⁵ According to Beekes, this difference came about by the split of a more primitive paradigm **CéC-n*, acc. **CC-én-m*: the hysterokinetic type arose by the generalization of the full grade of the suffix in stage A1. It resulted into a paradigm **CC-én*, **CC-én-m*. The amphikinetic type, on the other hand, generalized the *o* of

¹⁴ The word has been reconstructed as **krt-s*, **krt-s-n-os* (Rieken 1999: 391) and as **kert-sör* (Eichner 1974: 98; Kloekhorst 2008: 459).

¹⁵ Cf. Schindler 1976; Beekes 1985; Schaffner 2001:516f.

the suffix, which had arisen from unstressed **e*, viz. *CéC-on*, **CéC-on-m*. At a final stage, the vowels of the word-final nominative endings **-en* and **-on* were lengthened. This changed them into the attested forms **-ēn* and **-ōn*.



It is vital to realize that the amphikinetic and hysterokinetic types are only two of the possible modifications of the original paradigm **CéC-n*, **CC-én-m*. Several other types may have arisen at various stages.¹⁶ A variant **CéC-n*, **CoC-én-m*, for instance, could theoretically have been created by the introduction of unstressed *e* in the root of the accusative. Conversely, a variant **CoC-én*, **CC-én-m* can be explained from the analogical introduction of stressed *e* in the suffix of the nominative.

The proterokinetic inflection is somewhat isolated from the amphikinetic and hysterokinetic type. This is most probably the result of the simple fact that, unlike the other types, it did not have an accusative case. Another important difference is found in the genitive. While the amphikinetic and hysterokinetic types had a zero grade of the suffix in this case (**-n-és*, **-n-ós*), the proterokinetic type appears with a full grade (**-én-s*). In view of this very basic difference, Beekes adopted Pedersen's (1926) broader distinction between a hysterodynamic and proterodynamic inflection, the former category comprising both the amphi- and hysterokinetic type.

¹⁶ See Beekes (1985: 161) for a schematic overview.

2.3 The Proto-Germanic *n*-stems

2.3.1 The masculine *n*-stems

The far majority of the Germanic masculine *n*-stems directly continue the PIE amphikinetic type. The number of potentially hysterokinetic *n*-stems is extremely small in Germanic, the most prominent example being **uhsen-* ‘ox’ (see § 4.2.2.1), and they appear to have merged with the amphikinetic *n*-stems in most Germanic dialects. Hence, it is generally assumed that the Old Norse nominative in *-i*, as opposed to the rest of the case forms, continues a hysterokinetic ending. A “mixed” amphikinetic-hysterokinetic type can possibly be postulated on the basis of some ablauting *n*-stems with a stressed *o*-grade in the root (see § 9.3).

The original inflection of the amphikinetic type is best preserved in Gothic and Old High German, which preserve the *e*-vocalism in the genitive and dative singular, and the *a*-vocalism in the other cases. The *a*-grade became very much intrusive in all of the other Germanic dialects, most notably in Nordic and Anglo-Frisian, where it tended to spread to the oblique cases in both the singular and the plural.

The reconstruction of the nominative ending is debated, because the masculine ending has a different reflex than the feminine, e.g. OHG *-ō* vs *-a*, even though both must ultimately have developed from PIE **-ōn*. A widely acknowledged, but actually disputable solution to this problem consists of the reconstruction of a Proto-Germanic distinction between bimoric **-ō(n)* and trimoric **-ō̃(n)*.¹⁷ The postulation of this distinction is *ad hoc*, however, because it cannot be proven to have existed for any other long vowel. It is therefore left out of consideration here. Suffice it to say that, regarding the masculine endings, OHG *-ō* points to **-ōn* and ON *-i* to **-ēn*. The Gothic ending *-a* is derivationally ambiguous.

PGm.	Gothic	ON	OHG	OE
nsg. <i>*-ō(n), -ēn</i>	<i>guma</i>	<i>gumi</i>	<i>gomo</i>	<i>guma</i>
gsg. <i>*-enaz</i>	<i>gumins</i>	<i>guma</i>	<i>gomen, -in</i>	<i>guman</i>
dsg. <i>*-eni</i>	<i>gumin</i>	<i>guma</i>	<i>gomen, -in</i>	<i>guman</i>
asg. <i>*-anun</i>	<i>guman</i>	<i>guma</i>	<i>goman</i>	<i>guman</i>

¹⁷ It is generally assumed that plain *n*-stems in **-on-* and the Hoffmann formations in **-oHon-* were kept distinct in Germanic (cf. Jasanoff 1980, Harðarson 1989, and Olsen 2006).

npl. *-aniz	<i>gumans</i>	<i>gum(n)ar</i>	<i>gomon, -un</i>	<i>guman</i>
gpl. *-anan	<i>gumane</i>	<i>gum(n)a</i>	<i>gomōno</i>	<i>gumena</i>
dpl. *-ummiz	<i>gumam</i>	<i>gum(n)um</i>	<i>gomōm</i>	<i>gumum</i>
apl. *-anuns	<i>gumans</i>	<i>gum(n)a</i>	<i>gomon, -un</i>	<i>guman</i>

An important difference with the other Indo-European languages is that Germanic *n*-stems no longer have a zero grade variant of the suffix. In this way, the branch stands in contrast to e.g. Hittite, Sanskrit and Greek, which did preserve such zero grades in the weak cases. In Germanic, however, the gsg. ending *-*n-os* seems to have replaced by *-*en-os* (cf. Prokosch 1939: 252)¹⁸, the gpl. *-*n-om* by *-*on-om*.¹⁹ The Old Norse plurals with optional *n* do not necessarily attest to a zero grade, because they may have undergone syncope (cf. ON *himinn*, dat. *hifni* m. 'sky, heaven' < **heminaz*, **heminaī*). The zero-grade apl. ending *-*n-ns* was replaced by *-*anuns* < *-*on-ns*. The dpl. in *-*mis*²⁰, an ending which is close to the Lith. ipl. -*imis*, probably had a zero grade of the suffix as well, and most dialects point to *-*ummiz* < *-*ŋ-miz*²¹. However, here the zero-grade nasal was assimilated by the *m* of the ending.

2.3.2 The feminine *n*-stems

As opposed to the masculine and neuter *n*-stems, the feminine *n*-stems have no ablaut of the suffix. Instead, the lengthened grade *-*ōn-* is found throughout the paradigm. The generalization of this lengthened grade is not a reflection of the original PGm. situation. In view of the transfer of some old PIE *h*₂-stems to the feminine *n*-stems, e.g. Go. *qino* (cf. OCS *žena*, OIr. *ben* 'woman' < **g^wén-h*₂, **g^wn-éh*₂-s) and *tuggo* 'tongue' (cf. Lat. *lingua* < **d^hǵ^h-ueh*₂-), the loss of the ablaut can be ascribed to the

¹⁸ This ending can probably not directly be compared to the similarly analogical *n*-stem genitives Greek -ένοϛ and Arm. -in (Matzinger 2002: 69-70).

¹⁹ The discrepancy between Gothic -*e*, on the one hand, and ON, OE -*a*, OHG -*o* on the other, probably results from the loss of the original ending *-*an* in the dialects. This loss induced the analogical spread of the gpl. ending of other stem types. In Gothic, this was the gpl. *i*-stem ending -*e* < *-*ejan* < *-*ei-om* (Kortlandt 1978). ON -*a* and OHG -*o* continue the thematic ending *-*ōan* < *-*oHom*/*-*eh*₂-*om*.

²⁰ Cf. ON *tveim(r)*, OE *twæm* dpl. 'two' < **twai-miz* (Bammesberger 1990: 45).

²¹ Cf. Bammesberger (1990: 170). For the development *-*nm-* > *-*mm-*, cf. OHG *hamma*, OE *ham* f. < **kōnh*₂-*meh*₂- (see § 4.3.4) and perhaps also OFri. *omma* m. 'breath' < **amman-* < **h*₂*en-mon-* (cf. OIr. *animm*, *anman* 'soul').

Germanic amalgamation of the feminine *ōn*- and *eh₂*-stems.²² This seems to have occurred at a late stage, because, even in synchronic Gothic, there still are feminines that vacillate between the *ō*- and *ōn*-stems, e.g. *bandwo*, dsg. *bandwai* f. 'sign', *daura-wardo*, dsg. *daura-wardai* f. 'gatekeeper' (Streitberg 1910: 111; Van Hamel 1923: 96). The merger of Pre-Gm. **ā* and **ō*, by which the original nominatives **-ō* and **-eh₂* became identical, must be regarded as the *terminus post quem* to this development.²³ Another indication that the *ōn*-stems were created by the addition of an *n* to the *h₂*-stems comes from the Germanic *īn*-stems, which have arisen by the addition of the same suffix to the PIE *ih₂*-stems.²⁴

	PGm.	<i>ōn</i> -stems		PGm.	<i>īn</i> -stems
nsg.	<i>*-ōn</i>	<i>qino</i> 'woman'		<i>*-īn</i>	<i>bairandei</i> 'carrying'
gsg.	<i>*-ōnaz</i>	<i>qinons</i>		<i>*-īnaz</i>	<i>bairandeins</i>
dsg.	<i>*-ōni</i>	<i>qinon</i>		<i>*-īni</i>	<i>bairandein</i>
asg.	<i>*-ōnun</i>	<i>qinon</i>		<i>*-īnun</i>	<i>bairandein</i>
npl.	<i>*-ōniz</i>	<i>qinons</i>		<i>*-īniz</i>	<i>bairandeins</i>
gpl.	<i>*-ōnan</i>	<i>qinono</i>		<i>*-īnan</i>	<i>bairandeino</i>
dpl.	<i>*-ōmmiz</i>	<i>qinom</i>		<i>*-īmmiz</i>	<i>bairandeim</i>
apl.	<i>*-ōnuns</i>	<i>qinons</i>		<i>*-īnuns</i>	<i>bairandeins</i>

Since the feminine *n*-stems are of recent coinage, it must be assumed that, before the merger with the **eh₂*-stems, the feminine *n*-stems were formally identical to the masculine stems in **-ō*, also regarding the ablaut of the suffix.

2.3.3 The neuter *n*-stems

The Germanic neuters formally differ from the masculine *n*-stems only in the nominative and accusative. In the singular, they have the same

²² According to Harðarson (1989), the addition started in the weak adjectives.

²³ There is a parallel in Tocharian B, where some *ā*-stems (e.g. *kantwo* 'tongue' < **dnǵ^h-ueh₂-*) shifted to the *ōn*-stems, a development that was likewise facilitated by the merger of the nominatives **-ā* and **-ōn* into *-o* (cf. Hilmarsson 1988: 506).

²⁴ This extension may have taken place in the weak adjectives, where a weak ending had to be created to contrast with the strong endings. This probably happened according to the proportion **-os* : **-eh₂*/**-ih₂* = **-ēn*/**-ōn* : *x*.

ending **-ōn* that appears in the feminines; it is clearly distinct from the ending **-ō* that developed out of PIE **-ōn* (cf. Boutkan 1995: 285), and is likely to be an innovation. In all probability, it replaced the expected outcome **-un* from PIE **-n̥*²⁵. The original plural ending **-ōn*, most probably a collective suffix, was supplanted by **-ōn-eh₂* (cf. Skt. *-āṇi* < **-ōn+h₂*), giving Go. *-ona*.

The genitive ending of the neuter *n*-stems is identical to its masculine counterpart. This identity must be a recent phenomenon resulting from the replacement of amphi- and hysterokinetic gsg. **-n-ōs* by **-én-i* from the dative or the proterokinetic genitive **-én-(o)s*. The occurrence of the zero-grade suffix in Go. npl. *namna*, gpl. *namne*, dpl. *namnam* is remarkable and must be old. Perhaps, then, the *mn*-stems were static in the proto-language, having full grade of the root in the genitive. Alternatively, the Germanic paradigm may have merely been influenced by the influx of static heteroclitics into the neuter *n*-stems, cf. Go. *wato*, dpl. *watnam*, ON *vatn* n. 'water' < PIE **uód-r*, gen. **uéd-n-s*.

	PGm.	Gothic	ON	OHG	OE
nasg.	<i>*-ōn</i>	<i>augo</i>	<i>auga</i>	<i>ouga</i>	<i>ēage</i>
gsg.	<i>*-enaz</i>	<i>augins</i>	<i>augu</i>	<i>ougen, -in</i>	<i>ēagan</i>
dsg.	<i>*-eni</i>	<i>augin</i>	<i>augu</i>	<i>ougen, -in</i>	<i>ēagan</i>
napl.	<i>*-ōnō</i>	<i>augona</i>	<i>augu</i>	<i>ougun, -on</i>	<i>ēagan</i>
gpl.	<i>*-anan</i>	<i>augane</i>	<i>augna</i>	<i>ougōno</i>	<i>ēagena</i>
dpl.	<i>*-ammiz</i>	<i>augam</i>	<i>augum</i>	<i>ougōm</i>	<i>ēagam</i>

Like in the other IE languages, the neuter *n*-stems are relatively infrequent in Germanic: only few examples can be collected from the oldest dialects. The majority of the cases are found in Nordic, where they usually denote utterly non-agentive objects, cf. ON *bjúga* 'sausage', *hnoða* 'clew', *leika* 'plaything, doll', *lík-síma* (beside m. *sími*) 'bolt rope'. The same semantic generalization can probably be made about the small number of additional cases that can be gleaned from Gothic: *auga-dauro*

²⁵ PGm. **-ōn* has been identified as the collective ending PIE **-ōn*, comparable to e.g. Gr. *-ωρ* in *ὕδωρ* n. 'water' (Harðarson 2005: 217 fn.), but the retention of the final nasal into Proto-Germanic is a serious complication.

'window', *barnilo* 'child'²⁶, *kaurno* 'grain', *pairko* 'hole'. The class of neuter *n*-stems is relatively large in the Norwegian dialects, cf. *auka* 'sand bank', *hyrna* 'corner' < **hurnjōn-*, *ikorn(a)* 'squirrel' < **ikwernōn-*, *korna* 'grain' (= Go. *kaurno*), *kukla*, *nipla*, *nykla*, *nysta* 'clew', *lafta* 'log joint, log tip', *lette-drekka* 'light beer', *moro-leika* 'toy' (cf. Far. *leika* 'id.'), *oka* 'animal collar, corner beam of a fence' < **jukōn-*, *åkorn(a)* 'acorn' (cf. ON *akarn* 'id.', Go. *akran* 'fruit').

Despite the infrequency of the neuter *n*-stems in Germanic, the class nevertheless takes a prominent position, because it is well represented in the pan-Germanic names for body parts, e.g. Go. *augo* 'eye', *auso* 'ear', *hairto* 'ear', ON *auga*, *eyra*, *hjarta*, *lunga* 'lung', *nýra* 'kidney' (cf. OSw. *niura* < **neurōn*-²⁷), OHG *ouga*, *ōra*, *herza*, *wanga* 'cheek' etc.

Again, the class seems to have still been open in Old Norse. This is demonstrated by two different facts. First, the category contains body parts that have no parallels in the other Germanic languages, e.g. *flag-brjóska* 'breast-bone', *eista* 'testicle' (cf. Nw. dial. *eista*), *viðbeina* 'collarbone'; and second, the material reveals an apparently recent vacillation between neuters and masculines, cf. ON *strjúpa* n. 'throat' beside *strjúpi* m. 'id.', *økla* (cf. Nw. dial. *okla*) n. 'ankle' beside *økli* m. 'id.', Sw. *fot-bjälle* n. 'ankle' beside Icel. *bjalli* m. 'knoll, hill', Sw. *tumme* n. beside m. 'thumb'.²⁸ The neuters designating body parts are relatively well-attested in the Norwegian dialects, which beside common *auga*, *øyra* and *hjarta* also preserved *eista* and *okla*. Additionally, there are the uniquely Norwegian neuters *enna* 'fore-head' (for ON *enni* n. 'id.' < **anþja-*) and *jura* 'udder' (for ON *jú(g)r* n. 'id.'), which look like recent coinages. A form that must also be mentioned here is *hovda* 'hill', apparently continuing ON **hofða*, a derivative from ON *hofuð* 'head'.²⁹

²⁶ In the case of *barnilo*, the neuter gender of the diminutive probably reflects the gender of the base noun *barn*.

²⁷ The fronting of ON *nýra* is obscure, but has been explained as regular in the dative **neureni* < **neg^{wh}-r-en-i* (cf. Gr. νεφροί 'kidneys', Lat. *nefronēs* 'kidneys, testicles' < **neg^{wh}-r-*).

²⁸ Hellquist 1026.

²⁹ It further occurs in *kall-hovda* and *ve-hovda* 'supporting plank of the oven', cf. ON *karl-hofði* 'carved figurehead (on the prow of a ship)'. In the Pre-Christian era, these "wooden heads" probably had a religious function, cf. *ve-hovda* "holy head" < ON **vé-hofða* (Kiil 1960).

In spite of the apparently recent origin of the Nordic weak neuters, the productivity of the category as a whole seems to be an archaic feature inherited from Proto-Indo-European, as has been argued in the previous section. This follows from the fact that an association between the *n*-suffix and names for body parts can also be observed in a central Indo-European dialect group such as Indo-Iranian.

3 The Proto-Germanic geminates

The derivation of the Germanic *n*-stem endings is relatively unproblematic. A more intricate problem concerning the evolution of the *n*-stems consists of the origin of the remarkable root-final consonant alternations. These alternations, which have been described in the introduction, are unparalleled in the Indo-European languages, and therefore require an explanation. In order to be able to understand the rise of this specifically Germanic phenomenon, it is first necessary to examine the origin of the Proto-Germanic long stops itself.

3.1 Kluge's law

Unlike Germanic, Proto-Indo-European did not have geminates. It had a three-way opposition between e.g. **t*, **d* and **d^h*, but there are no indications whatsoever that it additionally had an opposition between short and long stops. On the contrary, when two identical PIE consonants collided alongside a morpheme boundary, the result seems to have been a single stop. A well-known example of this is the second person of the verb 'to be'. Morphologically, the PIE form must be analyzed as **h₁es-si*, with the root **h₁es-* and the ending **-si*. But as Skt. *ási* and Gr. *ἔῃ* show, the *s* was already shortened in the proto-language, since the expected outcome would otherwise have been Skt. ***ási* and Gr. ***ἔσσι*.³⁰ The conclusion must therefore be that consonantal length was not phonological in the Indo-European proto-language.

Germanic, on the other hand, is rife with geminates. Long consonants can occur almost anywhere, in nouns, verbs, adjectives, adverbs and prepositions. Two important hotspots can be recognized: the *n*-stems and the class 2 weak verbs:

- **skatta-*: Go. *skatts* m. 'money'
- **mannan-*: Go. *manna* m. 'man'

³⁰ In Hom. *ἔσσι*, apparently an Aeolic form, the long sigma has been restored.

- **smakkan-*: Go. *smakka* m. 'fig'
- **snittōn-*: MHG *snitzen* 'to chop'
- **hlakkōn-*: OFri. *hlakkia* 'to laugh'
- **hukkōn-*: G *hocken* 'to cower, squat'
- **wikkōn-*: OE *wiccian* 'to work magic'
- **kwerru-*: Go. *qairrus* 'mild'
- **uppai*: ON *uppi*, OE *uppe* 'up'
- **ferrai*: Go. *fairra*, ON *fjarri* 'far'

One of the first to touch upon the origin of the Proto-Germanic geminates was Bezzenberger. In a review in *Göttingische gelehrte Anzeigen* (1876: 1374-5), he drew attention to the lexical correspondence of PGM. "**lokka-*" 'hairlock' with Lith. *lūgnas* 'flexible'.³¹ The correspondence is actually false – ON *lokkr*, OE *loc* 'hairlock' etc. continue an internally Germanic derivation to the root found in OHG *liohhan* 'to pull, pluck' (see § 7.2.1) – but the idea that a nasal was involved in the creation of the Proto-Germanic geminates turned out to be a lucky strike. On the basis of material adduced by Paul (1880: 133 fn. 2), Osthoff (1882: 297-8) proposed to connect the Germanic *ōn*-verbs with geminates to the Sanskrit *nā*-class (see chapter 6), thus explaining the geminate of, for instance, G *hocken* from a Pre-Germanic sequence **-kn-*. The *-ō-* of the remaining part of the suffix he compared to the Skt. element *-ā-*.

Meanwhile, Kluge had started collecting more evidence in favor of nasal assimilation as the underlying principle of Proto-Germanic gemination. In 1884, eight years after the discovery of Verner's law, he published the article *Die germanische Consonantendehnung*. On the basis of a systematic survey of the material, which by then had reached considerable proportions, he concluded that the voiceless geminates had arisen by the assimilation of a stressed nasal suffix. The plosive geminates were thus brought in line with the long resonants, which on the basis of correspondences such as Go. *fulls* 'full' < **fullaz* with Skt. *pūrṇá-* < **p_lh₁-nó-s* had already been explained earlier by a similar

³¹ "Daß die von dem Hrn. Vf. S. 49 angeführte Erklärung von germ. l o k k a das k k unerklärt lasse, ist dem lit. l u g n a s gegenüber nicht zuzugeben. Die Aufklärungen, welche er uns an dieser Stelle in Aussicht stellt, entziehen sich einstweilen noch der Discussion; ich freue mich sehr auf sie, denn das »Allerweltsuffix« wird alsdann eine hübsches Pendant an einem »Allerweltnasal« haben."

assimilation of *n* (cf. Sievers 1878: 109 fn. 1). The following examples, primarily consisting of forms with double **l*, can be mentioned:

- Go. *wulla*, ON *ull* f. 'wool' < **wullō-* < **HulH-neh₂-* ~ Skt. *úrṇā-* 'id.'
- Go. *fairra*, ON *fjarri* 'far' < **perH-nōi* ~ Lit. *pérnai* 'last year'
- OE *hyl* 'hill' < **hulli-* < **kl(H)-ni-* ~ Lat. *collis* 'id.' < **kolH-ni-* or **kl-ni-*
- Go. *brut-fill* n. 'leprosy' < **fella-* < **pel-no-* ~ Lat. *pellis* 'id.' < **pel-ni-*
- OHG *wella* f. 'wave' < **uel-neh₂-* ~ Ru. *volná* f. 'id.' < **ul-neh₂-*
- Go. *alls* 'all' ~ Osc. *allo* (f.) 'all, entire' < **h₂el-nó-*

By comparing the Germanic evidence for geminates with the material from other Indo-European languages, it became clear that a Germanic long stop occasionally occurs where in Indo-European an original nasal suffix can be expected. Although the examples are not very numerous, they represent material of unambiguous Indo-European origin. The reliability of the evidence consequently does not suffer much from this scarceness. The following correspondences are in support of the link between Proto-Germanic geminates and Indo-European *n*-suffixes:³²

- OE *botm* m. 'bottom' < **butt-* ~ Skt. *budhná-*, Lat. *fundus* < **b^hud^h-no-*³³
- Go. *diups* 'deep' < **deup^a-* ~ OIr. *domain*, W *dwfn* 'deep' < **d^hub^h-no-*
- OE *friccea* m. 'herald' < **frekkjan-* ~ Go. *fraihnan* 'to announce' (Skt. *praśnín-* 'herald' < **prek^h-n-*³⁴)

³² Examples taken from Kluge (1884), Brugmann (1897: 383-4), Fick/Falk/Torp (1909), Franck/Van Wijk (1912), Hirt (1931: 92), Lühr (1988: 197).

³³ The form **buttma-* is a conflation of the PGm. nom. **budmēn* < **b^hud^h-mēn* (Gr. *πυθμήν*) and the gen. **buttaz* < **b^hud^h-n-ós* (Skt. *budhná-*). See § 4.2.2.3 for a more detailed analysis.

³⁴ An objection to the connection with *abhi-praśnín-* 'inquisitive person' is the productivity of the Sanskrit suffix *-in-* as an agent marker. According to Seebold (1989: 153), this makes a direct etymological link implausible. Alternatively, *friccea* can have been derived from a verbal stem **frekk-* with the suffix **-jan-* as in Go. *fiskja* m. 'fisherman', *timrja* m. 'carpenter'. This stem **frekk-* must be a further non-attested allomorph of **freh-* as in Go. *fraihnan*. To assume derivation

- Far. *hvøkk* s.v. 'to be startled; to diminish; to begin slowly' < **hwekkan-*, cf. OCS *čeznōti* 'to vanish, perish' < **k^weg^h-n-*
- EDu. *lappen* 'to lick' < **lappōn-*, cf. Lat. *lambō* 'to lick' < **lHb^h-néh₂-*
- OE *liccian* < **likkōn-*, cf. Gr. λιχνεύω, Lat. *lingō* 'to lick' < **liġ^h-n-*
- ON *lokka* 'to allure, caress' < **lukkōn-* ~ Lith. *lù(n)ginti* 'to wave, fondle, allure' < **lug^h-n(e)h₂-*
- Du. *mikken* 'to aim' (from older 'to peer') < **mikkōn-* < **mig^h-néh₂-*, cf. Lith. *mingù*, *migti* 'to fall asleep', Ru. *mignut'* 'to blink, wink' < **m(e)ig^h-n-*
- OE *roccian* 'to rock', MHG *rocken* 'to drag, jerk' < **rukkōn-* < **Hruk-néh₂-*, cf. Lat. *runcō* 'to weed' < **Hru-n-k-*
- MDu. *roppen*, MHG *rupfen* 'to pluck, tear off' < **Hrup-néh₂-*, cf. Lat. *rumpō* 'to break, tear' < **Hru-n-p-*
- MHG *stutzen* 'to bump' < **stuttōn-* < *(s)*tud-n-* ~ Lat. *tundō*
- Swi. *suckə* < **sukkōn-* < **suk^h-néh₂-*, cf. W *sugnaf* < **souk^h-néh₂-* (also cf. OCS *sъsati* 'id.')
- MDu. *stricken* 'to make fall' < **strikkōn-*, cf. Lat. *stringō* 'to draw tight; to compress' < **strig-n-*
- OE *þaccian* 'to pat' < **þakkōn-*, cf. Lat. *tangō* 'to touch' < **th₂g-n-* (rel. to Gr. Hom. τεταγών 'seizing')
- Du. *wit* < PGm. **hwitta-* ~ Skt. *śvítna-* 'white' < **kuit-no-*³⁵

On the basis of examples like the ones given above and the parallelism with the process of lengthening of the resonants, Kluge together with Bezenberger and Osthoff concluded that a PIE *n* was assimilated by any preceding stop, and ultimately resulted in a PGm. voiceless geminate. This development is nowadays known as Kluge's law.

from PIE **prek^h-nó-* > Skt. *praśná-* m. 'question' (Schaffner 2001: 398) is less attractive. The connection with Lat. *praeco* 'announcer', as suggested by Seebold (l.c.), is unlikely as well, because this word can be reconstructed as **prai-dikō* (De Vaan 2008: 169).

³⁵ Seebold (1989: 153) rejects this reconstruction in view of Go. *hveits* 'white' < **hwīta-*: "Nun ist Ablaut hochgradig unwahrscheinlich [...]; dagegen kommt eine Kürzung vor der Geminata sehr wohl in Betracht. Nur ist es keine Geminata aus *n*-Assimilation, sondern der Fortsetzer der alten neutralen NASg-Form (Heliand *huuitt*)." Still, this explanation does not explain why the root of Go. *hveits* 'white' < **hwīta-* has a *-t-* in the first place (see § 3.2).

3.2 Shortening in over-long syllables

The problem of the Proto-Germanic geminates is complicated by the fact that, after the operation of Kluge's law, geminates were shortened in over-long (superheavy) syllables, i.e. in syllables with long vowels and diphthongs. There are numerous examples of these shortened geminates (see below), and they can often be identified because their voiceless-occlusive articulation is in disagreement with etymologically related material:³⁶

- Go. *diups* 'deep' < **deup^pa-* to OIr. *domain*, W *dwfn* 'deep'
- OE *dīc* 'dam, pool' < **dīk^ka-* to Gr. *τείχος* 'wall'
- ON *gróp* f. 'ditch' < **grōp^pō-* to OCS *grobъ* m. 'grave'
- Swi. App. *heenxā* f. 'to (make) hang' < **hank^kjan-* vs MHG *hangen* 'to hang' < **hangan-*
- OE *huntian* 'to hunt' < **hunt^tōn-* to Go. *fra-hinþan* 'to capture'
- Go. *hveits* 'white' < **hwīt^ta-* to Skt. *śvetá-*, *śvítna-* 'white'³⁷
- ON *knútr* 'knot' < **knūt^ta-* to OHG *chnodo* 'id.' < **knuþan-*
- Go. *nota* m. 'stern' < **nōt^tan-* to Gr. *νότον* 'back' < **noHt-o-*³⁸
- OE *scāþp* 'sheep' < **skēp^pa-* to Go. *skaban* 'to shear'³⁹
- OE *tæcan* 'to show'⁴⁰ < **taik^kjan-* to Gr. *δείκνυμι* 'to show'⁴¹
- ON *vōttr* 'mitten' < **want^tu-* to PGm. **windan-* 'to wind'

The shortening of geminates was an essential change in Germanic phonology, as it reduced the array of possible syllable structures, leaving short syllables CŴ(C)-, long syllables CŴ(C)-, CVRC-, but no over-long syllables CŴCC- or CVRCC-. The fact that shortened geminates were not

³⁶ In order to avoid any confusion between old singulates and shortened geminates at the reconstruction of Proto-Germanic – a distinction that often appears to be critical – the latter will henceforth be given in superscript.

³⁷ Cf. Gamm (1973: 90).

³⁸ Cf. Fick/Falk/Torp 298: "Aus ig. nōtn-". Further cf. Lat. *natis* 'buttock' < **nĥ_{2/3t-}*.

³⁹ Connection first made by Woods (1919: 207).

⁴⁰ Unlike OE *tæcan*, Swi. Visp. *zeixu* 'to show' has no **jan*-suffix, because then we would expect the form to have been ***zeicku* (cf. *reicku* 'to smoke' < **raukjan-*). Thus, *zeixu* directly points to PGm. **taik^kōn-* from PIE **doiĥ^h-néh₂-*.

⁴¹ Lühr (1988: 340): "Da [...] eine Wurzelform **deiġ-* nur aufgrund des Germanischen angenommen werden müßte, empfiehlt sich eine innergermanische Herleitung des *k*-Lautes."

affected by Grimm's law is an indication that this process was posterior to this sound law. It thus seems to have formed the final step in the evolution towards Proto-Germanic phonology as we know it.⁴²

3.3 Exceptions to Kluge's law

Kluge's law did not operate under all circumstances. A number of forms appear to have resisted the assimilation of **-n-*. In many cases, these forms contain a cluster of a voiceless fricative plus a nasal. The material thus seems to indicate that either 1) Kluge's law only operated pretonically, or 2) Kluge's law only affected the PGM. voiced obstruents. The first explanation was offered by Kluge himself, the second was formulated by Lühr (1988: 195).⁴³

3.3.1 No gemination of voiceless fricatives

The following instances are in support of the conditioning as formulated by both Kluge and Lühr. They consist of formations with voiceless fricatives followed by an *n*-suffix:

- Go. *auhns*, OHG *ovan* m. 'oven' < **ufna-* < **úp-no-*
- Go. *aþn(s)* m./n. 'year' < **aþna-* < **h₂ét-no-* (cf. Lat. *annus*)
- Go. *rahnjan* 'to estimate' < **Hrók^w-n-* (?Skt. *racáyati* 'to order')
- OHG *lēhan* 'borrowed goods' < **laihna-* < **lóik^w-no-* (cf. Skt. *rékṇas-* 'inheritance, wealth')
- ON *svefn*, OE *sweƿ(e)n* m. 'sleep' < **sweƿna-* < **suép-no-* (cf. Skt. *svápna-*)
- ON *tafn* n. 'sacrifice, meal' < **tafna-* < **dh₂p-no-* (cf. Lat. *damnum* 'detriment', Gr. *δαπάνη* 'expenditure')

⁴² Beekes defined the syllabic interchange of *CŨCC-* ~ *CŨC-* as a substrate marker (cf. 1999: 15), but alternations of this type are, in fact, the result of fully regular Pre-Germanic sound laws.

⁴³ Lühr (1988: 192) rejected the accentual conditioning of the law: "[es] erscheint ratsam, den Akzent bei der Beschreibung der *n*-Gemination außer Betracht zu lassen, auch wenn sich mit Hilfe des Akzentes eine Reihe von Gegenbeispielen leichter erklären ließe."

There are some counter-examples that have voiced obstruents rather than voiceless fricatives. They potentially disprove nasal assimilation because they are in conflict with both Kluge's and Lühr's formulation of the conditioning. However, it was demonstrated by Gamm (1973: 92-3) and Lühr (1988: 330ff) that many of these counter-examples must have arisen secondarily. A number of cases consist of *na*-stems that are likely to be post-Proto-Germanic thematizations to older *n*-stems. As a result, they cannot be used as evidence against Kluge's law:

- MLG *brāgen* 'brain' < **brag(a)na-* to MLG *brēgen* < **bragina-* (cf. Gr. *βρεχμός* 'forehead, skull'⁴⁴)
- ON *hrafn* m. 'raven', OHG *raban* m. 'id.' < **hrabna-* to OHG *rabo* m. 'id.' < **hraban-*
- ON *hrogn* n. 'fish roe', OHG *rogan* m. 'id.' < **hrugna-* to OHG *rogo* m. 'id.' < **hrugan-*

Other supposed counter-examples can be explained away by assuming that the *n*-suffix was added to the root in late Proto-Germanic, i.e. after the great sound shifts, including Kluge's law. The *na*-suffix appears to have been quite productive. The following instances must be analyzed as having such a suffix.

- Go. *barn*, ON *barn*, OE *bearn*, OFri. *bern*, OS, OHG *barn* n. 'child' to **beran-* 'to carry'
- ON *gaupn* f. 'palm' < **gaupnō-* to OE *gēopan* 'to pick up' < **geupan-*
- ON *teikn*, OHG *zeihhan* n. 'sign' < **taik-na-* to OE *tācan* 'to show' < **taik^kjan-*
- G *trocken* 'dry' < **druk(k)na-* to G Bav. *tricken* 'to dry'⁴⁵ < **drukk(n)jan-*⁴⁶

⁴⁴ The suffix ablaut presupposes an old *n*-stem (Lühr 1988: 332).

⁴⁵ Bachmann (2000: 185).

⁴⁶ In view of Du. *droog* < **draugja-* and OE *dryge* 'dry' < **drūgja-*, the original root-final consonant must have been **k* or **g^h*. For this reason, *tricken* should probably be derived from **drukkjan-* (or **drukknjan-* with dissimilation). A form **drugjan-* would probably work as well, but seems less preferable in view of OS *drukno* adv. 'dry' < **drukkna-* and *druknian* 'to dry' < **drukkñō(ja)n-*.

Much of the remaining evidence against Kluge's law can be tackled by assuming that, in spite of Lühr's criticism, Kluge was right about his accentual conditioning, and that the assimilation of *n* was blocked by root stress. This explanation is particularly attractive in those cases that have a full grade of the root. Root stress must at any rate be assumed for words that originally had a static inflection in Proto-Indo-European, as the root of static nouns had a stressed full grade throughout the paradigm. The word for 'water', for instance, may have had a static paradigm **uód-r*, gen. **uéd-n-s* (cf. Beekes 1995: 188). In such nouns, the absence of geminates is expected in view of their original accentuation. Consider the following instances with full-grade roots:

- Go. *ibns*, ON *jafn*, OHG *eban* 'even, level' < **ebna-* < **h₁éb^h-no-*
- ON *skarn*, OE *scearn*, OFri. *skern* n. 'dung' < **skarna-* < **skór-n-*, cf. Gr. σκῶρ, σκατός n. 'id.' < **sk̑-ōr*, **sk̑-n-*
- Go. *rign* n. 'rain', OHG *regan* m. 'id.' < **regna-* < **Hrég^h-no-*?
- ON *vagn* m. 'wagon', OHG *wagan* m. 'id.' < **wagna-* < **uóg^h-no-*
- ON *vatn* n. 'water', Go. *wato*, gpl. *watne* n. 'id.' < **watōr*, **watn-* < **uód-(ō)r*, **uéd-n-s*

3.3.2 No gemination of sibilants

Another important exception to Kluge's law consists of **s* not being affected. This is evidenced by a number of cases that show the effects of Verner's law, but not of Kluge's law:

- Go. *razn* n. 'house' < **razna-* < **?h₁r_h₁-s-nó-* (to OHG *ruowa* 'rest', Gr. ἐρῶν 'id.' < **h₁roh₁-ueh₂-*)
- OE *lirnian* 'to learn' < **liznōn-* < **lis-néh₂-* (see § 6.1.5)
- OE *wisnan*, *weornan* 'to wizen' < **wis/znōn-* < **uis-neh₂-*
- OHG *zwirn* m. 'double thread' < **twizna-* < **duis-nó-*
- OHG *zwirnōn* 'to twine' < **twiznōn-* < **duis-néh₂-*
- ON *qnn* f. 'work' < **aznō-* < **h₂es-néh₂-*

The fact that **s* was not affected by Kluge's law has a bearing on the identification of the exact phonetic process that gave rise to the Proto-Germanic geminates. Probably, the phonetic motivation for the exception was that sibilants could not assimilate a following *n*, not even when they

were voiced by Verner's law. As PGm. **f*, **þ* and **h* remained untouched as well, the conclusion may be that Kluge's law did not affect fricatives. This, in turn, implies that PGm. **b*, **d* and **g*, which traditionally are reconstructed as the voiced fricatives **ḃ*, **ḡ* and **ǵ*, were, in fact, not fricatives at all, but voiced plosives. For the possible consequences of this hypothesis, see the next section.

3.4 Different configurations of Kluge's law

There are roughly three different variants of Kluge's law. The differences between these variants are centered around two issues. The first issue consists of the question of how exactly Kluge's law is to be interpreted phonetically: were the Proto-Germanic geminates caused by assimilation of the *n*-suffix, or did the nasal simply double a preceding obstruent before it was lost? The second issue is about chronology. Traditionally, Kluge's law is thought to have operated more or less between Grimm's law and Verner's law. However, it has been argued by Kortlandt some years ago, that Verner's law was anterior to Grimm's law. As a result, Kortlandt needed to formulate an alternative interpretation of Kluge's law in the Proto-Germanic chain of sound changes.

3.4.1 Kluge's traditional configuration

When Kluge formulated his law, he assumed that the process of Proto-Germanic gemination came about by the assimilation of a following *n*. On the basis of such exceptions as **swefna*- 'sleep', **ufna*- 'oven' and **aþna*- 'year', which do not show any sign of the operation of Verner's law, Kluge further argued that this process only took place pretonically. The case of **swefna*- is particularly strong, because its original barytonesis is supported by extra-Germanic evidence, viz. Skt. *svápna*-, Gr. *ὑπνοσ* and Alb. *gjumë*.

What Kluge basically observed was the concurrence of *n*-assimilation with Verner's law. This had an important chronological implication. Since both PIE voiceless and voiced aspirated stops merged into a PGm. voiceless geminate, Kluge supposed that Verner's law preceded the assimilation of *n*. Confirmation of this chronology was found in the formation **seuni*- 'image', as continued by e.g. Go. *siuns*, ON *sjón* f. 'id.' (Kluge 1884: 172-3 fn.2). Since this formation is clearly related

to the verb **sehwan-* 'to see' < **sé^wk^w-e-*, it shows that the development of **k^w* to **w* preceded the assimilation of *n* in pretonic position.⁴⁷ Kluge therefore situated the assimilation between the first and the second phase of Grimm's law, i.e. between the lenition of the PIE voiceless and voiced aspirated stops to fricatives and the devoicing of the PIE voiced unaspirated stops. The chronology thus became as follows:

	Du. <i>wit</i>	E <i>bottom</i>	MHG <i>stutzen</i>
PIE	<i>*k^{uit}-nó-</i>	<i>*b^hud^h-nó-</i>	<i>*stud-néh₂-</i>
Grimm's law 1	<i>*hwip^h-ná-</i>	<i>*bud^h-ná-</i>	<i>*stud-nó-</i>
Verner's law	<i>*hwið^h-ná-</i>	<i>*bud^h-ná-</i>	<i>*stud-nó-</i>
Assimilation	<i>*hwiðða-</i>	<i>*budða-</i>	<i>*studdō-</i>
Occlusivation	<i>*hwidda-</i>	<i>*budda-</i>	<i>*studdō-</i>
Grimm's law 2	<i>*hwitta-</i>	<i>*butta-</i>	<i>*stuttō-</i>
PGm.			

A possible objection to Kluge's chronology would be that it requires an additional occlusion rule for the change from **-dd̥-* to **-dd-*. A more critical difficulty is the phonetic improbability of a voiced fricative **ð* becoming a long voiced fricative **dd̥* by nasal *assimilation*. This scenario implies an intermediate stage with a nasalized voiced fricative **ð̃*, which would hardly result in the geminate **dd*. Kluge's chronology can, however, be bolstered against such criticism by assuming that the occlusivization occurred exactly by the nasalization of **ð̃*.

3.4.2 Lühr: *Verdoppelung* rather than assimilation

Kluge's law was formulated somewhat differently by Lühr in her monograph *Expressivität und Lautgesetz* (1988). Lühr accepted Kluge's chronology, but instead of nasal assimilation assumed unconditioned lengthening (G *Verdoppelung*) before a nasal that was subsequently lost.

⁴⁷ "Diese Assimilation [sic] hätten noch vor VERNER und GRIMM stattgefunden", Görtzen (1998: 434) summarizes, but this must be a mistake.

This approach, which involved the removal of the accent as a factor, was inspired by Murray (1982), who for the development of e.g. Skt. *svápna*- ‘sleep’ to Pāli *soppa*- had reconstructed an intermediate stage **svappna*-. Lühr thus reformulated Kluge’s law as follows: “Von der Verdoppelung waren, der westgermanischen Konsonantengemination vergleichbar, stimmlose Verschußlaute und stimmhafte Reibelaute betroffen, wobei die verdoppelten stimmhaften Reibelaute in der Geminatun zunächst zu Verschußlauten und dann wie im Oberdeutschen zu stimmlosen Verschußlauten wurden” (Lühr 1988: 196).

	Du. <i>wit</i>	E <i>bottom</i>	MHG <i>stutzen</i>
PIE	<i>*kuitnó-</i>	<i>*budná-</i>	<i>*studnéh₂-</i>
Grimm’s law 1	<i>*hwiþná-</i>	<i>*buðná-</i>	<i>*studnǫ́-</i>
Verner’s law	<i>*hwiðná-</i>	<i>*buðná-</i>	<i>*studnǫ́-</i>
Barytonesis	<i>*hwiðna-</i>	<i>*búðna-</i>	<i>*stúdnō-</i>
Doubling	<i>*hwiððna-</i>	<i>*buððna-</i>	<i>*studdnō-</i>
Nasal loss	<i>*hwidda-</i>	<i>*budda-</i>	<i>*studdō-</i>
Grimm’s law 2	<i>*hwitta-</i>	<i>*butta-</i>	<i>*stuttō-</i>
PGm.			

Although Lühr’s alternative is chronologically unproblematic, it raises a phonetic objection. If lengthening did take place before *n*, the question arises why this lengthening did not occur before **m* as well? A further objection against pure lengthening consists of the fact that *s* and *z* were not affected by Kluge’s law. This is perfectly understandable within the assimilation framework: PGm. **b*, **d*, **g* must have been occlusive prior to their assimilation of *n*, but since there is no way of occlusifying **z*, the *n* could not be assimilated. Such a solution is unavailable if we assume that *n* simply triggered lengthening of the preceding obstruent.

Another problem facing Lühr’s modification of Kluge’s law is that it remains difficult to explain why the voiced fricatives **b*, **d* and **g* were doubled, while the voiceless fricatives **f*, **þ* and **h* were not. Lühr (1988:

195) circumvented this problem by supposing that, in Germanic, the voiced fricatives had greater “consonantal strength” than the voiceless fricatives, thus being more susceptible to gemination.⁴⁸ A possible problem with this solution, however, is that it cannot account for the fact that *z just as much as *s remained unaffected by Kluge's law, even though it clearly must be analyzed as a voiced fricative.

3.4.3 A glottalic approach by Kortlandt

A radically different chronology was proposed by Kortlandt (1991). Kortlandt, who had already advocated in 1981 that Verner's law preceded Grimm's law, pointed at the improbability that the PIE voiced aspirates ever yielded voiced fricatives in Proto-Germanic. In Old High German, for instance, PIE *b^h, d^h, g^h are simply continued as b, d, g. In view of the supposed seniority of the plosives over the voiced fricatives in the Germanic dialects, Kortlandt argued that Verner's law preceded Grimm's law, postulating that PIE plain stops and the voiced aspirates merged into voiced stops at an early stage. The product of this merger remained distinct from the PIE voiced stops, because the latter had preglottalization, a feature that according to Kortlandt can be reconstructed for Proto-Germanic on the basis of, among other things, 1) the glottalization of voiceless stops in English, 2) the dialectal Danish *vestjysk stød*, and 3) the wide distribution of preaspiration in Nordic (cf. Far. *eta* ‘to eat’ = [e̞aː^hta]). With this in mind, Kortlandt (1991) concluded that Kluge's law occurred between Verner's law and Grimm's law:

“On the one hand, the rise of the new geminates was posterior to Verner's law because it affected the voiced reflexes of the PIE. voiceless plosives in the same way as the original aspirates. On the other hand, the devoicing of the geminates suggests that it was anterior to Grimm's law, or at least to the ‘Medienverschiebung’, as Kluge pointed out already. The

⁴⁸ “Vergleicht man [...] die Lautverhältnisse bei der westgermanischen Konsonantengemination, so sind gegenüber den Beispielen mit der Verdoppelung von ursprünglich stimmhaften Reibelauten nur ganz wenige mit stimmlosem Frikativ vorhanden. Das spricht für die Annahme, daß im Germanischen eine sprachspezifische Stärkerelation mit “voiced fricatives stronger than voiceless fricatives” geherrscht hat.” (1988: 195).

logical conclusion is that Verner's law preceded Grimm's law[...]" (Kortlandt 1991: 3)

Although Kortlandt's configuration hinges on the acceptance of the glottal stops for Proto-Germanic, it provides an elegant alternative to the traditional model, explaining the material by a minimum of sound laws:

	Du. <i>wit</i>	E <i>bottom</i>	MHG <i>stutzen</i>
PIE	<i>*k̥uitnó-</i>	<i>*budná-</i>	<i>*stuʔdnéh₂-</i>
Verner's law	<i>*kwidná-</i>	<i>*budná-</i>	<i>*stuʔdnó-</i>
Assimilation	<i>*kwidda-</i>	<i>*budda-</i>	<i>*stuʔddō-</i>
Grimm's law	<i>*hwiʔtta-</i>	<i>*buʔtta-</i>	<i>*stuʔttō-</i>
PGm.			

What can be inferred against Kortlandt's chronology, is that the glottalic articulation of the geminates must be secondary in those cases where no PIE glottalized stop is involved in the assimilation of **n*. An important argument in favor of Kortlandt's chronology, however, consists of the different susceptibility of the voiceless fricatives and the voiced obstruents to this assimilation. Since PGm. **b*, **d* and **g* were affected by Kluge's law, whereas the fricatives **s*, **z*, **f*, **p* and **h* were not (see § 3.3.1 and § 3.4.2), it is defensible to assume that **b*, **d* and **g* had a plosive rather than a fricative articulation. This is much in accordance with Kortlandt's reconstruction of Proto-Germanic phonology.

4 Kluge's law and the *n*-stems

In the preceding chapter, the focus has been on the extra-Germanic evidence for Kluge's law. There appears to be a limited number of cases in which Proto-Germanic geminates correspond to nasal suffixes in other Indo-European languages, e.g. PGm. **likkōn*- 'to lick' vs Lat. *lingō* 'id.' < **liġ^h-n*- and **lukkōn*- 'to allure, carress' vs Lith. *lūginti* 'to fondle, allure' < **lug^h-n*-. The relative paucity of the corpus of correspondences is sufficiently compensated by the reliability of the examples, some of them being ancient derivations, cf. PGm. **butt*- 'bottom' ~ Skt. *budhná*- 'id.', Lat. *fundus* 'id.' < **b^hud^h-n*- etc.

Perhaps even more important than the extra-Germanic evidence is the intra-Germanic evidence for Kluge's law. Critics of this law (see chapter 7) have always treasured the scarcity of the correspondences of Germanic geminates with nasal suffixes in other languages. These critics nevertheless seem to have failed to recognize the *internal* evidence for Kluge's law. The strongest proof for this sound law is, as it happens, provided by the Proto-Germanic *n*-stems themselves, because it is in exactly this morphological category that the geminates are particularly frequent. This was, in fact, already pointed out by Kluge himself in 1884:

“Was die theorie des in der gemination untergegangenen *n* zur gewissheit macht, ist die oben unter III B behandelte erscheinung wonach geminata in schwach flektierenden nominalstämmen besonders häufig auftritt.” (1884: 169).⁴⁹

In other words, the mere fact that long stops are significantly overrepresented in the *n*-stems clearly substantiates the supposed link between Proto-Germanic gemination and nasal suffixation elsewhere. The exact effects of this gemination on the morphology of the *n*-stems is investigated in the present chapter.

⁴⁹ Accepted Lühr (1988: 191), Kortlandt (1991: 1), Rasmussen (1989b: 252).

4.1 Gemination as *grammatischer Wechsel*

The fact that geminates are frequent in *n*-stems throughout the Germanic dialects implies that Kluge's law strongly affected the paradigm of this noun class. It did not affect all cases alike, however. Kluge convincingly showed that the suffixal nasal would have been assimilated in cases with a zero grade of the suffix, whereas the cases with full-grade suffixes remained untouched. As a result, Kluge's law gave rise to a paradigm with a *grammatischer Wechsel* between geminated and ungeminated forms. The individual daughter languages often generalized one of these forms in order to regularize the paradigm. This seems to be the source of many dialectal incongruencies in Germanic. Kluge formulated the process as follows:

"Wenn neben ahd. *chnoto* (*chnodo*) das ags. *cnotta* steht, so lässt sich unschwer erkennen, dass das ags. wort das *a* der schw. declination vom nominativ **cnoða* (acc. **cnoðan*) bezogen hat, da germ. *knudn-* in der schwächsten stammform der schw. declination (got. *áuhsnê*, *abnê*) zu einem cons. stamm *knutt-* hätte führen müssen." (1884: 169)

The idea that Kluge's law created allomorphic paradigms, and thus explains the root variation displayed by many *n*-stems, has much to recommend itself. It connects the internally Germanic evidence for the law to a linguistically falsifiable hypothesis regarding the cross-dialectal typology of the *n*-stems. As such, it provides an elegant model that efficiently predicts the actual data.

A decisive argument in favor of Kluge's explanation was furnished by Lühr. Lühr (1988: 191) pointed to the fact that *n*-stems with roots in stops and resonants were affected in the same way and – most importantly – in the same morphological environments.⁵⁰ In other words, the allomorphy of e.g. OHG *chnodo* : OE *cnotta* m. 'knot' < **ǵnúť-ōn*, **ǵnut-n-ós* should not be separated from the one of OFri. *stera* : OE *steorra* m. 'star' < **h₂stérōn*, **h₂st(e)r-n-ós*. This parallelism evidently corroborates the hypothesis that the gemination of stops is the result of

⁵⁰ "*n*-Stämme mit **ll* < **l-n*, **nn* < **n-n* verhalten sich morphologisch wie die *n*-Stämme mit Doppeltenuis".

the same process that led to the doubling of the resonants, cf. **fulla-* 'full' < **plh₁-nó-*, **wullō-* 'wool' < **h₂ulh₁-néh₂-*.

4.2 Reconstruction of the *n*-stem paradigm

In order to fully understand the allomorphy of the *n*-stems, it is vital to determine exactly which cases of the Proto-Germanic paradigm did, and which cases did not receive a geminate under Kluge's law.

The first condition for the operation of this law, of course, demands that the *n* of the suffix was in direct contact with the final obstruent of the root. The suffix, in other words, had to be in the zero grade. On the basis of the Indo-European reconstruction of the *n*-stems (see chapter 2), geminates can be expected in the genitive singular in **-n-ós*, the genitive plural in **-n-óm* and the accusative plural in **-n-ís*. The zero grade was also found in the dative plural in **-n-miz*, but because of the vocalization of the *n* in this ending, by which it developed into PGm. **-ummiz*, Kluge's law did not operate in this case.

The determination of the cases with original gemination does not rely on the Proto-Indo-European reconstruction alone. Germanic, too, offers some clues on where in the original paradigm a long stop can be anticipated, though it must be said that the evidence is marginal. The reason for the paucity of the evidence obviously lies in the operation of *n*-assimilation itself: it is difficult to locate the original zero-grade *n*-suffixes in Germanic exactly because these suffixes were regularly assimilated by Kluge's law in most contexts.

Nevertheless, there are at least two sources from which information can be obtained. The first source consists of formations that split off from an original *n*-stem paradigm; a number of secondary derivations can be identified that demonstrably continue one particular case form, such as the genitive singular or the accusative plural.

The second source consists of *n*-stems that for phonetic reasons responded to Kluge's law in a special way: Kluge's law did not affect **s* and **z*, for instance, and some old *n*-stems with roots ending in a sibilant therefore have preserved the Pre-Germanic zero-grade suffixes. With these pieces of information, valuable data can be obtained about where exactly in the paradigm Kluge's law can be expected to have operated.

4.2.1 Paradigmatic split-offs

A survey of the *n*-stems in the different Germanic dialects reveals that *n*-stems are often accompanied by other formations with the same meaning, such as *a*- or *u*-stems. These derivational variants are best explained by assuming that the *n*-stem paradigm occasionally spawned a derivational off-shoot, and that the stem of each particular off-shoot depended on the case from which it sprouted.

It is beyond doubt, for instance, that the stem of ON *hōttr* 'hat', i.e. **hattu-*, split off from an accusative plural case **hattuns* < **kHt-n-ŋs*. This follows from the fact that the stem combines a geminated root with a *u*-suffix, and thus displays all the characteristics that can be expected in the accusative plural of an *n*-stem. It thus becomes clear that there is a correlation between certain stem formations, such as the *u*-stems, and the presence of gemination in the root. Thanks to this correlation, essential intra-Germanic information can be collected on the original allomorphy of the *n*-stems.

The possibility of using the *n*-stem split-offs for the reconstruction of the *n*-stem paradigm was already suspected by Osthoff (1882) and Van Helten (1905), but a systematic analysis was performed only much later by Lühr (1988). In *Expressivität und Lautgesetz*, Lühr discussed most of the correlations in the section *Übertritt in andere Flexionsklassen* (C, III). The conclusions reached in the following sections are by and by large in agreement with Lühr's findings.

4.2.1.1 Genitive split-offs

The most prevalent *n*-stems off-shoots are doubtlessly the *a*-stems. This is pointed out by the following doublets, each one of them demonstrating a clear correlation between *n*- and *a*-stems:⁵¹

- MHG *swirre* m. 'post' ~ OE *swe(o)r* m. 'pillar'
- ON *hrími* m. 'hoarfrost' ~ *hrím* n. 'id.'
- OHG *scorro* m. 'rock' ~ MHG *schor* m. 'id.'
- Far. *snípi* m. 'pointy nose' ~ *snippur* m. 'tip'
- OE *twiga* m. 'twig' ~ *twig* n. 'id.'
- Far. *knúki* m. 'steep rock' ~ *knúkur* m. 'id.'

⁵¹ See also Van Helten 1905: 226.

- MDu. *kratte* m. 'crate' ~ OE *cræt* n. 'cart'
- Far. *labbi* m. 'paw' ~ Nw. dial. *labb* m. 'id.'
- MLG *tagge* 'twig' ~ OSw. *tagger* m. 'spike'
- ON *hroki* m. 'pile' ~ ON *hrokr* m. 'id.'
- MHG *klotze* m. 'lump' ~ MHG *kloz* m. 'id.'
- MDu. *knoppe* m. 'knot' ~ OHG *chnopf* m. 'id.'
- ON *koddi* m. 'pillow' ~ OE *cod* m. 'bag'

In view of the high frequency of geminates in this kind of doublets, it was already suggested by Osthoff (1882: 300fn) that a transfer from the weak to the strong declension took place on the basis of the genitive singular case. This case is indeed suitable as the locus for such a cross-over, because it originally had an ending **-n-ós*, which in Proto-Germanic gave rise to a geminated root ending in **-az*. Since the geminated genitive no longer had the appearance of an *n*-stem form, but rather looked like the nominative of an *a*-stem, the motivation for the subsequent thematization is obvious.

Lühr did not treat this particular source of thematization, because, in accordance with Schindler's ideas on PIE morphology, she reconstructed the genitive ending as **-iz* < PIE **-es*. For the same reason, Schaffner (2001: 549, 553, 565), derived the genitive as **-CC-iz* rather than **-CC-az*. There are two reasons, however, why the reconstruction of the ending **-az* must be preferred over **-iz*. First, if the genitival *n*-stem ending had been **-iz*, we should have seen more *i*-stem derivations with *i*-mutation. Second, the difference between the OHG genitive *hanen* and the dative *henin* must result from the original opposition between the PGm. genitive **-enaz* and the dative **-eni* in Proto-Germanic (Prokosch 1939: 252-253; Kortlandt 1993: 20; Boutkan 1995: 282-4).

In addition to the genitive singular, the genitive plural can also have served as a source for secondary *a*-stems. It was demonstrated by Kortlandt (1978; 2007) that Lith. gpl. *-ų*, OCS *-ѣ* and Skt. *-am* (in *asmákam* 'ours') go back to a PIE gpl. ending **-om* rather than **-ōm*, the latter representing the thematic ending **-oHom*. The ending **-om* developed into **-an* in Proto-Germanic, and was lost in the separate daughter languages. After that, it was replaced by *-e* < **-ejan* < **-ei-om* in Gothic, and by **-ōan* from **-oHom*/**-eh₂-om* in Northwest Germanic.⁵²

⁵² Cf. Boutkan 1995:140.

The original *n*-stem gpl. must consequently be reconstructed as **-n-óm*, giving rise to a PGm. ending **-an* preceded by a geminated root. Just like the singular, this plural genitive is likely to have been a source for thematic split-offs.⁵³

An important characteristic of the *a*-stem split-offs is that they tend to vacillate between the masculine and neuter gender. With this tendency, the *n*-stem split-offs are in stark contrast with the primary *a*-stems, that only rarely change their gender:

- MDu. *kratte* m. 'crate' : ON *kartr* m. 'id.' ~ OE *cræt* n. 'id.'
- ON *hrími* m. 'hoarfrost' : OE *hrīm* m. 'id.' ~ ON *hrím* n. 'id.'
- Da. *tvige* 'twig' : G *Zwick* m. 'id.' ~ OE *twig* n. 'id.'
- OHG *rogo* m. 'roe' : ON *hrogn* n. 'id.'
- OHG *rabo* m. 'raven' : ON *hrafn* m. 'id.', OE *hræmn* m. 'id.', OHG *raban* m. 'id.'

It is perhaps conceivable that the gender difference was called forth by the inflectional difference between the gsg. in **-az*, which looks like a masculine nominative, and the gpl. in **-an*, which is identical to the neuter nasg. The apparently arbitrary difference between ON *kartr* m. and OE *cræt* n. can possibly be explained in such a way. Certainly, not all neuter split-offs would have to be derived from the gpl. Different factors may have played a role at the determination of the choice between the masculine and neuter gender, such as the semantics of the particular word. In the case of ON *hrafn* m. 'raven' < **hrabna-*, for instance, the neuter gender would have been unsuitable. Still, it cannot entirely be excluded that, in other cases, the gender of a genitive split-off was assigned on the basis of the origin of the thematization – the singular or the plural.

4.2.1.2 Accusative split-offs

A different derivational correlation is implied by the frequent occurrence of *u*-stems beside *n*-stems, as was recognized by Van Helten (1905: 225;

⁵³ According to Osthoff (1882: 301), the *a*-stem ON *knútr* 'knot', which coexists with the *n*-stems OHG *chnodo* and OE *cnotta*, was created to the original gpl. *knúta* < **knūt-n-ón*, which resembles the thematic gpl., e.g. *daga*. Since, however, the ending was **-an* < **-om*, this type of analogy cannot be maintained.

also Lühr 1988: 200), e.g. *björn* m. 'bear' < **bernu*- beside OE *bera*, OHG *bero* < **beran*- and ON *ǫrn* m. 'eagle' < **arnu*- beside ON *ari* < **aran*-. Van Helten derived these *u*-stems from old *n*-stem accusatives in **-nuns* < **-n-ns*, reconstructing them as **b^hér-n-ns* and **h₂ér-n-ns*.

Additionally, there are some *u*-stems that show the operation of Kluge's law, e.g. ON *bqlkr* 'beam' < **balk^ku*-, ON *gqltr* 'boar' < **galt^tu*-, ON *hqttr* 'hat' < **hattu*-, ON *knqttr* 'ball' < **knattu*- and *svoppr* 'mushroom' < **swamp^pu*-. As Lühr rightly contends in *Expressivität und Lautgesetz* (1988: 200), these formations must have sprouted from the accusative plural case as well. In view of the geminates, they only seem to have differed from split-offs such as **bernu*- and **arnu*- in that they had a stressed ending. This type of accentuation can very well be old, especially since it seems to be confirmed by extra-Germanic pieces of evidence such as the Skt. acc.pl. *ukṣṇás* 'oxen'. It must therefore probably be concluded that the reconstructed accusative plurals **bernuns* < **b^hér-n-ns* and **arnuns* < **h₂ér-n-ns* arose under the influence of other cases with stressed full grades in the root.

4.2.1.3 Dative split-offs

There is marginal evidence for split-offs from the dsg. case in **-eni* < loc. **-én-i*. A relatively certain instance is ON *heðinn*, OE *heden* m. 'hood, chasuble' < **hadina*-. It is likely that this formation, with its combination of the **-in*- suffix and the operation of Verner's law, continues a dative **hadeni* < **h₂én-i* of an *n*-stem **haþan*- (cf. ON *hqttr* m. 'hat' ⇔ apl. **hattuns*).

Another example of such a dative-born formation is Go. *himins*, ON *himinn* 'heaven', which is based on the dative **hemeni* < **h₂kem-éni*- of the lost *mn*-stem **ahman*-, akin to Skt. *ásman*- m. 'stone, sky'. The pre-existence of this *mn*-stem is confirmed by the formation OE *he(o)fon*, OS *heþan* 'id.' < **hemna*-, which appears to have developed out of a genitive **hemnaz* (see p. 163).

Van Helten (1905: 225) pointed out that the dpl., which he reconstructed as **ummi* < **-ŋ^h-mís*, can have served as a potential source of *u*-stem derivatives. The vocalization of the *n* and the subsequent assimilation by the following *m* in this ending gave rise to a case form that no longer had the appearance of an *n*-stem. This is likely to have been the trigger for the transfer to the *u*-stems (cf. Lühr 1988: 200).

According to Van Helten, Go. *auhsau*, the oblique form of the expected *n*-stem nominative **auhsa* 'ox' (cf. Skt. *ukṣán-* 'id.'), is to be understood in such a way. The actual dative plural ending is not attested in the Gothic paradigm, but the ending **-ummiz* can nevertheless be retrieved from Old English. In this language, *oxum* occurs beside the more regular, and therefore probably more recent form *oxnum*.⁵⁴

Other possible examples are ON *stjǫlr* 'tail' < **stelu-* beside OE *ste(o)la* m. 'stalk' < **stelan-* and perhaps also ON *spjǫr-* 'spear' < **speru-* beside ON *sparri*, OHG *sparro* 'beam' < **sparran-*. An additional case may be represented by the etymological cluster of ON *lími* m. 'twig' < **līman-*, ON *limr* (apl. *-i, -u*) m. 'limb, twig' < **limu-*, ON *lim* nf. 'twig', Icel. *lim* n. 'foliage', OE *lim* n. 'limb, twig'. The different formations could theoretically all go back to an old *mn*-stem **līmō*, gsg. **limenaz*, dpl. **limummiz* < **léi-mŋ*, **li-mén-(o)s*, **li-mŋ-mis*.

4.2.2 Special cases

As has been mentioned above, the Germanic evidence for zero-grade suffixes is scarce. Nevertheless, some clues about the original presence of such suffixes can be collected from a number of special *n*-stems. One *n*-stem that escaped the removal of the zero grade, and thus provides information on its location in the original paradigm, is the old hysterokinetic word for 'ox'. In this paradigm, Kluge's law did not operate, simply because it did not affect root-final **s*. Then, there are a number of *mn*-stems that in spite of there suffix do show the effects of Kluge's law; it appears that this is the result of the loss of **m* in cases that had a zero-grade suffix **-mn-*. Finally, there is one *jan*-stem that bears clear signs of suffix ablaut, including a zero grade. All these pieces together offer some valuable information on where exactly in the paradigm gemination is to be expected.

4.2.2.1 PGm. **uhsan-* 'ox'

The pre-existence of zero-grade endings in Germanic does not have to be ascertained on the basis of the Proto-Indo-European state of reconstruction only. It can also be surmised on the basis of the paradigm

⁵⁴ Van Helten also mentions Go. *auhsum*, but this form was emended to *auhsnuns* by Ebbinghaus (1972: 10).

of PGm. **uhsan-* 'ox', an old hysterokinetic *n*-stem. The inflection of this word formed a sub-type of its own, something that is particularly clear in Gothic, Nordic and Anglo-Saxon. Remarkably, it appears to have completely generalized the zero grade of the suffix in the plural. This has nothing to do with the fact that **uhsan-* was of the hysterokinetic type, but rather with the fact that Kluge's law did not affect sibilants. As a result, the zero-grade suffix was regularly preserved in the weak cases. It must subsequently have spread to other cases, thus giving the word for 'ox' an appearance that radically differed from the usual hysterodynamic *n*-stems.

	PGm.	Gothic	ON	OE
nsg.	<i>*-ēn</i>	-	<i>uxi, oxi</i>	<i>oxa</i>
gsg.	<i>*-naz</i>	-	<i>uxa, -a</i>	<i>oxan</i>
dsg.	<i>*-(e)ni</i>	<i>auhsau</i>	<i>uxa, -a</i>	<i>oxan</i>
asg.	<i>*-(e)nun</i>	<i>auhsau</i>	<i>uxa, -a</i>	<i>oxan</i>
npl.	<i>*-niz</i>	-	<i>yxn, øxn, uxor</i>	<i>æx-, ex-, oxan</i>
gpl.	<i>*-nan</i>	<i>auhsne</i>	<i>yxnna</i>	<i>ox(e)na</i>
dpl.	<i>*-ummiz</i>	-	<i>oxnum</i>	<i>ox(n)um</i>
apl.	<i>*-nuns</i>	<i>auhsnuns</i>	<i>yxn, øxn, uxa</i>	<i>oxan</i>

In Gothic, the paradigm is incomplete, but the gpl. points to **-n-eiom* < **-n-om* and the apl. to **-nuns* < **-n-ns*.⁵⁵ The zero-grade gpl. ending **-nan* < **-n-om* can also be reconstructed for Old Norse and Old English; it seems unlikely, at least, that the Old Norse forms developed out of a full-grade ending by syncope, especially in view of Go. *auhsne*. The npl. can be reconstructed on the basis of umlauted forms in ON and OE, which point to **-niz* < **-n-es* (= Go. ***auhns*). This ending apparently replaced the usual ending **-eniz* (cf. Skt. *ukṣāṇas* < **uks-én-es*). The Gothic dasg. form *auhsau* has an *u*-stem ending, which must have been introduced analogically on the basis of the original dpl. **uhsummiz* < **uks-ṇ-mis* as continued by OE *oxum* (Hellquist 1905: 225; Lühr 1988: 200), for which see § 4.2.1.3.

⁵⁵ The expected outcome of **uks-n-ns* is not **uksṇs* > **uhsuns* (pace Bammesberger 1990: 170 fn.).

Everything taken together, the paradigm of ‘ox’ gives proof of a gpl. in **-nan* < **-n-om*, a dpl. in **-ummiz* < **-n-mis* and an apl. in **-nuns* < **-n-ns*, all with zero grade of the suffix. As a result, Kluge’s law can be expected to have given rise to geminates in the same cases in the paradigms of other *n*-stems.

4.2.2.2 An old *jan*-stem

In theory, the ablaut of the *n*-stems should also have applied to the *jan*-stems. This would have yielded paradigms with a suffix alternating between **-jō* < **-iōn* in the nominative, **-inaz* < **-in-ós* in the genitive and **-jeni* < **-ién-i* in the dative. In view of the marginal evidence for this type, Beekes (1985: 48-51) explicitly claimed that Proto-Indo-European did not have a *ion*-suffix. However, the stem variation of the West Germanic word for ‘juror’ is probably best explained by indeed reconstructing a *jan*-stem with ablaut in the suffix:

**skapjō*, **skapinaz*, **skapjeni* ‘juror’

- **skapjan*:- OHG *scepfō* ‘iuridicus, conditor’, MHG *schepfe* m. ‘juror’
- **skapina(n)*:- OHG *sceffin(o)* ‘scabinus, iudex’, MHG *scheffene* m. ‘juror’, G *Schöffe*, OLFra. *skepeno* ‘iudex’, MLG, MDu. *schepen(e)* m. ‘juror’⁵⁶ (= OFri. *skep(p)ena*), Du. *schepen* ‘id.’⁵⁷
- **skapjena(n)*:- MHG *schepfen(e)* m. ‘id.’

PGm. **skapjan*- is clearly derived from the verb **skap(j)an*-, cf. G *schaffen*, *schöpfen* ‘to create’. Apparently aiming at the vacillation between geminated and ungeminated variants, Seebold (2002: 822) called the formal details of the word obscure.⁵⁸ Still, this vacillation can well be explained with the help of regular West Germanic gemination before **j*, assumed that it operated in the nominative **skapjō*, but not in the genitive **skapinaz*. As a result, the PWGm. paradigm received an alternation of geminated and ungeminated forms: nom. **skapjō*, gen.

⁵⁶ Lübben 325; Verdam 517.

⁵⁷ Franck/Van Wijk 582.

⁵⁸ Kluge/Seebold 822: “Das Wort gehört wohl zu *schaffen*, *schöpfen* und könnte »der Andordnende« bedeuten; die morphologischen und semantischen Einzelheiten sind aber unklar.”

**skapinenaz*. These two cases presumably gave rise to the formations OHG *scepfō* < **skapjan*- and MDu. *schepene* < **skapina(n)*-. MHG *schepfene*, on the other hand, appears to have developed from a form **skapjenan*-, which could have split off from the dative case.

4.2.2.3 Old *mn*-stems with geminates⁵⁹

There are at least three Proto-Germanic *mn*-stems that show the effects of Kluge's law. In all of these instances, the zero-grade suffix *-*mn*- seems to have been reduced to *-*n*- in the weak cases, probably due to dissimilation against labial elements in the root. The remaining *n*-suffix was assimilated under Kluge's law, and thus gave rise to a geminate. Cases with geminated and ungeminated roots seem to have occasionally contaminated each other.

**budmēn*, **buttaz* 'bottom'

- **budma*-: OE *bodan* m. 'id.', OFri. *bodem* m. 'id.'
- **buttma*-: ON *botn* m. 'id', OE *botm* m. 'id.'
- ?**buþma*-: OHG *bodem* m. 'id.', G *Boden* 'id.'

Kluge already in 1884 suggested that the consonantal interchange of OFri. *bodem* < **budma*- with ON *botn*, OE *botm* < **bottma*- resulted from contamination of the *mn*-stem **b^hud^h-mēn* > Gr. πυθμήν with the *no*-stem **b^hud^h-nó*- > Skt. *budhná*-, Lat. *fundus*. This contamination is best understood by assuming that both forms originally belonged to the same paradigm, i.e. **budmēn*, **buttaz* > **b^hud^h-mēn*, **b^hud^h-mn-ós* (Kroonen 2006). A crucial process at the development of the root alternation in Germanic is the loss of the *m* in the genitive. The Sanskrit and Latin forms seem to indicate that the *m* was lost in the genitive at the Proto-Indo-European stage; this explains the **tt* of OE *botm* as well as the Latin and Sanskrit thematizations.

Incidentally, it is possible to think that support for the glottalic chronology 1) Verner, 2) Kluge, 3) Grimm as advanced by Kortlandt can be subtracted from the variant OHG *bodem* 'bottom' < **buþma*-, which with its **p* cannot be the regular outcome of PIE **b^hud^h-men*-. Since the variant OE *botm* < **buttma*- must be a conflation of the PGm. nominative

⁵⁹ This subsection is based on Kroonen 2006.

**budmēn* < **b^hud^h-mēn* with the genitive **buttaz* < *b^hud^h-(m)nó-s*, it can similarly be hypothesized that **buþma-* developed out of an earlier conflation **butma-* with a shortened geminate, as I have argued in my MA thesis (2002). This would mean that the fricativization of the PIE voiceless stops postdated Verner's law, Kluge's law, as well as the devoicing of the glottalized series (including the newly arisen geminates) under Grimm's law. Indeed, this potential piece of evidence for the Glottalic Theory and the pertaining chronology of the sound shifts was also recognized by Kortlandt (2007) some years later.

An important difficulty facing the glottalic interpretation of **buþma-*, however, is that the **þ* of OHG *bodem* can also be of Proto-West Germanic rather than of Proto-Germanic date. It may well be comparable to instances of **f* < **þ* in e.g. OHG *weval* : MHG *webel* n. 'weft' < **webla-* (< **ueb^h-lo-*) and *scūfla*, *scūvala* : *scūbla*, G *Schaufel* 'shovel' < **skūblō-*, on the basis of which it was argued by Kluge (1883: 98) and Bahder (1903: 258-65) that voiced fricatives were devoiced before resonants under certain circumstances.⁶⁰ Within the Glottalic Theory, however, "Bahder's law" is impossible, because it claims that Proto-Germanic never had voiced fricatives in the first place. Thus, the traditional and the glottalic theory prove to be completely incommensurable.

****hrīfmō*, **hrīp^aaz* 'rime'**

- **hrīma(n)-*: ON *hrím* n., *hrími* m. 'id.'⁶¹, OE *hrīm* m. 'id.', MDu. *rijm* m. 'id.'⁶², Du. *rijm* 'id.'⁶³, G Cimb. *raim* m. 'id.'⁶⁴
- **hrīp^aan-*: OHG *riffo* m. 'id.', G *Reif* 'id.'⁶⁵, Cimb. *raifo* m. 'id.'⁶⁶, OS *hrīpo* m. 'id.', MDu. *rīp(e)* mn. 'id.'⁶⁷, Du. *rijp* 'id.'

The original inflection of the Germanic word for 'rime' was similar to the one of 'bottom'. It, too, has a range of variants in the Germanic dialects,

⁶⁰ Cf. also Schaffner 2001: 263-4.

⁶¹ De Vries 1962: 256.

⁶² Verdam 495.

⁶³ Franck/Van Wijk 548.

⁶⁴ Schmeller/Bergmann 221.

⁶⁵ Kluge/Seebold 754.

⁶⁶ Schmeller/Bergmann l.c.

⁶⁷ Verdam 496.

e.g. ON *hrími* m. 'rime' < **hrīman-* vs OHG *rīffo* m. 'id.' < **hrīpan-*. The easiest way to explain this variation is to reconstruct a single paradigm for both formations, i.e. either an amphikinetic **hrīfmō*, **hrīp^haz* < **kréip-mōn*, **kr(e)ip-(m)n-ós* or a hysterokinetic **hrībmēn*, **hrīp^haz* < **kriHp-mén*, **kriHp-n-ós* (Kroonen 2006). It seems clear, at any rate, that the *m* was lost in the weak cases, the result being a form in which Kluge's law could operate. Conversely, the root final labial, i.e. **f* or **b* was assimilated by the *m*-of the suffix in the cases without geminates.

****piþmēn*, **pittaz* 'pith, root'**

- **piþman-* (and **pittman-*?): Du. dial. *pessem*, *pettem* 'root, field horsetail'⁶⁸, Du. *peem* 'root (of grasses)'⁶⁹
- **piþan-*: OE *piða* m. 'pith'⁷⁰, EDu. *pee* 'radix edulis', *peēn* 'agrostis, gramen nodosum', Du. *peen* 'carrot'⁷¹
 ⇒ **piþaka-*: MLG *ped(d)ik* m. 'pith'⁷², WFri. *pich*,
 piid, *piik* 'pith, stone'⁷³
- **pitta(n)-*: MLG *pit(te)* 'pith, core, strength'⁷⁴, MDu. *pit(te)* mf., *pit* n. 'pith, kernel', EDu. *pit(te)*, *pette* 'medulla arboris, nucleus', Du. *pit* 'seed, stone, kernel, spirit'⁷⁵, ?G Fra. *pfitze* f. 'pimple'⁷⁶

The co-existence of OE *piða* and MLG, MDu. *pitte* is suggestive of an *n*-stem **piþō*, **pittaz*. In view of Du. dial. *pessem*, *pettem*, Du. *peem*, it is further conceivable that this hypothetical *n*-stem sprouted from an even older hysterokinetic *mn*-stem with zero grade of the root throughout the paradigm. If this is correct, the *m* must have been dissimilated in the cases with zero grade of the suffix, like in the paradigms of **budmēn*, **buttaz* 'bottom' and **hrīfmō*, **hrīp^haz* 'hoarfrost'.

⁶⁸ Vercoullie 261; Weijnen 154; WLD I, 5, 121-2.

⁶⁹ Vercoullie 259.

⁷⁰ Bosworth/Toller 774.

⁷¹ Franck/Van Wijk 494.

⁷² Lübben 129.

⁷³ Zantema 1, 747.

⁷⁴ Lübben 276.

⁷⁵ Franck/Van Wijk 504: "Wsch. met *tt* uit idg. *tn*."

⁷⁶ Schunk 212.

The variation of Du. *pessem* and *pettem* points to a form **piþþman*- with West Germanic gemination before *m*, as *-þþ-* developed into both *-ss-* and *-tt-* in Dutch, depending on the dialect (cf. Du. *adem*, dial. *asem* ‘breath’ < **ēþma-*, Du. *klis*, *klit* ‘tangle’ < **kliþþōn-*, see p. 235). The variant *pettem*, on the other hand, can also have adopted the geminate **-tt-* of the oblique, just like OE *botm* must be a contamination of the nom. **budmēn* with the gen. **buttaz*. The reality of the root **pitt-* indeed seems to be corroborated by the Middle Franconian form *pfitze*, but only if its meaning ‘pimple’ actually developed out of the more general denotation ‘core’.

Finally, it is to be noted that EDu. *pee* ‘root’, Du. *peen* ‘carrot’ (with *-n* from the plural), which is generally assumed to be of obscure origin⁷⁷, is actually formally identical to OE *piðā* < **piþan-*.

****heuhmō*, **hukkaz* ‘pile’?**

- **heuhman-*: Go. *hiuhma* m. ‘pile’
- **hukka-*, *-ōn-*: MLG *hocke* f. ‘sheaf, pile of hay’⁷⁸, Tyr. *hock* m. ‘sheaf’⁷⁹

PGm. **hukka-* is derived from **kug-nó-* by Fick, Falk and Torp (p. 91), who connect it to Lith. *kúgis* ‘pile of hay’ and Lat. *cumulus* ‘pile’ (< **kug-*). Alternatively, it can be linked to Go. *hiuhma*, which e.g. Feist (1923: 191–2) compares to *huhjan* ‘to collect’ and *hauhs* ‘high’. By assuming an original paradigm **kéuk-mōn*, **kuk-(m)n-ós*, both formations can alternatively be analyzed as off-shoots from one single etymon. Again, the loss of the *m* in cases with the zero grade of the suffix may have triggered Kluge’s law, thus giving rise to a paradigm with a consonant alternation.

4.2.3 Evaluation

To sum up, the intra-Germanic evidence points to the genitive (singular and plural) and the accusative plural as the cases in which Kluge’s law operated. The author thus largely agrees to the analysis of the split-offs

⁷⁷ Cf. Franck/Van Wijk 494; Philippa/De Brabandere/Quak 518–9.

⁷⁸ Lübben 146.

⁷⁹ Schöpf/Hofer 270.

that was offered by Lühr (1988: 199). Some minor differences concern the reconstruction of the vowels in the endings. In view of the umlaut in root nouns of the type ON *bók*, gsg./npl. *bækr* 'book' < **bōkiz*, it must be assumed that PIE *-es became PGm. *-iz in both the genitive singular and the nominative plural (cf. Boutkan 1995: 260). However, the genitive singular in *-iz seems to have been confined to the inflection of the root nouns. The *n*-stems rather had the singular genitive ending *-az < PIE *-os in view of the gsg./dsg. ending OHG -en, which clearly points to *a*-umlaut. In the plural, the genitive ending is to be reconstructed as *-an < *-om rather than *-ōn < *-oHom (see § 4.2.1.1). Finally, although it is difficult to determine whether the accusative plural was *-uns or *-unz, the former variant is adopted in view of the vocalic apl. endings in the Old Norse *u*-stems, cf. *fjǫrðr*, npl. *firðir*, apl. *fjǫrðu* 'fjord'.

	Lühr	Kroonen	PIE
nsg.	CVC-ě/ǫn, -ō	CVC-ēn, -ōn	*-én, -ōn
gsg.	CVCC-(e/a)ne/az	CVCC-az	*-n-ós
dsg.	CVC-e/ani	CVC-eni	*-én-i
asg.	CVC-anun	CVC-e/anun	*-é/on-m
npl.	CVC-anez	CVC-e/aniz	*-é/on-es
gpl.	CVCC-(a)nōn	CVCC-an	*-n-óm
dpl.	CVC-u/a(n)mi/az	CVC-ummiz	*-ŋ-mis
apl.	CVCC-(a)nunz	CVCC-uns	*-n-ŋs

4.3 Paradigmatic analogy

As has been argued in the preceding sections, the *n*-stems were affected by Kluge's law in such a manner that they regularly developed a *grammatischer Wechsel* of singulates and geminates in their paradigm. The genitive singular and plural as well as the accusative plural received a voiceless long stop, the other cases preserved a singulate. With this paradigmatic alternation, it is possible to account for much of the cross-dialectal root variation as presented in the introduction:

- Swi. Visp. *toxxa* : OHG *tocha* f. 'doll' < **dukō*, **dukkaz*
- Icel. *hjari* : ON *hjarri* m. 'hinge' < **herō*, **herraz*
- MLG *strote* : *strotte* f. 'throat' < **strutō*, **struttaz*

- G *Truhe* : Swi. *trukxa* f. 'trough' < **pruhō*, **prukkaz*
- Sw. dial. *råga* : MDu. *roc* m. '(hay)stack' < **hrugō*, **hrukkaz*
- OE *piða* m. : MLG, MDu. *pitte* mn. 'kernel, core' < **piþō*, **pittan-*
- etc.

However, since Kluge's law only produced voiceless geminates, the frequent fricative and voiced geminates in the *n*-stems still remain unaccounted for. The Germanic dialects, however, offer a plethora of *n*-stems with such long fricatives and voiced stops. Consider the following instances:

- MHG *krebe* m. 'basket': *krepe* f. 'id.' < **kreban-*, **krebbōn-*
- OHG *chratto* : *chretzo* m. 'basket' < **kraddan-*, **krattan-*
- Go. *fauho* : OE *fogge* f. 'vixen' < **fuhōn-*, **fuggōn-*
- MHG *made* m. 'maggot' : *matte* f. 'moth' < **maþan-*, **maþbōn-*
- Icel. *rjúp-keri* : -*karri* m. 'male ptarmigan' < **kazan-*, **kazzan-*⁸⁰

In fact, the cross-dialectal variation can be even more complicated. A lot of *n*-stems actually exhibit more than two root variants. It is not uncommon, even, that as many as four different roots must be reconstructed for what seems to have been just one single Proto-Germanic etymon:

- OHG *chnabo*, OE *cnafa* : MHG *knappe* : *knapfe* : OFri. *knapa*, OE *cnapa*, MLG, MDu. *knape* 'boy' < **knab(b)an-*, **knap(p)an-*
- Du. *knaak* : dial. *knaag* : *knag* 'knob, big coin' < **knakan-*, **knag(g)an-*
- MHG *lade* m. 'plank' : *lat(t)e* f. 'lath' : MLG *late* f. 'sprout', OHG *latza* f. 'plank, twig' < **laþan-*, **laþbōn-*, **ladōn-*, **lat(t)ōn-* etc.

As Kluge's law only accounts for the variants with voiceless geminates in a regular way, the question is how the singulates in **knapan-*, **latōn-*, **knakan-* and the geminates in **knabban-*, **laþþan-*, **knaggan-* must be explained.

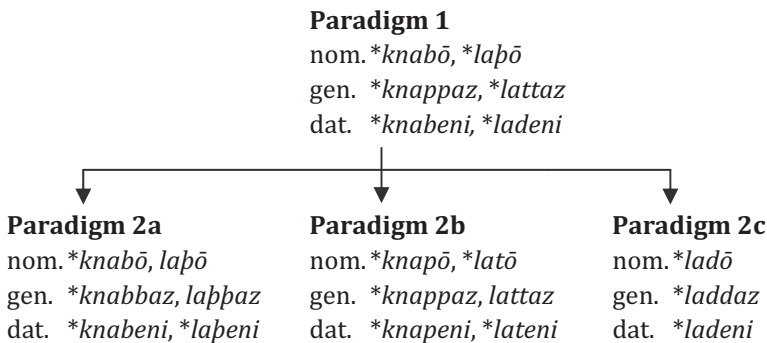
⁸⁰ With z-fronting only in the singulate form *keri* < **kazan-*?

4.3.1 Kluge's *Associationen*

A solution to the question of the wild root variation of the *n*-stems was first formulated by Kluge himself. Kluge (1884: 176) proposed to explain the irregular singulates and geminates by assuming that the regular root allomorphs contaminated each other in the original paradigm:

“Dass neben ahd. *chnabo* eine form *knapp-* (aus *knabn-*) denkbar ist, ergibt sich aus dem bisherigen. Diese doppelformen führten durch association zu zwei neuen formenpaaren: man bildete zu *knabo* eine neue geminationsform *knabba* oder zu der geminierten form *knapp-* im anschluss aus *knabo* eine form mit einfacher consonanz *knapa*: jenes ist das mhd. *knappe*, dies das ags. *cnapa*.”

Kluge's explanation, which with its combination of sound law and analogy was a showcase of the Comparative Method, turned out to be capable of predicting the complete amount of root variants. The introduction of the secondary singulates and geminates is best understandable by assuming that the original paradigm was split up into two new paradigms: one paradigm generalized the nominative consonantism by doubling it in the weak cases, the other generalized the genitive consonantism by shortening it in the nominative. It is even conceivable that paradigms containing a *grammatischer Wechsel* due to Verner's law could give rise to a third split-off based on the dative root. Paradoxically, the attempts to reduce the allomorphy by leveling the articulation of the consonants thus resulted in an overall increase of the potential amount of root variants:



Of course, it is not necessary to assume that all of the potential root variants actually existed beside each other in each and every dialect. On the contrary, the fact that different contaminations are found in the separate dialects clearly indicates that the original allomorphy was at least partially leveled out by the different dialects independently, i.e. well after the disintegration of the Proto-Germanic unity.

4.3.2 Reception of Kluge's *Associationen*

Initially, Kluge's configuration became widely accepted. It was included in many handbooks and can be found in e.g. Streitberg's *Urgermanische Grammatik* (1896: § 127A), Wright's *Old English Grammar* (1925: § 256)⁸¹, Hirt's *Handbuch des Urgermanischen* (1931: 1, § 60) and in *A comparative Germanic grammar* by Prokosch (1939: § 22). An especially fervent supporter was Bloomfield, who in 1925 devoted a complete article to the defense of Kluge's law. However, during the same time that Kluge's findings became adopted by quite a few mainstream Germanicists, opposition started to grow against the idea of the paradigmatic analogies. The main criticism of the opponents was that the contamination of fricative **b* and occlusive **pp* given their different articulations could never have led to the analogical consonantisms **bb* and **p*.

4.3.2.1 Kauffmann

One of Kluge's strongest opponents was Friedrich Kauffmann. Just three years after the publication of *Die germanische Consonantendehnung*, Kauffmann launched a strongly worded attack on Kluge's *Associationen*. In the article *Zur Geschichte des germanischen Consonantismus* (1887), he explicitly accepted the Proto-Germanic origin of the voiceless geminates, but at the same time rejected the analogical rise of long voiced and fricative obstruents, as he thought the contaminations proposed by Kluge to be in contradiction with the linguistic principles of analogy.

⁸¹ "Doubling of consonants by the assimilation of post-consonantal *n* to the preceding consonant also regularly took place in the weak declension of nouns, as sing. nom. **lapō, lappet*, acc. **lapan(un)*, beside gen.pl. **lapnō(n) > *lappō(n)* [...]. This interchange between the single and double consonants gave rise to levelling in a twofold direction, so that one or other of the forms was extended to all cases".

“Eine derartige formschöpfung halte ich für nicht vereinbar mit den allgemeinen anschauungen, die sich für die associationsbildungen der lebenden sprachen festgesetzt haben.” (p. 509)

By way of an alternative, Kauffmann proposed that the secondary geminates arose much later, by West Germanic gemination before *r*, *l*, and – allegedly – before *n* (1887: 531). Kauffmann assumed that in the original paradigm **knaþō*, **knappaz*, the zero grade of the suffix was at some point restored. This gave rise to a secondary genitive **knaþnesa*, which allegedly developed into PWGm. **knabbnes* by West Germanic gemination. Thus, the gemination of the voiced stops in the *n*-stems would be parallel to the gemination in, for instance, PGm. **akraz*, **akresa* ‘field’, which became PWGm. **akr*, **akkres* (cf. E *acre* : G *Acker*). The doubling of voiceless fricatives, such as in OE *moppe* ‘moth’ < **muppan-*, Kauffmann ascribed to the same process.

Kauffmann's alternative to Kluge's contaminations was accepted by some linguists, for example by Braune, who adopted it in his *Althochdeutsche Grammatik* (1891: § 96b). Soon, though, it became clear that Kauffmann's hypothesis contained critical fallacies. It was demonstrated by Van Helten (1905: 215-6) that the West Germanic gemination before *n* is disproved 1) by counter-examples such as OE *regn* ~ OHG *regan* ‘rain’, OE *wægn* ~ OHG *wagan* ‘wagon’, ON *hrafn* ~ OE *hræfn* ‘raven’ and 2) by the fact that many examples with voiced geminates have a clear Proto-Northwest Germanic origin, cf. ON *toddi* ‘tuft’ ~ Du. *tod(de)* ‘rag’, Nw. dial. *kodde* ~ MDu. *codde* ‘testicle’, ON *krabbi* ~ OE *crabba* ‘crab’ etc. Because of these counter-arguments, Kauffmann's hypothesis never reached a reasonable level of acceptance (cf. Hellquist 1905: 33; Luick 1964: 825; Lühr 1988: 197).

Kauffmann's explanation of the analogical singulates was even more fantastic. In order to account for the secondary singulates of e.g. ON *knapi* and OE *cnapa*, he assumed that interlexical contamination took place between the weak nominative **knaþō* and a hypothetical, strong genitive split-off **knappaz*, which supposedly developed into **knappz* by syncope and subsequent shortening of the geminate (1887: 532). This solution can obviously not be maintained either. First, the use of interlexical *Associationen* seems far more implausible than the paradigmatic contaminations proposed by Kluge. Second, the PGm.

ending **-az* did not develop into PWGm. **-z*, but rather into **-a* by the loss of the word-final sibilant. As a result, regular shortening of geminates cannot possibly have taken place before this ending.

In the end, none of the explanations advanced by Kauffmann can be uphold. His criticism of Kluge's paradigmatic contaminations, on the other hand, did make it into mainstream linguistics, as will be pointed out in the following sections.

4.3.2.2 Lühr

Another sceptic of Kluge's *Associationen* was Rosemarie Lühr. In *Expressivität und Lautgesetz*, which is basically a defense of Kluge's law, she argued against the analogical rise of the voiced and voiceless long fricatives, i.e. PGm. **bb*, **dd*, **gg* and **ff*, **hh*, **pp* (1988: 206-8). The analogical introduction of a secondary singulate (e.g. *knabō*, **knappaz* >> **knapō*, **knappaz*) was accepted by Lühr, because the co-existence of the alternations 1) *T: TT*, 2) **D: TT*; and 3) **p: TT* provided a model for such an analogy. The replacement of 1) **D: TT* by **D: DÐ* and 2) **p: TT* by **p: pp*, on the other hand, was rejected by Lühr in absence of a linguistically real proportion. As a consequence, Lühr had to infer that "bei diesen Lautungen nach einer nicht mit der *n*-Geminatation in Zusammenhang stehenden Erklärung gesucht werden muß" (p. 208). In practice, this meant that the long fricative of OE *pohha* m. 'bag' < **puhhan-* had to be explained as onomatopoetic (1988: 270), while the geminates of ME *latthe* 'lath', OE *moppe* f. 'moth' < **muppan-* and MDu. *clisse* f. 'burdock' < **klippōn-* were assumed to be continuations of PGm. clusters, viz. *-hp-* (p. 252, 255).⁸²

What can be brought against Lühr, is that it seems more economical to assume that analogy within the *n*-stem paradigm gave rise to double fricatives than to lexically isolate roots with long fricatives from the other variants. The alternation of OHG *chletha* 'burdock' < **klippōn-* and MDu. *clisse* 'id.' < **klippōn-*, for instance, is completely parallel to the

⁸² This view is adopted by Schaffner in *Das Vernersche Gesetz* (2001). While accepting Kluge's assimilation of the *n* in cases with the zero grade of the suffix (p. 534), Schaffner rejects the analogical doubling of voiceless fricatives. As a result, he has to resort to the reconstruction of an independent formation **hridjan-* in order to account for MHG *ritte* 'fever' < **hriþpan-/hriðdan-*, even though it is morphologically close to OHG *rido* 'id.' < **hriþan-* (p. 549-552).

length opposition that exists in many other *n*-stems. This formal parallelism is a strong indication that the origin of the long fricatives is to be found within the morphology of the *n*-stems itself. The fact that a linguistically real proportion for the introduction of these secondary geminates was lacking, does not have to be detrimental to Kluge's *Associationen*. On the contrary, one could conversely argue that the rise of fricative geminates simply proves that the length opposition had been grammaticalized (see § 4.3.3).

4.3.2.3 Van Helten – Rasmussen

Another important opponent of Kluge's *Associationen* was Van Helten. Although dissatisfied with Kauffmann's hypothesis on the secondary geminates, Van Helten (1905) agreed to the latter's criticism of Kluge's contaminations. In view of the different articulations of **b̥* and **pp* in e.g. **knabō*, **knappaz*, Kauffmann (1887: 508) judged it unlikely that they could form a proportion according to which the analogical paradigms **knabō*, **knabb̥az* (> **knabbaz*) and 2) **knapō*, **knappaz* would have been created. Like Kauffmann, Van Helten offered an alternative to Kluge's solution; whereas Kluge assumed that the *n*-stem **knabō*, **knappaz* 'boy' gave rise to analogical paradigms **knabō*, **knabb̥az* > **knabbaz* and **knapō*, **knappaz*, Van Helten proposed to push the chain of analogies back into the pre-Proto-Germanic stage. In this way, his solution became diametrically opposed to the one proposed by Kauffmann, who assumed that the analogies took place in West Germanic rather than in the Proto-Germanic period.

According to Van Helten, the contaminations leading to PGm. **knabban-* took place between the occlusivation of **-b̥b-* and the devoicing of PIE **b* > PGm. **p*. The original paradigm **knabō*, **knappaz* thus would have regularly developed out of **gnabō*, **gnabbaz*. The variant **knabban-*, on the other hand, Van Helten explained from an analogical paradigm **gnabō*, **gnabb̥az* that was created posterior to the occlusivation of old **-b̥b-*, but anterior to the regular devoicing under Grimm's law (1905: 217). Conversely, Van Helten took the variant **knapan-* to have arisen from a paradigm in which the fricative **b̥* of the nominative **gnabō* was analogically replaced by an occlusive **b* from the regular genitive **gnabbaz* (l.c., fn. 1).

	nominative	genitive
PIE	<i>*gnób^h-ōn</i>	<i>*gnob^h-n-ós</i>
Grimm's law 1		
	<i>*gnábō</i>	<i>*gnaǵnás</i>
Verner's law		
	<i>*gnaǵō</i>	<i>*gnaǵnaz</i>
Assimilation		
	<i>*gnaǵō</i>	<i>*gnaǵǵnaz</i>
Occlusivation 1		
	<i>*gnaǵō</i>	<i>*gnabbaz</i>
Cross-contamination		
	<i>*gnaǵō/*gnabō</i>	<i>*gnaǵǵnaz/*gnabbaz</i>
Grimm's law 2		
	<i>*knaǵō/*knapō</i>	<i>*knaǵǵnaz/*knappaz</i>
Occlusivation 2		
PGm.	<i>*knaǵō/*knapō</i>	<i>*knabbaz/*knappaz</i>

In a similar vein, Van Helten assumed that the voiceless fricative geminates arose by the analogical introduction of a long fricative into the genitive. This analogy would have been based on the opposition of single and double voiced fricatives that had arisen in other *n*-stems by the doubling of the consonants prior to the loss of the nasal (1905: 223-4). According to Van Helten, this analogy was to be understood from an accentual split of the original paradigm e.g. **glíp-ōn*, **glíp-n-és* into 1) **glíp-ōn*, **glíp-n-es* and 2) **glíp-ón*, **glíp-nés*. When, after Verner's law and the initial doubling phase of Kluge's law, the second paradigm had developed into **glidón*, **glidǵnés*, the first paradigm **glípōn*, **glípnes* would have accordingly been remodeled into **glípōn*, **glípǵnes*. The formation **klippōn-*, attested as OHG *chletta* and MDu. *clisse*, would then be the continuant of the genitive of this tertiary paradigm.

Although Van Helten's hypothesis has the disadvantage that it requires two different waves of Proto-Germanic occlusivation of **-ǵǵ-*, and even a third one for High German, in which – after all – PGm. **ǵ* is represented as *h*, it is theoretically capable of accounting for the whole set of allomorphs that must be reconstructed for the *n*-stems. Van Helten's modification of Kluge's configuration was therefore largely accepted by, for instance, Hellquist, author of the Swedish etymological

dictionary, in his treatment of the *Nordiska verb med mediageminata* (1908).⁸³

Another, much later advocate of Van Helten's approach is Rasmussen, who discussed Kluge's law and its effect on the *n*-stems in two articles published in 1989. In these articles, Rasmussen argued in favor of the same kind of cross-contaminations as proposed by Van Helten. Rasmussen makes no reference to Van Helten's article, however, and seems to have arrived at the same solution independently:

"In der germanischen Entwicklung wurde das ursprünglich nur nach Schwundstufensequenzen reguläre Suffixallomorph /-n-/ des Instr. verallgemeinert, so daß sich zunächst die normalisierte Flexion **d^hrub^h-ōn*, Gsg. **d^hrub^h-n-ós* (⇒ **-és*) ergab, woraus dann durch Lautwandel **drúbōn*/**drubnés* > **drúbōn*/**drubnéz*, weiterhin durch einen neuen Ausgleich **drúbōn*/**drubnéz* und neuen Lautwandel **drubōn*/**drubbiz*, das schließlich mit der Lautverschiebung zu urgerm. **drupōn*/**druppiz* wurde[...]" (1989b: 253)

An important objection to Van Helten's modification of Kluge's chronology is the relatively great time depth that it requires. If the paradigmatic contaminations really took place before the final phase of Grimm's law, i.e. the devoicing of the PIE voiced stops, the resulting leveling of the original paradigms should have been anterior to the rise of Proto-Germanic phonology as it is known. However, if this were the case, the Germanic dialects would never have displayed the rich root variation that is actually found, because many of the root variants would already have been removed before the disintegration of Proto-Germanic.

In conclusion, the main issue with Kluge's *Associationen* is that the contamination of voiced fricatives and voiceless geminate stops is deemed untenable by Kauffmann, Van Helten and Lühr. The alternative explanations offered by these scholars do not convince either, however;

⁸³ Hellquist nevertheless rejected Van Helten's view that the long voiceless fricatives, such as OHG *chletto* 'burdock' < **klippan-* and OHG *ritto* 'fever' < **hriþþan-*, arose by analogy in the *n*-stems, explaining them as deverbative from hypothetical **klipbōn-* 'to stick' and OE *hriðian* 'to have a fever' (1908: 44).

they are either demonstrably flawed, such as in the case of Kauffmann and Van Helten, or they isolate the forms with the secondary geminates from homonymic variants with regular consonantisms, as was done by Lühr. It therefore seems preferable to retain the original configuration that was developed by Kluge.

The problem of the *Associationen* may, in fact, not be a problem at all. The idea that the paradigmatic interchange of **b̥* and **pp* gave rise to **bb* and **p* fully predicts the allomorphic variation that is attested across the Germanic dialects, and this seems to be the most important thing. The phonetic difficulties regarding the contaminations can easily be overcome. It is fully possible, for instance, that **bb* regularly developed out of analogical **b̥b̥*. In the traditional reconstruction of Proto-Germanic phonology, such an occlusivation must be assumed anyway for the voiced geminates that arose from West Germanic gemination before **j*, cf. OS *ribbi*, OE *ribb* 'rib' < **riḅja-*, *-ō-*. It would therefore be illogical to deny the possibility of such a development in an older phase.

A more radical approach would be to reverse the whole argument against Kluge's *Associationen*. Possibly, the very fact that paradigmatic contaminations took place may simply prove that PGm. **b*, **d* and **g* were plosives at least at some stage in Proto-Germanic. Although, the occlusive reconstruction of the *mediae* is controversial in Germanic Studies, this way of reasoning would certainly be unproblematic in the configuration developed by the glottalicist Kortlandt, who assumed that PIE **T* and **D* (= **Dʰ*) first merged into PGm. **D* by Verner's law, and that PIE **T* and **Ḍ* were subsequently fricativized and devoiced to **p* and **ʔT* under Grimm's law (see § 3.4.3).⁸⁴

4.3.3 From allomorphy to consonant gradation

In Proto-Germanic, the original allomorphy theoretically consisted of three different sub-types, i.e. a long voiceless stop (**TT*) opposing 1) a voiced stop (**D*), 2) a voiceless fricative (**p*) and 3) a voiceless stop (**T*). It is important to notice that, while the former two types (**D:TT*, **p:TT*) constituted a phonetically complex opposition consisting of more than one articulatory feature, the third type (**T:TT*) consisted of a simple opposition of length only.

⁸⁴ For a response, see Woodhouse 2003.

The length opposition was not only simpler than the other oppositions, it was also more universal, as it occurred in roots in stops and resonants alike. PGm. *n*-stems with roots in resonants form a large category, e.g. Icel. *hjarri* : ON *hjarri* m. 'hinge' < **herō*, **herraz*, OFri. *throtbol(l)a* m. 'Adam's apple' < **bulō*, **bullaz* etc. Since resonants in Proto-Germanic did not have any voiceless or fricative alternants to form a complex opposition with, the only possible opposition that remained was one of pure length.

What Kluge's analogies basically entailed, was the spread of the simple length opposition at the expense of the other two, phonetically more complex oppositions. Or to phrase it more accurately, the introduction of the secondary types **D:DD* and **P:PP* shows that the pure length opposition became dominant over the more complex oppositions. As a result, the allomorphy became considerably simplified.

In the previous section, some objections against the rise of the secondary geminates have been discussed. It was argued by Lühr, for instance, that there was no linguistically real model for the introduction of **DD* (> **DD*) and **PP*, and that Kluge's *Associationen* therefore cannot be maintained (§ 4.3.2.2). This objection can be circumvented, however, by assuming that, at some point, the dominance of the pure length opposition resulted in the grammaticalization of a paradigmatic length alternation. It led, in other words, to the introduction of a new morphonological process in Proto-Germanic morphology, i.e. consonant gradation.

In the literature, the term *consonant gradation* is usually applied to the Finno-Ugric languages in order to describe the allomorphy of e.g. Fi. *tukki* : gen. *tukin* 'beam, log', *oppi* : gen. *opin* 'doctrine' and *nukun* 'am sleeping' : *nukkuu* 'is sleeping'. These paradigmatic consonant alternations, however, are highly comparable to the ones in Germanic after the occurrence of Kluge's *Associationen*. In Finnish and Germanic alike, the allomorphy can be defined as a systematic consonantal interchange that gives morphological value to a phonological opposition. The application of the term *consonant gradation* to the Germanic *n*-stems therefore seems justified.

Obviously, there is no *direct* correlation between Finnish and Germanic consonant gradation. The two languages are unrelated and they stem from very different areas. Still, the fact that consonantal strength is morphophonologically relevant in Finno-Ugric, Germanic and

probably in Celtic as well, does not have to be entirely coincidental. One way to interpret this communality is to regard it as a *Sprachbund* feature that arose through language contact in the prehistoric period (cf. § 7.3). This is a possibility that still needs further investigation.

4.3.4 Dating the rise of consonant gradation

While it is obvious that a morphological opposition of length already existed in the proto-language, i.e. in *n*-stems with roots ending in resonants (**R:RR*) and voiceless stops (**T:TT*), it is more difficult to pinpoint the exact moment when the spread of the paradigmatic length opposition took place.

There are some indications that the evolution into fully-fledged consonant gradation must be situated in the Northwest Germanic period. This seems to follow from the complete absence of long fricatives and voiced stops in the *n*-stems attested in Biblical Gothic. The language only has two *n*-stems with geminates, both of them voiceless, viz. *atta* m. 'father' < **attan-* and *smakka* m. 'fig'. This absence stands in stark contrast to an abundance of fricative and voiced geminates in the Northwest Germanic dialects, some of which can actually be given a Proto-Northwest Germanic reconstruction, as was already noticed by Van Helten (1905: 215-6), e.g. ON *toddi* 'tuft', Du. *tod(de)* 'rag', Nw. dial. *kodde*, MDu. *codde* 'testicle', ON *krabbi*, OE *crabba* 'crab'.

Proto-Northwest Germanic examples with secondary -zz- should especially be mentioned here. Cases such as Icel. *rjúp-keri*⁸⁵: -*karri* m. 'male ptarmigan' < **kazō*, **kazzaz* and Far. *knasi* m. 'gnarl, bump': Nw. dial. *knarre* 'stub', ME *knarre* 'gnarl' < **knasō*, **knazzaz* are in clear violation of Kluge's law, which did not affect **s* (see § 3.3.2). Their occurrence in both North and West Germanic proves that the grammaticalization of the length opposition must be dated back at least to the Proto-Northwest Germanic stage.

The dating of the consonant gradation to the Northwest Germanic stage further seems to be confirmed by the lack of analogically shortened geminates, such as the already mentioned **knapan-*, **latōn-*, **knakan-* in Gothic. Again, this is contrasted by a multitude of cases in Northwest Germanic. An especially elucidating case of analogical singulation in

⁸⁵ With z-fronting of *a* to *e* in the singulate forms.

Northwest Germanic is represented by the opposition of **hamō(n)-* > ON *hōm*, OHG *hama*, MLG *hame* with **hammōn-* > OHG *hamma*, MHG *hamme* f. 'ham'. The etymon is usually reconstructed as **konh₂-meh₂-* (cf. Gr. κνήμη f. 'shin', OIr. *cnáim* 'leg' < **knh₂-m-*), showing the regular development of **-nm-* to **-mm-*. Since it is etymologically unsatisfactory to separate the non-geminated stem **hamō(n)-* from this formation, the best way to deal with the singulate *m* is to ascribe it to analogical degemination in a secondary paradigm **hamō*, **hammaz*.

Likewise, the singulate of Nw. dial. *hjáre* m. 'brain' cannot be directly explained from the formation **hersō*, **herznaz* < **kérh₂s-ōn*, **kérh₂s-n-ós*, which regularly developed into e.g. ON *hjarsi*, *hjassi* m. 'crown' and *hjarni* m. 'brain'. Instead, it should probably be regarded as an analogical alternant to Nw. dial. *hjarre* m. 'brain' < **herzan-*.

Yet, the Proto-Northwest dating of consonant gradation is contradicted by the occurrence of primary and secondary geminates in the Gothic onomastic material. A large number of Gothic personal names can be gleaned from the historical records, and it is in this non-Biblical corpus that we find commandors with names such as *Ibba*, *Faffo*, *Mammo*, *Oppa*, *Riggo*, *Wacca* (cf. the prosopographical index by Amory 1997). Also here belongs the name of the Goth *Micca*, which is mentioned by Jordanes (*Romana* 281).⁸⁶

The structure of these and similar names is typical – if not archetypical – of the Germanic hypocorisms, i.e. pet names that were derived from full names. These hypocorisms, which are highly frequent in the Northwest Germanic languages, were created by coining an *n*-stem with a root consisting of 1) an onset, 2) a short vowel and 3) a geminate. Thus, we get G *Sicko* < **Sikkan-* to *Sigmar* < **Sigi(z)mēraz*, *Fritz* < **Fritta(n)-* to *Friedrich* < **Friþurikaz*, *Lutz* < **Hlutta(n)-* to *Ludwig* < **Hludawikaz* etc. The mechanism is moribund in most modern Germanic languages, but still productive in Icelandic, even in names of non-Germanic origin, e.g. *Gunna* to *Guðrún*, *Nonni* to *Jón*, *Magga* to *Margrét*, *Solla* to *Sólrun*, *Stebbi* to *Stefán* etc.⁸⁷

⁸⁶ The etymology of this name is highly disputed (cf. Lippold 1998: 272-3), but this is exactly the point, of course; it is notoriously difficult to determine the full name behind any Germanic hypocorism because of the simplification of the root structure: e.g. **Friþu-rīkaz* 'Frederic' ⇒ G *Fritz* < **Frittan-*.

⁸⁷ The mechanism can even be applied to plain nouns: *lög-regla-n* 'the police' can be personalized as *lögga* 'cop', and *Morgunblað-ið* 'the Morning Paper' as *Mogga-*

In spite of the late coinage of most hypocorisms, which as a rule cannot be projected into Proto-Germanic, the system as a whole seems to have evolved out of the old Indo-European tradition of giving nicknames consisting of personalizing *n*-stems, cf. Lat. *Catō*, *Varrō*, *Nerō* and Gr. Στράβων, Πλάτων etc. This is corroborated by that fact that Germanic has many personalizing or individualizing *n*-stems that are structurally identical with the hypocorisms, e.g. OHG *chresso* 'groundling' to *chresan* 'to crawl' (Kuryłowicz 1957: 136) and ON *farri* 'landlouper, vagrant' to *fara* (Martinet 1937: 178). It is only logical, for this reason, to link the geminates of the Germanic hypocorisms to Kluge's law, which operated in the weak cases of the *n*-stem paradigm (cf. Hirt 1931: 93).

Consequently, the fact that Gothic just as much as the other Germanic languages has hypocorisms of the type *Ibbo*, *Micca*, *Oppa* and *Wacca* proves that 1) this language did have primary and secondary geminates and that 2) the rise of consonant gradation occurred at the Proto-Germanic rather than the Proto-Northwest Germanic level.

This leaves the problem of the poor attestation of geminates in Biblical Gothic. A consideration concerning the scarcity of the long stops in Gothic was given by Kuryłowicz (1957: 140), who argued that Wulfila may have found geminates inappropriate in the Gothic translation of the Bible because the words in which they occurred had a colloquial or informal flavor. This is a defensible explanation, as it is clear from the Northwest Germanic evidence that many *n*-stems, in particular the hypocorisms, had such a connotation. The register of the *n*-stems should probably be compared to that of modern English words ending in *-ie* or *-y* as in *doggy*, *cookie*, *junkie*, which originally belonged to more or less informal contexts. From this perspective, it is not surprising that the Heliand, like the Gothic Bible, is virtually void of Proto-Germanic geminates as well (cf. Gamm 1973: 90). PGm. **tt* only occurs in the word *skatt* 'treasure, money', **kk* only in *likkon* 'to lick', and **pp* only in the directional cluster formed by *upp*, *uppa* and *uppan* (see chapter 5). Still, it is certain from later stages of the Saxon language that they must have already been out there in the oldest period.

n. Examples of this more general type can also be found in modern Swedish, cf. the manifestly recent formations *sosse* < ***sussan-* and *nasse* < ***nassan-* for *socialist* and *nasist*.

4.3.5 Gemination of **n*

In the context of Kluge's law, the rise of roots with double **n* poses a problem. Double **n* is found in a small number of *n*-stems and heteroclitics:

- ON *kona*, gpl. *kvinna* f. 'woman' < **kwenō*, gpl. **kwinnan* (cf. OIr. *ben*, gsg. *mná* f. 'woman' < **g^wén-h₂*, **g^wn-éh₂-s*)⁸⁸
- Go. *sauil* n., *sunno* f. 'sun', dsg. *sunnin* mn. 'id.', ON *sól*, *sunna* f. 'id.', OE *sunna* m. 'id.', *sunne* f. 'id.' < **sōel*, dsg. *sunneni* (cf. Gr. Dor. ἄελιος 'id.', Av. *huuarē*, gen. *x^vēng* 'id.' < **séh₂u_ḡ*, **sh₂uéns* / *sh₂unós*)
- Go., OE *brunna* m. 'spring', OHG *brunno* m. 'id.' (cf. Gr. φρέαρ, φρεῖατος 'well' < **b^hréh₂ur*, **b^hrh₂un(t)ós*)

In the literature, these geminates are usually explained as resulting from generalization of the oblique stem, to which a secondary nasal suffix was added in the oblique, viz. **kwenō*, **kwin-n-an*, **sunō*, **sun-n-az*⁸⁹, **brunō*, **brun-n-az*.⁹⁰ This solution clearly contains a paradox. On the one hand, the Proto-Indo-European simplification of **h₁es-si* 'you are' to **h₁esi* (cf. Skt. *ási*, Gr. *εἶ*) seems to imply that the creation of the sequence **-n-n-* cannot have happened before Kluge's law, as the resulting long *n* would have been simplified before that time limit. On the other hand, it is not plausible either that an *n* could have been added after Kluge's law, because the zero-grade suffix should for the most part have disappeared exactly by this law.

One way around the paradox is to ascribe the gemination of the *n* in the given instances to early (i.e. Proto-Germanic) consonant gradation. If this is correct, it is no longer necessary to assume that the heteroclitics first generalized their oblique stems, e.g. **sun-* and **brun-*, and then created new geminated oblique stems **sunn-* and **brunn-* on the basis of these. This is unlikely in the first place, because the material shows no traces of the hypothetical singulate forms ***sunō* and ***brunō*. On the contrary, the heteroclisys of at least **sōel*, **sunnaz* was actually preserved

⁸⁸ Harðarson (1989), on the other hand, reconstructs ON *kvinna* as **kwenōnōⁿ*, arguing that its ending is no different from the one in e.g. *gumna* < **gumanōⁿ*.

⁸⁹ Brugmann 1906: 303; Wessén 1914: 68; Feist 1939: 347.

⁹⁰ Cf. Franck/Van Wijk 94.

until after the breaking up of Proto-Germanic, only to be abandoned in synchronic Gothic, where the old nominative *sauil* and the new nominative *sunno* occur side by side.

And yet, it cannot entirely be excluded that double *n* did arise through the addition of an *n*-suffix. At least, this is what seems to have happened in the case of the dative directional **innai* 'inside', for instance, which is best analyzed as the preposition **in* < **h₁en* plus an ancient suffix **-noi* (see chapter 6). The possibility therefore remains that double *n* arose together with other resonants such as **rr* and **ll* by the addition of an *n*-suffix.

5 Kluge's law and the directionals⁹¹

The effects of Kluge's law are not only visible in the *n*-stems. There is substantial evidence that this law triggered the rise of consonant gradation in the *directional system* as well. This system consists of the tendency of the directional adverbs to form paradigms containing a locative, an allative and an ablative case. Some of these cases were affected by Kluge's law, because they received an *n*-suffix at an early pre-stage of Germanic, while others remained untouched. The resulting opposition between geminated and ungeminated cases triggered the rise of consonant gradation according to the well-known *Associationen*.

5.1 The Germanic directional system

In Germanic, a large part of the spatial adverbs systematically occur in locative, allative and ablative form. These three dimensions are expressed by the addition of certain suffixes. In Gothic, for instance, *-(a)r*, *-(a)d* and *-(a)pro* form a locative, ablative and allative case correspondingly when added to the root. Thus, we get paradigms such as *hvar* 'where', *hvaþ* ~ *hvað* '(to) where', *hvaþro* 'from where' or *jainar* 'there', *jaind* '(to) there', *jainþro* 'from there' etc.

An entirely parallel system is formed by the adverbial directionals that are of prepositional origin. Still, although the three dimensions remain the same, the endings are different. The allative forms have a zero ending across the different dialects; the locative ending was PGm. **-ai*, cf. Go. *-a*, ON *-i*, OHG *-e*; the ablative, attested as Go. *-ana* : ON *-an* : OHG *-ana*, continues PGm. **-a-n-* plus a vocalic element. This vocalic element is somewhat difficult to reconstruct as the result of the dialectal vowel reductions in word-final position (the "Auslautgesetze"), but there is no particular reason not to follow Schmidt (1962: 194), who systematically uses **-a-nē* on the basis of e.g. Skt. *vínā* 'without' <

⁹¹ This chapter is for the most part based on the article 'On Gothic *iup* and the Germanic directionals' as published in 2010.

**ui-neh*₁ (cf. Streitberg 1896: 188; Boutkan 1995: 376ff). Together, the allative, locative and ablative seem to have more or less behaved like cases of further defective nouns (cf. Blaisdell 1959: 52). This is reminiscent of the Proto-Indo-European situation, where many spatial adverbs demonstrably continue case forms of primitive root nouns. It was shown by Hrozný (1917: 21), for instance, that Gr. *ávτί* ‘opposed, facing’, Skt. *ánti* ‘before, facing’ < **h₂ent-i* started out as the locative of PIE **h₂ent-s*, cf. Hitt. *ḫanza* ‘forehead’.

The below sections contain a survey of Proto-Germanic ‘in’ and ‘out’ and ‘up’ across the Germanic dialects. Several observations can be made. To start, the formal variation of these three adverbial clusters differs greatly; while the root of ‘in’ is very stable, the paradigm of ‘up’ displays extensive allomorphy. An important feature of the adverbial directionals is, as Schmidt (1962: 179) already pointed out, the following: “[s]ie unterscheiden sich von dem Stamm der zugehörigen prp. durch ein grösseres Lautvolumen und veränderte, “verschärfte” Konsonanz [...]”. Indeed, *gemination* appears to be a characteristic belonging to the adverbs not so much as the pertaining prepositions.

a) ‘in’

	preposition	locative	allative	ablative
Go.	<i>in</i>	<i>inna</i>	<i>inn</i>	<i>innana</i>
ON	<i>í</i>	<i>inni</i>	<i>inn</i>	<i>innan</i>
OE	<i>in</i>	<i>inne</i>	<i>in</i>	<i>innan</i>
OS	<i>in</i>	<i>inna</i>	<i>in</i>	<i>innan(a)</i>
OHG	<i>in</i>	<i>inne</i>	<i>īn</i>	<i>innana</i>

The consonantal difference between the preposition **in*, akin to e.g. Gr. *έν*, poet. *έvi*, *ένί* < **h₁en(-i)*, and the adverbs **innai*, **inn* and **innan-* is clearly visible: the former has a single, the latter a double *n*. Apparently, the adverbial forms all received an *n*-suffix at some stage in Pre-Germanic.

There are no further complications except for the long *ī* in Old High German (cf. G *ein*), which probably arose by the secondary stressing of an unstressed form **in* that was degeminated. The *i* vocalism of the preposition can be due to a lost locative ending *-*i* (cf. Gr. *ένι*, *ένί*). Alternatively, it can be explained as analogical to the adverbial forms, where the raising was regular before a tautosyllabic nasal.

b) 'out'

	preposition	locative	allative	ablative
Go.	<i>us</i>	<i>uta</i>	<i>ut</i>	<i>utana</i>
ON	<i>or, ór, úr</i>	<i>úti</i>	<i>út</i>	<i>útan</i>
OE	<i>or-</i>	<i>ūte</i>	<i>ūt</i>	<i>ūtan</i>
OS	<i>ur-</i>	<i>ūta, -e</i>	<i>ūt</i>	<i>ūtana</i>
OHG	<i>ur-</i>	<i>ūzze</i>	<i>ūz</i>	<i>ūzzana</i>

As opposed to 'in', the oldest Germanic dialects do not reveal an opposition of consonantal length for 'out'. Instead, the preposition **uz* contrasts with the adverbs in that it has final **z* instead of **t*. It probably developed out of an old genitive **ud-s* (cf. Gr. $\acute{\epsilon}\xi < *h_1e\acute{g}^h-s$, Att. $\epsilon\acute{\iota}\varsigma$ 'to, in' $< *h_1en-s$) and is generally taken to be related to Skt. *ud* 'upwards, out of'. Evidence for a geminated root variant **utt-* is nevertheless provided by Modern Faroese, which has the ablative form *uttan*.

	preposition	locative	allative	ablative
Far.	<i>úr</i>	<i>úti</i>	<i>út</i>	<i>uttan</i>

This *uttan*, which seems to have gone unnoticed in the literature, cannot possibly be the regular reflex of ON *utan*, but rather presupposes an Old Norse variant **uttan*. As a matter of fact, this variant is not hypothetical: it can be retrieved from several Old Norse texts, such as the Gulaping law (*ef kona gipter sik uttan frónða raað*, cf. Flom 1937: 121), saga Hákonar Hákonarsonar (*fyrir uttan Þjórsá / uttan af Íslandi*, cf. *Fornmanna sögur* 114, 116), the Frostaping law (*uttan konongs umbodes mader / uttan uilia sialfra þæirra*, cf. Hægstad/Torp 1909: XV), etc. It therefore seems pertinent to postulate a Proto-Germanic ablative variant **uttanē*, which with its geminate is fully parallel to **innanē* and **uppanē* (see below).

In addition to the roots **ūt-* and **utt-*, there is unambiguous evidence for an ablative root variant **ut-* with a short vowel and a single stop. The West Germanic languages all have **ūtanē*, but, in spite of the fact that most Old Norse reference works give the form as *útan* with a long *ú*⁹², ON *utan* probably had a short vowel in view of the Modern Icelandic continuant *utan*. As such, it presupposes a North Germanic form **utanē* that contrasts with both **ūtanē* and **uttanē*.

⁹² E.g. Cleasby/Vigfusson 669; Zoëga 462; Faarlund 2004: 108.

c) 'up'

	preposition	locative	allative	ablative
Go.	<i>uf</i>	<i>iupa</i>	<i>iup</i>	<i>iupana</i>
ON	<i>of</i>	<i>uppi</i>	<i>upp</i>	<i>ofan</i>
OE	-	<i>uppe</i>	<i>up</i>	<i>ufan</i>
OS	-	<i>uppa</i>	<i>up</i>	<i>oþan(a), uppan</i>
OHG	<i>ob(a)</i>	<i>obe, ūffe</i>	<i>ūf</i>	<i>obana, ūffana</i>

All of the given forms are etymologically related to Gr. ὑπό 'under', Skt. *úpa* 'to, by, with' and OIr. *fo* 'under' < PIE **up-o*. As compared to 'in' and 'out', the root variation of 'up' is relatively extensive. There is evidence for three different roots across the different dialects. First of all, Gothic points to a root **eup^p*- occurring in all three case forms, and it arose from PIE **(h₁)eup-n-* (Lühr 2000: 190)⁹³. Apparently, the original allomorphy was leveled completely in this language.

The North and West Germanic languages add two other variants, i.e. **upp-* and **ub-*. While **upp-* is almost universally attested in the locative **uppai* and the allative **upp*, the root **ub-* has its base in the ablative form **ubanē*. This original distribution was partly obliterated, though, in Old Frisian and Old Saxon, which bear evidence for an alternative ablative variant **uppanē*.

Some innovations must have taken place in Old High German as well. Perhaps the most important one is that the allative *ūf*, which must have developed in stressed position from degeminated **upp* (cf. *īn* < **inn*), became intrusive in the locative and ablative. In this way, it ended up competing with the original ablative form **ubanē* and the uniquely German locative form **ubai*. The latter is probably based on the preposition **uba*, or otherwise on the ablative **ubanē*.

The situation is exceptionally complicated in Old Frisian, where a whole range of variants is found: *boppa*, *buppa*, *bobbe*, *(b)ova*, *up(pa)*, *opa*. Due to the widespread merger of the locative and ablative endings in Old Frisian, it is difficult to analyze the original distribution of the different root variants in this language. It is likely, however, that the variant *bobbe*, apparently from PGm. **bi-ubbanē*, arose as a

⁹³ "Ebenso ist g. *iup*, *iupana* aus idg. *eupn-* entstanden" (Johansson 1891: 242). Indeed, the final *p* continues a shortened geminate. There is no reason to reconstruct a PIE **b* (thus Feist 1923: 223) or to assume that Go. *iup* is unrelated to the PGm. root variants **ub-* and **upp-* (pace Hamp 1954).

contamination form of **bi-uppanē* and **bi-ubanē*. The variant *opa*, on the other hand, which Boutkan and Siebinga labeled “remarkable” (2005: 297), points to **upai* or **upanē*, and must be the result of the reverse analogy.⁹⁴ The resulting string of root variants, viz. **ub-*, **ubb-*, **upp-* and **up-*, reveals a type of allomorphy that is fully parallel to the consonant gradation of the *n*-stems.

5.2 The Pre-Germanic situation

With the dialectal variation and Kluge's explanation of the Proto-Germanic geminates in mind, it becomes possible to shed some light upon the directional system in the Pre-Germanic period. In order to do so, it is necessary to focus on the original distribution of the different root variants. Of the available material, the most pertinent evidence comes from the case of ‘up’, because the variation is maximal and therefore most informative here. Additionally, limited information can be obtained from ‘out’. The adverbial forms pertaining to ‘inn’, on the other hand, show no root alternations at all, and therefore tell us nothing about the original distribution.

It has already been mentioned before that there is a clear distribution between the root **upp-* and **ub-* in North and West Germanic, the former being associated with the locative and allative, the latter with the ablative. This distribution has been destroyed only in Gothic, where the otherwise unknown ablaut variant **eup^p-* was generalized. The fact that continuations of **upp-* became intrusive in the ablative in Old Frisian, Old Saxon and Old High German is irrelevant, because these languages have also preserved the original variant **ubanē*. It therefore seems reasonable to argue on the basis of the dialectal situation that the geminate arose in either the locative or the allative, but at any rate not in the ablative.

Of no less importance is the contrast of the geminates of the ablatives **innan-* and **uttan-* as opposed to the singulate in **uban-*. It follows from the different etymologies of the three different adverbs that this distribution is not coincidental: whereas **h₁en* and **ud* ended in a

⁹⁴ The same root **up-* is also found in the adjective ‘open’, cf. OFri. *epen*, OSw. *ypin* < **upina-* and ON *opin*, OHG *offan* < **upana-*. This adjective was obviously derived from the pertaining adverbial root.

consonant in Proto-Indo-European, **upó* had a vocalic ending. This means that, after the addition of the nasal suffix, Kluge's law could operate in the former two adverbials, but not in the latter. As a result, we can reconstruct the ablatives as **h₁en-neh₁*, **ud-neh₁* and **upó-neh₁*⁹⁵ in Pre-Germanic, and accordingly as **innē*, **uttē* and **ubanē* in Proto-Germanic. It can further be assumed that **innanē* and **uttanē* received their suffix from **ubanē*, which was apparently reanalyzed as **ub-anē*.

Further information can be obtained from the difference between ON *út*, *úti* and *utan*. The usually overlooked short vowel of *utan* directly points to an alternative proto-form **utanē* from PIE **ud-on-eh₁*. Its **u* was never lengthened to **ū*, because this process only affected the monosyllabic (proto-)allative **ud* > PGm. **ūt*. The pan-Germanic locative **ūtai* must be secondary because of the same principle. Presumably, it was created on the basis of the allative **ūt*. So, as opposed to **uppai* and **innai*, which are probably primary from **up-noi* and **h₁en-noi*, there is no Germanic evidence for a locative **uttai*⁹⁶ from **ud-noi*. It is possible, however, that OCS *vъně* 'outside' continues just this form. The suffix **-noi* further seems to be found in Lith. *pérnai* 'last year' < **perH-noi*.⁹⁷

The fact that the allative **ūt* seems to be primary in the case of 'out' raises the question whether the allatives **inn* and **upp* are primary as well. There is, of course, the possibility to derive these forms from **up-ne* and **in-ne*, as if they were parallel to e.g. Lat. *pōne* 'from behind' < **pos(t)-ne*, or from **up-no* and **in-no* if OCS *vъnъ* may serve as a *comparandum* (< **udnom*, Meillet 1894: 236). These reconstructions would certainly be able to explain the attested geminates, and the final vowels would also have been lost in the different dialects. It is equally attractive, however, to think that the allatives **upp* and **inn* were simply backformed to the locatives **uppai* and **innai* according to the proportion **ūtai* : **ūt*. Such a removal of the ending **-ai* explains the vowel length of OHG *ūf* (G *auf*) and *īn* (G *ein*), because it can only have arisen in degeminated **up* and **in*. This degemination, in turn, can exclusively have taken place word-finally, which means that **upp* and **inn* must have been monosyllables at an early stage.

⁹⁵ Cf. Boutkan (1995: 377): **up-ó-n-eh₁* vs Johansson (1891: 240): **upónā*.

⁹⁶ The reconstruction **ūt-nai* (Schmidt 1962: 194) seems like an artefact.

⁹⁷ It has been suggested that this form underlies PGm. **ferrai* 'far' as in Go. *fairra*, ON *fjarri* etc. (Reichelt 1901: 226).

In summarizing, the Pre-Germanic directional system can be reconstructed in the following way: the locatives **up-noi* and **h₁en-noi*, the ablatives **upo-neh₁*, **in-neh₁*, **ud-neh₁* as well as the allative **ud* can be primary. The locative **ūtai* is certainly secondary, and the same thing can perhaps also be said about the allatives **inn* and **upp*.

5.3 On the full grade of Go. *iup*

With the details of the Pre-Germanic directional system in mind, the question still remains whether the full grade of Go. *iupa*, *iup* and *iupana* can be clarified. The co-occurrence of a full grade and a zero grade is not unique for Germanic prepositions. Another instance that displays such ablaut is the doublet of Go. *pairh* and OHG *duruh* 'through', both of which continue **t(e)rh₂-* plus a velar suffix. Apparently, the PGM. system preserved the two ablaut variants within one and the same paradigm in some cases, something that clearly must be a reflection of the Proto-Indo-European situation. In this respect, the allomorphy of the prepositions should probably be compared with the main focus of this monograph, i.e. the nominal ablaut of the *n*-stems.

Outside Germanic, proof of prepositional ablaut can adduced from Hittite, which had a directional system of its own. In this system, a locative **-an* (from the PIE accusative **-m*) stands in opposition with an allative in **-a* < PIE **-o*. Interestingly, two of these directionals actually have an ablaut alternation, the zero grade being found in the allative, the full grade in the locative. Compare the following table adapted from Hewson/Bubeník (2006):

allative *-a* < **-o*

anda 'in' < **h₁end-o*
āppa 'afterwards' < **Hep-o*
katta 'downwards' < **knt-o*
parā 'forewards' < **pr-ó*
šarā 'upwards' < **sr-ó*

locative *-an* < **-m*

andan 'inside' < **h₁end-m*
āppan 'after' < **He/opm*
kattan 'below' < **knt-m*
pēran 'before' < **pér-m*
šēr 'above' < **šēr (n.)*

It would, of course, be interesting to see if full-grade accusatives such as *pēran* and *šēr* could be demonstrated for Pre-Germanic as well, as they may have served as a source for the full grades of **perh-* and **eup^p-*. The

problem, however, is that this does not seem to be the case. The Germanic directional system is radically different from the Hittite one. It was created by the addition of nasal suffixes plus the ending **-oi* in the locative and **-eh₁* in the ablative. This is very much unlike the Hittite adverbs, which arose as real cases to old root nouns. The Hittite adverbs must, in fact, be equated to the Pre-Germanic prepositions **upo*, **h₁en(i)* and **ud* rather than the adverbials.

It is nevertheless interesting to see that the zero grade of the Hittite allatives is neatly mirrored by the Pre-Germanic ablative **upó-neh₁*. The significance of this is demonstrated by the uniquely West Germanic ablative OHG *fan(a)*, *fon(a)*, OS *fan(a)*, OFri. *fan* 'from' < **fanē* < **h₂p-ó-neh₁*⁹⁸, which belongs to the preposition ON *af*, OE *of*, OHG *ab(a)* 'from' < **aba* < **h₂epó* (cf. Skt. *ápa* 'away', Gr. *ἄπο, ἀπό* 'away, from'). Although both the preposition and the ablative clearly continue the PIE allative in **-o*, only **fanē* preserves the original zero grade of the root. It is fully in accordance, in other words, with the zero grades of *parā* and *šarā*. Apparently, the adverbials were created at an extremely early stage, i.e. before the paradigmatic leveling of the full grade to the allative **h₂pó* and the subsequent evolution of this case into a preposition. In this way, the Germanic spatial adverbs appear to confirm the Hittite evidence after all.

⁹⁸ Cf. Kluge/Seebold 3: "wg. **fa-ne* (mit sekundäre Erweiterungen), aus ig. **po-ne* (einer Variante von ig. **apo* mit einem suffix zur Bezeichnung der Herkunft).

6 Consonant gradation in the verb⁹⁹

The consonant alternations displayed by the *n*-stems are not restricted to just one morphological category. As has been pointed out, they can furthermore be observed in the Germanic directionals. In addition, the weak *ōn*-verbs prove to be one of the most important hotspots.

The Germanic class 2 weak verbs form a mixed category. There is a clear bifurcation between the originally denominal and the truly verbal weak verbs.¹⁰⁰ The former sub-group is generally assumed to have arisen by the addition of the thematic suffix **-ie/o-* to the **h₂*-stems. It has a strong base in the West Indo-European languages, cf. Gr. *-άω*, Lat. *-āre*, OCS *-ajǫ*, Lith. *-óti*, and became a very productive type in Germanic as well, cf. Go. *salba* f. ‘salve’¹⁰¹ ⇒ Go. *salbon* ‘to enoint’, OHG *ahta* f. ‘heed’ < **ahtō* ⇒ OHG *ahtōn* ‘to heed’ < **ahtōjan-* etc. The truly verbal *ōn*-verbs, on the other hand, distinguish themselves by their iterative or frequentative semantics and, particularly, by a high incidence of geminates, e.g. OFri. *hlakkia* ‘to laugh’ < **hlakkōn-*, Du. obs. *jakken* ‘to rush’ < **jakkōn-*, ON *glotta* ‘to grin’ < **gluttōn-*, OHG *ritzōn* ‘to carve’ < **rittōn-*, MHG *snitzen* ‘to cuttle’ < **snittōn-* etc.

In view of the systematic occurrence of gemination, it was suggested by Hermann Osthoff (1882: 298) more than a century ago that the purely verbal sub-type of the *ōn*-verbs evolved out of the PIE *neh₂*-presents, cf. Skt. 3sg. *gr̥bhñáti*, 3pl. *gr̥bhñánti* ‘to seize’ < **g^hrb^h-néh₂-ti*, **g^hrb^h-ñh₂-énti*. He assumed that, in the singular of this paradigm, the nasal suffix would bear the accent, and thus trigger Kluge’s law. Moreover, the second part of the suffix, i.e. **-eh₂-*, offered a self-evident explanation for the Germanic **ō*-vocalism.

⁹⁹ This chapter was submitted for publication in the proceedings of the “Sound of Indo-European” congress held in Copenhagen in 2009.

¹⁰⁰ Cowgill, too, assumed a fusion of thematic and athematic *ōn*-verbs, but at the same time doubted that “Proto-Germanic – let alone West Germanic – still had athematic continuations of the Indo-European **nā/nə*-formation” (1959: 11fn.).

¹⁰¹ From PIE **solp-éh₂-*, cf. Alb. *gjalpë*, To. B *šalype* ‘butter’.

Osthoff's hypothesis has never become generally accepted (see chapter 7). The evidence nevertheless suggests that it must correct, as will be pointed out. The present chapter is therefore intended as a justification of his hypothesis.

6.1 Evidence for Osthoff's hypothesis

At least four arguments, both of formal and of semantic nature, can be advanced in favor of the idea that the Germanic iteratives derive from the PIE *neh₂*-presents. The formal arguments consists of 1) the internally Germanic parallelism of the root variation with the consonant gradation exhibited by the *n*-stems; 2) the fact that the majority of the iteratives has a zero grade of the root; and 3) direct correspondences of Germanic iteratives with nasal presents in other Indo-European languages. In addition, it turns out that 4) the iterative semantics of the *ōn*-verbs can very well be explained from the punctiliar aspect of aorist, the tense from which deverbal nasal presents were typically derived.

6.1.1 Internal reconstruction

A very important indication that the iteratives originate from the *neh₂*-presents is the internal evidence in Germanic. Osthoff formulated his hypothesis on the basis of iteratives with voiceless geminates only. The fact of the matter is that the iteratives, like the *n*-stems, display a wide variety of root-final consonantisms consisting of alternations between geminates and singulates, both voiced and voiceless:

- MDu. *drup(p)en* 'to sag, drip' : Nw. *drubba* 'to stoop', Du. dial. *drubben* 'to hang one's head' < **drubbōn*-, **drup(p)ōn*-
- Nw. *duppe* : *dubbe* 'to nod, bob', MDu. *dobben* 'to dunk' < **dubbōn*-, **duppōn*-
- OE *lōcian* 'to look' : OHG *luogēn* 'id.' < **lōk^kōn*-, **lōgēn*-
- LG *ribben* 'to scutch flax' : EDu. *reppen* 'id.' : Nw. dial. *ripa* 'to tear off', MLG, MDu. *repen* 'to tear off, scutch flax' < **ribbōn*-, **rip(p)ōn*-
- ON *rugga* 'to rock', ME *ruggen* 'id.' : OE *roccian* 'id.', MHG *rucken* : *ruchen* 'id.' < **ruggōn*-, **ruk(k)ōn*-

- EDu. *schobben* : *schoppen* 'to mock', OSw. *skoppa* 'id.' : ON *skopa* 'id.' < **skubbōn-*, **skup(p)ōn-*
- OE *stroccian* 'to stroke' : EDu. *stroocken* 'id.' < **struk(k)ōn-*
- MLG *wriggen* 'to twist' : ON *riga* 'to lift heavily' : ME *wricchen* 'to wiggle' : EDu. *wreken* 'to pry' < **wrig(g)ōn-*, **wrikkōn-*

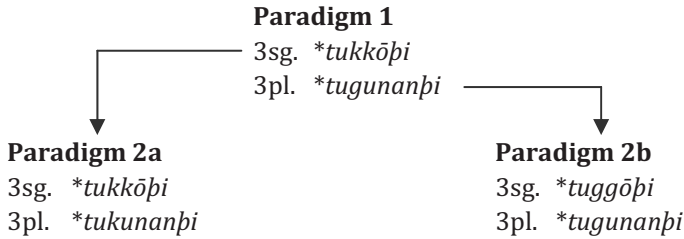
The remarkable typological parallelism of the class 2 weak verbs with the *n*-stems is reason enough to suspect the influence of a nasal suffix in the creation of the iteratives. The four-way alternation between short and long, voiced and voiceless consonants further implies an originally allomorphic paradigm that was dissolved according to the same *Associationen* that were assumed by Kluge to have occurred in the *n*-stems. Since the *neh₂*-presents, with their ablaut between the suffix and the ending, offer the exact preconditions that must be assumed for the rise of such a paradigm (cf. Skt. 3sg. *gr̥bhñāti*, 3pl. *gr̥bhñānti* < **g^hrb^h-néh₂-ti*, **g^hrb^h-ñh₂-énti*), the link with the Germanic iteratives seems like a perfect fit. Under Kluge's law, the inherited paradigm of the nasal presents would have received a geminate in the singular, where the suffix had the full grade (*-*neh₂*-), and a singulate in the plural, where the nasal of the zero-grade suffix was vocalized (*-*ñh₂*-). The verbal allomorphy thus became remarkably similar to the allomorphy of the *n*-stems. It only differed in one respect: the non-geminated roots were always affected by Verner's law:¹⁰²

Proto-Indo-European			Proto-Germanic	
	sg.	pl.	sg.	pl.
1p	CVC- <i>neh₂-mi</i>	CVC- <i>ñh₂-mé</i>	CVCC- <i>ōmi</i>	CVG- <i>umme</i>
2p	CVC- <i>neh₂-si</i>	CVC- <i>ñh₂-th₁é</i>	CVCC- <i>ōsi</i>	CVG- <i>unde</i>
3p	CVC- <i>neh₂-ti</i>	CVC- <i>ñh₂-énti</i>	CVCC- <i>ōpi</i>	CVG- <i>unanþi</i>

The allomorphy of the Proto-Germanic paradigm was obliterated by the same paradigmatic analogies that affected the *n*-stems. This is illustrated, for instance, by the cluster of iteratives belonging to PGm. **teuhan-* 'to pull' < PIE **déuk-e-*, i.e. ON *toga*, OHG *zogōn* 'to drag' < **tugōn-*, ME *toggen* 'to tug' < **tuggōn-*, OHG *zochōn* 'to jerk', MDu. *tocken* 'to strike' <

¹⁰² Van Helden (1905: 231): "Lange stimmlose Spirans kam den -*nā*-Bildungen ihrer ursprünglichen Accentuierung gemäss von rechts wegen nicht zu".

**tukkōn-*, MDu. *token* 'to push' < **tukōn-*. All four different root variants, **g*, **gg*, **kk* and **k*, are perfectly understandable from the occurrence of Kluge's *Associationen* and the splitting of the original paradigm:¹⁰³



There is a plethora of iterative verbs in the Northwest Germanic dialects for which the same scenario must be supposed. The below verbs all exhibit the kind of consonant alternations that can be expected from an **neh*₂-present with suffix ablaut:

- Nw. *duppe* : Nw. *dubbe* 'to bob, nod' : MDu. *dobben* 'to dunk, drown' < **duppōpi*, **dubunanpi*
- E *gloat* : ON *glotta* 'to grin' < **gluttōpi*, **glutunanpi*
- MLG, Du. *grabben* : LG *grappen* : MDu. *grapen* 'to grab' < **grappōpi*, **grabunanpi*
- OHG *jagōn*, Du. *jagen* : Du. *jakken* 'to rush, hunt' < **jakkōpi*, **jagunanpi*
- Nw. dial. *rige* : *rigge* 'to totter', MLG *wriggen* 'to wag' : Du. *wrikken* 'to pry' < **wrikkōpi*, **wrigunanpi*
- EDu. *schobben* : *schoppen*, OSw. *skoppa* : ON *skopa* 'to mock' < **skuppōpi*, **skubunanpi*
- ON *slafa-st* 'to slacken' : Icel. *slabba* 'to loaf around' : Icel. *slapa* 'to dangle' < **slappōpi*, **slabunanpi*
- MHG *snaben* 'to sniff' : EDu. *snabben* : Du. *snappen* : ON *snapa* 'to grab' < **snappōpi*, **snabunanpi*

The **u* of the plural ending **-unanpi* < **-nh₂-énti* has left no trace. Apparently, it was removed from the paradigm at an early stage. The

¹⁰³ Van Helten (1905: 229-232) assumed similar processes of analogy, but with a different chronological setting (see § 4.3.2.3).

motivation for this analogy probably was the lack of vocalization in iteratives with roots ending in a vocalic element, cf. ON *dvena* ‘to dwindle’ < **dwinōpi*, **dwinanpi* < **d^hg^{wh}i-néh₂-ti*, **d^hg^{wh}i-nh₂-énti* (cf. Gr. φθίνω), ON *gana* ‘to gape’ < **ganōpi*, **gananpi* < **g^hh₂-néh₂-ti*, **g^hh₂-nh₂-énti* (cf. Gr. χαίνω) and OS *hlinon*, OE *hleonian* ‘to lean’ < **hlinōpi*, **hlinanpi* < **klī-néh₂-ti*, **klī-nh₂-énti*. This analogical spread is confirmed by, for instance, the doublet OHG *storrēn* ‘to jut out’ : *stornēn* ‘to be rigid’, which probably continues an older *ōn*-verb **sturrōpi*, **stur(u)nanpi*.

6.1.2 The origin of the zero grade

The derivation of the iteratives from the *n*-presents also explains why so many Germanic iteratives have a zero grade of the root. This characteristic naturally follows from the fact that the stressed full grade shifted between the suffix in the singular and the ending in the plural in the PIE nasal presents, the root having the zero grade throughout the paradigm, cf. the Skt. *nā*-verbs, e.g. *badhnāti* ‘to bind, tie, fix, fasten’ < **b^hnd^h-néh₂-ti*, *grbhñāti* ‘to seize’ < **g^hṛb^h-neh₂-ti*, *skabhnāti* ‘to prop, support, fix’ < **skmb^h-néh₂-ti* etc. As a result, the zero grade of the Germanic iteratives can simply be regarded as a feature that was inherited from the proto-language.

6.1.3 Direct correspondences

An important part of the external evidence for Osthoff’s hypothesis – not to mention for Kluge’s law itself – comes from those Germanic iteratives that directly correspond to *n*-presents in other Indo-European languages. Quite a few cases can be identified, especially when one takes into account that Latin was affected by Thurneysen’s law. According to Thurneysen’s law, a nasal suffix became an infix by infection, i.e. by nasalization (and voicing) of the intermediary consonant, cf. *mungō* < **muk-n-* to Gr. μύσσομαι ‘to blow one’s nose’, *pandō* ‘to extend’ < **pt-ṇ-h₂-* to Gr. πίννυμι, and *pingō* ‘to paint’ < **piḱ-n-* to Skt. *piṃśati* ‘to adorn, carve out’ < **pi-n-k-* (Thurneysen 1883: 303). Extra-Germanic verbs with infixed *n* can theoretically be included into the corpus of correspondences as well, because the nasal infix is known to alternate with a suffix in instances like Gr. ζεύγνυμι ‘to yoke’ vs Skt. *yunākti* ‘id.’ (Gamm 1973: 98). All in all, a significantly large corpus can be collected from the Indo-European languages.

- MDu. *grobber* 'to scrape', E *to grub* 'id.' : MLG *gropen* 'to hollow out' < **gruppōþi*, **grubunanþi* (with analogical vocalization) < **g^hrb^h-néh₂-*, cf. Skt. *grbhṇāti* 'to seize'
- EDu. *labben* : *lappen* : OE *lapan* 'to lick' < **lappōþi*, **labunanþi*, cf. Lat. *lambō* 'to lick' < **lhb^h-néh₂-*
- OE *liccian* 'to lick' < **likkōn-*, cf. Lat. *lingō* < **liǵ^h-néh₂-*
- ON *lokka* 'to entice, stroke, caress', OHG *lochōn* : *lohōn* 'id.' < **lukkōþi*, **lukunanþi*, cf. Lith. *lù(n)ginti* 'to allure, fondle' < **lug^(h)-ŋ(é)h₂-*
- Du. *mikken* 'to aim' (from older 'to peer') < **mikkōn-* < **mig^h-néh₂-*, cf. Lith. *mingù*, *mìgti* 'to fall asleep', Ru. *mignut'* 'to blink, wink' < **m(e)ig^h-n*¹⁰⁴
- OE *roccian* 'to rock', MHG *rocken* 'to drag, jerk' : ON *rugga* 'to rock', ME *ruggen* 'id.' < **rukkōþi*, **rugunanþi* < **Hruk-néh₂-*, cf. Lat. *runcō* 'to weed' < **Hru-n-k-*
- MDu. *roppen*, MHG *rupfen* 'to pluck, tear off' < **Hrup-néh₂-* ~ Lat. *rumpō* 'to break, tear' < **Hru-n-p-*
- Swi. *sucka* : *sugā* : OE *socian*, MDu. *soken* 'to suck' < **sukkōþi*, **sugunanþi* < **suk^h-néh₂-*, cf. W *sugnaf* < **souk^h-néh₂-* (cf. OCS *sъsati* 'id.')
- MDu. *stricken* 'to make fall' (to **strikan-* 'to rub, stroke') < **strikkōn-*, cf. Lat. *stringō* 'to draw tight; to compress' < **strig-n-*
- OE *þaccian* 'to pat', EDu. *tacken* 'to pick up' : MDu. *taken* 'to grasp' < **þakkōþi*, **þagunanþi*¹⁰⁵, cf. Lat. *tangō* 'to touch' < **th₂g-néh₂-*.

Not all of the examples may be equally strong, but at least for **grubbōn-*, **labbōn-*, **likkōn-*, **strikkōn-* and **þakkōn-* an Indo-European derivation can safely be assumed. The parallelism of PGM. **lukkōþi*, **luk/gunanþi* with Lith. *lù(n)ginti* 'to allure, fondle' is actually perfect, the suffix *-in- having developed from **-ŋh₂-*. These correspondences are an important substantiation of Osthoff's hypothesis.

¹⁰⁴ Cf. Franck/Van Wijk 430: "*mikk- uit idg. *mig-n-of *migh-n-".

¹⁰⁵ The PGM. **t* is due to restoration of the reduplication when the present stem **te-th₂g-* (Gr. τεταγών) developed into **tedg-* by assimilation (Kortlandt 2000).

6.1.4 The iterative aspect

A final argument in favor of the link between the Germanic iteratives and the *n*-presents is of semantic nature. Verbs like MDu. *dobben* ‘to dunk’, LG *grappen* ‘to grab’, EDu. *labben* ‘to lick’, Nw. dial. *rigge* ‘to totter’, OE *roccian* ‘to rock’ have in common that they denote an action consisting of repeated sub-actions. For this reason, they are commonly referred to as frequentative, intensive or iterative verbs. The origin of this iterative aspect is an important question.

The debate on the original aspect of the PIE *n*-presents is very old, and several different attempts have been made to define it. The aspect was called “terminative” by Brugmann and Delbrück (1897: 15): “Terminativ ist eine Aktion, wenn ausgesagt wird, daß eine Handlung vor sich geht, doch so, daß ein Terminus in’s Auge gefasst wird, sei dieser nun der Ausgangs- oder Endpunkt, z.B. ῥηότι ὀρνυμι in Bewegung setzen, ἄρνυμι zerbrechen.” Strunk (1979: 244) analyzed it as *inflective-terminative*, and Meiser (1993: 295) adopted the concept “semantically transitive” from P. Hopper and S. Thompson (1980): “Proto-typische Nasalpräsentien sind demnach kurz gesagt transitive Handlungsverben, die ein – im Vergleich zur anders- oder uncharakterisierten Aktionsart – starkes Betroffensein des Objekts durch das intentional agierende Subjekt zum Ausdruck bringen.” Still, the definition that probably describes the semantic function most accurately, was given by N. van Wijk (1929: 255) in an article on the verbal aspect in Slavic. In this analysis, Van Wijk was principally referring to the Slavic aspect, and not to the Germanic second class weak verbs, but he nevertheless provided a good description of the Germanic aspect as well:¹⁰⁶

“En général, on peut dire que les verbes déterminés désignent des actions peu compliquées, menant directement à un but, tandis que les verbes indéterminés sont employés pour des actions se composant de plusieurs actes ou pour des actions prolongées ou répétées.”

¹⁰⁶ Cf. Kuiper (1937: 204): “Vergleichen wir nun den Inhalt des Begriffs “determinativ” (action déterminée) mit dem von Delbrück als “terminativ” bezeichneten, so ergibt sich, daß beide Bezeichnungen sich nahezu decken.”

From this perspective, one of the most important observations that have to be made is that the iterative aspect is not isolated to Germanic. It can, in fact, be found with nasal presents from all over the Indo-European language family, including the more central dialects such as Sanskrit. In this language, the following verbs seem to denote actions that are inherently iterative: *badhnāti* 'to bind, tie, fix, fasten' < **b^hṇd^h-neh₂*, *limpāti* 'to smear' < **li-m-p-é-*, *lunāti* 'to cut, sever, divide, pluck, reap' < **lu-n-eh₂*, *mṛṇāti* 'to grind' < **mṛ_ḷ-n-eH-*, *sināti* 'to tie, fetter' < **si-n-eH-*, *strṇāti* 'to spread out, strew' < **str_ṇ-n-eh₃-*, and *mathnāti* 'to stir, whirl round' < **mṇtH-neh₂*-. All these verbs denote actions consisting of sub-actions that typically need to be repeated in order to achieve the full denotation: to fasten is to bind several times, to stir is to move around repeatedly, to grind is to crush over and over again.

Additionally, there are clues for iterative nasal presents in Baltic. Lith. *kėlti* 'to lift' < **kelH-e-*, for instance, is flanked by an iterative Latv. *cilināt* 'to lift repeatedly' < **k_ḷH-n(é)h₂*-. The Latvian *-inā*-suffix is related to the elements *-in-*, *-no-* and *-ino-* in Lithuanian (cf. Schmalstieg 2000: 170, 195). It thus appears that Proto-Baltic still had an alternation between **-in-* from the zero grade **-ṇh₂*- and **-nā-* from the full grade **-néh₂*-, as the suffix *-inā*- can hardly be anything else than a conflation of the two different ablaut grades (cf. Kortlandt 1989: 107). Some of the Proto-Baltic iteratives appear to be exactly parallel, in other words, to the Germanic iteratives, both in meaning and in form.

In view of the dialectal communalities, including a striking Balto-Germanic *Sonderübereinstimmung*, little can be inferred against the assumption that the iterative aspect was already present in the proto-language at least in some sub-types of the nasal presents. Most probably, the "proto-iterative" sub-type consisted of nasal presents that were derived from primary aorists. It is generally accepted that these two formations were in close association with each other (cf. Strunk 1967). By assuming that the iterative nasal presents were derived from primary aorists, their semantic field can be regarded as reflecting the original aorist aspect.¹⁰⁷ The Indo-European aorist indicated a punctiliar or ingressive event, cf. Gr. ἀγοράζωσι 'they buy regularly' vs ἀγοράσωσι

¹⁰⁷ Cf. Bammesberger (1984: 23, fn. 3): "Die Intensiv-Iterativa vom Typ *ziehen* : *zucken* (vgl. ohne Intensivgemination an. *toga* und vom Formalen lat. *ē-duc-āre*) inkorporieren die aoristische Form der Wurzel, d.h. sind abgeleitet vom Aorist."

‘they buy once’, νοσεῖν ‘to be ill’ vs νοσῆσαι ‘to fall ill’ and ἀποθνήσκειν ‘to lie dying’ vs ἀποθανεῖν ‘to die’ (Wackernagel 1920, I: 173-5). When nasal presents were created to aorists, the iterative aspect could easily arise out of the punctiliar aspect that was made presentic, i.e. durative.

It follows from this analysis that there is no necessity to assume that the Proto-Indo-European nasal affix had a meaning of its own, i.e. “inflective-terminative” or “semantically transitive”. It is often claimed that the nasal present was causative-factitive. This is true, of course, for many cases, cf. Lith. *budėti* ‘to be awake’ ⇒ *bùdinti*, Latv. *budināt* ‘to wake up, incite’, Skt. *ríyate* ‘to flow’ ⇒ *riṇáti* ‘to make flow’, Lith. *añtras* ‘second’ ⇒ *añtrinti* ‘to repeat’, OIr. *rúad* ‘red’ ⇒ *rondid* ‘to make red’. Still, the causative and factitive aspects do not necessarily have to be inherent in the *n*-suffix itself. It seems more probable that the nasal affix originally was a non-specific presentic formant, and that the specific aspect of nasal presents, be it factitive, causative or iterative, simply depended on the derivational source. While factitives were created to adjectives, the aorist served as the basis for iteratives. Causatives are clearly related to the factitives, but differ from them in that they were created to verbal roots. So, the transitivity of the nasal presents was dependant of their original derivational source, not of their suffixation.

6.1.5 The “problem” of the inchoative verbs

Opponents of Kluge’s law have always adduced inchoative verbs such as Go. *fullnan* ‘to become full’, Go. *ga-waknan*, OE *wæcnian* ‘to wake up’ and closely related formations such as OHG *lernōn*, OS *līnōn*, OFri. *lirna*, *lerna*, OE *leornian* ‘to learn’ < **līznōn*- to testify against Kluge’s law.¹⁰⁸ The idea is that if the *nan*-verbs derive from the *neh*₂-presents, which is a generally accepted view, the iteratives with consonant gradation cannot possibly have the same origin.

A possible solution to this problem was given by Van Helten (1905: 38 fn.), who assumed that the forms with retained nasals had root stress so that Kluge’s law could not operate. Although this explanation is technically capable of accounting for the lack of gemination in the inchoatives, it seems more plausible, that the absence of geminates resulted from another morphological difference.

¹⁰⁸ Wissmann 1932: 160-1; Fagan 1989: 38-9; Hopper 1989: 247.

Clearly, the inchoatives cannot be separated from the PIE causative-factitives of the type Skt. *ṛíyate* 'to flow' $\Rightarrow$ *ṛiṇáti* 'to make flow', OIr. *rúad* 'red' $\Rightarrow$ *rondid* 'to make red' etc. However, the PIE factitives are transitive, while the inchoatives are not, neither in Germanic nor in Baltic, cf. Go. *bindan* 'to bind' $\Rightarrow$ *and-bundnan* 'to become loose', ON *rauðr* 'red' $\Rightarrow$ *roðna* 'to become red', Lith. *budėti* 'to be awake' $\Rightarrow$ *bundù, bùsti* 'to wake up', *plikas* 'bald' $\Rightarrow$ *plinkù, plikti* 'to become bald'. It has therefore been suggested by Meiser (1993: 292) and Kortlandt (1995) that the inchoatives continue medial factitives.¹⁰⁹ This starting point is especially salient in **liznōn-* 'to learn', which is semantically better understandable as a medial causative 'to make oneself know' (cf. Go. *lais* 'I know') than as an inchoative 'to start knowing'.

With this in mind, the lack of gemination in the Germanic inchoatives can simply be explained from the fact that the present middle forms had the zero grade of the suffix in the larger part of the paradigm. For instance, the middle paradigm of the Sanskrit verb *gr̥bhñáti* 'to grab', namely 3sg. *gr̥bh-ñī-té*, 3pl. *gr̥bh-ñ-áte*, indirectly continues **g^hrb^h-nh₂-toi*, **g^hrb^h-nh₂-ntoi*.¹¹⁰ The zero grade of the suffix caused the nasal to become vocalized, as it was positioned directly in front of a consonantal laryngeal. In this way, it precluded the operation of Kluge's law. Like in the active *néh₂*-presents, the vowel of the vocalized suffix **-un-* < **-ñh₂-* was probably removed on the basis of inchoatives with roots ending in a vocalic element.

6.1.6 An alternative hypothesis by Lühr

A different explanation of the Germanic iteratives was offered by Lühr (1988: 345-77). Lühr, as a proponent of Kluge's law, argued that these verbs, with their characteristic geminates, started as adjectives in **-nó-*, which, like the Hittite factitives in *-aḫ-* (cf. *neuaḫḫi* 'to renew' < **neu-eh₂-mi*), received an athematic **-eh₂-*suffix.¹¹¹ Within this framework, G *locken* 'to entice' < **lukkōn-*, MDu. *bocken* 'to bend over' < **bukkōn-* and Nw. *duppe* 'to dip' < **duppōn-* can be directly connected with Lith. *lūgnas*

¹⁰⁹ Kortlandt suggested that the class 4 weak verbs were derived from the middle of the root aorist, which in Germanic must have had root stress, cf. OE *cūðe* 'could' < **kunþa* < **h₁e-ǵǵh₃-to*, ūðe 'granted' < **h₁e-h₃ǵh₂-to*.

¹¹⁰ The nasal must have been restored in the expected 3pl. ***grbhāte*.

¹¹¹ Accepted: Kortlandt (1991: 2).

‘pliable’, Skt. *bhugná-* ‘bent’ and OIr. *domain* ‘deep’ < **d^hub^h-nó-*, respectively.

In spite of these outer-Germanic connections, however, Lühr’s hypothesis seems difficult to maintain, because it cannot account for the consonant alternations displayed by the iteratives. In practice, the derivation from the **nó*-adjectives is indeed capable of clarifying iteratives with regular voiceless geminates, e.g. Du. *wrikken* ‘to pry’, G *zucken* ‘to jerk’, but not for alternants with different consonantisms, e.g. Nw. dial. *rig(g)e* ‘to stagger’, ME *toggen* ‘to tug’, of MDu. *token* ‘to push’. The consonant variation, which is highly reminiscent of the allomorphy of the *n*-stems, can only be understood by supposing a paradigm with a regular alternation of singulates and geminates. The subsequent rise of forms with voiceless singulates and voiced geminates was again triggered by the usual paradigmatic cross-contaminations.

6.2 The iterative system

Within the verbal system, the Germanic iteratives are far from isolated. They typically occur beside a non-iterative verb, usually of the strong conjugation. The pattern that emerges from these verbs is very pervasive, and can best be understood within the context of the classical Proto-Germanic aspectual system consisting of derivationally productive pathways between 1) strong verbs, 2) statives in *-*ēn-*, 3) causatives in *-*jan-*, and 4) the factitive/inchoatives in *-*nan-*, cf. ON *vaka* ‘to be awake’, *vekja* ‘to (make) wake up’ and *vakna* ‘to wake up (intr.)’.

The list below consists of cases that demonstrate the iterative system. Each case consists of a strong verb that is flanked by one or more related iteratives. The iterative formations often have a geminated root, but they usually display the kind of consonant gradation that can be expected from the original *-*neh*₂-conjugation. Not seldomly, as many as four different root variants can be found.

- Go. *sneip̃an* ‘to cut’ : G *snitzen* ‘to cuttle’
- MHG *fliegen* : *flocken* ‘to fly’
- ON *fljóta*, OE *flēotan* ‘to flow’ : MDu. *vlot(t)en* ‘to flow, float’ ~ OE *flotian* ‘to float’, ON *flota* ‘id.’
- Go. *liugan* ‘to lie’ : OHG *lochōn* ~ *lohhōn* ‘to entice’

- ON *rjúfa*, OE *rēofan* ‘to break’ : MHG *ropfen* ‘to pluck’ ~ Icel. *rubba* ‘to scrape’
- Go. *tiuhan* : OHG *zogōn* ‘to drag’ ~ ME *toggen* ‘to tug’ ~ MDu. *token* ‘to strike’ ~ MDu. *token* ‘to push’
- Go. *skiuban* ‘to shove’ : MHG *schopfen* ~ *schoppen* ‘to stuff’
- OE *dūfan* ‘to duck, sink’ : EDu. Flem. *doppen* ‘*intingere*’ ~ Nw. dial. *dubba* ‘to bob’
- Nw. dial. *guva* ‘to blow (of dust, smoke); to swing’ : Icel. *gufa* ‘to steam’ ~ MHG *gupfen* ‘to swing’ ~ Icel. *gubba* ‘to gush’
- MHG *snūfen* ‘to sniff’ : MLG *snoppen* ‘to blow your nose’
- ON *stinga*, OE *stingan* ‘to stick, sting’ : OHG *stunchōn* ‘to stuff’
- Go. *gawigan* ‘to move’ : MHG *wagen* ~ *wacken* ‘to stagger’
- ON *steka*, OHG *stehhan* ‘to stab’ : OHG *stehhōn* ~ *stechōn* ‘to stick’
- Go. *tekan*, ON *taka* ‘to take’ : EDu. *tacken* ‘*apprehendere*’ ~ MDu. *taken* ‘to grasp’
- Go. *hlahjan* ‘to laugh’ : OFri. *hlakkia* ‘id.’¹¹²
- Du. *stuiten* ‘to stop, bump’ : OHG *stotzōn* ‘to tremble’¹¹³

Some cases clearly indicate that the iterativization mechanism remained productive until after the Germanic sound shifts. The iterative verbs in question seem to have simply adopted the root-final voiced stop of the strong verb, and doubled it at a recent stage. Apparently gemination was “derivationalized”, i.e. used as a derivationally salient instrument. In this respect, its function can be compared to the role geminates in the hypocorisms (see § 4.3.4). For obvious reasons, it is unlikely that old **neh₂*-formations can be stipulated on the basis of these secondary iteratives.

- OE *rēodan* ‘to kill’ : G *aus-rotten* ‘to exterminate’
- OE *scūdan* ‘to rush’ : G *schotten* ‘to shake’¹¹⁴
- Go. *trudan*, OHG *tretan* ‘to tread’ : OHG *trettōn* ‘to trample’, MHG *trot(t)en* ‘to trot’¹¹⁵

¹¹² Van Helten 231: **klok-néh₂*.

¹¹³ For a discussion of most of these iteratives, see Wissmann 1932: Chapt. 6. *ō-Verba mit Geminata*.

¹¹⁴ Grimm 15, 1612.

Additional evidence for the permanent productivity of the iterative system is furnished by those cases that have an analogical zero grade. Some of these verbs are of the so-called *tudáti*-type, which originally had a zero grade root in the present. This characteristic led to the situation that the iterative, which usually has the zero grade as well, was only distinguished from the strong verb by its geminate. The ablaut opposition between the strong verb and the iterative was then “restored” by the introduction of the productive zero-grade marker **u*.

- Go. *graban* ‘to dig’ : E *grub* ‘id.’, MDu. *grobben* ‘to scrape’¹¹⁶ ~ MLG *gropen* ‘to hollow out’¹¹⁷
- ON *skaka*, OE *sceacan* ‘to shake’ : MHG *schocken* ‘id.’
- OHG *stehhan* ‘to stick, sting’ : MHG *stocken* ‘to coagulate’ ~ G *stoehen* ‘to poke’

The case of MHG *stocken* < **stukkōn*- to OHG *stehhan* < **stekan*- is especially elucidating. The strong verb **stikan*- is clearly related to Lat. *instīgō* ‘to urge, incite’, and must be reconstructed as a zero-grade present **stig-é*.¹¹⁸ Formally, it is parallel to other strong *tudáti*-verbs, such as Go. *digan* ‘to knead’ (pret. *daig*) < **d^hig^h-é*, OHG *redan* ‘to sieve’ < **hriþan*- (cf. Gr. κρίνω ‘to separate’ < **kri-n-ie/o-*) and ON *vega* ‘to fence, fight, kill’ < **wigan*- < **uik-é*- (cf. Lat. *vincō* ‘to conquer’). The creation of the secondary iterative **stukkōn*- probably took place after the transfer of the verb into the fourth (OHG *stehhan*) and fifth (ON *steka*) class. This, in turn, was triggered by the lowering of *i* to *e* by *a*-mutation in Northwest Germanic. The original iterative is preserved as OE *stician*, MLG *sticken* ‘to stick’ < **stik(k)ōn*-. Note that the variation of the consonantism and vocalism in nouns such as Go. *stiks* m. ‘sting’, OE *stecca* m. ‘stick’, ON *stjaki* m. ‘id.’, ON *stokkr* m. ‘post’ is due to their derivation from the verbal complex at different moments in time.

¹¹⁵ The strong verb is of the *tudáti*-type, i.e. a primary aorist with root ablaut, viz. 3sg. **tred^h-t*, 3pl. **trd^h-ént*. In Germanic, the expected *Schwebeablaut* was resolved into a root alternation **tred-* ~ **trud-* (cf. Bammesberger 1984).

¹¹⁶ Verdam 230.

¹¹⁷ Lübben 130; Franck/Van Wijk (p. 213): “De secundaire basis met *p* kan haar uitgangspunt gehad hebben in klankwettige vormen met *pp* uit idg. *bhn*.”

¹¹⁸ Cf. Prokosch § 54c.

6.3 Evidence for de-iterativization

Importantly, there is compelling evidence in support of a reverse derivational process from the iteratives to the strong verbs, i.e. a process that could be called *de-iterativization*. Such a mechanism is evinced by the fact that a large number of strong verbs demonstrably adapted their root final consonantism to the iterative geminate. The evidence consists of strong verbs with roots in **-p-*, **-t-* and **-k-* corresponding to intra-Germanic or extra-Germanic cognates that point to a PIE final plain stop or voiced aspirate. Since such correspondences can only be maintained by assuming that this **p*, **t* or **k* resulted from a shortened geminate, they are likely to be formed on the basis of an iterative.

- MLG *knīpen* ‘to pinch’ : Du. *knippen* ‘to cut’ < **gni(H)b^h*- (cf. ON *knífr* ‘knife’)
- OE *snican* ‘to creep’ : G *schnecken* ‘id.’ (cf. MLG *snigge* ‘snail’)
- Go. *dis-hniupan*, OE *hnēopan* ‘to tear (off)’ : OE *hnoppian* ‘to pluck off’ < **knup-* (cf. MLG *nobbe* f. ‘tuft’)
- ON *drjúpa* ‘to drip, droop with the head’ : Nw. *drubba* ‘to walk with a stoop, fall over’, Du. dial. *drubben* ‘to hang one’s head, be downcast’¹¹⁹ ~ MLG, MDu. *drupen*, *druppen* ‘to sag, drip’ < **d^hrub^h*-¹²⁰
- ON *hrjóta* ‘to snore’, OHG *riozan* ‘to cry’ : G *rotzen* ‘to cry, lament’ < **krut-* (cf. ON *hroði* m. ‘(lump of) spit’)
- ON *krjúpa* ‘to crawl’ : Cimb. *kruppen* ‘id.’ < **grub^h*-
- ON *strjúka* ‘to stroke’ : OE *stroccian* ‘id.’ ~ EDu. *stroocken* ‘id.’ < **strug^h*- (cf. OCS *strъgati* ‘to shave, shere’)
- OE *sūpan* ‘to sip’ : OE *soppian*, Du. *soppen* ‘to sop, dunk’, WFri. *sobje* ‘to suck’ (cf. Skt. *sūpa-* m. ‘broth’)¹²¹
- OFri. *stapa* ‘to go’ : OHG *stapfōn* ‘to tramp’ < **stop-* (cf. OCS *stopa* ‘footstep’)

¹¹⁹ Boekenoogen 109.

¹²⁰ The consonant variation of ON *dropi*, OHG *tropfo*, *troffo* m. ‘drop’ < **drup(p)an-* does not have to be due to its inflection as an *n*-stem (Rasmussen 1989b: 253), but is more likely to be a reflection of the verbal alternations (cf. **sti/ek(k)ōn-* ⇒ *sti/ek(k)an-* ‘stick’). Nw. *drubba* proves that the original root was **d^hreub^h*- rather than **d^hreub-*.

¹²¹ Cf. already Osthoff 1882: 298.

- Go. **mimpan-* ⇒ Go. *bi-mampjan* ‘to mock’ (cf. Gr. μέμφομαι ‘to disapprove’ < **me(-)mb^h-*)
- OHG *laffan* (pret. *luaf*) ‘to lick’ : EDu. *labben* ~ *lappen* ~ OE *lapan* ‘id.’ < **lHb^h-*
- OE *sceacan* ‘to shake’ : MHG *schocken* ‘id.’ ~ Du. Limb. *schoggen* ‘id.’¹²², ME *shoggen* ‘id.’ < **skok/g^h-*
- Go. *slepan* ‘to sleep’ : Icel. *slafa-st* ‘to slacken’ ~ *slabba* ‘to hang’ ~ *slapa* ‘to slack’ < **slob^h-*
- ON *sópa* ‘to sweep’ : E *swab* ~ *swap* ‘id.’

The spread of geminates from the iteratives to the strong verbs was discussed by Lühr (1988: 351ff) in a discussion of the doublet ON *rifa*, OFri. *rīva* ‘to tear’ : OE *rīpa* ‘to harvest’. According to Lühr, the latter verb adapted its consonantism to the iterative ON *rippa* ‘to rip up’, which she analyzed as a derivative from the past participle in **-no-* (see § 6.1.6). The derivation of the consonantism from the iterative seems to be appropriate, but in view of the cognates Nw. dial. *ripa* ‘to tear off’, G Pal. *reffen* ‘to reap with a scythe’, MLG *repen* ‘to scutch flax’, MDu. *repen* ‘to tear’ < **ripōn-*, EDu. *reppen* ‘*rapere, capere*’ < **rippōn-*, G obs. *ribben* ‘*cortices lini decutere*’¹²³ and ON *rifa* ‘to sew up’ < **ribōn-*, the iterative is preferably reconstructed with an allomorphic paradigm **rippōpi*, **ribunanpi* < **Hrip-néh₂-ti*, **Hrip-nh₂-énti*.

As to **rīpan-*, it is probably better to assume that this verb did not merely adopt the consonantism from the iterative, but that it was, in fact, *derived* from the iterative. What speaks for such a derivation is the semantic difference between OFri. *rīva* ‘to tear’ and OE *rīpan* ‘to harvest’. The latter meaning is best analyzed as a continuing act of repeated reaping or tearing. OE *rīpan*, in other words, represents a *durative* formation derived from the *iterative* formation **rippōpi*, **ribunanpi*, which was in turn created to the semantically more primitive OFri. *rīva* < **rīfan-* < PIE **Hréip-e-*.

The productivity of the de-iteratives accounts for many other doublets in the Germanic dialects. Since the doublets are frequently accompanied by an iterative, it seems unlikely, at any rate, that they arose from independent **nu-*presents such as OE *bannan* ‘to order’ <

¹²² WLD II.3, 113.

¹²³ Grimm 14, 1033.

**b^h(e)h₂-néu-ti*, **b^h(e)h₂-nu-énti*¹²⁴ or Go. *winnan* ‘to suffer’ (cf. Skt. *vanóti* ‘to want, win’) < **uen-néu-ti*, **uen-nu-énti*. The following examples can be mentioned:

- ON *rífa*, OFri. *rīva* ‘to tear’ < **rīfan-* : OE *rīpa* ‘to harvest’ < **rīp^aan-* vs EDu. *reppen* ‘to pluck’ : Nw. *ripa* ‘to tear off’, G Pal. *reffen* ‘to reap’, MDu. *repen* ‘to tear’ : (L)G obs. *ribben* ‘*cortices lini decutere*’ < **ripp/bōn-*
- ON *vífandi* (ptc.) ‘arriving as by chance’ < **wīfan-* : MHG *wīfen* ‘to sway’ < **wīpan-* vs OHG *wipfōn* ‘to lose one’s way’ < **wippōn-*
- MDu. *vliegen* ‘to fly’ < **fleugan-* : *vlieken* ‘id.’ < **fleuk^kan-* vs MHG *flocken* ‘*volitare*’ < **flukkōn-*
- SFri. *smugen* ‘to be misty, drizzle’ < **smūgan-* : OE *smēocan*, MDu. *smūken* ‘to smoke’ < **smeuk^kan-/smūk^kan-* vs Du. obs. *smokken* ‘to snuff, put out’ < **smukkōn-*
- OE *smūgan* ‘to sneak’ < **smūgan-* : MLG *smūken* ‘to sneak’ < **smūk^kan-* vs MHG *smucken* ‘to slip into’ < **smukkōn-* (cf. OCS *smučati* ‘to crawl’ < **smuk-*)
- ON *s(fj)úga*, OE *sūgan* ‘to suck’ < **seugan-*, **sūgan-* : *sūcan* ‘id.’ < **sūk^kan-* vs Swi. App. *sukkə* : *sugə*¹²⁵ : OE *socian* ‘to suck’ < **sukk/gōn-*
- G *zaufen* ‘to pull back’ : G *zupfen*, obs. *zopfen* ‘to pluck, pick’ ~ G dial. *zobeln* ‘to pull someone’s hair, tousle’¹²⁶
- OE *cringan* ‘to perish’ < **kringan-* : *crincan* ‘id.’ < **krink^kan-* vs EDu. *kronken* ‘to wind’ < **krunk^kōn-* (cf. Lith. *grėžti* ‘to turn’ < **greng^h-*)
- OE *scrimman* ‘to shrivel’ < **skrimban-* vs MHG *schrumpfen* ‘to shrink’ < **skrimp^aan-* vs MHG *schrumpfen* ‘id.’ < **skrump^a/bōn-* < (quasi) **skrṃb^h-n(é)h₂-* (cf. Lith. *skrembù*, *skrėbti* ‘id.’ < **skre(m)b^h-*)
- OE *þrindan* ‘to swell’ < **þrindan-* : OE *þrintan* ‘id.’ < **þrint^aan-*

¹²⁴ Cf. Fick/Falk/Torp 256.

¹²⁵ Vetsch 159.

¹²⁶ Cf. Grimm 31, 397: “z. liegt dem Intensivum zuppen zoppen zurückgehn, zurückziehen und zupfen, nd. tuppen zerren, ruckweise reizen zu grunde [...]”

- MHG *schreven* 'to scratch' < **skrefan*- : OE *screpan* 'to scratch' < **skrepan*- vs MDu. *schraven* : *schrabben* : *schrappen* : *schrappen* 'id.' < **skrapp/bōn*-
- MDu. *treken* 'to pull, drag' < **trekan*- : MDu. *trecken* 'id.' < **trekkan*- vs MDu. *trocken* 'id.' < **trukkōn*- (perhaps related to OCS *dr̥gati* 'to shudder', Ru. *děrgat* 'to jerk' < **dr(u)g^h*-)
- Sw. dial. *dimba* 'to fog' < **dimban*- : MHG *dimpfen* 'to smoke' < **dimp^pan*- vs MLG *dumpen* 'to choke, extinguish' < **dump^pōn*-
- OE *slingan* 'to wind, slink' < **slingan*- : *slincan* 'to crawl, slink' < **slink^kan*- vs G *schlunken* 'to bag, be loose' < **slunk^kōn*-
- OHG *chlimban* 'to climb' < **klimban*- : MHG *klimpfen* 'id.' < **klimp^pan*-
- OHG *bahhan* 'to bake', Swab. *bacheⁿ* 'id.'¹²⁷ : OHG *bachan* 'id.' : OHG *bachōn* 'id.' (cf. Gr. φάγω)¹²⁸

The impact of de-iterativization mechanism should not be underestimated. It probably forms the answer to the question why the strong verbs with roots in **-p-*, **-t-* or **-k-* have such a high representation in Proto-Germanic, even though in Proto-Indo-European voiced stops occurred much less frequently than other stops, the phoneme **b* even being absent. Shortened geminates can indeed be postulated for the following strong verbs:

- Go. *greipan* 'to seize' : MHG *gripfen* 'to grab' ~ G Als. *grippen* 'to steal', Du. *gribbe(le)n* 'to grab'¹²⁹ < **g^hreib^h*- (cf. Lühr 1988: 356)
- Go. *dis-skreitan* 'to tear apart' : G Bav. *schritzen* 'to tear'
- Go. *ga-smeitan* 'to smear' : OE *smittian* 'to befile'
- ON *ríta* 'to carve, write' : OHG *retzōn*, *ritzōn* 'to scratch'
- ON *fjúka* 'to blow', EDu. *fuycken* 'to thrust': MDu. *vocken* 'id.' ~ MHG *fochen* 'id.'
- OHG *riuhhan* 'to smoke' : Cimb. *rucken* 'id.'

¹²⁷ Fischer/Taigel 55.

¹²⁸ Kuryłowicz (1957) analyzed **bakkan*- as a contamination form of the strong verb **bakan*- and the iterative **bakkōn*-.

¹²⁹ De Jager 1875: 160.

- MHG *spriezen*, OFri. *sprūta* ‘to sprout’ : EDu. *sproten* ‘id.’
- MHG *striefen* ‘to strip’ : MHG *strupfen* ‘id.’, MDu. *strop(p)en* ‘to strip; to skin; to rob’
- OE *scēotan* ‘to shoot’ : G *schutzen* ‘to swing, rock’¹³⁰
- MDu. *hūken* (w.v.) ‘to cry’ : G Cimb. *hocken* ‘id.’¹³¹
- WPhal. *hiuken* ‘to squat’ : G *hocken* ‘id.’ ~ ON *hoka* ‘id.’ (cf. Pol. *kuczeć* ‘id.’ < **kuk-*, Uhlenbeck 1897b: 539)
- MLG, MDu. *dūken* ‘to duck, dive’ : MDu. *docken, ducken* ‘to duck’
- MHG *slūchen* ‘to swallow’ : G *schlucken* ‘id.’, Du. *slokken* ‘id.’
- Go. *ana-trimpan* ‘to press upon’ : MHG *trumpfen* ‘to walk, toddle off’
- OFri. *stapa* ‘to step’ : OHG *stapfōn* ‘id.’ ~ Nw. dial. *stabba* ‘to stumble’¹³²

6.4 De-iterativization in Gothic

One of the problems concerning Kluge’s law in Germanic is that, while geminates abound in the other Germanic dialects, they are conspicuously rare in Gothic. This problem has been put into perspective in § 4.3.4, where it was argued that geminates must have existed in Gothic, too, because they are widespread in Gothic hypocorisms. Across the Germanic dialects, hypocorisms are created by forming *n*-stems to full names, and it is unsurprising, for that reason, that they more often than not appear to contain geminates.

The scarcity of long stops in Biblical Gothic nevertheless remains somewhat enigmatic. As opposed to a multitude of words with long resonants, there are only four words that have Proto-Germanic geminates in Gothic, viz. *sakkus* ‘sack’, *skatts* ‘money’, *atta* m. ‘father’ < **attan-* and *smakka* m. ‘fig’. The case of *sakkus* can be discarded, because it is a loanword from Lat. *saccus*. This leaves only three inherited words that directly show the effects of Kluge’s law. It must be stressed,

¹³⁰ Grimm 15, 2128.

¹³¹ Schmeller/Bergmann 193.

¹³² Lühr (1988: 360): “Die aus dem Stamm **stapp-* des Iterativ-Durativs hervorgegangene Lautung **stap-* bildet auch die Grundlage für nominale Ableitungen wie ahd. *stafel*”.

however, that the Gothic Bible in this respect does not differ much from the Old Saxon Heliand, where Proto-Germanic geminates occur in just a few words, i.e. three. The fact that they are ever-present in Middle Low German, on the other hand, proves that geminates for unclear reasons must have been underrepresented in the Heliand. The same probably holds true for Biblical Gothic.

Nevertheless, the lack of geminates in Gothic has led to some wild assumptions, such as the idea that the geminates did not arise in Proto-Germanic, but much later, in Proto-Northwest Germanic (Kuryłowicz 1957). This idea was promoted in a more recent publication by Fagan (1989), who in view of “the absence of geminates in Gothic verbs” (1989: 39) argued that Kluge’s law could not possibly have been of Proto-Germanic date. Fagan assumed that the geminates arose through “expressive gemination” (see chapter 7) no earlier than the Northwest Germanic period (1989: 54). The case of *smakka* Fagan explained away as a loanword from Slavic (cf. OCS *smoky* ‘fig’), while *atta* and *skatts* were discarded for the reason of their unclear etymology.¹³³

However, the scarcity of geminates in Biblical Gothic turns out to be fully compensated by the demonstrable presence of shortened geminates in the strong verbs, as can be concluded on the basis of the previous section. It was convincingly argued by Lühr (1988: 352) that the Gothic strong verbs *dis-hniupan* ‘to tear off’, *sliupan* ‘to crouch’ and *slepan* ‘to sleep’ must have adopted their consonantisms from the pertaining iteratives, e.g. OE *hnoppian* ‘to pluck’, OHG **slopfōn-* ⇒ OHG *slopfāri* ‘itinerant monk’, Icel. *slabba*, *slappa*, *slapa* ‘to slack, laze’ etc. Indeed, *greipan* must, too, be derived from **greip^p-* in view of both Du. *gribben* and Als. *grippen* < **gribbōn-*. Go. *bi-mampjan* ‘to mock’ can probably be added here as well: if it is really related to Gr. μέμφομαι (< **me-mb^h-?*), it can only be derived from a geminated root **mamp^p-*. Further cases that come into question are Go. *dis-skreitan* ‘to tear apart’, *ga-smeitan* ‘to smear’ and Go. *ana-trimpan*. In view of the pertaining iteratives, G Bav. *schritzen* ‘to tear’, OE *smittian* ‘to befle’ and MHG *trumpfen* ‘to walk,

¹³³ Fagan’s explanation of *smakka* as a loanword is not generally accepted (see Gamm 1973: 90). It has been suggested that OCS *smoky*, like *skotā* ‘cattle’, was adopted from Germanic (Vasmer 1953-8, II: 674). It is possible, for instance, that *smakka* is related to the verb **smakōn-* as in OFri. *smakia* ‘to taste (good)’. Moreover, the fact that a geminate coincides with an *n*-stem inflection in *smakka* as well as in *atta* can hardly be attributed to coincidence.

toddle off', it seems likely that these verbs, too, once contained a geminate.

To sum up, there is indirect but nevertheless decisive evidence for the existence of geminates in Gothic. It cannot be claimed, for that reason, that gemination arose in Northwest Germanic, let alone that there was no Kluge's law at all (see chapter 7).

6.5 The rise of PGm. **ū* as a full-grade marker

The continuous derivational flow from strong verbs to iteratives and vice versa does not only clarify the overrepresentation of root-final **p*, **t* and **k* in Germanic. It furthermore seems to offer an explanation for the abundance of class 2 strong verbs with **ū*-vocalism. In this verbal context, the origin of the **ū* has always been problematic. The problem is that the only phonetically regular IE source of PGm. **ū* is **uH*, whereas the only morphologically expectable vocalism is **eu*, strong verbs having the full grade in the present forms.

What makes the problem particularly complex is that **eu* and **ū* typically occur side by side in one and the same strong verb, sometimes even within one and the same dialect. Moreover, the distribution is not the same throughout the Germanic linguistic area; it differs from language to language. Regarding the cross-dialectal distribution, two important observations can be made: 1) both **eu* and **ū* occur in all of the Germanic dialects, including Gothic, but 2) Old Norse, Old High German and – to a much lesser extent – Old English, have a preference for the **eu* vocalism, whereas **ū* has the strongest representation in Old Frisian, Middle Low German and Middle Dutch.

In general, it must be stressed, however, that the distribution of **eu* and **ū* differs from verb to verb, even in the dialects that have a strong inclination towards either variant. The table below, which is an adaptation of Perridon (2001: 32), consists of a sample of the dialectal variation in the Northwest Germanic dialects. Although the problem requires a more extensive survey of the complete material, the sample is still instructive enough to show that the balance between **eu* and **ū* is different in each and every dialect, and that, while **eu* is strongest in Old Norse and Old High German, **ū* prevails in Low German and Dutch. This distribution should be the starting point for any further analysis.

	ON (Nw.)	(O/M)HG	OE/ME	OFri.	(M/E)Du./MLG
‘to bend’	-	<i>biogan</i>	<i>būgan</i>	-	<i>būgen</i>
‘to drip’	<i>drjúpa</i>	<i>triufan</i>	<i>drēopan</i>	<i>driāpa</i>	<i>drēpen</i>
					<i>drūpen</i>
‘to rush, to hiss’	<i>fjúka</i>	<i>fauchen</i>	-	-	<i>fijcken</i>
					<i>fuycken</i>
‘to roar’	<i>hrjóta</i>	<i>rūzzan</i>	-	<i>hrūta</i>	<i>rūten</i>
‘to cleave’	<i>kljúfa</i>	<i>klioban</i>	<i>clēofan</i>	-	<i>klieven</i>
					<i>klūven</i>
‘to crouch’	<i>krjuka</i>	<i>kriohhan</i>	<i>crouchen</i>	-	-
‘to creep’	<i>krjúpa</i>	-	<i>crēopan</i>	<i>kriāpa</i>	<i>krēpen</i>
					<i>krūpen</i>
‘to smoke’	<i>rjúka</i>	<i>riohhan</i>	<i>rēocan</i>	<i>riāka</i>	<i>rē/ieken</i>
					<i>rūken</i>
‘to push’	-	<i>skioban</i>	<i>scēofan</i>	<i>skūfa</i>	<i>skūven</i>
			<i>scūfan</i>		
‘to sneak’	-	<i>sliufan</i>	<i>slūpan</i>	<i>sliāpa</i>	<i>slūpen</i>
‘to close’	-	<i>sliozzan</i>	-	<i>slūta</i>	<i>slūten</i>
‘to sneak’	<i>smjúga</i>	<i>smiogan</i>	<i>smūgan</i>	<i>smūga</i>	<i>smūgen</i>
‘to sniff’	-	<i>schniefen</i>	-	-	<i>snūven</i>
		<i>snūfen</i>			
‘to sprout’	-	<i>spriozzan</i>	<i>sprēotan</i>	<i>sprūta</i>	<i>sprēten</i>
			<i>sprūtan</i>		<i>sprūten</i>
‘to dash’	-	<i>stioban</i>	-	-	<i>stūven</i>
‘to howl’	<i>þjóta</i>	<i>diozzan</i>	<i>þēotan</i>	-	<i>dieten</i>
			<i>þūtan</i>		

It can further be shown that the balance between the two vocalisms in the individual dialects has shifted over time. This is demonstrable, at least, for Old Norse during its evolution into Modern Icelandic. In Old Norse, the class 2 strong verbs include doublets such as *súga* ~ *sjúga* ‘to suck’ and *lúka* ~ *ljúka* ‘to close’, but the variants with **ū* are by far the most frequent ones. This distribution has been completely reversed in Modern Icelandic, where the doublet *lúka* ~ *ljúka* still exists, but *ljúka* has become the dominant variant. The doublet *súga* ~ *sjúga* is not even a doublet anymore, because *sjúga* has fully replaced *súga*.

An evolution in the opposite direction can be observed in the Low German speech area as represented by Middle Low German, Middle Dutch and contemporary Dutch. The verb ‘to smell, smoke’, for instance, goes back to **reukan-* in Old Norse, Old High German, Old English and Old Frisian alike. Middle Low German, on the other hand, has both *rēken* and *rūken*, and in Middle Dutch, too, we find the doublet *rieken* ~ *rūken*. In Modern Dutch, the doublet still exists, but *ruiken* has clearly become the default form, *rieken* being restricted to archaic and figurative speech, e.g. *dat riekt naar censuur* ‘that smacks of censorship’. This distribution indicates that *ruiken* is the invasive variant ousting older *rieken*.

And there is more evidence for the replacement of **eu* by **ū* in Dutch. None of the MLG doublets *drēpen* ~ *drūpen*, *krēpen* ~ *krūpen* and *sprēten* ~ *sprūten* have a counterpart in this language; it only exhibits the variants pointing to **ū* vocalism, viz. *druipen*, *kruipen* and *spruiten*.

In conclusion, three important observations can be made on the basis on the distribution of **eu* and **ū* across the Germanic dialects: 1) **eu* and **ū* were morphologically isofunctional; 2) since all the Germanic dialects have both variants, though in different proportions, the variation must find its origin in Proto-Germanic; and 3) the distribution of the two variants was probably unstable in Proto-Germanic times already, and drifted toward **eu* in some dialects, and to **ū* in others. Now how can these observations be used to find an explanation for the origin of PGm. **ū* in the strong verbs?

The most widely accepted hypothesis regarding the problem was formulated by Campbell (1959: 303), who argued that **ū* was created analogically on the basis of **ī* in the class 1 strong verbs:

“The reason for the intrusion of *ū* into the present of this class is uncertain, but may be no more than analogy with class 1 in

Germanic: after *ei* > *i*, since verbs with *ai* in the past had *i* in the present system, those with *au* in the past might develop long *ū* in the present system”.

This view is a variation to Prokosch (1939: 150), who argued for a similar analogy, though sticking to the old idea that **ū* arose in the *tudāti*-verbs or aorist presents with stressed thematic vowels and zero grade of the root:¹³⁴

“Probably the forms are analogical, following the proportion
steigan (*stīgan*) *staig stigum stigan* = *lukan lauk lukum lukans*.
 An analogical leveling of **lūkan* to **lūkan* is required to make
 the parallelism complete.”

The connection of the **ū*-verbs with the *tudāti*-presents goes back to Osthoff (1882), who assumed that these verbs acquired an analogical full grade **eu* in Germanic that was monophthongized to **ū* before the accent (thus also Boer 1924: § 94). Streitberg (1896: 292), on the other hand, started from the idea that the *ū* regularly developed out of “long *u*-diphthongs”, i.e. roots containing *-ēu-*, in unstressed position. Both explanations depend on *ad hoc* sound laws, however, and cannot be maintained.

Campbell’s explanation has not remained unchallenged either, because the analogical introduction of **ū* is by some held to be implausible (cf. Vine 1985: 62-3 following Schmitt-Brandt 1967; Perridon 2001: 31). It was alternatively argued by Perridon (2001) that **eu* did regularly develop into **ū*, but that this sound law did not reach the whole of the lexicon. This situation would then be comparable to the difference between the British and American English pronunciation of *duke* [djuuk] : [duuk] and *news* [njuuz] : [nuuz] (cf. Phillips 1981). Though the Proto-Germanic problem of the distribution of **eu* and **ū* is indeed reminiscent of the English variation of [juu] and [uu], an important objection to Perridon’s approach is that it does not account for the productiveness of **eu* in Old Norse, where many instances of old *-ú*-have demonstrably been replaced by younger *-jú*-.

¹³⁴ Cf. Go. *digan* ‘to knead’; Go. *trudan*, ON *troða* ‘to tread’; Go. *wulan* ‘to seethe’; ON *koma*, OHG *chuman* ‘to come’; ON *knoða* ‘to knead’.

Another phonologically regular explanation was furnished by Vine (1985). Vine argued that the **ū*-type arose regularly in *tudáti*-verbs of the structure **TReH-u-* or **TeRH-u-*, which in the zero grade would be metathesized to **TRuH-*. Although it cannot be excluded that indeed some laryngealic *tudáti*-verbs ended up in class 2b of the strong verbs, the verbs of the structure **TRuH-* would be very small a basis for the rise of the notoriously productive **ū*-vocalism.

The most likely approach therefore still consists of the view that the **ū*-vocalism arose analogically, as was claimed by Campbell. Although the analogy with the **ī : *i* ablaut of the class 1 strong verbs may indeed have been too weak to introduce the **ū : *u* alternation in the second class, the analogical pressure required for the rise of **ū* on the basis of **ī* is more likely to have been exerted by the iterative system. The iterativization mechanism created a highly dynamic derivational interaction between strong verbs and iteratives. Within such a system, it is plausible that the **ī : *i* opposition triggered the analogical introduction of **ū : *u* next to regular **eu : *u*. Thus, the iterative system provides additional support of Campbell's hypothesis.

Importantly, there is a clear link between the **ū*-vocalism in the strong verbs and the presence of a related iterative (thus also Gamm 1972: 99). To start, it is striking that 10 out of the 15 strong verbs with **ū*-variants shown in the table above have roots ending in voiceless stops. This can hardly result from anything else than the infection of the geminates that arose in the pertaining iteratives.

Then, there are other formal arguments that point to the same link. The verb **sūgan-*, for instance, which apart from the secondary Icelandic *sjúga* has no variant **seugan-*, could theoretically continue a primary thematic present. But if this the verb were the continuation of a thematic present **séuk-e-*, the expected form would have been **seuhan-* (with **eu* and root-final **h*), not **sūgan-* or **seugan-*. This clearly not being the case, it is much more probable that both **sūgan-* and **sūk^kan-* were derived from the iterative **sukkōpi*, **sugunanpi* by the introduction of long **ū* in either of the two different root allomorphs. The required iterative is well attested (see p. 98).

Many other verbs can be mentioned here. It is clear, for instance, that G *zaufen* 'to pluck' < **tūp^pan-* in lack of a variant **teup^pan-* can only be analyzed as a late derivation from the iterative **tuppōpi*, **tubunanpi* (p. 108, 344). Moreover, the root-final consonantism, i.e. **p* < **pp*,

unambiguously reveals its secondary origin as a de-iterative. PGM. **tūp^{an}*- thus is derivationally similar to e.g. OE *sūpan* 'to sip' < **sūp^{an}*-, which is not accompanied by a variant ***seupan-* either, but only by an old iterative OE *soppian*, Du. *soppen* 'to sop, dunk' < **suppōn-* (p. 106). Apparently, this verb, too, was derived from the iterative root **supp-* by the introduction of **ū*.¹³⁵

To sum up, it turns out that the old connection of the class 2b strong verbs with the Sanskrit *tudāti*-presents was unmotivated right from the beginning (cf. Perridon 2001) and can safely be forgotten. One should rather call these verbs "iterative-presents", because many **ū*-verbs appear to be secondary derivations from primary iteratives. Apparently, these iteratives evolved out of nasal presents that were created to primary aorists with a punctiliar aspect, e.g. 'take a zip' or 'pluck off'. They only acquired a strong present when the iterative system became productive enough.

¹³⁵ But perhaps Skt. *sūpa-* m. 'broth' points to a root **suHp-*.

7 A life without Kluge's law?

The idea that the morphology of the Germanic weak nouns and iteratives directly evolved out of the PIE *n*-stems and verbs in **-neh₂-* by the operation of Kluge's law was initially accepted and adopted by Streitberg in *Urgermanische Grammatik* (1896: § 127A), Wright in his *Old English Grammar* (1925: § 256) and Prokosch in his *Comparative Germanic Grammar* (1939: § 22), as has been pointed out in § 4.3.2. Yet however succesful in accounting for the actual data, the Assimilation Theory has been seriously challenged throughout the 20th century. It has been criticized to such an extent even, that it nowadays borders on the uncanonical in both Indo-European and Germanic linguistics. Initially, only the analogical mechanisms as proposed by Kluge were criticized by Kauffmann, Van Helten and Hellquist, who thought the paradigmatic cross-contaminations to be impossible (see § 4.3.2). Later on, the discussion came to be increasingly focused around the reality of Kluge's law itself.

7.1 The Expressivity Theory

The most important criticism of Kluge's law and its consequences was raised by the proponents of the Expressivity Theory. This theory revolves about the idea that consonantal length, in both the nominal and verbal domain of the lexicon, was somehow connected with the charged semantics of the word concerned. This idea, which stems from the time before the rise of the Neogrammarian doctrine of *Ausnahmslosigkeit der Lautgesetze*, was formulated by Gerland in his 1869 monograph *Intensiva und Iterativa und ihr Verhältnis zu einander*. According to Gerland, the Germanic geminates served as a way of indicating the shortness and intensity of the act signified by a root. One of the most frequently mentioned and generally accepted examples of such "psychological" gemination given by Gerland is the German verb *placken* 'to tease', which appears to belong to the more current *plagen* with the same meaning. Geminates, in other words, were not considered to be semantically

empty phones, but rather extra-linguistic instruments that enabled the speaker to randomly adjust the meaning of lexemes.¹³⁶

Gerland's approach became redundant after the formulation of the more falsifiable theory of *n*-assimilation by the Neogrammarians, but was later reanimated by Trautmann, a fierce opponent of Kluge's law. According to Trautmann (1906: 66), iteratives such as OHG *zochōn* 'to jerk' < **tukkōn*- and *lechōn* 'to lick' < **likkōn*- should not be compared to the class 9 verbs in Sanskrit. The lengthening of the root-final consonants would rather be triggered by the "intensive" meaning of these verbs.

In a similar vein, Wissmann (1932) claimed that no correspondences of Germanic iteratives with *n*-presents in other Indo-European branches would be available¹³⁷, and that, as a result, the link between this class and the PIE *neh*₂-type was to be rejected. In Wissmann's view, a verb such as *hüpfen* 'to hop' denoted "eine wiederholte kurze (und oft energische) Handlung: *hüpfen* is nicht einfach 'sich wiederholt im Gelenk biegen (und springen)', sondern, wie es das *Deutsches Wörterbuch* 4, 2, 1954 umschreibt, 'sich in kurzen weiten Sprüngen bewegen' (1932: 172-3). In order to convey this iterativity, the verb would have been given an expressive geminate.

The Expressivity Theory was accepted by a whole range of renowned linguists, including Meillet (1908-9: 355-7¹³⁸, 1928: 166ff., 1937), Martinet (1937), Trnka (1940 [1982]) and – most notably – Pokorny, who systematically applied it in his *Indogermanisches etymologisches Wörterbuch*. As a consequence, it became the dominant opinion in Indo-European and Germanic Studies.

¹³⁶ In contemporary linguistic terminology, this comes down to a breach of Martinet's *double articulation*. According to this principle, independent phones are meaningless, and can only become meaningful by being strung together with other phones. Onomatopoeias form a clear exception to this principle.

¹³⁷ According to Wissmann "gibt es [...] keinen Fall, in dem ein germ. Verbum mit geminiertem Verschlusslaut einem *n*-Präsens einer andern idg. Sprache entspräche" (p. 160), but this is a misconception (see section 3.1 and 5.1.1).

¹³⁸ "Le type intensif à consonne gémignée intérieure, dont lat. *lippus*, delph. λεκχω etc., fournissent des exemples, a certainement tenu beaucoup de place en indo-européen, et il est largement représenté en germanique [...]. L'arm. *lakem* 'je lèche' repose sur **lakk*-, tandis que le *k* simple de lit. *lakù* 'je lèche' [...]; le germanique a de même la gémignée dans le synonyme v. h. a. *lecchōn*; cf. aussi v. irl. *sluccim* 'j'avale', v. h. a. *slucko* 'glouton' [...]."

The acceptance of the Expressivity Theory basically entailed a restoration of the pre-Neogrammarian order, and this can clearly be observed in the literature. While Kluge's law is applied as often as 94 times by Fick, Falk and Torp in *Wortschatz der germanischen Spracheinheit*, the same geminates are as a rule labeled as "intensive" by Pokorny. Pokorny claims, for instance, that MHG *zecke* 'tick' (p. 187-8) has "Intensivschärfung", while Fick, Falk and Torp propose "germ. kk aus ġhn-". Similarly, Pokorny (p. 227) calls OE *tæppa* 'tip' a "*mot populaire* mit intensiver Konsonantenschärfung", thus referring to Meillet's distinction between the phonetically regular *mots savants* and the supposedly expressive *mots populaires*. Somewhat differently, ON *kløpp* 'bridge' is derived from **klampō-* by Fick/Falk/Torp (p. 57), whereas Pokorny (p. 356-64) explicitly ascribes the geminate to "intensive Konsonantenschärfung".

Ever since its incorporation of the Expressivity Theory into Pokorny's dictionary, it has remained a persistent axiom. It is frequently encountered in Seebold's *Vergleichendes etymologisches Wörterbuch der germanischen starken Verben* (1970) and *Etymologisches Wörterbuch der deutschen Sprache* (2002). The Expressivity Theory has further been advocated by Moulton (1972: 166), Van Coetsem (1972: § 3.7.4), Hopper (1989), Fagan (1989) and – more recently – by Ringe (2006). On the whole, it has gained a strong position in Germanic philology. Thus, the state of historical Indo-European linguistics appears to have been more progressive in 1910 than one century later, in 2010.

7.2 Evaluation of the argumentation

The proponents of the Expressivity Theory have brought up numerous arguments against the Assimilation Theory. The remainder of this section consists of an evaluation of these arguments.

7.2.1 No evidence for Kluge's law?

The most important reason for Trautmann and his followers to reject Kluge's law seems to have been the scarcity of extra-Germanic material with *n*-suffixes corresponding to Germanic geminates. Trautmann himself accepted only two pieces of evidence for Kluge's law, i.e. only OFri. *hwit* 'white' ~ Skt. *śvitná-* (sic) and ON *lokkr* 'lock' ~ Lith. *lūgnas*

'lithe'¹³⁹, and he adduced a much larger collection of forms that according to him decisively falsified the law, e.g. ON *botn* 'bottom', ON *logn* n. 'calm', *hrogn* n. 'roe', Go. *rign* n. 'rain', OE *swefn* n. 'sleep' etc. In view of these instances, Trautmann considered it a proven fact that "die heutzutage geltende und blindgläubig angenommene Theorie einer *n*-Assimilation überhaupt falsch und daher aufzugeben ist" (p. 63).

A survey of the literature shows that this argument has been repeated over and over again. According to Wissmann, "gibt es [...] keinen Fall, in dem ein germ. Verbum mit geminiertem Verschlußlaut einem *n*-Präsens einer andern idg. Sprache entspräche" (p. 160). Kuryłowicz in his article *Morphological gemination in Keltic and Germanic* (1957), wrote that "[t]here are [...] quite a number of Germanic verbs with *-nō*-suffix corresponding to the *-nā*-verbs of other IE languages. But no Germanic verb with final geminated stop corresponds to a *-nā*-verb of another language" (p. 133 fn.). In 1989, Fagan stated that "the only possible evidence for assimilation of *n* to a preceding stop is ON *lokkr* 'lock of hair' : Lit. *lūgnas* 'flexible', where the IE accent can be inferred, and OHG *lechōn* 'lick' : Gr. λικνός" (p. 38). In the same year, Hopper started an particularly fierce agitation against Kluge's law in a reaction to Rasmussen (1989a), who had argued in favor of nasal assimilation as the source for the Proto-Germanic geminates:

"There is virtually no evidence from within or outside Germanic for an {*n*-} suffix in any of the geminated forms, nor is there any indication that the progressive assimilation *-dn* > *-dd*- ever occurred. [...] The whole unbelievable complex sequence, whose only empirical stage is the final one (viz. *-tt*-), is to my mind an artefact of the obsession with preserving the Germanic sound shift theory – the very theory for which this bizarre and purely hypothetical train of events is now adduced as evidence." (1989: 247)

¹³⁹ The original meaning of PGM. **lukka-* must have been 'pluck' (cf. Cimb. *lock* 'flock of wool, snow flake' (Schmeller/Bergmann 205)), and seems to be derived from an iterative **lukkōn-* 'to pluck'. The pre-existence of this unattested iterative is supposed by the shortened geminate of **leuk^kan-* ~ **lūk^kan-* 'to pull, pluck', cf. OE *lūkan*, OFri. *lūka*, OHG *liohhan*. The old link with Lith. *lūgnas*, which was suggested by Bezzenger (1876: 1374-5), must therefore be rejected.

Even more recently, the same argument was repeated by Ringe (2006) in his monograph *From Proto-Indo-European to Proto-Germanic*:

"The problem with Kluge's suggestion is simply that the etymologies are unconvincing in detail: the best examples are assembled at Brugmann 1897: 383-4, and not one *must* reflect a form with *-n-. On the other hand, perusal of the numerous examples scattered throughout Seebold 1970 strongly suggests that they have been generated by some sort of sound symbolism ('Intensiv-Gemination'), and that is still perhaps the most widely accepted explanation." (2006: 115)

Still, in spite of this argument being rehearsed time and again over more than a century, the statement that Kluge's law is not sufficiently supported by extra-Germanic cognates with *n*-suffixes is simply incorrect. It is, in fact, an audacity in view of relatively reliable examples such as OE *botm* with **butt-* = Skt. *budhná-*, Du. *wit* ~ Skt. *svítna-* 'white' < **kuit-n-*, EDu. *lappen* 'to lick' ~ Lat. *lambō* < **IHb^h-n-*, Lat. *lingō* 'to lick' ~ Gr. *λιχνεύω* < **lig^h-n-*, MHG *stutzen* 'to bump' ~ Lat. *tundō* < **(s)tud-n-*, OE *þaccian* 'to pat' ~ Lat. *tangō* < **th₂g-n-* etc.

More importantly, the rejection of Kluge's law always seems to be coupled with the failure to recognize the *internal* evidence for Kluge's law in Germanic, which is implied by the strong representation of the geminates in the *n*-stems, as Kluge already pointed out himself (see chapter 4). The fact that it is the internal evidence that decides the issue was also acknowledged by Lühr, who put it as follows: "Die Doppelobstruenten treten vor allem in *n*-Stämmen auf, was in der Flexion dieser Stämme begründet ist" (1988: 191).¹⁴⁰

Lühr (1988: 191) further rightly contended that many *n*-stems with consonant gradation have meanings that cannot possibly be labeled expressive: "Die Bedeutung der meisten Nomina mit Doppeltenuis oder Konsonantenwechsel läßt keine expressive, lautnachahmende oder Intensität beziehungsweise Iteration ausdrückende Lautgebung vermuten." Both these arguments were also advanced by Rasmussen,

¹⁴⁰ Lühr further convincingly argued that the *n*-stems with geminate resonants (cf. **skelō*, **skel-n-ós* ⇒ OHG *scelo*, MHG *schel(l)e* m. 'breeding stallion') are completely parallel to the ones with geminate stops, so that Kluge's law must be assumed to have affected resonants and consonants alike.

who similarly emphasized the importance of the intra-Germanic evidence, and at the same time delicately pointed to the lack of expressiveness of many *n*-stems:

“Daß alle Geminatenwörter als expressiv zu erklären wären, ist aber nicht wahrscheinlich, und daß es so gut wie keine Anhaltspunkte für *n*-haltige Suffixbildung in den einschlägigen Wörtern gebe, ist einfach nicht wahr. Eine sehr bedeutsame Klasse umfaßt *n*-stämmige Substantiva ohne erkennbare ‘expressive’ Bedeutung” (1989b: 252).

In conclusion, the material leaves no room for downplaying the amount of evidence for Kluge’s law, whether internal or external.

7.2.2 Expressive gemination vs analogical singulation

Another counter-argument against Kluge’s law that was featured by Wissmann is that “es den Vertretern der Assimilationstheorie nicht gelungen ist, das Nebeneinander von Bildungen mit Doppeltennis und solchen mit Doppelmedia einigermaßen glaubhaft zu erklären” (p. 161), thereby implicitly rejecting the attempts by Kluge and Van Helten to explain this by paradigmatic contaminations. It is obvious, however, that the Expressivity Theory does not offer an explanation for the voiced geminates either. This was, in fact, admitted by Trautmann himself: “Wie wir uns freilich das Nebeneinander von z. B. *kk-gg-k-g* zu erklären haben, weiss ich nicht” (1906: 66).

The only existing theory that is powerful enough to explain such root variations, is the one that acknowledges consonant gradation and the underlying mechanism of the paradigmatic contaminations. The co-occurrence of ON *riga* ‘to lift heavily’ : MLG *wriggen* ‘to twist’ : ME *wriccen* ‘to wiggle’, for instance, implies two different expressive formations within the Expressivity Theory, the choice between a voiced and voiceless geminate being arbitrary, erratic, or, in other words, scientifically unfalsifiable. By reconstructing a paradigm **wrikkōpi*, **wrigunanþi* < **urík-néh₂-ti*, **urík-nh₂-énti*, on the other hand, the only irregular root form is **wrigg-*, which can readily be explained by contamination of **wrig-* and **wrikk-*.

The occurrence of *analogical singulation* is especially detrimental to the Expressivity Theory. Unetymological singulates must be

reconstructed in many cases, such as Du. *teek*, Cimb. *zecho* < **diǵ^h*- (see p. 254), and also in iterative variants such as MDu. *token* 'to push' < PIE **duk-*, ON *skrapa* 'to scrape' < **skrop/b^h*-, EDu. *stroocken* 'to stroke' < **strug^h*- etc. Within the framework of Kluge's Assimilation Theory, these secondary singulates can easily be accounted for by assuming a paradigmatic split, according to which, for instance, the original paradigm **tukkōpi*, *tugunanpi* 'to jerk' would have been bifurcated into 1. **tukkōpi*, **tukunanpi* (= MDu. *tocken* : *token*) and 2. **tuggōpi*, **tugunanpi* (= E *tug* : *tow*). The Expressivity Theory, on the other hand, offers no solution whatsoever. On the contrary, if one assumes that geminates were introduced on semantic or psychological grounds, long stops being more expressive than short stops, the idea that at the same time a secondary (un-expressive?) singulate should have been inserted, makes no sense. This clearly is the most direct objection against the Expressivity Theory.

To sum up, not one of the objections against Kluge's law can be maintained, in spite of the fact that they have been repeated over and over again. Moreover, the even older, but reanimated Expressivity Theory fails to clarify the systematic nature of the consonantal root variation in the *n*-stems and the iteratives, and therefore implies a serious "Rückschritt", to quote Bloomfield (1925: 92), in comparison to the Assimilation Theory.¹⁴¹ Moreover, the Expressivity Theory seems to contain a critical theoretical fallacy. It is *a priori* implausible that a completely new range of phonemes (i.e. geminates) could be introduced into a linguistic system by extra-linguistic factors such as charged semantics. In this respect, some versions of the Expressivity Theory are truly comparable to what in biology is known as Aristotle's *generatio spontanea* hypothesis (cf. Von Friesen 1897: 14, 17), which revolved around the idea that living organisms, such as flies and eels, come about spontaneously in decaying corpses.

Needless to say that not all the iteratives mentioned by Wissmann and other advocates of the Expressivity Theory must go back to PIE *neh₂*-verbs. Clearly, instances such as ON *klappa* 'to clap', OSw. *kratta* 'to scratch', Nw. *tikka* 'to tap', OE *cluccian* 'to cluck', OFri. *kloppa* 'to knock', ON *okka* 'to sigh', ON *skvakka* 'to make a gurgling sound', are of strong

¹⁴¹ Lühr 1988; Rasmussen 1989b; Kortlandt 1991.

onomatopoeitic nature. The mere existence of onomatopoeias, however, cannot be used as an argument against Kluge's law. A balanced take on the issue was presented by Hellquist in the article *Nordiska verb med mediageminata* (1908).¹⁴² Hellquist accepted Van Helten's (1905: 229-232) adaptation of Kluge's configuration¹⁴³, but nevertheless resisted Von Friesen's inclination to project verbs of the type Sw. dial. *bobba*, Icel. *babba*, *drabba*, *kvabba* back into Proto-Indo-European in spite of their pertinent sound symbolic nature ("Allting skulle vara indoeuropeiskt!"). He endorsed the view expressed that the iteratives ultimately had sprung from the PIE *neh₂*-present, but at the same time stressed that the resulting geminates could have come into use in expressive words. Hellquist's solution was adopted by Prokosch (1939: 71), who delicately summarized that "[o]nce geminates had been established by assimilation, they could easily become the instrument of sound symbolism." By far, this seems to be the most sensible approach to the matter.

7.3 The Leiden Substrate Theory

In the preceding sections, criticism has been directed towards the Expressivity Theory, which according to the author is for the larger part based on an incorrect rejection of Kluge's law and its effects on Proto-Germanic morphophonology. A similar criticism can be brought up against the so-called Leiden Substrate Theory as it was developed towards the end of the 20th century by comparative linguists such as Beekes, Schrijver and Boutkan. This theory was formulated as a way to account for that part of the Germanic lexicon – often estimated around one third – that does not have an Indo-European etymology. Germanic, after all, had been under suspicion of harboring a substrate from the very beginning of Indo-European comparative linguistics, when Sir

¹⁴² The article is a strong attack on von Friesen, who in *De germanska Mediageminatorna* (1897) erroneously tried to explain all the Germanic geminated iteratives as secondary derivations from *n*-stems. Hellquist (1908: 40): "v. Friesen har som bekant i hög grad förenklat problemet för sig genom att afleda dem samtliga ur urgermanska *n*-stammar".

¹⁴³ As has been pointed out, Van Helten retained the derivation of the iteratives from the *neh₂*-presents, but pushed back the paradigmatic cross-contaminations until before the devoicing phase of Grimm's law.

William Jones spoke of “the Gothick” as “blended with a very different idiom” (1798: 423).

7.3.1 The “Language of the Geminates”

It was the Indologist and Indo-Europeanist Kuiper who gave the initial impetus to the formulation of a new methodology. Kuiper, who had studied the Munda loanwords in Sanskrit, decided to apply his experience to the Germanic situation. An important difference between the Sanskrit and Germanic situation, however, is that while the Dravidian and Munda languages are still spoken, the language (or languages) that preceded the Germanic branch became extinct in prehistoric times. Kuiper's way around this problem was to focus on phonetic alternations in Germanic that could not possibly be attributed to any Proto-Indo-European process, so as to isolate non-Indo-European from inherited material. By doing so, Kuiper basically devised a method that could be applied not only to Germanic, but, in fact, to any language of which the parent language's phonology is more or less known.

Two of the most important Germanic substrate features (layer “A2”) that were proposed by Kuiper were 1) root-final consonant variation and 2) prenasalization. This idea was inspired by the typologically similar alternation of intervocalic *-m-*, *-mb-* and *-b-* in Mundari, a North Munda language. The variation, for instance, of Mundari *haba'*, *hamba'* and *hama'* ‘up to, as far as, during’, Kuiper compared to the alternations of **dūb-*: ON *dúfa* ‘to immerse’, **dubb-*: Nw. dial. *dubba* ‘to stoop’, MDu. *dubben* ‘immerse’, **dūp-*: Du. *duipen* ‘to hang one's head’, **dupp-*: Nw. dial. *duppa* ‘to nod’ and **dump-*: SFri. *dumpen* ‘to dive’. The substrate allegedly lying at the basis of this variation was conveniently dubbed the “Language of Geminates” (Schrijver 2001; 2003).

7.3.2 *Hamartia* in Germanic Studies

With the use of the new methodology, Kuiper's colleagues published a considerable number of articles on Kuiper's substrate in Germanic. They added new words and substrate features, many of which are convincing, such as the case of Go. *magus* ‘boy’, *megs* ‘son-in-law’, OIr. *mug* ‘boy’ and OIr. *macc* ‘son’.¹⁴⁴ The root variants pertaining to this etymon cannot be

¹⁴⁴ Boutkan 1998; 2003a.

traced back to a single (Indo-European) proto-form, and this incongruity provides a firm basis for the hypothesis that the word results from some kind of pre-historic language contact.

It should nevertheless be recognized that the Leiden Substrate Theory is weak at a vital point, namely the interpretation of the Proto-Germanic geminates. Kuiper and his followers were not aware, or at least not sufficiently aware of the fact that the alleged substrate-born consonant variation primarily occurred in the *n*-stems and the iteratives. This distribution alone should have been reason enough to doubt the alien origin of such variation, because it begs the question why other morphological categories should have remained untouched by the substrate.

Unfortunately, no such questions were raised. Kuiper, in fact, explicitly mentioned the root alternation of **knaban-*: OE *cnafa*, **knabban-*: OHG *chnappo*, **knapan-*: OE *cnapa* and **knappan-*: OFri. *knappa* 'boy' or 'young man', apparently ignoring the fact that exactly this *n*-stem had been used to illustrate the effects of Proto-Germanic *n*-gemination by Kluge himself. As a result, it appears that many consonant alternations that were staged by Kuiper and his followers as symptoms of substrate influence in reality must be attributed to Kluge's law and its morphophonemic consequences.

Furthermore, one of Kuiper's other prime examples of supposed substrate alternations, the variation of **dūb-*: ON *dúfa* 'to immerse', **dubb-*: Nw. dial. *dubba* 'to stoop', MDu. *dubben* 'immerse', **dūp-*: Du. *duipen* 'to hang one's head' and **dupp-*: Nw. dial. *duppa* 'to nod', can be explained by sound law and analogy as well. By postulating an old opposition of a strong verb **dūban-* and an iterative **duppōpi*, **dubunanpi* < **d^hub^h-néh₂-ti*, **d^hub^h-nh₂-énti*, related to e.g. Lith. *dubùs* 'deep' < **d^hub^h-u-*, the complete set of root variants can be accounted for. The iterative was split-up into 1) **duppōpi*, **dupunanpi* and 2) **dubbōpi*, **dubunanpi*, and thus gave rise to Nw. *duppa*, *dubba*, MDu. *dubben* etc. The consonantism of Du. *duipen*, with final **p* instead of **b*, finds its origin in the iterative geminate; either the strong verb **dūban-* attracted the geminate from **duppōn-*, a kind of contamination that occurred frequently, or **duppōn-* itself served as the basis for the strong verb itself (see §§ 6.3-6.5).

The feature of prenasalization cannot be maintained either, at least not in the root **dump-*, cf. SFri. *dumpen* 'to dive'. In this case, the nasal

can very well continue the Proto-Indo-European nasal infix, which also occurs in many other verbal stems, e.g. Du. *blinken* 'to shine' < **blinkan-* vs *blijken* 'to appear' < **blīkan-* (< **b^hleig-*, cf. Lith. *blizgėti* 'to shine' < **b^hlig-sk-*) and OE *climban* 'to climb' < **klimban-* vs ON *klífa* 'to climb' < **klīban-*, Go. *ga-geigan* 'to gain' < **g^hei-g^hh₂-* (with full reduplication, cf. Uhlenbeck 1905:283) vs OHG *gingēn* 'to desire', Go. *us-keinan* 'to germinate' < **ǵei-n-H-* vs *us-kijanata* 'germinated' and possibly also ON *svimma* 'to swim' < **sue-n-m-(i)e-* vs *symja* 'id.' < **suem-ie-*. In addition, there are the presents with nasal suffixes, such as Go. *fraiñnan* (pret. *frah*), OE *wæcnan* (pret. *wōc*) 'to wake' < **waknan-*.¹⁴⁵

Some of the typically Germanic vowel alternations were added to the substrate armamentarium as well. For instance, the alternation **ū* ~ **u* as in *duipen* and *duppen* was held to be equally indicative of substrate influence as the consonant alternations displayed by this etymological cluster. The problem with this procedure obviously is that the ablaut **ū* : **u* can very well have arisen analogically within the prehistory of Proto-Germanic morphophonology itself. It is indeed non-Indo-European in the sense that it came about in the Germanic branch after the dissolution of the Indo-European dialect continuum, but at the same time, it does not in any way point to language contact.

On top of it all, the ablaut of **ō* with **a* was, too, analyzed as resulting from substrate influence, even though it can well be explained from PIE *-*eh_{2/3}-* : *-*h_{2/3}-* ablaut. According to Boutkan (2003: 248), "[a]ll instances with **ō* : **a* ablaut concern (North) European substrate material and are likely to represent a vowel vacillation that somehow finds its origin in the donor languages." A case that was analyzed as a substrate word on the basis of this criterion was Go. *lofa* 'palm', ON *lófi* 'id.', OHG *lappo* 'id.', *laffa* 'id.', Far. *labbi* 'id.', Icel. *löpp* f. 'paw'¹⁴⁶. First, Boutkan (2003: 247-8) assumed that the consonant variation displayed by this cluster was due to irregular sound substitution: "the borrowed substrate items displayed consonants that were not available in the PGmc. phoneme inventory[...]. This may have led to hesitation and, subsequently, to variation".¹⁴⁷ On the basis of this analysis, Boutkan was

¹⁴⁵ The weak inchoative **waknan-* originally had a medial inflection.

¹⁴⁶ Boutkan 1999b.

¹⁴⁷ Boutkan (1999b: 17): "we could explain kk- as the result of Kluge's Law, but not the voiced stops [...] -gg-."

forced to assume that the $*\bar{o} : *a$ ablaut was not Indo-European either. The reconstruction of an ablauting n -stem $*l\bar{o}f\bar{o}$, $*lappaz$, $*labeni$ (see p. 309) thus became out of the question.

All in all, it seems clear that, even though the isolation of substrate features is a legitimate approach to the investigation of contact with unknown languages, this methodology was focused on the wrong features in the case of Germanic. As such, the Substrate Theory can hardly be seen as anything else than a tragic injury against Germanic Studies. It may be theoretically correct to focus on phonological irregularities in search of substrate elements, but this strategy can only become successful by the incalculable of the specifically Germanic changes that altered the Indo-European dialect into a language with a spirit of its own.

On the whole, it is a simplification to regard linguistic change as a mere series of sound laws making their way through the lexicon. Linguistic change revolves about the transformation of old phonological and morphological systems into new phonological and morphological systems with new distinctions and new oppositions. To quote Bloomfield:

“Jedenfalls dürfte es sicher stehen, daß man das neugerm. Wortmaterial nicht unmittelbar aus dem Jahrtausende weit entfernten uridg. zu erklären hat, und vor allem nicht durch phantastische Erfindungen, sondern daß es unter Beachtung der bisher gewonnenen sprachwissenschaftlichen Prinzipien aus den neugerm. Sprachverhältnissen als normale Entwicklung verständlich ist.” (1925: 106)

In Germanic, the rise of long obstruents by Kluge's law had an important impact on the phonology, because it gave the language a new, characteristic feature that was absent in Proto-Indo-European: phonological consonant length. The operation of Kluge's law in the n -stems and the n -presents affected Proto-Germanic morphology in an important manner, as it transformed the typically Proto-Indo-European ablaut opposition of the suffix into a new opposition of consonant length. Consequently, the language acquired both nominal and verbal paradigms with an allomorphy based on consonant length, a development that truly reshaped the face of Proto-Germanic morphology.

From this perspective, the identification of substrate words on the basis of gemination turns out to be a methodological instrument that must be reconsidered: anyone who accepts that geminates arose regularly by the assimilation of **n*, cannot at the same time use them as an unambiguous substrate feature. It is very well possible that Kluge's law itself was triggered by the absorption of speakers of a local language into the Indo-European dialect that ultimately was to become known as Germanic. In view of the morphological role of gemination in Celtic and Finno-Ugric, it can probably even be defined as an areal feature. The way in which Kluge's law operated, however, was fully regular and thus cannot be treated differently than, for instance, Grimm's law.

8 Root ablaut in the *n*-stems

When in 1887, Kauffmann published his article *Zur Geschichte des germanischen Consonantismus*, his main aim was to refute the way in which Kluge had dealt with the consonant alternations in the Germanic *n*-stems. Kluge ascribed the rise of irregular, voiceless singulates and fricative geminates to paradigmatic *Associationen*, but this stance called forth strong criticism from Kauffmann, who preferred to explain these geminates by regular sound change in the West Germanic period (see § 4.3.2.1). In the final pages of his article, Kauffmann touched upon a very different issue, namely the vocalic alternations that are often found in the roots of the same *n*-stems. These are the topic of the next chapter.

8.1 Kauffmann and nominal ablaut

According to Kauffmann, alternations such as ON *flík* : OHG *flecho* ‘patch’, ON *flóki* ‘tangle’ : OHG *flocho* ‘flake’, ON *fraukr* : OE *frocca* ‘frog’, OE *clēat* ‘pittacium’ : MHG *klotz* : G Hess. *klüte* ‘lump’, OHG *chratto* : *chretzo* ‘basket’, OHG *chreta* : *chrota* ‘toad’, Go. *lofa* : OHG *laffa* ‘palm of the hand’, and OE *hōc* : OHG *hācco*, OE *haca* ‘hook’ proved that the ablaut of the Proto-Indo-European *n*-stems had at least partly remained intact in Proto-Germanic. He formulated this observation as follows:

“Zweifellos war auch noch die alte vocalische abstufung des ablaufs der wurzel lebendig und wir sind berechtigt, die verschiedenen vocalstufen, die wir historisch auf etymologisch identische aber meist nach der bedeutung differenzierte nomina verteilt sehen, in einem und demselben urgerm. paradigma zu vereinigen.” (1887: 544)

Not all of Kauffmann’s examples are tenable within the present state of reconstruction. Since, for example, short **o* is no longer accepted as a Proto-Germanic phoneme, the alleged ablaut of ON *flóki* and OHG *flocho* ‘flake’ can no longer be maintained. Similarly, the vowel alternation of

OHG *chratto* and *chretzo* must be attributed to umlaut rather than ablaut, as will be demonstrated in chapter 10. Still, some of the *n*-stems that were mentioned by Kauffmann seem to have been correctly identified as originally apophonic, e.g. Go. *lofa* : OHG *laffa* 'palm' (see p. 309), G Hess. *klüte* : MHG *klotze* 'lump' (see p. 272).

Kauffmann's idea that the Germanic *n*-stems had retained the root ablaut from the Indo-European parent language did not attract a lot of attention, and never made it into the handbooks. After Otto von Friesen's *De germanska mediageminatorna* (1897), in which a number of *n*-stems with a vowel alternation **ū ~ *u* were referred to as apophonic, the idea seems to have been abandoned for more than a century. Three cases were recently identified by Schaffner, who pointed to the vowel alternations of OSw. *val-mōghe* ~ OHG *magō*, *maho* 'poppy' < **mōgō*, **mageni*, OHG (Notker) *rīdo*, dat. *rīten* 'fever' < **hrīpō*, **hrideni* and OE *mūha* 'pile, bunch' ~ MHG *mocke* 'lump' in his study of Verner's law (2001). Scrutiny of the Germanic lexicon reveals that there are many more *n*-stems as well as some *mn*-stems, *m*-stems and *r/n*-stems that have preserved the old ablaut of the root.

8.2 The ablaut types

On the basis of the linguistic data, several different types of ablaut can be reconstructed for the Proto-Germanic *n*-stems. The types can roughly be divided into two categories: 1) stems with qualitative ablaut, and 2) stems with quantitative ablaut.

The most straightforward category is the one exhibiting qualitative **e : *u* ablaut. It appears to continue Proto-Indo-European **e : zero* ablaut in its purest form, and can be reconstructed on the basis of *n*-stems such as:

- ON *bjalki* m. 'beam' < **belk^kan-* : OE *bolca* m. 'beam, plank' < **bulk^kan-*
- MHG *krebe* m. 'basket' < **kreban-* : MHG *krupfe* 'id.' < **kruppōn-*
- Far. *breddi* m. 'board' < **brezdan-* : OHG *borto* 'id.' < **burzdan-*

- OE *grēofa* m. 'pot' < **greuban*- : MLG *groppe* m. 'pot, cauldron' < **gruppan*-
- G *Zimpe(n)* m. 'tip, stub' < **timban*- : MHG *zumpfe* m. 'id.' < **tump^pan*-

The existence of a rather marginal, but closely related sub-type is supported by a few words that display **e* : **a* ablaut. The most important example is the word for 'hare':

- Nw. dial. *jase* m. 'hare' < **hesan*- : OE *hara* m. 'id.' < **hazan*-

A third qualitative ablaut pattern is supported by a group of *n*-stems with PGm. **a* ~ **u* alternations. It seems to have evolved out of an older **o* : *zero* opposition, and can be abstracted from a couple of very old formations. Although the ablaut **o* ~ *zero* has no real counterpart in any other Indo-European *n*-stem, it is occasionally found in other stem types, cf. Av. *paṇtā*, gen. *paθo* 'road' < *pónt-éh₁*, **pnt-h₁-ós*, and may therefore be old. The most important examples are:

- OHG *lado* 'stake, pole' < **laþan*- : OHG *sumar-lota* f. 'summer shoot' < **ludōn*-
- Go. *maþa* m. 'maggot' < **maþan*- : ON *motti* m. 'moth' < **muttan*-
- OHG *rato* m. 'rat' < **radan*- : MLG *rotte* f. 'id.' < **ruttōn*-

Within the quantitative category, **ī* : **i* ablaut is one of the more prominent types. It developed out of PIE **ei* : **i* ablaut by the regular monophthongization of **ei* to **ī*. This development must have taken place within Proto-Germanic, because the type seems to have triggered the rise of other, secondary quantitative ablaut types, including the **ū* : **u* type.

- Nw. *bie* f. 'bee' < **bīōn*- : G *Biene* m. 'id.' < **binan*-
- OHG *rīdo* m. 'fever' < **hrīþan*- : OHG *ritto* m. 'id.' < **hriþþan*-
- OE *clīðe* f. 'burdock' < **klīþōn*- : OHG *chletta* f. 'id.' < **klīþþōn*-
- Du. dial. *tijg* 'tick' < **tīgan*- : MHG *zecke* m. 'id.' < **tikkan*-
- G *Reihen* m. 'instep' < **wrihan*- : Du. obs. *wreeg* 'id.' < **wriɡan*-

The second quantitative type consists of *n*-stems displaying $*\bar{u} : *u$ ablaut. Like in the strong verbs, the $*\bar{u}$ as a full grade marker probably arose analogically on the basis of the $*\bar{i} : *i$ ablaut. It thus seems to replace the phonetically regular $*eu : *u$ ablaut.

- Icel. *hró* n. 'pile' < $*hr\bar{u}ha-$: MDu. *roc* 'stack' < $*hrukka-$
- Swab. *knaupe* m. 'knob' < $*kn\bar{u}bban-$: OE *cnoppa* m. 'knob' < $*knuppan-$
- Icel. *hnúði* m. 'knob' < $*kn\bar{u}\bar{\theta}an-$: OE *cnotta* m. 'knot' < $*knuttan-$

Another originally quantitative type is represented by *n*-stems with an $*\bar{o} \sim *a$ alternation. This type appears to have arisen in stems with $*h_2$ (and $*h_3?$) in the root. The full grade $*eh_2$ regularly developed into PGm. $*\bar{o}$, the zero grade into $*a$.¹⁴⁸ In synchronic Proto-Germanic, the ablaut is no longer purely quantitative, but it may very well have been purely quantitative prior to the merger of Pre-Gm. $*\bar{a}$ and $*\bar{o}$ into $*\bar{o}$.

- Go. *lofa* m. 'palm of the hand' < $*l\bar{o}fan-$: OHG *lappo* m. 'id.' < $*labban-$
- OSw. *val-mōghe* m. 'poppy' < $*m\bar{o}gan-$: OHG *maho*, *mago* m. 'id.' < $*mahan-$, $*magan-$
- OHG *zuogo* m. 'branch' < $*t\bar{o}gan-$: MDu. *tac(ke)* 'id.' < $*takkan-$

A small, closely related sub-type is the one with $*\bar{o} : *\bar{u}$ ablaut. It occurs in two heteroclitics and one *n*-stem, and seems to have arisen from the alternations $*eh_2/3u : *h_2/3u$ by the regular loss of $*u$ after $*\bar{o}$:

- Go. *fon* : gen. *funins* n. 'fire' < $*f\bar{o}-$, $*funan-$
- Go. *sauil* n. 'sun' < $*s\bar{o}el-$: OHG *sunno* m. 'id.' < $*sunnan-$
- OHG *chruog* m. 'jug' < $*kr\bar{o}ga-$: OFri. *krocha* m. 'scuttle' < $*kruhhan-$

Another quantitative type is borne out by a number of Northwest Germanic *n*-stems with an $*\bar{a} \sim *a$ alternation. The ablaut pattern can

¹⁴⁸ For the vocalization, cf. PGm. $*magra-$ 'lean' < $*mh_2k-r\bar{o}-$ (Beekes 1988).

theoretically have arisen in roots with $*h_1$, as the opposition of $*eh_1$ and $*h_1$ would regularly develop into PGm. $*ē : a$ ablaut. Since, however, there is no extra-Germanic evidence for this type, it is likely to have been created analogically after the $*ī : *i$ and $*ū : *u$ types. Given the dialectal distribution of the particular cases, the ablaut was probably introduced in Proto-Northwest Germanic after the lowering of $*ē$ to $*ā$.

- OHG *hācco* m. 'hook' < $*hēggan-$: OE *haca* m. 'id.' < $*hakan-$
- ON *snákr* m. 'snake' < $*snēk^kan-$: OE *snaca* m. 'id.' < $*snakan-$
- OHG *chrācco* m. 'crook' < $*krēggan-$: G *Krack* 'id.' < $*krakka-$

It is important to stress that not all of the ablauting *n*-stems can be fit into the sub-types mentioned here. The categorization into different types is merely intended as description of most important tendencies, not as a completely hermetic system.

8.3 Methodology

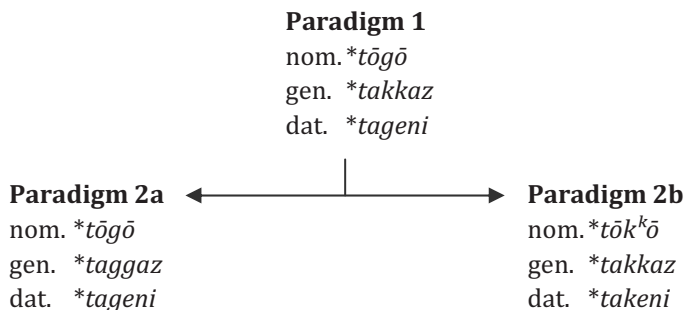
The possibility of ablaut in the *n*-stems was discussed by Lühr (1988) in her treatment of the correlation between Nn. *ferē* 'narrow field, earthen ridge' < $*ferhan-$ and OE *furh* f. 'furrow' < $*furhō-$. Lühr, who was primarily focusing on the Proto-Germanic geminates, took up an agnostic position: "Aus einem derartigen Nebeneinander könnte nun ein ablautender *n*-stämmiger Typ gewonnen worden sein, eine theoretisch zwar mögliche, aber nicht weiter belegbare Vermutung" (1988: 318-9). With this observation, Lühr explicitized an important problem: although the ablaut of some *n*-stems is self-evident in a number of cases, as Kauffmann has shown, it is difficult to *prove* it in the case of Nn. *ferē* vs OE *furh*, because the two words may just as well represent two independent formations. The reconstruction of an ablauting paradigm $*ferhō$, $*furhaz$ is therefore not necessarily wrong, but will at the same time never become more than speculative.

There is, however, a way around the problem. When the *n*-stems exhibit both consonant and vowel gradation, the paradigmatic ablaut is often betrayed by the consonantal analogies. It is somewhat unfortunate, in this respect, that Kauffmann was unable to correctly analyze the *Associationen* called forth by Kluge's law.

8.3.1 Ablaut betrayed by Kluge's *Associationen*

Kluge's paradigmatic *Associationen* play a vital role at identifying paradigmatic ablaut in the *n*-stems, because they often can be shown to have preceded the leveling of the paradigmatic ablaut. As a result, full-grade roots in the separate dialects often appear with geminates that originally belonged to the zero-grade root of the genitive.

For instance, the dialectal variation of MLG *tagge* 'twig, branch' < **taggan*-, MLG *tacke* 'id.' < **takkan*- and LG *take* 'prickle' < **takan*- alone is reason enough to reconstruct a paradigm **tagō*, **takkaz* that split up into both **takō*, **takkaz* and **tagō*, **taggaz*. It can further be surmised on the basis of the clearly related forms such as OHG *zuogo* 'branch', OS *tōgo* 'twig', Du. dial. *toege* 'branch' < **tōgan*-, that the paradigm was, in fact, apophonic. In Proto-Germanic, it would then have to be reconstructed accordingly as **tōgō*, **takkaz* < **déh_{2/3}g^h-ōn*, **dḡ_{2/3}g^h-n-ós* (see p. 317). This reality of this reconstruction is only confirmed, however, by a formation such as Du. dial. *toek(e)* 'branch (with leaves)' < **tōk^kan*-; with its shortened geminate, this formation clearly demonstrates that Kluge's *Associationen* took place before the leveling of the ablaut.



On the basis of the original distribution of the ablaut and the geminates, certain expectations can be made about the frequency of particular root variants in the different dialects. It is to be expected, for instance, that geminated roots more often than not have a zero grade of the root, as Kluge's law primarily operated in the genitive. Indeed, the geminated forms of the word for 'branch, prickle' occur mostly in combination with a zero grade, **takkan*- and **taggan*- being highly frequent forms in the dialects. The formation **takan*-, in contrast, is only marginally attested, and, again, this is exactly what can be expected on the basis of the

original situation. Similarly, the good attestation of the variant **tōgan-* as opposed to the marginality of **tōk^kan-* directly follows from the fact that the full grade was linked to ungeminated roots in the original paradigm.

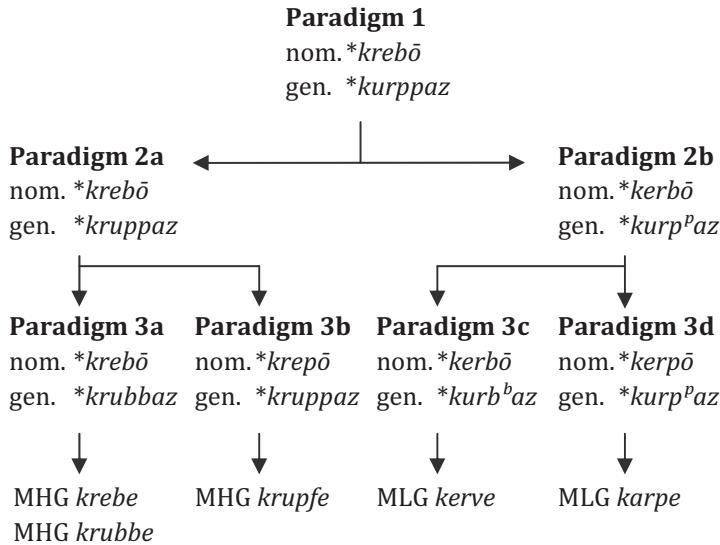
A similar distribution is found with the Germanic word for ‘maggot’ or ‘moth’ (see p. 218), which shows a large number of root variants, viz. Go. *maþa* ‘worm’ < **maþan-*, MHG *matte* ‘moth’ < **maþþōn-*, OE *moppe* ‘id.’ < **muþþan-* and ON *motti* ‘id.’ < **muttan-*. The formal variation of this etymon points to an original paradigm **maþō*, **muttaz* that gave rise to a secondary inflection **maþō*, **muþþaz* by the generalization of the **þ*, and later to a paradigm **maþō*, **maþþaz* by the leveling of the ablaut. In spite of all these secondary developments, the original distribution of ablaut and gemination has at least partly been preserved; throughout the different dialects, the full grade primarily occurs in combination with a singulate **þ*, and – conversely – the zero grade emerges exclusively with a geminate **tt* or **þþ*. Both of these facts are a clear reflection of the proportions that existed in the original paradigm.

8.3.2 Resolution of the *Schwebeablaut*

Additional proof of ablaut is furnished by the *Schwebeablaut* that is displayed by some *n*-stems with a resonant in the root. One of the Germanic words for ‘basket’, which can be reconstructed on the basis of the co-occurrence of e.g. MHG *krebe* m. ‘basket’ < **kreban-* and *korbe* m. ‘id.’ < **kurban-*, seems to be such a case (see p. 179).

	PIE		PGm.
nom.	<i>*gréb^h-ōn</i>		<i>*krebō</i>
gen.	<i>*grb^h-n-ós</i>	→	<i>*kurp^paz</i>
loc.	<i>*grb^h-én-i</i>		<i>*kurbeni</i>

It follows from apparently secondary forms such as MLG *kerve* m. ‘fish trap’ < **kerban-* and MHG *krupfe* f. ‘basket’ < **kruppōn-* that the *Schwebeablaut* was resolved by the leveling of the vowel slot. A new root **krupp-* was created by inserting the zero-grade vocalism into the full-grade ablaut slot. Conversely, a secondary variant **kerb-* was introduced by the insertion of the full grade into the zero-grade slot. As a result, the two new paradigms were split up in many different ways, leading to a staggering amount of root variants. Again, the material seems to imply that this leveling occurred prior to the *Associationen*:



Needless to say that it is not necessary to assume that all of these four different paradigms actually existed side by side, at least not within one and the same language system. The analogical inflections described here are to be interpreted as *possible* pathways of analogy. The actual analogies probably differed from dialect to dialect, and it seems impossible to predict which pathway is used by which language; there are, at any rate, no unambiguous indications for a dialectal distribution regarding the different analogical pathways. The conclusion must therefore probably be that the analogies to a large extent were random.

8.3.3 Over-long syllables in Upper German

In § 3.2, evidence is presented for the claim that geminates were shortened in superheavy syllables. However, the pan-Germanic date of this shortening is actually debated. Over-long syllables occur quite abundantly in the Upper German dialects and it has therefore been claimed that this is proof of their preservation in this sub-group:

- MHG *tāpe*, Swi. App. *đwōppə* 'paw' < **dēbban-*
- OHG *hācco*, Visp. *haacko* 'hook' < **hēggan-*
- OHG *chrācco* 'crook' < **krēggan-*
- OHG *chrāppo* ~ *chrāpfo* 'crook' < **krēbban-*, **krēppan-*

- G Bav. *kauzen* m. 'bundle of flax' < **kūttan*-
- G Swab. *knaupe* m. 'bump, knot' < **knūbban*-
- G *Raupe* f. 'caterpillar' < **rūbbōn*-
- G Pal. *schnaupe* f. 'spout'¹⁴⁹ < **snūbbōn*-
- G *Schnauze* f. 'snout' < **snūttōn*-
- MHG *zūpe*, G *Zaupe* f. 'bitch' < **tūbbōn*-
- Swi. App. *gniippə*, G Swab. *kneip(e)* mf. 'large knife'¹⁵⁰ < **knībba/ōn*- (cf. ON *knífr* < **knība*-)

In his analysis of the material, Kluge himself seems to have had trouble explaining the long stops. "[N]ach langer silbe musste das hd. der treue bewahrer [...] der urgerm. gemination sein", Kluge (1884: 178) first writes in his *Consonantendehnung*. Yet on p. 183, he already withdraws this claim in view of e.g. OHG *wīz*, G *weiß* < **hwīt^taz* < **ķueit-nó*-s. In order to be able to explain the long stops of OHG *hācco* > G *Haken*, Kluge therefore proposed that the paradigmatic alternations in the above cases were analogically reintroduced from *n*-stems with a short vowel, like **knabō*, **knappaz*.¹⁵¹

Kluge's explanation immediately provoked a negative reaction from Kauffmann (1887: 509 fn.), who deemed it "nur auf dem papier denkbar". Lühr, too, dismissed the analogy (1988: 214) and referred to the older notion that "außerhalb des Althochdeutschen in den germanischen Sprachen Doppelobstruenten nach langer Silbe grundsätzlich vereinfacht wurden".¹⁵² Lühr thus claimed that the shortening of geminates in heavy syllables did not affect Upper German, although – as Kluge already pointed out – this is in conflict with the shortening of e.g. **hwīt^ta*- 'white' in e.g. Swi. *wīss*,¹⁵³ and many other examples such as OHG *tiof*, G *tief* 'deep' < **deup^aa*- < **d^heub^h-nó*-.

It is possible, though, to avoid the wholesale reintroduction of consonant gradation from *n*-stems with light syllables to the ones with heavy syllables, as Kluge proposed, and at the same time retain the pan-Germanic date for geminate shortening. If the ablaut of the affected *n*-

¹⁴⁹ Christmann 5, 1265.

¹⁵⁰ Vetsch 143; Fischer/Taigel 279.

¹⁵¹ Accepted: Van Helten (1905: 229).

¹⁵² Cf. Görtzen 1998: 433.

¹⁵³ Vetsch 184.

stem remained intact long enough, it is conceivable that the geminate of e.g. OHG *hācco* was adopted from the zero-grade oblique **hakkaz*, where the geminate should never have been lost. The original paradigm **hēgō*, **hakk(en)az*, *hageni* (see p. 327), for instance, may have been changed into Proto-Alemannic **hāggō*, **haggen*, **hagin*. Similarly, the geminate of Swab. *knaupe* can be explained by assuming that an original paradigm **knūbō*, **knupp(en)az* was remodeled into **knūbbō*, **knubben* in Proto-Alemannic.

Phonologically, the reintroduction of geminates to heavy roots may have been enabled by the effects of West Germanic *j*-gemination. By this gemination, superheavy syllables were re-introduced in Proto-West Germanic, and unlike in the other West Germanic dialects, these new geminates were never shortened in Upper German. Thus we find nouns such as G *Weizen*, Visp. *weitz* 'wheat' < **hwaitja-* and Swi. *zōukx* 'bitch' < **taukjō-* and late de-iteratives such as G *schnauzen* 'to snap' < **snüttōn-*.

In summarizing, there are three possible methodological procedures that can be used to prove that the *n*-stems were apophonic: 1) Kluge's *Associationen*, 2) the resolution of *Schwebeablaut* and 3) the possible re-introduction of superheavy roots in the Upper German *n*-stems. The last option is disputable, however, because the shortening of over-long syllables in Upper German is not generally accepted. In the discussion of the material in chapter 9, the first two procedures are therefore given priority.

8.4 Reconstructing the ablauting paradigm

The reconstruction of the Proto-Germanic ablaut is based on two sources of information. First, there is the evidence from the other Indo-European languages. On the basis of the paradigm e.g. of Skt. *ātmán-* 'breath', the Proto-Indo-European full grade of the root is usually located in the nominative and the accusative, the zero grade in the other cases (see § 2.1.1). This distribution is more or less confirmed by the internal evidence from Germanic. Although it is generally difficult to assign the different vowel grades to a particular case on the basis of the Germanic material alone, it has been shown in § 8.3.1 that it is nevertheless possible to establish a correlation between zero-grade roots and

geminates. Although this correlation is not equally strong in each particular *n*-stem, it clearly points to the genitive case as the point of origin, because this case originally combined a zero-grade root with a zero-grade suffix and stressed full-grade ending (**CC-n-ós*). As such the cross-dialectal distribution of the *n*-stem variants appears to be a clear reflection of the Proto-Indo-European situation.

Germanic further offers some potentially interesting information on the presence of an *o*-grade in the paradigm. A considerable number of *n*-stem clusters are accompanied by forms with an *o*-grade in the root. In many cases, these are thematic *a*-stems which, given the productivity of this formation, could easily qualify as independent derivations from a primary verbal root. In other cases, however, no such primary verbal root is extant. This could, of course, simply mean that the original verb was lost after the creation of the nominal derivations. Yet, in view of the strikingly nominal character of some clusters, the *o*-grade can alternatively be attributed to old paradigmatic ablaut. The corpus of thematic *o*-grades consists of the following cases:

- **brezdō*, **burzdnaz* ‘edge, board’ : **brazda*- = Far. *breddi*, OHG *borto* : OHG *brart* (p. 149)
- **elm*, **ulmaz* ‘elm’ : **alma*- = OHG *elm(o)*, OE *ulm-trēow* : ON *almr* (see p. 155)
- **kelkō*, **kulk^kaz* ‘mandible’ : **kalk^ka*- = Icel. *kjálki*, Da. dial. *kulk* : Icel. *kálkur* (see p. 172)
- **reumō*, ?**rūm^maz* ‘cream’ : **rauma*- = Icel. *rjómi*, ?Swi. *ruumme* : MHG *roum* (see p. 186)
- **timbō*, **tump^paz* ‘stub, penis’ : **tamp^pa*- = G *Zimpe(n)*, MHG *zumpf(e)* : Du. *tamp* (p. 192)
- **gīmō*, **gimenaz* ‘aperture’ : Icel. *gíma*, ON *gima* : Icel. *geimur* < **gaima*- (p. 231)
- **sīlō*, **sillaz* ‘trace, horse harness’ : **saila*- = G *Seilen*, ON *seli* : G *Seil* (p. 243)
- **skīmō*, **skimenaz* ‘shine, shade’ : **skaima*- = Go. *skeima*, MLG *scheme* ‘shade’ : MHG *scheim* (p. 246)
- **swīmō*, **swīmenaz* ‘dizziness’ : **swaima*- = Icel. *svími*, *svími* : ON *sveimr*, *sveim* (p. 251)
- **hrūhō*, **hrukkaz* ‘pile’ : **hrauk^ka*- = Icel. *hró*, MDu. *roc* : ON *hraukr* (p. 268)

- **klūþō*, **kluttaz* ‘clot’ : **klaut^a-* = MHG *klūde*, MHG *klotze* : OHG *chlōz* (p. 272)
- **knūþō*, **knuppaz* ‘knob’ : **knaup^a-* = Swab. *knaupe*, OE *cnoppa* : MHG *knouf* (p. 297)
- **hēgō*, **hakkaz* ‘hook’ : **hōk^a-* = OHG *hācco*, OE *haca* : OE *hōc* (see p. 327)
- **snēgō*, **snakkaz* ‘snake’ : **snōk^a-* = ON *snákr*, OE *snaca* : Sw. *snok* (see p. 333)
- **krēgō*, **krakkaz* ‘crook’ : **krōk^a-* = OHG *chrācco*, G *Krack* : ON *krókr* (see p. 331)

The majority of the *o*-grade forms given here are probably independent thematizations. There are some indications, however, that the *o*-grade did exist within the original *n*-stem paradigm. This can be concluded on the basis of the vocalism of a few old accusative plural split-offs.

The word for ‘beam’, for instance, has preserved two vowel grades in Old Norse: an *e*-grade in ON *bjalki* < **belk^an-* and an *o*-grade in *bōlkr* < **balk^u-*. In view of the clear distribution of these two ablaut grades, it is possible to postulate a paradigm **belk^ōō*, gsg. **bulk^aaz*, apl. **balk^uuns* from **b^hélǵ^h-ōn*, **b^hlǵ^h-n-ós*, **b^holǵ^h-n-ís* (see p. 148).

Similarly, the ablaut of Icel. *hnúði* ‘knob, hump’ < **knūþan-*, Icel. *hnotti* ‘tussock, ball’, OE *cnotta* ‘knot’ < **knuttan-* and ON *knōttr* ‘ball, knob’ < **knattu-* could theoretically be attributed to a paradigm **knūþō*, gsg. **knuttaz*, apl. **knattuns*. This particular case, however, must be secondary in two different ways. After all, the **ū* : **u* ablaut of the nominative and the genitive and the **a* : **u* ablaut of the accusative and the genitive can only indirectly continue PIE **eu* : **u* and **o* : *zero* respectively.

In spite of the scarcity of the evidence, the triple **e* : **o* : *zero* ablaut can very well have had a linguistically real basis in the Indo-European parent language. As a matter of fact, the existence of the **e* : **o* ablaut was already predicted by Beekes (1985: § 95). Beekes assumed that the original hysterodynamic paradigm **CéC-R*, acc. *CC-ér-m* would give rise to a secondary type **CéC-ōR*, acc. **CoC-ér-m* by the leveling of the ablaut and the development of unstressed **e* to **o* (see § 2.2). In view of this explanation, it is conceivable that at least some of the Germanic *a*-grades, in spite of their thematic suffixation, split off from the accusative case of an original *n*-stem paradigm. It may not be coincidental, then, that many

of these cases have a geminated root, as can be observed from the list above.

9 The evidence

The present chapter is an attempt to provide a survey of the Germanic *n*-stems that potentially qualify as apophonic. In addition to these *n*-stems, some related stem formations, i.e. *mn*-stems, *m*-stems and *r/n*-stems, have been included as well, because they are morphologically and typologically similar.

The survey is set up rather maximalistically. It contains not only those *n*-stems of which the reconstruction of the ablaut is beyond doubt, but also the less certain cases. Although this approach has the disadvantage that the reader's effort sometimes remains unawarded, the advantage is that little relevant material is left out. Moreover, the inclusion of rejected items hardly detracts from the evidentiality of approved items. On the contrary, they to a large extent elucidate the kind of considerations that are to be made while studying the ablaut of the *n*-stems.

The material has been split up into three categories: 1) formations with plausible ablaut, 2) formations with questionable ablaut, and 3) formations of which ablaut is rejected. These classifications are put behind the reconstructed Proto-Germanic paradigms. Plausible cases are left unmarked, questionable cases are marked with a question mark (?), and rejected cases are indicated with a dagger (†).

The material is arranged according to the nature of the actual ablaut patterns. Ten different subtypes have been set up in order to arrange the evidence as logically as possible. Most of the below ablaut types have been described in § 8.2.

9.1 The **e ~ *u* type

The **e ~ *u* alternations displayed by the *n*-stems below belong to the most straightforward type, continuing PIE **e* : *zero* ablaut. The evidence for this type is limited in comparison to, for instance, the **ī ~ *i* type, but, still, the material is substantial. Beside the usual *n*-stems, it also contains

two ablauting *m*-stems: **elm*, **ulmaz* ‘elm’ and **helm*, *hulmaz* ‘cane, blade (of grass)’.

****belkō*, **bulk^kaz* ‘beam’**

- **belk^kan-*: ON *bjalki* m. ‘beam’¹⁵⁴, OSw. *biælke* m. ‘id.’
- **balk^kan-*: OE *bealca* m. ‘id.’, E *balk*, *bawk*, OFri. *balka* m. ‘id.’, OS *balko* m. ‘plank’, MLG *balke* m. ‘beam’, MDu. *balk(e)* m. ‘id.’, Du. *balk*, OHG *balcho* m. ‘id.’, MHG *balke* m. ‘id.’, G *Balken*
- **balk^ku-*: ON *bþlkr* m. ‘partition’¹⁵⁵, OSw. *balker* m. ‘beam’
- **bulk^kan-*: OE *bolca* m. ‘gangway, duckboard’¹⁵⁶, OHG *bolcho* m. ‘gang board’¹⁵⁷

The individual Germanic dialects contain evidence for three different ablaut grades for this PGM. *n*-stem. An *e*-grade is found in ON *bjalki*, which displays regular *a*-breaking of *-e-* to *-ja-*. The zero grade **bulk^k* occurs in OE *bolca* and OHG *bolcho* (5x), which bear the slightly differentiated meaning ‘duckboard’. In West Germanic, the *a*-grade is the dominant ablaut form, being represented by the wide-spread *n*-stem **balk^kan-*. The *a*-grade is not restricted to West Germanic, however, as is shown by the ON *u*-stem *bþlkr* < **balk^ku-*.

The consonantism is stable throughout the Germanic dialects.¹⁵⁸ This could mean that the root-final **k* regularly continues PIE **ǵ* (thus Stang 1972: 11), but it is possible, too, that PGM. **k* reflects an oblique geminate that was generalized at an early stage. In that case, the original articulation of the root-final consonant cannot be determined on the basis of the Germanic evidence. Indeed, the Balto-Slavic correspondences indicate that the PIE root was **b^holǵ^h* rather than **b^holǵ-*, as follows from the accentuation of e.g. Lith. *balžienas* m. ‘cross-beam’ and Ru. *bólozno* ‘thick plank’¹⁵⁹. The only way to reconcile the Balto-Slavic

¹⁵⁴ De Vries 1962: 38.

¹⁵⁵ De Vries 1962: 70.

¹⁵⁶ Holthausen 1934: 30.

¹⁵⁷ EWA 229: “Viell. ist das erst spät bezeugte ahd./mhd. Wort aus dem Ae. entlehnt?”

¹⁵⁸ Note that the case of **hnekkōn* ‘neck’ (see p. 167) is comparable in this respect.

¹⁵⁹ Derksen 2008: 54.

material with the Germanic *n*-stem is to derive the root-final **k* from a geminate produced by Kluge's law.

The North Germanic stem **balk^ku-* sheds more light on the exact inflection of the original *n*-stem. Formally, it is completely parallel to other *u*-stems with geminates, such as *knǫttr* 'ball' and *hǫttr* 'hat', which all evolved out of old plural accusatives in **-n-ŋs* (Lühr 1988: 208). It follows that the stem **balk^ku-* probably originates from the same case, and that this case may have been the source of the *o*-grade. Given the fact that *bǫlkr* occurs in combination with the *e*-grade *bjalki* in Old Norse, the original paradigm can tentatively be reconstructed as **belk^kō*, gsg. **bulk^kaz*, apl. **balk^kuns* from older **b^hélǵ^h-ōn*, **b^hlǵ^h-n-ós*, **b^holǵ^h-n-ŋs*.

A couple of etymological dictionaries¹⁶⁰ raise the question whether PGm. **bluka-* 'block' belongs here. This is questionable, however, because the Balto-Slavic evidence show that the original root was **b^helǵ^h-* rather than **b^hleǵ^h-*.

****brezdō, *burzdeni* 'edge, board'**

- **brezda(n)-*: Far. *breddi* m. 'edge, side'¹⁶¹, OSw. *brædder* m. 'id.', Nw. *bredd*, dial. *bredde* m. 'id.'
- **bruzda(n)-*: ON *broddr* m. 'tip, edge, shoot'¹⁶², Nw. *brodd* m. 'tip, shoot, sting, elk hair', Nw. *brodde* m. 'tip', OE *brord* m. 'tip, shoot, blad', OHG *brort* m. '*corona, labium, prora*', MHG *brort* m. 'id.'¹⁶³
- **burzda-*: ON *borð* n. 'edge, table, (ship)board'¹⁶⁴, OE *bord* n. 'board, plank', OHG *bort* '*sculptorium*', MHG *bort* mn. 'edge, board'¹⁶⁵, OS *bord* 'board, shield'
 ⇒ **burzdan-, -ōn-*: ON *borði* m. 'tapestry'¹⁶⁶, OHG *borto* '*auriphrygium, ligatura, limbus*', MLG *borde*, OE *borda* m. 'seam, embroidery', *borde* f. 'tab'

¹⁶⁰ Vercoullie: 40; De Vries/Tollenaere: 86; Franck/Van Wijk: 73.

¹⁶¹ Poulsen 140.

¹⁶² De Vries 1962: 58.

¹⁶³ Lexer 1, 359.

¹⁶⁴ De Vries 1962: 50.

¹⁶⁵ Lexer 1, 329: "*durch Ausfall des r aus ahd. prort, Rand, Vorderteil des Schiffes.*"

¹⁶⁶ De Vries 1962: 50.

- **brazda-*: Icel. *bradd* n. 'edge'¹⁶⁷, Nw. dial. *bradd* mf. 'shore, side', OHG *brart* 'labium, prora', MHG *brart* m. 'edge, board', OE *brerd*, *breard*, *breord* m. 'brim, margin, border'
- **barzda-*: ON, Icel. *barð* n. 'edge, prow', Nw. *bard* m. 'side, edge'

Forms such as Far. *breddi* < **brezdan-* and ON *borð* < **burzda-* are in clear ablaut relation to each other.¹⁶⁸ It further follows from the leveling of the *Schwebeablaut*, which gave rise to contamination forms such as ON *broddr*, OE *brord*, OHG *brort* < **bruzda-*, that the two ablaut variants must have been in close contact with each other. It seems justified, for these reasons, to consider the reconstruction of a paradigm **brezdō*, **burzdeni* < **b^hrésd^h-ōn*, **b^hrsd^h-én-i*.¹⁶⁹

As in many other cases, an *o*-grade root is found in some closely related thematic formations, viz. Icel. *bradd* n. 'edge'¹⁷⁰, OHG *brart* 'id' < **brazda-* and ON, Icel. *barð* n. 'edge, prow' < **barzda-*. The first formation is strikingly similar to OIr. *brot* 'prickle' < **b^hrozd^h-o-*, and can therefore be old. The latter formation, **barzda-*, seems to have been adapted to the vowel slot of the zero-grade root **burzd-*. This could mean that the formation split off from the *n*-stem at a late stage, i.e. after the leveling of the *Schwebeablaut*. It cannot be excluded, however, that the *o*-grades were created to a lost strong verb **brezdan-*.

An important question is whether the formations under discussion are related to the Germanic word for 'beard', cf. ON *barð*, OE *beard*, OFri. *berd*, OHG *bart* m. 'beard'. This is not at all implausible in view of the relatively small semantic difference between the original meaning 'prickle' (cf. OIr. *brot*) and 'beard'. Admittedly, the reconstruction of the word as **barzda-* has rather great consequences. It implies, for instance, that Lith. *barzdà* and OCS *brada* 'beard', which apparently reflect **b^horzd^h-eh₂-*, are loanwords from Germanic, the vowel slot of **barzd-* being a purely Germanic innovation. The same can be said about Lat. *barba*, which cannot be derived from **b^horzd^h-eh₂-* anyway, because the

¹⁶⁷ Böðvarsson 98.

¹⁶⁸ Note that Fick/Falk/Torp (1909: 264, 266) already tentatively suggest that PGm. **burda-* 'side, board' etymologically belonged to the cluster of **brezd-*.

¹⁶⁹ In view of the lack of geminated forms, however, it is possible, too, that the word was originally inflected as an ablauting root noun **brezd-z*, **burzd-iz*.

¹⁷⁰ Böðvarsson 98.

outcome would have been ***forba*. It is therefore not improbable that the Latin word indeed is a loanword. However, it is unclear how and why the Germanic word should have spread to Balto-Slavic and Italic at such an early stage.

Etymologically, the element **brezd-* can be an extension of the PIE root **b^hrs-* as found in Skt. *bhṛṣṭí-* f. 'tip, edge' and cognates¹⁷¹, but this word is usually reconstructed as **b^hrk-ti-*. Within Germanic, Kluge and Mitzka (1967: 99) give G *Brett* < PGm. **breda-* 'board' as "eine ablautende Nebenform zu *Bord*". Holthausen (1934: 33), on the other hand, considered this formation to be related to **braidā-* 'broad', cf. OHG *breta* 'palma', OE *hand-brede* f. 'palm of the hand' < **bridōn-* (cf. OHG *brit* 'tabula, laterculus'). It seems more probable, then, that it is a dissimilatory form that came about in the plural **Brerter* < **brezd-izō*.

****dimbō, *dump^aaz 'haze'†***

- **dimbōn-*: OSw. *dimba*, *dimma* f. 'mist', Sw. dial. *dimma* f. 'id.'¹⁷²
 - **dumbōn-*: ON *dumba* f. 'dust'¹⁷³, Icel. *dumba* f. 'mist, dust'¹⁷⁴, Far. *dumba* f. 'chaff'¹⁷⁵, Nn. *dumbe* f. 'dust, chaff'
 - **dump^aa-*: MDu. *domp* m. 'haze', EDu. *domp* 'vapor, exhalatio'
⇒ **dumpjan-*: MHG *dümpfen* 'to extinguish'
-
- **damba-*: OSw. *damb* n. 'steam, haze', Nw. *damb* n. 'dust, chaff'
⇒ **dambjōn-*: Icel. *demba* f. 'shower', Nw. dial. *dembe* f. 'thin overcast'
 - **damp^aa-*: OHG, MHG *dampf*, MLG, MDu., Du. *damp* m. 'haze'
⇒ **dampjan-*: OHG *tempfen* 'to extinguish', MHG *dempfen* 'to choke', MLG *dempen* 'to suppress', EDu. *dempen* 'to choke, extinguish', Du. *dempen* 'to temper'

¹⁷¹ Pokorny 109-110.

¹⁷² Hellquist 92; De Vries: 87; Pokorny 247-248.

¹⁷³ De Vries 1962: 87.

¹⁷⁴ Böðvarson 151.

¹⁷⁵ Poulsen 203.

The co-occurrence of OSw. *dimba* ‘mist’, ON *dumba* ‘dust’, and MDu. *domp* ‘haze’, in spite of their slightly different meanings, could theoretically be accounted for by reconstructing an ablauting *n*-stem **dimbō*, **dump^aaz* < **d^hémb^h-ōn*, **d^hmb^h-n-ós*. Given the complete absence of gemination in Nordic, however, the material basis for this paradigm remains weak. More probably, the nominal ablaut is to be explained as resulting from a strong verb, e.g. Sw. dial. *dimba*. This is further supported by the strikingly deverbative *o*-grade form OSw. *damb* ‘steam’.

It is to be noted that, while the Nordic root variants all end in a singulate **b*, West Germanic exclusively has geminated roots. There is the aforementioned MDu. *domp* < **dump^aa-* and the widespread OHG *dampf*, Du. *damp* < **damp^aa-*. The Swedish strong verb *dimba* is furthermore mirrored by MHG *dimpfen* ‘to smoke’, MLG *gedumpen* ‘choked’¹⁷⁶, MDu. *bedompen* ‘covered with condense’ < **dimp^aa-*. The verb probably received its geminate from an iterative formation **dump^aōn-*, cf. EDu. *dompen* ‘to haze, smoke’.

The *n*-stems *dampan-*: OHG *dampfo* m. ‘cold’, MHG *tampfe* m. ‘cloggedness’, MLG *damp(e)* m. ‘shortness of breath’, **dampjan-*: OHG *dempfo* m. ‘cold’, MHG *dempfe* f. ‘shortness of breath’ and **dumpa(n)-*: MHG *dumpfe* m. ‘tuberculosis’, MLG *dumpe* m. ‘asthma’ appear to be derived from **damp^aōn-*, **damp^ajan-* and **dump^aōn-* correspondingly. The meaning ‘cloggedness’ developed in early West Germanic.¹⁷⁷

***drenō, *durraz ‘drone’**

- **drena(n)-*, *-ōn-*: OHG *treno* ‘apis, fucus’, MHG *tren* m. ‘drone, bee’¹⁷⁸, Swi. App. *tree* f.¹⁷⁹, Ja. *trenə* m. ‘id.’¹⁸⁰, OS *dreno* ‘fucus’, MLG **drene* (= EDa. obs. *drene* ‘drone’¹⁸¹), Du. dial. *drene* ‘drone’¹⁸²

¹⁷⁶ EWA 576.

¹⁷⁷ Franck/Van Wijk 105.

¹⁷⁸ Lexer 2, 1503.

¹⁷⁹ Vetsch 105

¹⁸⁰ Stucki 123 = § 69, 2: “Die nasalierten *e*-Laute erscheinen alle als *ɛ*”.

¹⁸¹ Kalkar 380.

¹⁸² Weijnen 36; WLD II.6, 5.

- **drana(n)-, -ōn-*: OE *dran, drane, dræn* ‘fucus’¹⁸³, ME *drane*, E dial. *drane*, OS *drano* ‘fucus’, *drani* ‘fuci’¹⁸⁴, G obs. *Tran*
- **druna-, -ōn-*: MLG *drone, drane* m. ‘drone, slacker’¹⁸⁵, G *Drohne*¹⁸⁶, MDu. *darne, dorne* f. ‘some kind of bee’¹⁸⁷, Du. *dar* ‘drone’¹⁸⁸, SFri. *droane* f. ‘id.’, E *drone*
- **duran-*: OE *dora* m. ‘bumble-bee’¹⁸⁹, ME *dorre* ‘drone’, E obs. *dor* ‘buzzing bee’¹⁹⁰

The Germanic word for ‘drone’ displays a number of ablauting root variants. At least four different variants with three different vowel grades must be reconstructed.

An *e*-grade is found in OHG *treno*, MHG *tren(e)* and in forms from the Swiss dialects. The Appenzell form *tree* is of some importance, because this dialect retained the distinction between [ɛ] < PGm. **e* and the primary and secondary umlaut products [e] < OHG **ā₁* and [æ] < OHG **ā₂* < PGm. **a*. According to Vetsch’s historical grammar, App. [æ] and [ɛ] were raised to [ɛ] and [e] before a nasal, which means that *tree* points to PGm. **drenōn-* with **e* rather than umlautet **a*. The formation is not attested in Middle Low German, but the Trier gloss *dreno*, the Dutch Limburgian form *dreen* and the obsolete Danish form *drene* ‘drone’ provide sufficient evidence for the continuation of PGm. **drenan-* in the Low German area.

A zero-grade form **drunan-, *drunōn-* is found in MLG *drone, drane*, MDu. *darne, dorne*, SFri. *droane* and E *drone*. In MLG, the vacillation between *a* and *o* is the usual outcome of PGm. **u* in open syllables. MDu. *darne* goes back to the zero grade as well, the shifting vocalism being the result of the common methathesis of *r*, cf. MDu. *barne, borne* ‘spring’ < **brunnan-* and MDu. *starte, storte* ‘throat’ < **strut(t)an-* (Van Loey § 58). Vercoullie (p. 60) and Philippa/De Brabandere/Quak (p. 521) assume

¹⁸³ OEC 0614, 0043, 0562.

¹⁸⁴ Gallée 47.

¹⁸⁵ Lübben 84.

¹⁸⁶ Kluge/Seebold 216.

¹⁸⁷ Verdam 148.

¹⁸⁸ Vercoullie 60; Philippa/De Brabandere/Quak 520-1.

¹⁸⁹ Bosworth/Toller 209.

¹⁹⁰ Cf. also EMoE *dorre* ‘drone’ (P. Levens (1570): *Manipulus Vocabulorum*).

that modern Dutch *dar* developed from this **darne* by assimilation of the *n*. Given the late 19th century attestations of the plurals *darns* and *darnen*, however, such a phonetic explanation seems unwarranted. Instead, *dar* can be explained as a backformation from an apocopated form **darn*, which would have received epenthetic *a* between the *r* and the *n*. The resulting **dārān* would then have been interpreted as a plural form with the suffix *-en* (cf. *ui* 'onion' < **uien* = Fr. *oignon*).

Another zero-grade is evidenced by OE *dora* 'bumble-bee', ME *dorre* 'drone' < **duran*.

The OE glosses *dran(e)* and *dræn* are often assumed to have had long vowels, i.e. *drān* and *drǣn* < PGm. **drēn(i)-* or **drain(i)-*. The problem, however, is that the root **drain-*, with its diphthong, makes no sense etymologically. Furthermore, the root **drēn-* would have developed into OE ***drōn* with rounding before *n* as in *mōna* m. 'moon' < **mēnan-*. The OED therefore reconstructs PGm. *dran-* with a short vowel, by which also ME and E dial. *drane* receive a natural explanation.

Just like the Old English forms, the OS glosses *dran* (sg.) and *drani* (pl.) are often cited with long vowels.¹⁹¹ The reason for this is that G *Drohne* is believed to have developed out of PGm. **drēn-* with the incidental rounding of *ā* as in *Mond* 'moon' < **mēna-* and *Ton* 'clay' = MHG *dāhe*, -n f. < **panhōn-*. In view of the initial *d*, however, it is more likely that *Drohne* was borrowed from Low German *drone* < **drunan-*. The form **drēn-* is at any rate excluded by the Saterlandic Frisian form *droane* < **drunan-*. Had the root been **drēn-*, this dialect would have shown the form ***drāine* (cf. *äil* 'eel' < **ēla-*). Consequently, the Old Saxon and Old English evidence only points to **dran-* and **drun-*.

In summarizing, the following stem variants must be supposed: **drenan*, **drana(n)-*, **drunan-* and **duran-*. One possible way to account for this polymorphism is to reconstruct the original paradigm as **d(é)rōn*, **durraz*, **dreni*, **dranun* from a paradigm PIE **d^hér-ōn*, **d^hr-n-ós*, **d^hr-én-i*, **d^hr-ón-m*. Such a paradigm can account for the variants **dren-* and **dran-* directly, because they occurred in the original locative and accusative case. The genitive **durraz* may furthermore have given rise to OE *dora* < **duran-*, possibly through the creation of a secondary

¹⁹¹ Fick/Falk/Torp 211; Pokorny 255-256; Kluge/Mitzka 143; Philippa/De Brabandere/Quak 520-1; Tiefenbach 2010: 58.

paradigm **durō*, **durraz*. The remaining variant **drun-* can be explained as a secondary zero grade to the roots **dren-* and **dran-*.

It is possible, too, however, that the root was **d^hren-* rather than **d^her-* and that, accordingly, the paradigm should be reconstructed as **d^hrén-ōn*, **d^hryn-n-ós*, **d^hryn-n-éni*. This paradigm does not explain the root variant **dran-*, however, unless one assumes an accusative plural **d^hron-n-ns*.

The Greek material is relatively complex and does not clarify much. The simplest form is Laconic θρῶναξ ‘bee’ (Hsch.). Then there are the reduplicated forms τενθρήνη ‘hornet’ (Nic.) and τενθρήνιον (Arist.), which perhaps presuppose an unreduplicated form *θρήνη. The form ἀνθρήνη ‘bee, wasp’ (Ar., Arist.) is influenced by ἄνθος ‘flower’. This is clear from ἀνθηδών ‘bee’, which synchronically can be analyzed as ἀνθ- with the suffix -ηδών as in ἀ-ηδών ‘nightingale’, τερ-ηδών ‘shipworm’, Κηλ-ηδόνες ‘Sirens’, ἀχθ-ηδών ‘load’, ἀλγ-ηδών ‘sorrow’, ἐδ-ηδών ‘tumor’.¹⁹² Further contaminations are ἀνθρηδών ‘hornet’ and τενθρηδών (Arist., Dsc.). Still problematic is πεμφορηδών ‘wasp’, handed down to us by Nicander of Colophon.¹⁹³

The Balto-Slavic material has an unexpected initial **t*: Lith. *trānas*, Latv. *tran(i)s*¹⁹⁴ < **tron-*, Ru. *trúten* ‘m. ‘drone, parasite’, SCr. *trūt* ‘wasp’, Slov. *trôť* ‘parasite’ < **tron-t-*.

****elm*, **ulmaz* ‘elm (tree)’**

- **elma-*: OHG *elm(o)* m. ‘id.’, OHG, MHG *elm-boum* ‘id.’¹⁹⁵, MLG *elm* ‘id.’¹⁹⁶ (= Da. *elm*¹⁹⁷), OE *elm* m. ‘id.’¹⁹⁸, E *elm*
 ⇒ **elmjō-*: OHG *ilma* f. ‘id.’, MHG *ilme* f. ‘id.’¹⁹⁹ (= Ru. *ílem*)

¹⁹² Schwyzer 529 fn.

¹⁹³ The variation between θρηνη- and φρηνη- does not imply that the original root was **g^{wh}rēn-*. It is more probable that πεμφορηδών is a more recent coinage, perhaps a derivation of Gr. **πεμφορος* (cf. Skt. *bambhara-* m. ‘bee’) with the same suffix -ηδών.

¹⁹⁴ Latv. *dran(i)s* may be influenced by Low German (Fraenkel 1010-1).

¹⁹⁵ Lexer 1, 541.

¹⁹⁶ Lübken 95.

¹⁹⁷ Falk/Torp 21: “Im dän. is der Vokal aus dem kollektiven anord. *elmi* n. (Sw. dial. *älme*) entlehnt [...]. Oder die Form ist entlehnt dem mnd. *elm*[...]”

¹⁹⁸ Bosworth/Toller 247.

- **ulma*:- OE *ulm-trēow* 'id.'²⁰⁰, MHG *ulm-boum* 'id.'²⁰¹, G *Ulme*²⁰², MLG *olm* 'id.'²⁰³, MDu. *olme* 'id.'²⁰⁴, Du. *olm*²⁰⁵
- **alma*:- ON *almr* m. 'id.'²⁰⁶, Icel., Far. *álmur* m. 'id.'²⁰⁷, Nw., Sw. *alm* m. 'id.'²⁰⁸
 - ⇒ **almja*:- ?ON (top.) *Elmi-kjarr*²⁰⁹, Sw. dial. *älme* n. 'alm grove'²¹⁰ (= Gutn. *älmä* 'id.'^{211?})
 - ⇒ **almjō*:- Sw. dial. *älm* f. 'elm'²¹²

The combined attestation of OHG *elm(o)*, MLG *elm* < **elma(n)*- and OE *ulm-trēow* < **ulma*- provides evidence for an ablauting paradigm. It is not self-evident that the word in question was inflected as an *n*-stem, however. The vacillation of OHG *elm(o)* between an *n*-stem and an *a*-stem is perhaps in favor of such a reconstruction, and this is, in fact, the solution that we find in EWA (p. 1059): "Sofern daneben für das Germ. eine Ablautstufe **!mo*- anzunehmen ist, könnte diese aus einem *n*-stämmigen vorurgerm. **elm-on*-, *!m-n*- hervorgegangen sein, wobei zu *!m-n*- über *!m-on*- sekundär ein *o*-Stamm rückgebildet werden konnte [...]." Since, however, the original zero-grade genitive *(*h*₁)*lm-n-ós* of this paradigm should have regularly given PGm. **lummaz* (cf. ON *luma* 'to let go', Nw. dial. *luma* 'to relax', Lith. *lìmti* 'to succumb' < **!m̥H-*), the reconstruction of an old *m*-stem **elm*, **ulmaz* seems more appropriate. It

¹⁹⁹ Benecke 1, 429.

²⁰⁰ Bosworth/Toller 1088.

²⁰¹ Lexer l.c.

²⁰² Kluge/Seebold 940: "In dieser Form bezeugt seit dem 15. Jh. [...], und zwar entlehnt aus l. *ulmus* [...]."

²⁰³ Lübben l.c.

²⁰⁴ Verdam 391.

²⁰⁵ Franck/Van Wijk 468: "Uit lat. *ulmus* [...] of uit ofr. *olme*, bijvorm van *orme* (uit lat. *ulmus*)."

²⁰⁶ De Vries 1962: 7: "daneben abl. ae. *ulm-treow*, mhd. *ulmboum*, nhd. *ulme*, mnd., nnl. *olm*."

²⁰⁷ Böðvarsson 23; Poulsen 71.

²⁰⁸ Falk/Torp l.c.; SAOB A1123.

²⁰⁹ Heggstad 124.

²¹⁰ Rietz 845.

²¹¹ Klintberg/Gustavson 1791.

²¹² Rietz l.c.

implies a Proto-Indo-European paradigm $*(h_1)él-m$, $*h_1l-m-ós$ comparable to e.g. $*h_2érh_2-m$, $*h_2rh_2-m-ós$ ‘arm’ (cf. Lat. *armus* ‘upper arm, shoulder’, *rāmus* ‘branch’, Skt. *īrmá-* ‘arm’).

The reconstruction of an ablauting *m*-stem offers an explanation for the unexpected formation W *llwyf* ‘elm’ < $*leim-$ in Celtic. This *llwyf* is best understood as a secondary full grade that arose after the vocalization of the *l* in the zero-grade $*lim-$ > Mlr. *lem*.²¹³ In other words, the paradigm $*h_1él-m$, $*h_1l-m-ós$ developed into PCelt. $*elm$, $*limos$ and was subsequently remodeled into $*leim$, $*limos$. The motive for this remodeling seems to have been the removal of the *Schwebeablaut*.²¹⁴

$*eudur$, $*ūdraz$ ‘udder’

- $*eudra-$: ON *júr*, *júgr* n. ‘udder’ (< $*júðr$), Icel. *júfur*, *júgur* n. ‘id.’²¹⁵, MLG *jēder* n. ‘id.’, OFri. *jāder* ‘id.’, WFri. *jaar* n. ‘id.’²¹⁶ (= Du. Holl. *jaar*, *jadder* ‘teat’²¹⁷)
- $*eldra-$: E dial., Du. dial. *elder* ‘id.’²¹⁸
- $*ūdra(n)$:²¹⁹ OHG *ūter*, *ūtro* ‘uber’, MHG *iuter*, *ūter* mn. ‘id.’, G *Euter*²²⁰, Bav. *auter* n. ‘id.’²²¹, Swi. App. *uuttər* ‘id.’²²², Visp. *üütter*²²³ n. ‘id.’, MDu. *uder* m. ‘id.’, Du. *uier* ‘id.’²²⁴, OE *ūder* n. ‘id.’, E *udder*

²¹³ It is difficult to find additional evidence for the vocalization of $*l$ to $*li$ before $*m$ in Proto-Celtic, but the parallel development of $*r$ to $*ri$ in OIr. *cruim* ‘worm’ < PCelt. $*k^{w}rimi-$ < PIE $*k^{w}rm-i-$ (cf. Matasović 2009: 181) may suffice.

²¹⁴ According to Schrijver (1997: 311), the ablaut of $*alm-$, $*elm-$, $*l̥m-$, $*lim-$ and $*leim-$ points to a substrate origin.

²¹⁵ Böðvarsson 472.

²¹⁶ Zantema 1, 453.

²¹⁷ Weijnen 1996: 82.

²¹⁸ Weijnen 1996: 43

²¹⁹ Cf. Fick/Falk/Torp 29: $*eudar(a) \sim *ūdar(a)$; Kluge/Seebold 263: $*eudara- \sim *ūdara-$.

²²⁰ Kluge/Seebold 263: “Man erklärt dies [i.e. the Indo-European vowel alternations] durch einen alten Ablaut *ēu/ōu/ū*, doch hat diese Annahme nicht viel Wahrscheinlichkeit für sich.”

²²¹ Grimm 1, 1044.

²²² Vetsch 76.

²²³ Wipf 36.

²²⁴ Franck/Van Wijk: 717.

The ablaut of the Germanic word for ‘udder’ is unambiguous. An *e*-grade is ascertained by a range of forms found in both North and West Germanic. It is present in MLG *jēder* and OFri. *jāder*, the latter showing the usual Frisian development of **eu* to **iā*. The anomalous form *elder*, which occurs in an area that unifies some Dutch and English dialects, is certainly no reflex of **alipra-* < **h₂el-i-tro-* ‘feed-organ’, as has been claimed,²²⁵ but rather continues the form **eudur* with a (dissimilatory?) change of **eud-* to **eld-*.²²⁶

In North Germanic, the *e*-grade is supported ON *júgr*, a form that is generally assumed to have developed out of **júðr* < **eudra-*. The velarization of the dental fricative is paralleled by instances such as *fjogur* n. ‘4’ < **fjōður* < PGm. **fedwōr* < PIE **k^wetuōr*, and seems to have been triggered by an adjacent labial vowel. As a matter of fact, the velarization may have required two surrounding labial vowels. If this is correct, *júgr* must represent the original nominative form **eudur* < **h₁eu(H)d^h-r* rather than a secondary thematic form **eudra-*. In modern Icelandic, *júgr* developed into *júgur* and subsequently into *júfur* by the regular loss of fricative *g* after rounded vowels. The resulting hiatus was filled up with a labial glide, which in the orthography is represented as *f*.

Theoretically, it could be argued that the *e*-grade of ON *júgr* is secondary, because **ū* was often replaced by **eu* in West Norse (see § 6.5). No such explanation is available for OFri. *jāder* and MLG *jēder*, however, as these dialects usually display the reverse replacement of **eu* by **ū*. The conclusion must therefore be that PGm. **eudra-* is primary.

The zero-grade form **ūdr-* can be reconstructed on the basis of OE *ūder*, MDu. *uder*, OHG *ūter*, Bav. *auter*, Swi. *uuttar* etc. MHG *iuter* and G *Euter* are opaque, as they can be derived from either **eudr-* or **ūdr-* with umlaut. A possible source for this umlaut is the original dative case in **-eri*, which developed out of the PIE locative **-er-i-*. The gloss *utir* in all likelihood preserved this element, and thus appears to be formally parallel to the heteroclitic split-off OHG *fuir*, G *Feuer* ‘fire’ < **fueri* < **ph₂u-er-i* (see p. 320).

Another possible vestige of the original Indo-European heteroclisys may be the *n*-stem inflection of OHG *ūtro*, dat. *ūtrin* < **ūdran-*. Bloomfield

²²⁵ OED, s.v. *udder*.

²²⁶ A similar change can be observed in Swi. *Višp. hālfpa* f. ‘hip, wild rose’ < OHG *hiufa* f. ‘id.’ < **heupōn-*. It shows the reverse development of *l* > *ʃ* as in e.g. Polish.

(1891: 4), for instance, assumed that “in Ohg. dative sg. *ūtrin* there is a mixture of *r*- and *n*-stems in reversed order from that which has taken place in Lat. *iecinoris*”. Also to be noted is Nw. dial. *jura* (n.) < **judrōn*-.

With the two ablaut variants **eudra*- and **ūdra(n)*-, the Germanic material seems to continue a paradigm **eudur*, **ūdraz* < **h₁éu(H)d^h-r*-, **h₁u(H)d^h-r-ós*.²²⁷ In view of cognates such as Skt. *údhar*, *údhnas* < **h₁u(H)d^h-r/n*-, Lat. *uber* and Gr. οὔθαρος, οὔθατος < **h₁ou(H)d^h-r/n*-, a static paradigm **h₁óuHd^h-r*, **h₁(é)uHd^h-n-s* has been reconstructed by Schindler (1975: 8). Since the neuter heteroclitics did not have a separate accusative case that can have harbored the *o*-grade that is found in Greek, this indeed seems to be one of the few ways in which the triple ablaut can be accounted for.

**finkō*, **funk^kaz* ‘spark’†

- **finka*-: MHG *vinc* m. ‘id.’²²⁸
- **fankan*-: MHG *vanke* m. ‘id.’²²⁹
- **funka(n)*-: OHG *funcho* m. ‘id.’, MHG *funke* m. ‘id.’²³⁰, G *Funke(n)*²³¹, MLG *vunke* ‘id.’, MDu. *vonke* ‘id.’, Du. *vonk* ‘id.’, ME *fonke*, *funke* ‘id.’, E *funk*

In Middle High German, two or possibly three different nominal stems meaning ‘spark’ can be found, viz. *vink*, *vanke* and *vunke*. According to *Deutsches Wörterbuch*, “in diesen drei formen zusammengenommen nun treten vollständig laut und ablaute der von Jacob Grimm unter *fink* angenommenen wurzel *finken*, *leuchten*, *glänzen*” (p. 593-613).

The zero grade **funkan*- is the form with the oldest attestations and the widest distribution. It first occurs as OHG *funcho* and is still in use in the modern West Germanic languages as G *Funke(n)*, Du. *vonk* and E *funk*. A more limited form is **fankan*-, occurring as MHG *vanke*. Its root, **fank*-, is further supported by the causative verb **fankjan*- as in MHG *venken*, MDu. *ont-fenken* ‘to kindle’.

²²⁷ The second laryngeal is perhaps redundant: it is possible that methathesis took place in the zero grade **h₁ud^h-*.

²²⁸ Benecker 4, 318.

²²⁹ Lexer 3, 19.

²³⁰ Lexer 3, 568.

²³¹ Kluge/Seebold 322.

Finally, there is some marginal evidence for a form **finka-*, attested as MHG *vinc*. It occurs only once in *Wolfdietrich* in the phrase “*er mohte niht entwîchen des heizen vinc*” (745, 3).

Etymologically, the zero grade **funkan-* has been derived from the weak stem of PIE **péh₂-ur*, **ph₂u-n-ós* > PGm. **fōr*, **funaz* ‘fire’²³² with a velar suffix **-go-*. This solution works well for **funkan-*, but it does not explain the ablaut of **finka-* and **fankan-*. The root form **fank-* is by Kluge and Seebold assumed to have arisen through “bloße Lautabwandlung”²³³, while Pokorny (p. 828) assumes an ablauting stem **puon-* in which the **u* was lost. Beekes (1996), in view of all the formal difficulties, assumed that the entire cluster was adopted from a non-Indo-European substrate language.

A very different solution was advanced by Lexer (3, 357), who suggested that the ablaut of the three different root forms implies derivation from a strong verb MHG **vinken*. This seems like an attractive hypothesis, but the problem is that no such verb is actually attested (though G obs. *finkeln* ‘to burn, ache’²³⁴ may be a continuation of it). The question therefore arises whether **fink-* and **fank-* can be explained as analogical full-grade forms to **funk-* < **ph₂un-go-*. The required ablaut pattern may then have been adopted from *n*-stems such as **belkō*, **bulk^kaz* ‘beam’ (see p. 148) and **skinkō*, **skunk^kaz* ‘shank’ (see p. 187).

An element the importance of which has not yet been fully assessed is the morphology of the causative, which, unlike the hypothetical strong verb, is reasonably well-attested. This causative has two forms: MHG *vengen*, MDu. *ont-vengen* ‘to kindle, ignite’²³⁵ < **fangjan-* (< **ponk-éie-*) and MHG *venken* MDu. *ont-fenken* ‘to kindle’²³⁶ < **fank^kjan-*. The alternation between the consonantism of **fang-* and **fank^k-* is in all probability due to the influence of the iterative formation **funk^kōn-* (< **pnk-néh₂-*), which can be retrieved from MHG *vunken*, MDu. *vonken* ‘to shimmer, sparkle; give off sparks’²³⁷ (only partly **funkōjan-*). The

²³² Kluge/Mitzka 224; Kluge/Seebold 322.

²³³ Cf. Kluge/Seebold: “Die mhd. Variante *vanke* setzt eine *o*-stufe voraus, die nach dem paradigmatischen Ablaut nicht zu erwarten wäre. Vielleicht handelt es sich bei ihr um eine bloße Lautabwandlung.”

²³⁴ Grimm 3, 1664.

²³⁵ Lexer 3, 64; Verdam 430.

²³⁶ Lexer 3, 65; Verdam l.c.

²³⁷ Lexer 3, 569; Verdam 732.

primary **fangjan-* < **ponk-éie-*, in other words, must have been remodeled into **fank^kjan-* under the influence of the iterative.

All in all, it seems unlikely that the nominal roots **fink-*, **fank-* and **funk-* go back to an ablauting *n*-stem **finkō*, **funk^kaz*. It is not plausible either that **funkan-* derives from **ph₂un-go-n-* ‘fire’. Instead, the central derivational role of the iterative seems to demonstrate that the whole cluster was indeed derived from a verbal complex, as was claimed by Lexer.

****greubō*, **gruppaz* ‘pot’**

- **greuban-*: OE *grēofa* m. ‘id.’²³⁸
- *?*greup^pō-*: OE *gripu* f. ‘cauldron’²³⁹
- **gruppan-*: MDu. *groppe(n)* m. ‘iron pan’²⁴⁰ (= MHG *grop(p)e* ‘id.’, G *Groppen* ‘iron pan, cauldron’²⁴¹)
- **grupan-*: OE *gropa* m. ‘pan’²⁴², MLG *grobe(n)*, *grape(n)* m. ‘pot’²⁴³ (= EDu. Sax. *grape*, *grobe* ‘chytra, lebes’), MDu. *grobe*, *groop* m. ‘vase, cauldron’²⁴⁴

The variation of OE *gropa*, MDu. *grobe* < **grupan-* and MDu. *groppe* < **gruppan-* unambiguously points to an *n*-stem with consonant gradation, viz. **grupō*, **gruppaz*. The single **p* of this paradigm is likely to be secondary, as has already been claimed by Lühr (1988: 243-4), in view of the consonantism of e.g. OE *grēofa* ‘pot’. This form additionally points to a full-grade formation **greufan-* or **greuban-*. Together with the zero-grade variants, it seems to demonstrate an apophonic *n*-stem, e.g. **greubō*, **gruppaz* < **g^hréub^h-ōn*, **g^hrub^h-n-ós*.

The position of the OE gloss *gripu* ‘cauldron’ is unclear. It looks like a light-syllable *ō*-stem (**gropō*?), but the derivation of the word is not transparent, not in the least because of its sparse attestation (2x). If it

²³⁸ Bosworth/Toller 488; Holthausen 1934: 137.

²³⁹ Bosworth/Toller 490; Holthausen 1934: 138.

²⁴⁰ Verdam 232.

²⁴¹ Lexer 1, 1093; Grimm 9, 445-6.

²⁴² Holthausen 1934: 138.

²⁴³ Lübben 130.

²⁴⁴ Verdam 231, 232.

represents **grīopu*, it can theoretically be derived from a formation **greup^pō-* that combines a full grade with a geminate.

The etymology of the word is relatively clear. In view of correspondences such as Sw. dial. *grjopa* ‘to hollow out’ < **greup^pan-*,²⁴⁵ ON *greypa*, MLG *gröpen* ‘to scoop’ < **graup^pjan-*, ON *gryffa* f. ‘hole’ < **grubjōn-* and Nw. dial. *grove* f. ‘hole’ < **grubōn-*, it seems plausible that the *n*-stem **greubō*, **gruppaz* originally denoted ‘a vessel hollowed out of wood’.

****helm*, **hulmaz* ‘blade, cane, reed’?**

- **helma(n)-*: ON *hjalmr* m. ‘helm, tiller’, OE *helma* m. ‘helm’, MLG, MDu. *helm* ‘id.’²⁴⁶
- **helma-*: ON ?*hjalmr* m. ‘plant name’²⁴⁷, Sw. dial. *hjel* m. ‘ear’²⁴⁸, EDu. *helm* ‘sedge’, Du. *helm* ‘marram grass’²⁴⁹
- **halma-*: ON *halmr* m. ‘straw’²⁵⁰, OHG *halm* m. ‘blade’, OE *healm* m. ‘id.’
 ⇒ **halmjōn-*: ON *ax-helma* f. ‘stalk and ear of grain’²⁵¹, Icel. *helma* f. ‘stalk’²⁵², Nw. dial. *helme* f. ‘grain stub’

Germanic, as well as the other Indo-European languages, unambiguously points to an ablauting *m*-stem for the word for ‘cane’. Three different vowel grades are found. An *e*-grade **kelh₂-m-* must be reconstructed for Lith. *kėlmas* m. ‘tree-trunk’²⁵³, ON *hjalmr* m. ‘helm, tiller’, OE *helma* m. ‘helm’, and probably also for Du. *helm* ‘marram grass’²⁵⁴. Gr. καλάμη,

²⁴⁵ Lühr (1988: 244 fn.) regarded the strong verb **greup^pan-* as a variant of more primary, but unattested **greuban-*, which she took to be the source of the full-grade *n*-stem **greuban-*.

²⁴⁶ Lübben 140.

²⁴⁷ De Vries 1962: 231.

²⁴⁸ Rietz 280.

²⁴⁹ De Vries/Tollenaere 249; Franck/Van Wijk 244.

²⁵⁰ De Vries 1962: 206.

²⁵¹ De Vries 1962: 221.

²⁵² Böðvarsson 360.

²⁵³ For expected ***šėlmas*. The **š* was depalatalized by the following *l* in the zero grade.

²⁵⁴ Lübben 140.

κάλαμος ‘cane’, on the other hand, has a zero grade of the root and a full grade of the suffix: **k₁lh₂-em*²⁵⁵. W *calaf* f. ‘reed, stalk’ may derive from the same stem, but it is also possible that it was adopted from Latin *calamus*²⁵⁶, which in turn is a loanword from Greek. The genuine Latin form *culmus* m. ‘blade’ as well as Latv. *saļms* ‘straw’ and ON *halmr*, OHG *halm* reflect PIE **k₁olh₂-mo-*. The *o*-grade is also present in the Slavic feminine OCS *slama*, Ru. *solóma* < **k₁olh₂-m-eh₂-*.

All the evidence taken together, it seems best to start from a PIE paradigm nsg. **k₁élh₂-m*, gsg. **k₁lh₂-m-ós*, lsg. **k₁lh₂-ém-i*. The origin of the *o*-grade is not fully clear: it can theoretically have spread from the accusatives **k₁olh₂-m-m* and **k₁olh₂-m-ns*, but, as in many other cases, it cannot entirely be excluded that it arose independently in a thematic *mo*-derivation.²⁵⁷ The reconstruction **k₁ólh₂-m*, asg. **k₁lh₂-ém-m*, as proposed by Beekes (1985: 43-4), must be rejected because it discards the *e*-grades in Germanic and Lithuanian.

****hemō, *humnaz* ‘heaven’**

- **hemina-*: Go. *himins* m. ‘id.’, ON *himinn* m. ‘id.’
- **hemna-*: OS *heban* m. ‘id.’, OE *he(o)ffen* m. ‘id.’
- **hemila-*: OHG *himil* m. ‘id.’, OS *himil* m. ‘id.’, OFri. *himul*, *himel* m. ‘id.’
- *?*humila-*: OHG *humel* m. ‘id.’²⁵⁸

The PGm. word for ‘heaven’ at first sight does not look like an ablauting *n*-stem, but its apophonic nature is revealed by the different suffixations of Go. *himins*, ON *himinn* < **hemina-* and OE *he(o)ffen*, OS *heban* < **hemna-*. The two formations apparently continue the original dative and genitive of an *n*-stem **hemō, *hemnaz, *hemeni*.

²⁵⁵ Not from **κόλαμος* by assimilation (pace Pokorny 612).

²⁵⁶ Pokorny 612.

²⁵⁷ Similarly, the *o*-grade of OHG *hama* f. ‘ham’ < **k₁onh₂-m-eh₂-*, related to Gr. κνήμη f. ‘shinbone’, OIr. *cnáim* m. ‘bone’ < **k₁nh₂-meh₂-*, can be due to thematization. At any rate, Beekes’s reconstruction **k₁ónh₂-m*, **k₁nh₂-ém-m* must likewise be replaced by **k₁énh₂-m*, **k₁nh₂-m-ós*, **k₁nh₂-ém-i*.

²⁵⁸ Noreen 1894: 62; Schützeichel 83. Pokorny (556-557) calls the form *humel* “mitteldeutsch”, a characterization that is based on the mixture of High and Low German features that is displayed by the manuscript in which *humel* occurs.

The etymology of PGM. ‘heaven’ points to old ablaut, too. The word is usually connected to Skt. *ásman-* m. ‘stone, sky’, Gk. *ἄκμων* m. ‘anvil, meteorite, sky’, Lith. *akmuõ* m. ‘stone’ < **h₂eḱ-mon-*.²⁵⁹ The problem with this connection is that the PGM. full grade is not where it is expected, as it seems to continue quasi-PIE **h₂ḱem-on-* rather than **h₂eḱ-mon-*. Since, however, the similarities between the Germanic and extra-Germanic forms are too great to be discarded, it is likely that the unexpected full grade arose through some kind of analogy that was motivated by the irregular outcome of the paradigm in Proto-Germanic.

Assuming that the original inflection of the word had an amphikinetic ablaut pattern, i.e. **h₂eḱ-mōn*, **h₂ḱ-(m)n-ós*, **h₂ḱ-mén-i* (cf. Skt. *ás mā́, áśnaḥ, áśman(i)*)²⁶⁰, the phonetically regular outcome of the paradigm would in Proto-Germanic be **ahmō*, **humnaz*, **hmeni*. The irregularity of this paradigm may have been resolved by reshaping it into **hemō*, **humnaz* by introducing the full grade in the zero-grade slot of the genitive.²⁶¹ This hypothetical zero-grade root can perhaps be retrieved from OHG *humel*,²⁶² a variant of the usual OHG form *himil*. However, the original vowel quality of these forms is ambiguous, because the grapheme <u> may also have been used to indicate a secondarily rounded front vowel [y], cf. Cimb. *hūm(m)el* m. ‘heaven’.²⁶³

It has been claimed that the *l*-suffixed forms, such as OHG *himil*, *humel*, in combination with the *n*-suffixed stems **hemna-* and **hemina-* point to an old heteroclitic *l/n*-paradigm.²⁶⁴ Since, however, such an *ml/n*-stem is unparalleled, it is probably better to assume that the *l*-forms are secondary, i.e. due to the influence of **sō(e)l*, **sun(n)az* ‘sun’.²⁶⁵

²⁵⁹ Cf. Reichelt 1913; Maher 1973.

²⁶⁰ Lühr (2000: 70): **h₂akmō̃, *h₂ḱ-mn-és, *h₂ḱ-mén(-i), *h₂ak-món-ṛ̃*.

²⁶¹ Differently Wachter (1997: 18 fn.): “Das Paradigma lautete wohl etwa Nom. **h₂eḱ-mōn*, Gen. **h₂ke-mn-ós*, und von hier aus würde sich **kemen-os* mit der v.a. bei germanischen Thematisierungen üblichen *e*-Stufe [...] leicht verstehen lassen.”

²⁶² It appears twice in the Cambridge Songs manuscript Carmen XXVII, *Clericus et Nunna*: 8) *hoc evanescet omne | also uuolcan in themo humele; solum Christi regnum | thaz bilibit uns in evun*; 9) *quod ipse regnat credo | in humele so scono; non recusat dare | thaz geleistit her ze uuare*.

²⁶³ Schmeller/Bergmann 1855: 132 [194].

²⁶⁴ Pedersen 1893: 145, Noreen 1894: 142.

²⁶⁵ Kluge (1886:332) already assumed an analogical origin. Braune (1891:94) proposed dissimilation of **himin-* to **himil-*, which is an attractive idea. Wachter

It must be stressed, in this respect, that the *l*-form *himil* seems to be of High German origin. In the Old Saxon Heliand, for instance, *heþan* only occurs as the first member of compounds (e.g. *heþan-cuning*) or in fixed clauses (e.g. *heþenes kuning*), whereas *himil* occurs freely both in compounds and as a simplex. The simplest way to account for this distribution is to assume that in Old Saxon *heþan* was in the process of being supplanted by *himil*, but that it was able to hold ground in bound position. The intrusion of *himil* was obviously posterior to the Anglo-Saxon emigration to Britain, because Old English only has **hemna-*. It may have entered the Low German dialects along with the Christianization of Northern Europe.

The position of ON *hamarr* m. 'hammer, back of an axe, crag', OHG *hamar*, OE *hamar* (etc.) < **hamar-* is unclear. PIE did have *mr/n*-stems, e.g. Gr. τέκμαρ, -ωρ 'sign' < **k^wek-mōr*, -*mr* or **g^heh₂-mr* 'palate' (see. p. 302), and it is therefore theoretically possible to assume that it developed out of a form **h₂k-mor-* by metathesis²⁶⁶, i.e. **k_h-mor-*. Such a conjecture is nonetheless difficult to falsify: since Skt. *aśmará-* 'made of stone' probably reflects **h₂ek-m_h-ró-* rather than **h₂ek-mer-ó-*, the indications for a heteroclitc paradigm remain strictly Germanic. This means that, in the end, little can be said in favor of a reconstruction **h₂ék-mōr*, **h₂k-mn-ós*, **h₂k-mén-i*.

****hersō*, **hurznaz* 'brain'?**

- **hersan-*: ON *hjarsi*, *hjassi* m. 'crown', Nn. *hjasse* 'id.', Sw. *hjässa* 'id.', ODa. *jessæ*, Da. *isse* 'skull, crown'²⁶⁷
- **herzan-*: Nw. dial. *hjar(r)e* m. 'brain'
 ⇒ **(ga-)herznja-*: OHG *hirni* n. 'id.', MHG *hirn(e)* n. 'id.'²⁶⁸, G *Gehirn*, *Hirn* 'id.', MLG *herne*, *harne* nf. 'id.'²⁶⁹ (= East MDu. *herne* nf. 'id.'²⁷⁰)

[1997: 18]: "Für den nur im Germanischen bezeugten, *l*-haltigen Stamm **himila-* aber genügt es vollkommen, eine Analogie zum alten Wort für 'Sonne', germ. **sāwil(a)-*, anzunehmen zu einer Zeit, da dessen *l/n*-Wechsel im Sprachbewußtsein der frühen Germanen noch lebendig war."

²⁶⁶ Cf. OCS *kamy* 'stone' < **keh₂-mōn*.

²⁶⁷ Falk/Torp 469.

²⁶⁸ Lexer 1, 1303.

²⁶⁹ Lübben 143.

²⁷⁰ Verdam 248.

- **hers(n)an*-. MDu. *hersene*, *harsen* pl. 'id.'²⁷¹, EDu. *herssen* 'id.', Du. *hersenen*, -*ens* pl. 'id.'²⁷²
- **herzna(n)*-. ON *hjarn(i)* m. 'id.'²⁷³, Nw., Da. *hjerne* 'id.', Sw. *hjärna* 'id.', ME *hernes* pl. 'id.', E *harns*
- ?**hurzna*-. Du. *hoorn-dol*, *hoorn-woedig* 'crazy'²⁷⁴

The PIE root **kerh₂s-* 'head' is inflected as an *n*-stem in Germanic (**hersan-*). The material contains both forms with and without the effects of Verner's law²⁷⁵, which implies that the word had an accentually mobile paradigm. According to Benediktsson (1968: 110), "each of the alternative stem forms has been generalized to form an *n*-stem paradigm of its own".

On the one hand, there is ON *hjarsi*, representing the original nominative **hersō* < **kerh₂s-ōn*. ON *hjarni*, on the other hand, clearly generalized the oblique stem as in, for instance, the gen. **herznaz* < **k(e)rh₂s-n-ós*. All other formations are due to analogy. Nw. *hjarre* < **herzan-* looks like a nominative **hersō* that adopted the **z* from the oblique. Conversely, Du. *hersens* < **hersnan-* is best explained from an oblique form **herznaz* that assumed the **s* from the nominative. G *Gehirn* and *Hirn* continue a collective formation *(*ga-*)*herzn-ja*-²⁷⁶.

There is only marginal evidence for a zero grade **hurzn-*, but it can theoretically be established on the basis of Du. *hoorn-dol* 'frenzied'. Superficially, the word looks like a compound of *hoorn* 'horn' and *dol* 'mad', which would refer to animals poking with their horns, but the new *Etymologisch woordenboek van het Nederlands* hints at the possibility that this association is due to folk etymology. The dictionary points to the parallelism with MHG *hirn-wüetec* 'delirious'²⁷⁷, i.e. "brain-raging", which makes sense in view of the symmetrical opposition of Du. *hoorn-*

²⁷¹ Verdam 249.

²⁷² Franck/Van Wijk 248.

²⁷³ Falk/Torp 410.

²⁷⁴ Vercoullie 137; WNT s.v. *hoorndol*.

²⁷⁵ Cf. Schaffner 2001: 546-9.

²⁷⁶ They do not represent a substantivized adjective **herznja-* 'belonging to the skull', as has been claimed by Nussbaum (1986: 192).

²⁷⁷ Lexer 1, 1304.

woedig and G *hirn-toll* ‘frantic’.²⁷⁸ It is conceivable, from this perspective, that the first elements of *hoorn-dol* and *hoorn-woedig* indeed continue the original zero-grade allomorph **hurzna-* to **hersō* ‘brain’. This is all the more likely in view of similar formations such as EDu. *herssen-woedig* ‘*phreneticus, cerebrosus*’ and ME *brain-wōd* ‘frenzied’.

It was suggested by Nussbaum (1986: 191-4) that the Germanic masculine *n*-stem **hersan-* sprang from the oblique cases of the irregular neuter paradigm, which is preserved as Sanskrit *śírah*, gen. *śírṣṇáh*, loc. *śírśán* ‘head’ < **kṛh₂-os*, **kṛh₂-s-nós*, **kṛh₂s-én*. This, of course, raised the problem why the Germanic *n*-stem has an *e*-grade, and not simply a zero grade. In order to explain this, Nussbaum referred to the apparently innovatory full grades of the kind found in OS *ambo* ‘stomach’ < **h₃emb^h-on-* and Lat. *homō* ‘man’ < **d^hg^hem-on-*. This suggestion was elaborated by Schaffner (2001: 549), who assumed that the *e*-grade could have been introduced analogically after the model of other PIE ablauting paradigms. The alternative is to assume that the ablaut of a paradigm **kérh₂s-ōn*, **kṛh₂s-n-ós* was actually preserved by Germanic, which, according to the author, is the most straightforward solution; the accentual mobility presupposed by the opposition of **hersan-* : **herzan-* points to old ablaut anyway, and it is furthermore possible, as has been pointed out, that the old zero grade survived in Du. *hoorn-dol*.

****hnekkō, *hnukkaz* ‘neck’**

- **hnekkān-*: OE *hnecca* m. ‘id.’, E *neck* ‘id.’, OFri. *hnekka* m. ‘id.’, SFri. *näkke* f. ‘id.’, MLG *necke* ‘id.’, MDu. *necke* ‘id.’, Du. *nek*, dial. *näk* ‘id.’²⁷⁹
 ⇒ **ga-hnekkja-*: G *Genick* n. ‘id.’
- **hnakka(n)-*: ON *hnakki* m. ‘id.’, Far. *nakki* m. ‘id.’, *nakkur* m. ‘steep rock’, Nw. *nakke* m. ‘neck, peak, hook’, *nakk* n. ‘peak’, OHG *hnach* m. ‘summit, crown, neck’, G *Nacken* ‘neck’²⁸⁰, G Tyr. *genagge, gnaggn* n. ‘neck’²⁸¹, MLG *nacke* m. ‘id.’

²⁷⁸ Cf. Cutter 1879: 113; Höfler (1899: 738): “*haupt-töbig* = *hirntoll* im Gegensatze zum Muttertoben oder Furor uterinus”.

²⁷⁹ De Bont 1962: 32.

²⁸⁰ Kluge/Seebold 643.

²⁸¹ Schatz/Finsterwalder 216.

- **hnukka(n)*:- ON *hnokki* m. 'iron hook', Far. *nokki* m. 'crook, bar in the loom, top of the yard', Nw. *nokk(e)* m. 'top of the yard, metal books on a bobbin', OE *hnoc* m. 'hook', MLG *nocke* 'notch on an arrow tip', LG *nock(e)* 'tip', Tyr. *nock* m. 'knoll, rock'²⁸², MDu. *nocke* mf. 'tip', EDu. *nocke* 'collar beam, neck, spine', Du. *nok* c. 'roof ridge'

The ablaut of ON *hnakki* and OE *hnecca* is widely acknowledged,²⁸³ and Kauffmann (1887: 515) already himself mentioned the pair as a possible example of an ablauting *n*-stem. This was rejected by Lühr (1988: 219), however: "da die *e*-stufigen Wörter nicht mit den *a*-lautigen Bildungen unter einem Paradigma vereinbar sind, ist eine Verbalwurzel **χnek-*, 'zusammendrücken' zu erwägen, von der urgerm. **χnekkan-* sein **e* bezogen haben könnte." In view of the presence of a third ablaut variant **hnukkan-* (cf. EDu. *nocke*), it nevertheless seems meaningful to consider the reconstruction of an *n*-stem with paradigmatic ablaut.

Of the three vowel grades, the *a*-grade is dominant, being attested across the Northwest Germanic languages, cf. ON *hnakki* 'neck', Far. *nakkur* 'steep rock', Nw. *nakke* 'peak, neck, hook', OHG *hnach* 'summit, neck'. The meaning '(overhanging) protrusion' appears to be old, because the zero-grade forms have it as well: ON *hnokki* 'hook', Tyr. *nock* 'knoll'²⁸⁴, OE *hnoc* 'hook', EDu. *nocke* 'collar beam', Du. *nok* 'roof ridge, tip'. Of the zero-grades, only EDu. *nocke* is attested with the meaning 'neck'. The anciennity of this meaning is nevertheless demonstrated by the Romance loanwords Fr. *nuque*, It., Spa. *nuca* f. 'nape of the neck'.²⁸⁵

The *e*-grade forms, all meaning 'neck', predominantly occur in the Ingvaenic languages, cf. OE *hnecca*, OFri. *nekka*, MLG, MDu. *necke* 'neck'. An *e*-grade is further found in the collective **ga-hnekk-ja-* underlying MHG *genic(ke)* and G *Genick*.

In view of the triple ablaut displayed by this cluster, a comparison can be made with the paradigm **belkō*, gsg. **bulk^kaz*, apl. **balk^kuns* 'beam' < **b^hélǵ^h-ōn*, **b^hǵ^h-n-ós*, **b^holǵ^h-n-ǵs* (see p. 148). Reconstructing a parallel paradigm **hnekkō*, gsg. **hnukkaz*, apl. **hnakkuns* poses several

²⁸² Schatz/Finsterwalder 454.

²⁸³ Brugmann II, 1, 307; Van Wijk 1912: 461; Vercoullie 1925: 422-3.

²⁸⁴ Taken from Lühr 1988: 219.

²⁸⁵ Falk/Torp 769; Vercoullie 242-3.

problems, however, because the underlying PIE reconstruction, **knék-ōn*, **knk-n-ós*, **knok-n-ŷs*, would have given **hnehō*, **hunk^kaz*, **hnakkuns*. This problem can perhaps be solved by assuming that the original paradigm was restructured at some point in Proto-Germanic. This would require 1) the generalization of the geminate, and 2) the removal of the *Schwebeablaut* in the zero grade.

Some evidence can be adduced for the original genitive form **hunk^kaz*. It possibly has some vestiges in the form of MDu. *honc* 'corner, base', Du. *honk* 'id.', WFri. *honk* 'id.', SFri. *hunk* 'id.', G *Hunke* 'hillock'.²⁸⁶ In view of the Dutch and German meanings, it must be assumed that the word originally denoted a small hill or – more specifically – a hillock that was used as a boundary mark. This is close to the meaning of e.g. Tyr. *nock* 'knoll'.

Outside Germanic, OIr. *cnoc* m. 'hill', W *cnwch* m. 'id. < **knokko-* or **knukko-* are generally taken to be related²⁸⁷. Since, however, the Celtic geminate is difficult to explain²⁸⁸, whereas the Germanic geminate is the logical outcome of the *n*-stem paradigm, it seems probable that the Celtic word was borrowed from Germanic. A Celtic origin is further unlikely, because the formation **hnukka-* is part of a very elaborate derivational cluster in Germanic. This leaves only To. (A) *kñuk* 'neck' as a possible cognate (Pedersen 1944: 29).²⁸⁹ Since this *kñuk* may continue an *n*-stem **knek-on-* (Michaël Peyrot, p.c.), it can theoretically be reconciled with the Germanic forms.

****hnellō, *hnullaz* 'nape, crest'**

- **hnella(n)-, -ōn-*: OHG (*h*)*nel* 'sinciput, posterior pars testis', *aftirnel* 'occiput', *nilla* 'vertex', MHG *nel(le)* m. 'peak, top', G

²⁸⁶ The German word is found in e.g. Hietzinger's *Statistik der militärgrenze des österreichischen Kaiserthums* (1817: 54): "Beinahe überall wo das Gebiet der Militärgrenze abgeschlossen ist, sind die Gränzmarken genau bestimmt, und in Ermangelung natürlicher, durch die Kunst, grösentheils durch Hügel (H u n k e n) bezeichnet."

²⁸⁷ Cf. Kluge/Seebold 643: "Außergermanisch wird verglichen air. *cnocc*, kymr. *cnwch* »Buckel, Hügel«, toch. A *kñuk* »Hals, Nacken«."

²⁸⁸ Whitley Stokes's (1893) suggestion of a Kluge's law in Celtic cannot be maintained.

²⁸⁹ Hilmarsson (1996: 162-3) derived To. (A) *kñuk* from either **Kneuk-o-* or **Knekw-o-*, both of which he thought to be irreconcilable with PGm. **hnVkk-*.

Car. *(n)élle* n. 'nape'²⁹⁰ (= **hnel-līn-?*), Cimb. *(n)ello* m. 'id.'²⁹¹, Tyr. *nalle* f. neck'²⁹²

- **hnulla(n)-*: Icel. *hnúllóttur* 'round, fat', Nw. dial. *null(e)* m. 'small ball, bundle', OHG *(h)nol* 'culmen, vertex, sinciput', *nollo* 'collis', G *Nollen* 'mountain crest'²⁹³, MDu. *nol(le)* mf. 'back of the head, tip of a dike, dune', MHG *nol* m. 'peak, top', *vude-nol* m. 'mons veneris', OE *hnoll* m. 'crown', ME *nol* 'back of the head, nape of the neck, pole'

The Germanic dialects contain potential evidence in favor of the reconstruction of an ablauting *n*-stem **hnellō*, **hnullaz*.

The zero-grade root **hnull-* is most frequent. It occurs throughout the West Germanic dialects, cf. OHG *nollo*, MDu. *nolle*, OE *hnoll*, the meaning ranging from 'crest' to 'crown'. The appurtenance of Nw. *null(e)* 'ball, bundle' and Icel. *hnúllóttur* 'round' is less certain because of the different semantics.

In the Upper German dialects, there are additional forms with *e*-vocalism,²⁹⁴ among which OHG *nel* 'crown, hill', *nella* 'crown' and MHG *nelle* 'peak'. Given the limitation of the root **hnell-* to the High German speech area, the question arises whether the *e* represents unrounded OHG **ō*. This **ō* may have arisen in the plural, where secondary umlaut was productive (see § 10.1). However, the attestation of *nello* in the Cimbrian dialects, where unrounding has never taken place, proves that such a scenario is unlikely in this particular case. Moreover, the unrounding of **ō* probably is too recent to explain the *e* of OHG *nel*. Like Cimb. *nello*, it therefore rather presupposes a real root variant **hnell-*. Tyr. *nalle*, which apparently represents **nālle* with high *a* from OHG *e*, is basically the same form as OHG *nella* < **hnellōn-*.

Together, the stems **hnellan-* and **hnullan-* can theoretically be used to reconstruct an *n*-stem with paradigmatic ablaut, e.g. **hnellō*, **hnullaz*. There are some formal problems with this paradigm, however. The Germanic reconstruction implies an older paradigm **knél-ōn*,

²⁹⁰ Lexer 1862: 198.

²⁹¹ Schmeller/Bergmann 149.

²⁹² Schöpf/Hofer 458.

²⁹³ Grimm 13, 879.

²⁹⁴ Fick/Falk/Torp 98.

**knl-n-ós*, but the regular outcome of such a paradigm would probably be **hnelō*, **hnullaz*.²⁹⁵ Since the material offers no support for a variant **hnelan-* with a singulate, it must be assumed that the geminate was generalized at an early stage. In this respect, the *n*-stem could be similar to **belk^kō*, **bulk^kaz* ‘beam’ (p. 148) and **hnekkō*, **hnukkaz* ‘neck’ (p. 167).

The word has no evident etymology. Some dictionaries compare the zero grade **hnullan-* with **knulla(n)-* as in ON *knollr* m. ‘knoll’, OE *cnoll* m. ‘id.’, MHG *knolle* m. ‘lump’, EDu. *knolle* ‘id.’²⁹⁶, but the original meaning of **hnulla(n)-* is ‘overgrown hill-top’, cf. MHG *vude-nol* ‘mons veneris’.

**hrēhō*, **hrugeni* ‘fish roe’?

- **hrēhan-*: Icel. *hrái* m. ‘swarm of water fleas’²⁹⁷
- **hrugan-*: ME *rowe* ‘fish roe’, MDu. *roge* ‘id.’, OHG *rogo* m. ‘id.’
- **hrugna-*: ON *hrogn* n. ‘fish roe’, MLG *rogen* ‘id.’, OHG *rogan* m. ‘id.’, G *Rogen* ‘id.’

The formations **hrugan-* and **hrugna-*²⁹⁸ are usually compared to Balto-Slavic cognates such as Lith. *kurkulaĩ* m.pl. ‘frogspawn’, Latv. *kuřkulis* ‘id.’, Slov. *krāk* ‘id.’ < **krh₁k-* and Slov. *krék* ‘id.’ < **kreh₁k-o-*²⁹⁹ (cf. Pokorny IEW: 619; Stang 1972: 30). The Slavic material is more extensive, however, and poses serious formal problems. Ru. *krjak*, for instance, must belong here, too, but as the phonetically regular proto-form of this variant would be PSl. **krękǣ*, it must be regarded as a dialectal form for expected ***krek* in order to save the comparison. The formal irregularities have otherwise been resolved by assuming a non-Indo-European origin for the word (Polomé 1986: 661).

If the word was inherited from dialectal Proto-Indo-European after all, the Germanic root **hrug-* could theoretically continue the zero-grade root **krh₁k-*. The vocalization as PGm. **hrug-* rather than expected

²⁹⁵ The alternative, **hunulaz*, would surely have been removed at an early stage.

²⁹⁶ Kluge/Mitzka 384; Franck/Van Wijk 326.

²⁹⁷ Böðvarsson 404.

²⁹⁸ The derivational correlation is comparable to e.g. OHG *rabo* ‘raven’ < **hraban-* vs ON *hrafn*, OHG *raban* ‘id.’ < **hrabna-*.

²⁹⁹ The Lith. variant *krakulaĩ* seems younger than *kurkulaĩ*, because the latter has a correspondence in Latvian.

***hur-** requires an explanation, however. Given the full grade of Slov. *krék*, it seems possible to assume that the original zero-grade **hurg-* was remodeled into **hrug-* on the basis of a full-grade form, so as to remove the *Schwebeablaut*. Within Germanic, this full grade may be attested as Icel. *hrái* ‘swarm of water fleas’, which can have developed out of either **hrahan-* or **hrēhan-*. The latter form is preferable from the Balto-Slavic perspective.

In view of the material presented here, it is possible to tentatively reconstruct a PGm. paradigm **hrēhō*, **hurgeni* < **kréh₁k-ōn*, **krh₁k-én-i*. The expected genitive **hurk^kaz* < **krh₁k-n-ós* is not attested, unless one would be willing to include the semantically remote Nw. *hork* f. ‘ruff’ (a small fish). Its absence may be due to the removal of the *Schwebeablaut* at an early stage.

****kelkō*, **kulk^kaz* ‘jaw, throat’**

- **kelka(n)-*: ON *kjalki* m. ‘jaw, sledge’, Icel. *kjálki*, -ur m. ‘jaw, bar (on a sledge or loom)’³⁰⁰, Far. *kjálki* m. ‘cheek-bone’³⁰¹, Nw. *kjelke* m. ‘small sledge, dial. Adam’s apple’, dial. *kjelk* m. ‘cheek’, Sw. *kälke* ‘sledge’³⁰², OHG *chelah* ‘gibbus, struma’, MHG *kelch* m. ‘crop, double chin’
- **kulka-*: Sw. dial. *kolk*, Da. *kulk* ‘gullet, dial. throat, Adam’s apple’³⁰³
 ⇔ **kulkō(ja)n-*: Far. *kulka* ‘to gulge, swallow’³⁰⁴
- **kalka-*: Icel. *kálkur* m. ‘sledge, bar on a sledge’³⁰⁵

The North Germanic dialects provide enough potential evidence for the reconstruction of an apophonic *n*-stem **kelkō*, **kulk^kaz* with a meaning ‘jaw’ or ‘throat’. A full-grade stem **kelkan-* is supported by ON *kjalki* ‘jaw,

³⁰⁰ Böðvarsson 497.

³⁰¹ Poulsen 590.

³⁰² SAOB K3612.

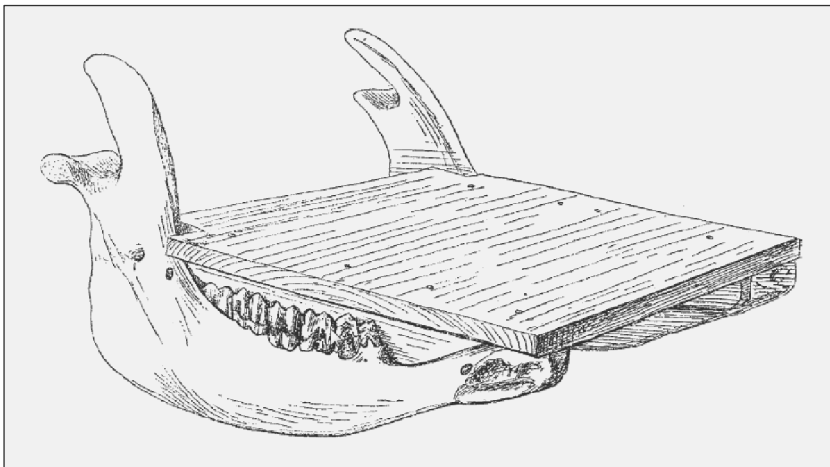
³⁰³ Perhaps also MLG *kolk*, *kulk* m. ‘water hole’, G *Kolk* ‘hole’, MDu. *colc* m. ‘water hole’, Du. *kolk* ‘whirl’, OFri. *kolk* m. ‘hole, pit’, OE *wīn-colc* m. ‘wine barrel’, *ōden-colc* ‘hole in the floor’.

³⁰⁴ Poulsen 642-3.

³⁰⁵ De Vries 1962: 311; Böðvarsson 479.

sledge', Icel., Far. *kjálki* 'jaw, cheek, runner', Nw. *kjelke* and Sw. *kälke* 'sledge'³⁰⁶. A thematic formation with the same vocalism must be reconstructed on the basis of Icel. *kjálkur* 'jaw, runner', Nw. dial. *kjelk* 'cheek'. Icel. *kálkur*, bearing the same meaning as the *e*-grade forms, presupposes an *o*-grade **kalka-*. As is often the case, this *o*-grade is restricted to a thematic formation, which again raises the suspicion that its occurrence depended on the thematization. Finally, a zero-grade formation is supported by Sw. dial. *kolk*, Da. *kulk* 'gullet, dial. Adam's apple'.

The different forms with *e*- and *a*-grade show a remarkable semantic split between 'jaw' and 'sledge'. One of the most attractive ways to deal with this problem is to assume that cattle jaws were used as sledge runners.³⁰⁷ Such use of animal mandibles is confirmed by Stopp and Kunst (2005), who on the basis of archaeological and ethnological data argue that jaw-sledges were employed in that way from Late Iron Age Switzerland to 19th century Prussia (see image). The semantic evolution of the Nordic etymon suggests that this practice was known in the North as well. Presumably, the jawbone skids became the



A 19th century depiction of a Pomeranian sledge with runners made of cattle mandibles ('Kieferschlitten') from Stopp/Kunst, p. 194.

³⁰⁶ Cf. Falk/Torp 516; De Vries 1962: 310-11.

³⁰⁷ Cf. De Vries 1962: 311.

Benennungsmotiv for the sledge in which they were used. The meaning 'sledge' must then be regarded as a *pars pro toto* formation in order to be able to explain why the semantic starting point 'jawbone' was preserved as well. In this respect, the semantic parallelism of Nw. *kjelke*, which dialectally also means 'Adam's apple', with the Danish zero-grade form *kulk* 'id.' can be called remarkable.

The word has no extra-Germanic etymology. The closest formal match is OHG *chelah* 'crop, tumor in the neck', which looks like a *k*-diminutive to OE *ceole*, OHG *chela* f. 'throat' < PGm. **kelōn*-.³⁰⁸

**kernō*, **kurnaz* 'grain'?

- **kernan*:- ON *kjarni* m. 'core', MHG *kerne* m. 'id.', G *Kern* 'id.'
 ⇒ **kernjōn*:- ON *kirna* f. 'churn', OE *cyren* f. 'id.', MLG
 kerne, *kirne* 'id.'
- **kurna*:- Go. *kaur̥n* n. 'grain', ON *korn* n. 'grain (kernel and cereal)', OE *corn* n. 'id.', OHG *chorn* n. 'id.'
 ⇒ **kurnila*:- OE *cyrnel* 'kernel'
- **kurnōn*:- Go. *kaur̥no* n. 'kernel', Nw. dial. *korna* n. 'id.'

There is a wealth of evidence, in the Indo-European languages, for the reconstruction of a formation **ǵrh₂-nó-*, cf. Skt. *jīrṇá-*, OCS *zr̥no*, Lat. *grānum*, OIr. *grán*, W *grawn* 'kernel'. In Germanic, the same form is substantiated by Go. *kaur̥n* n. 'grain', ON *korn* n. 'grain (kernel and cereal)', OE *corn* n. 'id.', OHG *chorn* n. 'id.' < **kurna*-.

Germanic further offers evidence for a different nominal formation. This is PGm. **kernan*-, as supported by ON *kjarni*, MHG *kerne* etc. There are two arguments to think that this form continues an old *n*-stem, i.e. 1) the full grade, and 2) the double *n*-suffix. The full grade has no counterpart in any other Indo-European language, but in combination with the double *n*-suffix, it is reminiscent of forms such as ON *hjarni* 'brain' < **hersnan*- < **ǵérh₂-s-ōn*, gen. **ǵérh₂-s-n-ós* (see p. 165) and ON *stjarna*, OHG *sterno* 'star' < **sternan*-, **nōn*- < **h₂ster-on*-. In view of these parallels, it is possible that **kernan*- continues the full-grade root of an old *n*-stem, i.e. either amphikinetic **ǵérh₂-ōn*, **ǵrh₂-n-ós* or proterokinetic **ǵérh₂-n*, **ǵrh₂-én-os*. Another option would be to reconstruct a more primitive paradigm **ǵérh₂-n*, **ǵrh₂-n-ós*. This would

³⁰⁸ Hellquist 25.

make the word inflectionally parallel to **dēi-n*, **di-n-ós* ‘day’, a paradigm reconstructed by Beekes (1985: 46-47) on the basis of e.g. Lith. *dienà* ‘id.’ < **dei-n-eh₂-* and OCS *дѣнь* ‘id.’ < **di-n-i-*.

Theoretically, Go. *kaurno* and Nw. dial. *korna* – an ancient Northeast Germanic isogloss – could be included in the reconstruction of a Proto-Germanic paradigm **ker(n)ō*, **kurnaz* < **gérh₂-ōn*, **grh₂-n-ós*. It seems more likely, however, that this neuter *n*-stem **kurnōn-* was derived from the primary **kurna-*, perhaps as some sort of singulative, i.e. “a single corn kernel” (cf. § 2.3.3).

****keudō*, **kuttaz* ‘bag’**

- **keuda(n)-*: OHG *chiot* ‘marsupium, sacculum, pecunia’, OE *cēod(a)* m. ‘bag’³⁰⁹
 ⇒ **keudila-*: G *Keutel* m. ‘cod-net (bag-shaped fishing net), bowel, dewlap’³¹⁰, MLG *kūdel* m. ‘bag’, MDu. *cudel(e)*, *cuil*, Du. *kuil* ‘cod-net’³¹¹
- **kudda(n)-*: ON, Icel., Far. *koddi* m. ‘pillow, scrotum, clava’³¹², Nw. *kodd(e)* m. ‘cushion, scrotum, testicle’, MLG *kode* ‘testicle’, OE *cod* m. ‘bag, husk’³¹³, ME *cod* ‘bag, cod-net, husk, throat, belly, scrotum’³¹⁴, EDu. *kodde* ‘testicle’, Du. *kode* ‘backside, tail’³¹⁵
- **kuttan-*: G Swab. *kozze* mf. ‘blister, pimple’³¹⁶

The Northwest Germanic dialects provide a reasonable amount of evidence for an ablauting *n*-stem **keudō*, **kuttaz* meaning ‘bag’.

A West Germanic full-grade root **keud-* is attested in OHG *chiot* ‘bursa’ and OE *cēoda* ‘bag’, which is found in Anglo-Saxon law as *man sceal habban [...]* *cisfæt*, *ceodan*, *wilian* [...] ‘one should have [...] cheese-vessels, bags, baskets [...]’ and in the gloss *ceodas* ‘marsupia’, where it is

³⁰⁹ Holthausen 1934: 46.

³¹⁰ Grimm 11, 655-6.

³¹¹ Verdam 316; Franck/Van Wijk 356.

³¹² Böðvarsson 510; Poulsen 612.

³¹³ Holthausen 1934: 56.

³¹⁴ MED, s.v. *cod*.

³¹⁵ De Vries/Tollenaere 341.

³¹⁶ Fischer/Taigel 283.

thematic. The full grade is further supported by the derivation **keudila-* as in MHG *kiutel* m. 'crop, dewlap', G *Keutel* 'cod-net, bowel', and MDu. *cudel(e)*, *cuil* 'cod-net'.

A zero-grade root **kudd-* is found in both North and West Germanic, e.g. ON *koddi* 'pillow, scrotum' and EDu. *kodde* 'colon, testicle'. The fact that this root combines a zero grade with a geminate, can be an indication that it developed out of the genitive. With this in mind, the paradigm can be set to nom. **keudō*, gen. **kuddaz* for the Proto-Northwest Germanic period. Except perhaps from Swab. *kotze* 'blister', there are no traces of the expected genitive **kuttaz*.³¹⁷

Other possible cognates are Icel. *koðri* m. 'scrotum', G *Koder* 'dewlap' < **kupra(n)-* and EDu., Du. *kossem* 'dewlap' < **kupma-*.³¹⁸

**klewō*, **klunaz* 'clew'

- **klewa(n)-*: ON *klé*, obl. *kljá* m. 'loom weight', Icel. *klé* m., *kljá* n. 'loom weight, bob'³¹⁹, Far. *klíggja-steinur* 'loom weight, stone for weighting haystacks'³²⁰, Nw. *kljá(-stein)* m. 'loom weight, bob'
- **klewōn-*: OHG *chliuwa* 'ball, clew' (⇒ **klewō-kīn-*: Swi. Visp. *xlüüxji* 'id.')
⇒ **klewila-*: MHG *kliuwel* n. 'id.', G *Knäuel* 'id.'³²¹
- **klewīn-*: OE *clēowen*, *clīowen*, WS *clī(e)wen* n. 'clew, ball, strand'³²², OS *kleuwin* 'offa'³²³, MDu. *clouwen*, *clu(w)en* n. 'clew', Du. *kluwen*, dial. *klouwen*, *kloen* 'id.'³²⁴ (= Da. *klyne* 'lump (of peat)'³²⁵), OHG *chliuwi* 'glomus', MHG *kliuwe* n. 'id.'

³¹⁷ The semantic difference between 'bag' and 'pimple' is trivial, cf. OE *pocca* m. 'bag', *poc* m. 'pock' etc.

³¹⁸ De Vries/Tollenaere (p. 353) sets the reconstruction to **kup-sma-*, but the sibilant probably stems from a form with West Germanic gemination before *m*, i.e. **kupþm-*, cf. Du. dial. *pessem* 'root' < **peþþm-*.

³¹⁹ Böðvarsson 502, 504.

³²⁰ Jacobsen/Matras 182-3; Poulsen 598.

³²¹ Kluge/Seebold 502.

³²² Bosforth/Toller 158-9; Holthausen 1934: 51.

³²³ Gallée 178.

³²⁴ Franck/Van Wijk 321.

³²⁵ Falk/Torp 539; ODS s.v. *klyne*.

- **kluni-*: OE *clyne* m. 'lump (of metal)'³²⁶

The Germanic dialects offer sufficient support for the reconstruction of an ablauting *n*-stem **klewō*, **klunaz* 'clew, lump'.

The *e*-grade is most prevalent in the Northwest Germanic languages. Old Norse has ON *klé*, obl. *kljá* m. 'loom weight', which directly points to a formation **klewan-*.³²⁷ It is formally close to OHG *chliuwa* 'ball, clew' < **klewōn-*, which formed the basis for the diminutive **klew-īn(a)-*, attested as OE *clēowen*, *cliowen*, WS *clī(e)wen*³²⁸, E *clew*, OS *kleuwin* and Du. *kluwen*. The modern German form *Knäuel* developed out of MHG *kliuwel*, another diminutive to the same formation, by the dissimilation of the first *l*.

A possible zero grade of the same *n*-stem is encountered in the form of OE *clyne* 'lump' < **kluni-*.³²⁹ On the basis of the underlying root **klun-*, Fick, Falk and Torp (p. 58) reconstructed a paradigm **kluwan*, **klūniz*. This paradigm can be reconciled with the full grades **klewan-* and **klewōn-* by reconstructing an ablauting *n*-stem **klewō*, **klunaz*.

The Germanic forms are clearly related to OCS *žely*, *žbyly* 'tumor' < **gel(H)-uh₂*, **gl(H)-uéh₂-s* (Derksen 2008: 557) and Skt. *gláu-* f. 'ball, lump' < **gleHu-* (Mayrhofer 1, 511). The paradigm underlying the Germanic material is probably best reconstructed as **gléuH-ōn*, **gluH-n-ós* (with laryngeal metathesis?). The short vowel of the root **klun-* must then be explained from older **klūnós* by Dybo's law. Support for an unshortened vowel is found in the undoubtedly related formations MLG *klūs* 'lump', EDu. *kluyksen loocks* 'garlic head, clove' < **klūsa-* and **klū-þan-* 'lump' (see p. 272).

³²⁶ Holthausen (1934: 53) mentions Sw. *kluns*.

³²⁷ Far. *klavi* m. 'piece of rope', seemingly from an *o*-grade form **klawan-*, is a loanword from MLG *klove*, *klave* 'cleft, clew' < **kluban-*. PGm. **klawan-* would have yielded Far. **klái*.

³²⁸ The vowel length in *cliwen* and *cleowen* is called uncertain by the OED, but long diphthongs must be supposed here. PGm. **-ew-* developed into **-euw-*, **-iuw-* in West Germanic, emerging as either *-īo-* or *-ēo-* in the Old English manuscripts. In West Saxon the diphthong was affected by front mutation (Wright 52), which explains the form *clī(e)wen*. Similarly, we find WS *hīew*, *hīw* 'hew' < PGm. **hewja-* as opposed to *hēow*, *hīow* elsewhere.

³²⁹ The often adduced form Sw. *klunn* (Fick/Falk/Torp 58; SAOB K1420) does not belong here. It has a variant *klund* and should therefore be reconstructed as **klunda-*.

Another possibility would be to reconstruct the PIE word as a *u*-stem **gél-ōu*, **gl-u-ós*. Skt. *gláu-* can then be analyzed as a contamination form **gl-ōu-* rather than as laryngealic **gleHu-*. OCS *žely*, *žbly* is similarly derivable from a paradigm **gél-uh₂*, **gl-uh₂-ós*.

****klimbō*, **klump^aaz* ‘lump, hillock’†**

- **klimpa(n)-*: ON *kleppr* m. ‘plummet, lump’, Nw. *klepp* m. ‘lump, chunk, cliff, block’, OSw. *klimper* m. ‘lump’, Sw. *klimp* ‘id.’³³⁰, Da. (*jord*-)*klimpe* ‘clod of earth’, *klimp* ‘lump’, LG *klimpe* ‘chunk’³³¹
 - **klimbō(n)-*: MHG *klimme* f. ‘elevation’³³²
 - **klumpa(n)-*: MHG *klumpe* m. ‘lump’³³³, G *Klumpen*³³⁴, MDu. *clompe*, Du. *klomp* (= Nw. *klump* ‘id.’, Da. *klump(e)* ‘chunk’)
 - **klumbōn-*: ON *klumba* f. ‘club’, *klumbu-fótr* ‘club-foot’
-
- **klamp^a-*: Nw. *klamp* m. ‘block of wood’, Sw. *klamp* ‘wooden leg’
 - **klamp^ō-*: ON *kløpp* f. ‘duckboard, clapper bridge’, Da. *klamp(e)* ‘lump, chunk, block of wood’, MHG *klampe* ‘chunk’³³⁵, MLG *klampe* ‘plank bridge’, MDu. *clamp(e)* ‘pile of hay’

Theoretically, the formal contrast between MHG *klimme* and G *Klumpen* could be reason enough to assume an ablauting *n*-stem **klimbō*, **klump^aaz*, **klumbeni*. ON *kleppr* < **klimpa-* can then be regarded as a full-grade form with an analogical geminate. Since, however, there is a strong verb **klimp^aan-*, attested as MHG *klimpfen* ‘to compress’³³⁶, there is a real possibility that the ablaut of **klimp^a-*, **klimba-*, **klump^aan-* and **klamp^aan-* is of verbal origin. The preservation of the original consonantism by **klimba-* does not necessarily point to an *n*-stem either,

³³⁰ Hellquist 318.

³³¹ Doornkaat-Koolman 2, 260.

³³² Lexer 1, 1623.

³³³ Lexer 1, 1636.

³³⁴ Kluge/Seebold 500.

³³⁵ Lexer 1, 1605.

³³⁶ Lexer 1, 1624.

because the verb **klimp^{an}*³³⁷ seems to be an iterativicized form of **klimban-*, cf. G *klimmen* ‘to climb; to squeeze’³³⁸. This means that the root **klimpⁿ*- can very well be due to the influence of a pertaining iterative **klumpⁿōpi*, **klumbunanpi*. As a result, the reconstruction of an ablauting *n*-stem remains doubtful.

****krebō*, **kurp^{az}* ‘basket’**

- **kreb^{an}*:- MHG *krebe* m. ‘crib’, G *Krebe*³³⁹, Swab. *kreb^o* [ĕ] m. ‘wicker basket, wicker car carriage, sty’³⁴⁰, SFri. *krääf*, *krääwe* m. ‘trough, crib’
- **kreb^{bōn}*:- MHG *krepp^e* f. ‘id.’³⁴¹
 ⇒ **kreb(b)jō(n)*:- OHG *chrippa* ‘basket, crib’, G *Krippe*, Swi. App. *xrep* ‘id.’³⁴², OS *kribbia* f. ‘id.’, Du. *krib(be)* ‘manger, crib’³⁴³
- **krepp^{an}* ⇒ **kreppjō(n)*:- OHG *chripfa*, MHG *kripfe*, Swi. Visp. *xripfa* f. ‘crib’
- **kerba(n)*-, *-ōn*:- ON *kjarf*, *kerf* n. ‘bundle’, OSw. *kærve* m. ‘id.’³⁴⁴, MLG *karve* (= Icel. *karfa* f. ‘basket, hamper’³⁴⁵), *kerve* f. ‘creel’³⁴⁶
- **krup^{pa}*-, *-ōn*:- MHG *krupfe*, G *Krupfe* f. ‘basket’³⁴⁷
 ⇒ **kruppjō(n)*:- G *Krüpfe* ‘id.’
- **krubbōn*:- Icel. *krubba* f. ‘jug, pen, sty’³⁴⁸, Far. *krubba* [f.] ‘compartment; crib, feed bin’, Nw. dial. *krubbe* f. ‘box, small sledge’, MHG *kroppe*, *kruppe* f. ‘crib’³⁴⁹

³³⁷ Fick/Falk/Torp 57.

³³⁸ Grimm 11, 1163-69.

³³⁹ Lexer 1, 1714; Grimm 11, 2126.

³⁴⁰ Fischer/Taigel 285.

³⁴¹ Lexer 1, 1722, 1734.

³⁴² Vetsch 63.

³⁴³ Franck/Van Wijk 348: “echter is *grēbh-*, ablautend met *grebh-*, waarschijnlijk-lijker.”

³⁴⁴ De Vries 1962: 311.

³⁴⁵ Böðvarsson 482.

³⁴⁶ Schiller/Lübben 456.

³⁴⁷ Lexer 1, 1684; Grimm 11, 2471.

³⁴⁸ Böðvarsson 527.

³⁴⁹ Lexer 1, 1757.

⇒ *krub(b)jō(n)-: G *Krüppe*, OE *cryb* f. 'crib'

- *kurba(n)-, -ōn-: OHG *chorb*, *churb* m. 'basket', MHG *korb(e)*, *karb* m. 'id.'³⁵⁰, Cimb. *korba* f. 'id.'³⁵¹, MDu. *corf* m. 'basket, cage'³⁵², Du. *korf* 'basket'³⁵³

This etymon meaning 'basket' (or anything for which baskets are used) displays a wide variety of forms. As has been argued in § 8.3.2, these can all be explained by reconstructing a paradigm **krebō*, **kurp^{az}* < **grēb^h*-ōn, **grb^h*-n-ós that was split up according to the usual *Associationen*. A additional analogy that occurred in this *n*-stem was the resolution of the *Schwebeablaut*, by which the roots **kerb*- and **kurpp*- gave rise to the secondary variants **kerb*- and **krupp*-. It follows from the latter variant that the resolution of the *Schwebeablaut* was prior to the shortening of geminates in superheavy syllables.

The material provides evidence for at least seven different root variants, among which several dissimilar full and zero grades.

A full-grade root **kreb*- is supported by MHG *krebe*, a masculine *n*-stem, and by MHG *krebbe*, which points to **krebbon*-. OHG *chripfa* gives proof of a further unattested root **krepp*-. MLG *kerve* 'creel' has a full-grade, as well, but the position of the vowel is analogical to the regular zero-grade root **kurb*-.

The regular zero grade **kurba(n)*- is attested as OHG *chorb*, MHG *korb(e)*, MDu. *corf*. It has been suggested that these words were adopted from Lat. *corbis*,³⁵⁴ but since **kurba(n)*- is perfectly understandable from within the Germanic context, it is more probable that the Germanic word was adopted by Latin.³⁵⁵ The other zero-grade forms MHG *krupfe* < **kruppōn*- and MHG *kroppe* < **krubbōn*- must be secondary formations, because the position of the vowel slot seems to be analogical to the original nominative **krebō*.

³⁵⁰ Lexer 1, 1679, 1684.

³⁵¹ Schmeller/Bergmann 200.

³⁵² Verdam 307.

³⁵³ Franck/Van Wijk 339.

³⁵⁴ E.g. Franck/Van Wijk 339.

³⁵⁵ Similarly, G *Korb* was adopted by Slavic at an early date, i.e. before the rise of *polnoglasi*: Pol. *korb*, Ru. *kórob* (Fraenkel 220-1). These Slavic forms are again the source for Lith. *kařbas* 'basket'. Similarly, Fi. *karpio* 'bushel' is from Slavic **korbъja*, cf. Ru. *korob'ja* 'basket, chest' (Kylstra et al. II, 50).

There are a number of forms with *a*-vocalism, but these are all later developments. There is probably no evidence for **karbōn*- “als eine echte Nebenform von vorgeschichtlichem Alter”, as was claimed by Grimm (11, 1797). MLG *karpe* with its *p* seems to continue a root **karp^p*-, but it only occurs in “*veer grote tunnen werxs und twee carpen mit werke*”³⁵⁶ and may be borrowed from MHG *karb*, *karp*. These forms, in turn, are etymologically identical to MHG *korb*, reflecting the delabialization of *o* in the South German dialects, such as in early Bavarian *darf* ‘Dorf’, *wart* ‘Wort’, *tachter* ‘Tochter’ and indeed also *karb* ‘Korb’³⁵⁷. MLG *karve*, on the other hand, is from older *kerve* with lowering of *e* to *a* before *r* as in *karke* ‘church’, *wark* ‘work’ and *hart* ‘heart’.³⁵⁸ This *karve* is almost certainly the source for Icel. *karfa* ‘basket’. Similarly, late ON *korf* f. ‘id.’ has been analyzed as a loanword from MLG *korf*³⁵⁹, which seems probable.

The root polymorphy of the *n*-stem is neatly mirrored by some *jō*-stem derivations: e.g. OHG *chripfa*, Swi. Visp. *xripfa* < **kreppjō*-³⁶⁰, G *Krippe* < **krebbjō*-, G *Krüpfe* < **kruppjō*-, and G *Krüppe* < **krubbjō(n)*-. This parallelism may indicate that, when the *jō*-stems were derived from the *n*-stem,³⁶¹ there was some hesitation about what allomorph to use as a base. This means that the allomorphy must still have been intact at that time. It is probably not necessary to reconstruct two separate *n*-stems **krebō*, **kreppaz* and **krubō*, **kruppaz* (Lühr 1988: 250-1) in order to explain the root variation of the *jō*-stems.

In spite of the straightforward reconstruction of **krebō*, **kurp^paz*, no clear etymology is available. The connection with Gr. γρῖπος, γρῖφος ‘basket, fish net’³⁶² is uncertain because of the Greek consonantal irregularities. ON *hrip* n. ‘pannier’³⁶³ has been compared, and if this link

³⁵⁶ Schiller/Lübben 431.

³⁵⁷ Tauber 1993: 69.

³⁵⁸ Lasch 1914: § 76.

³⁵⁹ De Vries 1962: 326.

³⁶⁰ Kluge/Seebold (p. 540) ascribe the difference between OHG *chrippa* and *chripfa* to “intensity” in the latter form.

³⁶¹ There may also have been an ablauting *jō*-stem **gréb^h-ihz*, **grb^h-iéhz-s* > **krebja*, **kurbjōz*, but this reconstruction does not account for the stems **krebān*- and **kurba(n)*-.

³⁶² Pokorný 385-390.

³⁶³ = Allgäu German *reaf* ‘hölzernes Rückentraggestell’.

is correct, the word must be of non-Indo-European origin, as has been argued by Kuhn (1959: 39).³⁶⁴ The problem with these etymologies, however, is that the meaning ‘basket’ is secondary in Germanic. At least, this is what must be concluded on the basis of the closest intra-Germanic cognates ON *kerf*, *kjarf* n. ‘bundle (of twigs)’ and OSw. *kærve* m. ‘id.’ < **kerba(n)-*.

****leuhmō, *l(a)uhmenaz* ‘flash’**

- **leuhman-*: ON *ljómi* m. ‘flash of light, radiance’, OE *lēoma* m. ‘(ray of) light, splendor’³⁶⁵, OS *liomo* m. ‘ray, glare’
- **leuhna-*: Nw. *lyn*, dial. *ljon* n. ‘lightning’³⁶⁶, EDa. *ljun* n. ‘id.’³⁶⁷
- **l(a)uhmunjō-*: Go. *lauhmuni* f. ‘lightning’
- **l(a)uhumnja-*: ME *levene* n. ‘lightning’, E poet. *levin* ‘flash, lightning’

The above words for ‘ray of light’ and ‘lightning’ are in clear ablaut correlation with each other and may therefore point to an apophonic *mn*-stem derived from the Proto-Indo-European root **leuk-* ‘shine’.

An *e*-grade is found in ON *ljómi*, OE *lēoma*, OS *liomo* < PGm. **leuhman-*. The same vocalism is pointed out by the Scandinavian word for ‘lightning’, viz. Nw. *lyn*, EDa. *ljun* < **leuhna-*. It is possible that this formation actually split off from the *mn*-stem. It must then be assumed that the *m* was lost in a genitive form **leuk-mn-os*, for which the paradigm of **b^hud^h-mēn*, **b^hud^h-mn-ós* (see p. 65) can be compared. The parallelism is imperfect, however, because **leuhna-* clearly was not affected by Kluge’s law. Perhaps, then, the genitive **leuk-(m)n-os* was barytone.

The original vocalism of *lauhmuni* is uncertain because of the ambiguity of the Gothic grapheme {au}, which can continue both PGm. **u* and **au* in the position before *h*. The form must accordingly be reconstructed as either **luhmunjō-* or **lauhmunjō-* < **l(o)uk-mn-ieh₂-*.

³⁶⁴ Theoretically, ON *hrip* can also be interpreted as a loanword from a hypothetical Proto-Celtic form **kribi-*, which can be postulated on the basis of Lat. *corbis* < **k_ṛb^h-i-* (cf. De Vaan 2008: 135).

³⁶⁵ Bosworth/Toller 633.

³⁶⁶ Torp 384-5.

³⁶⁷ Fick/Falk/Torp 373; Kalkar 817-8.

Pogatscher (1902: 234-5) supposed a diphthong in view of ME *levene* 'lightning', which he derived from OE **lēahufne* or **lēhifne*³⁶⁸. Both of these reconstructions were taken to be possible continuations of a PGm. neuter *ja*-stem **lauhmunja-* by Pogatscher, the latter variant representing a form affected by *chain umlaut*. However, the Middle English form is probably not as decisive as Pogatscher claimed it to be. Similar formations such as ME *heven* 'heaven' < OE *he(o)fen* and ME *stev(e)ne* 'voice' < OE *stefn* show a development that is identical to the one of *levene*, and neither of them had a PGm. diphthong. The reconstruction of a diphthong therefore does not seem compelling.

In fact, there probably is no objection against deriving *levene* from OE **lyhifne* and ultimately from PGm. **luhumnja-* (again with chain umlaut). This variant closely approaches Go. *lauhmuni*, the only difference being that *levene* continues a stem **l(o)uk-ṃn-* with vocalization of the *m*, whereas *lauhumni* presupposes **l(o)uk-mṃ-* with vocalization of the *n*. This is only a minor problem, however, because both variants are reflexes of the same suffix.³⁶⁹

In summarizing, the Germanic evidence unambiguously points to an ablauting paradigm. There is no way to determine the original vocalism of *lauhmuni* and *levene*, however, so that it remains impossible to decide between a hysterokinetic **léuk-mōn*, **luk-mn-ós*, a proterokinetic **léuk-mn*, **luk-mén-s*, or even a static paradigm **lóuk-mn*, **léuk-mn-s*. Perhaps, the latter option finds some support in the form of **leuhna-*, but only if it developed out of a barytone genitive **léuk-mn-s*.

****melhmō*, **mulhnaz* 'cloud'?**

- **melhman-*: Go. *milhma* m. 'cloud'
- *?*mulhna-*: Sw. *moln* n. 'cloud, darkness'³⁷⁰

The correlation between Go. *milhma* and Sw. *moln* is such that it can be explained by the reconstruction of an apophonic *mn*-stem. The Gothic word would in that case represent the original full grade. Sw. *moln*, on

³⁶⁸ Cf. Pokorny 687-690; Lehmann 228.

³⁶⁹ In Gothic, the variant *-ubni/-ufni* < **-ṃn-io/hz-* seems to have been productive, cf. *witubni* n. 'knowledge' < **uid-ṃn-io-* and *fraistubni* f. 'temptation' from **proist-ṃn-ihz*.

³⁷⁰ SAOB M1285.

the other hand, could be a zero-grade genitive **mulhnaz* continuing a Pre-Germanic form **mulk-mn-os* with dissimilation of the second *m*.

A problem regarding this reconstruction is that the genitive lacks the gemination that is displayed by other *mn*-stems with dissimilation of the labial nasal, e.g. **budmēn*, **buttaz* < **b^hud^h-mēn*, **b^hud^h-(m)n-ós* (see p. 65); the expected outcome of the genitive **mlk-mn-ós* would in Germanic be **mulk^kaz*. Since such a form is not attested, the reconstruction of the ablauting paradigm remains difficult to prove. It may well be, then, that *moln* is to be seen as a substantivation of an adjective *mulen* ‘shady, overcast’, as has been claimed by Hellquist (1922: 483).

**melm*, **mulmaz* ‘sand’?

- **melma(n)-*: OHG *melm* ‘pulvis’, MHG *melm* m. ‘sand, dust’³⁷¹, OS *melm* mn. ‘dust’, MDu. *melm* m. ‘dust, dry sand’, *melme* f. ‘id.’
- **mulma*-³⁷²: G dial. *mulm* m. ‘dust, moldered wood’, MDu. *mol(e)m*, *molle(n)* n. ‘dust, dry earth’³⁷³, EDu. *molm* ‘wood rot, ore’, Du. *molm* ‘wood rot’³⁷⁴
- **malma(n)-*: Go. *malma* m. ‘sand’, ON *malmr* m. ‘ore, metall’, OSw. *malMBER* m. ‘ore’, Sw. *malm* ‘id.’³⁷⁵, OE *mealma-stān* ‘sandstone’, E *malm* ‘limestone’

Bammesberger (1990: 71) lists ON *malmr* (etc.) under the *ma*-stems, but in view of its variants OHG *melm* and MDu. *molm*, it is possible, too, that the starting point of this cluster was an athematic formation, i.e. an *mn*-stem. The is especially attractive in view of the formal similarity with Lith. *melmuõ* m. ‘kidney stone’, which points to PIE **mélh₁-mōn*. In view of the limited attestation of the Germanic forms as *n*-stems, it may be preferable, on the other hand, to reconstruct an ablauting *m*-stem

³⁷¹ Lexer 1, 2096. The weak form *melme* that is mentioned by Lexer is marginal.

³⁷² The semantics of the continuants of **mulma*- was influenced by G *Ulm* (OHG *olmoht* ‘moldered’), Du. *olm* ‘moldered wood’.

³⁷³ Verdam 367.

³⁷⁴ De Vries/Tollenaere 452.

³⁷⁵ Hellquist 452.

**mélh₁-m*, **mlh₁-m-ós*, comparable to e.g. **h₂érh₂-m*, **h₂rh₂-m-ós* ‘arm’ (cf. Go. *arms* ‘id.’, Lat. *armus* ‘upper arm’ : Skt. *īrmá-* ‘arm’, Lat. *rāmus* ‘branch’). The reality of the ablaut of this type is ascertained for Germanic by **elm*, **ulmaz* ‘elm (tree)’ < **h₁él-m*, **h₁l-m-ós* (see p. 155), as has been pointed out above.

Another way to deal with the ablaut of the different nouns is to assume that they are independent formations based on a verbal complex. Thus, Go. *malma* and ON *malmr* can be analyzed as being derived from the strong verb **malan-* < **mólh₁-e-*, MDu. *molm* (as well as *molsem* m. ‘dry earth, wood rot’) as a derivation from the iterative **mullōpi*, **mulunanpi* < **ml-néh₂-ti*, **ml-nh₂-énti*, cf. MDu. *molen* ‘to decay, molder’, EDu. *be-mullen* ‘to sprinkle, cover with dust’. In lack of a strong verb ***melan-*, however, it remains unclear what kind of verbal formation could have been the source of the *e*-grade of **melma-*.

****rehhō*, **ruhhaz* ‘ray’?**

- **rehhōn-*: OE (*h*)*reohhe* f. ‘fannus (= ray)’, ME *reihe*, *rezge*, *righe*, *raie*, *raize* ‘id.’
- **ruhhan-*, *-ōn-*: OE *ruhha* m. ‘id.’, MLG *roche*, *ruche* m. ‘id.’, MDu. *roche*, *rogghe* f. ‘id.’, EDu. *roch* ‘id.’, Du. *rog* ‘id.’

The evidence for an ablauting *n*-stem **rehō*, **rukkaz* is not overwhelming. The root **ruh-* is attested in all the North Sea Germanic languages, e.g. OE *ruhha*, MLG, MDu. *roche*. The full grade, on the other hand, is only supported by three Old English glosses ostensibly representing OE *hrehhe*. Formally, this full grade poses no real problems, however. Since the short diphthong *eo* represents PGm. **e* that was broken before **h*, the form *reohhe* can only continue **(h)rehhōn-* (not **reuhhōn-*). This form, the predecessor of ME *reihe*, *rezge*, *righe* ‘ray’, should be separated from the Old French loanword *raie*, *raize* ‘id.’, which ultimately goes back to Lat. *rāia* ‘ray’.

The variants **(h)rehhōn-* and **(h)ruhhan-* are clearly in ablaut relation to each other. Given their *n*-stem inflection, it is theoretically possible to explain the variants out of an ablauting paradigm **hrehō*, **hruhhaz* replacing older **hrehō*, **hrukkaz*. The evidence for such a paradigm is very limited, however, in absence of an actual variant **hrukkan-*.

****reumō*, ?**rūm*^maz ‘skin, cream’**

- **reuman*:- Icel. *rjómi* m. ‘cream’³⁷⁶, Far. *rómi* m. ‘id.’³⁷⁷, Nn. *rjome* m. ‘id.’, Nw., Da. *rømme* ‘id.’³⁷⁸, Sw. *römme* ‘id.’³⁷⁹, OE *rēama*, *rēoma* m. ‘membrane, meninx’³⁸⁰
- **rauma*(*n*):- OE *rēam* m. ‘cream’³⁸¹, E obs. *ream* ‘id.’, WFri. *rjemme* ‘id.’³⁸², MHG *roum* m. ‘id.’³⁸³, G *Rahm* ‘id.’³⁸⁴, Swi. Rhtl. *roomm* ‘id.’³⁸⁵, MLG *rōm(e)* ‘id.’³⁸⁶, MDu. *room*, *rome* ‘id.’³⁸⁷, Du. *room* ‘id.’³⁸⁸, Limb. *room* ‘skin’³⁸⁹
- ?**rūm*(*m*)*ōn*:- Swi. *ruum(m)e* f. ‘skin (on milk or butter), crusty skin’³⁹⁰

The Germanic word for ‘cream’ or ‘skin (on milk)’ appears with at least two different root vocalisms. The modern Nordic languages, e.g. Icel. *rjómi*, Far. *rómi*, Nw., Da. *rømme*, Sw. *römme*, OE *rēoma*, *rēama*³⁹¹ continue PGm. **reuman*- with an *e*-grade³⁹². The rest of the West Germanic dialects have forms that point to formations with an *o*-grade: OE *rēam*, MHG *roum* ‘cream’ < **rauma*- and WFri. *rjemme*, MLG, MDu.

³⁷⁶ Bððvarsson 799.

³⁷⁷ Poulsen 956.

³⁷⁸ Falk/Torp 935.

³⁷⁹ SAOB R4410.

³⁸⁰ Bosworth/Toller 791: “*se reóma ðes brægenes*”.

³⁸¹ Bosworth/Toller 788.

³⁸² Zantema 1, 823.

³⁸³ Lexer 2, 516.

³⁸⁴ Kluge/Seebold 741: “Die neuhochdeutsche Form beruht auf einer Mundart, die mhd. *ou* zu *ā* entwickelt hat.”

³⁸⁵ Berger 56.

³⁸⁶ Lübben 306.

³⁸⁷ Verdam 499, 500.

³⁸⁸ Franck/Van Wijk 559; De Vries/Tollenaere 590.

³⁸⁹ WLD I, 11: 128.

³⁹⁰ Schweizerisches Idiotikon 915.

³⁹¹ WS *ēo* = North. *ēa* (cf. Wright § 137).

³⁹² De Vries 1962: (p. 449) gives *rjúmi* m. ‘rahm’, but this form only occurs in the nickname *rjúma-rauðr* (cf. Heggstad 544), and can be discarded. Falk and Torp (p. 935) give ON *rjómi*, but this form does not exist but in modern Icelandic.

rome < **rauman*-.³⁹³ The evidence for a Proto-Germanic form with a long **ū*, which is reconstructed by Pokorny (p. 873) on the basis of Sw. *rūm* (in *Schweizerisches Idiotikon* lemmatized as *ruum(m)e*), is marginal. It can theoretically be reconstructed with a zero grade, but the reality of this proto-form is relatively uncertain.

Etymologically, the connection with Avestan *raoyna-* n. 'butter'³⁹⁴ gives the word an Indo-European base. As a result, the formation can tentatively be reconstructed as **Hreug^h-men-* or – if Lith. *ráugas* m. 'sourdough'³⁹⁵ is related – as **HreuHg^h-men-*. The Avestan word may continue a form **Hreu(H)g^h-mno-* with dissimilation of the *m*.³⁹⁶ In Germanic, the root-final stop was lost before **m* as in e.g. **drauma-* 'dream' < **d^hroug^h-mo-* and **hrīman-* 'rime' < **kriHp-men-* (see p. 66).

The distribution of the different ablaut grades is not entirely in agreement with the usual pattern, according to which the *e-* and zero grades are found in roots inflected as *n*-stems, the *o*-grade being restricted to thematic derivatives. In this case, the *o*-grade is attested in an *n*-stem in Frisian, Middle Low German and Middle Dutch. Perhaps, then, the original paradigm can be reconstructed as **Hréu(H)g^{wh}-mōn*, gsg. **Hru(H)g^{wh}-mn-ós*, apl. **Hrou(H)g^{wh}-mn-ŋs* giving PGm. **reumō*, **rūm^maz*, **raum^muns*. Alternatively, the triple ablaut can be derived from a static paradigm **HróuHg^h-mn*, **HréuHg^h-mn-s*. Indeed, the ablaut of **reuman-*, **rauman-* and **rūman-* is parallel to that of the supposedly static word **h₁eu(H)d^h-r-*, **h₁ou(H)d^h-r-*, **h₁u(H)d^h-r-* 'udder' (p. 157).

**skinkō*, **skunk^kaz* 'shank'

- **skinkan-*, -ōn-: OHG *scincho* m., *scincha* f. 'ham', MHG *schinke* m. 'id.', G *Schinken*, Car. *schinke(n)* m. 'shank, leg, ham'³⁹⁷, Cimb. *schinko* m. 'id.'³⁹⁸, ?Swi. Visp. *šeixo* 'id.', MLG *schenke*, *schinke* m. 'ham'³⁹⁹, ?Du. dial. *schenk*, *schink(e)* 'id.'⁴⁰⁰

³⁹³ Franck/Van Wijk reconstructs the different ablaut variants as **reugman-* and **raugma(n)-*.

³⁹⁴ Schwyzler 1907: 180-3; Pokorny 873.

³⁹⁵ Thus Fraenkel: 705-6; Franck/Van Wijk: 559.

³⁹⁶ Cf. Av. *asman-*, gen. *aśnō* m. 'stone, meteorite, sky' = Skt. *áśmā*, gen. *áśnaḥ* < **h₂ék-mōn*, **h₂(e)k-mn-ós*.

³⁹⁷ Lexer 1862: 218.

³⁹⁸ Schöpf/Hofer 166.

³⁹⁹ Lübben 329.

⇒ **skinkja*:- OFri. *ber-skinze* ‘nudiped’⁴⁰¹

- **skankan*:- OE *sc(e)anca*, *sconca* m. ‘shank, shin, upper part of the leg’⁴⁰², E *shank* ‘shin, shaft’, LG *shanke* ‘leg’ (= Far. *skankur* m. ‘leg’⁴⁰³, Nw. *skank* ‘ham, hollow of the knee’, Sw., Da. *skank* ‘shinbone’⁴⁰⁴)

⇒ **schankila*:- G *Schenkel* m. ‘shank’⁴⁰⁵, Du. *schenkel* ‘id.’⁴⁰⁶

- **skunka(n)*:- OFri. *skunka* m. ‘shank’, WFri. *skonk* m. ‘leg’⁴⁰⁷, LG *schunke* ‘thigh, ham’, Du. *schonk* ‘bone’⁴⁰⁸, G Car., Swab. *schunke* m. ‘ham, leg’⁴⁰⁹, Bav. *schunkchn* m. ‘ham’⁴¹⁰

The usual way of dealing with the formal variation of OHG *scincho* ‘shank, leg’, OE *sc(e)anca* ‘shank’ and Du. *schonk* ‘bone’ is to reconstruct a three-way ablaut opposition **skink*- : **skank*- : **skunk*-.⁴¹¹

The *e*- and *a*-grades are beyond doubt, the former being demonstrated by e.g. OHG *scincho*, *scincha*, MHG *schinke*, G *Schinken*, the latter by OE *sc(e)anca*, E *shank* ‘shin, shaft’, LG *shanke* ‘leg’ etc. In addition, the etymological dictionaries posit a zero-grade root **skunk*-. Still, this root can not be established on the basis of the Anglo-Frisian forms OE *sconca* and OFri. *skunka*, because these can have developed out of **skankan*- with regular rounding (“Verdumpfung”) before nasals. LG *schunk* and Du. *schonk* are stronger indications of the zero grade, but there is a true risk that these forms are Frisianisms. Better evidence for **skunkan*- comes from Swabian *schunke* ‘shank, leg’, but the reality of even this seemingly certain zero grade has been questioned. In Carinthian, *schunke* occurs beside *schinke* ‘shank, leg’. For this reason, it

⁴⁰⁰ Kocks/Vording 1069.

⁴⁰¹ Richthofen 627; Hofmann/Popkema 35.

⁴⁰² Bosworth/Toller 823; Holthausen 1934: 271.

⁴⁰³ Poulsen 1030.

⁴⁰⁴ Hellquist 727-8, ODS, s.v. *skank*; Falk/Torp 984-5.

⁴⁰⁵ Kluge/Seebold 799.

⁴⁰⁶ De Vries/Tollenaere 614.

⁴⁰⁷ Zanterma 1, 901.

⁴⁰⁸ Franck/Van Wijk 591; De Vries/Tollenaere 623.

⁴⁰⁹ Lexer 1862: 218; Fischer/Taigel 386.

⁴¹⁰ Kranzmayer/Lessiak 1983: 136.

⁴¹¹ Kauffmann 544 fn.; Fick/Falk/Torp 450; Pokorny 930; Kluge/Seebold 804.

has been claimed by Kranzmayer/Lessiak (l.c.) that the *u*-vocalism arose in a “mißverständener Sing.-Bildung zum pl. *šijnkxe*, dessen *-i-* man als Umlaut-*ü* auffaßte”, but this is doubtful in view of the large area in which it occurs (cf. Deutschrut *šunkxn*).

In fact, the reality of the *e*-grade root **skink-* has been questioned, too. According to Århammar (2004), it is uncertain whether the vowel of WFri. *skinke* reflects OFri. *i* or *e*.⁴¹² The vacillation of MLG, MDu. *schenke* ~ *schinke* could equally well be ascribed to a formation **skankjan-* with front mutation. Such a formation indeed seems to underly Visp. *šeixo*, as this dialect distinguishes between *-eix-* < OHG **-ānch-*, *-iix-* < **-inch-* and *-āix-* from **-anch-*, cf. *šeixu* ‘to give’ < **skankjan-* vs *triixu* ‘to drink’ < **drinkan-* and *bāix* ‘bench’ < **banka-*. It is possible, too, however, that this *šeixo* continues quasi-OHG **scāncho* with analogical umlaut from the plural **scānchon* (see chapter 10 on “pseudo-ablaut”). Whatever the case may be, none of the mentioned forms disprove the existence of the full grade **skinkan-* as supported by e.g. OHG *scincho*.

In view of the absence of any related verbal formations, the reconstruction of an ablauting paradigm nsg. **skinkō*, gsg. **skunk^kaz*, apl. **skank^kuns*, the *a*-grade accusative being modeled after the paradigm of **belkō*, **bulk^kaz*, **balk^kuns* ‘beam’ (see p. 148), has something to recommend itself. The etymon can be related to Gr. σκάζω ‘to limp’ < **skng-^{ie/o-}*, OIr. *scendim* ‘to jump’, and maybe also to Skt. *sákthi-*, Av. *haxti-* ‘leg, ham’.⁴¹³

****sterō*, **sturraz* ‘infertile animal’?**

- **stera(n)-*: OHG *stero* ‘aries, vervex’, MHG *ster(e)* m. ‘id.’⁴¹⁴, G Stär ‘ram’⁴¹⁵
- **sterran-*: MHG *sterre* m. ‘ram’⁴¹⁶

⁴¹² “Der Stammvokal von wfr. *skinke* (mfr. *schin(c)ke*, 1614-1782) kann ein afr. *-i-*, aber auch *-e-* (> spätawfr. *-i-*) enthalten. Am wahrscheinlichsten ist wohl afr. **skinka* mit Parallelen in ahd. *skinco*, mhd. *schinke* > [114] mhd. *Schinken*, as. *skinka*, mnd. *schinke* sowie mnl. *schinke* neben *schenke*, falls letzteres *-e-* < *-i-* enthält (aber wegen mnd. *schenke*, das allerdings neben *schinke* nur selten vorkommt, vielleicht doch < *a* + *i*-Umlaut).“

⁴¹³ Pokorny 930.

⁴¹⁴ Lexer 2, 1177.

⁴¹⁵ Grimm 18, 2389-91.

⁴¹⁶ Lexer l.c.

- **sturran*:- G *Storre* m. 'gelded stallion'⁴¹⁷, Du. dial. *storre* 'small person or animal, piglet'⁴¹⁸

The vacillation of MHG *stere* and *sterre* points to an old *n*-stem with consonant gradation. The word is usually connected to Go. *stairo* f. 'barren one'⁴¹⁹, which is acceptable in view of the obvious cognate G *Stärke* f. 'heifer (= cow that has not yet calved)' < **starikō*-.⁴²⁰ A more closely related formation is G *Storre* 'gelded stallion', probably to be linked with dialectal Dutch *storre* 'small animal or person'. Although G *Storre* more generally means 'stump', a meaning that may well have been used metaphorically to designate a castrated stallion, there is a good possibility that both **ster(r)an*- and **sturran*- once belonged to a single paradigm **sterō*, **sturraz* < **stér-ōn*, **str-n-ós*. This paradigm must then be based on the root **ster*- 'infertile', which is found in e.g. Skt. *starī*- f. 'infertile cow', Gr. στεῖρα f. 'infertile cow, woman', Alb. *shtjérrë* f. 'lamb, kid' < **steri*-, **ster-en*-⁴²¹ and Lat. *sterilis* 'infertile'.

****telgō*, **tulkʰaz* 'twig'**

- **telga(n)*-, -*ōn*:- OE *telga* m. 'branch, bow'⁴²², MHG *zelch*, *zelge* m. 'twig'⁴²³, G *Zelge* f. 'twig, shoot'⁴²⁴, MLG *telch* m. 'twig'⁴²⁵, MDu. *tel(e)ch*, *telgh(e)* mn. 'twig, shoot, arm'⁴²⁶, EDu. *telghe* 'ramus', Du. *telg* 'scion'
 ⇒ **telgra(n)*:- MLG *telgere* pl. 'branches', EDu. *telgher* 'twig', OE *telgor*, *telgra* m. 'shoot, twig'⁴²⁷

⁴¹⁷ Grimm 19, 423.

⁴¹⁸ Kocks/Vording 1190.

⁴¹⁹ Grimm 18, 2389; Fick/Falk/Torp 486; Pokorny 1031; Lehmann 322; Kluge/Seebold 786.

⁴²⁰ The OED lumps OE *stierc* n. 'calf', E *stirk* together with EDu. *stierick* 'iunex' < **steuraka*- 'little bull', but the reconstruction **starika*- works too.

⁴²¹ Demiraj 1997: 377.

⁴²² Bosworth/Toller 975; Holthausen 1934: 344.

⁴²³ Lexer 3, 1052.

⁴²⁴ Kluge/Seebold 1007.

⁴²⁵ Lübben 401.

⁴²⁶ Verdam 600.

⁴²⁷ Bosworth/Toller 975; Holthausen 1934: 343, 344.

- **telgōn*:- ON *tjalga* f. 'thin twig'⁴²⁸, MHG *zelge* f. 'third "pillar" in the three-field system'⁴²⁹, G *Zelge* f. 'id.'⁴³⁰, ?OE *telge* f. 'rod'⁴³¹, E *tellow* 'shoot'⁴³²
- **tulga(n)*:- OE *tungan tulg* 'root of the tongue'⁴³³, G *Zolch* m. 'twig, nozzle', Hess. *zulch*⁴³⁴, *Zungenzolch*, Swi. *zolgge* 'nozzle'⁴³⁵
 ⇒ **tulk^kra*:- MHG *zolcher, zolker* m. 'branch'⁴³⁶
- *?*tulk^ka(n)*:- Du. *tolk* 'small stick'⁴³⁷

The connection of OE *telga*, ON *tjalga*, EDu. *telghe* 'branch' with Du. *tolk* 'small stick' has not yet been made. This is probably because *tolk* is usually analyzed as a diminutive (**tullaka*?) to PGm. **tulla*-, cf. MHG *zoll* m. 'peg'⁴³⁸. In view of G *Zolch* 'twig, nozzle' < **tulga*-, it nevertheless seems meaningful to consider the reconstruction of a paradigm **telgō*, **tulk^kaz* < **dēlg^h-ōn*, **dlg^h-n-ós*. Du. *tolk* may then continue the original genitive root.

It is to be noted that Du. *tolk* and G *Zolch* cannot be directly equated with each other. Since the expected outcome of PGm. **-lk-* is *-lk-* in the non-Alemannic dialects, *Zolch* can only be identical to Du. *tolk* if it continues **tullaka*-, cf. *Milch* < **meluk-* vs *Wolke* < **wulkō-*. It is more likely, however, that *Zolch* represents a Middle German form with *-lg-* > *-ly-* (cf. Hess. *zulch*⁴³⁹), especially in view of the fact that MHG *zilch*, too, may be a Middle German form of *zelge* (cf. Rhnl. *telg* [tɛlɐχ] m. 'twig'⁴⁴⁰). Swi. *zolgge* 'nozzle' < **tulggan-* further seems to substantiate this.

⁴²⁸ De Vries 1962: 591.

⁴²⁹ Lexer 3, 1052.

⁴³⁰ Kluge/Seebold 1007.

⁴³¹ Bosworth/Toller (p. 975) calls the form corrupt. Not so Holthausen 1934: 344.

⁴³² OED, s.v. *tiller*.

⁴³³ Bald's Leechbook Ch. 42, § 1; Fick/Falk/Torp 160.

⁴³⁴ Grimm 32, 31.

⁴³⁵ Ibid.; Hunziker 311.

⁴³⁶ Lexer 3, 1148.

⁴³⁷ Vercoullie 350; De Vries/Tollenaere 378.

⁴³⁸ Ibid.

⁴³⁹ All the Hessian dialects have fricativization (cf. Schirmunski 1962: 331).

⁴⁴⁰ Franck/Müller 8, 1130.

The reconstruction **tulg-* is also supported by the compound G *Zungen-zolch* ‘root of the tongue’⁴⁴¹, which is attested in Höfler’s *Krankheitsnamenbuch* (p. 857). This compound can be ancient, as it reappears as the Old English syntagm *tungan tulg* ‘tongue’ in *Bald’s Leechbook*.

All the material taken together seems to provide a reasonable amount of evidence for a paradigm **telgō*, **tulk^kaz*. Etymologically, this *n*-stem can be related to ON *telgja* ‘to prune’, OIr. *dlongid* ‘to split’, Lith. *daĩgis* ‘scythe’ < **d^holg^h*.⁴⁴² Another possible set of cognates consists of Lith. *dilgùs* ‘stinging’, *dilgė* f. ‘nettle’, OIr. *delg* ‘thorn’⁴⁴³, and especially *delgae* < **delg-en*.⁴⁴⁴, but this root can also be reconstructed as **d^helg-* in view of ON *dálkr* ‘pin, dagger’ < **dalka-*.⁴⁴⁵

****timbō*, **tump^aaz* ‘stub’**

- **timba(n)-*: G *Zimp*, *Zimpe(n)* m. ‘tip (of bread)’⁴⁴⁶
⇒ G *Zimpel* ‘tip, penis’⁴⁴⁷, Pal. *zimpel* f. ‘mane, strand of hair’⁴⁴⁸
- **timp^aan-*: MLG *timpe* m. ‘tip, nozzle’⁴⁴⁹, MDu. *timp(e)* mf. ‘tip, toe’⁴⁵⁰, Du. *timp* ‘long stick’⁴⁵¹
- **tumban-*: OHG *zumpo* ‘veretrum’, MHG *zump(e)* m. ‘id.’⁴⁵², G *Zump*, *Zumpe(n)* ‘penis, stub’⁴⁵³
- **tump^aa(n)-*: MHG *zumpff(e)* m. ‘penis’⁴⁵⁴, G *Zumpf* ‘id.’⁴⁵⁵, MLG *tumpe* m. ‘stub’⁴⁵⁶, Du. dial. *tomp*, *tump(e)* ‘tip, corner’⁴⁵⁷, E dial. *tump* ‘hillock, clump of trees’⁴⁵⁸

⁴⁴¹ Grimm 32, 31.

⁴⁴² Cf. Pokorny 194-6.

⁴⁴³ Holthausen 1934: 344.

⁴⁴⁴ Stüber 173-4.

⁴⁴⁵ Pokorny 247.

⁴⁴⁶ Grimm 31, 1360-1.

⁴⁴⁷ Grimm l.c.

⁴⁴⁸ Christmann 6, 1617.

⁴⁴⁹ Fick/Falk/Torp 164; Lübben 404.

⁴⁵⁰ Verdam 606.

⁴⁵¹ Franck/Van Wijk 694; Vercoullie 348; De Vries/Tollenaere 1991: 376.

⁴⁵² Lexer 3, 1174.

⁴⁵³ Grimm 32, 541-2; Martin/Lienhart 2, 904b.

⁴⁵⁴ Lexer 3, 1174; Benecke/Müller/Zarncke 4, 949.

- **tamp^a-*: Du. *tamp* ‘rope end, penis’⁴⁵⁹ (= Nw., Sw., Da. *tamp* ‘rope end’⁴⁶⁰), G Zarz *zampf* [tsɔmpf] m. ‘tuft, tassel’⁴⁶¹
- *?*tamba-* ⇒ **tambla-*: G Pal. *zambel* m. ‘shag, nap (of a skirt)’⁴⁶²

The material presented here may point to an original paradigm **timbō*, **tump^aaz* that was split up in the usual way. At least two new paradigms can be reconstructed: 1) G *Zimpe(n)* and MLG *timpe* point to generalization of the *e*-grade **timbō*, **timp^aaz* and 2) MHG *zumpe* and *zumpfe* presuppose a zero-grade *n*-stem **tumbō*, **tump^aaz*. A similar variation is displayed by the *a*-grade, which is found in Du. *tamp*, Zarz *zɔmpf* < **tamp^a-* and Pal. *zambel* ‘shag’ < **tambla-*. The question arises whether this *a*-grade points to the influence of a related strong verb. The existence of such verb cannot be pointed out, however.

The meaning ‘penis’ is frequently found with this cluster of cognates, and seems to be quite old. The original meaning of the word probably ranged from ‘stub’ to ‘penis’ in prehistoric times already. G *Zimpel* not only means ‘tip’, like its derivational source *Zimpen*, but also designates the male organ. The same goes for OHG *zumpo*, MHG *zumpe* and *zumpfe*. Dutch *tamp* is cited by the dictionaries as a technical shipping term meaning ‘rope end’, in which sense it was apparently adopted by the Scandinavian languages. It is nevertheless better known as a colloquial word meaning ‘prick’. Although Franck/Van Wijk and WNT call the etymology of *tamp* uncertain, the word must clearly be connected with its West Germanic ablaut variants.

Etymologically, the word is often associated with **tippa-* ‘tip’. Fick, Falk and Torp, for instance, treat **timp-* under **tippa-* (p. 164), while Franck and Van Wijk call it a nasalized form of the same root under *tepel*

⁴⁵⁵ Grimm 32, 541-2; Schatz/Finsterwalder 736.

⁴⁵⁶ Schiller/Lübben 630.

⁴⁵⁷ Franck/Van Wijk 694; WLD I, 3, 36; WBD I, 7, 1309/II, 6, 1829; Weijnen 211; Kocks/Vording 2, 1265.

⁴⁵⁸ OED, s.v. *tump*.

⁴⁵⁹ Franck/Van Wijk 687; Vercoullie 344; De Vries/Tollenaere 1991: 370.

⁴⁶⁰ Falk/Torp 1245; Hellquist 952.

⁴⁶¹ Kranzmayer/Lessiak 181.

⁴⁶² Christmann 6, 1531.

'nipple'. Grimm (32, 541), too, assumes nasalization, and even includes a whole range of allomorphs pertaining to **tabō*, *tappaz* (see p. 341). It remains unclear, though, what morphological process could have inserted the nasal into the paradigm **tabō*, **tappaz*. It certainly cannot have been a verbal *n*-infix, because there is no verb **timban-* or **timp^pan-*. It therefore seems better to etymologically separate the two *n*-stems from each other, although they will doubtlessly have become associated with each other in many dialects at various moments.

What is clear, at any rate, is that no PIE **b* can be assumed on the basis of the Germanic material. De Vries (1962), for instance, reconstructs PIE **dumb-* on the basis of the Old High German form *zumpfo*.⁴⁶³ The evidence rather points to Pre-Germanic **demb^h-*, however. Perhaps, then, the word can be connected with Lith. *demblys* 'ear'⁴⁶⁴ (< **demb^h-* or **d^hemb^h-*).

****wekō*, **wukkaz* 'wick'?**

- **weuka(n)-*, *-ōn-*: OHG *wiohha* 'twirled yarn', *wioh* mn. 'wick', MHG *wieche*, *wicke* mf. 'wick, cotton fibres', G *Wieche*, *Wieke*⁴⁶⁵, dial. *wicke* 'wrap of flax', MLG *wēke* mf. 'wick, bandage'⁴⁶⁶ (= Da. *væge*, Sw. *veke*⁴⁶⁷), MDu. *wieke* 'wick, bandage, mill vane, wing', EDu. *wiecke* '*ala*, *ellyphnium*, *linamentum*', Du. *wiek* 'wing, mill vane'⁴⁶⁸, Flem. dial. *wiek(e)* 'wick'⁴⁶⁹, WFri. *wjuk(ke)* 'wing', SFri. *juuke* m. 'wing', OE *wēoce* f. 'wick', E *wick*
- **wekkan-*: OE *wecca* m. 'wick', MLG *wecke* m. 'wick, bandage'⁴⁷⁰
- **wukkan-*: OS *wokko* '*cincindila*, *colus*'⁴⁷¹, MLG *wocke* m. 'distaff'⁴⁷², *wocken-blät* 'rag to fix the flax on', G *Wocken*⁴⁷³,

⁴⁶³ Cf. Sütterlin (1894: 93): Av. *dūma-* 'tail' < **d^(h)umb^(h)-mo-*.

⁴⁶⁴ Fraenkel 88.

⁴⁶⁵ Kluge/Seebold 987.

⁴⁶⁶ Lübben 569.

⁴⁶⁷ Hellquist 1108; Törnqvist 1977: 109.

⁴⁶⁸ De Vries/Tollenaere 834.

⁴⁶⁹ WBD III/2.1, 271.

⁴⁷⁰ Lübben 569.

⁴⁷¹ Gallée 393.

MDu. *wocke* m. 'distaff'⁴⁷⁴, EDu. *wocke* '*funiculus* (= slender rope)'

- ?**wukan*-. Nw. dial. *oke* m. 'frill'⁴⁷⁵, EDu. Holl. *woack* 'burial shroud'⁴⁷⁶

The material contains evidence for at least three different roots forms. The root **weuk*- is well attested and can be found in e.g. OHG *wiohha*, OE *wēoce* (= E *wick*) and WFri. *wjukke*.⁴⁷⁷ A second root is reconstructed on the basis of MLG *wecke* < **wekkan*-. OS *wokko*, MLG, MDu. *wocke* are probably to be traced back to PGm. **wukkan*-⁴⁷⁸; theoretically, these forms may also continue **wekkan*- with labialization of *e* after *w* (cf. MLG *wepse* ~ *wopse* 'wasp', *webbe* ~ *wobbe* 'web'), but the Westphalian opposition of *wocke* vs *webbe* (Woeste 1882) seems to decide the issue in favor of **wukkan*-. EDu. *woack* 'winding sheet' is formally obscure, and can hardly be interpreted as reflecting **wukan*-. In view of Nw. *oke*, however, this reconstruction gains some credibility.

In order to explain the vocalic alternation of **e* and **eu* in MLG *wecke* an OHG *wiohha*, it has been suggested that **weukōn*- is a reduplicated formation **ue-ug*-.⁴⁷⁹ The problem with this explanation is that it fails to account for the potential third root **wukk*- as in MLG *wocke*, and – more importantly– for the consonant gradation of **k* and **kk*. It therefore seems that the reconstruction of a reduplicated formation is *ad hoc*.

In view of the overwhelming number of *n*-stems among these words, the vocalic alternations are better explained as resulting from paradigmatic ablaut. The reconstruction of an original paradigm **wekō*, **ukkaz* from **uég-ōn*, **ug-n-ós* would at any rate clarify the different root forms. OE *wecca* suggests that the paradigm was transformed into **wekō*, **wekkaz* in the prehistoric dialect underlying Anglo-Saxon. OS *wokko*, on

⁴⁷² Lübben 591.

⁴⁷³ Kluge/Seebold 995.

⁴⁷⁴ Verdam 806.

⁴⁷⁵ Falk/Torp 1400-1.

⁴⁷⁶ Kilianus 1599: 677.

⁴⁷⁷ Note that the meaning appears to have shifted from 'wick' to 'bandage' and 'wing' in Frisian and Dutch.

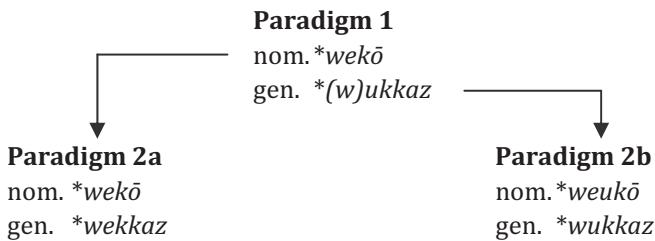
⁴⁷⁸ Grimm 30, 965; Fick/Falk/Torp 381.

⁴⁷⁹ Fick/Falk/Torp: 381; Hellquist 1108; Franck/Van Wijk: 793; Pokorny 1117.

the other hand, can be derived from a paradigm **wekō*, **wukkaz*, with the analogical introduction of the **w* from the full grade. In this way, the suggestion by Pokorny (p. 1117) that **wukkan-* is from **ug-* “mit Übernahme von *w-* aus den hochstufigen Formen” becomes perfectly understandable.⁴⁸⁰

In more or less the same way, the root **weuk-*, as supported by OHG *wiohha* and OE *wēoce*, may have arisen as a secondary full grade to the root **wukk-*. Obviously, this analogy must have taken place after the introduction of **w* in the oblique, which led to the reinterpretation of the **u* of **wukk-* as belonging to the root.

If the doubtful stem **wukan-* really existed, it can be explained from a secondary paradigm **wukō*, **wukkaz*. The evidence for this particular form is extremely sparse, however.



All the different vowel alternations could be given an alternative explanation by assuming that the various, ablauting roots were derived from a verbal complex, cf. MHG *wickeln*, EDu. *wikkelen* ‘to wrap’ < **wekkljan-*, MDu. *wocken* ‘id.’ < **wekkōn-* or **wukkōn-*. It is not entirely clear, however, how this could have worked, but it is defensible to think that there was an iterative **wekkōn-* (cf. MHG *ver-wicken* ‘to wrap’ < **wekkjan-*) or – with a secondary zero grade – **wukkōn-* that gave rise to a de-iterative strong verb **weuk^kan-*. This verb can then have served as the derivational base for the *n*-stem **weuk^kōn-*.

Etymologically, the whole cluster of forms seems to be related to the root **ueg-* or **ueg^h-* as in Mlr. *figid*, W *gweu* ‘to weave’.⁴⁸¹ The connection with OE *wōcig* ‘noose, snare’⁴⁸² is more doubtful.

⁴⁸⁰ Also Grimm 30, 965: “mit Übertragung des *w* von der Hochstufe”.

⁴⁸¹ Cf. Hellquist; Pokorny; De Vries/Tollenaere; Kluge/Seebold.

⁴⁸² Franck/Van Wijk: 796.

9.2 The **e ~ *a* type

There are four *n*-stems that potentially display an **e ~ *a* alternation: 1) **aulō / *eulō* ‘hollow stalk’, 2) **fesō, *faznaz* ‘fiber’, 3) **hesō, *haznaz* ‘hare’ and 4) **kekō, *kaweni* ‘jaw’. Since it is difficult to see how this alternation could reflect real **e : *o* ablaut, it is more probable that it arose as a result of the regular **e : zero* ablaut that is found in other paradigms. This leaves two possibilities: either the alternations continue old PIE **h₁é : *h₁ablaut* (with vocalization of the laryngeal in the zero grade), or the full grade was introduced analogically in Germanic. Since the evidence for the **e : *a* ablaut is extremely limited, it is difficult to determine which of the two solutions is correct in the individual cases.

**eulō / *aulō* ‘hollow stalk’

- **eula(n)-*: ON *hvan-n-jóli* m. ‘stalk of angelica’⁴⁸³, Icel. *hvan-n-jóli* m. ‘id.’, *njóli* m. ‘sorrel, stalk, cigar’⁴⁸⁴, Far. *hvan-n-jóli, -ur* ‘stalk of angelica’, *jólur* ‘stalk (of angelica)’⁴⁸⁵, Nw. dial. *jol* m. ‘angelica’, *kvann-jol* m. ‘cane, stalk (of angelica)’⁴⁸⁶
- **aula(n)-*: Nw. dial. *aul* m. ‘stalk of angelica’, *geit-aule* m. ‘wild angelica’, *kvann-aule* m. ‘id.’

The angelica plant was used in Scandinavia throughout the Middle Ages as an herb and vegetable. The Vikings took the plant to every land on which they set foot, and, as a consequence, it has become indigenous on the Faroe Islands, Iceland and Greenland.

Angelica was known for its hollow stalks. This becomes particularly clear from the application of it by Ólaf Tryggvason on a mission to Christianize the north of Norway. At some point, Ólaf captures the pagan Rauð, and demands him to be baptized. When Rauð refuses, Ólaf becomes infuriated, and decides to kill him by feeding him a snake through an angelica tube that was pushed down his throat. The explicit use of angelica for its hollowness confirms the generally accepted connection of Nw. *aul* with Gr. *αύλος* m. ‘tube, flute’, Lith. *aūlas* m. ‘boot

⁴⁸³ De Vries 1962: 292.

⁴⁸⁴ Böðvarsson 429, 471, 688.

⁴⁸⁵ Poulsen 500, 556.

⁴⁸⁶ Torp 250: “paafaldende avlydsform til a u l.”

leg'⁴⁸⁷ < **h₂eulo-* and furthermore Ru. *úlej*, gen. *úl'ja* m. 'bee hive', Lith. *aulýs* m. 'id.' < **h₂eul-io-*⁴⁸⁸.

In addition to the forms with **aul-*, the West Norse dialects have an *e*-grade root **eul-*, cf. Icel. *-jóli*, Far. *jólur*, Nw. dial. *jol*. This variant is problematic, because the PIE root **h₂eul-* can account for PGM. **aul-*, but not for **eula-*, initial **e* being impossible after a laryngeal. By way of a solution, Pokorny (p. 88-89) derives the root from a PIE lengthened grade, i.e. **h₂ēul-*, assuming that the laryngeal did not modify this long vowel (Eichner's law). The problem with this solution is twofold: 1) it is rather tricky to reconstruct a lengthened grade for Proto-Germanic or even Proto-Indo-European on the basis of a West Norse vowel alone, and 2) it is unclear what kind of morphological process could have given rise to a lengthened grade, not in the least because the formation is otherwise identical to **h₂eul-o-*. For these reasons, the reconstruction **h₂ēul-* must be rejected.

In view of the limitation of the root **eul-* to Germanic, it is more likely that the *e*-grade is a post-Indo-European innovation. Possibly, it was triggered by the creation of an *n*-stem to an inherited thematic form **aula-*. This may follow from the fact that the attested *n*-stems predominantly occur in compounds, e.g. Nw. dial. *aul* vs *kvann-aule*, Far. *jólur* vs Far., Icel. *hvan-jóli*. Icel. *njóli* is a simplex, but its initial *n-* must be due to reanalysis of *hvan-jóli* as *hvan(n)-njóli*. In view of this distribution, it is conceivable that the *e*-grade arose in an *n*-stem that was created in order to form a compound with **hwannō-* 'angelica'.

In conclusion, the etymon described here does not directly attest to an ablauting *n*-stem, because a zero grade ***ullaz* < ***h₂ul-n-ós* does not seem to have ever existed. It nevertheless points to paradigmatic ablaut indirectly, because it proves that the *e*-grade must have been productive in the Germanic *n*-stems. In this respect, it can perhaps be compared to, for instance, the formation **kernan-* as presupposed by ON *kjarni*, OHG *cherno* m. 'kernel, grain' (see p. 174). It is not entirely clear, however, whether this **ǵerH-n-on-* represents an independent Germanic formation or that it continues the full-grade form of an old neuter *n*-stem, cf. Go. *kaurno* n. 'grain' < **ǵrH-n-ōn-*, Lat. *grānum* n. 'grain, seed' < **ǵrH-no-* and Lith. *žirnis* 'pea' < **ǵrH-ni-*.

⁴⁸⁷ Cf. Torp 9.

⁴⁸⁸ Derksen 2008: 508.

***fesō, *faznaz ‘fiber’†**

- *fesō-: ODa. *f(j)þs* ‘thread, fiber’, Sw. dial. *fös* ‘id.’, *fjas* ‘down’
- *fesōn-, -na-: OHG *fesa* f. ‘chaff’⁴⁸⁹, MHG *vese* f. ‘id.’⁴⁹⁰, G Car. *fēse* ([ɛ] = *e) f. ‘pod’⁴⁹¹, Swi. Rhtl. *fēasə* ([ɛə] = *e) ‘chaff’⁴⁹², MLG *vese(n)* mf. ‘chaff, fiber, fringe’⁴⁹³, MDu. *vese* f. ‘frill, border, fiber’⁴⁹⁴
 ⇒ *fasila- or *fesla-: MHG *vesel* n. ‘chaff’⁴⁹⁵, Du. *vezel* ‘fiber’
- *fasan-, -ōn-: OHG *faso* m. ‘fiber, fringe, border’, *fasa* f. ‘id.’⁴⁹⁶, MHG *vase*, G *Faser* f. ‘frill’⁴⁹⁷, E *feaze*, MDu. *vase* f. ‘fiber, seam’⁴⁹⁸
- *fasa-: OE *fæs* n. ‘fringe, border’, MDu. *vas* n. ‘cervical muscle, hair of the head’⁴⁹⁹
- ?*fus-: E *fuzz* ‘fluff’ ⇒ *fozy* ‘fluffy’

The alternation of OHG *fesa*, Swi. Rhntl. *fēasə* < **fesan-* with OHG *fasa*, MDu. *vase* may theoretically point to an ablauting paradigm **fesō*, **faznaz* < **ph₁és-ōn*, **ph₁s-n-ós*. This reconstruction finds little support outside the Germanic languages, however. The possible cognates Ru. *pásmo* ‘strand’ and Latv. *puosma*, *puōsms* ‘strand of flax’⁵⁰⁰ point to a proto-form **poHs-mo-* in which the full grade precedes the laryngeal rather than the other way around. The question therefore arises whether the Germanic *e*-vocalism can be secondary, i.e. introduced analogically in the *n*-stem **fasan-*.

⁴⁸⁹ EWA 182.

⁴⁹⁰ Lexer 3, 324.

⁴⁹¹ Lexer 1862: 94.

⁴⁹² Berger 33.

⁴⁹³ Lübben 477.

⁴⁹⁴ Verdam 710.

⁴⁹⁵ Lexer l.c.

⁴⁹⁶ EWA 80-1.

⁴⁹⁷ Kluge/Seebold 277: “Offenbar zu ig. (w/oer.) **pes-* (älter **pwes-* ‘wehen, reinigen’) in russ. *pachát* ‘wehen, fegen’, l. *pūrus* ‘rein’.”

⁴⁹⁸ Verdam 643.

⁴⁹⁹ Verdam l.c.

⁵⁰⁰ Fraenkel: 640.

A late origin must at any rate be assumed for E *fuzz* – if this form is related at all. With an earliest attestation in 1674 (OED, s.v. *fuzz*) it is extremely unlikely that the form continues a zero-grade formation **fusan-*. The double *zz* rather indicates that it is a recent formation that is at least partly of onomatopoetic origin.

****hesō*, **haznaz* ‘hare’**

- **hesan-*: Nn. *jase* m. ‘id.’
- **hezan-*: Icel. *héri* (= *hjeri*) m. ‘id.’
- **hasan-*: OHG *haso* m. ‘id.’, MHG *hase* m. ‘id.’, G *Hase*, MLG *hase* m. ‘id.’, MDu. *hase* ‘id.’, Du. *haas*⁵⁰¹, OFri. *hasa* m. ‘id.’
- **hazan-*, -*ōn-*: ON *heri* m. ‘id.’, OSw. *hare*, *hære* m. ‘id.’, Sw., Nw., Da. *hare* ‘id.’⁵⁰², OGutn. *heri* ‘id.’, Far. *hara* f. ‘id.’, OE *hara* m. ‘id.’

The word for ‘hare’ shows a number of different root variants across the Northwest Germanic dialects. This is mainly due to two Proto-Germanic alternations. First, a clear Verner alternation is evinced by e.g. OHG *haso* < **hasan-* vs OE *hara* < **hazan-*. It seems to demonstrate that the accentuation of the *n*-stem was still mobile when Verner’s law started to operate. Second, there is strong evidence for an ablaut alternation in view of the unambiguous *e*-grades in Nordic, among which most notably Nw. dial. *jase* < **hesan-*. Together, these two alternations point to an ablauting paradigm **hesō*, **haznaz*.

The distribution of the Verner variants is as follows. With the exception of OE *hara* < **hazan-*, all West Germanic dialects have forms that go back to PGm. **hasan-*, e.g. OHG *haso*, MDu. *hase*, OFri. *hasa*. This Verner alternation was projected back into the Proto-Germanic paradigm by Schaffner (2001: 544-6), who convincingly argued that the original paradigm **hasō*, **hazeni* was leveled as both 1) **hasō*, **haseni* and 2) **hazō*, **hazeni* in the West Germanic dialects. He explained the accentual mobility by reconstructing an amphikinetic paradigm nom. **ḱásō*, gen. **ḱas-n-és*, loc. **ḱas-én-i*.

In addition to the interchange of **s* and **z*, the North Germanic evidence shows a salient interchange of *e* and *a* in the root. OSw. *hare*

⁵⁰¹ De Vries/Tollenaere 230.

⁵⁰² SAOB H440.

and Far. *hara* reflect **hazan-* and **hazōn-* with PGm. **a*, but Nn. *jase* unambiguously points to a proto-form **hesan-* (cf. Pokorny 533), as it has *a*-breaking of *-e-* to *-ja-*. The *e*-grade must additionally be reconstructed for Icel. *héri*.

In Icelandic orthography, the initial phone [ç] is usually represented as *hj*. However, in front of *é* [je], the *j* is omitted, cf. *hér* 'here' = [çe:r]. Since the usual derivation of ON and Icel. *é* from PGm. **ē₂* is impossible in the case of *héri*, it can hardly be analyzed as something else than a "wrong" spelling for *hjeri*. In this form, the word can have regularly developed out of PGm. **hezan-* by 1) *a*-breaking of **e* to *-ja-*, 2) *z*-fronting of **-az-* to *-ez-*, and 3) the change of **z* (*R*) to **r*. It cannot possibly be derived from **hazan-*, as Schaffner (2001: 545 fn.) explicitly claims, because this would have become Icel. ***heri* (cf. *ker* 'tub' < **kaza-*).⁵⁰³

The consequence of deriving Icel. *héri* from < **hezan-* is that the Old Norse form *heri* becomes ambiguous; it may represent either *heri* < **hazan-* or *héri* < **hezan-*. In view of the contemporary form, *héri* is probably preferable. It would then differ from OSw. *hare* and modern Sw. *hare*, which are the expected outcomes of PGm. **hazan-*. The similarity of ON *heri/héri* with OSw. *hære* and OGutn. *heri* is deceiving; the latter forms probably received their front vowels as a result of vowel harmony.

All things considered, the four different stems **hesan-*, **hezan-*, **hasan-* and **hazan-* point to a paradigm **hesō*, **haznaz*, **hazeni* with ablaut of the root and the suffix. This paradigm fits relatively well into the Proto-Germanic system of the ablauting *n*-stems.

The reconstruction of the Proto-Indo-European paradigm, on the other hand, is disputed. Lat. *cānus* 'hare' < **kāsno-*, MW *ceinach* 'female hare' < **kāsniḱā-*, OPru. *sasins* and Skt. *śása*.⁵⁰⁴ are usually reconstructed with a root **kās-* with **a* (Cf. Pokorny 533). This **a* is problematic, not just because it was a marginal phone in PIE, but more particularly because the ablaut **e* ~ **a* cannot possibly have been Proto-Indo-European. As an alternative, Lubotsky (1989: 56-7) proposed a stem **kh₁-s-*. This stem is indeed capable of explaining the *a* of the Italo-Celtic

⁵⁰³ One could alternatively argue that ON *he-* regularly developed into Icel. *hé-*, as in ON *heðan* > Icel. *héðan* 'from here'. This would imply that *héri* likewise regularly continues *heri* < **hazan-*. It is more probable, though, that the onset of Icel. *héðan* is analogical after *hér* < **hiar* < **ki-or*.

⁵⁰⁴ From **śása-* by assimilation of the second **s* to the preceding *ś*.

forms (cf. Schrijver 1991: 91). It also allows for the reconstruction of the Germanic *n*-stem as **kh₁és-ôn*, **kh₁s-n-ós*, **kh₁s-én-i*.⁵⁰⁵

The *n*-stem formation can be considerably old since the root **kh₁s-* is attested with an *n*-suffix in Germanic, Baltic and Italo-Celtic. Traditionally, the *n*-stem is derived from an adjective meaning 'grey', i.e. OHG *haso*, ON *hǫss* 'grey' < **kh₁s-uo-* and Lat. *cānus* 'grey' (~ OHG *hasan* 'polished?') < **kh₁s-no-*⁵⁰⁶ (cf. Lith. *pilkas* 'grey' ⇒ *pilkšis* 'hare, horse', and with similar meanings: *šiřvas* ⇒ *šiřvis*⁵⁰⁷). However, Lat. *cānus* 'grey' can just as well be derived from the *n*-stem. Similarly, ON *hǫss*, OHG *haso* 'grey' may represent a derivative from the word for 'hare', the color suffix *-*wa-* being productive in Germanic.

**kekō*, **kaweni* 'jaw'?

- **keukōn-*: MLG *keke* f. 'jaw', OFri. *ciāke* f. 'id.'⁵⁰⁸, WFri. obs. *tsjeak* 'id.'⁵⁰⁹, SFri. *sooke* f. 'cheek', NFri. Wdh. *sīk* f. 'id.'⁵¹⁰, OE WS *ceoce*, Angl. *cece* f. 'jaw', ME *ch(e)oke* 'id.', E *choke* 'fleshy parts under the jaws'
- **ke(u)kōn-*: MLG *kēke* f. 'throat, gill, jaw' (⇒ *keken* w.v. 'chatter'), LG *keke* 'mouth'
- **kekan-*: Nw. *kjake* m. 'jaw, cheek', OSw. *kiæke* mf. 'jaw', Sw. *käke* 'id.'⁵¹¹, ODa. *kiæge* 'id.'⁵¹², Da. dial. *kaje* 'jaw'
- **kekō-*: Nw. dial. *kjok*, pl. *kjakir* f. 'id.', Sw. dial. *kjåk* 'id.'⁵¹³
- **kakōn-*: MLG *kake* f. 'jaw, cheek, gill, throat' (= G dial. *kaken* f.pl. 'yellow sides of a bird's beak'), MDu. *kak(e)* f. 'jaw', Du. *kaak* 'id.', OE *ceace* 'jaw, cheek', ME *ch(i)eke*, *chik* 'jaw(bone), cheek, mouth', E *cheek*

⁵⁰⁵ The alternative is to assume that **hesan-* is "eine Ablautsneubildung", as Pokorny states. Either way, we end up with Germanic ablaut, because the latter solution implies that the ablaut was still productive in North Germanic.

⁵⁰⁶ Cf. Heidermanns 1993: 283-4.

⁵⁰⁷ Fraenkel 591, 989-990; Derksen 1996: 88.

⁵⁰⁸ Richthofen 861-2; Holthausen 1925: 134; Hofmann/Popkema 518.

⁵⁰⁹ Zantema 1086.

⁵¹⁰ Jensen 481.

⁵¹¹ Hellquist 385.

⁵¹² Falk/Torp 513.

⁵¹³ Hellquist 315; GM, s.v. *kjok* II.

- **kuka-*, *-ō-*: ON *kok* f. 'throat', Icel. *kok*, *kók*, *kvok* n. 'pharynx', Nw. dial. *kok* n. 'throat'
- **kukan-*: Far. *koka* f. 'cavity in the rectum of livestock'

Of all the material involved here, the Nordic forms can be analyzed relatively easily. Three different roots must be identified. First, there is the *n*-stem **kekan-* as evidenced by Nw. *kjake*, OSw. *kiæke*, ODa. *kiæge*. These attestations presuppose a further unattested form **kjaki* for Old Norse. Similarly, dialectal Nw. *kjok* and Sw *kjåk* imply that Old Norse had a form **kjok* which developed out of **keku* < **kekō-* by *u*-breaking.

The establishment of the third formation is more challenging. Fritzner, Heggstad, De Vries and Fick/Falk/Torp cite an Old Norse form *kók* f. 'mouth, throat'⁵¹⁴, which is taken to have developed out of PGM. **kōkō-*. The same word re-appears in the Norwegian Nordmøre dialect in the expression *dæ sto fast i kokje* 'it got stuck in the throat' (*Grunnmanuskriptet* explicitly identifies the vowel as *ó*). Contrarily, Modern Icelandic mainly uses the form *kok* n. 'throat', which does not seem to continue **kōkō-*, but rather **kuka-*. In spite of the semantic distance, the same root can be retrieved from Far. *koka* f. 'cavity in the rectum of livestock'⁵¹⁵ < **kukōn-*. Íslensk Orðabók (p. 511) lists two additional forms corresponding to *kok*, viz. *kvok* and *kók*. The derivation of the former variant is unclear to me. The latter variant ostensibly supports the reconstruction of a long vowel in ON *kók*. However, the conspicuous synonymy with *kók* n. 'cough'⁵¹⁶ and *kóka upp* 'to cough up'⁵¹⁷ opens the possibility that the originally feminine *kok* was adapted to the neuter *kók*. One may wonder, in fact, whether the whole complex of forms is not simply onomatopoeic, like e.g. Du. *kok-halzen* 'to retch'.

The West Germanic material requires at least three different reconstructions. The Frisian material is relatively easy to account for. OFri. *ciāke* corresponds with SFri. *sooke* and NFri. *sīk*, and in view of such a correlation as NFri. *jūp*, OHG *hiufo*, OE *hēopa* 'rose hip' (< **heupan-*) it can be reconstructed as PFri. **ciak-* < **keukōn-*. The seemingly unpalatalized Old Frisian form *keke* is best explained as a loanword from

⁵¹⁴ Heggstad 375; De Vries 1962: 324.

⁵¹⁵ Jacobsen/Matras 187; Poulsen 612.

⁵¹⁶ Böðvarsson 510.

⁵¹⁷ Benediktsson 44.

Low German *kēke*, and indeed the North Frisian dialect of Wiedingharde has native *sīk* beside *kēk* ‘mouth’⁵¹⁸ from Low German.

It is furthermore possible that Nw. dial. *kjuke* f. ‘hemp-nettle’ < **keukōn*- belongs to the Frisian word; the flower of this plant bears resemblance to a ‘beak’, and is therefore categorized under the *lamiaceae*, the “lip-flowers”. It is therefore conceivable that, in Norwegian, the flower was named after its beak-like shape.

In the Low German area, MLG, MDu. *kāke*, EDu. *kaecke* and Du. *kaak* furnish evidence for another variant, viz. **kakōn*- or **kēkōn*-. The literature disagrees on the original vocalism of all these forms. Fick, Falk and Torp (p. 33) reconstruct **kakōn*-. The OED links it with OE *ceace*, and derives both forms from **kēkōn*-. This interpretation is accepted by *Etymologisch woordenboek van het Nederlands* (2, 592), but De Vries and Tollenaere (1991: 290) split up the different attestations into several proto-forms, deriving OFri. *ciāke* from **keukōn*-, OE *cēace* from **kaukōn*- or **kēkōn*-, and Du. *kaak* from **kēkōn*-.

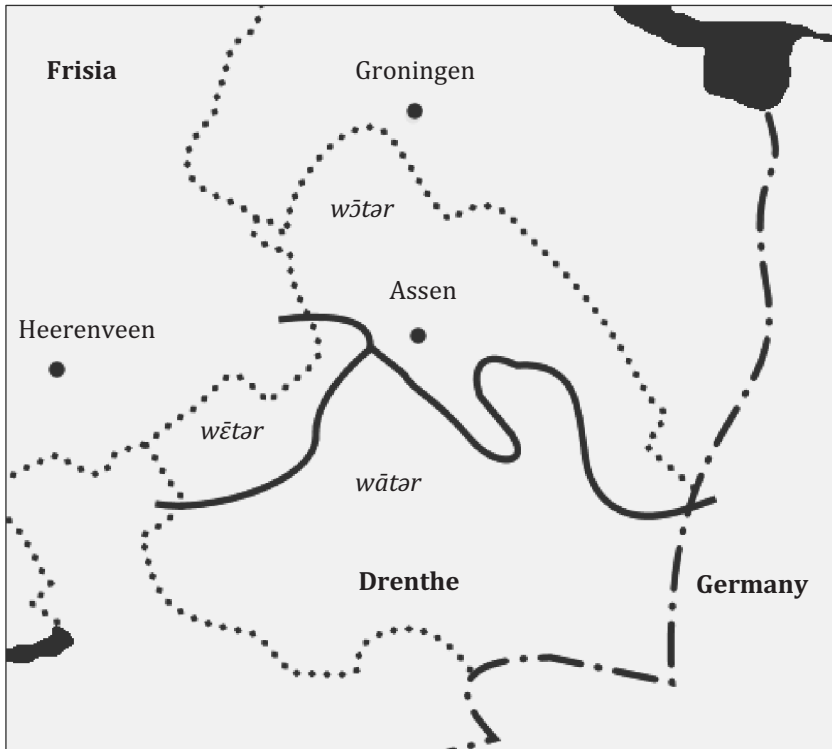
All these accounts, however, discard the evidence furnished by the modern Saxon dialects that have upheld the distinction between Proto-Germanic lowered **ē* and lengthened **a*. In the Dutch province of Drenthe, for instance, the dialectal distribution of *kēk* : *kāk* : *kōk* ‘jaw’ exactly matches the one of *wētār* : *wātār* : *wōtār* ‘water’ < PGm. **wātra*-⁵¹⁹ (see map on p. 205). The small patch with **a* > *ē* is part of the larger Stellingwerven dialect area, which borders with Frisian in the Northwest. This dialect has *kēke*, as opposed to e.g. *skōp* ‘sheep’ < **skēpa*- and *jōr* ‘year’ < **jēra*-. Evidently, the reconstruction **kēkōn*- cannot be upheld, and must be replaced by **kakōn*-.

The situation is most complex in Old English, where three different forms are found, i.e. *ceace*, *cece* and *ceoce*. The last form *ceoce* f. ‘jaw’ and its continuants ME *ch(e)oke* and E *choke* can probably be unified with OFri. *ciāke* < **keukōn*-.⁵²⁰ The prevailing Old English form, however, is *ceace*, underlying ME *cheke* and E *cheek*. Since the length of the diphthong is unknown, it can be read as either *cēace* or *cāace*. As a result, there are no less than three possible ways of reconstructing it: **kakōn*-, **kēkōn*-, and **kaukōn*-.

⁵¹⁸ Jensen 259.

⁵¹⁹ Kocks/Vording 505.

⁵²⁰ Already Noreen 1894: 222.



The development of PGm. **a* in the word for 'water' across the Saxon dialects of the Dutch province of Drenthe.

The pre-form PGm. **kakōn-*, which is well attested for Low Germanic, would regularly develop into *čěace* by the diphthongization of *æ* after *c, g* and *sc* (Wright 1925: § 72) and by the palatalization of the velar like, for instance, *ceaf* 'chaff'. Under the same conditions, i.e. after velars, PGm. *kēkōn-* would have given *čěace* in West-Saxon, and PGm. **kaukōn-* would give the same outcome according to the usual sound laws.

The problem is further complicated by yet another variant *cece*, which, as opposed to West Saxon *ceace*, is labeled Anglian by the OED (s.v. *cheek*). According to the OED, *cece* occurs only once in the Lindisfarne gloss to the Gospel of Luke, which is in the Northumbrian dialect. Anglian *ē* can have four different sources in this environment: 1. **ē*, which developed into West-Saxon *ēa* after *c, g*, and *sc*, but became *ē* in the other dialects (Wright 1925: § 124); 2. **au*, yielding *ēa* in (early)

West-Saxon, as opposed to *ē* in Northumbrian and *ǣ* (late *ē*) in Anglian proper; 3. **eu* became *ēo* in West Saxon, but merged with **ēa* from **au* in Northumbrian, where it changed into *ē* before velars (Wright 1925: §§ 137, 189); 4. **e* usually became *ěo* in Anglian and *ěa* in Northumbrian by back-mutation before single consonants, but not if the consonant was a velar (Wright 1925: § 93).

All in all, the Old English material does not seem to be easily analyzable, especially when the entire Germanic context is taken into consideration; WS *ceace* (**cēace*) and *cece* (**cēce*) could be unified into either **kakōn-* or **kaukōn-*, but neither of these forms is found in the other dialects. It is therefore safer to subordinate the English material to the non-English forms that are less opaque, i.e. to lump together OE *ceace* (**cēace*) and MDu. *kake*, on the one hand, and OE *cece*, *ceoke* (**cēoce*), ME *ch(e)oke* and OFri. *keukōn-*, on the other. Note that the only compelling evidence for a root **kek-* comes from Scandinavian, because MLG *kēke* can theoretically stem from both **kekōn-* and **keukōn-*.

It follows from the analysis given here, that at least the existence of the roots **kak-*, **kek-* and **keuk-* cannot be denied. The roots **kēk-* and especially **kauk-* may have existed as well, but the evidence is not compelling.

When the root **keuk-* are compared to **kek-* and **kak-*, it is clear that the former two variants must be younger than the latter, because they can be derived from the verb **kewwan-* ‘to chew’ (cf. ON *tyggva*, OHG *chiuwan*, OE *cēowan*) with a **k-* suffix (cf. OHG *chelah* ‘throat’ to *chela* ‘id.’). In the roots **kek-* and **kak-*, on the other hand, the final labial of the root **keu-* < **ǵieuH*.⁵²¹ is conspicuously absent, as if it was replaced by a voiceless velar. This situation is reminiscent of the development PIE **-Hu-* > PGm. **-k-* as in OE *tācor* ‘brother-in-law’ < **daHiwer-* < **deh₂iuer-* as suggested by Kortlandt (1988: 356)⁵²², and the question arises whether this change cannot also be applied to the word for ‘jaw’.⁵²³ A paradigm nom. **ǵéh₁u-ōn*, loc. **ǵh₁u-én-i*, for instance, would, according to Kortlandt’s rule, regularly develop into PGm. **kekō*, **kaweni*.

⁵²¹ With the regular change **ǵiV-* > **ǵV-*.

⁵²² Kortlandt also mentioned **kwikwa-* ‘vivid’ in this context, but this may in fact be a reduplicated adjective **g^{wi}-g^{wH}-o-* that by dissimilation developed into **g^{wi}?uH-ó-* and Go. *qius* (< **g^{wi}uó-* with Dybo’s law).

⁵²³ Compare Seebold (1982: 174-6): PIE *-Rwu-* > **-Rgu-* > PGm. **-Rku-*.

A paradigm **kekō*, **kaweni* would expectably lead to a number of analogies that can explain for most of the root variants attested in the Germanic dialects. The root form **kak-*, for instance, could have arisen as the result of a contamination of **kek-* with **kaw-*. Conversely, the root **keuk-* can be analyzed as the nominative root **kek-* that adopted the labial from the locative **kaweni*. The root of this locative is possibly continued by OHG *chowe*, MHG *kouwe*⁵²⁴, ?EDu. *kauwe*, *kouwe* ‘jaw, throat’ < **kawōn-*.

The most important obstacle at the reconstruction of the paradigm **ġeh₁u-ōn*, **ġh₁u-én-i* is that it requires laryngeal metathesis, the Germanic evidence otherwise pointing to a root **ġuH-* instead of **ġHu-*, cf. MLG *kuse*, EDu. *kuyse* ‘molar’ < **ġuH-s-*. The accent of Lith. *žiáunos* f.pl. ‘jaws’ < **ġ(i)euH-neh₂-* as well as OCS *žьvati* ‘to chew’ < **ġiuH-* and To. (B) *śuwam* ‘to eat’ < **śəwa-* seem to confirm the root **ġiuH-*.

Another problem is that the verb **kewwan-* may have given rise to several derivations at different points in time, such as, for instance, the form **kewōn-*, which must be reconstructed on the basis of e.g. OE *cian*, *ciun* f.pl. ‘gills’, OHG *chiuwa* ‘branchia, edulia, mandibula, palatus’, MHG *kewe*, *ki(u)we*, G *Käu* f. ‘jaw’, MLG *kewe*, *kiwe* f. ‘gill’, MDu. *kieuwe* f. ‘jaw, gill’ and Du. *kieuw* ‘gill’. This productivity blurs the distinction between possible *n*-stem split-offs and secondary deverbative derivations.

Nevertheless, the peculiar form of the well-attested root **kek-* seems to leave room for postulating an originally nominal formation **ġ(i)eh₁u-on-*.

9.3 The **a ~ *u* type

A considerable group of *n*-stems show a potential **a ~ *u* alternation. In the present section, the following cases are discussed: 1) **barsō*, **burznaz* ‘perch’; 2) **brahsmō*, **bruhs(m)az* ‘bream’; 3) **dabō*, **duppaz* ‘puddle’; 4) **galdō*, **gultaz* ‘castrated boar’; 5) **lapō*, **luttaz* ‘shoot, lath’; 6) **mapō*, **muttaz* ‘moth’; 7) **radō*, **ruttaz* ‘rat’; 8) **swambō*, **swumpaz* ‘sponge, mushroom’ and 9) **tadō*, **tuttaz* ‘tuft’. With the full grade being represented by PGm. **a* instead of **e*, the type seems to have formed a category of its own.

⁵²⁴ Lexer 1, 1591.

Regarding the Proto-Indo-European situation, the fact that the *o*-grade forms appear to have been stressed implies a mixed amphikinetic-hysterokinetic inflection **CóC-én*, gen. **CC-n-ós*. The variation of Go. *maþa* ‘worm’ and OE *moþþe* ‘moth’, for instance, points to a PGM. paradigm **maþō*, **muttaz*, which in turn presupposes PIE **mót-én*, **mt-n-ós*. Similarly, The variation of MDu. *baerse* ‘pike’ < **barsan-* and OSw. *agh-borre* ‘pike’ < **burzan-* is best explained as continuing a paradigm **barsō*, **burznaz* < **b^hórs-én*, **b^hrs-n-ós*.⁵²⁵

In the framework developed by Beekes (1985: § 94), the “mixed” type can be given a place by assuming that the original hysterodynamic inflection **CéC-n*, **CC-én-m* developed into **CeC-én*, **CC-én-m* by generalization of the full grade of the suffix, and consequently into **CoC-én*, **CC-én-m* by the change **e > *o* in unstressed position. Additional barytonesis of the *o*-grade of the root must have ultimately yielded a paradigm **CóC-én*, **CC-n-ós*.

With the exception of the PIE word for ‘path’, cf. Skt. *pánthāḥ*, gen. *patháh*, Av. *paṇtā*, gen. *paθō* < **pónt-eh₁*, **pnt-H-ós*, the “mixed type” is not known from any other Indo-European language. In Germanic, it is nevertheless strongly supported by the resolution of the *Schwebeablaut* in the typologically similar cluster of **laþō*, **luttaz* ‘shoot’, **maþō*, **muttaz* ‘maggot, moth’ and **raþō*, **ruttaz* ‘rat’. In all of these *n*-stems, the regular zero grades **ult^taz*, **unt^taz* and **urt^taz* seem to have been replaced by **luttaz*, **muttaz* and **ruttaz* on the basis of the full-grade forms. This could only have happened at a time when the apophony of the paradigm was still intact.

It is important to realize, in this context, that the **u* as a zero-grade marker was not confined to the *n*-stems. It can, for instance, also be found in the word for ‘nose’, an old *s*-stem. On the basis of Skt. *nāsá-* f.du. ‘nose’, Lith. *nósis* f. ‘id.’, Nn. *nos* f. ‘snout’ < **neh₂-s-*, OCS *nosъ* m. ‘nose’ < **nh₂-es-* and ON *nps*, OHG *nasa* f. ‘id’ < **nh₂-s-*, Beekes (1995: 180) reconstructed the original PIE paradigm as **néh-s*, **nh₂-s-ós*, **nh₂-és-m*. In Germanic, this would yield a paradigm **nōz*, **nazaz*, **nasun*, which is able to account for both ON *nps*, OHG *nasa* < **nasō* and Nn. *nos*, but not for OE *nosu*, OFri. *nos(e)*, Du. *neus* ‘nose’ < **nusō-*. Just like the *n*-stems with zero-grade **u* vocalism, this **nusō-* must therefore have received an analogical zero grade (cf. Griepentrog 1995: 330).

⁵²⁵ Schaffner (2001: 341) reconstructs PGM. **burzēⁿ*.

***barsō, *burznaz ‘perch’?**

- **barsa(n)-*: OHG *bars* ‘turonilla, perca’, MHG *bars*, *bers(e)* m. ‘id.’, G *Barsch*, OE *bærs*, *bears* m. ‘id.’, E *bass*, MLG *bars* ‘id.’, MDu. *ba(e)rse* ‘id.’, Du. *baars* ‘id.’
- **burzan-*: Nw. *abbor*, *åbor* m. ‘golden redfish (sebastes norvegicus)’⁵²⁶, OSw. *agh-borre* m. ‘id.’, Sw. *abborre* ‘id.’⁵²⁷, ODa. *ag-borræ* m. ‘id.’, Da. *aborre* ‘id.’

The North and West Germanic words for ‘perch’ clearly are in ablaut relation to each other. Most of the West Germanic material, e.g. OHG *bars*, OE *bears*, points to PGm. **barsa-*, MDu. *baerse* providing some evidence for an *n*-stem **barsan-*. In North Germanic, on the other hand, there only is a zero grade **burzan-*, which occurs in a compound with ON *ǫgr*, Nw. dial. *au(g)ur* ‘golden redfish’, MHG *ag* m. ‘perch’ (< PGm. **agura-*), viz. OSw. *agh-borre*, ODa. *ag-borræ* m. ‘perch’ etc. On the basis of these forms, it is theoretically possible to reconstruct a paradigm **barsō*, **burznaz* < **b^hórs-ēn*, **b^hrs-n-ós*.

There are some reasons to reject the possibility of an ablauting *n*-stem, however. The Nordic compound, for instance, can synchronically be analyzed as from ON *ǫgr* < **agu-* (< **h₂ék-ú-*) and OSw., Nw. *borre*, Da. *borre*, *burre* ‘burdock’ < **burza-*. In view of the dialectal Norwegian meaning of *borre*, i.e. ‘silver brooch’, it is conceivable that the compound arose as a specifically Nordic creation that originally meant “perch-prickle” and refers to the prickly fin on the back of the fish.⁵²⁸

Even if the compound **agu-burzan-* is not such a late North Germanic compound, it still does not have to prove that there was a paradigm **barsō*, **bursznaz*. It is possible, for instance, that the zero grade of **burzan-* was dependent on the accentuation of the compound, e.g. **h₂ék-u-b^hrs-on-*.

It must at the same time be stressed, however, that the formal correlation of **barsan-* and **burzan-* fits perfectly into the ablaut and stress pattern of the “mixed” type that must be assumed for other words.

⁵²⁶ Torp (p. 9) isolates *augur* from the rest of the material: “vistnok avledning av *auga* paa grund av de utstaaende øine”.

⁵²⁷ Hellquist 1.

⁵²⁸ The old age of this meaning is further supported by ON *barr* n. ‘pine needle’ < **barza-*.

***brahsmō, *bruhs(m)naz 'bream'**

- **brahsmān-*, *-ōn-*: OHG *brahsma* f. 'id.', *brahsmo* m. 'id.', MHG *brahsem*, *brasme*, *bresme*, *bresmo* m. 'id.', *prasma*, *bresma* f. 'id.', MLG *brassem*, *brasme*, *bres(s)em*, *bresme* 'id.', MDu. *brasem*, *braessem*, *bressem*, *bresen*, Du. *brasem*⁵²⁹
- **brahsan-*, *-ōn-*: OHG *brahsa*, *brehsa* f. 'id.'⁵³⁰, G *Brachsen* m., *Brachse* f. 'id.'⁵³¹
- **brahsnjō-*: OHG *brahsina*, *prehsin* (= **brāzhsāna*) f. 'id.'
- **brahsmōn-*: ON *brosma* f. 'fish of the cod-kind'⁵³², Nw., Sw. *brosme* f. 'torsk, tusk'

In the West Germanic dialects, the word for 'bream' is represented by a number of different formations, the most wide-spread one being the *mn*-stem **brahsmān-*, *-ōn-*: OHG *brahsma*, *-o*, MHG *brahsem*, MLG *brassem*, (M)Du. *brasem*. This *mn*-stem served as the basis for the **jan-* and **jōn-* stems that are evinced by a number of umlauted forms, e.g. MHG *bresme*, MLG *bresme*, MDu. *bressem* < **brahsmjōn-*. The addition of the suffix fits into a recurrent pattern of fish names ending in **-jan-* and **-jōn-*, e.g. OHG *stur(i)o* m. < **sturjan-*, ON *styrja*, OE *styria* f. < **sturjōn-* and MHG *asche*, *esche*, G *Äsche* f. 'greyling' < **askjōn-*.

A similar feminine formation **brahsnjōn-* is presupposed by OHG *brahsina* and *brehšina*. The alternation of *a* and *e* points to secondary ablaut of **a*, and this umlaut seems to have been indicated by the *i* in the second syllable. Phonetically, this *i* probably represented a shwa that arose through epenthesis. Ultimately, both **brahsmjōn-* and **brahsnjōn-* may go back to a form **brahsmnjōn-*.

A different formation is found in North Germanic, viz. ON *brosma* f. (etc.) < **bruhs-mōn-*. It represents what looks like the zero grade of the WGM. **brahsmōn-*.⁵³³ The most straightforward way to account for this alternation is to reconstruct a PGm. paradigm **brahsmō*, **bruhs(m)naz*, **bruhsmeni*, which in view of the reversed zero grade probably replaced older **brahsmō*, **burhs(m)naz*, **burhsmeni*. Given the **a : *u* ablaut, one

⁵²⁹ Franck/Van Wijk 90.

⁵³⁰ EWA 280-2.

⁵³¹ Kluge/Seebold 144.

⁵³² De Vries 1962: 59.

⁵³³ Cf. Torp (p. 43): "*brosma* kunde være avlydende til *b r a s m e*."

way of projecting the Gm. evidence back into the IE proto-language would be to reconstruct a “mixed” paradigm **b^hróks-mēn*, **b^hrks-(m)n-ós*, **b^hrks-mén-i*. In the genitive, the *m* may have been lost at an early stage due to dissimilation against the initial labial (cf. **budmēn*, **buttaz* < **b^hud^h-mén*, **b^hud^h-(m)n-ós*, see p. 65).

There are no indications whatsoever that the word was adopted from a European substrate language, as has been claimed.⁵³⁴

***dabō, *duppaz ‘puddle’?**

- **daban-*: Nw. dial. *dave* m. ‘draw-well’
- **dabban-*: Nw. dial. *dabbe* m. ‘id.’, Du. dial. *dabbe* ‘mud, hare’s den’⁵³⁵
- **dapan-*: ON *dapi* m. ‘pool, puddle’, Nw. *dape* m. ‘pond, draw-well’
 - ⇒ **dapila-*: ON *leir-depill* ‘loam-pit’, Icel. *depill* m. ‘dot, spot, puddle in a wetland’, Nn. *depel* ‘puddle’
 - ⇒ **dapja-*: Nw. dial. *dep* n. ‘waste pit’
- **duban-*: Nw. dial. *dove* m. ‘muddy spot, quagmire’
- **dubbōn-*: MLG *dobbe* f. ‘pool’⁵³⁶, Du. dial. *dobbe* ‘puddle, hole, pit’⁵³⁷
- **duppa(n)-*: Du. dial. *dop* ‘hare’s den’⁵³⁸
- **dupan-*: Nw. dial. *dope* m. ‘puddle’
 - ⇒ **dupla-*: Nw. dial. *dopel* m. ‘puddle’

A potential apophonic *n*-stem with a strong representation in North Germanic is represented by Nw. *dave*, *dabbe*, *dape* and the ablauting forms *dove* and *dope*. In itself, the forms *dave*, *dabbe* and *dape* already form a strong case of consonant gradation: the variation may point to a paradigm **dabō*, **dappaz* that was split up into 1) **dabō*, **dabbaz* and 2) **dapō*, **dappaz*. With the ablauting forms *dove* and *dope*, the paradigm can at least theoretically be reconstructed as **dabō*, **duppaz*.

⁵³⁴ Boutkan (1999) assumed a substrate origin because “it is unlikely that three [sic] ablaut grades would have survived in a single Gmc. fishword.”

⁵³⁵ WZD I, 153; WBD III, 4.2, 62.

⁵³⁶ Schiller/Lübben 527.

⁵³⁷ Kocks/Vording 205.

⁵³⁸ WBD III, 4.2, 62.

It is interesting to see that in Nordic, the different roots variants have given rise to different derivations. Nw. *depel* (= ON *depill*), *dypel* and *dopel*, all meaning 'puddle', represent the diminutive formations **dapila-*, **dupila-* and **dupla-*, which were derived from two different roots. Nw. dial. *dep* can be reconstructed as **dapja-*. The etymologically obscure ON *dof* f. 'rump', Icel. *döf* f. 'loin' < **dabō-* can be connected to Nn. *dov* f. 'crotch, rump, waving ground on soft mud', assuming that the meaning 'loin' developed out of 'soft spot'. Nw. dial. *dembel* m. 'puddle' does not belong here, but is derived from *dam* 'dam, pool', viz. **dammila-*.

In West Germanic, the same consonant and vowel alternations re-emerge in the Low German area. Especially the Dutch dialects provide some important reflexes, i.e. *dabbe* 'mud, hare's den', *dobbe* 'puddle, hole', *dop* 'hare's den'. These forms, too, may point to a paradigm **dabō*, **duppaz*, thus giving the paradigm a Proto-Northwest Germanic horizon.

The reality of the ablauting paradigm becomes more doubtful, however, when the etymological context is taken into account. It seems clear that the nouns are related to the iterative verb **dabbōn-* as in Nw. *dabbe* 'to hit (with the feet)', Sw. dial. *dabba* 'to soil', G *tappen* 'to hit', MDu. *dabben* 'to toddle', EDu. *dabben* 'subigere, suffodere', E *dab* 'to strike, peck, obs. fish by dipping the bait in the water' and **dabblōn-*: ON *dafla*, EDu. *dabbelen* (= EDu. *dabben*), E *dabble* 'to splash' (see also **dēbō*, **dappaz* 'paw', p. 327). Next to this **dabbōn-*, a related zero-grade iterative must have existed in view of MDu. *dubben* 'to submerge; to dig out', Nw. dial. *duppa* 'to push under water' < **duppōpi*, **dubunanpi*. It is therefore possible that the ablaut of the iteratives was exported to the nominal level.

****galdō*, **gult'az* 'gelding'**

- **galt'an-*: ON, Far. *galti* m. 'boar', Nw. *galte* '(castrated) boar'
- **galt'u-*: ON *goltr*, Icel. *göltur* m. 'boar', Far. *gøltur* 'id.', Nw. *galt* m. '(castrated) boar', Da. *galt*⁵³⁹, OE *gealt-bearg*, -*borg* m. 'pig'
- **galt'ōn-*: OHG *galza* f. 'young sow', MHG *galze* f. 'castrated sow', G *Galz(e)* f. 'id.', Bav. *galz*⁵⁴⁰ 'id.', Swi. *galz* f. 'id.'⁵⁴¹

⁵³⁹ Falk/Torp 298.

⁵⁴⁰ Schmeller 2, 46.

⁵⁴¹ Stalder 1, 418.

- **galt^tjō-*: OHG *gelza*, MHG *gelze*, G *Gelze* f. 'gilt, castrated sow'⁵⁴², MLG *gelte* f. 'castrated sow', MDu. *gelte* f. 'id.'
- **gult^ti-*: ON *gyltr* m. 'pig', Nw. dial. *gylt* m. 'id.'
- **gult^tjō(n)-*: ON *gyltr* f. 'sow', ON, Icel. *gylta* f. 'id.'⁵⁴³ (= OE *gilte* f. 'young sow', E *gilt*)

The *n*-stem **galdō*, **gult^taz* is clearly derived from the root **gald-* as found in ON *gelda* 'to castrate' < **galdjan-*, ON *geldr* 'milkless', OE *gieldē* 'infertile' < **galdja-*, OSw. *galder*, OHG *galt*, G Crn. *galt*⁵⁴⁴ 'not giving milk' < **galda-*. The semantic gap between ON *galti* 'boar' and *gelda* 'to castrate' is regarded as problematic by Kluge/Seebold (l.c.), but Nw. *galt(e)* 'castrated boar' seems to preserve the semantic link between the two meanings.

In view of the link with **galdjan-*, the consonantism of *galti* and parallel forms must be explained from a shortened geminate (**galt^tan-*). This geminate mechanically follows from the attested *n*-stem inflection. Apparently, there was a paradigm **galdō*, **galt^taz*, in which the geminate became generalized at an early stage. Geminatization most probably also took place in the gpl. **galt^tan* < **g^hold^h-n-óm* and the apl. **galt^tuns* < **g^hold^h-n-ŋs*. The Old Norse formation *goltr* < **galt^tu-* appears to have directly sprouted from the latter case.⁵⁴⁵

A very old formation **gult^tjō-* can be established on the basis of ON *gylta*, *gyltr* 'sow'. It contains the feminizing suffix **-īz*, gen. **-jōz* from PIE **-ih₂-s* < **-iéh₂-s*, which is also found in e.g. ON *ylgr*, OSw. *ylva* 'she-wolf' < **ulkw^w-ih₂-s*, gen. **ulkw^w-iéh₂-s*. In the case of *gylta*, the suffix must have been added to a zero-grade root with gemination. This clearly points to a genitive **gult^taz* < **g^hld^h-n-ós*.

A parallel derivational history must be supposed for G *Ricke* 'doe'⁵⁴⁶, which through **rikkī-* stems from **Hrik^k-n-ih₂-*. In addition, this formation, too, was derived from an *o*-grade *n*-stem, viz. OE *rāh(a)*, OHG *rēh(o)* m. 'deer' < **raiha(n)-*.

⁵⁴² Kluge/Seebold 343.

⁵⁴³ Falk/Torp 298: < **g^hldī-*.

⁵⁴⁴ Lexer 1862: 108.

⁵⁴⁵ The link with Skt. *hṛdu-* m. 'ram' (Fick/Falk/Torp 131) must at any rate be rejected.

⁵⁴⁶ Grimm 14, 908-9.

Alternatively, one could isolate **g^hld^h-n-ih₂-* and **Hrik^h-n-ih₂-* from the masculine *n*-stems, and assume that their zero grades were triggered by the **nī*-suffix. This solution must be rejected, however, in view of forms such as ON *birna* f. 'she-bear' < **bernjōn-* and Nw. dial. *yrkne*, Sw. dial. *ynn* 'she-ptarmigan'⁵⁴⁷ < ON **yrna* < **urznī-*, which were created to the original *n*-stems ON *beri* < **beran-* and *orri* < **urzan-*. Apparently, the feminizing suffix did not require a particular ablaut grade; it was simply added to the root that was found in the masculine form. No doubt, the same procedure was followed when *gyltr* was created to **galdō*, **gult^haz*.

OHG *galza* < **galt^hōn-* an OHG *gelza*, MDu. *gelte* < **galtjō-* are more recent, purely West Germanic formations. In the latter case, the **jō*-suffix was again used to coin a feminine formation, this time to the full-grade stem **galt^h-*. Once more, there is a striking parallelism with **raihan-* 'deer', which in addition to **rikkī-* has a West Germanic **jō*-stem based on the *o*-grade root: OHG *rēia*, OE *ræge* < **raihjō(n)-*.⁵⁴⁸

Finally, Franconian German has a form *gelte* f. 'infertile cow'⁵⁴⁹ < **galdjō-*, which looks like an even younger derivation, probably from the adjective *gelt* 'passed the fertile age (of a cow)'.

****lapō*, **luttaz* 'shoot'**

- **lapān-*, *-ōn-*: **ladōn-*: OHG *lado*, *lada* 'asser', MHG *lade* f. 'shoot, plank, stand, store', G *Laden* m. 'board, hatch, store', MLG *lade* f. '(off)shoot'⁵⁵⁰, MDu. *lade* f. 'runner, twig, lath, bar'⁵⁵¹, EDu. *laede* 'board, bar'
- **lappan-*, *ōn-*: OHG *laddo*, *lat(t)o* 'asser', *ladda*, *latta* 'tignum', MHG *lat(t)e* f. 'lath', *sumer-lat(t)e* f. 'one-year-old shoot'⁵⁵², G *Latte* f. 'lath, sprout'⁵⁵³, *Sommer-latte* f. 'one-year-old

⁵⁴⁷ Rietz 293; Hellquist 533.

⁵⁴⁸ Not **raigjōn-* (pace Fick/Falk/Torp 332; Pokorny 859) with Verner's law, because then the loss of the **g* in OHG *rēia* remains unexplained. The *g* in *ræge* represents a glide like in OE *blæge*, akin to MLG, MDu. *bleie* 'gudgeon' from **blai(h)jōn-*, not **blaijōn-* (Fick/Falk/Torp 287).

⁵⁴⁹ Brückner 1996: 71.

⁵⁵⁰ Lübben 195.

⁵⁵¹ Verdam 318.

⁵⁵² Lexer 1, 1839.

⁵⁵³ Grimm 12, 279-80.

- shoot⁵⁵⁴, MDu. *latte* f. 'lath'⁵⁵⁵, EDu. *latte* 'small bar', Du. *lat* 'id.'⁵⁵⁶, ME *latthe* 'id.', E *lath*
- **ladōn*:- OHG *lata* 'asser', *sumar-lata* f. 'summer shoot', OS *sumar-lada* 'id.'
 - **latta*:- G dial. *latz* m. 'plank'⁵⁵⁷
 - **lattō(n)*:- OHG *latza* 'tignum, tigillum', G dial. *latz(e)* f. 'plank, twig'⁵⁵⁸, OE *læt* f. 'lath'⁵⁵⁹, E dial. *lat* 'lath'⁵⁶⁰
 - **latōn*:- MLG *late* f. 'shoot'⁵⁶¹, WFri. *leat* '(off)shoot, blade (of grass)'⁵⁶²
 - **ludōn*:- OHG *sumar-lota* f. 'summer shoot', G *Lote*⁵⁶³, OS *sumer-loda* 'virgultum, palmes'⁵⁶⁴, MLG *lode* 'shoot, twig'⁵⁶⁵, MDu. (*somer*-)*lode* f. 'runner'⁵⁶⁶
 - **lutta*:- Du. poet. *duimelot* 'thumb', *langelot* (= WFri. *lange leat*) 'middle finger'
 - **lutōn*:- MDu. *lote* f. 'twig, sprout'⁵⁶⁷, EDu. *loote* 'twig', Du. *loot* 'shoot', WFri. *loat* '(off)shoot'⁵⁶⁸
 - ?**luddan*:- WFri. *lod(de)* 'spade'⁵⁶⁹

The word for 'shoot' displays an impressive amount of formal variation with possibly as many as nine different root variants. All the different variants can be unified into one single Proto-Germanic paradigm **laþō*, **luttaz* with consonant and vowel gradation.

⁵⁵⁴ Grimm 16, 1540-1.

⁵⁵⁵ Verdam 324.

⁵⁵⁶ Franck/Van Wijk 371.

⁵⁵⁷ Grimm 12, 284.

⁵⁵⁸ Venema 1997: 320.

⁵⁵⁹ Holthausen 1934: 193.

⁵⁶⁰ Wright 1869: 625.

⁵⁶¹ Lübben 199.

⁵⁶² Zantema 561.

⁵⁶³ Kluge/Seebold 579, 583.

⁵⁶⁴ Gallée 1903: 311.

⁵⁶⁵ Lübben 209.

⁵⁶⁶ Verdam 336.

⁵⁶⁷ Verdam 338.

⁵⁶⁸ Zantema 582.

⁵⁶⁹ Buitenrust Hettema 1891: 244; Zantema 583.

Many of the difficulties surrounding the etymon were discussed by Lühr (1988: 251-2), who focussed on the cross-dialectal consonant alternations. Lühr reconstructed three different root variants, viz. **lap-*, **lapp-* and **latt-*. The first root is supported by OHG *lado*, MHG *lade* ‘shoot, plank’ and similar forms in the Low German area. The stem **lappōn-* is found throughout the West Germanic continuum, e.g. OHG *ladda*, *latta* and ME *latthe*, E *lath*. The variant **latt-* is evinced by OE *læt*, E dial. *lat*, OHG *latza* and Rhinelandish *latz(e)* ‘lath, twig’, which can be found as far North as Dutch Limburg (WLD II/12, 9).⁵⁷⁰ Then, there are the additional variants **lad-*, **lat-* as implied by OHG *lata*, MLG *late* and WFri. *leat*.

The variants **lap-* and **latt-* were derived by Lühr from a paradigm **lapō*, **lattaz*. The remaining roots, **lapp-*, **lad-* and **lat-* can be explained by assuming that this primary paradigm was split up into 1) **lapō*, **lappaz*, 2) **latō*, **lattaz* and 3) **ladō*, **laddaz* (see § 4.3.1).

It is not necessary to derive the long fricative of **lapp-* from a cluster **-hþ-*, as has been suggested by Lühr (1988: 525). On the basis of the alternation of OE *mopþa* with Nrth. *mohþa*, she argued that many cases of West Germanic **-þþ-* in reality continue older **-hþ-*, assuming that “die Assimilation von **χþ > *þþ* erst einzelsprachig eingetreten ist”. It is more likely, however, that *mohþa* developed out of **mopka* (see p. 218ff).

A number of additional roots can be added to the corpus. Many of these root variants not only display the expected consonant alternations, but also a vowel alternation **a ~ *u*. The interchanges are particularly clear in the West Germanic compound meaning ‘summer shoot’, i.e. a one-year-old twig: OHG *sumar-lata*, *-lota*, MHG *sumer-late*, *-latte*, G *Sommer-latte*, *-lote*, *-lotte*, OS *sumar-lada*, *-loda*, MLG *som(m)er-lade*, *-late*, MDu. *somer-lade*, *-lode*, Du. spec. *zomer-lat* ‘lath for mending the floor of a boat’, *zomer-lot* ‘vertical tree-shoot’. This variation unambiguously points to an ablauting paradigm **lapō*, **luttaz*.

The ablaut seems to have been leveled in different ways in the separate dialects. MLG *late*, MDu. *lote* and Du. *loot* have single **t*. This clearly points to a secondary paradigm **latō*, **luttaz* that in turn was

⁵⁷⁰ Given all the evidence, it is unlikely that the affricate of *latz(e)* is due to a *pseudo-Verschiebung*, i.e. a hypercorrect High Germanization of *Latte*, as has been proposed by Goossens (1968).

split up into 1) **latō*, **lattaz* and 2) **lutō*, **luttaz*. It is by no means necessary to assume that the *t* of Du. *loot* results from **lood* by *Auslautsverhärtung*, as was claimed by Franck and Van Wijk, a phenomenon that occasionally occurs in Dutch *singularia tantum*, cf. Du. *riet* < **hreuda-* and *schoot* ‘lap’ < **skauda-*.

The zero-grade root **lutt-* or **lupp-* is preserved in Du. *zomer-lot* ‘summer shoot’ in fruit pruning jargon. It further occurs in two compounds existing in a children’s song about the five fingers, in which the thumb is featured as *duimelot*, the middle finger as *langelot*.⁵⁷¹ It is interesting to see, in this context, that the meaning ‘finger’ is also attested for the form WFri. *leat*. This word cannot possibly be a zero grade, nor does it continue **laut-*, as Franck and Van Wijk assumed on the basis of Du. *loot*. In view of e.g. *leane* ‘lane’ < PGm. **lānō-*, it must rather contain a full-grade root **lat-*.

Etymologically, the full grade and the zero grade root have always been separated from each other, and it is a common place in the literature to derive G *Lote* from PGm. **leudan-*, Go. *liudan* ‘to grow’ < PIE **Hleud^h-*.⁵⁷² Seebold adopted this separation, and concluded that both variants merely influenced each other in such forms as *Sommer-lot(t)e* and *Sommer-lat(t)e*. Grimm simply calls *Latte* a “Verstümmelung von ursprünglichem *Lote*”. None of these approaches succeed at explaining the formal variation of the word, however.

The Germanic word is close to OIr. *slat* ‘rod, lath, twig’ and W *llath*, *ystlath* ‘rod’, which continue PCelt. **slattā-*; PCelt. *sl-* remained in Old Irish, but became W *ll-* in lenited position (Schrijver 1995: 431-3). It may be worthwhile considering a further connection with Lith. *lazdà*, dial. *lazà* ‘stick’, Latv. *lazda* ‘hazel’ and Sl. **loza* ‘vine’, to which Fraenkel (p. 827) added Lith. *slastaĩ*, Latv. *slasts*, *slazds* ‘animal trap’.⁵⁷³ Since **zd*

⁵⁷¹ In view of *langelot*, which happens to be completely parallel to WFri. *lange leat* ‘middle finger’ < **latōn-*, it is unlikely that *duimelot* is derived from *duim* ‘thumb’ with a French diminutive suffix *-lot*, as the 1915 article of the WNT claims. The suggestion by Boekenogen (1949), that *lot* is from the obsolete verb *lotten* ‘to suck’, does not explain *lange leat* either. It is therefore more probable that the further unattested simplex *lot* means ‘finger’, a metaphorical use of the original meaning ‘lath’ or ‘shoot’.

⁵⁷² Kluge/Seebold 579; Franck/Van Wijk 398-9.

⁵⁷³ For a semantic parallel, cf. OHG *dona* f. ‘twig’ and G *Dohne* ‘animal trap’ < **tnh₂-eh₂-*.

regularly becomes *th* in Welsh and voiceless *t* in Irish⁵⁷⁴, a European root **slazd*^(h)- could indeed account for the Celtic forms.⁵⁷⁵ E *slat* is either adopted from Old Irish or – as is argued by the OED – from OFr. *ésclat*, Fr. *éclat*. The French word, in turn, may be a loanword from Gaulish.⁵⁷⁶

It is further possible that MHG *slāte* f. ‘reed’ and MHG *slāt*, *slōt*, G *Schlot* ‘chimney’⁵⁷⁷ somehow belong here, too⁵⁷⁸, especially in view of the gloss *slat* ‘*novellum*’.⁵⁷⁹ The forms can theoretically be derived from **slazd*^h- by assuming that the *z was lost with compensatory lengthening of the preceding vowel, cf. MHG *miete*, G *Miete* ‘rent’, Go. *mizdo* f. ‘payment’ < PGm. **mizdōn*- < PIE *mizd*^h*eh*₂-.

**maþō*, **muttaz* ‘moth’

- **maþan*-, *-ōn*:- Go. *maþa* m. ‘worm’, OE *maða* m., -u f. ‘grub, worm, maggot’⁵⁸⁰, OS *matho* ‘*vermis*, *teredo*’, OHG *mado* m. ‘id.’, Du. *made* ‘maggot’
- **maþþōn*:- MHG *matte* f. ‘moth’⁵⁸¹, MDu. *matte* f. ‘id.’
 ⇒ **maþ(i)ka(n)*- (= Fi. *matikka* ‘worm’): ON *maðkr* m. ‘maggot’ (= E *mawk*⁵⁸²), ME *maðek*, E *maddock*, *maggot* (with metathesis)⁵⁸³, MLG *maddike*, *med(d)ek(e)* ‘earthworm’
- **muþþōn*:- OE *moppe* f. ‘moth’⁵⁸⁴, E *moth*, MLG *mutte* f. ‘id.’, MDu. *mot(te)* ‘id.’, Du. *mot* ‘id.’, MHG *motte*, *mutte* f. ‘id.’⁵⁸⁵

⁵⁷⁴ Cf. PIE **nizdos* ‘nest’ > W *nyth*, OIr. *net*.

⁵⁷⁵ The reconstruction **slatnā*- (Fick/Falk/Torp 359; Kluge/Mitzka 425; Kluge/Seebold 559), which is based on Whitley Stokes’s (1893) idea that Celtic had a Kluge’s law of its own, must at any rate be rejected.

⁵⁷⁶ It. *latta*, Fr. *latte* and Ru. *lotók* ‘groove’ are probably Germanic loanwords, cf. Franck/Van Wijk 371; Kluge/Seebold 425.

⁵⁷⁷ Kluge/Seebold 811.

⁵⁷⁸ Lühr 1985: 311; 1988: 252.

⁵⁷⁹ Grimm 15, 501.

⁵⁸⁰ Bosworth/Toller 671.

⁵⁸¹ Lexer 1, 2062.

⁵⁸² OED, s.v. *mawk*; Holthausen 1917: 101.

⁵⁸³ Lith. *mākatas* ‘midge’ may a loanword from a similarly (Low German?) metathesized form, quasi PGm. **makaþan*- < **maþakan*- (= ME *maðek*).

⁵⁸⁴ Bosworth/Toller 699.

⁵⁸⁵ Lexer l.c.

⇒ **muþkōn*:- OE *mohþe* f. 'id.', ME *mohthe* 'id.',
Scot. *mogthe* 'id.'

- **mutta(n)*:- ON *motti* m. 'moth', Nw. *mott* m. 'id.'

On the basis of the material, several different stem variants must be reconstructed for the Germanic word for 'maggot, moth'. First, Go. *maþa*, OE *maða* and OHG *mado* continue a stem **maþan*-. In MHG and MDu. *matte*, the same root reappears with a geminate **-þþ*-. The forms OE *mohþe*, MHG *motte*, MDu. *motte* also have a geminate, but a different root vowel, viz. **u*. The same vocalism occurs in ON *motti* and Nw. *mott*, but here the geminate seems to have been plosive, i.e. **-tt*-.

The different roots are closely related to each other. It has long been suspected that the forms with **u* form the "Schwundstufenbildung zur Vollstufenform MHG *matte*", as stated by Streitberg (1896: 68).⁵⁸⁶ This ablaut, as well as the apparent consonant alternations, are best understood from an apophonic *n*-stem **maþō*, **muttaz*, which was remodeled into **maþō*, **muþþaz* in Proto-West Germanic or Proto-Northwest Germanic. The variant **maþþōn*-, as evinced by MHG *matte*, points to a further leveling of the paradigm into **maþō*, **maþþaz*; apparently, the original zero grade was removed from this paradigm.

An important aspect of the ablauting paradigm is that it must to a certain extent have been remodeled: the underlying "mixed" paradigm **mót-ēn*, **mt-n-ós* would give PGm. **maþō*, ***untʰaz* rather than **maþō*, **muttaz*. This difficulty can be resolved by assuming that the *Schwebeablaut* of *maþō*, **untʰaz* was leveled. The anciennity of the *a*-vocalism is at any rate confirmed by a number of Slavic cognates, e.g. Ru. *motýl'* 'maggot'⁵⁸⁷, and also by Nw. dial. *mår(e)* m. 'woodworm' < **maþra(n)*-, *mære* m. 'mite' < **maþrjan*- (with **-aþr-* > **-ār-* as in ON *hvárr* 'which of the two' < PGm. **hwaþeraz* < PIE **kʷoteros*).

The origin of the medial cluster of Northumbrian *mohþe*, ME *muhthe*, Scot. *mogthe* (ostensibly from PGm. **muþþan*-) is debated. Kluge and Mitzka (1967: 490) doubted that *mohþe* could be related to *moððe* at all, and connected it to **mugjō*- 'mosquito'. Lühr, on the other hand,

⁵⁸⁶ Cf. Noreen 1894: 223; Kluge/Mitzka 489-90.

⁵⁸⁷ Slov. *metúlĵ* 'butterfly' and SCr. *mètĵlĵ* 'intestinal worm' were probably borrowed from MHG *medel* n. 'vermiculus' (Benecke/Müller/Zarncke 2, 18) < **maþlīn*-, or perhaps from its Old High German precursor **mātheli*/*mādeli*.

retained the link with **maþan-*, assuming that *mohþe* developed out of a diminutive **muþ-han-* < **mut-ko-* by metathesis. Such a metathesis is conceivable, but it is probably better to reconstruct the original form as **muþ-(V)kan-* with a suffix **(V)ka(n)-* in view of ON *maðkr*, MLG *maddike*, *med(d)ek(e)* and ME *maðek* < **maþ(V)ka(n)-*. Most probably, **muþkan-* became **mukþan-* by metathesis. In the resulting cluster, the *k* was subsequently fricativized under the influence of the *þ*. This development is paralleled by the vacillation of OE *bīecþ* vs *bīehþ* ‘beacon’ < **baukiþō-*.

The developments assumed here are confirmed by the remarkably parallel evolution of PGm. **piþ(V)ka(n)-* ‘pith’ (cf. MLG, MDu. *ped(d)ik*) in Anglo-Frisian. In Scottish, this formation developed into *picht* ‘pith, force’⁵⁸⁸, a form that presupposes a metathesized Nrth. form **pihþa*. In addition, there is the polymorphism of WFri. *pich*, *piid*, *piik* ‘pith, stone’, which has gone unnoticed in the literature. The form *piid* appears to be identical to OE *piða* m. ‘pith’ < **piþan-*, but *pich* and *piik* seem to have bifurcated from a diminutive **piþ(V)ka(n)-*. The bifurcation happened as follows: while *piik* continues regular **piþVk-* through loss of the dental between vowels, *pich* can only have developed out of **pihþ-* from **piþk-* by a metathesis. This *pich*, in other words, is fully parallel to Scot. *picht*.

The difference between MLG, MDu. *medik*, *pedik*, WFri. *piik*, on the one hand, and MLG *maddik*, *meddik*, *peddik*, WFri. *pich*, on the other, is probably to be explained from paradigms in which some cases were affected by syncope, while others were not, e.g. **piþikaz*, gen. **piþikesa* > **piþik*, **piþkes*. This syncope also explains the lack of umlaut in MLG *maddik*, which with its double *-dd-* must have developed out of a syncopated root **maþk-*. Nrth. *mohþe* developed out of a similar syncopated form **muþk-*. ME *maðek*, on the other hand, seems to continue *maþak-*, or perhaps **maþik-* with analogical removal of the umlaut after the syncopated cases.⁵⁸⁹

The conclusion that OE *mohþe* developed out of a metathesized form begs the question whether the geminates OE *moþþe* as well as MHG *mutte*, ON *motti* (etc.) developed out of the same cluster, as has been proposed by Lühr (l.c.). Although this solution does not seem unlikely in

⁵⁸⁸ Jamieson 1818, s.v. *picht*.

⁵⁸⁹ Note that E *maggot* developed out of *maddock* (< **madaka-*?) by a strange swap of the articulation place of *d* and *k*.

the case of OE *moppe*, it can probably not be maintained. First of all, the metathesis is a purely Anglo-Frisian development: there are no indications whatsoever that the syncopated variants **mapk-* and **pīpk-* ever metathesized to ***makp-* and ***pīkp-* in the Franconian and Saxon dialects. Second, the mechanism of consonant gradation removes the necessity to explain fricative geminates from clusters. As a matter of fact, consonant gradation seems to be the only way to clarify the long fricatives of e.g. **klippōn-* ‘burdock’ (see p. 235) and **rappōn-* ‘rat’ (see p. 221). In the end, the conclusion must therefore be that the allomorph **mupb-* resulted from paradigmatic analogy, and not from the assimilation of **-hp-* to **-pp-*.

Etymologically, the only plausible extra-Germanic cognates are the Slavic words given above. Other connections must be rejected. Falk and Torp (p. 700-1), for instance, separate **mup-* from **map-*, linking the former to Lat. *mutilus* ‘mutilated’ and the latter to Lat. *mateola* ‘club’. The connection with Skt. *matkuṇa-* ‘bug’ (Falk/Torp l.c.) is semantically more appropriate, but the strange morphology of the Sanskrit word and the parallel form *utkuṇa-* ‘louse’ (suffix ***kuṇa-*?) conspicuously point to a non-Indo-European origin. It has further been suggested by Kallio (2000) that PGm. **maþan-* was adopted from Finnic **mato* ‘worm, maggot’, while **mupþan-* was borrowed from Saamic **muočē*. The vowel and consonant alternations of the Germanic *n*-stem are too regular, however, to be caused by language contact. Finally, the link with Arm. *mat’il* ‘louse’ was already doubted by Polomé (1986), who pointed at Kartvelian **ma-ṭl-* ‘worm’ as a possible source.⁵⁹⁰

****rapō, *ruttaz* ‘rat’**

- **radan-*, *-ōn-*: OHG *rato* ‘sorex’, MHG *rat(e)* mf. ‘rat’⁵⁹¹
- **rappōn-*: OHG *radda*, *ratta* ‘glis’, MHG *radde*, *ratte* f. ‘rat’⁵⁹², G *Ratte*⁵⁹³

⁵⁹⁰ In this language, the word is analyzable as a derivation of the root **ṭl-* ‘to eat up’ (Klimov 190).

⁵⁹¹ Lexer 2, 346.

⁵⁹² Ibid.; Benecke 2, 584.

⁵⁹³ Grimm 14, 204-5; Kluge/Seebold 745.

- **ratta(n)-, -ōn-*: OHG *ratza* ‘*glis*’, MHG *ratz(e)* m. ‘*rat*’⁵⁹⁴, G *Ratz* m. ‘*id.*’, Bav. *ratze* f. ‘*rat*, *polecat*’⁵⁹⁵, OS *ratta* ‘*glis*’⁵⁹⁶, MLG, MDu. *ratte* f. ‘*rat*’⁵⁹⁷, Du. *rat*⁵⁹⁸, OE *ræt* m. ‘*id.*’, E *rat*
- **ruttōn-*: MLG *rotte* f. ‘*id.*’⁵⁹⁹ (= Icel. *rotta*, Sw. *råtta*, Nw., Da. *rotte*⁶⁰⁰), MDu. *rot(te)* f. ‘*id.*’⁶⁰¹, Du. *land-rot* ‘*landlubber*’

A paradigm **raþō*, **ruttaz* is implied by material from the North and West Germanic dialects. The consonant variation is particularly rich in the High German dialects, cf. MHG *rate*, *radde*, *ratte*, *ratze*. The Low German dialects, on the other hand, have preserved the ablauting variant *rotte*.

The full grade **a* is found in three different root variants. A root **rad-* is implied by OHG *rato* and MHG *rate*. The variant **rattōn-* is more frequent, and can be reconstructed on the basis of OHG *ratza*, MHG *ratze*, MLG, MDu. *ratte*. OE *ræt* contains the same root, but represents a thematic formation **ratta-*. Yet another variant is evidenced by OHG *radda*, *ratta*, MHG *radde*, *ratte* and G *Ratte*. Lühr (1988: 284) reconstructed the underlying root as **radd-*, but **raþþ-* seems more appropriate in view of the fact that, while WGM. **dd* changed into OHG *tt* from the earliest sources, the development of WGM. **þþ* into *dd* and *tt* occurred within the historic period.⁶⁰²

The consonant variation has led to a great deal of confusion in the literature. It is often assumed that the word for ‘*rat*’ was adopted from Romance **rattu-* (It. *ratto*, Sp. *rato*, Fr. *rat*), which is sometimes taken to be from Lat. *rapidus* ‘tearing away’ (Brøndal 1917: 117-9). This solution does not explain, however, “warum neben *Ratte* auch *Ratze* auftaucht” (Kluge/Seebold 745). This particular problem was addressed by Uhlenbeck (1897a: 196), who tried to explain OHG *radda*, *ratta* as Low

⁵⁹⁴ Lexer 2, 353.

⁵⁹⁵ Grimm 14, 209-10; Kluge/Seebold 746.

⁵⁹⁶ Gallée 247.

⁵⁹⁷ Lübben 293; Verdam 486.

⁵⁹⁸ Franck/Van Wijk 536.

⁵⁹⁹ Lübben 308.

⁶⁰⁰ Falk/Torp 913.

⁶⁰¹ Verdam 486, 501.

⁶⁰² Braune § 164, § 167, fn. 10.

German loanwords, labeling *Ratze* as the regular High German form. Still, there seems to be no reason to regard the dialectal consonant variation as problematic. It is, in fact, typical of the inflection of the *n*-stems (cf. Franck/Van Wijk 536; Falk/Torp 913). All the different variants can be explained from a paradigm **rapō*, **rattaz*, **radeni* that was split up according to the *Associationen* that were proposed by Kluge. It follows that the Germanic word must rather have been adopted by the Romance languages as well as by Celtic, cf. Ir. *rata*, Bret. *raz* < **ratt-* (Lühr 1988: 285), rather than the other way around.

In addition to the forms with **a*-vocalism, an ablauting stem **ruttōn-* is furnished by MLG, MDu. *rotte*, and this is the form that was borrowed into the Nordic languages. Modern Dutch has more or less retained the doublet: the full grade *rat* is the default word for 'rat', but the zero grade is still in use in the compound *landrot* 'landlubber'.

The zero grade can be reconciled with the other forms by reconstructing an ablauting paradigm **rapō*, **ruttaz*. Diachronically, it seems to continue a paradigm **Hrót-ēn*, **Hrt-n-ós*, which seems to be a mixture of the amphikinetic and hysterokinetic type. The original genitive of this paradigm must have been subjected to some remodeling, as its expected outcome in Proto-Germanic would have been **urt^taz*. Apparently, the *Schwebeablaut* was removed on the basis of the nominative.

Etymologically, the old link with Skt. *rāditi* 'to scratch, gnaw'⁶⁰³ must be abolished, because it suggests PIE **Hrod-*, whereas Germanic points to **Hrot-*. In view of G *Ratz(e)* 'polecat', it is plausible that the Germanic word originally denoted a different animal, and that it "auf die später auftretende ratte übertragen worden ist" (Falk/Torp l.c.).

****swambō*, **sump^paz* 'sponge, mushroom'?**

- **swamba-*: OHG *swamp* m. 'mushroom'
- **swamma(n)-*: Go. *swamm* asg. 'sponge', OE *swom* m. 'mushroom', OHG *swam* m. 'sponge', MHG *swamme* m. 'mushroom, sponge', G *Schwamm* m. 'sponge, dry rot'⁶⁰⁴, EDu. *swamme* 'spongia, tuber, panus', Du. *zwam* 'mushroom'

⁶⁰³ Cf. Uhlenbeck 1901: 306; Pokorny 845.

⁶⁰⁴ Kluge/Seebold 830.

- *swamp^{pu}:- ON *soppr* m. 'sponge, ball', Icel. *sveppur*, gsg. *svepps*, †*svappar*, npl. *sveppar*, -ir m. 'mushroom, fungus'⁶⁰⁵, OSw. *swamper* m. 'mushroom, sponge', Sw., Da. *svamp* 'mushroom'⁶⁰⁶
- *s(w)ump^a-⁶⁰⁷: ON *soppr* m. 'ball'⁶⁰⁸, Icel. *soppur* m. 'ball, float of a net' (also *soppa* f., *soppi* m. 'float')⁶⁰⁹, Far. *soppur* m. 'tuft, fungus, mushroom'⁶¹⁰, Nw., Da., Sw. *sopp* 'mushroom'⁶¹¹

The consonant alternations of OHG *swamp* < *swamba- and ON *soppr*, OSw. *swamper* < *swamp^{pu}- can be explained in the usual way by reconstructing a Proto-Germanic *n*-stem with a nominative *swambō and an accusative plural *swamp^{uns} continuing *suómb^h-ēn, *suomb^h-n-ís. This *u*-stem is fully parallel to the formations ON *hōttr* 'hat' < *hattu- (see p. 304) and *knōttr* 'ball' < *knattu- (see p. 299), which, too, seem to have sprouted from old plural accusatives. ON *soppr* is continued as Icel. *sveppur*, which is formally based on the dsg. and npl. of the original paradigm *soppr*, gsg. *svappar*, dsg. *sveppi*, asg. *sopp*, npl. *sveppir*, gpl. *svappa*, dpl. *soppum*, apl. *soppu*.

In addition to the roots *swamb- and *swamp^p-, a root *swamm- is presupposed by Go. *swamms* (and probably also by OHG *swam*, MHG *swamme* and EDu. *swamme*). In this third variant, the labial stop has disappeared. Consequently, it can neither be explained from *suomb^h-, nor from *suomb^h-n-, as these root forms in would have developed into *swamb- and *swamp^p-. It is possibly, though, that the variant *swamm- continues a suffixed form with root stress, viz. *suómb^h-n-, in which Kluge's law did not operate. The labial disappeared between the two nasals, thus giving rise to a long *m*, viz. *swambna- > *swamma-. The formation would then be formally comparable to e.g. OHG *hunno* m. 'centurion' < *hunþnan- < *d̥kmt-n-, OHG *zinna* f. 'merlon' < *tinþnōn- < *h₃d-ent-n- and OHG *channa*, *chanta*⁶¹², MHG *kanne*, *kante* f. 'jug' <

⁶⁰⁵ Böðvarsson 1006.

⁶⁰⁶ Falk/Torp 1209.

⁶⁰⁷ Falk/Torp (p. 1209): *swumpa-.

⁶⁰⁸ De Vries 1962: 530.

⁶⁰⁹ Böðvarsson 930.

⁶¹⁰ Poulsen 1106.

⁶¹¹ Falk/Torp 1108.

⁶¹² Cf. App. *xq̃ntā* (Vetsch 111).

**kand-(n)ōn-*. Morphologically, the barytone stem **suómb^h-n-* seems to be parallel to **ster-n-* as in Go. *stairno*, ON *stjarna* f. ‘star’ < **h₂stér-n-*.⁶¹³

Beside the different roots with *a*-vocalism, there was a root **sump^p-* with *u*-vocalism. This variant must be reconstructed on the basis of Icel. and Far. *soppur*, which cannot reflect ON *soppr*.⁶¹⁴ The easiest way to explain the root variant **sump^p-* is to assume that it stems from the genitive case with a zero grade, i.e. *gsg. *sump^paz*, gpl. *sump^pan* < **sumb^h-n-ós*, **sumb^h-n-óm*.⁶¹⁵ The **a* ~ **u* alternation is further mirrored by the word for ‘swamp’ in West Germanic, cf. MHG *sumpf*, MLG *sump*, MDu. *somp*, *sump*, SFri. *sompe*, E *sump* < **swump^p-* vs E *swamp* < **swamp^p-*.

Outside Germanic, PGm. **swamban-* is clearly connected with OCS *goba* f. < **g^(h)omb^(h)-*, Lat. *fungus* < **g^{wh}o/ung^(h)-*, Gr. σπόγγος and σφόγγος ‘sponge’ < **sb^(h)ong-*, Arm. *sunk* < **suong^{wh}-*. The irregularities of the correspondences are suggestive of a non-Indo-European word⁶¹⁶, something that is, too, confirmed by its occurrence in the Finno-Ugric languages, cf. Mordv. *pango* ‘fungus, lichen’, Mari *ponḡō* ‘fungus, lichen, blight’, Mansi *pōōṅk* ‘toadstool’, Khanty *paṅk* ‘id.’ (cf. Flattery/Schwartz 1989: 121-129). The Germanic allomorphy is not likely to have resulted from prehistoric borrowing processes, however, but must have arisen as a consequence of internal sound changes.

**tadō*, **tuttaz* ‘tuft’?

- **tadan-*, -*ōn-*: OHG *zato* m. ‘villus’, *zata* ‘saeta, villus’
- **taddōn-*: OHG *zatta* ‘flax’, MHG *zatte* f. ‘swath’⁶¹⁷, G *Zatte* f. ‘windrow, sheaf’⁶¹⁸

⁶¹³ Van Heltten (1905: 224) reconstructed **stérnō* (beside **sternó* > **sterrō*), which he assumed to have arisen as an analogically root-stressed form that arose before Verner’s law and Kluge’s law.

⁶¹⁴ It is difficult to say whether ON had both variants *soppr* and *soppur*, because the manuscripts do not always differentiate between *q* and *o*.

⁶¹⁵ It is perhaps significant that the plural accusative **swamp^puns* < **suomb^h-n-ŷs* has an *o*-grade rather than a zero grade.

⁶¹⁶ Kluge/Seebold (p. 830): “Doch ist in Anbetracht des lautlich ähnlichen gr. *spóngos* »Schwamm«, l. *fungus* »Pilz«, die als Lehnwörter aus einer unbekannten Sprache gelten, nicht mit einem Erbwort zu rechnen.”

⁶¹⁷ Lexer 3, 1154.

⁶¹⁸ Grimm 31, 320.

⇒ **tad(d)la-*: G *Zattel* 'rag', LG *taddel* 'id.'⁶¹⁹, G Als.

Zat(t)el 'cluster, grape'⁶²⁰

⇒ **tad(d)ila-*: MHG *zettel*, G *Zettel* m. 'warp of a loom'⁶²¹

- **tat-* ⇒ **atura-*: ON *tǫturr*, pl. *tǫtrar* m. 'tatter, rag' (= E *tatter*)⁶²²
- **tatt-* ⇒ **tattaka-*: OE *tættec* m. 'rag'⁶²³
- **tuddan-, -ōn-*: ON *toddi* m. 'little piece', Icel. *toddi* 'tuft of grass', MHG *zotte* mf.⁶²⁴, G *Zotte* f. 'topknot, tuft of hair'⁶²⁵ (⇒ *Zottel* m. 'small wisp', Swab. *Zotter*, pl. *Zetter* m.⁶²⁶), Du. *tod(de)* 'rag, tatter'⁶²⁷, SFri. *todde* 'bundle'
 ⇒ **tud(d)lōjan-*: MLG *toddelen* 'to break down into tufts'⁶²⁸
- **tudōn-*: OHG *zota* 'villus, fimbria, coma', MHG *zote* mf. 'rag, fluff'⁶²⁹, G *Zote* f. 'tuft', Tyr. *zoute* f. 'id.'⁶³⁰
- **tuttōn-*: G (Mainz) *zotze* f. 'fag end, tip'⁶³¹, Swab. *zotze* f. 'tuft, brush'⁶³² (⇒ *zotzlen* pl. 'fuzz'⁶³³), Tyr. *zutzn* m. 'tuft'⁶³⁴

The consonant and vowel alternations in this material have not yet received a satisfactory explanation. Kluge and Seebold (p. 1016) called

⁶¹⁹ Grimm 31, 321.

⁶²⁰ Martin/Lienhart 2, 916a.

⁶²¹ Kluge/Seebold 1009.

⁶²² De Vries 1962: 604.

⁶²³ Bosworth/Toller 970; Holthausen 1934: 342.

⁶²⁴ Lexer 3, 1154.

⁶²⁵ Kluge/Seebold 1016.

⁶²⁶ Fischer/Taigel 1999: 422. The singular Swab. *Zetter* [e] m. 'cluster, twig with berries' (Fischer/Taigel 439) has **ä₂*, and sprouted from the delabialized plural to *Zotter*. Similarly, the late and sparsely attested G *Zette* f. 'leafy twig' (Grimm 31, 814) hardly presupposes PGM. **teddōn-*, but rather *Zäztte*.

⁶²⁷ Franck/Van Wijk 699.

⁶²⁸ Lübben 406.

⁶²⁹ Lexer 3, 1154.

⁶³⁰ Schatz/Finsterwalder 733.

⁶³¹ Schramm 1966: 280.

⁶³² Fischer/Pfleiderer 6/1, 1270.

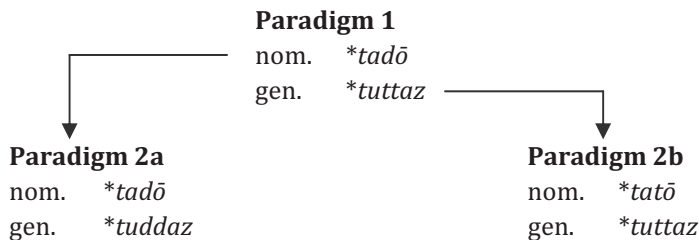
⁶³³ Fischer/Taigel 506.

⁶³⁴ Schatz/Finsterwalder 738.

the origins of the word unclear. In spite of the early attestation in Old High German, Franck and Van Wijk (l.c.) assumed the cluster of words to be of recent coinage. Fick, Falk and Torp (p. 150) went even further and denied the etymological link between OHG *zata* and *zota* on the whole. The vowel and consonant alternations of this paradigm are nevertheless in accordance with *n*-stems such as **maþō*, **muttaz* (see p. 218) and **raþō*, **ruttaz* ‘rat’ (see p. 221), and can therefore at least theoretically be explained as reflecting a paradigm **tadō*, **tuttaz*.

The different alternations are especially clear in Upper German, even in the oldest period. Thus, we find OHG *zata*, *zatta*, *zota* and MHG *zate*, *zatte*, *zotte*, *zote*. In Modern High German, however, *Zotte* < **tuddōn*- has prevailed over the other variants, and the same root is attested as ON, Icel. *toddi* ‘piece, wisp’, SFri. *todde* ‘bundle’, Du. *tod(de)* ‘rag’ < **tuddan*-. The semantic development from ‘wisp’ to ‘rag’ apparently went through an intermediate meaning ‘bundle’ or ‘frill’.

The roots with **d* and **dd* are dominant throughout the North and West Germanic dialects. They seem to prove that the introduction of voiced geminates through paradigmatic analogy took place at an early stage. Swabian *zotze*, on the other hand, preserves the root of the original genitive **tuttaz*, pl. **tuttan*. The same consonantism, though with a different ablaut grade, is also found in the OE *tættec* ‘rag’ < **tattka*-.⁶³⁵ ON *tōturr* m. ‘rag’ < **tat-ura*- contains a root **tat*- with an analogical singulate. All the different variants receive an explanation by assuming the usual paradigmatic split.



The *n*-stem may be cognate with ON *teōja* ‘to dung, manure’, G *zetten*,⁶³⁶ Visp. *zettu* ‘to spread the math’ < **tadjan*-. The link with MLG *tas* ‘corn-

⁶³⁵ Not with “expressives *tt*” as per Pokorny 175-9.

⁶³⁶ Grimm 31, 823-4.

stack', MDu. *tas* m. 'pile of hay' < **tassa-* is less certain, because it can be a Celtic loanword, cf. OIr. *daiss* f. 'heap of hay or peats'. Borrowing in the opposite direction, however, is not inconceivable either in view of PGm. **hrauk^ka-* 'haystack' emerging as OIr. *crúach* f. 'stack of corn, rick' (see p. 268). ON, Icel., Far. *des* f. 'haystack' may given the purely West Norse distribution conversely have been adopted from Old Irish.⁶³⁷

Alternatively, the *n*-stem **tadō*, **tuttaz* can be derived from an iterative **tuttōpi*, **tudunanpi*, as in MHG *zoten* 'to go slowly'⁶³⁸, Du. dial. *tooien* 'to drag, carry' < **tudōn-*, Du. dial. *todden* 'id.'⁶³⁹ < **tuddōn-*. If the original meaning of the *n*-stem was 'to pull, pluck', it can be connected with the verb by starting from a meaning 'to pull'. The question remains whether the iterative had variants with *a*-vocalism, i.e. ***tattōpi*, ***tadunanpi*, because this might explain the origin of the nominal formations with the same ablaut.

9.4 The **ī ~ *i* type

The *n*-stems with **ī ~ *i* alternations probably represent the most prominent apophonic type. It evolved out of the PIE ablaut **ei : *i* by regular sound change. It may thus have served as the model for other types with quantitative ablaut, viz. the **ū ~ *u* type and the **ā ~ *a* type.

**bīō*, **binaz* 'bee'

- **bīōn-*: Nw. *bie* f. 'bee', Gutn. *bāiā* f. 'id.'⁶⁴⁰, OHG *bīa* 'id.'⁶⁴¹, MHG *bīe* f. 'id.'⁶⁴², G dial. *beie* 'id.', Cimb. *paia* f. 'id.'⁶⁴³ (⇒ **bī-līn-*: Swi. App. *biili* 'id.'⁶⁴⁴, Visp. *biiji* n. 'id.'), OE *bīa* m. 'id.', *bīo* f. 'id.', Du. *bij* 'id.'⁶⁴⁵

⁶³⁷ Bugge 1905: 257; contra De Vries 1962: 75.

⁶³⁸ Lexer 3, 1154. The second meaning '*in zotten niederhangen*' points to a denominal **tudōjan-* rather than primary **tudōn-*.

⁶³⁹ Weijnen 206-8.

⁶⁴⁰ Klintberg/Gustavson 39.

⁶⁴¹ EWA II, 69.

⁶⁴² Lexer 1, 266.

⁶⁴³ Schmeller/Bergmann 214.

⁶⁴⁴ Vetsch 85.

⁶⁴⁵ Franck/Van Wijk 64.

- **bīnōn-*: MHG *bīn(e)*, *beine* f. 'id.'⁶⁴⁶
 - **binōn*: OHG *bina* f. 'id.' (⇒ dim. *bini* n. 'id.'), G *Biene* f. 'id.', Swab. *bine* f. 'id.'⁶⁴⁷, MLG *bēne* f. 'id.'⁶⁴⁸
-
- **bīja-*: Icel. *bý* n. 'id.'⁶⁴⁹, OSw. *bi*, *by* n. 'id.', Sw. *bi* n. 'id.'⁶⁵⁰, Da. *bi* c. (dial. n.) 'id.'⁶⁵¹, MHG *bīe* n. 'bee swarm'
 ⇒ ON **bī-fluga*: Icel., Far. *bý-fluga* f. 'bee'⁶⁵², Nn. obs.
bi-fluga. 'id.'⁶⁵³, Sw. dial. *bi-fluga* 'id.'⁶⁵⁴, Da. obs.
bi-flue 'horse fly'⁶⁵⁵

The material provided by the Germanic dialects implies that the Proto-Germanic word for 'bee' was an ablauting *n*-stem. This was first recognized by Lühr (2000: 98), who reconstructed the original paradigm as nom. **bīōn*, gen. **bīnes*.

The full grade **bīōn-* is ascertained by OE *bīo*, OHG *bīa*, MHG *bīe*, G *beie*, and Du. *bij* in West Germanic, and by Nw., Sw. *bie* f. in Scandinavian. The zero-grade stem is implied by OHG *binen* m.pl. < **binan-* as recorded by Notker, and its feminine equivalent **binōn-*, which is extant as MLG *bēne* and G *Biene*. Together, the variants point to a PGm. *n*-stem **bīō*, **binaz* < **b^héi-ōn*, **b^hi-n-ós*. The variants MHG *bīn(e)*, G Bav. *bein* < **bīnōn-* and OHG *bian* < **bian-* are secondary forms that evolved out of this paradigm by contamination.

The derivation of the Notker form *bini*, *pini* n. 'bee' is debated. It is usually analyzed as stemming from PGm. **binja-*, but the question then remains why the *j* did not cause doubling of the preceding nasal, as would be the expected effect of West Germanic gemination. Lühr (l.c.) reconstructed *bini* as PGm. **bini-*, suggesting that its creation would be derivationally comparable to the formation of Skt. *nīḍī-* 'housemate' to

⁶⁴⁶ Lexer 1, 277.

⁶⁴⁷ Grimm 1, 1122.

⁶⁴⁸ Lübben 39.

⁶⁴⁹ Böðvarsson 119.

⁶⁵⁰ Hellquist 41; SAOB B2368.

⁶⁵¹ Falk/Torp 71.

⁶⁵² De Vries 1962; Böðvarsson 119; Poulsen 171.

⁶⁵³ Collet 1877.

⁶⁵⁴ Möller 1928.

⁶⁵⁵ Cf. Fabricius (1804, p. 262, 565): *biflue* '*tabanus groenlandicus*'.

nīḍa- 'lair'. It seems more likely, however that *bini* arose within Old High German itself as a regular diminutive in **-īn*, cf. OHG *chizzi(n)* n. 'young goat' < **kittin(a)-* and Go. *gaitein* n. 'little goat' < **gaitīna-*. It must, in other words, be reconstructed as **bin-īn(a)-* with the zero-grade stem of the ablauting **bīō*, **binaz* plus the aforementioned diminutive suffix **-īn(a)-*.

Still unexplained is the exact derivation of ON *bý* n. 'bee', which is not an *n*-stem, but a thematic neuter. The most important problem consists of the origin of the rounded vowel. In order to explain it, a form **bīwa-* has been proposed⁶⁵⁶, as a *w* would cause labial mutation of *í* to *ý* in Old Norse before its deletion (cf. *Týr* < **tīwaz*). The problem is that there is no additional evidence for this *w*, which makes the reconstruction **bīwa-* rather *ad hoc*.

It has further been suggested that the *í* was rounded in the plural of a formation **bīa-* (or **bija-*).⁶⁵⁷ This plural **bīō* would have developed into Proto-Norse **bīu*, and further into ON *bý* by the required rounding. Still, this explanation cannot be maintained either, because Proto-Norse **bīu* would result in ON **bjú* rather than *bý*. This follows, for instance, from *þrjú* n. '3' < **þrīō* < ***trei-eh₂* and *hjú* n. 'inmate' < **hīwō* < **kei-u-ōn*.

Since all the older explanations are demonstrably incorrect, a different solution needs to be found. Perhaps the rounded vowel of *bý* can be explained by assuming that the original Old Norse form was a neuter **bí* < **bī(j)a-* that was influenced by *mý* n. 'mosquito' < **muwja-*. This seems probable, because the two words 1) have a comparable meaning, 2) are both neuter, and 3) both occur as the first member of a compound with *fluga* f. 'fly', cf. Icel., Far. *bý-fluga* f. 'bee', *mý-fluga* f. 'mosquito'.⁶⁵⁸ The reality of primary **bí* is ascertained by the neuter Sw. *bi*, MHG *bie*, and by the compounds Sw. dial. *bi-fluga* 'bee' and Da. obs. *bi-flue* 'horse fly'.

Extra-Germanic cognates are Lat. *fūcus* m. 'drone' < **b^hoi-ko-*⁶⁵⁹, OIr. *bech* 'bee', W *begegryr* 'drone' < **b^hi-ko-*, OCS *bъčela*, Ru. *pčelá*, SCr *pčèla* f.

⁶⁵⁶ Franck/Van Wijk 64.

⁶⁵⁷ Kock 1894: 297; Falk/Torp 71; Lühr 2000: 98; EWA II, 3.

⁶⁵⁸ Böldvarsson 664.

⁶⁵⁹ Pokorny (IEW: 163) isolates Lat. *fūcus* and OE *bēaw* m. 'horsefly' from OIr. *bech*, and reconstructs **b^houk^w-os*. Yet, Lat. *ū* can very well have developed out of PIE **oi*, so that there is little against reconstructing **b^hoi-ko-*.

'bee' < **b^hi-k-el-eh₂-* and Lith. *bitė* f. 'id.', OPru. *bitte* 'id.' < **b^hit-en-*. Just like the Germanic *n*-stem, they seem to be extensions to a root **b^hi-*.

****gīmō*, **gim(e)naz* 'open space'?**

- **gīmōn-*: ON, Icel. *gíma* f. 'aperture'⁶⁶⁰, Nw. dial. *gime* f. 'id.', Sw. dial. *gjäim* 'id.'⁶⁶¹
 - **gimōn-*: ON *gima* f. 'id.', Nw. dial. *gjeme* 'id.'
 - **gim(e)na-*: OE *geofon*, *gifen* n. 'sea'⁶⁶², OS *geþan* 'id.'
-
- **gaima-*: Icel. *geimur* m. 'expanse, space, sea'⁶⁶³
 - **gaiman-*: ON poet. *geimi* m. 'sea'⁶⁶⁴

The North Germanic languages provide potential evidence for the existence of two alternating *mn*-stems **gīmōn-* and **gimōn-*. Given the semantic and morphological similarities of these formations, it is theoretically possible that they both go back to an ablauting *mn*-stem **gīmō*, **gim(e)naz* that was created to the root **ǵ^hei-* as in ON *gjá* f. 'cleft' < **ǵī(w)ō-* and Lat. *hiāre* 'to be open'.⁶⁶⁵

The vowel length of the *i* of ON *gima* is ambiguous. De Vries (p. 176) gives *gíma*, following Björkmann's (1900-2: 309) analysis of the Middle English loanword *gime*. This vocalism is corroborated by Icelandic *gíma* and Sw. dial. *gjäim* (with regular diphthongization). Fritzner and Heggstad (p. 211), on the other hand, cite *gima* with a short vowel. This does not have to be erroneous, because the reality of both ON **gíma* and **gīma* is supported by the Norwegian dialects, for which *Grunnmanuskriptet* sets up two different entries, viz. *gíme* and *gime*.⁶⁶⁶

The Nordic forms have a cognate in the "Saxonic" dialects, i.e. OE *geofen*, *gifen* and OS *geþan* 'sea'.⁶⁶⁷ The root vowel of this formation

⁶⁶⁰ Böðvarsson 283.

⁶⁶¹ Lindblom 1988: 79.

⁶⁶² Bosworth/Toller 24.

⁶⁶³ Böðvarsson 275.

⁶⁶⁴ De Vries 1962: 161.

⁶⁶⁵ This **ǵ^hei-* may be a corruption of an *i*-present to the root **ǵ^hh₂-* 'to be open'.

⁶⁶⁶ The form *gime* is ascertained by the Telemark attestation *gjēme*, which has lowering and consecutive lengthening of ON **ī*.

⁶⁶⁷ The parallelism of OE *geofenes strēam* and OS *geþenes strōm* 'the ocean's flow' allows for the reconstruction of a poetic syntagm for "Proto-Saxonic".

must, without question, have been short (Kluge 1883: 87). The original form of the suffix, though, is less clear. Superficially, the attestations seem to continue PGm. **gimna-*. It is unclear, however, why Kluge's law did not operate in a form that appears to have developed out of PIE **ǵʰi-mn-ós*. It is not inconceivable, for this reason, that OE *geofen*, *gifen* and OS *geþan* actually developed out of **gimena-* by syncope of the **e*. If this is correct, it must be assumed that the underlying genitive adopted the ablaut of the original locative **gimeni* < **ǵʰi-mén-i*.

As in many other cases, an *o*-grade is found in a closely related thematic form, Icel. *geimur* '(open) space' < PGm. **gaima-* < **ǵʰoi-mo-* (the *n*-stem ON *geimi* 'sea' < **gaiman-* occurs in poetic contexts only, and may be a late nonce form). This thematic formation may be old, and could indicate that the ablaut of all the forms discussed here arose by independent derivation from a verbal complex. If this is correct, the about of the *n*-stem **ǵimān-* must be rejected.

****hrīpō, *hrittaz* 'fever'**

- **hrīpan-*: OHG *rīdo* m. 'fever', EDu. *rijde* 'febris'
 ⇔ **hrīpō(ja)n-*: OHG *rīdōn* 'to shiver', MHG *rīden* 'id.'
- **hriþa-*: OE *hrið* m. 'fever'
 ⇔ **hridō(ja)n-*: OE *hridian* 'to shiver'
- **hridan-*: OHG *rito* m. 'fever', MHG *rite* m. 'id.'⁶⁶⁸, OS *rido* 'id.', MLG, MDu. *rede* m. 'id.', EDu. *rede* 'id.'
- **hridan-* or **hriþpan-*: OHG *ritto* m. 'id.', MHG *ritte* m. 'id.', G *Ritte(n)*⁶⁶⁹, MDu. *ridde* m. 'id.', EDu. *redde*, *ridde* 'id.'
- **hrittān-*: MHG **ritze* m. 'id.' (= EDu. NRhnl. *ritse*) ⇒ G Swab. *ritze-rot* 'crimson'⁶⁷⁰
 ⇒ **hrittīga-*: G dial. *ritzig* 'rutting, in heat'⁶⁷¹ (= EDu. *ritsigh*, Du. *ritsig* 'in heat'⁶⁷²)

The pattern displayed by the different Germanic formations meaning 'fever' is suggestive of an originally apophonic *n*-stem in Proto-Germanic.

⁶⁶⁸ Lexer, 2, 463.

⁶⁶⁹ Grimm 14, 1051; Kluge/Seebold 767.

⁶⁷⁰ Grimm 14, 1086.

⁶⁷¹ Haas 1998: 851.

⁶⁷² Vercoullie 286.

At least four stem variants must be reconstructed. OHG *rīdo* and EDu. *rijde* unambiguously point to a full-grade form **hrīpōn-*. A zero-grade variant **hridan-* is ascertained by OHG *rito*, MHG *rite* and MLG, MDu. *rede*. A third stem **hriþpan-* or **hriddan-* occurs in OHG *ritto*, MHG *ritte* and MDu. *ridde*.⁶⁷³ Finally, EDu. NRhnl. *ritse* and Swab. *ritze-rot* ‘crimson’ point to a variant **hrittān-*. On the basis of these forms, the original paradigm can be reconstructed as **hrīpō*, **hrittaz*, *hrideni* < **kréit-ōn*, **krit-n-ós*, **krit-én-i*.

Remarkably, it was discovered by Schaffner (2001: 549-551) that the Verner variation as well as the ablaut of this paradigm were still intact in Notker’s Old High German idiolect, in which the nominative *rīdo* < **hrīpō* is accompanied by a dative *ríten* < **hrideni*. This means that, at least in this particular case, the Proto-Indo-European ablaut must have survived at least until the end of the first millennium.

OHG *ritto*, G *Ritte(n)* have traditionally been reconstructed differently. It was claimed by Grimm (l.c.), for instance, that it continues PGm. **hridjan-*. Similarly, Kluge/Mitzka (p. 602) reconstructed **hriþjan-* for both the geminated and the non-geminated forms (e.g. *rido*). According to the *Reallexikon der germanischen Altertumskunde* (9, 6), the problem of the geminate is “mit der Annahme von geminiertem *þ* aus westgerm. *þj* zu lösen.” Finally, also Schaffner (2001: 551) reconstructed **hridjan-*.

It is unsatisfactory, however, to isolate OHG *ritto* (etc.) from the other forms by reconstructing a separate *jan*-stem. There is no semantic motivation to do so, and, moreover, the alleged **hriþjan-* or **hridjan-* would presumably have left some traces of the **j* in the oldest stages of Old High German, viz. OHG ***(h)ritteo*. This not being the case, the geminate of *ritto* must rather be explained from an analogical paradigm **hrīpō*, **hriþpaz*, **hriþeni* or **hridō*, **hriddaz*, **hrideni*.

Parenthetically, it has been claimed by Schaffner (l.c.) that the root of the original genitive **hrittaz* < **krit-n-ós* is not attested. As a matter of fact, it can be recovered from EDu. NRhnl. *ritse* ‘fever’, which ostensibly represents a High German form **Ritze*. It was argued by Venema (1997: 347) that this *ritse* is an instance of *Pseudolautverschiebung*, because it is found North of the area in which **-tt-* shifts to **-tz-*. Since, however,

⁶⁷³ Note that MDu. *ridde* excludes the reconstruction **hriþpan-*, because, depending on the dialect, this would have become ***ritte* or ***risse*.

Swabian to the South has a compound *ritze-rot* 'crimson'⁶⁷⁴, as in the sentence *Es [= Mädchen] ward ritzerot* 'she flushed'⁶⁷⁵, the form *ritze* must at least partly be genuine. It further re-occurs in the dialectal German adjective *ritzig* 'in heat', which was borrowed into Early Modern Dutch as *ritsig(h)* 'id.'

Etymologically, the *n*-stem **hrīþō*, **hrittaz* is related to the verbs OHG *rīdōn* 'to shiver' < **hrīþō(ja)n-*, OE *hridian* 'to shiver' < **hridō(ja)n-* and to ON *hrīð*, OE *hrīð* f. '(snow)storm' < **hrīþō-*. It furthermore has a semantically apt parallel in Mlr. *crith* and W *crydd* 'fever' < PCelt. **kriti-* or **kritu-*, which is a derivation from the nasal present attested as W *crynu* 'to shiver'.

****kībō*, **kippaz* 'basket'**

- **kībōn-*: MHG *keibe* f. 'peddler's pack'⁶⁷⁶
- **kīp^oōn-*: MLG *kīpe* f. 'basket'⁶⁷⁷, LG EFri. *kiepe* 'peddler's pack'⁶⁷⁸, G WPhal. *kīpe* f. 'wicker basket, peddler's pack'⁶⁷⁹ (= G *Kiepe*⁶⁸⁰, *Keipe* f. 'id.'⁶⁸¹), MDu. *kijp* 'pack, bundle', OE *cīpan* m.pl. 'basket', E dial. *kipe* 'id.'
- **kippōn-*: ON *korn-kippa* f. 'basket for corn', Sw. dial. *kippa* 'bundle, pack'⁶⁸², EDa. *kippe* 'dying vat', Swi. *kipf* f. 'wine measure'⁶⁸³, MLG *kip* 'pack'⁶⁸⁴
- **kibbōn-*: Du. *kib(be)* 'basket'⁶⁸⁵

⁶⁷⁴ Cf. Grimm (14, 1085/6) *ritz(e)roth*: "gewöhnlich erklärt man 'roth wie ein Ritz in der menschlichen Haut, der das Blut sehen lässt' [...]"

⁶⁷⁵ Fischer/Keller/Pfleiderer 379.

⁶⁷⁶ Lexer 1, 1535.

⁶⁷⁷ Lübben 174.

⁶⁷⁸ Byl/Brückmann 65.

⁶⁷⁹ Woeste 126.

⁶⁸⁰ Kluge/Seebold 487.

⁶⁸¹ Grimm 11, 685-6.

⁶⁸² Rietz 321.

⁶⁸³ Grimm 11, 780.

⁶⁸⁴ Lübben 174.

⁶⁸⁵ Vercoullie 162.

At least four different roots can be reconstructed for the word for 'basket'. Together they form a pattern that is consistent with an old *n*-stem with ablaut.

MHG *keibe*, with its diphthong, points to PGm. **kībōn-*. ON *kippa* and Swi. *kipf*, on the other hand, support a Northwest Germanic root variant **kipp-*. Hence, it is possible to reconstruct an ablauting paradigm **kībō*, **kippaz*. This paradigm may have been regularized in several different ways. OE *cīpa* and MDu. *kijp* contain a root **kīp^p-*, which may have come about through a secondary paradigm **kīpō*, **kippaz*. Conversely, the root of Du. *kib* < **kibbōn-* can very well have arisen in an analogical paradigm **kībō*, **kibbaz*. The position of MLG *kīpe* and G *Kiepe* is not entirely clear. These forms could be reconstructed as either **kīp^pōn-* or **kīpōn-*. Westphalian *kīpe*, however, unambiguously points to a root with **ī*, as **kīpōn-* would have yielded ***kiæpe* in this dialect.

Lühr (1988: 235) explained the formations **kībōn-* and **kīpōn-* as primary derivations from a verb **kīban-/kīpan-*, thus disconnecting it from ON *kippa*, Sw. dial. *kippa*, Da. *kippe*. The close semantic match between all the different stems, however, could well point to a shared origin, i.e. an *n*-stem **kībō*, **kippaz*. It nevertheless remains possible to uphold the link with **kippōn-*: OE *cippian*, G dial. *kipfen* 'to cut', as suggested by Lühr, by assuming that the *n*-stem originally referred to a container hollowed out of wood. Alternatively, it is possible to start from an original meaning 'pack, bundle' as conveyed by Sw. dial. *kippa*.

****klīpō*, **klittaz* 'burdock, tangle, clay'**

- **klīpōn-*: OE *clīðe* f. 'burdock'⁶⁸⁶, E obs. *clithe* 'cleavers'
- **klīt^tōn-*: OE *clīte* f. 'coltsfoot, butterbur'⁶⁸⁷, E *clite* 'cleavers, goose-grass', G *Kleise* f. 'dodder'⁶⁸⁸
- **klīpōn-*: OHG *chletha* 'lappa'
- **klidōn-*: OHG *chleta* 'lappa, philanthropos', *deni-chleta* 'agrimonia', MHG *klete* f. 'burdock'
- **klitōn-*: ME *clēte* 'id.', G *Kliese* f. 'burr'^{689,690}

⁶⁸⁶ Holthausen 1934: 52.

⁶⁸⁷ Ibid.

⁶⁸⁸ Grimm 11, 1133.

⁶⁸⁹ Schottelius (1663: 64) apud Grimm (11, 1163): "solche worte fallen ins herze, wie die kliesen an die wolle".

- **klitta-*: G dial. (Brandenburg) *klitz* ‘burdock’⁶⁹¹
 - **klittōn-*: G Tyr. *?kletze* ‘id.’⁶⁹², MLG *kletze* f. ‘down’⁶⁹³
 - **klippan-, -ōn-*: OHG *chleddo, chletto, chledda, chletta* ‘lappa, philanthropos’, G *Klette* f. ‘id.’⁶⁹⁴ (⇒ *Baum-klette* ‘treecreeper’), Swi. Ja., Visp. *xlätta* f. ‘id.’⁶⁹⁵, MDu. *clesse, clisse, clitte* f. ‘burdock, tangle, clay’⁶⁹⁶, Du. *klis, klit* ‘tangle, burdock’⁶⁹⁷
-
- **klaitōn-*: OE *clāte* f. ‘clotbur’⁶⁹⁸, ME *clōte*, E *clote* ‘id.’⁶⁹⁹

“Welcher Reichthum der Entwicklung bei einem so geringen Dinge,” Grimm writes under *Klette*. Indeed, the formal variation found with this etymon is bewildering: the root vocalism shifts between **ī*, **i* and **ai*, while the final consonantism varies between **p*, **t(t)*, **d* and **pp*. Both the consonantal and vocalic interchanges can be attributed to the inflection of an ablauting *n*-stem **klīpō*, **klittaz*, **klideni*. This paradigm appears to have given rise to a plethora of contamination forms.

Regarding the full grade, it must be stressed that the evidence for the vocalism **ī* is limited. OE *clīde* possibly represents the original nominative **klīpō*, but it can just as well be reconstructed as **klīpōn-*, because the length of the *i* is uncertain. Perhaps, the form **klīpōn-* is even preferable, because it must be supposed anyway for OHG *chletha*.

Unambiguous evidence for a long vowel, on the other hand, comes from OE *clīte* ‘coltsfoot’ and modern English *clite* [k^hlait] ‘cleavers’ < **klītōn-*, the latter word containing a diphthong. Du. *klijt* ‘clay’, though semantically more remote, probably belongs here too, especially in view of the meanings of MDu. *clisse* ‘burdock, tangle, clay’. It is further

⁶⁹⁰ Also compare EDu. *klijt(e)*, Flem. *klijte* ‘clay’ (Willems 8, 182; WVD I, 1, 40).

⁶⁹¹ Taken from Grimm 11, 1152.

⁶⁹² Datenbank zur deutschen Sprache in Österreich, s.v. *Kletze*.

⁶⁹³ Lübben 176.

⁶⁹⁴ Grimm 11, 1151-3; Kluge/Seebold 495-6.

⁶⁹⁵ Wipf 34.

⁶⁹⁶ Verdam 295.

⁶⁹⁷ Franck/Van Wijk 317.

⁶⁹⁸ Bosworth/Toller 158.

⁶⁹⁹ Holthausen 1934: 51.

probable that also G *Kleise* continues **klīt^tōn-*. The *s* instead of *β* is unexpected (perhaps dialectal), but the diphthong *ei*, at any rate, points to PGm. **ī*.

The creation of the variant **klīt^tōn-* may have taken place when the geminate of the original genitive **klittaz* spread to the nominative **klībō*. The root **klitt-*, however, is extremely sparse. Grimm makes mention of a Brandenburg dialect form *klitz*, which on the surface seems to support PGm. **klitta-*. Yet Brandenburg is in the Low German speech area, where *-tt-* never changed into *-tz-*. Alternatively, it has been claimed that *klitz* was imported by 12th century Dutch-speaking settlers.⁷⁰⁰ The problem is that **tt* does not become **tz* in Dutch either. The form *klits* is sporadically found in the modern dialects of Limburg and Brabant, but not in Flanders, where the settlers originated from.⁷⁰¹ Even in Brabant and Limburg, *klits*⁷⁰² almost exclusively occurs in areas where *klis* and *klit* are found side by side. This raises the suspicion that *klits* is a contamination form. Whether this form was actually taken to Brandenburg by Dutch-speaking colonists remains doubtful.⁷⁰³ The form *klitz* can equally well be a High German intrusion into the Low German speech area⁷⁰⁴, especially since this must probably be assumed for MLG *kletze* ‘lanugo (downy hair)’, too. As such, the word confirms the pre-existence of a variant **klittōn-*.

The reason for the paucity of the root **klitt-* is easy to give: the original genitive **klittaz* < **glit-n-ós* must have been replaced by **klībbaz* at an early stage. The variant **klībb-* is first of all found in OHG *chledda*, G *Klette*. Middle Dutch has both *clisse* and *clitte*, which is expected, since double **-bb-* regularly developed into *-ss-* in many Dutch dialects. In Modern Dutch, too, both *klit* and *klis* occur side by side, predominantly with the meaning ‘tangle’. The analogical replacement of **klittaz* by **klībbaz* is paralleled by other *n*-stems such as **lappōn-* ‘lath’ (p. 214) and **mupbōn-* ‘moth’ (p. 218). There are no indications that **-bb-*

⁷⁰⁰ Kluge/Mitzka 337; Teuchert 1944: 204-5.

⁷⁰¹ PLAND, s.v. *klit*.

⁷⁰² Additionally, *klits* frequently bears the meaning ‘poppy’ in the Limburgian dialects, which is conspicuously close to *klats* ‘id.’, cf. G *Klatsch-mohn* ‘poppy’.

⁷⁰³ Afrikaans *klits-gras* ‘bur bristle grass’ seems to provide a parallel.

⁷⁰⁴ Cf. Grimm 11, 1152.

developed out of **-hþ-*, as was argued by Lühr (1988: 255), or resulted from West Germanic gemination in a form **klīþjōn-*.⁷⁰⁵

At least two more root variants can be distinguished. OHG *chleta* contains the stem **klidōn-* with an allomorph displaying the operation of Verner's law. The combination of a zero-grade root with a stressed suffix may point to a locative **klideni* < **glit-én-i*. Finally, a root **klitōn-* can be reconstructed on the basis of ME *clēte*⁷⁰⁶ and G *Kließe*. These forms appear to have a secondary singulate that must have arisen through the creation of an analogical paradigm **klitō*, **klittaz*.

A difficult form is OE *clāte* 'burdock', which with its long *ā* (< **aī*) secures an *a*-grade. The length of the vowel is ascertained in two ways, i.e. by the fact that the geminate **-tt-* would not have been shortened if the *a* was short, and simply because the vowel of Modern E *clote* 'id.' can only have developed out of OE *ā*. The root can thus be reconstructed as **klait^t-*.⁷⁰⁷ It possibly arose in an apl. case **klait^tuns* < **gloit-n-ŋs*. Alternatively, it could have arisen as an *o*-grade to a strong verb such as OE *clīðan* 'to stick' < **glēi-t-e-* or perhaps rather to a de-iterativized variant **klīt^tan-*. It is conceivable, for instance, that G *kletz* 'sticky' < **klitta-* (< **glit-nó-?*) was derived from such a verbal cluster.

In addition to the forms with **i-* and **ī-*vocalism, there is a limited number of variants with **a*-vocalism in the Low German/Dutch, i.e. originally Frisian area, cf. MLG *klatte* f. 'rag'⁷⁰⁸, MDu. *classe* f. 'burdock, dirt'⁷⁰⁹, EDu. *kladde* 'macula, (Holl.) lappa', Du. dial. *klad(d)e*, *klarre* 'burdock, reed mace, bag, blot, smudge'⁷¹⁰, WFri. *kladde* 'burdock, stain slur, bag'.⁷¹¹ This vocalism is problematic, because it disrupts the normal ablaut pattern. Since the **a*-variants often carry the meaning 'smudge', the *n*-stem **klīþō*, **klittaz* probably became associated with the cluster of G Swab. *klatteren* 'das Kleid mit Dreck beschmutzen'⁷¹², MLG *kladder*⁷¹³,

⁷⁰⁵ Pace Kluge/Mitzka 337.

⁷⁰⁶ Usually reconstructed as **klaitjōn-*, cf. MED: OE **clāte*.

⁷⁰⁷ From **gloit-n-* (Fick/Falk/Torp 58), not from **gloid-* (thus OED, sv. *clote*; Pokorny 356-364).

⁷⁰⁸ Lübben 175.

⁷⁰⁹ Verdam 292.

⁷¹⁰ Kocks/Vording 550.

⁷¹¹ Zantema 1, 495.

⁷¹² Fischer/Taigel 476.

⁷¹³ Lübben 174.

MDu. *cladden*, *clatten*⁷¹⁴, Du. *kladden* ‘to smudge’⁷¹⁵ (see Lühr 1988: 279ff.), which may go back to an iterative **klattōpi*, **kladunanþi* or – as Lühr (l.c.) suggests – to a primary *n*-stem **klaþō*, **klattaz* ‘Schmutzklumpen’.

Etymologically, the *n*-stem **klīþō*, **klittaz* together with OE *clīðan* belongs to the root found in e.g. Gr. γλία f. ‘glue’, Lat. *glūs*, *-tis* n. ‘id.’, and Lith. *gliejù*, *gliėti* ‘to smear’ < PIE **glei-*. Other well-known Germanic cognates are **klaja-*: OE *clæg*, Du. *klei* ‘clay’, and the sub-group of G *kleben* ‘to stick’ < **klibōn-*, cf. SCr. *glib* ‘filth’ < **glei-b^ho-*. A very old formation must be the *tudāti*-verb OHG *chlenan* ‘to stick, glue’ (cf. OIr. *glenaid* ‘to adhere’ < **gli-néh₂-*), which was transferred to the fourth class of the strong verbs as a result of the *a*-umlaut of **i*.

****rīhō*, **rikkaz* ‘stringing pole, line’**

- **rīhōn-*: OGutn. *ri* f. ‘pole’, Gutn. *rāj* f. ‘bar’⁷¹⁶, Da. *ri(e)* ‘long bar, measuring rule’, MHG *rīhe* f. ‘line’⁷¹⁷, G *Reihe*⁷¹⁸, MDu. *rie* f. ‘slat, measuring rule, line, row’⁷¹⁹, Du. *rij*
- **rīhan-*: Nw. dial. *rjå* m., Sw. dial. *rie* m. ‘pole on which grain is placed to dry’⁷²⁰, Swi. Visp. *reijo* m. ‘row’
- **rīgōn-*: OHG *rīga* f. ‘line’, MLG *rīge* f. ‘line, series of houses’, MDu. *rige* f. ‘row, ridge, plank’⁷²¹, EDu. *rijghe* ‘line’
⇒ EDu. *rijchel* ‘bar, slat’, Du. *richel* ‘ledge’
- **rīgōn-*: OHG *riga* f., G *Riege* ‘line, row, squad’⁷²², MLG *rege* f., EDu. *reghe* ‘line’, Du. dial. *reeg* ‘line, series’⁷²³
⇒ OHG *rigil* m. ‘bolt’, G *Riegel*, MLG *regel* ‘crossbeam, rail’, MDu. *reghel* m. ‘plank, slat, ruler’
- **riggōn-*: MDu. *regghe*, *rigghe* f. ‘line, row, slat’

⁷¹⁴ Verdam 291, 292.

⁷¹⁵ Franck/Van Wijk 310.

⁷¹⁶ Klintberg/Gustavson 927 apud Schlyter 1877: 511.

⁷¹⁷ Lexer 2, 430.

⁷¹⁸ Kluge/Seebold 754.

⁷¹⁹ Verdam 494.

⁷²⁰ Falk/Torp 895.

⁷²¹ Verdam l.c.

⁷²² Grimm 14, 992.

⁷²³ WNT, s.v. *reeg*.

- **rikka(n)-, -ōn-*: Gutn. *räckå* f. 'post'⁷²⁴, MHG *ric* m. 'horizontal bar on which to put things', G *Reck, Rick* mn. 'stake, row', *Recke* f. 'row, series'⁷²⁵, dial. *ricke* m. 'line'⁷²⁶
 - **rikōn-*: MDu. *reke* f. 'line, row'⁷²⁷
 - **rihōn-*: MDu. *ree* f. '(guide)line, building line, marcation line'⁷²⁸
-
- **raihō-*: Nw. dial. *rå* f. 'border marcation'
 - **raiga-*: Nw. *reig* m. 'border line'

The comparison of G *Reihe* 'line', *Recke* 'series' and *Riege* 'line, row, squad' reveals that the German standard language alone offers sufficient evidence for the reconstruction of an ablauting *n*-stem **rīhō*, **rikkaz*, **rigeni*. *Reihe* (cf. Du. *rij*) with its combination of a full grade and original root stress clearly continues the original nominative form **rīhō*. *Recke*, on the other hand, combines a zero grade with a geminate, and thus can be traced back to the singular and plural genitives **rikkaz* and **rikkam*. Then, there is the additional form *Riege*, which with its coupling of a zero grade and a **g* resulting from Verner's law points to the original locative case **rigeni*.

As has been pointed out, Modern High German itself already offers sufficient evidence for the reconstruction of a full-fledged ablauting *n*-stem paradigm with consonant gradation. Still, the variation is even more extensive in the older dialects of West Germanic, most notably in Old High German and Middle Dutch.

In Old High German, the Notker form *rīga* clearly points to PGm. **rīgōn-*, and the same formation is evidenced by EDu. *rijghe*. From the perspective of Kluge's *Associationen*, it must probably be understood as a contamination of the nominative **rīhō* with the locative **rigeni*.

A different analogy must be responsible for the creation of the Middle Dutch forms *regghe* and *rigghe* 'line', which go back to PGm.

⁷²⁴ Klintberg/Gustavson 980.

⁷²⁵ Grimm 14, 444.

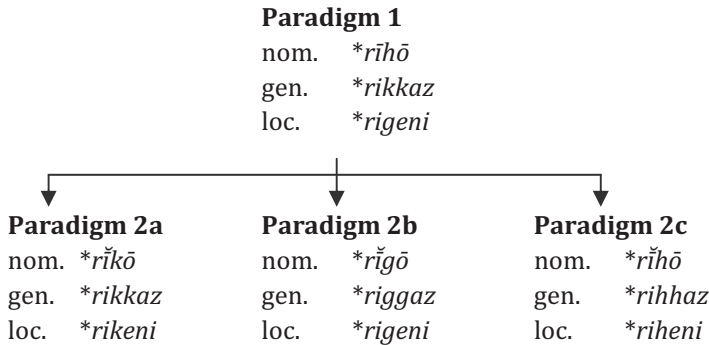
⁷²⁶ Grimm 14, 907-8.

⁷²⁷ Verdam 490.

⁷²⁸ Verdam 488.

**riggōn-*. In this case, it is most probable that the formation arose due to the interference of the original genitive **rikkaz* with the locative **rigeni*.

Yet another interesting secondary form is MDu. *reke*, which appears to have a secondary singulate. This consonantism can be explained by reconstructing a secondary paradigm **rikō*, **rikkaz*, **rikeni*.



A different explanation for the *grammatischer Wechsel* of **rīhōn-* and **rīgōn-* is given by Schaffner (2001: 403), who tentatively compares the accentual difference of Skt. *rekhá-* ‘stripe, line’ < **(H)reik(H)-éh₂-* and *lékhā-* ‘stripe, furrow’ < **(H)réikh₂-eh₂-*. In theory, it is therefore possible to reconstruct a *h₂*-paradigm with ablaut, e.g. **(H)réik-h₂-*, **(H)rik-h₂-ós* > PGm. **rīhō*, **rīgōz*. Such a paradigm, however, does not account for the geminates of G *Recke* and MDu. *regghe*, *rigghe*. The latter forms are therefore reconstructed as **rigjō(n)-* by Schaffner, but it seems preferable to ascribe the voiced geminates to Kluge’s *Assoziationen*.

The *n*-stem also has reflexes in North Germanic, i.e. Nw. *rjā* ‘corn stick’, Gutn. *rāj* ‘bar’. Nw. *rjā* can theoretically have developed out of both **rīhan-* and **rīhan-* through the West Nordic accent shift of **-īā-* to **-iā-* (cf. ON *ljá* ‘lend’ < **līa* < **līhwan-*, ON *ffjá* ‘to hate’ < **fīa* < **fijan-* (Go. *fī(j)an*). The vowel length is nevertheless confirmed by the Gutnish form *rāj* ‘bar’, which shows regular diphthongization of OSw. *ī*.

The semantic discrepancy between the North and West Germanic material is somewhat problematic. Whereas the West Germanic forms signify both ‘line’ and ‘stick’, the meaning ‘line’ is completely absent in Nordic. This could indicate that ‘stick’ is the original meaning. On second thought, however, this possibility must be rejected, because it defies the evident link with the strong verb **rīhan-* (e.g. MDu. *rijen* ‘to string’). This



Photograph of a field with *rjās* in Rodal, Norway (1953).

paradox can be resolved, however, by starting from the meaning of Nw. *rjā*, i.e. 'to stick on which bundles of grain are pierced to dry'. Perhaps, then, the more general meaning 'pole' developed out of 'stringing stick' (see image).

Etymologically, the *n*-stem belongs to the already mentioned strong verb **rīhan-*, which also gave rise to Nw. dial. *rā* f. 'border marcation' < **raihō-* and *reig* m. 'border line' < **raiga-*. It has already been mentioned that the Germanic etymon has been linked to Skt. *rekhā-* 'rift, line'⁷²⁹, but this old etymology is implausible from the semantic perspective: Skt. *rikhāti* does not mean 'to string', but 'to scratch'. As such, it is clearly cognate with Gr. *ἐρείω* 'to tear', Lith. *riēkti* 'to cut bread' and Lat. *rīma* f. 'rift'. Consequently, the Sanskrit meaning 'line' must have developed only secondarily out of older 'rift', which makes a direct connection with the Germanic word improbable.

Other suggestions are equally problematic. OE *ræw* f. 'row' has been connected with Lith. *rievė*, *reivė* f. 'stripe'⁷³⁰, but both words are

⁷²⁹ Grimm l.c.; Fick/Falk/Torp 343; Pokorny 857-9; WNT, s.v. *rij*.

⁷³⁰ Falk/Torp 895; Pokorny 857-859; Fick/Falk/Torp 343; Holthausen 1934: l.c.

irreconcilable with a root **Hreik-*. Fick, Falk and Torp isolate Nw. *rjǎ* and MHG *ric* from the rest of the material, connecting it to Lith. *rikė* ‘post, plank’⁷³¹, but this is a loanword from Low German, cf. East Frisian *rick*.⁷³² Pokorny proposes a link with ON *reigjask* ‘stretch’, *rígr* ‘stiff’ and Icel. *riga* ‘to waver’ under a semantic category ‘to stretch, stumble’, but the given words belong to the root **uroik-* ‘to twist, sprain’. Kluge and Seebold (p. 754) call the etymology unclear.

****sīlō*, **sillaz* ‘strap, horse harness’†**

- **sīlan-*: G *Seilen* m. ‘horse harness’⁷³³, Du. dial. *zijl(e)* ‘trace, rope’⁷³⁴
 ⇒ **sīljan-*: Icel. *síla*, *-di* ‘to tie together’⁷³⁵
 - **sila(n)-*: ON *seli*, *sili* m. ‘harness’, Nw. *sele* m. ‘harness, suspender’⁷³⁶, OSw. *sele*, *sile*, Sw., Da. *sele*⁷³⁷, OFri. *sil-rāp* m. ‘trace’, WFri. *sile* ‘hames’, SFri. *siele* mf. ‘id.’, NFri. *selle* f. ‘id.’⁷³⁸, OHG *silo* m. ‘rope’, MHG *sil(e)* m. ‘strap, trace, harness’, G *Siele* ‘id.’⁷³⁹, Swi. Visp. *silo* m. ‘plow-trace’, MLG *sele* m. ‘harness, trace’
 ⇒ **siljōn-*: Nw. dial. *silje* f., Sw. *silja*, *silla* ‘harness’⁷⁴⁰
 (= G *Sille* f.^{741?})
-
- **saila-*, *-ō-* ‘rope’: ON *seil* f.⁷⁴², Far. *seil* f. ‘band, cow harness, scarf’, G *Seil* n. ‘rope, noose’⁷⁴³, OS *sēl* ‘rope’, MDu. *seel* n. ‘id.’,

⁷³¹ Fraenkel 733.

⁷³² Byl/Bückmann 106.

⁷³³ Grimm 16, 221.

⁷³⁴ Ter Laan 1929: 1259.

⁷³⁵ Böðvarsson 830.

⁷³⁶ Falk/Torp 956.

⁷³⁷ Hellquist 704; ODS, s.v. *sele*.

⁷³⁸ Zantema 1, 861; Jensen 475. Cf. Århammar 2004.

⁷³⁹ Grimm 16, 953-6; Kluge/Mitzka 708; Kluge/Seebold 847.

⁷⁴⁰ SAOB 1808.

⁷⁴¹ Grimm 1058.

⁷⁴² De Vries 1962: 468.

⁷⁴³ Grimm 208; Kluge/Mitzka 700; Kluge/Seebold 839.

Du. *zeel* n. 'id.'⁷⁴⁴, OFri. *wind-sēl* n. 'certain rope used at a sailing boat'⁷⁴⁵, OE *sāl* mf. 'rope, tether', E *sole*
 ⇒ Go. *in-sailjan* 'to rope up', OE *sælan* 'id.', OFri. *sēla*
 'id.', MLG *sēlen* 'id.', MHG *seilen* 'id.'

The co-occurrence of G *Seilen* < **sīlan-*, ON *seli*, *sili*, OHG *silo* < **sīlan-* may possibly point to an old apophonic *n*-stem. One objection to the reconstruction of such a paradigm could be that, even if Icel. *síla* < **sīljan-* was really derived from a nominal root **sīl-*, the full grade is only marginally attested. Another objection consists of the thematic *o*-grade form ON, OHG *seil*, OE *sāl* < **saila-*, which could point to a lost strong verb **sīlan-*. This strong verb, if it existed, may very well have served as the basis for both **sīlan-* and **sīlan-*.

Alternatively, Kluge and Seebold (p. 847) considered the possibility that G *Siele* is an "alter *l*-Stamm oder ablautende Zugehörigkeitsbildung". If this is correct, the formations discussed here should perhaps be unified into a paradigm **séi-l*, **si-l-ós*.

The etymon is closely related to Lith. *seilas* 'band, tie'⁷⁴⁶ < **sei-l-o-*, a derivation from Lith. *siėti* 'to bind'. This verb is akin to Skt. *syāti* 'id.'⁷⁴⁷, which, in turn, appears to be an *i*-present to a root **seh₂-* 'to bind'.

****skīō*, **skinaz* 'shinbone'**

- **skīa(n)-* and **skīōn-*: OE *scīa* m. 'shinbone', E dial. *shy* 'pole'⁷⁴⁸, MHG *schīe* mf. 'fence post', G *Scheie* f. 'id.'⁷⁴⁹, Swi. Visp. *šīija* f. 'leg splint, stick'
- **skinō-*: OE *scīnu* f. 'shin'⁷⁵⁰, OHG *scena*, *scina* f. 'shinbone, strip, needle', MHG *schin(e)* f. 'strip, shin(plate)', G *Schiene* f. 'shin, strip'⁷⁵¹, MLG *schēne* f. 'shin(plate), strip' (= Nw. dial.

⁷⁴⁴ Franck/Van Wijk 813.

⁷⁴⁵ Hofmann/Popkema 588.

⁷⁴⁶ Fraenkel 770-1.

⁷⁴⁷ Pokorny 891-2.

⁷⁴⁸ Bosworth/Toller 830; Holthausen 1934: 276.

⁷⁴⁹ Lexer 2, 723; Grimm 14, 2418.

⁷⁵⁰ Bosworth/Toller 834; Holthausen 1934: 279.

⁷⁵¹ Lexer 2, 746; Grimm 15, 15-8.

skine, skjene, Sw. *skena*, Da. *skinne* ‘shin, strip, stave’⁷⁵²), MDu. *schene* f. ‘shin(plate), hollow bone, strip’, EDu. *scheene* ‘*tibia*, *crus*’, Du. *scheen* ‘shinbone’

The etymological dictionaries usually treat the variants **skīan-* ‘stick’ and **skinō-* ‘shinbone’ as two separate formations, but given the remarkable morphological parallelism with the paradigm of **bīō*, **binaz* ‘bee’ (G *Beie* : *Biene* = *Scheie* : *Schiene*), it seems preferable to explain them as the offshoots from a single *n*-stem **skīō*, **skinaz*. The full-grade nominative allomorph **skīō* is supported by most of the West Germanic languages, cf. OE *scīa* and Visp. *šijja*. The oblique zero-grade stem **ski-n-* is attested in OHG *scena*, OE *scinu* etc. The fact that the two variants mean both ‘stick’ and ‘shinbone’ is an additional argument to trace them back to a single formation. Possibly, the Vispertermin form *šijja* ‘leg splint’ preserved the missing link between the two different meanings.

The etymon is often connected to the root **ski-* as in Skt. *chyáte* ‘prune’.⁷⁵³ Lubotsky (2001: 232-3) recently proposed a semantically more straightforward link with Ru. *cévka* ‘tube, shin of a horse’, Cz. *céva* ‘reed, tube’, Lith. *šaivà*, *šeivà* ‘tube, net, needle, spool’⁷⁵⁴ and with the second member of Av. *ascūm* asg. ‘shank’ and Skt. *aṣṭhivá(nt)-* ‘shinbone’ < **h₃esth₁-(s)kīH-uo-* ‘bone-tube’. The difference between PBSl. **koi(H)u-* and **koi(H)u-* was explained by Lubotsky as resulting from the influence of the *s*-mobile⁷⁵⁵ that can be reconstructed on the basis of the Germanic forms.

Lubotsky further derived OE *scīa* from **skīHu-o-*. Given the absence of a labial in OE *scīa* (cf. *spīwan* ‘to spit’ < **spīwan-*, OE *gīw* m. ‘vulture’ < **gīwa-*) as well as in **skinō-*, it seems better still to analyze the **u* in the other Indo-European languages as a suffix. PIlr. **Hast-čīua-*, Lith. *šaivà* and *šeivà* may then point to an old ablauting *u*-stem **ké/oiH-u*, **kīH-u-ós*. The Germanic *n*-stem, on the other hand, continues **ské(h₁)i-ōn*, **sk(h₁)i-n-ós* directly, or PIE **skéiH-ōn*, **skiH-n-ós* with Dybo’s law in the oblique cases.⁷⁵⁶ This formation cannot be directly related to Gr. κίων, Myc. *ki-wo*

⁷⁵² Hellquist 733.

⁷⁵³ Cf. Franck/Van Wijk; Holthausen 1934; Pokorny 919-22.

⁷⁵⁴ Cf. Pokorny 919-22.

⁷⁵⁵ Kortlandt 1978: 238.

⁷⁵⁶ Lubotsky 2001: 323 fn.

and Arm. *siwn* ‘pillar’⁷⁵⁷, because these forms must be derived from **kiHu-ōn*.

Within Germanic, there can be a link with ON *skíð*, OHG *scīt*, OE *scīd* n. ‘wooden bar’ < **skīda-*, EDu. *schie(de)r*, *schie(de)rken houts*, Flem. *schier* ‘wooden fragment’ < **skīd-ra*-⁷⁵⁸, and OFri. *skidel* m. ‘spoke-bone’⁷⁵⁹, WFri. *skyl*⁷⁶⁰, NFri. *skidjel* ‘piece of wood used for making nets’, MLG *schēdel* m. ‘bone in the arm’⁷⁶¹, which Århammar (2004) derives from **skīd-la-*. It is not entirely inconceivable, however, that these words were formed to the verb **skīpan-* as in e.g. MHG *schīden* ‘to split’.

****skīmō, *skimenaz* ‘shine’†**

- **skīma(n)-*: Go. *skeima* m. ‘torch’, Icel. *skími* m. ‘glimmer, gleam’⁷⁶², OHG *scīmo* m. ‘shine, gleam’, MHG *schīm(e)* m. ‘id.’⁷⁶³, OS *dag-skīmo* m. ‘daylight’, MDu. *schime* m. ‘shine’⁷⁶⁴, OE *scīma* m. ‘splendor, brightness’⁷⁶⁵
 ⇒ **skīmla-*: Du. dial. *schijmel* ‘shade’
 - **skīma(n)-*: ?ON *skimi* m. ‘gleam, shine’⁷⁶⁶, OE *scīma* m. ‘shadow’⁷⁶⁷, MHG *scheme* m. ‘shade’ m. ‘id.’⁷⁶⁸, G *Schemen*, OS *skimo* ‘umbra’, MLG *scheme* m. ‘shade’, MDu. *scheme* ‘shine, shade’⁷⁶⁹ (⇒ EDu. *schemel* ‘umbra’)
-
- **skaima-*: MHG *scheim* m. ‘gleam’⁷⁷⁰

⁷⁵⁷ K. Praust apud Lubotsky 2001: 323 post scriptum.

⁷⁵⁸ Not **skī-ra-*, Franck/Van Wijk: 577.

⁷⁵⁹ Hofmann/Popkema 440.

⁷⁶⁰ Zantema 890.

⁷⁶¹ Franck/Van Wijk 557.

⁷⁶² Böðvarsson 862.

⁷⁶³ Lexer 2, 742.

⁷⁶⁴ Verdam 521.

⁷⁶⁵ Bosworth/Toller 832.

⁷⁶⁶ De Vries 1962: 492.

⁷⁶⁷ Bosworth/Toller l.c.; Holthausen 1934: 279.

⁷⁶⁸ Lexer 2, 698, 742.

⁷⁶⁹ Verdam 516.

⁷⁷⁰ Lexer 2, 687.

On the basis of the evidence, it is theoretically possible to reconstruct an ablauting *mn*-stem. The full-grade form **skīman-* is found in Go. *skeima*, Icel. *skími*, the zero grade **skiman-* in MHG, MLG, MDu. *scheme*.

Determining the vowel length is problematic in Old Norse, because length is not (systematically) indicated in this language. De Vries and Fritzner, as a result, give ON *skimi* ‘shine, light’ with a short vowel. However, the vocalism of Icel. *skími* rather suggests that the **i* was long.

For West Germanic, the handbooks traditionally differentiate between **skīman-* and **skiman-* on the basis of the semantic difference between MHG *schīme* ‘shine’ and *scheme* ‘shade’. This difference was projected back into Old Saxon by, for instance, Sehrt (1950). Bosworth and Toller similarly differentiated between OE *scīma* ‘splendor’ and *scima* ‘shadow’. It can accordingly be assumed that **skīman-* and **skiman-* are independent deverbal derivations from the strong *n*-present **skīnan-* ‘to shine’ (cf. Go. *keinan* ‘to germinate’, past ptc. *kijans*). This is further supported by the occurrence of the apparently deverbal *o*-grade MHG *scheim* < **skaima-*.

Etymologically, all words are related to e.g. Gr. σκιά f. ‘shade’ < **skh₁-i-h₂-* and Skt. *chāyā-* f. ‘gleam’ < **skéh₁-i-*.

****snībō, *snippaz* ‘pointy nose, snipe’?**

- **snīp^{ōn}-*: ON, Icel., Far. (*mýri-*)*snípa* f. ‘snipe’, Nw. *snipe* f. ‘snipe, small boat, dial. bill, northern pike’, ME *snipe* ‘snipe’
- **snīp^{a(n)}-*: Icel. *snípur* m. ‘penis, clitoris’⁷⁷¹, Far. *snípi* m. ‘pointy nose’⁷⁷²
- **snippa(n)-, -ōn-*: Far. (*nasa-*)*snippur* m. ‘tip (of the nose)’⁷⁷³, Nw. *snipp* m. ‘long tip, collar’, OHG *snepfō* m., -a f. ‘snipe’, MHG *snepfē*, G *Schnepf* m. ‘snipe, tip, edge’⁷⁷⁴, *Schnepfe* f. ‘snipe, tip’⁷⁷⁵, OS *snippa* f. ‘id.’, MLG *snippe* ‘snipe, shoe tip’⁷⁷⁶ (= Da. (*myre-*)*sneppe*, *snippe* ‘snipe, snout, longspine

⁷⁷¹ De Vries 1962: 525; Böðvarsson 920.

⁷⁷² De Vries l.c.; Böðvarsson l.c.; Poulsen 1097.

⁷⁷³ Poulsen l.c.

⁷⁷⁴ Grimm 15, 1335.

⁷⁷⁵ Grimm 15, 1313-4; Kluge/Seebold 819.

⁷⁷⁶ Lübben 360.

bellowfish'⁷⁷⁷), MDu. *sneppe*, *snippe* f. 'snipe', EDu. *sneppe* 'id.', Du. *snip* 'id.'⁷⁷⁸

- **snipan-*: Du. *sneep* 'carp'⁷⁷⁹
 ⇒ **snepila-*: MLG *sne(p)el* m. (= G *Schnäpel*, *Schnepel*) 'whitefish'⁷⁸⁰
- **snibbōn-*: MLG *snebbe*, *snibbe* f. 'bill'⁷⁸¹, G *Schneppe*, *Schnibbe*, *Schnippe* 'bill, tip, snipe'⁷⁸², EDu. *snebbe* 'rostrum avis', Du. *sneb* 'bill'⁷⁸³

A comparison of Germanic words for 'snipe' reveals a sharp division between "Anglo-Nordic" and the German dialects. ON *mýri-snípa* 'moor-snipe' and ME *snīpe* 'snipe' point to a form **snīpōn-*, OHG *snepfō*, *snepfā*, MLG, MDu. *sneppe*, *snippe* to **snippan-*, *-ōn-*. The OED (s.v. *snipe*) calls the relation between the two different forms "not clear", but Lühr (1988: 320) considered the possibility that they sprang from a single, ablauting paradigm. In the end, however, she rejected this possibility, separating the Anglo-Nordic form **snīpōn-* from the German **snippōn-*.⁷⁸⁴

It is generally accepted that the meaning 'snipe' evolved out of an older word meaning 'pointy nozzle' or 'bill', which is one of the most prominent features of the bird. A semantic parallel can be adduced from French, where *bécasse* 'snipe' is indeed derived from *bec* 'bill'.⁷⁸⁵ The two meanings can certainly hardly be separated from each other in the Germanic material, cf. Nw. *snipe* 'snipe, long bill, northern pike'.

Importantly, the most primitive meaning is also conveyed by cognates that preserved different consonantisms. MLG *snibbe* 'bill', for instance, proves that the single **p* of *snīpa* represents a shortened

⁷⁷⁷ Falk/Torp 1093.

⁷⁷⁸ Franck/Van Wijk 633.

⁷⁷⁹ WNT, s.v. *sneep*; Franck/Van Wijk 631.

⁷⁸⁰ Lübben 359; Grimm 15, 1311-12.

⁷⁸¹ Lübben 359, 360.

⁷⁸² Grimm 15, 1312, 1316-18, 1335; Mensing 1927: 646.

⁷⁸³ Vercoullie 320.

⁷⁸⁴ Lühr alternatively proposed to derive **snīpōn-* from a strong verb **snīpan-* as in Nw. dial. *snipa* 'to snatch', which is mentioned by Fick/Falk/Torp (p. 523). This *snipa*, however, is absent from the exhaustive *Grunnmanuskriptet* database, and may be a ghost word.

⁷⁸⁵ Cf. Franck/Van Wijk 633; Falk/Torp 1093.

geminate. The same conclusion follows from *sneb(be)* ‘carp’⁷⁸⁶ < **snibban-*, a dialectal variant of Du. *sneep* ‘id.’ < **snipan-*. This fish was apparently named after its prominent nose⁷⁸⁷ (cf. G *Näsling*, *Schnabel* ‘*chondrostoma nasus*’).⁷⁸⁸ With all these different forms, it seems defensible to at least theoretically consider the reconstruction of a paradigm **snībō*, **snippaz*.

In view of OE *snite* f. ‘snipe’, PGM. **snībō* can be analyzed as deriving from a root **snei-* with a labial suffix, possibly PIE *-b^hon-.⁷⁸⁹ A parallel proportion is found in the juxtaposition of OE *clīte* f. ‘colt’s foot’ < **klītōn-* and *clīfe* f. ‘burdock’ < **klībōn-* (see p. 235ff.).

Alternatively, MLG *snebbe*, *snibbe* and Du. *sneb* have sometimes been derived from **snabja-*⁷⁹⁰, and further compared to OHG *snabul*, OFri. *snavel* m. ‘id.’ < **snabla-* and OFri. *snabba* m. ‘mouth’.⁷⁹¹ Although the two roots **snīb-* and **snab-* will certainly have been associated with each other, their origins must ultimately be different. PGM. **snab-* is found in the iterative MLG, MDu. *snappen* ‘to gasp, grasp, snap’⁷⁹² < **snappōn-*, ON *snapa* ‘to bite, snap’ < **snapōn-*, MHG *snaben* ‘to snap, sniff, smack’, MLG *snaven* ‘to stotter, stumble’⁷⁹³ < **snabōn-* and in Du. *snoepen* ‘to nibble’⁷⁹⁴ < **snōp^pōn-*. It has a different ablaut pattern and, unlike **snei-b^hon-*, can hardly be broken down into more basic elements. It is probably related to Lith. *snāpas* ‘bill’ and *snapēlis* ‘nozzle’⁷⁹⁵, if these words were not adopted from Low German in the first place.

⁷⁸⁶ WBD III 4.2, 83.

⁷⁸⁷ Boutkan (1999: 21 fn. 15) argued that *sneep* belongs to the family of ON *snákr*, OE *snaca* ‘snake’, because “variation of labials and velars is also a frequent characteristic of European substrate words”. Since, however, the *Benennungsmotiv* “nose fish” has excellent parallels, the derivation from **snībō*, **snippaz* must be preferred.

⁷⁸⁸ The same consonant can perhaps be established on the basis of Icel. *sniff(a)*, Nw. *snive*, Da. *snive*, *snibe* ‘equine nose condition’. Yet the Danish doublet probably indicates that the word is identical to ON *snípa* (in the sense of ‘nose?’), and that the Danish form *snive* spread to the other Nordic languages.

⁷⁸⁹ Cf. Vercoullie (p. 321): **sneip-*.

⁷⁹⁰ Vercoullie 320; De Vries 1962: 525.

⁷⁹¹ Hofmann/Popkema 451.

⁷⁹² Franck/Van Wijk 629.

⁷⁹³ Grimm 15, 1070; Lexer 2, 1022; Lübben 359.

⁷⁹⁴ Franck/Van Wijk 634.

⁷⁹⁵ Fraenkel 851-2.

The extra-Germanic connection with Ir. *naosga* (or rather *naoscach*) f. ‘snipe’, which has been derived from **snoip-sk-eh₂*-⁷⁹⁶ is improbable, since an initial *s* is normally not lost before *n* in Irish.

****strīmō, *strimenaz* ‘stripe, streak’?**

- **strīman-*: OHG *strīmo* ‘tractus’, MHG *strīme*, *streime* m. ‘stripe, streak’⁷⁹⁷, G *Strieme*⁷⁹⁸, Swi. Visp. *štriimo* m. ‘streak’, MLG *strīme* m. ‘streak, stripe’⁷⁹⁹, MDu. *strieme* m. ‘stripe, streak’⁸⁰⁰, Du. *striem* ‘streak’⁸⁰¹
- **strīman-*: MLG *streme* m. ‘streak, lash’ (⇒ *stremel* m. ‘strip of cloth, paper’)⁸⁰², EDu. *streme* ‘linea, filum, tractus’

The Germanic dialects provide evidence for two ablauting *mn*-stems meaning ‘streak, stripe’, viz. OHG Notk. npl. *strīmen*, dpl. *strīmon*, Visp. *štriimo* < **strīman-* vs MLG *streme* < **strīman-*. On the basis of these and other forms, a paradigm **strīmō, *strimenaz* < **strēi-mn, *stri-mén-s* can at least theoretically be reconstructed. There is a risk, however, that both the full-grade and the zero-grade *mn*-stems were created independently to a lost verb **strīan-*. Etymologically, the root of this hypothetical verb can be related to Lat. *stria* f. ‘furrow, channel’.

Some of the attested forms are difficult to interpret. The combination of MHG *strīme, streim(e)*, G *Strieme*, for instance, is problematic for its vocalism. At first sight, G *Strieme* seems to point to a zero grade **strīm-*. However, a short **i* should have produced reflexes with *e*-vocalism in this environment (cf. MHG *scheme* < **skīman-*). It has been argued, for this reason, that the German as well as the Dutch forms with *-ie-* go back to a lengthened grade **ēi* (cf. Franck/Van Wijk l.c.), yielding a vowel that merged with **ē²*, but this is all very improbable.

In view of the identical wavering of the vowel length in OHG *chīmo*, MHG *kīme, kieme*, G *Keim*, EDu. *kieme, kijme*, Du. *kiem* < PGm. **kīman-*

⁷⁹⁶ Falk/Torp 1093; Fick/Falk/Torp 523.

⁷⁹⁷ Lexer 2, 1230.

⁷⁹⁸ Grimm 19, 1601-9; Kluge/Seebold 891.

⁷⁹⁹ Lübben 386.

⁸⁰⁰ Verdam 583.

⁸⁰¹ Franck/Van Wijk 676.

⁸⁰² Lübben 385.

'germ', it is probably best to assume that the long **ī* was shortened before *m* in dialectal German and Dutch. The phonetic rationale for this shortening is the inherent length of the phone *m*, which due to the required labial closure takes more time to pronounce than, for instance, a dental nasal. Thus, the intrinsic length of the *m* explains word pairs such as MHG *gumme* vs *guome* 'palate', Du. *blom* vs *bloem* 'flower', as well as the lack of open syllable lengthening in G *kommen* < **kuman-*, *Himmel* < **hemila-* etc.

The form *streime* occurs only in late Middle High German, Modern Bavarian and Swabian (Grimm 19, 1304), and probably exhibits early diphthongization of **ī*. There is therefore no evidence for an *o*-grade variant **straiman-*.⁸⁰³

****swīmō, *swimenaz* 'dizziness'†**

- **swīman-*: ON, Icel. *svími*, *svimi* m. 'dizziness' (in *vaða í villu og* ~ 'to be on the wrong track')⁸⁰⁴, OS *swīmo* m. 'giddiness', Du. *zwijsm* 'swoon', OFri. *swīma* m. 'unconsciousness', OE *swīma* m. 'dizziness, giddiness'⁸⁰⁵
- **swaima-*: ON *sveimr* m. 'stir', *sveim* n. 'id.', Far. *sveim* n. '*tað að sveima*', MHG *sweim* m. 'floating, sway'⁸⁰⁶

The different formations Icel. *svími*, Du. *zwijsm* < **swīman-* and Icel. *svimi* < **swiman-* are in clear ablaut correlation with each other, and can therefore be adduced to reconstruct an *mn*-stem **swīmō*, **swimenaz* < **suéi-mn*, **swi-mén-s*. In view of Icel. *svía* 'to diminish, abate'⁸⁰⁷, it is more probable, however, that the different formations were created independently to a verb **swīan-*. Similarly, ON *sveimr* < **swaima-* is best analyzed as an independent *mo*-stem to the same verb.

The Germanic forms are most probably related to a range of Celtic formations, e.g. W *chwil* < **swi-lo-* 'turning', *chwyf* m. 'movement' < **swi-*

⁸⁰³ Pace Fick/Falk/Torp 500.

⁸⁰⁴ De Vries 1962: 570; Böðvarsson 1009.

⁸⁰⁵ Bosworth/Toller 957.

⁸⁰⁶ Lexer 2, 1353.

⁸⁰⁷ Böðvarsson 1007.

*mo*⁸⁰⁸ etc. Kümmel and Rix (2001) further assume the root to be an extension of a more primary base **sueh₁-*, which can be reconstructed on the basis of MLG *swāien* ‘to swing, sway’ and Ru. *xvéjat’ sja* ‘to move’. Still, the Du. Stw. form *zwaaien* cannot regularly have developed out of **swējan-* with PGM. **ē*, because the verbs *mi’jen* ‘to mow’ < **mē(j)an-*, *ni’jen* ‘to sew’ < **sē(j)an-* and *dri’jen* ‘to turn’ < **prē(j)an-* demonstrate that this should have become ***swi’jen*. Instead, it could be reconstructed as **swanhjan-*, which is possibly akin to **swingan-* ‘to swing’.

****swīrō, *swirraz* ‘neck, mooring-mast’**

- **swīran-*: ON *svíri* m. ‘neck, ship’s beak’⁸⁰⁹, Far. *svíri* m. ‘thick neck’⁸¹⁰, Sw. obs. *svire* ‘pig’s neck; ship’s beak’⁸¹¹
- **swiran-*: OE *swe(o)ra*, *swura* m. ‘neck’, Swi. Visp. *šwiro* ‘post, stake’
 ⇔ **swirō(ja)n-*: OHG *swirōn* ‘to fasten’
- **swira-*: MHG *swir* m. ‘mooring mast’⁸¹², G dial. *schwier* ‘bridge post’⁸¹³, OE *swe(o)r* m. ‘column, pillar’⁸¹⁴
- **swirra(n)-*: MHG *swirre* m. ‘mooring-mast’⁸¹⁵, G dial. *schwir(re)n* ‘post, bridge post’⁸¹⁶

There are strong indications that the above forms go back to an *n*-stem **swīrō, *swirraz* with consonant and vowel gradation.

The full-grade stem **swīran-* is attested in North Germanic, e.g. ON *svíri* ‘neck (esp. of an ox), curled ship’s beak’. With the exception of Faroese, where *svíri* means ‘thick neck (esp. of cattle)’, the word is absent from the modern Nordic languages. Sw. *svire* has gone out of use, but according to SAOB, it meant ‘pig’s neck’ and ‘ship’s beak’, which is close to the semantic field of the Old Norse word. The zero grade is evinced by

⁸⁰⁸ Pokorny 1041-2.

⁸⁰⁹ De Vries 571.

⁸¹⁰ Poulsen 1187.

⁸¹¹ SAOB S15202.

⁸¹² Lexer 2, 1318.

⁸¹³ Grimm 15, 2619.

⁸¹⁴ Bosworth/Toller 949.

⁸¹⁵ Lexer 2, 1318.

⁸¹⁶ Grimm 15, 2716.

MHG *swirre* ‘mooring mast’, which goes back to a geminated stem **swirran-*. The additional MHG form *swir* ‘id.’ has a singulate and a thematic inflection. It seems to be close to Visp. *šwiro* ‘post’ < **swiran-*, which preserved the *n*-stem inflection. The same root is further found in the Old High German verb *swirōn* ‘to fasten’ < **swirō(ja)n-*.

The semantic bifurcation between ‘neck’ and ‘ship’s beak’ is explained by Fritzner as from an original sense ‘mooring mast’, either on a boat or along the shore⁸¹⁷. In view of MHG *swirre* ‘mooring-mast’, which preserves this semantic primitive, Fritzner’s interpretation can be accepted. A semantic parallel can be adduced from the Celtic languages, where Mlr. *farr* f. ‘post’ corresponds to W *gwar* f. ‘neck’. Both words can be traced back to a proto-form **urs-eh₂-*, which may belong to the root **uers-* ‘high’ as in Skt. *varṣmán-* m. ‘height’ (cf. Pokorny 1151-2).

An important issue is the position of the Old English forms *swe(o)ra* ‘neck’ and *swe(o)r* ‘pillar’. It is generally acknowledged that these words correspond to the North and West Germanic material, but there is widespread disagreement over the vowel length, which is not indicated in the Old English manuscripts. Pokorny (p. 1050) and Holthausen (1934: 335) reconstruct long diphthongs in *swēora* and *swīor*. Fick/Falk/Torp (p. 550) have *swēora* ‘neck’ as opposed to *swēor* ‘post’, and, conversely, De Vries (p. 571) gives OE *swīora* vs *swēor*. Bosworth and Toller (p. 949), on the other hand, give short vowels in both instances, i.e. *swēora* and *swēor*. The variant form *swura*, mentioned by Bosworth/Toller, is omitted from the etymological dictionaries, but seems to be of crucial importance for determining the original vowel length. It clearly displays the late Kentish change of *-weo-* to *-wu-*, as in *sweoster* > *swuster* ‘sister’, *sweord* > *swurd* ‘sword’ (cf. Wright 1925: § 94), and since this change applies to short *eo* only, the corresponding form *sweora* must likewise have had a short vowel (pace Mitchell/Robinson 2001: 376). Hence, OE *swe(o)r(a)*, in spite of its meaning ‘neck’, turns out to be formally identical to Visp. *šwiro* ‘post’ < **swiran-*.

All things considered, it seems that the original meaning of the ablauting *n*-stem **swirō*, **swirraz* was ‘mooring mast’, and that the semantic development into ‘neck’ is an “Anglo-Nordic” isogloss. For Indo-European, the inflection **swīro*, **swirraz* presupposes earlier **swéir-ōn*, **swir-n-ós*. Older reconstructions such as **swerhjan-* and

⁸¹⁷ Cf. Bugge 1879: 110.

**sweriha-*, which are found in all the etymological dictionaries, were inspired by the alleged link with Lat. *surculus* 'twig' and Skt. *svāru-* 'post', which can no longer be maintained.

****tīgō, *tikkaz* 'tick'**

- **tīgan-*: Du. dial. (Kumtich) *tijg* 'id.'⁸¹⁸
- **tīk^kan-*: OE **tīca* (= *ticia*) 'id.', ME *tīke* 'id.', E obs. *tyke* 'sheep-tick', Du. dial. (Fijnaart) *schape-tijk* 'id.' (= Fr. *ticque*?)⁸¹⁹
- **tikan-, -ōn-*: OHG *zehho* '*ricinus, cimex*', MHG *zeche* m. 'id.', G Cimb. *zecho* m. 'spider'⁸²⁰, Swi. Visp. *zäxxo* m. 'tick', MLG *teke* 'id.', EDu. *teecke* 'id.', Du. *teek* 'id.', WFri. *tyk* 'id.'⁸²¹, SFri. *tieke* f. 'id.', NFri. *teg* f. 'id.'⁸²², ME *teke*, E *tick*
- **tikka(n)-, -ōn-*: Nw. dial. *tikk* m., *tikke* f. 'id.', MHG *zecke* m. 'id.', G *Zecke* 'id.'⁸²³

The word for tick displays the typical features of an ablauting *n*-stem. The etymological dictionaries distinguish between three different stem forms, i.e. **tikkan-*, **tikan* and **tīkan*⁸²⁴, but do not seek to clarify the relationship between these forms. Only Fick, Falk and Torp (p. 163) mention the possibility that the root variation can be the result of the *n*-stem inflection.

The reconstruction of the three different root variants is relatively straightforward. The first variant **tikkan-* can be mechanically reconstructed on the basis of MHG *zecke*, G *Zecke* etc. A second variant **tikan-* is evidenced by OHG *zecho*, MHG *zeche*, MLG, MDu., ME *teke*⁸²⁵. It

⁸¹⁸ WBD III, 4, 2.

⁸¹⁹ Wartburg (1966: 329): "Gam[milscheg] Germ 1, 245 möchte aus fr. *ticque* ein anfrk. **tīka* erschliessen. Doch ist diese Form wenig wahrscheinlich, da das mndl. nur *teke*, *teecke* kennt, das auf *ī* weist."

⁸²⁰ Schmeller/Bergmann 181.

⁸²¹ Zantema 1, 1050.

⁸²² Jensen 618.

⁸²³ Kluge/Mitzka 876-7.

⁸²⁴ Cf. Pokorny 187-8; Franck/Van Wijk 690; Kluge/Mitzka 876-7; OED, sv. *tick*.

⁸²⁵ According to the OED, English *tick* can have developed out of ME *teke* by a similar shortening as found in *sick* < OE *sēoc* < **seuka-*.

is interesting to see that, in Cimbrian, both variants occur side by side as *zecko* and *zecho*⁸²⁶. The retention of two forms resulted from a semantic split in this dialect, where *zecko* and *zecho* mean ‘tick’ and ‘spider’ respectively⁸²⁷.

The variant **tīk^kan-* can only be obtained with some effort, because the attestations on which it is based are generally obsolete, dialectal or ambiguous. The Old English gloss *ticia*, for instance, can be read as either *ticca* or *tiica*⁸²⁸, which renders it indecisive. Similarly, the Saterland Frisian form *tieke* can represent PGm. **tīk^kan-* just as much as **tikan-*.⁸²⁹ In the end, the dictionaries seem to rely solely on ME *tike* and E *tyke* ‘sheep-tick’⁸³⁰ for their reconstruction of **tīkan-*. The long vowel, however, finds additional support in the Dutch form (*schape*)-*tijk* ‘(sheep)-tick’, which occurs in the dialects of Brabant and Limburg⁸³¹. With the addition of these forms, the basis for the reconstruction of **tīk^kan-* becomes sufficiently reliable.

Having arrived at a range of three forms, i.e. **tīk^kan-*, **tikan-* and **tikkan-*, it is not difficult to recognize the root variation that is so typical for the ablauting *n*-stems; there is a clear opposition between a full-grade **ī* and a zero-grade **i*, and the opposition between singulate and geminate consonants is in agreement with Kluge’s *Associationen*. Yet, the etymological dictionaries do not establish a link between the consonantal and vocalic interchanges on the one hand, and the inflection of ‘tick’ as an *n*-stem on the other. Franck/Van Wijk (p. 690), Kluge/Mitzka (p. 876) and Falk/Torp (p. 1311) fail to explain the geminate of **tikkaz*, and Pokorny (p. 187-8) expectedly attributes it to “Intensivgemination”. The only dictionary that mentions the possibility of nasal assimilation is Fick/Falk/Torp (p. 163), but even this dictionary hesitates between reconstructing PIE **dīg^h-n-* and **d(e)īg-*.

⁸²⁶ If Nw. *tikk(e)* is not a loanword from Low German, it proves that the word occurred in North Germanic as well.

⁸²⁷ Schmeller/Bergmann 181.

⁸²⁸ OED, s.v. *tick*; Franck/Van Wijk 690; Falk/Torp 1311.

⁸²⁹ Cf. *uut-wieke* ‘evade’ < **wīkan-* vs *stiekel* ‘prickle’ < **stīkila-*.

⁸³⁰ MED, s.v. *tīke*; Wright 1869: 988.

⁸³¹ The exact forms are not included in the printed versions of WBD and WLD, but can be looked up in the source material on which these publications are based. The source material is available online at www.ru.nl/dialect/wbd and www.ru.nl/dialect/wld.

Indeed, the forms **tīkan-* and **tikan-* outwardly point to a PIE root **d(e)ig-*. The problem is that PIE phonology did not allow roots with two voiced (glottalized?) stops. Moreover, the reconstruction of the root as **deig-* is conflicted by Arm. *tiz* 'tick', which together with Mlr. *dega*, asg. *degaid* 'stagbeetle' points to PIE **d(e)ig^h-*. By way of a solution, Falk and Torp parenthesize the aspiration, supposing a double root **deig^(h)-*. Franck and Van Wijk even go so far as to completely reject the link between the Germanic and Armenian word. It is clear, however, that the consonantism of PGm. **tikan-* and **tikan-* is secondary; the single **k* was most probably introduced analogically on the basis of the genitives **tikkaz* and **tikkan* < **diǵ^h-n-ós* and **diǵ^h-n-óm*.

Direct proof of a PGm. variant **tīg-* is furnished by the *Woordenboek van de Brabantse dialecten*, which in a recent volume on animal names makes mention of the variant *tijg* 'tick' (p. 234). Although this variant is isolated, it directly corresponds to its Arm. cognate *tiz*. This is an additional indication that the original PGm. paradigm was **tīgō*, **tikkaz* rather than **tīkō*, **tikkaz*. For Indo-European, the paradigm must accordingly be reconstructed as **deig^h-ōn*, **diǵ^h-n-ós*, **diǵ^h-én-i*.

****twīgō*, **twikkaz* 'twig'?**

- **twīga(n)-*: OHG *zwīg* mn. 'branch', MHG *zwīc* m. 'id.', G *Zweig* 'id.', obs. *Zweige*⁸³², MLG *twīch* n. 'id.', MDu. *twijch* mn. 'id.', Du. *twijg* 'id.', SFri. *twiech* m. 'id.'
- **twīga(n)-*: EDa. *tvege*, *tvige* 'branch, two-pronged fork'⁸³³, Da. *tvege* 'forked twig'⁸³⁴, OE *twig* n., *twiga* m. 'twig, sprout'⁸³⁵
- **twiggōn-*: Da. *tvegge* f. 'branch'⁸³⁶, OE *twigge* f. 'id.'⁸³⁷, LG *twig* 'id.'⁸³⁸
- **twikka(n)-*: OHG *zwech(o)* 'clavus, paxillus', MHG *zwec* m. 'nail, bolt, twig'⁸³⁹, G *Zweck* m. 'nail, bolt, aim', *Zwick*⁸⁴⁰ m.

⁸³² Grimm 32, 1036ff.; Kluge/Mitzka 895.

⁸³³ Kalkar 490.

⁸³⁴ Falk/Torp 1302.

⁸³⁵ Holthausen 1934: 357.

⁸³⁶ ODS, s.v. *tvege*.

⁸³⁷ Holthausen 1934: 357.

⁸³⁸ Rosemann/Klöntrup 329.

‘plug, flagellum, sprout’ (⇒ G *Zwickel* ‘wedge’⁸⁴¹), Swi. Visp. *zwäck* ‘hobnail’, LG WPhal. *twick* m. ‘twig’⁸⁴²

- **twikkōn*–: G *Zwecke*, *Zwicke* f. ‘nail, plug, sprout’⁸⁴³
- **twikōn*–: LG WPhal. *twiak* f. ‘twig’⁸⁴⁴

A close inspection of the predominantly West Germanic word for ‘twig’ yields a number of different root variants that together point to an apophonic *n*-stem **twīgō*, **twikkaz*.

A full grade is found in OHG *zwīg*, G *Zweig*, MDu. *twijch*, Du. *twijg* < **twīga*–. The *n*-stem inflection of the full-grade root is only marginally attested as obsolete G *Zweige*.

The zero grade is found in a number of formations with different consonantisms. In Anglo-Nordic, the root ends in a voiced stop, cf. OE *twig(a)*, Da. *tvege* < **twigan*–. WFri. *twige*, *twiich* may, too, belong here, but the original vowel length is uncertain. A root with a voiced geminate is supported by OE *twigge*, E *twig* < **twiggōn*–. It has a correspondence in LG *twig* and possibly also in Da. *tvegge*, if this word is not borrowed from Low German. Most German dialects have a voiceless geminate, cf. OHG *zwech(o)*, G *Zwecke*, *Zwicke* ‘nail, plug’⁸⁴⁵, WPhal. *twick* ‘twig’. Finally, WPhal.n *twiak* (with lengthened **i*) combines a zero grade with an analogical singulate **k*.

Regarding the etymology of the word, the literature agrees on the link with the number ‘2’, and usually Skt. *dviká*– ‘twofold’ is compared⁸⁴⁶. Franck/Van Wijk stressed that the PGm. **g* can be from both PIE **k* and **g*^h, and indeed Pokorny (IEW: 228-232) chooses to reconstruct **duei-g*^h– in view of Lith. *dveigỹs* ‘two year old animal’ and Alb. *degë* f. ‘branch’ <

⁸³⁹ Lexer 3, 1204.

⁸⁴⁰ Grimm 32, 1109-10.

⁸⁴¹ Grimm 32, 1112-4; Kluge/Mitzka 896.

⁸⁴² Woeste 377.

⁸⁴³ Grimm 32, 964; Grimm 32, 1111.

⁸⁴⁴ Woeste 1882: 277.

⁸⁴⁵ These meanings are secondary, and have developed out of the more original meaning ‘twig’. In order to illustrate this, Kluge and Mitzka (p. 894) cite from Rollenhagen’s *Froschmeuseler* (1595), in which a raven sits down on “ein durren zweck”, i.e. ‘a dry twig’. According to Grimm (32, 1110), the meaning ‘sprout’ is also attested for *Zwick*.

⁸⁴⁶ Fick/Falk/Torp 173; Franck/Van Wijk 716; Kluge/Mitzka 895.

**dueig^h*- or **duōg^h*-⁸⁴⁷. The Germanic material contains no evidence for PIE **k*, which makes the reconstruction **duéig^h-ōn*, **duig^h-n-ós* all the more straightforward. The association with OHG *zuogo* 'branch'⁸⁴⁸ is a persistent misconception, and must be reconsidered. This formation actually belongs to a different ablauting *n*-stem, i.e. **tōgō*, **takkaz* < **déh_{2/3}g^h-ōn*, **dh_{2/3}g^h-n-ós*, for which see p. 317.

****wīō*, **wiwaz* 'kite'**

- **wīwan*:- OHG *wī(w)o* 'milvus, asida', MHG *wī(w)e* m. 'harrier'⁸⁴⁹, G *Weihe* f. 'id.', MDu. *w(o)uwe(r)* 'kite, harrier'⁸⁵⁰, Du. *wouw* 'kite'⁸⁵¹
- **wiwan*:- OHG *weho*, *wewo* 'ibis, laoficus', MHG *wehe*⁸⁵², *wewe*⁸⁵³ m. 'harrier', Cimb. *bibo* m. 'id.'⁸⁵⁴

Scrutiny of the West Germanic dialects shows that the word for 'kite, harrier, ibis' qualifies as an ablauting *n*-stem. The predominantly masculine formation appears both as a full-grade stem **wīwan*- and as a zero-grade stem **wiwan*-. The original paradigm can therefore hypothetically be reconstructed as **wīwō*, **wiweni* from older **uéiu-ōn*, **uiiu-én-i*. The regular Proto-Germanic outcome of the genitive **uiiu-n-ós*, i.e. **ujunaz*, would probably have quickly been removed from the paradigm.

The long **ī* of the full-grade form **wīwan*- is most clearly visible in MHG *wī(w)e* and German *Weihe*, the vowel length of the Old High German attestations being uncertain. It is further ascertained by the Low Franconian evidence, viz. MDu. *wouwe* and Du. *wouw*, which show regular rounding of **ī* to **ū* before a contiguous labial glide, cf. MDu. *w(o)uwere* 'pond' < Lat. *vīvārium* (cf. G *Weiher*) and *sp(o)uwen* 'to vomit'

⁸⁴⁷ Demiraj 125.

⁸⁴⁸ Cf. Franck/Van Wijk 716; Fick/Falk/Torp 173; Pokorny 228-232.

⁸⁴⁹ Lexer 3, 876.

⁸⁵⁰ Verdam 811.

⁸⁵¹ Franck/Van Wijk 804.

⁸⁵² Benecke 4, 548.

⁸⁵³ Lexer loc. cit. = Michael Beheim (1416-±1476): "der adelar wil sich verkêren und neuen - - er ist worden zuo einem w e w e n".

⁸⁵⁴ Schmeller 111.

< **spīwan-* (cf. Go. *speiwan* 'to spit').⁸⁵⁵ The intervocalic **w* was regularly lost in OHG *wīo*, the apparent precursor of Modern German *Weihe*.⁸⁵⁶

The zero-grade form **wīwan-* is less frequent in the German dialects, but it already emerges alongside the full-grade form as *wanne-weho* 'kestrel' in the Old High German period. It lives on as Middle High German *wannen-wehe*⁸⁵⁷ and Modern German *Wannenweher*⁸⁵⁸, both with the same meaning. The simplex appears as OHG *weho*, *wewo* 'ibis', MHG *wehe*, *wewe* 'harrier' and – remarkably – as Cimb. *bibo* 'id.'. This *bibo* shows the regular change of MHG *w* > Cimb. *b*. Apparently, intervocalic **w* was not lost in this dialect.⁸⁵⁹

The *n*-stem **wīwan-* is usually connected with a Nordic word for 'auk' or 'murre', a fishing bird of the family that also includes the puffin genus, e.g. Icel. *lang-víi* m., -*vía* f. 'murre'⁸⁶⁰, Nw. *lang-vi*, dial. -*vie* m. 'id.'. Superficially, the word even seems to mirror the West Germanic ablaut of long and short **i* in view of ON *lang-vé* m. 'auk'⁸⁶¹, Nw. *lang-ve* m. 'murre' < **wewan-* (cf. *klé* 'warp weight' < **klewan-*) < **wīwan-*. The link with **wīwan-* nevertheless poses some problems. First, the semantic difference between 'auk' and 'harrier' is quite a gap to bridge. Second, it follows from instances such as ON *ýrr* 'ivy' < **īwa-* and *Týrr* 'Tyr' < **tīwa-* that the regular outcome of **wīwan-* should have been ***ýi* (with loss of initial *v* before a rounded vowel) rather than *víi*. The Nordic evidence, in other words, points to **wīan-* and **wiwan-*.

In order to resolve the cross-dialectal incongruities, the original paradigm can be reconstructed as **wīō*, **wieni* on the basis of Icel. *víi*. This does not help much, however, because the *w* of OHG *wīwo*, MHG *wewe* and Cimb. *bibo* must then be intrusive. There seem to be no parallels to such a development, however. Therefore, a different solution must be found in order to reconcile the North and West Germanic formations.

⁸⁵⁵ Cf. Franck/Van Wijk s.v. *wouw*: "voor 't vocalisme vgl. s p u w e n".

⁸⁵⁶ Cf. Braune 1891: § 110, n. 1: "So findet sich *grâêr*, *êa*, *sêes*, *spīan* statt *grâwer*, *êwa*, *sêwes*, *spīwan*, auch im Lehnwort *wīwari* und *wīari* (Vivarium, Weiher)."

⁸⁵⁷ Benecke loc. cit.

⁸⁵⁸ Grimm 27, 1908.

⁸⁵⁹ It follows that the reconstruction **wīhan-* as proposed by Fick/Falk/Torp (p. 407) cannot be maintained.

⁸⁶⁰ Böðvarsson 559.

⁸⁶¹ De Vries 345-6.

Perhaps, both the forms with and without intervocalic *w* are old. There are some Indo-European bird names with *u*-suffixes, often alternating with variants with *n*-suffixes, cf. Lat. *corvus* m. 'raven' < **korH-u-* beside Lat. *cornīx* f. 'crow' < **korH-n-* and SCr. *žērāv* 'crane' < **đerH-ōu* beside Gr. γέρην 'id.' < **đerH-ēn*. The 'crane' word is especially interesting in this context because it demonstrates the presence of ablauting *u*-stems in Indo-European (see p. 307), in this case **đerH-ōu*, **grH-u-ós* (with depalatalization of **ǵ* before *r*). The ablaut of the suffix was preserved, at least, by Balto-Slavic, as is revealed by the co-occurrence of SCr. *žērāv* < **gerH-ōu-* and Lith. *gėrvė* < **gerH-u-ieh₂-*. With this in mind, it is theoretically possible to assume that the Germanic word for 'harrier' continues an ablauting *u*-stem as well, viz. **uēi-ōu*, gen. **ui-u-ós*. This paradigm could have regularly developed into PGm. **wīō*, **wīwaz* by the loss of **u* after long **ō*. On the basis of the nominative, this paradigm may further have gone over to the *n*-stems, just like PIE **népōt*, gen. **np-t-ós* 'grandson, nephew'⁸⁶² after the loss of the final dental (cf. ON *nefi*, OE *nefa*, OHG *nevo*). Both the roots with and without final **w* can be explained from the paradigm **wīō*, **wīwaz* by assuming leveling of either the nominative or the genitive root. In this way, **wīwan-* may have arisen by the adoption of the genitive root **wīw-*.

Outside Germanic, **wī(w)an-* has been linked to Lat. *avis* m. 'bird', Skt. *vé-*, *ví-* m. 'id.' < PIE **h₂éu-i-*, **h₂u-éi-* and OIr. *fíach* 'raven' < **uei-ko-*⁸⁶³, but this all seems very doubtful. The Nordic word can perhaps be connected with the Icelandic verb *vía* 'to guard, spy', to which Böðvarsson (p. 1147) adds the illustrative phrase: *örninn víar yfir hræinu* 'the eagle is preying on the carcass'. If this is correct, **wī(w)an-* must ultimately have denoted "prowler". Compare for this sense also the doubtlessly related Icel. *ví* n. 'fly egg, swarm of flies or birds surrounding a cadaver'.⁸⁶⁴ The verb *vía* can further be linked with Skt. *véti* 'to turn to, strife for', Lith. *výti* 'to chase, hunt' (etc.) for which the *Lexikon der indogermanischen Verben* reconstructs **ueih₁-* 'sein Augenmerk richten auf'.

⁸⁶² The alpha of Gr. ἀνεψιός 'cousin' is probably comparable to the one of ἀδελφός 'brother'.

⁸⁶³ Cf. Fick/Falk/Torp loc. cit.

⁸⁶⁴ Theoretically, this *ví* can also be reconstructed as **h₂u-éi-(i)o-*, so as to align it with **ajja-* 'egg' < **h₂ōu-ió-*.

***wrihō, *wrigeni ‘instep’**

- *wrihan-: OHG *rīho* m. ‘hollow of the knee, instep’, MHG *rihe* m. ‘instep’⁸⁶⁵, G *Reihen*, Als. *rih(ə)* m. ‘instep, coupling of the wagon pole’⁸⁶⁶
- *wriġan-: MDu. *wrijch*, *wrijf*, *wrijghe* m. ‘instep’⁸⁶⁷, EDu. *wrijf des voets* ‘id.’, Du. obs. *wrijg* ‘id.’⁸⁶⁸
- *wrihan-: MHG *riche* m. ‘id.’⁸⁶⁹, Swi. Rhtl. *reahā* m. ‘id.’⁸⁷⁰, Du. dial. *wree* m. ‘id.’⁸⁷¹
- *wriġan-: Du. obs. *wrege*, *wreeg* ‘id.’⁸⁷², Du. *wreef* ‘id.’⁸⁷³

The West Germanic word for ‘instep’ was discussed by Schaffner in his study of the effects of Verner’s law. Schaffner ascribed the *grammatischer Wechsel* to the shifting accent of an old *n*-stem (2001: 573-4), i.e. *wriġō, loc. *wriġeni < *urēik-ōn, *ureik-én-i. This paradigm explains the interchange of e.g. G *Reihen* < *wrihan- and MDu. *wrijghe* < *wriġan-. In addition, there is evidence for a zero grade in Swiss and Dutch; Rhtl. *reahā*, Du. dial. *wree* unquestionably continue a formation *wrihan-, which further seems to be supported by the MHG hapax *riche*. In addition, Du. obs. *wrege* must go back to *wriġan-. The original paradigm thus seems to have been apophonic, and can accordingly be reconstructed as *wrihō, *wrigeni < *urēik-ōn, *urik-én-i. It is formally and semantically close to Lith. *riėša(s)* ‘wrist, instep, knuckle’ < *ureik-o-.⁸⁷⁴

⁸⁶⁵ Lexer 2, 431.

⁸⁶⁶ Martin/Lienhart 2, 244b-245a.

⁸⁶⁷ Verdam 810.

⁸⁶⁸ Vercoullie 398.

⁸⁶⁹ Lexer 2, 416.

⁸⁷⁰ Berger 76.

⁸⁷¹ WLD II/10, 23-4; Van Es 1989, 139.

⁸⁷² Vercoullie 398.

⁸⁷³ Franck/Van Wijk 805: “Evenals Kil. *wrijf* ‘wreef’ een jongere vorm, in de plaats gekomen voor mnl. *wriē”.

⁸⁷⁴ The co-occurrence of masculine *riėšas* and feminine *riėša* suggests that the acute accent is due to *métatonie rude*, and that the original form was an oxytone neuter *o*-stem, cf. Lith. *várna*, *vařnas* and – analogically – *várnas* ‘crow/raven’ (Derksen 1996: 253-256). There is therefore no compelling reason to reconstruct a form *urēik-o- with a lengthened grade (pace Schaffner 2001: 574).

The material is especially polymorphic in Middle and Modern Dutch, which in addition to the already mentioned full and zero grades have opaque variants ending in the labio-dental fricative *f*. The different variants seem to have competed with each other through the ages. In Middle Dutch, there are three forms, i.e. *wrijch*, *wrijf* and *wrijghe*, but Kilian only gives *wrijf van de voet*. In the 19th century, *wreeg* appears to have temporarily prevailed over *wreef*, which is called dialectal and obsolete.⁸⁷⁵ In modern Dutch, however, *wreef* has become the only existing form.

The origin of the *f* is not entirely clear. Usually, the *f* is considered to be due to the influence of *wrijven* ‘rub’.⁸⁷⁶ It is more likely, though, that the change of final [x] into [f] resulted from some kind of assimilation at the time when initial [wr-] changed into [vr-] and [fr-]. Another *terminus post quem* for the rise of *f* is the Middle Dutch apocope of final *a*, which led to the devoicing of *g* [ɣ] to *ch* [x].

Etymologically, **wrihō*, **wrigeni* can be compared to ME *wrāh* ‘wrong, stubborn’, Du. *wreeg* ‘stiff’ < **wraiga-* and especially to the iterative verb **wrikkōpi*, **wrigunanþi* (< **urik-n(e)h₂-*): Icel. *riga* ‘to move to and fro’, OE *wrigian* ‘to turn’, OFri. *wrigia* ‘to stumble’, MLG *wriggen*, *wricken* ‘to twist, turn’, Du. *wrikken* ‘to pry, tug’. The meaning ‘wrist’ is also attested for **wrihsti-* > ON *rist*, OSw. *vrist*, OFri. *wrist*, OE *wrist*, *wyrst* f. ‘wrist, instep’ and **wrihtja-* > MHG *riste* n. ‘instep’, Du. *gewricht* n. ‘joint’. It thus seems that the original meaning of the *n*-stem probably was ‘twist’ or ‘joint’. Consequently, the meaning ‘coupling of the pole’ of Als. *rih(ə)* may be old.

Further Indo-European cognates are Gr. *ῥοικός* ‘bent’, Av. *uruuisiia-* ‘to turn’, *uruuāsa-* m. ‘bend’. The meaning ‘to turn’ apparently developed into ‘to wrap’ in many Indo-European languages, cf. Lith. *rišti* ‘to bind, tie’, OPru. *perrēist* ‘to link’, OHG *int-rīhan* ‘to disclose’, OE *wrēon*, *wrīon* ‘to cover’ < **wrihan-*.

9.5 The **ai* ~ **ĩ* type

There are only two *n*-stems that potentially have **ai* : **ĩ* ablaut. These are the Germanic words for ‘squirrel’ and ‘heron’. Both items are special

⁸⁷⁵ De Jager 1837: 471.

⁸⁷⁶ WNT, s.v. *wreef*; Kluge/Mitzka 592.

cases, because the cross-dialectal vowel alternations are probably not based on root ablaut, but rather on the ablaut of the reduplication syllable. Both cases, in other words, appear to be old reduplicated nouns that became *n*-stems at a relatively late stage.

***aikwernō / *ikwernō ‘squirrel’**

- **aikwerna(n)-*: OHG *eihhorn(o)*, *eihhurno* mn. ‘id.’, MHG *eichhorn* n. ‘id.’, OE *ācurna*, *ācwe(o)rn(a)* m. ‘id.’, ME *aquerne* ‘id.’, MLG *ēkern(e)*, *ēk-horn(e)* m. ‘id.’, MDu. *ee(n)coren* mn. ‘id.’, Du. *EEK-hoorn(tje)* ‘id.’, WFri. *iik-hoarntsje* ‘id.’, NFri. *ik-hörn* n. ‘id.’⁸⁷⁷
- **ikwerna(n)-*: ON, Icel., Far. *íkorni* m. ‘id.’, Nn. *ikorn*, dial. *ikorn(a)* n. ‘id.’
- *?*ikwern-*: OSw. *ekorne*, *ikorne* ‘id.’, Sw. *ekorre* ‘id.’⁸⁷⁸, EDa. *egerne* ‘id.’⁸⁷⁹, Da. *egern* n. ‘id.’, Nw. *ekorn* mn. ‘id.’

At least two different proto-forms must be reconstructed for the Germanic word for ‘squirrel’. OE *ācwe(o)rna* and ME *aquerne* straightforwardly point to **aikwernan-*, ON and Icel. *íkorni* to **ikwernan-*. A third variant **ikwernan-* with short **i* could theoretically be reconstructed on the basis of OSw. *ekorne*, EDa. *egerne* and Nw. *ekorn*, but these forms can equally well have developed out of **aikwern-*. The neuter gender of both Nw. (dial.) *ikorn(a)* and Da. *egern*, which is unexpected on the basis of the Old Norse masculine *n*-stem *íkorni*, must have arisen under the influence of Nw. *åkorn(a)*, Da. *agern* n. ‘acorn’.

Unlike OE *ācwe(o)rna* and ME *aquerne*, the West Germanic forms cannot be used to determine the form of the second syllable – whether it was **-wer-* or **-ur-* in Proto-Germanic. OHG *eihhorno* and *eihhurno* superficially seem to continue to a Proto-Germanic variant **aikurnan-*. However, the adjective *nachot*, *nachut* ‘naked’ < **nakwed-* (cf. ON *nǫkkviðr*) provides strong evidence for a reduction of **-we-* in unaccentuated syllables, so that *eihhorno* and *eihhurno* are actually ambiguous. Likewise, late West Saxon *ācurna* probably does not

⁸⁷⁷ Zantema 433; Jensen 226.

⁸⁷⁸ Hellquist 116.

⁸⁷⁹ Kalkar 446.

continue **aikurnan-* either, but rather represents a reduced form of older *ācwe(o)rna*.

ON *íkorni* poses a similar problem. On the one hand, the already mentioned *nǫkkviðr* proves that **-we-* was not necessarily simplified word-internally in Old Norse. This means that the Old Norse form can theoretically be reconstructed as **ikurnan-* with a zero grade. The case of *nǫkkur* ‘someone’ < **nakwer-* (from older **ne veit ek hvern*), on the other hand, clearly shows a reduction of **-we-* in middle syllables. It therefore seems best to stick to the reconstruction **ikwernan-* even though a reconstruction **ikurnan-* cannot entirely be excluded.

OHG *eichorno* and *eihhurno* are continued by MHG *eichhorn* and G *Eichhörnchen*. Just like MLG *ēk-horn(e)* and Du. *eek-hoorn(tje)*, these forms have an unetymological *h*. Apparently, the word was reanalyzed as a compound of **aik-* ‘oak’ and **hurna-* ‘horn’ in many dialects, a development that seems to have depended on the *sandhi* deletion of *h* after consonants. The popular etymology of **aikwernan-* as **aikhurnan-* is probably also the reason why the word became neuter in some of these languages.

Etymologically, WGM. **aikwernan-* looks like an old compound. Falk & Torp (p. 186) analyze the word as **aik-wernan-* from **aik-* ‘oak’⁸⁸⁰ and **werna-* ‘weasel’ (or rather ‘squirrel’). This *werna-* has various cognates, predominantly in the West Indo-European languages, cf. 1) Lith. *voverė*, *vóverė* ‘squirrel’, Latv. *vāvere* ‘id.’; 2) OCS *věverica* ‘id.’, Lith. *vaiveris* ‘polecat’; 3) Ir. *feoróg*, Gae. *feòrag* ‘squirrel’, 4) Lat. *viverra* f. ‘ferret’, W *gwiwar* ‘squirrel’; 5) Pers. *varvarah* ‘id.’.⁸⁸¹ The great variety of forms makes it difficult to reconstruct the PIE word, but it seems clear that we are dealing with a reduplicated noun.⁸⁸² In all likelihood, the original paradigm was similar to the one of the Indo-European word for beaver, viz. **b^hé-b^hr*, **b^hi-b^hr-ós*.⁸⁸³ On the basis of this model, it can be

⁸⁸⁰ The connection with ON *eikinn* ‘vivid’, Skt. *ējati* ‘move quickly’ < PIE **h₂zeig-* (De Vries 1962: 283; Hellquist 116; Pokorny 13-4) is implausible.

⁸⁸¹ Little can be said about modern Gr. σκίουρος (= Lat. *sciūrus*, MLat. *squiriolus*, *spiriolus*, *asp(e)riolus*, Fr. *écureuil*, Wall. *skiron*, *spirou* (Grandgagnage 1857: 10). It may have contained the element **uer-*, but synchronically it looks like a compound of σκιά f. ‘shadow’ and -ουρος, ‘tailed’ < οὐρά f. ‘tail’. Perhaps the original form of the word was altered by popular etymology.

⁸⁸² Cf. Bailey 1979: 209; RLGA 6, 536.

⁸⁸³ According to Beekes (1995: 190), this paradigm developed out of older **b^hé-b^hr*, **b^he-b^hr-ós* by the raising of pretonic **e* to **i* in the genitive.

reconstructed as **h₂ué-h₂ur*, gen. **h₂ui-h₂ur-ós*. The paradigmatic alternations were probably leveled in the different daughter languages. Thus, Lith. *voverė*, *vóverė* and Latv. *vāvere* generalized a stem **h₂ue-h₂uer-*, while Lat. *vīverra* together with W *gwiwar* looks like the continuation of **h₂ui-h₂uer-*.⁸⁸⁴ The material further suggests that the reduplication process differed from language to language. In Balto-Slavic, the reduplication often occurred with **o* or **oi* instead of **e*, cf. Lith. *bėbras*, *bābras*, Ru. *bobr* ‘beaver’ < **b^he-b^hr-*, **b^ho-b^hr-* and Lith. *gaĩgalas* ‘drake’, OPru. *gegalis* ‘kind of fishing bird’ < **g^hoi-g^hol-o-*.⁸⁸⁵ This explains the variant Lith. *vaĩveris*, ORu. *věverica* < **h₂uoi-h₂uer-*. Lith. *vėveris*, on the other hand, seems to point to a formation **h₁ue-h₁uer-* with *h₁* rather than *h₂*. It is possible, though, that the word was borrowed from Slavic (cf. Huld 2009: 137). Finally, Pers. *varvarah-* goes back to **h₂uer-h₂uer-* with ‘broken’ reduplication.

The reconstruction of a reduplicated noun enables us to explain a large part of the formal variation in the different Indo-European languages. It, too, offers a solution to the remarkable alternation of WGm. **aikwernan-* and NGm. **ikwernan-*. Pokorny (IEW: 116), in his analysis of this problem, hesitates between “alter Ablaut oder Schwächung aus *aik-* im Nebenton”. Since, however, the weakening of pretonic *ai* to *í* is unparalleled in Old Norse, old ablaut seems the more probable solution. This means that the word probably was no compound of **aik-* ‘oak’ and **wernan-* ‘squirrel’, as has been claimed by, for instance, Falk and Torp, because the PIE did not have paradigmatic ablaut in compounds.⁸⁸⁶

The problem can be solved by assuming that the Germanic **k* of both **aikwernan-* and **ikwernan-* developed secondarily. Seebold (1982) suggested that the West Germanic word continues “**woiwr-*” through an alleged change of PIE **-Viu-* to PGm. **-Vkw-*. It seems preferable, then, to use Kortlandt’s laryngealistic reformulation of this rule (1988: 356),

⁸⁸⁴ It has been claimed that Lat. *vīverra* was borrowed from Gaul, or that conversely the Latin word was borrowed into British. For a discussion, see Huld 2009: 132.

⁸⁸⁵ Endzelīns/Schmalstieg/Jegers 1971: 85.

⁸⁸⁶ Alternatively, it is possible to assume that ON *íkorni* was borrowed from Frisian in the wake of the Frisian fur trade, which was centered in Birka (Singleton 1998: 16). Indeed, PGm. **aik-* ‘oak’ developed into *īk* and *iik* in North and West Frisian correspondingly, which could explain the Nordic vocalism. Still, it is questionable that the Vikings would borrow the name of an animal that they already were familiar with.

which entails the velarization of a laryngeal between a resonant and **u*, cf. OE *tācor* ‘brother-in-law’ < **taik(w)er-* < **daiHuer-* < PIE **deh₂i-uer-*. Seebold’s reconstruction would then translate as **h₂uoi-h₂ur*, a form that in Germanic should have developed into **waikwur-*.

An important problem with this outcome, as Seebold pointed out himself, is that it contains an initial **w*. It therefore seems better to leave the labial glide out of the reduplication process, and reconstruct the proto-forms as **h₂ei-h₂uer-* and **h₂i-h₂u(e)r-*.⁸⁸⁷ These reconstructions would regularly yield PGm. **aikwer-* and **īkwer-* (**īkur-*). If this is correct, the original paradigm should – fully parallel to **b^hé-b^hr*, **b^hi-b^hr-ós* – be reconstructed as **h₂é-h₂ur*, gen. **h₂i-h₂ur-ós*.

Although the proposed reduplication process is uniquely Germanic, this does not necessarily detract from its validity. After all, most of the Indo-European languages show an unexpected form of reduplication as well. Perhaps it can even be argued that the Germanic form displays the most archaic type of reduplication, since it clearly dates back to a stage prior to the loss of the initial laryngeals. When initial laryngeals were lost in the separate daughter languages, the reduplication may have been restored on the basis of the new onset. In fact, this loss is probably the very reason why such remarkable diversity in the reduplication syllable is found across the Indo-European dialects.

****h(r)aigrō / *higrō* ‘heron’**

- **hraigra(n)-*: OHG *hreigir*, *reigar* ‘id.’, G *Reiher* ‘id.’⁸⁸⁸, OE *hrāgra* m. ‘id.’, MLG *rēger* m. ‘id.’, MDu. *rēger*, *reiger* ‘id.’, Du. *reiger* ‘id.’
- **haigra(n)-*: Nw. dial. *heigre* m. ‘id.’, OHG *heigar*, *heigiro* m. ‘id.’, MHG *heiger* m. ‘id.’, OS *hēgro* m. ‘id.’, MDu. *hēger*, *heiger* ‘id.’
- **higra(n)-*: ON, Icel., Far. *hegri* m. ‘id.’, Nw. *hegre* m. ‘id.’, *higre* m. ‘id.’, OSw. *hægher* m. ‘id.’, Sw. *häger* ‘id.’⁸⁸⁹, Da. *hejre* ‘id.’

⁸⁸⁷ The derivation of Ir. *iora* ‘squirrel’ is unclear to me, but it seems to lack an initial **u-* as well.

⁸⁸⁸ Kluge/Mitzka 593.

⁸⁸⁹ SAOB H1904; Hellquist 255.

Three different stems can be established for the word for ‘heron’. The full grade is evidenced by OHG *hreigir*, OE *hrāgra* < **hraigra(n)-* and by Nw. dial. *heigre*, OHG *heigiro*, OS *hēgro* < **haigran-*. The zero grade is found in ON *hegri* < **higran-*. Both **haigran-* and **higran-* seem to have lost their first **r* as a result of dissimilation (cf. De Vries 1962; Schrijver 1997: 301).

According to Schrijver (1997: 302), the vocalism of ON/Icel. *hegri* is due to borrowing from Danish, where **ai* regularly becomes **ē*. It remains unclear, however, why exactly this bird name should have been adopted from Danish. The interchange of **h(r)aigran-* and **higran-* is therefore rather to be interpreted as resulting from an ablauting paradigm. In view of the parallelism with the word for ‘squirrel’, it can tentatively be reconstructed as an originally reduplicated noun **hraigur*, gen. **hrigraz* continuing PIE **krói-kr*, gen. **kri-kr-ós*. The *o*-grade of the reduplication syllable would then be comparable to the one of e.g. Ru. *bobr* ‘beaver’ < **b^ho-b^hr-*. In fact, the whole paradigm appears to be remarkably similar to **b^hé/ó-b^hr*, gen. **b^hi-b^hr-ós* (see p. 263-266).

9.6 The **ū ~ *u* type

A large group of *n*-stems display an **ū ~ *u* alternation. The first to seriously address this alternation was Osthoff (1882: 299 fn.), who happened to stumble over it in his article on the ablaut of the class 2b strong verbs. Following a personal comment by Kluge, who at that time was collecting evidence for his publication on nasal assimilation, Osthoff argued that the shortening of geminates in superheavy syllables implied the “ursprüngliche paradigmatische einheit von an. *knút-r* m., ags. *cnotta* m., ahd. *knotho*, *chnodo* und *knoto*, mhd. *knode* und *knote* m. ‘knoten’.”⁸⁹⁰ A key question facing this reconstruction was raised by Kauffmann (1887: 529): “Wie ist aber *ū* zu erklären?”

In contrast to Osthoff’s old theory of the “nebentonige Tiefstufe”, which explains PGm. **ū* as the regular outcome of unstressed **eu*, and Perridon’s hypothesis of an uncompleted sound change **eu > *ū* in dialectal Proto-Germanic (see § 6.5), the solution that is preferred here is not based on regular sound change. Given the parallelism with the class

⁸⁹⁰ Cf. the reconstruction of rudimentary paradigm **knópan-*, **knudén*, **knutt’* : **knütt’* by Noreen (1894: 164).

2b strong verbs, it does not seem unlikely at all that the **ū : *u* alternations in the *n*-stems arose analogically after the model of the phonetically regular **ī : *i* ablaut. In fact, other ablaut types may have played a role as well. The ablaut of the **ō ~ *a* type, for instance, at least partly continues the purely quantitative **ā : *a* ablaut (PIE **eh₂ : *h₂*) that must have existed right until the merger of Pre-Gm. **ā* and **ō*. The introduction of a secondary length opposition is therefore perfectly understandable within the morphophonological framework of Pre- and Proto-Germanic.

In addition, a small but linguistically real basis for the rise for the **ū : *u* opposition may have been created independently by Dybo's law, the development underlying the pretonic shortening of e.g. PIE **suHnús* to PGm. **sūnuz* 'son'.⁸⁹¹ By this law, an originally non-ablauting *mn*-stem with a root ending in **-uH-* should have regularly developed into a paradigm with quantitative ablaut **ū : *u*. Such regular ablaut can possibly be reconstructed for the Germanic word for 'thumb', which on the basis of the attested forms OHG *dūmo* 'thumb' < **pūman-* and OSw. *pume* 'id.' < **puman-* seems to go back to a paradigm **pūmō, *pūmenaz* < **túH-mōn, *tuH-mén-(o)s*. It can therefore be argued that Dybo's law created an additional starting point for the otherwise secondary **ū : *u* opposition.

****hrūhō, *hrukkaz* 'pile'**

- **hrūha-*: Icel. *hró* 'hillock'⁸⁹², Far. *rógv* n. 'stack'⁸⁹³
- **hrūgōn-*: ON, Icel. *hrúga* f. 'pile'
- **hrūkōn-*: Icel. *hrúka* f. 'small pile'⁸⁹⁴, Nw. dial. *hruke* f. 'pile, haystack', Sw. dial. *ruka* f. 'hillock, pile', E dial. *ruck* 'stack'
- **hrūk^a-*: Nw. dial. *ruk* m. 'haystack, potato row'
- **hrugan-, -ōn-*: Sw. dial. *råga* 'stack'⁸⁹⁵, Gutn. *rugā* m. 'load'⁸⁹⁶
- **hruggan-*: Sw. *rugge* 'bush'⁸⁹⁷

⁸⁹¹ Dybo's law only operated through resonants, cf. **hūdiz* < **kuHtís* (Kortlandt 1975).

⁸⁹² Böðvarsson 412.

⁸⁹³ Benediktsson 1960: 137.

⁸⁹⁴ Böðvarsson 415.

⁸⁹⁵ Hellquist 659.

⁸⁹⁶ Klintberg/Gustavson 979.

- **hrukka-*: MDu. *roc* m. 'haystack'⁸⁹⁸, EDu. *rock* 'id.'
- **hruka(n)-, hrukōn-*: ON *hroki*, -*r*, Icel. *hroki*, -*ur* m. 'pile'⁸⁹⁹, Far. *roki* m. 'pile on a waggon'⁹⁰⁰, Nw. dial. *roke* m. 'haystack', Gutn. *rukā* f. '(dung) heap'⁹⁰¹
- **hrauk^ka-*: ON *hraukr* m. 'pile'⁹⁰², Icel. *hraukur* m. 'stack, big guy'⁹⁰³, Far. *reykur* m. 'bird's crest'⁹⁰⁴, OE *hrēac* m. 'heap, stack, rick'⁹⁰⁵, Du. *rook* 'haycock, rick'

The interchange of ON *hrúga* and OE *hrēac* is mentioned by Kauffmann (1887: 515) as a possible case of paradigmatic ablaut. Hellquist (p. 680), too, acknowledges Icel. *hrúka* as an ablaut variant of **hrauk-*, but calls the consonant alternation "ej fullt klart". In order to be able to obtain a clear picture of the correlation between the different root variants, the complete material has to be assessed. In total, as many as eight different root variants can be identified.

The full-grade vocalism **ū* is found scattered through the Nordic dialects, and is accompanied by several different consonantisms, e.g. ON *hrúga* < **hrūg-*, Icel. *hrúka*, Sw. dial. *ruka* < **hrūk^k-*. Icel. *hró* and Far. *rógv* probably continue **hrūh-*, although **hrüh-* is possible, too (cf. ON *þó* 'though' < **þuhwe* < **tu-k^we*). Since the full grade is likely to have occurred in stressed position, however, the variant **hrūh-*, continuing a Pre-Germanic nominative allomorph **krūk-*, is preferable.

The zero grade is attested in a wide variety of formations, e.g. Sw. dial. *råga* < **hrugōn-*, Sw. *rugge* < **hruggan-*, ON *hroki* < **hrukan-* and MDu. *roc* < **hrukka-*. The latter of these variants clearly developed from the original genitive **hrukkaz* < **kruk-n-ós*.

⁸⁹⁷ Hellquist 659.

⁸⁹⁸ Verdam 499.

⁸⁹⁹ De Vries 1962: 259; Böðvarsson 413.

⁹⁰⁰ Jacobsen/Matras 296.

⁹⁰¹ Klintberg/Gustavson 979.

⁹⁰² De Vries 1962: 252.

⁹⁰³ Böðvarsson 405.

⁹⁰⁴ Poulsen 932.

⁹⁰⁵ Bosworth/Toller 556.

Together, all the different forms point to a PGm. paradigm **hrūhō*, **hrukkaz*, **hrugeni* that was split up in accordance with Kluge's *Associationen*. It seems clear, for instance, that Sw. *rugga* is a contamination of the genitive **hrukkaz* with the dative **hrugeni*. Similarly, ON *hroki* can be analyzed as a formation deriving from a secondary nominative **hrukō*, or otherwise from an analogical dative **hrukeni*.⁹⁰⁶

The position of the thematic *o*-grade PGm. **hrauk^ka-* is not entirely clear. It could be an independent formation to a strong verb ***hrúk^kan-*, but no such verb is attested. The alternative is to assume that it split off from an apl. form **hrauk^kuns* < **krouk-n-ŋs*.

Interestingly, PGm. **hrauk^ka-* is formally and semantically close to OIr. *crúach* f. 'stack of corn, rick', W *crug* 'id.' < PCelt. **krouk*-⁹⁰⁷. Given the fact that the Germanic word is derivationally transparent and embedded in a broader etymological context, whereas the Celtic word is lexically isolated, the direction of borrowing must have been from Germanic to Celtic. Other connections, such as Lat. *crux*⁹⁰⁸ and Skt. *kruñcati* 'to bend' are more uncertain, but the appurtenance of PGm. **hrugja-* 'ridge' is perhaps not inconceivable.

**hūfō*, **huppaz* 'heap'?

- **hūpan-*: OHG *hūfo* 'strues, tumulus, cumulus', MHG *hūfe* m. 'id.', G *Haufen*⁹⁰⁹, Swi. Visp. *hüüfo* m. 'id.', MLG *hūpe* m. 'id.'⁹¹⁰
- **hubbōn-*: G Tyr. *huppe* f. 'hill'⁹¹¹, LG *hobbe* 'hillock'⁹¹², EDu. *hobbe* 'big cheese'
 ⇒ **hubla-*: Swi. Visp. *hubol* m. 'hill', EDu. *hobbel* 'node, bump', Du. *hobbel* 'bump'⁹¹³

⁹⁰⁶ Noreen 1894: 164; Falk/Torp 866.

⁹⁰⁷ Hellquist (p. 680): "F.ö. urbesl. med ir. *chrúach* (av **krouko-*) [...]"; Falk/Torp 866: "Außerhalb des Germ. entspricht air. *crúach* (von **kroukā-*), kymr. *crûg* „haufe, heudieme“.; De Vries 1962: 252.

⁹⁰⁸ Cf. Pokorny 935-8.

⁹⁰⁹ Kluge/Seebold 396: "Außergermanisch stehen am nächsten (mit Auslautvariationen) lit. *káupas* »Haufen«, akslav. *kupŭ* »Haufen«."

⁹¹⁰ Lübben 154.

⁹¹¹ Schöpf/Hofer 282.

⁹¹² Doornkaat-Koolman 2, 89.

⁹¹³ De Vries/Tollenaere 259.

- **huppōn-*: OE *hoppe* f. ‘capsule’
- **haup^a-*: OHG *houf* ‘strues, tumulus’, OS *hōp* m. ‘id.’, MLG *hōp* m. ‘id.’⁹¹⁴, OE *hēap* mf. ‘pile, host’⁹¹⁵, OFri. *hāp* m. ‘heap, crowd’⁹¹⁶

It was Kauffmann (1887: 518) himself who in the 19th century suggested a paradigm **haufō*, **hūp^aaz* in order to explain the vocalic and consonantal alternations found in the material. In laryngealistic terms, the underlying reconstruction would translate as **kéHup-ōn*, **kHup-n-ós* (with laryngeal metathesis). The root structure of this formations seems to be confirmed by Lith. *káupas* and SCr. *kŭpa* ‘hill’ < **keHup-o-*, *-eh₂-*.

It follows from the reconstruction given here that there is no compelling reason to reconstruct the PIE root with **b* on the basis of the Germanic material, as has been suggested by, for instance, Kluge/Seebold (p. 396) and Boutkan/Siebinga (p. 152). Von Friesen (1897: 51) already correctly emphasized that the consonant alternations of **hūpan-* and **huppan-* are fully understandable as resulting from Kluge’s law and the subsequent paradigmatic analogies. The original consonantism was preserved by OHG *hovar* ‘gibbus’, MHG *hover* m. ‘hump’⁹¹⁷, OE *hofer* m. ‘id.’⁹¹⁸ < **hufra-* < **kúp-ro-*.

An alternative way to reconstruct the word is to bring the original paradigm in line with other *n*-stems with **ū ~ *u* ablaut, e.g. ON *hrúga* ~ MDu. *roc* ‘haystack’. A parallel paradigm **hūfō*, **huppaz*, **hubeni* would at least partly be analogical, however, because either the long **ū* or the short **u* must then be secondary. The stem **haup^a-*, on the other hand, becomes analyzable as a thematic *o*-grade morphologically comparable to e.g. ON *hraukr* ‘haystack’ < **hrauk^a-*. It can also be explained, on the other hand, as the continuant of the apl. **haup^auns* < **kouHp-n-ŋs* (< *koHup-n-ŋs?*). This is perhaps particularly attractive in view of the geminated coda of the root.

⁹¹⁴ Lübben 297.

⁹¹⁵ Bosworth/Toller 521.

⁹¹⁶ Boutkan/Siebinga 152: “the ablaut form **hūp-* (< **kuH-b-?* [...]) is problematic”.

⁹¹⁷ Lexer 1, 1365.

⁹¹⁸ Bosworth/Toller 548.

The Balto-Slavic and Germanic words are related to Mr. *cúan* f. 'group, pile' < **k(o)Hup-n-eh₂*.⁹¹⁹ The appurtenance of Av. *kaofa-* m. 'mountain' with its conspicuous *f* can only be maintained if we reconstruct the word as **koHup-H-o-* (cf. Av. *raθa-* m. 'wain', Skt. *rátha-* m. 'id.' < **Hrot-h₂-o-*). Balto-Slavic points to **kHup-* rather than **kuHp-*⁹²⁰, however. Alb. *qipí* f. 'pile' < **kūp-iā-* is a loanword from Slavic.⁹²¹

****klūpō*, **kluttaz* 'clot'?**

- **klūpōn-*: MHG *klūde* f. '(stone used as) weight for wool'⁹²², Du. dial. *kloede* 'lump'⁹²³
- **klūda-*: OE *clūd* m. 'pile, rock'⁹²⁴, *stān-clūd* 'rock'⁹²⁵, E *cloud*⁹²⁶
- **klūt^ta(n)-*: MLG *klūt(e)* m. 'clod'⁹²⁷, EDu. *kluyte* 'clod, floe'⁹²⁸, OE *clūt* m. 'rag, piece of metal'⁹²⁹ (= ON *klútr* m. 'rag'⁹³⁰), E *clout*⁹³¹
- **klutta(n)-*: MHG *kloz*, *klotze* m. 'lump'⁹³², G *Klotz*⁹³³, MDu. *clot(te)* m. 'ball, lump'⁹³⁴, EDu. *klot(te)* 'ball, clod', OE *clot* 'lump'⁹³⁵, E *clot*

⁹¹⁹ Cf. Pokorny 588-592.

⁹²⁰ Derksen 2008: 256.

⁹²¹ Demiraj 1997: 341.

⁹²² Grimm 11, 1157; Lexer 1, 1635. Contra Venema 1997: 283.

⁹²³ Ter Laan 1929: 1081.

⁹²⁴ Bosworth/Toller 160; Holthausen 1934: 53.

⁹²⁵ Bosworth/Toller 910.

⁹²⁶ Barnhart 181.

⁹²⁷ Lübben 178.

⁹²⁸ EDu. NRhnl. *kloet* is not identical to *kloot* 'globus' (Franck/Van Wijk 317-8), but to *kluyte*, which in the dialects in the East of the Netherlands was not fronted to [kly:tə]. There is no compelling evidence for a PGm. root ***klōt^t*.

⁹²⁹ Bosworth/Toller 160; Holthausen 1934: 53.

⁹³⁰ De Vries 1962: 318.

⁹³¹ Barnhart 182.

⁹³² Lexer 1, 1634.

⁹³³ Grimm 11, 1248-53; Kluge/Seebold 499.

⁹³⁴ De Vries/Tollenaere 332.

⁹³⁵ Holthausen 1934: 53.

- ?**klupþōn*:- MDu. *clos(se)*, *clotte* f. 'ball, lump'⁹³⁶, EDu. *klos* 'globus', Du. *klos* 'clew'
 - **kludda(n)*:- OE *clod* m. 'clod', EDu. *klodde* 'clew, prop'
⇒ **klud(d)ra*:- Du. *klodder* 'blotch'
-
- **klaut^ta*:- OHG *chlōz* 'massa, pila, sphaera', MHG *klōz* m. 'lump, clew, knob'⁹³⁷, G *Kloß*⁹³⁸, MLG *klōt* m. 'lump, ball'⁹³⁹ (= ON *klót* n. 'sword knob', G *Klōten* 'testicles'⁹⁴⁰), MDu. *cloot* m. 'ball, clod, bullet'⁹⁴¹, Du. *kloot* 'ball'⁹⁴², OFri. *klāt* m. 'pile, clod'⁹⁴³, E *cleat* 'wedge-shaped piece'⁹⁴⁴
⇒ **klaut^tjan*:- MHG *klæzen* w.v. 'to split'⁹⁴⁵

In the literature, the different formations belonging to the derivational cluster given here are usually separated from each other and projected back into Indo-European without doing justice to the mechanisms of Proto-Germanic morphophonology. Thus, the Barnhart Dictionary of Etymology (p. 181-2) reconstructs four different PIE roots: 1) **gloud-* for E *cleat*, 2) **glūd-* for *clout*, 3) **glud-* for *clot-* and 4) **glūt-* for *cloud*. The same approach is found in Pokorny (p. 356-364), Kluge/Seebold (p. 499), Franck/Van Wijk (p. 319) and the OED, all supposing a long **ū* and a root extension **d* for Proto-Indo-European.

The problems with this procedure are legion. First of all, the separation of the semantically and formally closely related Germanic forms is artificial, especially when the effects of Kluge's law are taken into account. Secondly, the only extra-Germanic evidence in support of a root extension **d* comes from Ru. *glýda* f. 'clod', which is a very small basis. Since Slov. *glûta*, *glúta* f. 'gnarl, lump' is clearly connected (cf. Vasmer 1, 415-6), the Russian *d* is probably unreliable. This must be

⁹³⁶ Verdam 296.

⁹³⁷ Lexer 1, 1633.

⁹³⁸ Grimm 11, 1244-8; Kluge/Seebold 499.

⁹³⁹ Lübben 177.

⁹⁴⁰ Kluge/Seebold 499.

⁹⁴¹ Verdam 296.

⁹⁴² Franck/Van Wijk 319.

⁹⁴³ Holthausen 1925: 58; Hofmann/Popkema 274.

⁹⁴⁴ Barnhart 178.

⁹⁴⁵ Lexer 1, 1634.

concluded anyway on the basis of the peculiar variants *glyba* ‘clod’ and *glýza* ‘id.’. Apparently, several different words became formally and semantically associated with each other in Russian.

In Germanic, the reconstruction of an ablauting *n*-stem **klūpō*, **kluttaz* could account for much of the attested root variation. This *n*-stem is likely to be formed to the root of the more primary *n*-stem **klewō*, **klunaz* ‘lump’ < PIE **gleu(H)-* (see p. 176) by the addition of a **t*-suffix. The same suffix seems to be present in Gr. γλουτός ‘bottom’ < **glou(H)-to-* and the already mentioned Slov. *glūta*, *glūta* f. ‘gnarl, lump’.

The thematic *o*-grade **klaut^a-* can theoretically be derived from a hypothetical apl. case **klaut^tuns* < **glou(H)-tn-ńs*. Since there is no proof of a *u*-suffix, however, the assumption of such a split-off remains doubtful. Alternatively, **klaut^a-* can be regarded as an independent formation, perhaps to a further non-attested strong verb **kleut^aan-* or **klūt^aan-*.⁹⁴⁶ This may in turn be an iterativicized form of **kleuþan-* or **klūþan-*. It is problematic, though, that material offers no support for such a strong verb. There does seem to have been an iterative **kluttōþi*, **kludunanþi*, on the other hand, cf. G *klotzen* ‘conglobare’ < **kluttō(ja)n-*, E *to clod / clot* ‘to coagulate’ < **kluddōn-* / **kluttōn-*. Perhaps, this verb could be the source of PGm. **klaut^a-* as well as the other formations.

****krūmō*, **krumenaz* ‘crumb’?**

- **krūman-*, *-ōn-*: OE *crūma* m. ‘crumb’, MDu. *crume* m. ‘inside of a bread, chunk’, Du. *kruim(el)* ‘crumb’⁹⁴⁷
 - **kruma(n)-*, *-ōn-*: ?Icel. *krumur* m. ‘gut’, OE *croma* m. ‘crumb’⁹⁴⁸, MHG *krume* f. ‘id.’, MDu. *crome* f. ‘id.’⁹⁴⁹, MLG *krume* f. ‘id.’⁹⁵⁰
-
- *?*krauma-*: Icel. *kraumur* m. ‘core, marrow’⁹⁵¹

⁹⁴⁶ Possibly, RhnL. *kliezen* ‘to split’ (Franck/Müller 4, 720-723) can be a vestige of such a verb.

⁹⁴⁷ Franck/Van Wijk 354.

⁹⁴⁸ Bosworth/Toller 172.

⁹⁴⁹ Verdam 314.

⁹⁵⁰ Lübben 190.

⁹⁵¹ Böðvarsson 522, 528.

Theoretically, the alternation of Du. *kruim* < **krūman*- and OE *croma* < **kruman*- can be attributed to an ablauting paradigm. The appurtenance of Icel. *kr(a)umur* is more uncertain because of the semantic differences, although most etymological dictionaries consider them as unproblematic.⁹⁵²

Outside Germanic, Alb. *grimë* f. 'crumb' and Lat. *grūmus* m. 'heap (of earth)'⁹⁵³ have been connected to the etymon in question. The involved root **gruH-* could in an *mn*-stem **grúH-mōn*, **gruH-mén(o)s* theoretically have resulted in a paradigmatic **ū* : **u* alternation under Dybo's law.

It is perhaps more conceivable, though, that the derivation of **krūman*- and **kruman*- happened much later, within Germanic itself. Given the meaning of OE *crimman* 'to crumb(le); to insert' < **krimman*-, it is at least theoretically possible that the *n*-stems were derived from the expected (but non-attested) iterative **krummōn*-. These *n*-stems must clearly be secondary, then, since the **u* of **krummōn*- does not continue real **u*, but rather a vocalized resonant to a root **krim(m)-* < **grem*-. The verbal nature of the root **krem*-/**krum*- is confirmed, at any rate, by formations such as OE *crymmian* 'to crumble', MHG *krümmen* 'to squeeze' < **krumjan*- and the *a*-grade iterative OE *crammian* 'to cram'.

**kūpō*, **kuttaz* 'tuft'

- **kūta(n)-*: G Bav. *kauzen* 'bundle of flax', Swab. *kauzen* 'entangled thread'⁹⁵⁴, Rhnl. *kūz* m. 'ball of yarn, tangle', *kützche* (dim.) 'tuft of hair, bird's crest'⁹⁵⁵, Swi. *kuuz* m. 'pelt wool, female bush, knotty hair' (⇒ Swi. *kuuzig* 'shaggy, hirsute'⁹⁵⁶)
- **kūddōn*- or **kūttōn*-. MHG *kūte* f. 'bunch of flax'⁹⁵⁷, G *Kaute* f. 'bundle of flax'⁹⁵⁸

⁹⁵² Cf. Fick/Falk/Torp 54; Franck/Van Wijk 354; Falk/Torp 583-4; Pokorny 385-390.

⁹⁵³ Cf. Holthausen: 61; Franck/Van Wijk 354.

⁹⁵⁴ Grimm 11, 363.

⁹⁵⁵ Franck/Müller 4, 349-50.

⁹⁵⁶ Grimm 11, 372.

⁹⁵⁷ Lexer 1, 1803-4.

⁹⁵⁸ Grimm 11, 1902-3; Haas 265.

- **kuddan*:- Du. *kodde* 'tail'⁹⁵⁹, G Rhnl. *kudden-tol* 'mixed up'⁹⁶⁰, MLG *kuddeken* n. 'small pile'⁹⁶¹
- **kutta(n)*:- Nw. dial. *kott* n. 'small clew', OHG *chotzo* 'paenula, laena, birrus', OS *kott* m. 'woolen rug, coat'⁹⁶² (= Icel. *kot* n. 'waistcoat', Far. *kot* n. 'woolen vest'^{963?}), MHG *kotz(e)* m. 'woolen rag' (⇒ MHG *kotzeht* 'shaggy'⁹⁶⁴), G *Kotze* 'woolen cloth, rugged cloth', dial. Zips/Spiš *kotzen* 'knotty hair', E *cot* 'matted lock', *cot-gare* 'refuse wool'⁹⁶⁵ (⇒ *cotted*, *cotty* 'matted, entangled'⁹⁶⁶)

On the basis of the material, it is possible to consider the reconstruction of an *n*-stem **kūpō*, **kuttaz* with two different vowel grades.

The zero-grade root **kutt-* has quite a large distribution. It is attested in Old High German as the masculine *n*-stem *chotzo* 'woolen coat, rug', and it is found with the same meaning as Old Saxon *kot* in the Freckenhorst and Werden tax scrolls. It is clear from other, more peripheral sources, that the word originally had a more restricted semantic field. In the Bohemian German dialect of Zips, for instance, *kotzen* signifies 'knotty hair'. Similarly, the obsolete English term *cot* 'matted lock' and *cot-gare* 'refuse wool' point to an original meaning 'woolen tuft' or simply 'tuft'. The semantic reconstruction is further corroborated by North Germanic in the form of Nw. *kott* 'small clew'. Semantically less attractive is the connection with Sw. *gran-kotte* 'fir-cone' as found in *Grunnmanuskriptet* and Hellquist (p. 348).

English *coat*, and Icel., Far. *kot* 'vest' are probably loanwords from Old French. This word was in turn adopted from a Germanic source, as Harper suggests in his *Online Etymological Dictionary*. The source may be identical with OS *kott* and the related G *Kotze*, both meaning 'woolen cloth or coat'.

⁹⁵⁹ EWN 408; WNT, s.v. *kodde*.

⁹⁶⁰ Franck/Müller 4, 1656.

⁹⁶¹ Schiller/Lübben 590.

⁹⁶² Gallée 182; Fick/Falk/Torp 47.

⁹⁶³ Böðvarsson 520; Poulsen 624.

⁹⁶⁴ Lexer 1, 1691.

⁹⁶⁵ Wedgewood 1859: 380; Wright 1869: 345.

⁹⁶⁶ Grimm 11, 1901-3.

A long vowel is found in the root **kūtt-*, which is supported by Bavarian *kauzen* ‘bundle of flax’, Swi. *kuuz* ‘pelt wool, knotty hair’ and Rhnl. *kūz* ‘ball of yarn’.⁹⁶⁷ The diminutive Rhnl. *kützche* ‘tuft, crest’ is semantically more primitive.

The **ū*-vocalism further occurs in MHG *kūte* and G *Kaute*. At first sight, these forms seem to have a different consonantism. From the High German perspective, they must reflect PGm. **kūd-*. However, both Lexer and Grimm acknowledge that the word only occurs in the Middle German area, e.g. in the dialects of Bohemia and Thuringia, Göttingen and Hesse. Perhaps, then, the *t* of *Kaute* is best explained as a continuant of **tt* in those Middle German dialects where the shift to *tz* did not take place. The *appel/apfel*-isogloss, for instance, runs to the south of the Hesse dialect area, whereas the *dorp/dorf*-isogloss lies north of it. It is not entirely impossible, however, that a proto-form **kūddōn-* did exist, as the consonantism of this form is paralleled by Du. obs. *kodde* ‘tail’ and Rhnl. *kudden-tol* ‘mixed up’ < **kuddan-*.

The consonantisms of both **kūttan-* and **kūddōn-* are clearly secondary. With the combination of a long vowel and a long stop, they defy the Proto-Germanic shortening of geminates after long vowels. Such roots are quite frequent in the High German dialects, however, especially in words that are inflected as *n*-stems (see § 8.3.3). Possibly, these formations generalized both the full grade and the geminate of an original paradigm **kūpō*, **kuttaz*, **kudeni*. The original consonantism further follows from G *Kauder* m. ‘rope, refuse hamp or wool’, Swi. *k(x)uuder* ‘refuse hamp’⁹⁶⁸, which reflect PGm. **kūpra-*. Similarly, G Rhnl. *kuddel* ‘muddle’⁹⁶⁹ may represent **kupla-*.

Etymologically, the whole cluster of forms can be reconstructed as a **ton*-extension to the root of ON *kárr* m. ‘curl’ < **gouero-*, Nw. dial. *kaure* m. ‘curl, lock of wool’, *kaur* n. ‘fine, curly wool’, Lith. *gaūras* m. ‘hair, down, tuft, flax fiber’⁹⁷⁰ < **gouro-*, Mlr. *gúaire* ‘hair’ < **gourio-* and Av. *gaona-* n. ‘hair’ < **gouno-* (cf. Pokorny 393-8).⁹⁷¹

⁹⁶⁷ Cf. Nw. *kott* ‘clew’ with a similar meaning.

⁹⁶⁸ Grimm 11, 306-7; Kluge/Mitzka 398.

⁹⁶⁹ Kluge/Mitzka 410; Franck/Müller 4, 1656.

⁹⁷⁰ Fraenkel 140.

⁹⁷¹ The improbable connection with Gr. *βεῦδος* n. ‘woman’s dress’ from a supposed root **gweud-* (Fick/Falk/Torp 47; Kluge/Mitzka 298) must be

****mūhō*, **mukkaz* ‘bunch’⁹⁷²**

- **mūhan*–: OE *mūwa* (*mūha*, *mūga*) m. ‘mow, heap’⁹⁷³ (⇒ OHG *mu(l)-werf*, MHG *mū(l)-werf*, *molt-werf(e)*⁹⁷⁴, G *Maul-wurf* m. ‘mole’⁹⁷⁵), E *mow* ‘stack’
- **mūga(n)*–, *-ōn*–: ON (*al*-)*múgi*, *-mugr* m. ‘swath, crowd’⁹⁷⁶, Icel. *múgi* m. ‘pile, crowd’⁹⁷⁷, Far. *múgvi* m., *múgva* f. ‘crowd’⁹⁷⁸, OSw. (*al*-)*moghe* m. ‘crowd, people’, Gutn. *måuä* m. ‘pile, stack’⁹⁷⁹
- **mūkōn*–: MLG, MDu. *mūke* ‘blade of grass’⁹⁸⁰
- **mukōn*–: Nw. dial. *moke* f. ‘pile’
- **mukka*–, *-ōn*–: Nw. dial. *mukke* f. ‘pile’⁹⁸¹, Sw. Gutn. *måckå* f. ‘id.’⁹⁸², Du. dial. *mok* ‘whisp’
- **muggan*–: Nw. dial. *mugge* f. ‘stack of 10 sheafs of corn’

The etymon under discussion has been mentioned as an ablauting *n*-stem by Schaffner (2001: 563-5). The ablaut pattern consists of a quantitative opposition of long and short **u* in the strong and weak cases. In combination with the consonantal variation, it points to a Northwest Germanic paradigm **mūhō*, **mukkaz*, **mugeni*.

The full and zero grades are both combined with several different consonantisms. Long **ū* occurs in e.g. OE *mūwa* < **mūhan*–, ON *múgi* < **mūgan*– and MDu. *mūke* < **mūk^kan*–. Short **u* is found in e.g. Nw. dial. *moke* < **mukan*–, Du. dial. *mok* < **mukka*– and Nw. dial. *mugge* < **muggōn*–. The recombination of the ablaut and the consonant opposition

abolished. Lubotsky (2008) convincingly showed that βεῦδος is a loanword from Old Phrygian *bevδος* ‘statue, image’.

⁹⁷² RLGA 20, 268-9.

⁹⁷³ Bosworth/Toller 700.

⁹⁷⁴ Lexer 1, 2195.

⁹⁷⁵ Kluge/Seebold 606-7.

⁹⁷⁶ De Vries 1962: 7, 394.

⁹⁷⁷ Böðvarsson 659.

⁹⁷⁸ Poulsen 794.

⁹⁷⁹ Klintberg/Gustavson 713.

⁹⁸⁰ Lübben 237; Verdam 371.

⁹⁸¹ Nw. dial. *mukka* m., as given by Schaffner (2001: 563, 564), cannot be retrieved from *Grunnmanuskriptet*.

⁹⁸² Klintberg/Gustavson 711.

implies that the original paradigm was split up into several different secondary ones, e.g. 1) **mūkō*, **mukkaz*, 2) **mūgō*, **muggaz* etc.

Within Germanic, the *n*-stem is related to ON *mostr* f. 'pile, bunch' < **muhstrō*.⁹⁸³ Beyond the Germanic horizon, the etymon has no cognates except for the remarkably close Hesychius gloss μύκων 'pile'.⁹⁸⁴ Unfortunately, the length of the upsilon is unknown, so that it remains uncertain whether the root must be reconstructed as **muk-* or **muHk-*. Since the Germanic ablaut type **ū* : **ǔ* is completely analogical, there is no compelling reason, at any rate, to assume that the original root contained a laryngeal.

****mūhō*, **mukkaz* 'chunk'**

- **mūk^kōn-*: MHG *mūche* f. 'malanders', G *Mauke*, *Mauche* f. 'id.'⁹⁸⁵, MLG *mūke* 'id.'⁹⁸⁶, MDu. *mūke* f. 'id.'⁹⁸⁷, Du. *muik* f. 'malanders, chunk'⁹⁸⁸
- **mukkan-*, *-ōn-*: MHG *mocke* m. 'chunk, fat person'⁹⁸⁹, G *Mocke* 'id.', MLG *mucken* pl. 'dried sods'⁹⁹⁰, Du. obs. *mok* f. 'equine condition, cooky, piece of wood'⁹⁹¹, dial. *mok* 'sod'⁹⁹², NFri. *mok* 'Mauke'⁹⁹³
- **muggan-*: MLG *mugge* m. 'equine condition'⁹⁹⁴, Du. dial. *mugge* 'whipping top'⁹⁹⁵

The combined occurrence of MHG *mūche* < **mūk^kōn-* and MLG *mugge*, both denoting an equine condition called 'malanders', could theoretically

⁹⁸³ De Vries 393: "Weiterbildung zur Wzl von *múgi*."

⁹⁸⁴ Cf. Pokorny 752.

⁹⁸⁵ Grimm 12, 1771, 1781; Kluge/Seebold 606.

⁹⁸⁶ Lübben 237.

⁹⁸⁷ Verdam 371.

⁹⁸⁸ Vercoullie 234; De Vries/Tollenaere 451.

⁹⁸⁹ Grimm 12, 2434.

⁹⁹⁰ Lübben 236.

⁹⁹¹ WNT, s.v. *mok* 4, 5; Vercoullie 230; De Vries/Tollenaere 451.

⁹⁹² WLD I/18, 8-9.

⁹⁹³ Löfstedt 2, 74.

⁹⁹⁴ Schiller/Lübben 131.

⁹⁹⁵ Kocks/Vording 763.

point to an ablauting *n*-stem **mūhō*, **mukkaz*, **mugeni*. The word has by some been connected to the word for ‘weak’, viz. Go. *muk-* (in *mukamodei* ‘gentleness’), Swi. *mauch* ‘id.’ (cf. Kluge/Seebold 606), but this is unlikely in view of the obviously related Du. *muik* ‘lump’. Malanders is the condition characterized by eczema around the knee of a horse’s foreleg. Apparently, the original usage of the word in the meaning of ‘chunk’ was extended to the disease in the Proto-West Germanic period.

The word has no good etymology, but it may be identical to the previous *n*-stem **mūhō*, **mukkaz* ‘bunch’ (see p. 278).

**pūpō*, **puttaz* ‘pout’?

- **pūpa-*, -*ōn-*: G Swab. *pfaude* f. ‘toad’⁹⁹⁶, MDu. *puut* m. ‘frog’⁹⁹⁷, Du. dial. *puid* ‘id.’⁹⁹⁸, *poede* ‘tadpole, eelpout’⁹⁹⁹
- **pūt^ta-*, **pūt^tōn-*: OE *æġl-pūte* f. ‘capito’, EDu. *puyt-ael*, *ael-puyt* ‘eelpout’, Du. *puut-aal* ‘eelpout’
- **pūpan-*: Du. *poon*, dial. *poo*, *pooi* ‘sea robin’¹⁰⁰⁰
- **puddōn-*: MDu. *podde*, *puddē* f. ‘toad, flab’¹⁰⁰¹, EWFri. *puddē* ‘*mustela piscis*’, SFri. *budde* f. ‘eel larva’, Du. dial. *podde* ‘mud, ooze, toad’, *puddē* f. ‘frog’, WFri. *budde* ‘burbot’¹⁰⁰²
 ⇒ **pudaka-*: OE *puduc* m. ‘crop, tumor’¹⁰⁰³, Scot. *puddock*¹⁰⁰⁴, LG. *puddēk* m. ‘lump, pudding, sausage’
- **puttan-*: LG *aal-putte* ‘eelpout’, Du. dial. *putte-kol* ‘tadpole (lit. “toad-head”)’

The large cluster of formations denoting ‘toad’ or ‘frog’ is etymologically obscure, and the initial **p* makes that the word is unlikely to be of Indo-European origin. The ablaut pattern is nevertheless compatible with

⁹⁹⁶ Fischer/Taigel 76.

⁹⁹⁷ Verdam 478.

⁹⁹⁸ WVD III, 3, 114-121.

⁹⁹⁹ Kocks/Vording 952.

¹⁰⁰⁰ Philippa/De Brabandere/Quak 576.

¹⁰⁰¹ Verdam 469.

¹⁰⁰² WNT *podde*, *puddē*.

¹⁰⁰³ Holthausen 1934: 250.

¹⁰⁰⁴ Jamieson 1825: 245.

other *n*-stems with $*\bar{u} \sim *u$ alternations. It is therefore at least theoretically possible that the word belonged to this apophonic subgroup. The question therefore arises whether the original paradigm can be reconstructed as $*p\bar{u}b\bar{o}$, $*puttaz$, $*pudeni$.

A form with long $*\bar{u}$ is supported by MDu. *puut*, Du. dial. *puid*, *poede*. The word seems to have a close correspondence in Swab. *pfaude* ‘toad’, a form that extends the spread of $*p\bar{u}b\bar{o}n$ - to the Upper German area. A long vowel is also present in OE *æġl-pūte* ‘capito’ as well as Du. *puut-aal* ‘eelpout’, but here it is combined with a (shortened) geminate. Geminata is furthermore found in MDu. *podde* < $*pudd\bar{o}n$ - and its direct cognates, but the primary voiceless geminate is only preserved by LG *āl-putte* ‘eelpout’ and Du. dial. *putte-kol* ‘tadpole’.

Du. *poon*, dial. *poō*, *poōi* ‘sea robin’ is generally assumed to be without an etymology¹⁰⁰⁵, but since the fish makes a frog-like sound when caught¹⁰⁰⁶, there are no strong objections against connecting it with Swab. *pfaude* and MDu. *podde*.¹⁰⁰⁷ Formally, the word can safely be reconstructed as MDu. $*pode$ < PGm. $*pu\bar{b}an$ -. Intervocalic *d* was regularly lost in most Dutch dialects, and the resulting hiatus was often resolved by the insertion of a palatal glide, thus rendering *poōi* (on the former island of Urk). In the dialects where this did not happen, the outcome would have been monosyllabic, e.g. *pao* ([p̄]) in the coastal dialect of Katwijk. The final *n* of the Standard Dutch form is analogical from the oblique, e.g. acc. $*pudanun$, or – as in *teen* ‘toe’ < $*taihw\bar{o}$ - – from the plural.

In addition to the roots with $*\bar{u}$ - and $*u$ -vocalism, which point to a paradigm $*p\bar{u}b\bar{o}$, $*puttaz$, there is the common formation $*padd\bar{o}n$ -, cf. ON *padða*, OE *padde*, MLG, MDu. *padde* f. ‘toad’, and an additional $*j\bar{o}$ -stem $*paddj\bar{o}n$ -, cf. LG WPhal. *pedde*¹⁰⁰⁸, MRhnl. *ped(de)*, MLG, MDu. *pedde*¹⁰⁰⁹, Du. dial. *pedde* f. ‘toad’. Since, however, these formations never show

¹⁰⁰⁵ Franck/Van Wijk 516; Vercoullie 270; De Vries/Tollenaere 290.

¹⁰⁰⁶ Cf. Philippa/De Brabandere/Quak 3, 576: “De rode poon wordt ook wel *knorhaan* genoemd vanwege het knorrende geluid dat hij maakt als hij uit het water wordt gehaald.”

¹⁰⁰⁷ Or perhaps the semantic field of MDu. *puude* ‘flab’ and OE *puduc* ‘crop, tumor’ points to an original meaning ‘flab’, a *Benennungsmotiv* for toads that occurs more often, cf. EDu. *quabbe* ‘toad, frog’, Du. *kwab* ‘flab’.

¹⁰⁰⁸ Woeste 1882: 196.

¹⁰⁰⁹ Lübben 272; Verdam 461.

consonant gradation, they can hardly be related to the hypothetical *n*-stem **pūbō*, **puttaz*. Instead, **paddōn-* and **paddjōn-* must be regarded as late derivations from the verb **paddōn-*: LG, Du. dial. *padden* ‘to crawl’.¹⁰¹⁰

****rūbō*, **ruppaz* ‘caterpillar’**

- **rūbbōn-*: MHG *rūp(p)e* f. ‘eelpout, caterpillar’¹⁰¹¹, G *Raupe* f. ‘caterpillar’, *Aal-raupe*, Pal. *raupe* f. ‘id.’¹⁰¹²
- **rūp^pōn-*: MLG *rūpe* ‘hairy maggot’, EDu. *ruype* ‘caterpillar’, Du. dial. *ruip* ‘id.’¹⁰¹³, WFri. *rûp* ‘id.’
- **rubbōn-*: MHG *ruppe* f. ‘caterpillar, eelpout’¹⁰¹⁴, G *Ruppe* f. ‘eelpout’¹⁰¹⁵, Pal. *Ool-rapp*, *-ropp*, *-rupp* ‘id.’¹⁰¹⁶, *Ruppe* f. ‘id.’¹⁰¹⁷, Thur. *roppe*, *ruppe* ‘caterpillar’

The word for ‘caterpillar’ shows the kind of formal variation that is typical of the ablauting *n*-stems. The material gives proof of a vocalic interchange of **ū* with **u* and of a consonantal interchange of **bb-* with **pp-*. This type of variation can be accounted for by reconstructing a paradigm **rūbō*, **ruppaz*.

The variant **rūp^pōn-* is found in the Low German speech area, and is supported by MLG *rūpe*, EDu. *ruype*, Du. dial. *ruip* and WFri. *rûp*. It superficially resembles the High German form *Raupe*, which therefore has been regarded as a Low German intrusion.¹⁰¹⁸ The geminate of MHG *rūppe* nevertheless shows that *Raupe* must have developed out of **rūbbōn-*, which with its combination of a long vowel and a geminate looks like a typically High German *n*-stem, cf. Swab. *kauzen* m. ‘entangled thread’ < **kūttan-*, Pal. *schaupe* f. ‘forelock’ < **skūbbōn-* etc. It can, at any

¹⁰¹⁰ Cornelissen 3, 932.

¹⁰¹¹ Lexer 2, 554.

¹⁰¹² Christmann 5, 415-6.

¹⁰¹³ Van Es 1989: 110.

¹⁰¹⁴ Lexer 2, 554.

¹⁰¹⁵ Grimm 14, 1533: “*das Wort stammt aus lat. rubeta*”.

¹⁰¹⁶ Christmann 1, 4: “*rubēta* = ahd. **rupta*; dieses mit Assimilation von pt zu pp in mhd. *Ruppe*”.

¹⁰¹⁷ Christmann 5, 662.

¹⁰¹⁸ Cf. Benecke (2, 821) on *rūpe*: “*wohl eig. niederdeutsch*.”

rate, not be derived from **rūpōn-* or **rūbōn-*, because these forms would have yielded ***Raufe* and ***Raube* respectively. If interdialectal borrowing ever did take place, the direction must therefore have been from High to Low German, not the other way around.

Finally, G *Ruppe*, with its correspondences in e.g. Palatinate and Thuringian, seems to point to a variant **rubbōn-* with a short **u*.

The attested polymorphism can be interpreted as deriving from a paradigm **rūbō*, **ruppaz* that was split up into 1) **rūpō*, **ruppaz* and 2) **rūbō*, **rubbaz*. Most probably, it was derived from the IE root **reup-*, which in Germanic gave rise to a large verbal complex including an iterative opposition, cf. ON *rjúfa*, OE *rēofan* ‘to break’ < **reufan-* vs MHG *ropfen* ‘to pluck’ ~ Icel. *rubba* ‘to scrape’, Als. *roppen* ‘to pull, pluck’ < **ruppōpi*, **rubunanpi* (cf. García García 2005: 104). The original meaning of the West Germanic *n*-stem therefore probably was “plucker”.¹⁰¹⁹

A slightly different etymology was proposed by De Vaan (2000). De Vaan argued that, given the widely attested meaning ‘rough maggot’, the *Benennungsmotiv* for the word must have been “rough one”. He further connected MDu. *robbe* ‘seal, rabbit’, EDu. *robbe(ken)* ‘rabbit’, Du. *rob* ‘seal’, MLG *rubbe*, LG *rabbe* m. ‘seal’, WFri. *robbe* ‘id.’, G *Robbe* mf. ‘id.’ < PGm. **rubba/ōn-*, because these animals are “rough-haired” as well.

Finally, Boutkan and Kossmann (1999) attempted to explain the formal variation as being the result of substrate influence. On the basis of Lat. *rēpō*, Lith. *réplióti* and Latv. *rāpāt*, all meaning ‘to creep, crawl’, they hypothesized that a non-Indo-European root **rū/āp-* ‘to crawl’ entered the given languages at a relatively late date. The same root would also have been borrowed into Germanic, ultimately to surface as **rūp/bb-* ‘caterpillar’, i.e. “crawler”. This explanation, however, fails to do justice to the intra-Germanic morphophonological innovations.

****skūbō*, **skuppaz* ‘brush’**

- **skūba(n)-*: ON *skúfr* m. ‘tassel’, Icel. *skúfur* m. ‘tassel, tuft’¹⁰²⁰, Far. *skú(g)vur* m. ‘id.’¹⁰²¹, Nw. dial. *skuv(e)* m. ‘brush, tuft’

¹⁰¹⁹ Note that the presence of consonant gradation in the verbal complex opens the possibility that the polymorphism of ‘caterpillar’ is not due to its inflection as an *n*-stem, but rather to its derivation from the iterative. This explanation, however, probably has the disadvantage that the *n*-stem would need to have been coined several times to several different verbal roots.

¹⁰²⁰ Böðvarsson 887.

- **skūbbōn*:- G Pal. *Schaupe* f. 'forelock'¹⁰²²
 - **skuban*:- MDu. *schove* m. 'sheaf, bundle'¹⁰²³
 - **skubban*:- MLG *schobbe* m. 'sheaf'¹⁰²⁴, G *Schuppen* m. 'tuft, shelter, barn'¹⁰²⁵, Swi. Visp. *šuppo* m. 'bunch'¹⁰²⁶
 - **skuppa(n)*:- OHG *scopf* m. 'lean-to', MHG *schopf(e)* m. 'hair of the head, shackle'¹⁰²⁷, G *Schopf*, *Schupfe* m. 'hair, shelter'¹⁰²⁸, Du. dial. *schop* 'lean-to'¹⁰²⁹, OE *sceoppa* m. 'shop, booth, shed'¹⁰³⁰, E *shop*
 - **skupa*:- OHG *scof* 'shed', MHG *schuff* m. 'forelock'¹⁰³¹
 - **skupinō*:- OE *scypen* f. 'cowshed'¹⁰³², E *shippon* 'id.'¹⁰³³
-
- **skauba*:- ON *skauf* n. 'sheaf', OHG *scoup* m. 'id.', OE *scēaf* m. 'id.'

The contrast between Pal. *Schaupe*, MHG *schopfe* and Visp. *šuppo* is archetypical of the variation found with ablauting *n*-stems. Consequently, it is possible to postulate a hypothetical paradigm **skūbō*, **skuppaz* that was split up into 1. **skūbō*, **skubbaz* and 2. **skūp^oō*, **skuppaz*. Notably, the Palatinate dialects seem to contain much of the complete amount of formal variation, cf. *schopf* m. 'forelock'¹⁰³⁴ < **skuppa*-, *schupp(en)* m. 'forelock'¹⁰³⁵ < **skubban*- with a zero grade, and *schaupe* f. 'forelock' (ib.) < **skūbbōn*- with a full grade.

¹⁰²¹ Poulsen 1068.

¹⁰²² Christmann 5, 901.

¹⁰²³ Verdam 524, 527.

¹⁰²⁴ Lübben 330.

¹⁰²⁵ Grimm 15, 2019.

¹⁰²⁶ Wipf 90.

¹⁰²⁷ Lexer 2, 770.

¹⁰²⁸ Grimm 15, 1527-52; 15, 2005-6; Kluge/Seebold 823; Christmann 5, 1408-9.

¹⁰²⁹ Kocks/Vording 1079.

¹⁰³⁰ Bosworth/Toller 839.

¹⁰³¹ Lexer 2, 770.

¹⁰³² Bosworth/Toller 847-848.

¹⁰³³ OED, s.v. *shippon*.

¹⁰³⁴ Christmann 5, 1408-9.

¹⁰³⁵ Christmann 5, 1497.

According to Lühr, OE *scypen* ‘cowshed’ provides some evidence for an additional allomorph **skup-*, which may have sprouted from an analogical paradigm **skupō*, **skuppaz*: “Die Variante mit einem **p* bildet die Grundlage von ae. *scypen* < **skupiniō-*” (1988: 239). It is possible, as well, that it directly continues the locative **skupeni* to the same *n*-stem **skūpō*, **skuppaz*. It must then be assumed that the original locative **skubeni* < **skub^h-én-i* was replaced by **skupeni*. The root **skup-* may also be attested in the OHG gloss *scof* ‘shed’ (Lühr 1988: 238).

Parenthetically, the dictionaries often differentiate between **skuppan-* ‘hair, tuft’, on the one hand, and **skuppan-* ‘shed’ on the other.¹⁰³⁶ Etymologically, there is no reason for such a separation, as both meanings can easily be reconciled with each other. Most probably, the oldest meaning ‘tuft’ or ‘brush’ developed into ‘underbrush’, ‘shelter’, ‘cowshed’ and ‘barn’ in West Germanic. The physical context that gave rise to this semantic chain must have been the keeping of cows or other grazing animals in the open field, where a roof of foliage provided the only shelter against the elements.¹⁰³⁷

Within Germanic, there are a number of cognates, viz. OHG *scobar* m. ‘haystack’, MHG *schober* m. ‘bush, tuft’, G *Schober* < **skubra-* and OHG *scubil* m. ‘bundle’ < **skubila-*. Similar formations are represented by OE *scyfele* f. and ON *skupla* f. ‘woman’s hood hiding the face’, Icel. *skupla* f. ‘scarf’ < **skubilōn-/skupilōn-*. A thematic formation is the pan-Germanic **skauba-*, which can be retrieved from e.g. ON *skauf*, OHG *scoup*, OE *scēaf* ‘sheaf’. This thematic *o*-grade could point to a verbal origin of the ablauting *n*-stem, e.g. **skūban-*, **skeuban-* ‘to shove’.¹⁰³⁸

Fick, Falk and Torp (p. 470) further compare *s*-less forms, such as Nw. *koppe* ‘crest’, OE *coppod* ‘crested’, Du. *kuif* ‘crest’, Flem. *kobbe* ‘plumage, hair’, OHG *chuppa*, *chupfa*, which form a very similar pattern, suggestive of a paradigm **kūbo*, **kuppaz* ‘crest’. Finally, there is Go. *skuft*, ON *skoft* and OHG *scuft* n. ‘hair’.

Possible extra-Germanic cognates are Ru. *čubъ*, *čupъ*, Cz. *čub*, *čup*, SCr. *čūpa*, Cz. *čupa* ‘shock’¹⁰³⁹, which point to both **keub-* and **keup-*.

¹⁰³⁶ Cf. Fick/Falk/Torp 469-70.

¹⁰³⁷ Alternatively, it can be assumed that sheafs of hay were used as shelter (Kluge/Seebold 823), but this seems less evident.

¹⁰³⁸ Cf. Kluge/Seebold 822.

¹⁰³⁹ Pokorny 956.

Given the vacillation of the *b* and *p*, however, it is more likely that the word was borrowed from Germanic, where the consonant gradation is innate.

****stūfō*, **stuppaz* ‘stub’†**

- **stūf/ba(n)*:- ON *stúfr* m. ‘stub’¹⁰⁴⁰, Nw. dial. *stuv(e)* m. ‘trunk, tree-stump’, MLG *stūve* m. ‘stub’¹⁰⁴¹
- **stūp^hōn*:- MLG, MDu. *stūpe* f. ‘pillary’¹⁰⁴²
- **stuf/ban*:- MDu. *stoof*, *stove* ‘tree-stump’¹⁰⁴³
- **stubna/ō*:- ON *stofn* n. ‘stub’¹⁰⁴⁴, OE *stofn* f. ‘tree-stump, shoot’¹⁰⁴⁵
- **stubba(n)*:- ON *stubbi*, *stubbr* m. ‘tree-stump, small piece’¹⁰⁴⁶, Nw. *stubb(e)* m. ‘id.’, MLG *stubbe* m. ‘stub’¹⁰⁴⁷, OE *stub*, *stȳb* m. ‘stump’, MDu. *stobbe*, *stubbe* m. ‘tree-stump’¹⁰⁴⁸
- **stuppōn*:- MHG *stupfe* f. ‘stubble’¹⁰⁴⁹, MLG, MDu. *stoppe* ‘stubble’

⇒ OHG *stopfela*, *stupfula* f. ‘prickle’, MLG *stoppel* m. ‘id.’¹⁰⁵⁰ (= G *Stoppel*¹⁰⁵¹), MDu. *stoppel(e)* mf. ‘stubble’¹⁰⁵²

The cross-dialectal consonant variation displayed by the Germanic word for ‘stub’ can be satisfactorily explained by reconstructing an *n*-stem **stubō*, **stuppaz* < **stúb^h-ōn*, **stub^h-n-ós* (cf. Fick/Falk/Torp; Lühr 1988: 246-7). OHG *stopfela*, MLG, MDu. *stoppel*, which have been regarded as loanwords from Late Latin *stipula* (> ?**stupula* > It. *stoppia*, OFr.

¹⁰⁴⁰ De Vries 1962: 555.

¹⁰⁴¹ Lübben 389.

¹⁰⁴² Lübben 388; Verdam 586.

¹⁰⁴³ Verdam 580.

¹⁰⁴⁴ De Vries 1962: 550.

¹⁰⁴⁵ Bosworth/Toller 923-924.

¹⁰⁴⁶ De Vries 1962: 555: “das -bb- ist lautmalende Geminatio”.

¹⁰⁴⁷ Lübben 387.

¹⁰⁴⁸ Verdam 585.

¹⁰⁴⁹ Lexer 2, 1274.

¹⁰⁵⁰ Lübben 382.

¹⁰⁵¹ Kluge/Seebold 887.

¹⁰⁵² Verdam 581.

(*e*)*stuble*) ‘ear’¹⁰⁵³ seem to be diminutives to the stem **stuppōn-* (Lühr 1988: 247). Likewise, E *stubble* does not have to continue OFr. *estoble*, *estouble*, as stated by the OED, but may be a similar diminutive to the stem **stubba(n)-*.

The reconstruction of paradigmatic ablaut is more precarious. Lühr touches upon the issue in her discussion of the frequent interchange of *ū* and *u* in pairs such as ON *stúfr* and *stubbi* ‘tree-trunk’, arguing that “das lange *ū* sich wahrscheinlich analogisch ausgebreitet hat”. Lühr (1988: 20) nevertheless rejects the possibility that the two variants once belonged to one and the same paradigm: “die jeweiligen u- und *ū*-Lautungen [dürften] kaum einem gemeinsamen Paradigma angehört haben, da man dann auch bei Wörtern mit Wurzelsvokal **ī* ein solches Nebeneinander erwarten würde.” Although it can, in fact, be demonstrated that such **ī* : **i* ablaut existed (see § 9.4), the reconstruction of an ablauting paradigm in this particular case relies on very limited evidence. The direct paradigmatic appurtenance of ON *stúfr* m. ‘stub’ to the zero grades is difficult to prove, and the only other full-grade form that potentially gives proof of Kluge’s *Associationen* is MLG, MDu. *stūpe* ‘pillary’ < **stūp^pōn-*. This word has very different meaning, however.

All in all, it seems preferable to ascribe the formal variation of the cluster to influence from a verbal complex. This is especially attractive in view of the presence of an iterative **stuppōpi*, **stubunanpi*, cf. Nw. dial. *stubba* ‘to cut (wood) into pieces’, MHG *stuppen* ‘to chop off, shorten’ < **stubbōn-*.

The most appropriate outer-Germanic cognates are Gr. *στόπος* ‘stick’, Latv. *stups* ‘broom stump’ and Ru. *stópka* ‘peg’¹⁰⁵⁴.

****pūmō, *pūmenaz* ‘thumb’**

- **pūman-*: OHG *dūmo* m. ‘thumb’, MHG *doume* m. ‘id.’, G *Daumen*, Swi. Visp. *dūūmo* m. ‘id.’, MDu. *dume* m. ‘id.’, Du. *duim* ‘thumb, inch’¹⁰⁵⁵, OFri. *thūma* m. ‘id.’, OE *pūma* m. ‘id.’
⇒ **pūmila-*: OE *pūmel* m. ‘thimble’

¹⁰⁵³ Franck/Van Wijk 672; OED, s.v. *stubble*, Kluge/Seebold l.c.

¹⁰⁵⁴ Cf. Fick/Falk/Torp 496; Pokorny 1032-1034; Frisk 2, 813-814.

¹⁰⁵⁵ Franck/Van Wijk 141.

- **puman-*: OSw. *pume* m. 'thumb, inch', Sw. *tumme* 'id.'¹⁰⁵⁶, ODa. *thumæ* m. 'id.'¹⁰⁵⁷, Da., Nw. *tomme* 'id.', Far. *tummi* m. 'inch'¹⁰⁵⁸
 ⇒ **pumala-*: ON *pumall* m. 'thumb'¹⁰⁵⁹, Icel. *pumall* m. 'thumb (of a glove)'¹⁰⁶⁰, Far. *tummil* m. 'id.'¹⁰⁶¹, Da., Nw. *tommel* 'id.'
 - **puma-*: OSw. *thum* n. 'inch', G dial. *dum* 'thumb', EDu. *dom* 'pollex'
-
- **pauma-*: MHG *doum* m. 'peg, chock'

The alternation of OE *pūma*, OFri. *thūma*, OHG *dūmo* < **pūman-* with OSw. *pume*, ODa. *thumæ* < **puman-* seems to point to an old paradigm **pūmō*, **pūmenaz*. The long **ū* has been ascribed to "expressive Dehnung im Westgermanischen"¹⁰⁶², but this explanation is difficult to falsify. The **ū ~ *u* alternation can, in fact, be due to the operation of Dybo's law, by which any long vowel was shortened before a resonant when the next syllable was stressed. A paradigm **tūH-mn*, **tuH-mén-s*, for instance, would have regularly developed into PGm. **pūmō*, **pūmenaz*.

Etymologically, the word for 'thumb' is generally derived from a root **tuH-* 'to swell', because the thumb is the biggest finger.¹⁰⁶³ It is nevertheless difficult to see how the clearly related *o*-grade MHG *doum* 'chock' < **pauma-* could be fit into this etymology. It seems preferable to connect this *doum* to the MHG verb *doumen* 'to stuff', since the meaning 'chock' can easily be explained from it (cf. E *plug*).

The semantic path from 'to stuff' to 'thumb' is more tricky, but the intermediate meaning may have been 'to push with the thumb', i.e. what is done in the act of stuffing. Icel. *puma* 'to feel, finger, knit'¹⁰⁶⁴ (whence

¹⁰⁵⁶ Hellquist 1126.

¹⁰⁵⁷ Falk/Torp 1270.

¹⁰⁵⁸ Poulsen 1274.

¹⁰⁵⁹ De Vries 1962: 626.

¹⁰⁶⁰ Böðvarsson 1215.

¹⁰⁶¹ Poulsen 1274.

¹⁰⁶² Kluge/Seebold 182.

¹⁰⁶³ Cf. Falk/Torp 1270; De Vries 1962: l.c.; Franck/Van Wijk 141.

¹⁰⁶⁴ Böðvarsson 1215.

Icel. *þum(a)* f. ‘thumb hole’) can then be regarded as the missing link between the two meanings.

Perhaps, then, the root **þǔ-* ‘to push (with the thumb)’ is related to OE *þýwan*, OHG *dūhen*, MDu. *duwen* ‘to push’. The underlying root is usually reconstructed with a velar, e.g. **þunhjan*-¹⁰⁶⁵ or **þūh(w)jan*-¹⁰⁶⁶, but **þūjan-* would probably work as well. Alternatively, it can be supposed that the velar was lost before *m* in the formation **þǔhman-* (cf. **hrifman-* > **hrīman-* ‘hoarfrost’, see p. 66).

**þūhō, *pukkaz* ‘bag’?

- **puhhan-*: OE *pohha* m. ‘purse’, E *pough* ‘bag’
- **pukan-*, *-ōn-*: ON, Icel. *poki* m. ‘bag, sack’¹⁰⁶⁷, E *poke* ‘bag’, MDu. *poke* ‘bag (for wool)’¹⁰⁶⁸, EDu. *poke* ‘hairshirt, crop’, Du. *pook* ‘bag’¹⁰⁶⁹, G *Pfoch* ‘id.’
- **pukōn-*: ME *pouk(e)*, *powk(e)* ‘blister, sty’, E *pouk* ‘id.’¹⁰⁷⁰, *Pfoche* f. ‘blister’
- **pukka(n)-*, *-ōn-*: OE *pocca* m. ‘bag’, *poc* m. ‘pock’¹⁰⁷¹, MLG, MDu. *pocke* f. ‘pimple, blister’¹⁰⁷², G *Pocke* f. ‘pock’
⇒ **pukkila-*: EDu. *pockel*, *puckel*, Du. *pukkel* ‘zit’

In the oldest Germanic dialects, this word for ‘bag’ displays three different stem variants, viz. **pukkan-*, **puhhan-* and **pukan-*. Together, they point to an original paradigm **puhō, *pukkaz*¹⁰⁷³ that was split up into either **puhō, *puhhaz* (= OE *puhha*) or **pukō* (= ON *poki*), **pukkaz* (= OE *pocca*). There is no reason to assume that the geminate **kk* is due to “intensivity”, as has been suggested by Kluge/Seebold (p. 557), or that the fricative geminate **hh* had a “lautnachahmende Funktion” (Lühr 1988: 271).

¹⁰⁶⁵ Pokorny 1099-1100.

¹⁰⁶⁶ Franck/Van Wijk 114.

¹⁰⁶⁷ De Vries 1962: 427; Böðvarsson 736.

¹⁰⁶⁸ Verdam 470.

¹⁰⁶⁹ De Vries/Tollenaere 539.

¹⁰⁷⁰ OED; Halliwell 1850: 641.

¹⁰⁷¹ Holthausen 1934: 248.

¹⁰⁷² Verdam 470.

¹⁰⁷³ Cf. Fick/Falk/Torp (p. 219): “pukk- aus ig. būkn²”; Franck/Van Wijk (p. 514): “De *kk* gaat op vóórgerm. *qn* of *gn* terug”.

The paradigm **puhō*, **pukkaz* seems to imply an etymologically obscure root **buk-*. It is often assumed that the word ultimately derives from a PIE root **bŭk-* ‘inflate’: W *bugad* ‘bellowing’, Lat. *bucca* ‘inflated cheek, mouthful’, Pol. *buczyć się* ‘puff oneself up’ and, with an onomatopoetic geminate, Skt. *búkkati* ‘bark’¹⁰⁷⁴. Within Germanic, however, it is hard to disconnect Go. *puggs*, ON *pungr*¹⁰⁷⁵, OHG *pfung*, OE *pung* m ‘pouch’ < **b(u)nk-í-*, even though these forms contain an unexplained nasal. Feist (1923: 290) therefore assumed the formation to be a “gemeingerm. Lehnwort aus unbekannter Quelle”, which is not unlikely in view of the initial **p*. It must be stressed, however, that the consonant gradation can very well have arisen within Germanic itself.

Prenasalization has been interpreted as a substrate feature in Germanic (Kuiper 1995). Accordingly, one could set up a substrate root **buk-* ~ **bunk-*. In this particular case, however, there is a different solution to the vacillating nasal. If the Pre-Gm. root was **bunk-*, the *n*-stem paradigm **bunk-ōn*, **bunk-n-ós* should have regularly become PGM. **pūhō*, **punk^kaz*, with nasalization of the vowel before **h*. It is theoretically possible that this otherwise regular paradigm was absorbed by the larger group of *n*-stems with *ū ~ u* ablaut after the analogical removal of the nasal in those cases where it had remained: **pūhō*, **punk^kaz* >> **pūhō*, **pukkaz*.

It is unclear whether MDu. *pūc* n. ‘(high quality) sheet(ing)’, MDu. *puik-goet* ‘fine stuff’ and Du. *puik* adj. ‘fine’ belong to the same etymon. Kluge and Seebold (p. 702) derive it from MDu. *pūcken* ‘to pick’, assuming an intermediate meaning “selected”, but Franck and Van Wijk (p. 526) call the etymology of *puik* unclear. It seems difficult, however, to disconnect it from MLG *pūche*, *pūghe* f. ‘blanket, cover(ing)’, LG *pūch* ‘bed’¹⁰⁷⁶ < **pūgōn-* and G dial. *pugge* f. ‘cradle’¹⁰⁷⁷ < **puggōn-*. There also seems to be a link with EDu. *poke* ‘hairshirt, bag, crop’ and Nw. dial. *poka* f. ‘pigskin, sward, fatty layer under the skin’. Together, these forms point to a variation **pūk-*, **pūg-*, **puk-* and **pug-*, which is close to the aforementioned variants of the paradigm **puhō*, **pukkaz*. It is therefore possible that all of the material discussed here originally belonged to one

¹⁰⁷⁴ Pokorny 98-102; Zantema 514.

¹⁰⁷⁵ De Vries 1962: 429.

¹⁰⁷⁶ Cf. Mensing 1927: 342.

¹⁰⁷⁷ Haas 1994: 263.

and the same etymon. The original meaning of the word would then have been ‘animal skin’ or ‘bag made of skin’.

****pūsō*, **pusnaz* ‘purse’?**

- **pūsa(n)*:- ON *púss* m. ‘pouch’¹⁰⁷⁸, Icel. *púsi* m. ‘bag’¹⁰⁷⁹, Nw. *pus* m. ‘protuberance’
- **pusan*:- ON *posi* m. ‘pouch’¹⁰⁸⁰, Icel. *posi* m. ‘small bag’¹⁰⁸¹, Far. *posi* m. ‘id.’, Nw., Da. *pose*, Sw. *påse* ‘id.’¹⁰⁸², OHG *pfoso* ‘marsupium, byrsa’, MHG *pfose* m. ‘purse’¹⁰⁸³, OE *posa* m. ‘bag’¹⁰⁸⁴

The vowel alternation of ON *púss*, Icel. *púsi* < **pūsa(n)*-, ON *posi*, OE *posa*, OHG *pfoso* < **pusan*- is in accordance with other ablauting *n*-stems of the **ū* ~ **u* type. Thus, the material may point to an original paradigm **pūsō*, **pussaz*. This reconstruction would certainly account for the forms given above. It also creates some important problems, however.

For a start, the etymology of the word is unclear. In spite of the customary connection with the root **pūs*- ‘to blow’ (cf. MHG *pfūsen* ‘to sniff’¹⁰⁸⁵), the only semantically attractive connection outside Germanic seems to be OIr. *búas* ‘pouch, belly’ as given by e.g. De Vries (1962: 429). As a consequence, the Germanic *n*-stem can be regarded as a loanword from PCelt. **bousto*- (or Proto-British **bōss*-?), just like **tassa*- ‘haystack’ was borrowed from the precursor of OIr. *daiss* ‘id.’ < ?**dasti*-. However, if this is correct, it must be assumed that the zero-grade root **pus*- was introduced analogically. All together, this seems like a long shot, especially since the root **pus*- is found in North and West Germanic, while **pūs*- occurs in West Norse only. It is further to be noted that the etymon is conspicuously similar to **pūhō*, **pukkaz* ‘bag’, which may indicate that the two words influenced each other.

¹⁰⁷⁸ De Vries 1962: 429.

¹⁰⁷⁹ Böðvarsson 744.

¹⁰⁸⁰ De Vries 1962: 427.

¹⁰⁸¹ Böðvarsson 737.

¹⁰⁸² Falk/Torp 844.

¹⁰⁸³ Lexer 2, 261.

¹⁰⁸⁴ Holthausen 1934: 248.

¹⁰⁸⁵ Cf. Falk/Torp l.c.; Pokorny l.c.

****snūfō*, **snuppaz* ‘sniffing, cold’†**

- **snūfa(n)*:- MLG *snūf*, *snūve* m. ‘cold’¹⁰⁸⁶ ⇒ EDu. *snuyfelen* pl. ‘asthmatic condition’
- **snufa(n)*:- MLG *snove* m. ‘cold, smell’¹⁰⁸⁷, MDu. *snof* m. ‘cold’¹⁰⁸⁸, EDu. *snof*, *snuf* ‘sniffing, cold’
⇒ **snufla*:- OE *snofl* ‘snot’
- **snuppan*-, -*ōn*:- MHG *snupfe* m. ‘cold’¹⁰⁸⁹, G *Schnupfen* ‘id.’¹⁰⁹⁰, MLG *snope* m. ‘snot’¹⁰⁹¹, MDu. *snope* m. ‘cold’¹⁰⁹²

The co-existence of three different *n*-stems meaning ‘cold’, i.e. MLG *snūve* < **snūf/ban*-, MLG *snove* < **snuf/ban*- and MLG *snope* < **snuppan*-, could theoretically be interpreted as resulting from an old PGm. *n*-stem nom. **snūfō*, gen. **snuppaz*, dat. **snubeni* related to MHG *snūfen*, G *schnauben*, *schnaufen*, MLG, MDu. *snūven*, Du. *snuiven* ‘to sniff’ < **snūfan*- (**snūban*-) and G *schniefen* ‘id.’ < **sneufan*-/**sneup^han*-.¹⁰⁹³

It nevertheless seems preferable to derive the different variants from a verbal complex. It is clear from G *schnupfen* ‘to sniff’, MDu. *snoppen* ‘id.’, Sw. dial. *snooppa* ‘to snuff’ that the strong verb **sneufan*- ~ **snūfan*- was accompanied by an iterative formation **snuppōn*- < **snuppōpi*, **snubunanpi* < **snup-néh₂-ti*, **snup-nh₂-énti*.¹⁰⁹⁴ Franck and Van Wijk further point to the alternation of OHG *snoffizen*, *snoffizen* < **snup(p)atjan*-, which with the suffix *-*atjan*- that is often added to original iteratives demonstrates an analogical paradigm **snuppōpi*, **snupunanpi*. Conversely, E dial. *snoob* ‘to sob’, Du. dial. *snobben* ‘to suck’¹⁰⁹⁵ must be derived from an equally secondary paradigm **snubbōpi*,

¹⁰⁸⁶ Lübben 361.

¹⁰⁸⁷ Ibidem.

¹⁰⁸⁸ Verdam 553.

¹⁰⁸⁹ Lexer 2, 1046.

¹⁰⁹⁰ Grimm 14, 1387-88.

¹⁰⁹¹ Lübben 360.

¹⁰⁹² Verdam 553.

¹⁰⁹³ It has been claimed that the strong conjugation of *schnauben*, which is now obsolete in German, is secondary (Kluge/Seebold: 817), but this can hardly be the case for *schniefen* < **sneufan*-.

¹⁰⁹⁴ Grimm (15, 1388) on *schnupfen*: “mit schnaufen, schnauben verwandt (ähnliche verhältnisse liegen vor bei rupfen, raufen, rauben.”

¹⁰⁹⁵ Kocks/Vording 1135.

**snubunanþi*. It is therefore more likely that the nouns under discussion are all independent formations to the different verbal forms, than that they continue an old ablauting *n*-stem.

****sprūtō, *spruttaz ‘sprout’†***

- **spreuta-*: OE *sprēot* m. ‘stake’¹⁰⁹⁶, MLG *sprēt* n. ‘id.’, MDu. *spriet* m. ‘stake, prong’, Du. *spriet* ‘blade, antenna’¹⁰⁹⁷
- **sprūtō(n)-*: MLG *sprūte*, MDu. *sprute* f. ‘sprout’, Du. *spruit* ‘shoot’
- **spruta(n)-, -ōn-*: ON *sproti* m. ‘twig’, OE *sprota* m. ‘shoot, nail’, *sprot* n. ‘sprout, plug’, OHG *sprozzo* m., MHG *sproz(ze)*, *spruz(ze)* m ‘shoot’¹⁰⁹⁸, G *Spross(e)* ‘shoot, rung’¹⁰⁹⁹
- **sprutōn-*: MHG *sprozze* f. ‘rung’¹¹⁰⁰, MLG *sprote* f. ‘id.’, MDu. *sporte, sprote* ‘id.’, Du. *sport* ‘id.’¹¹⁰¹
- **spruttōn-*: G Swi. *šprotza* ‘rung’¹¹⁰²

The formations **spreuta-*, **sprūtō(n)-*, **sprutan-* and **sprutōn-* are clearly in ablaut relationship with each other, and it can therefore be hypothesized that the vowel alternations result from an old *n*-stem. It remains problematic, however, that the expected consonant gradation is so marginal: the overwhelming majority of forms contains a single **t*, a geminate **tt* only being supported by Swi. *šprotza*. An additional, critical argument against reconstructing an ablauting paradigm is the derivational approximity of the strong verbs MHG *spríezen* < **spreut^tan-* and OFri. *sprūta* < **sprūt^tan-*, which show the characteristic competition of **eu* and **ū* as full-grade markers. It is therefore likely that the different formations discussed here represent independent derivations from the strong verbs or from the pertaining iterative *spruttōn-*, cf. EDu. *sprossen* ‘to bud out, sprout’.

¹⁰⁹⁶ Cf. Holthausen 1934: 313.

¹⁰⁹⁷ Franck/Van Wijk 652.

¹⁰⁹⁸ Lexer 2, 1122.

¹⁰⁹⁹ Grimm 17, 150-6.

¹¹⁰⁰ Lexer 2, 1120.

¹¹⁰¹ Franck/Van Wijk 650.

¹¹⁰² Grimm 17, 154.

****strūpō*, **struppaz* ‘throat’†**

- **strūpan-*, **streupan-*, -*ōn-*: ON *str(j)úpi* m., *strjúpa* n. ‘(cut) throat’¹¹⁰³, Icel. *strjúpi* m. ‘id.’¹¹⁰⁴, Far. *ranga-strúpi* m. ‘wrong throat’¹¹⁰⁵, Nw. *strupe* m. ‘throat, small inlet’, Sw. *strupe* ‘throat’, Da. *strube* ‘id.’¹¹⁰⁶
- **strūpa-*: Nw. dial. *strup* m. ‘narrow hole’
- **strupan-*: Nw. dial. *strop* n. ‘mouth of a river’, *strobe* m. ‘throat’, Sw. dial. *stråpe* ‘id.’¹¹⁰⁷

The material contains at least three different stems, i.e. **streupan-*, **strūpan-* and **strupan-*. This could point to an apophonic paradigm.

The stems **streupan-* and **strūpan-* can hardly be isolated from each other, as the correlation between ON *strúpi* < **strūpan-* and *strjúpi* < **streupan-* is clarified by the more general tendency in West Norse (Old Icelandic) to replace *ú* by *jú*, cf. ON *súga* ~ *sjúga* vs Icel. *sjúga* or Icel. *hnúkur* ~ *hnjúkur* (see § 6.5). It follows from this development that *strúpi* is the original form, as also Nw., Sw. *strupe* and Da. *strube* seem to confirm. Far. *strúpi* is actually ambiguous, because ON *jú* normally loses the palatal glide after consonants in this language (cf. Far. *rúka* = ON *rjúka* ‘to smoke’). The **eu* diphthong reappears, however, in the neuter form ON *strjúpa*. Apparently, the original masculine form *strjúpi* was incorporated into the class of neuter *n*-stems denoting parts of the body (see § 2.3.3), cf. *hjarta* ‘heart’, *lunga* ‘long’, *eyra* ‘ear’, *auga* ‘eye’.

A zero-grade stem **strupan-* emerges as Sw. and Nw. *strobe*, for which *Grunnmanuskriptet* gives the expression *svelgja seg i stropa* and *eta seg i stropa* ‘to have something go down the wrong way’. This is remarkably parallel to the Faroese expression *fáa eitthvørt í rangastrúpan*, which has the same meaning.

Morphologically, the alternation of *strúpi* and *strobe* can be explained by assuming an ablauting *n*-stem, e.g. nom. **strūpō*, loc. **strupeni*. This solution is especially attractive in view of the semantic match between the two different ablaut grades.

¹¹⁰³ De Vries 1962: 554; Jóhanesson 1956: 877.

¹¹⁰⁴ Böðvarsson 982.

¹¹⁰⁵ Poulsen 912.

¹¹⁰⁶ Falk/Torp 1183.

¹¹⁰⁷ Hellquist 882-3.

An objection to reconstructing an apophonic *n*-stem is that the expected consonant gradation is lacking. It thus becomes attractive to derive both formations from a strong verb **strūpan-* as supported by Nw. *strupe* 'to squeeze (of clothes), strangle' (cf. Nn. *stropen* 'choking'). An additional reason to assume that the verb is primary, is that it bears the more general meaning 'to squeeze', which is inexplicable if one assumes that the verb was derived from the *n*-stem 'throat'. Importantly, the Norwegian verb also shifts between *strupe* and dial. *strjupa* (Sogn). This is remarkably parallel to the variation of **strūpan-* and **streupan-*.

Other verbs pointing to a verbal complex are Nw. *strøype* 'to strangle' < **straupjan-* and Nw. *strypa* 'id.' < **strūpjan-*. Semantically primitive is Nw. dial. *strype* n. 'narrow spot' < **strupja-*. Probably, this form, too, points to an original meaning 'to squeeze' or something similar¹¹⁰⁸.

The etymological dictionaries usually connect a whole range of West Germanic forms, e.g. MHG *strūben* 'to jut out', G *struppig* 'rough', MHG *struppe* 'shrub', Du. *struif* 'contents of an egg'. This is all uncertain on the semantic side, however. Proponents of this etymology usually derive *strūpi* from a meaning 'to jut out', because the throat is a protrusion of the neck. But this suggestion must be rejected in view of the original meaning 'to squeeze'. Semantically, only the link with G *strupfen* 'to writhe'¹¹⁰⁹ can perhaps be maintained. It is possible, for instance, that it represents an old iterative **struppōn-* to the strong verb **streup^aan-/strūp^aan-*. Other alleged extra-Germanic connections, such as Gr. στρῦφνός 'bitter, crusty' and Lith. *strūbas* 'short', are even more doubtful, and Mlr. *srub* 'snout' is an obvious loanword from Old Norse.

**strūtō / *prūtō, *struttaz / *pruttaz 'throat'?*

- **prūta-*: ON *prútr* m. 'snout', Nw. *trut* m. 'mouth'
- **strūta-*: ON *strútr* 'pointed hood', Far. *strútur* m. 'spout, nozzle, snout', Nw. *strut* m. 'id.'
- **strūtō(n)-*: OFri. *strot-bolla* 'Adam's apple', OS *strota* (asg. *strótun*) f. 'tuba'¹¹¹⁰, MLG *strote*, *strate* f. 'throat'¹¹¹¹, MDu.

¹¹⁰⁸ Torp (1919: 731): "kanske egtl. «trang aapning»".

¹¹⁰⁹ Grimm 20, 137.

¹¹¹⁰ Gallée 308.

¹¹¹¹ Lübben 387.

strote f. 'id.'¹¹¹², Du. dial. *stroot* 'id.'¹¹¹³, MHG *strozze* f. 'id.'¹¹¹⁴,
G *Strosse*¹¹¹⁵, Rhdl. *strosse* f. 'pharynx, throat'¹¹¹⁶

⇒ **strutōjan*:- OS *stroton* (= pres. ptc. *stróthóndion*
'*oris garruli vox inquieta*') 'to prattle'¹¹¹⁷

- **prutōn*:- OE *prote* f. 'throat', E *throat*, OFri. *throt-bolla*
'Adam's apple', OHG *drozza* f. 'throat', MHG *drozze* mf. 'id.'¹¹¹⁸

⇒ **prutla*:- E *throttle* 'throat (of a bottle), larynx', G
Drossel 'windpipe'¹¹¹⁹

- **struttōn*:- MLG *strotte* f. 'throat'¹¹²⁰, MDu. *starte*, *sterte*,
strot(te) f. 'id.'¹¹²¹, Du. *strot* 'id.'¹¹²²

The opposition of ON *prútr* 'snout' with OE *prota* 'throat', Far. *strútur* 'spout, snout' and OE *strotta*, MLG *strotte* 'throat' could theoretically point to a paradigm **prūtō*, **pruttaz* or **strūtō*, **struttaz* (with *s* movable).

An objection to the reconstruction of this ablauting paradigm is that the full-grade vocalism is restricted to thematic formations. An additional difficulty consists of the fact that the etymology of the word is unclear. There can be a correlation with the root **prūt*- 'to bloat'¹¹²³, as in ON *prútinn* 'swollen' and OE *prūtian* 'to puff up' < **prūtējan*-¹¹²⁴, but it is also possible to connect the word to Lat. *strūma* f. 'crop' (< **stre/oud-meh₂*- or **struHd-meh₂*-). Neither of the two possibilities are self-evident, however.

In conclusion, the reconstruction of the paradigms **prūtō*, **pruttaz* and **strūtō*, **struttaz* is relatively certain, but the evidence for paradigmatic ablaut is too slight.

¹¹¹² Verwijs/Verdam 585.

¹¹¹³ WBD III, 217.

¹¹¹⁴ Lexer 2, 1251.

¹¹¹⁵ Kluge/Seebold 892.

¹¹¹⁶ Franck/Müller 8, 868-9.

¹¹¹⁷ Gallée 309.

¹¹¹⁸ Lexer 1, 469.

¹¹¹⁹ Kluge/Seebold 217.

¹¹²⁰ Lübben 387.

¹¹²¹ Verwijs/Verdam 585.

¹¹²² Franck/Van Wijk 679.

¹¹²³ Pokorny 1022-1027.

¹¹²⁴ Cf. Lühr 1988: 256ff.

9.7 The **ū ~ *u ~ *a* type

The *n*-stems in this section are a subcategory of the former type with **ū ~ *u* alternations. They largely display the same quantitative ablaut, but are in addition characterized by the recurring incidence of related forms with unexpected *a*-vocalism. The origin of this unexpected vowel grade is not clear, but there are strong indications that it must be secondary. Since all the concerned *n*-stems have a root structure **knu-* plus consonant, and furthermore share the meaning ‘knot’ or ‘knob’, it is highly probable that the roots are extensions of the Proto-Indo-European element **ǵnu-* ‘knee, node’. Apparently, the **u* of this element was interpreted as a zero grade **u* at some point, and then replaced by **a* under certain circumstances. It is conceivable that this process marked the last convulsions of the triple ablaut that is also reconstructable for e.g. nsg. **belk^kō*, gsg. **bulk^kaz*, apl. **balk^kuns* ‘beam’ < **b^hélǵ^h-ōn*, **b^hlǵ^h-n-ós*, **b^holǵ^h-n-ŋs* (see p. 148). If this is correct, the **ū*, **u* and **a* must stem from the Proto-Germanic nominative, genitive and accusative case respectively.

**knūbō, knuppaz* ‘knob’

- **knūban-*, *-ōn-*: Icel. *hnúfa* f. ‘knob, stub’¹¹²⁵, Nw. *knuv* m. ‘bump’, G Swab. *knaupe* m. ‘bump, knot, gnarl’¹¹²⁶, Swi. Bern. *xnuupa* ‘swelling’¹¹²⁷ (= **knūbbōn-*), SFri. *knuufe* m. ‘lump’
- **knub-* ⇒ **knubla-*: MDu. *cnovel* m. ‘joint, ankle’¹¹²⁸
- **knubba(n)-*: Far. *knubbi*, *-ur* m. ‘tip, bud, stub’¹¹²⁹, Nn. *knubb* ‘stub’, MLG *knobbe*, *knubbe* ‘gnarl, bump’, E *knob*
- **knuppa(n)-*: Nw. *knupp* m. ‘sprout’, OE *cnoppa* m. ‘bunch’, OHG *chnopf* m. ‘knot, knob’, G *Knopf*, MDu. *knoppe* m. ‘knot, bunch, bud’, *knop* m. ‘knob, knag’, OFri. *ers-knop* m. ‘coccyx’, E *knop*

¹¹²⁵ Böðvarsson 393.

¹¹²⁶ Fischer/Taigel 279.

¹¹²⁷ Cf. Kluge 1884: 178 fn.

¹¹²⁸ Verdam 298.

¹¹²⁹ Poulsen 609.

- **knaup^a*:- MHG *knouf* m. 'knob', MLG *knōp* m. 'knot, knob, gag', MDu. *cnoop* m. 'knot, knob'
- **knaban*:- Sw. dial. *knave* 'clasp, knob'¹¹³⁰, G dial. *knabe* m. 'peg'
- **knabba(n)*:- Far. *knabbi* m. 'tip, knob'¹¹³¹, Nw. *knabb(e)* m. 'stub'
- **knapan*:- Nw., Sw. dial. *knape* m. 'peg'
- **knappa(n)*:- ON *knappr* m. 'button', Far. *knappur* m. 'tip (of a stick)'¹¹³², Nw. *knapp* 'knob', Sw. dial. *knappe* 'peg', OE *cnæp* m. 'top, broche', OFri. *knep* m. 'button'

Von Friesen (1897: 61) reconstructed an ablauting *n*-stem **knūban-* on the basis of the opposition between the short **u* of e.g. Far. *knobbi*, OE *cnoppa* and the long **ū* of Swab. *knaupe* < **knūbban*-¹¹³³. He further adduced ON *knýfill* 'short horn' < **knūbila-* to demonstrate the Proto-Germanic nature of the full grade **knūb-*. The original vowel length of Nw. *knuv* and SFri. *knuufe*, on the other hand, is difficult to determine, and cannot be used to substantiate Von Friesen's reconstruction. By adding Icel. *hnúfa* f. 'knob' to the evidence, however, the paradigm **knūbō*, **knuppaz* indeed gains some credibility.

In addition, the occurrence of the *o*-grade thematization **knaup^a*- is neatly paralleled by other *n*-stems with an **ū ~ *u* alternation, cf. **hrūhō*, **hrukkaz* vs **hrauk^a*- 'pile' (p. 268), **klūpō*, **kluttaz* vs **klaut^a*- 'clod' (p. 272) etc.

The reconstruction of **knūbō*, **knuppaz* is in conflict, however, with the occurrence of forms with *a*-vocalism, e.g. Sw. dial. *knave* 'knob', Far. *knabbi* 'tip, knob', Nw. *knape* 'peg', OE *cnæp* 'top'. It is possible to think that these forms arose due to interference from the **a ~ *u* type. The *a*-vocalism may then have arisen secondarily in the nominative case, so as to give a paradigm **knabō*, **knuppaz*. Alternatively, it can be surmised that the *a*-vocalism was introduced in the *apl.* case. The paradigm must then be reconstructed as **knūbō*, **knuppaz*, **knappuns*.

¹¹³⁰ SAOB K1582.

¹¹³¹ Poulsen 605.

¹¹³² Ibid.

¹¹³³ Von Friesen falsely reconstructs **knūppan-*.

****knūþō*, **knuttaz* ‘knot’**

- **knūþa(n)*:- Icel. *hnúði*, -ur m. ‘knob, hump’¹¹³⁴
 - **knūt‘ōn*:- Icel. *hnúta*, Far. *knúta* f. ‘bone’¹¹³⁵
 - **knūt‘a*:- ON *knútr* m. ‘knot, knag’, Icel. *hnútur* m. ‘knot’¹¹³⁶, Far. *knútur* m. ‘knot, lump’¹¹³⁷
 - **knuttan*:- Icel. *hnotti* m. ‘tussock, ball’¹¹³⁸ (⇒ *hnjóta* ‘to stumble’ ⇒ *hnjóti*, -ur m. ‘bump’¹¹³⁹), MLG *knutte* m. ‘knot (of flax)’, MDu. *knutte* m. ‘knot of flax’, OE *cnotta* m. ‘knot’
⇒ **knuttjan*:- OE *cnyttan* w.v. ‘knot’, E *knit*
 - **knupān*-, -ōn:- Icel. *hnoði* m., *hnoða* n. ‘ball, clew’¹¹⁴⁰, OHG *chnodo* m. ‘knuckle’, Swi. Ja. *xnōdā*¹¹⁴¹, Visp. *xnodo*¹¹⁴² m. ‘id.’
 - **knupþan*:- G Cimb. *knotto* m. ‘rock’, EDu. *knodde* ‘nodus, nexus’
 - **knutōn*:- Icel. *hnota* ‘clew, vertebra’, Far. *knota* f. ‘bone’
 - **knudan*:- OHG *chnoto* m. ‘knot’, G *Knoten* ‘id.’
-
- **knattu*:- ON *knōttr* m. ‘ball, knob’

Most of the material points to a paradigm **knūþō*, **knuttaz*, **knudeni*, which seems to be derived from PIE **ǵnu*- with the same *ton*-suffix that must be reconstructed for e.g. **klīþō*, **klittaz* ‘burdock’ (p. 235) and **klūþō*, **kluttaz* (p. 272). The original nominative **knūþō* is directly continued by Icel. *hnúði* ‘knob’, the genitive **knuttaz* by Icel. *hnotti* ‘tussock, ball’ and OE *cnotta* ‘knot’. This original genitive was replaced by **knupþaz* in a secondary paradigm that underlies Cimb. *knotto* ‘rock’ and EDu. *knodde* ‘node’. Finally, OHG *chnoto* ‘knuckle’ seems to preserve the consonantism of the locative **knudeni*.

Fully parallel to other **ǵnu*-derivatives, the paradigm of **knūþō*, **knuttaz* further seems to have contained an *a*-graded root variant, viz.

¹¹³⁴ Böðvarsson 393.

¹¹³⁵ Poulsen 609.

¹¹³⁶ Böðvarsson 394.

¹¹³⁷ Poulsen 610.

¹¹³⁸ Böðvarsson 393.

¹¹³⁹ Böðvarsson 392.

¹¹⁴⁰ Ibid.

¹¹⁴¹ Stucki 70.

¹¹⁴² Wipf 41.

ON *knōttr* ‘ball, knob’ < **knattu-*. In view of the combination of the *a*-ablaut with a geminate and a *u*-suffix, it is attractive to think that it split off from the apl. case **knattuns* < **ġnot-n-ġs*. The derivational pathway at any rate runs parallel to similar split-offs such as ON *bōlkr* ‘partition’ < **balk^kuns* < **b^holġ^h-n-ġs*.

****knūsō, *knuzzaz* ‘gnarl’**

- **knūsa-*: G Swab. *knaus* m. ‘knobbly bump’¹¹⁴³, Swi. *xnuus* m. ‘messy pile’¹¹⁴⁴
 - **knūza(n)-*: MHG *knūr(e)* m. ‘knob, gnarl, summit’¹¹⁴⁵, G *Knauer* m. ‘hard lump of stone, knob’¹¹⁴⁶
 - **knuzzan-*: MHG *knorre* m. ‘bump, cartilage’¹¹⁴⁷, MLG *knorre* m. ‘knob, bump’¹¹⁴⁸, G *Knorre(n)* m. ‘gnarl’¹¹⁴⁹, MDu. *cnor(re)* f. ‘bump’¹¹⁵⁰, EDu. *cnorre* ‘tuber’, Du. *knor* ‘bump’¹¹⁵¹, ME *knorre, knurre*, E *knur* ‘gnarl’
-
- **knausa-*: ON *knauss* m. ‘round summit’¹¹⁵², Far. *kneysur* m. ‘cliff’¹¹⁵³, Nw. *knaus* m. ‘small summit’, Sw. dial. *knös* m. ‘hillock, gnarl, protuberance’¹¹⁵⁴, Da. *knøs* ‘hill(top), skerry’
-
- **knasan-*: Far. *knasi* m. ‘gnarl, bump’¹¹⁵⁵
 - **knazza(n)-*: Nw. dial. *knarre* m. ‘stub’, LG *knar(re)* ‘lump, stump’, Du. *knar* ‘skull, old person’, ME *knarre*, E *knar* ‘gnarl’

¹¹⁴³ Fischer/Taigel 279.

¹¹⁴⁴ Weber/Bechtold 1961: 46.

¹¹⁴⁵ Lexer 1, 1656.

¹¹⁴⁶ Grimm 11, 1365-6.

¹¹⁴⁷ Lexer 1, 1653.

¹¹⁴⁸ Lübben 180.

¹¹⁴⁹ Kluge/Seebold 505: “Alles Bildungen mit der Bedeutung »verdickter Gegenstand« und Anlaut *kn-*.”

¹¹⁵⁰ Verdam 298.

¹¹⁵¹ Franck/Van Wijk 327.

¹¹⁵² De Vries 1962: 320.

¹¹⁵³ Poulsen 608.

¹¹⁵⁴ Rietz 342.

¹¹⁵⁵ Poulsen 606.

PGm. **knūsō*, **knuzzaz* seems to be yet another *n*-stem derived from PIE **ǵnu-* ‘node’, this time with an *s*-suffix. The pertaining material fully patterns with the other derivatives **knūpō*, **knuttaz* and **knūbō*, **knuppaz*; a nominative allomorph **knūsō* is supported by Swi. *xnuus* ‘gnarl’, while MHG *knorre* ‘bump’ presupposes a geminated genitive **knuzzaz*. It must be stressed that this long **-zz-* cannot be regular, as Kluge’s law did not affect PIE **s* (cf. ON *ǫnn* f. ‘harvest’ < **aznō-* < **h₂es-néh₂-*). This means that the introduction of the long voiced sibilant must be completely analogical, a development that can only be understood as pointing to the grammaticalization the paradigmatic length opposition.

A thematic *o*-grade **knausa-* is represented by ON *knauss* m. ‘round summit’ and related forms in the Nordic languages.

Like the other **ǵnu*-derivatives, **knūsō*, **knuzzaz* is accompanied by related *n*-stems with *a*-vocalism, e.g. Far. *knasi* ‘gnarl, bump’ < **knasan-*, LG *knar(re)* ‘stump’ < **knazzan-*. It can be surmised that these variants arose under the influence of the *n*-stem **knagō*, **knakkaz*, based on Sw. *knagg(e)* ‘pin, knob’¹¹⁵⁶, Da. *knag* ‘knob, handle’¹¹⁵⁷, MLG *knagge* ‘knob, piece of wood’¹¹⁵⁸, Du. *knaak*, *knag* ‘big coin’¹¹⁵⁹, dial. *knaag*, *knag(ge)* ‘notch on a stick’¹¹⁶⁰. Alternatively, there is the option of reconstructing a paradigm with triple ablaut, viz. **knūsō*, gsg. **knuzzaz*, apl. **knazzuns*.

9.8 The **ō* ~ **a* type

The group of *n*-stems with **ō* ~ **a* ablaut is relatively small, but contains a number of strong examples. The old age of the type is supported by the correspondence of **mōhō*, **mageni* ‘poppy’ with Gr. *μήκων* < **méh₂k-on-*. Given this clear etymology, it seems certain that the type evolved in *n*-stems with a laryngealic root: in the full-grade cases, **-éh_{2/3}-* became PGm. **ō*, while in the zero-grade the laryngeal was vocalized to PGm. **a*. Before the merger of Pre-Gm. **ā* and **ō*, the nature of the ablaut is likely to have been purely quantitative, viz. **ā* ~ **a*, in roots containing **h₂*.

¹¹⁵⁶ SAOB K1535.

¹¹⁵⁷ Falk/Torp 543.

¹¹⁵⁸ Lübben 178.

¹¹⁵⁹ WNT, s.v. *knag*, *knaak*.

¹¹⁶⁰ Kocks/Vording 571.

****gōmō*, **ga(w)umnaz* ‘palate’?**

- **gōma(n)*:- ON *gómr* m. ‘roof or floor of the mouth, finger-tip’, Icel. *gómur* m. ‘id.’¹¹⁶¹, Far. *gómi* m. ‘oral cavity’¹¹⁶², *finfur-gómur* m. ‘finger-tip’¹¹⁶³, Nw. *gom(me)* ‘palate, gum’, OSw. *gōme* m. ‘upper or lower part of the mouth’, Sw. *gomme* ‘oral cavity, gum’¹¹⁶⁴, Da. dial. *gumme* ‘id.’¹¹⁶⁵, OE *gōma* m. ‘id.’, E *gum*, OHG *guomo* m. ‘throat’, MHG *guome* m. ‘id.’, EDu. Sax. *gumme* ‘palatum’, G obs. *gomme*, *gumme(n)* ‘id.’¹¹⁶⁶, Pal. *gummen* m. ‘mouth, pl. lips’¹¹⁶⁷
- **gauma(n)*:- OHG *goumo* m. ‘throat’, MHG *goum(e)* m. ‘id.’, G *Gaumen*¹¹⁶⁸, Cimb. *gaumo* m. ‘id.’¹¹⁶⁹
- †**geuman*:- OHG *giumo* (= nsg. *giumo* ‘palatus’, npl. *giumen* ‘fauces’) m. ‘throat’
- ?**gumman*:- OHG *gommo* (= gpl. *commono* ‘faucium’) m. ‘id.’

The formal variation of the word for ‘palate’ is difficult to interpret. The material provides clear evidence for **gōma(n)*- > ON *gómi*, OE *gōma*, OHG *guomo* and **gauman*- > OHG *goumo*, G *Gaumen*, but the correlation between the two root variants is not straightforward. Finally, OHG *giumo* has been derived from an *e*-grade **geuman*-, but this reconstruction is erroneous, as will be shown below.

What seems beyond doubt, is that the Proto-Germanic paradigm represents an *mn*-stem related to ON *gana* (pret. *ganda*) ‘to gape, yawn’ < **ganējan*-, Gr. χαίνω ‘to yawn’ < **g^hh₂-n*-, χήμη f. ‘yawn’ < **g^heh₂-meh₂*-. In the literature, the *mn*-stem is generally assumed to have been created to an extension of the underlying root **g^heh₂*-, namely **g^heh₂-u*- (cf.

¹¹⁶¹ Böðvarsson 299.

¹¹⁶² Poulsen 374.

¹¹⁶³ Poulsen 264.

¹¹⁶⁴ SAOB G759.

¹¹⁶⁵ Falk/Torp 361: “Formen **ghô^hmon* und *ghaumon*, von der Wurzel **ghôu*-, **ghau*-”.

¹¹⁶⁶ Grimm 4, 1576-81.

¹¹⁶⁷ Christmann 3, 73: “Die F. *gumæ* geht auf mhd. *guome* [...] zurück, wobei jedoch für dieses Wort auch in der südl. VPf Kürzung von *ū* < *uo* angenommen werden muß (vgl. Blume).”

¹¹⁶⁸ Kluge/Seebold 334.

¹¹⁶⁹ Schmeller/Bergmann 186.

Pokorny 449), but the origin of this **u* is uncertain. It is further plausible, in view of Lith. *gomurė* ‘palate’, and Latv. *gāmurs* m. ‘windpipe, larynx’¹¹⁷⁰, that the Proto-Indo-European word originally was a heteroclitc. The original inflectional type is difficult to determine, but if the options consist of amphikinetic **g^héh₂(-u)-mr*, gen. **g^hh₂(-u)-mn-ós*,¹¹⁷¹ proterokinetic **g^héh₂(-u)-mr*, gen. **g^hh₂(-u)-mén-s* and perhaps also static **g^hóh₂(-u)-mn*, gen. **g^héh₂(-u)-mn-s*.

Regarding the Germanic material, the most important issue is to determine what kind of inflection offers the best preconditions for the rise of the two variants **gōman-* and **gauman-*. In view of the Baltic forms, it is attractive to start from a heteroclitc that developed into a proterokinetic *mn*-stem in Germanic. The proterokinetic paradigm **g^héh₂-mōn*, **g^hh₂-mén-s* would regularly develop into PGm. **gōmō*, **gamenaz*. With this outcome, the stem **gōman-* receives a good explanation, but **gauman-*, on the other hand, does not. The extended variant **g^héh₂-u-mōn*, **g^hh₂-u-mén-s* would develop into **gōmō*, **gumenaz*. Again, **gauman-* cannot be straightforwardly accounted for.

The hysterokinetic paradigm **g^héh₂-mōn*, **g^hh₂-mn-ós*, **g^hh₂-mén-i* seems to be a better starting point, as it would result into PGm. **gōmō*, **gumnaz* (**gummaz*), **gameni*. This triple root alternation can account for the stem **gōman-* directly, and may have led to the second variant **gauman-* by contamination of **gummaz* (= OHG *commono*?) with **gameni*. Thus, the hysterokinetic paradigm seems to offer more favorable preconditions for the attested variation of **gōman-* and **gauman-* than the proterokinetic variant.

Alternatively, it is possible to resort to a hysterokinetic reconstruction with the extended root **g^héh₂-u-*. The nominative of this paradigm, **g^héh₂-u-mōn*, would have regularly produced the expected form **gōman-* by Indo-European loss of the **u* before **m*. Perhaps, then, the variant **gauman-* can be explained as the continuant of a regular genitive **gawumnaz* < **g^héh₂-u-mn-ós*.

It is furthermore possible that a static paradigm **g^hóh₂(-u)-mn*, gen. **g^héh₂(-u)-mn-s* explains the alternation of **gōman-* and **gauman-*. The key question then probably must be whether the long Pre-Gm. diphthongs **ōu* and **āu* could have behaved differently, i.e. developed

¹¹⁷⁰ Pokorny 449; Fraenkel 161.

¹¹⁷¹ Mallory/Adams 387: **g^héh₂(u)-mr*, *-mn-ós*.

into **ō* and **au* by the loss of **u* and Osthoff's law respectively. However, Go. *sauil* < **sōel* < **seh₂uel-* seems to preclude that the loss of **u* was prior to the merger of Pre-Gm. **ā* and **ō*.

Finally, the OHG alternant *giumo* needs to be explained. It is based on only two attestations in Notker, but has been projected back into PGm. as **geuman-* and even into PIE as **g^hēh₂u-mon-* or **g^hh₂éu-mon-* with a lengthened grade (Pokorny 449; Rasmussen 1999: 401 fn.). The Old High German grapheme <iu>, however, does not necessarily indicate the diphthong [iu] from PGm. **eu*. In view of its occurrence in the plural *giumen*, it is far more likely that it represents OHG *goumo* with analogical umlaut, i.e. **göumen* (see chapter 9). This explanation is more plausible than to assume that these two forms miraculously preserved an Indo-European lengthened grade, not in the least because Notker is known for incidentally indicating front mutation, e.g. *hût*, pl. *híute* 'skin' < **hūdi-*, *líuten* 'to sound' < **hlūdjan-*.¹¹⁷²

****hōdō*, **hattaz* 'hood'†**

- **hadina-*: ON *heðinn* m. 'jacket', OE *heden* m. 'robe, hood, chasuble'
 - **hatta-*: ON *hattr*, OE *hæt* m. 'hat'
 ⇒ **hattjōn-*: ON *hetta* f. 'hood, cape', Nw. *hette*, Sw. *hätta*, Da. *hætte* 'cowl'
 - **hattu-*: ON *hōttr* m. 'hat'
-
- **hōda-*: OHG *huot* m. 'hood, hat', OS *hōd* m. 'hat', OE *hōd* m. 'hood', OFri. *hōd* m. 'hat'

The etymological link between OE *hōd* 'hood', *hæt* 'hat' and *heden* 'robe' (and cognates) is generally recognized¹¹⁷³, but the possibility that the three different forms can be traced back to one single paradigm has not yet been investigated. It nevertheless seems appropriate to do just that, because Lühr (2000: 266) already reconstructed an *n*-stem **hadō*, gsg. **hattaz*, apl. **hattuns* on the basis of ON *hattr* < **hatta-* and *hōttr* < **hattu-*. This analysis effectively explains the origin of the geminates of

¹¹⁷² Cf. Braune 1891: 29.

¹¹⁷³ Fick/Falk/Torp 69; Franck/Van Wijk 254; Pokorny 516; Falk/Torp 384-5; Holthausen 1934: 282.

these stems, which otherwise must be ascribed to random *no-* and *nu-* suffixes.¹¹⁷⁴ In reality, these suffixations automatically follow from the case forms of the original paradigm, viz. gsg. **kHt-n-ós*, apl. **kHt-n-ŋs*.

Additional proof of an old *n*-stem comes from ON *heðinn* and OE *heden* < **hadina-*.¹¹⁷⁵ The etymological appurtenance of **hadina-* was already tentatively suggested by Holthausen.¹¹⁷⁶ Its exact origin is best understood by assuming that it came into existence as the original dative **hadeni*, continuing a locative **kHt-én-i*. This derivation is attractive in view of its consonantism, as the **d* regularly follows from the operation of Verner's law in this case form. It further gains probability because there is a similar dative off-shoot to another old (*m*)*n*-stem, viz. Go. *himins*, ON *himinn* 'heaven' < **hemina-* to PIE **h₂eḱ-mon-* (see p. 163).

In view of the strong evidence in favor of an *n*-stem with the case forms gsg. **hattaz*, dsg. **hadeni*, apl. **hattuns*, the question arises whether the paradigm was originally apophonic. This was, in fact, already suggested by Kauffmann (1887: 544), who attempted to explain the ablaut of OE *hōd* and *hæt* in this way. Indeed, the reconstruction of a paradigm **hōpō*, **hattaz*, **hadeni* from older **kéh_{2/3}t-ōn*, **kh_{2/3}t-n-ós*, **kh_{2/3}t-én-i* is able to account for this vocalic alternation.

In the end, however, there seem to be critical drawbacks to the reconstruction of an ablauting paradigm. First of all, the nominative **kéh_{2/3}t-ōn* would have resulted in a root ***hōp-*, not **hōd-*. An additional problem is that the root **hōd-*, as opposed to e.g. **lōfan-* 'palm' and **mōgan-* 'poppy', is never inflected as an *n*-stem. It is therefore probably best to analyze **hōda-* as a thematic *o*-grade **koHt-ó-* that existed alongside an otherwise non-ablauting *n*-stem **hapō*, **hattaz*.

Etymologically, the Germanic words are usually compared to Lat. *cassis* 'helmet', which has led to the reconstruction of a root **kat-* or **kad^h*.¹¹⁷⁷ The second variant **kad^h* has been lumped together with

¹¹⁷⁴ Cf. Fick/Falk/Torp; Franck/Van Wijk; De Vries 1962.

¹¹⁷⁵ This formation has been interpreted as a loanword from Gr. κίθων, χιτών (Fick/Falk/Torp 90), but this is difficult on the formal side. The consonantism is unstable in Greek itself and a PGM. reconstruction **hidina-* would rather have given ON ***hiðinn*.

¹¹⁷⁶ Holthausen 1934: 153; rejected: Lühr 1988: 121.

¹¹⁷⁷ Pokorny 516; Lühr 2000: 266; Falk/Torp 382; Franck/Van Wijk 254; Kluge/Mitzka 322-3.

Lith. *kuōdas* ‘aigrette’¹¹⁷⁸, which superficially points to a proto-form **kōd^h-o-*. It has been suggested, however, that *kuōdas* is a recent backformation from *kuodēlis* ‘lap, tuft’, which in turn is alleged to be a loanword from BRu. *kudel’* ‘lap, distaff’.¹¹⁷⁹ Alternatively, it could be a Germanic loanword from **hōdaz*. At any rate, it seems better to refrain from reconstructing a root **kat-* or **kad^h-*, because Lat. *cassis* with its genitive *cassidis* points to a peculiar stem **kassid-*. The connection with Av. *kata-* ‘room, cellar’ and Go. *heþjo* f. ‘room’¹¹⁸⁰ is even more vague.

A slightly more attractive etymology consists of the connection with OHG *hadara* f. ‘patch, goat skin’, MHG *hader*, G *Hader* f. ‘rag’ < **haprō(n)-*. It is possible that the original meaning of the *n*-stem **hapō*, **hattaz* was ‘cover made of goatskin’, and that it later developed into ‘hood’ and ‘hat’. Nw. *hette* f. ‘cowl’, a derivative of **hatta*-¹¹⁸¹, may then provide the semantic link between ‘hood’ and ‘hat’. Other related formations are ON *haðna* f. ‘young goat’ < **hapnōn-*, MHG *hatele* f. ‘id.’, Mlr. *cadla* ‘goat’, Lat. *catulus* m. ‘young animal’ < **kHt-(e)l-*. The verb **hōdjan-* > OHG *huoten*, OE *hēdan* ‘to guard’ is probably derived from the noun **hōda-*.

***kōkō, *kakkaz ‘cake’†**

- **kōka(n)-, -ōn-*: OHG *chuohho* m. ‘cake’, Swi. Visp. *xūoxo* ‘id.’, MLG *kōke* ‘id.’, MDu. *coeke* ‘id.’, Du. *koek* ‘id.’¹¹⁸², Nw. dial. *kok(e)* m. ‘lump, ball, pile (of dung)’, Sw. *kok* m. ‘lump (of earth)’, (*jord-*)*koka* f. ‘id.’¹¹⁸³
 ⇒ **kōkila-*: OE *cācil* ‘tortum’¹¹⁸⁴
- **kakōn-*: ON *kaka* f. ‘cake’, Du. *kaakje* n. ‘cookie’

The vowel alternation of OHG *chuohho* and ON *kaka*¹¹⁸⁵ can theoretically be accounted for by reconstructing an ablauting *n*-stem, e.g. nsg. **kōkō*,

¹¹⁷⁸ Falk/Torp 384; Franck/Van Wijk 254.

¹¹⁷⁹ Fraenkel 311.

¹¹⁸⁰ Franck/Van Wijk 254.

¹¹⁸¹ Falk/Torp 450.

¹¹⁸² De Vries/Tollenaere 341-2.

¹¹⁸³ SAOB K1802; Hellquist 335.

¹¹⁸⁴ Bosworth/Toller 120.

¹¹⁸⁵ Nw. dial. *kōke* ‘lump’ is not from **kukan-* or **kōkan-*, but just like Far. *køka* has generalized the oblique stem with *u*-mutation, cf. ON *kaka*, obl. *køku*.

gsg. **kakkaz*. In absence of any consonant alternations, however, this reconstruction remains doubtful or at least hypothetical.

The word has no good Indo-European etymology. The proposed link with Lith. *gúogas* ‘skull’ < **gog-*¹¹⁸⁶ is semantically far from evident. Nonetheless, there is no compelling reason to ascribe the vocal alternation to substrate influence, as has been proposed by Boutkan (1999b: 19).

****krōn*, **kranaz* ‘crane’†**

- **krōna-*, *-ō(n)-*: MHG ?*kruone* ‘id.’¹¹⁸⁷, MLG *krōn* m. ‘id.’¹¹⁸⁸, LG *kroune* f. ‘id.’¹¹⁸⁹, SFri. *krouns-bäie* ‘cranberry’
- **krana(n)-*: ON *trani* m. ‘crane’¹¹⁹⁰, OE *cran* m. ‘id.’¹¹⁹¹, OHG *chrano*, MLG *kran* m. ‘id.’¹¹⁹², MDu. *craen*, *cran(e)* m. ‘id.’¹¹⁹³, Du. *kraan-vogel* ‘id.’¹¹⁹⁴
 - ⇒ **kranaka(n)-*: OE *cranoc*, *cornuc* m. ‘crane’¹¹⁹⁵, OHG *chranih*, *-oh*, *-uh* m. ‘id.’, MHG *kran(e)ch(e)*, *kren(i)ch*, *kreneche*, pl. *kreniche* m. ‘id.’¹¹⁹⁶, G *Kränich*¹¹⁹⁷, MLG *kranekes-snavel* ‘geranium’¹¹⁹⁸

The Germanic dialects contain two roots meaning ‘crane’. First, there is the widespread **kran-*, which is mostly attested as an *n*-stem, cf. OHG *chrano*, OE *cran* and – with irregular anlaut – ON *trani*. In addition, there is the more marginal root **krōn-*, predominantly attested in Low and

¹¹⁸⁶ Pokorny 349.

¹¹⁸⁷ = Frankfurter Baumeisterbuch *krone*, Lexer 1, 1709.

¹¹⁸⁸ Lübben 190.

¹¹⁸⁹ Rosemann/Klöntrup 1982-4: 452-3.

¹¹⁹⁰ De Vries 1962: 596: “Das auffallende *t*- statt *k*- hatt man sehr unbefriedigend durch den Einfluss des gar nicht sinnverwandten Wortes *trami* [‘troll’] erklären wollen”.

¹¹⁹¹ Bosworth/Toller 169; Holthausen 59.

¹¹⁹² Lübben 187.

¹¹⁹³ Verdam 311.

¹¹⁹⁴ Franck/Van Wijk 342.

¹¹⁹⁵ Holthausen 59.

¹¹⁹⁶ Lexer 1, 1709.

¹¹⁹⁷ Kluge/Seebold 534-5.

¹¹⁹⁸ Lübben 187.

Middle German: MHG *kruon*, MLG *krōn*. Both roots seem to have merged into the tautological compound Du. dial. *kroene-krane*, LG *krune-krane*, a word that occurs in a famous nursery rhyme.

The Indo-European word for ‘crane’ cannot be captured by a single proto-form. The material gives proof of a considerable number of roots, and they can be traced back to at least two different stem formations, i.e. a *u*-stem and an *n*-stem.

The *u*-stem is based on the Balto-Slavic and Latin evidence. With Lith. *gėrvė* f., Latv. *dzērve* f. and OPru. *gerwe*, the Baltic languages point to a proto-form **gerh₂-u-*. SCr. *žěrāv* and Ru. dial. *žórav*, on the other hand, point to a lengthened grade of the suffix, i.e. **gerh₂-ōu*. Lat. *grūs*, gen. *gruis* has a zero grade in the root as well as the suffix, and probably continues **gruh₂-* from **grh₂-u-* with laryngeal metathesis.¹¹⁹⁹ Together, the different stem forms are suggestive of a paradigm **gérh₂-ōu*, **grh₂-u-ós* as reconstructed by Kortlandt (1985: 120). The plain velar results from depalatalization of **ǵ* before *r*¹²⁰⁰ in the zero grade **grh₂-*, from where it could spread to the full-grade root.

There is substantial evidence for an *n*-stem, too. Gr. Hsch. γέρην ‘γέφανος’ is attested as such, and can be reconstructed as **gérh₂-ēn*. The thematic form Gr. γέφανος, on the other hand, must be derived from either **gérh₂-n-* or **gérh₂-en-*. The latter reconstruction is also supported by W *garan*, as **grh₂-n-* would have given ***grawn*. However, in this case the *a* can also reflect **e* by Joseph’s rule (**-eRa- > *-ara-*) as was argued by Schrijver (1995: III.3.1.1). In spite of the fact that it can be expected on morphological grounds, there consequently is no compelling evidence for an ablauting *n*-stem **gérh₂-ōn*, **grh₂-én-i*.

It is tempting to connect the PIE *n*-stem with the one found in Germanic, especially since both formations may have had ablaut of the root. Still, the connection turns out to be impossible on formal grounds. The paradigm **gérh₂-ōn*, **grh₂-n-ós*, **grh₂-én-i* would regularly have yielded PGm. **kerō*, **kurraz*, **kureni*, but certainly no root **kran-* or **krōn-*. Together, these two roots rather seem to point to a paradigm nom. **kr-ōn*, acc. **kr-an-un* from older **gr-ōn*, **gr-on-m*, but the lack of the laryngeal in this paradigm is difficult to explain.

¹¹⁹⁹ Schrijver 1991: 246.

¹²⁰⁰ Kortlandt 1978: 237.

Given the more general tendency of thematicized forms to introduce the *o*-grade, it is also possible to regard **krōna-* as a split-off from a further non-apophonic *n*-stem **kranan-*. Such a derivational path is not unique, as is evident from the correlation between OHG *hano* m. 'rooster' < **hanan-* and OHG *huon* n. 'fowl' < **hōna(z)-* (cf. Darms 1978: 134-8). This still does not clarify the derivation of **kranan-*, however.

****lōfō, *lappaz* 'palm of the hand'**

- **lōfan-*: Go. *lofa* m. 'id.', ON *lófi* m. 'open hand, palm of hand'¹²⁰¹, ME *lōve* 'palm', EDu. *loef*, *loeve* 'oar peg, thole pin', Du. *loef-zijde* 'windward side'¹²⁰²
- **labba(n)-*: OHG *lappo* '*palmula, extrema latitudo remi*', G Als. *lappe*ⁿ m. 'rudder blade'¹²⁰³, Far. *labbi* m. 'paw, open glove'¹²⁰⁴, Nw., Sw. *labb* m. 'paw, big hand'¹²⁰⁵, Da. *lab(be)* 'id.'
⇒ **labbōjan-*: Icel. *labba* 'to walk'¹²⁰⁶
- **lappō-*: Icel. *löpp* f. 'paw'¹²⁰⁷
- **lapōn-* or **laffōn-*: OHG *laffa* '*palmula (remi)*', MHG *laffe* f. 'id.'¹²⁰⁸

The ablaut of Go. *lofa* and OHG *laffa* was first mentioned by Kauffmann in his article *Zur Geschichte des germanischen Consonantismus* (1887: 544). A lot of additional material can be adduced in favor of an ablauting paradigm, which most probably looked like **lōfō, *lappaz, *labeni* (cf. Lühr 1987: 68). The evidence can be collected from all Germanic dialects, including Gothic.

The full grade **lōfan-* is relatively well-attested and can be retrieved from three different branches, cf. Go. *lofa*, ON *lófi*, ME *lōve*, EDu. *loeve*. It does not occur, however, in Old High German. The zero grade is found throughout North and West Germanic in combination with several

¹²⁰¹ De Vries 363.

¹²⁰² Franck/Van Wijk 393.

¹²⁰³ Martin/Lienhart 1, 600b.

¹²⁰⁴ Poulsen 660.

¹²⁰⁵ SAOB L2: "i avljudsförh. till got. *lofa*, flat hand".

¹²⁰⁶ Böðvarsson 549.

¹²⁰⁷ Böðvarsson 613.

¹²⁰⁸ Lexer 1, 1812.

different consonantisms. One of the best represented variants is **labban-*, which is implied by e.g. OHG *lappo* ‘palmula’, Als. *lappē* ‘rudder blade’ and Far. *labbi* ‘paw’. The attestation in both North and West Germanic seems to indicate that it is old, and it can probably best be analyzed as a contamination of the genitive **lappaz* with the dative **labeni*. The original genitive root seems to be preserved by Icel. *löpp* ‘paw’¹²⁰⁹ < **lappō-*.

Semantically similar to OHG *lappo* is the formation OHG *laffa*, which can continue either **lapōn-* or **laffōn-*. The former variant could continue a secondary paradigm **lapō*, **lappaz*, the latter **lafō*, **laffaz*. It seems difficult, however, to determine which of the two variants actually existed.

Together, the different root variants point to a paradigm that was split up in several different ways in the separate daughter languages. The co-occurrence of ON *lófi* and Icel. *löpp* indicates that the paradigm **lōfō*, **lappaz* was retained by Proto-Norse. In West Germanic, too, both the consonant and the vowel alternations seem to have been preserved up to a late stage.

The Germanic *n*-stem receives a good etymology with the connection of the Balto-Slavic word for ‘paw’, viz. Lith. *lōpa*, Ru. *lāpa* f. ‘paw’ < **leh₂p-eh₂-*.¹²¹⁰ On the basis of this etymology, the paradigm underlying the Germanic *n*-stem can be reconstructed as **léh₂p-ōn*, **lh₂p-n-ós*, **lh₂p-én-i*. This reconstruction would have regularly developed into PGm. **lōfō*, **lappaz*, **labeni* by the vocalization of the laryngeal in the cases with zero-grade roots.¹²¹¹

There is no compelling reason to analyze the interchange of **ō* with **a* as a substrate feature, as has been done by Boutkan (1999: 19-20). It has been argued in the above (§ 7.3) that this is a misunderstanding induced by a flawed interpretation of the consonant alternations in the *n*-stems.

¹²⁰⁹ The semantically close ON, Icel. *lappa* f. ‘paw’ is unrelated. De Vries (p. 366) derives it from PGm. **lumpōn-*.

¹²¹⁰ The Slavic forms with initial dentals, e.g. Cz. *tlapa*, dial. *dlapa*, Pol. dial. *dtapa* ‘paw’, were no doubt influenced by OCS *dlanъ*, Cz. *dlaň*, Pol. *dłoń* ‘palm of the hand’ < PSL. **dōlnъ*.

¹²¹¹ Compare the following examples: MLG *lak* ‘limp’ < *lh₂g-o-* to Gr. *λαγάρως* ‘weak’, OE *læccan* ‘to seize’ < **lakjan-* to Gr. *λάζομαι* ‘id.’ < **lh₂g-ie/o-* and Go. *lats* ‘sluggish’ < **lh₁d-o-* to Gr. Hsch. *ληδεῖν* ‘to be tired’.

***mōhō, *mageni ‘poppy’**

- *mōgan-, -ōn-: OSw. *val-mōgha* f. ‘id.’, -mōghe m. ‘id.’, Sw. *vall-mo* ‘id.’, Gutn. *vall-moge* f. ‘id.’, Nw. dial. *vall-mo(g)* m. ‘Lily-of-the-Valley’, *kvit-mo(ge)* m. ‘melancholy thistle’, ODa. *val-mu(gh)æ* ‘poppy’, Da. *valmue* ‘id.’ (= Far. *valmua*, Icel. *valmúi*, Nw. *valmue*)
- *mahan-: OHG *maho* m. ‘id.’, MHG *mahen*, *mān*, *mōn* m. ‘id.’, G *Mohn*, OS *maho* ‘papaver’, EDu. *maen* ‘id.’, Du. *maan-zaad* ‘poppy seed’
- *magan-: OHG *mago* m. ‘poppy’, MHG *mage(n)* m. ‘id.’, Cimb. *mago* m. ‘id.’¹²¹², Swab. *mag^e* m. ‘id.’¹²¹³, E *maw-seed* ‘poppy seed’, OS *magon-hōbūd* ‘poppy head’

The form of the Germanic word for ‘poppy’ is rather changeable across the different dialects. Several different stem variants need to be reconstructed, viz. *mōgan-, *mahan- and *magan-.¹²¹⁴ Together, these stems point to the pre-existence of an ablauting *n*-stem.

The North Germanic stem *mōgan- can be retrieved without great effort. The word is not attested in Old West Norse, but in East Norse it emerges as the second member of a compound with *val-* ‘sleep’, viz. OSw. *val-mōghe*, ODa. *val-mu(gh)æ*, Gutn. *vall-moge*. The dialectal Norwegian compound *kvit-mo(ge)* belongs here too, but refers to a thistle rather than a poppy. This can be due to the physical similarities between the burr of the thistle and the poppy seed box. Note that the vocalism of ODa. *val-mu(gh)æ* is opaque, because Old East Norse *-ugi* and *-ōgi* merged into Old Danish *-u(gh)æ* (cf. ODa. *albuæ* ‘elbow’, ON *alnbogi* < PGM. **bugan-*).¹²¹⁵

The dialectal situation is more complex in West Germanic, especially in view of the Old High German variants *maho* and *mago*. The problem is not so much the clear *grammatischer Wechsel* displayed by these forms, but rather the difficulties concerning the reconstruction of the vowel length.

¹²¹² Schmeller/Bergmann 207.

¹²¹³ Fischer/Taigel 310.

¹²¹⁴ Cf. Schaffner 561-2.

¹²¹⁵ Icelandic *valmúi*, Far. *valmua* and Nw. *valmue* were adopted from Danish, and have no further relevance in this context.

In Old High German, vowel length is usually not indicated systematically. Even if a vowel is marked with an accent, this may very well indicate syllable stress rather than vowel length. This ambiguity has led to a confusing divergence regarding the reconstruction of the vowel length of both OHG *maho* and *mago*. For example, Fick, Falk & Torp (p. 303) and Kluge & Mitzka (p. 484) cite *māho* and *māgo* with a long vowel. Similarly, Lexer assumes length for all extant Middle High German forms, i.e. *māge*, *māgen*, *māhen*, *mōn*. Pokorny (p. 698), on the other hand, differentiates between OHG *māho* and *māgo*, implying that Proto-Germanic had both **mēgan*- and **magan*-. Kluge and Seebold (p. 627) conversely give MHG *māhen* vs OHG *mago* from **mēhōn*- and **magōn*-.

The main problem concerning the Old High German vowel quantity is that the dictionaries usually emend length on the basis of the modern German form *Mohn* 'poppy', which shows the occasional rounding of *ā* to *ō* in the standard language (esp. in nasal environments, cf. *Mond* 'month' < **mēnōp*-). This line of thinking is incorrect, however, because the originally long **ā* of G *Mohn* arose through contraction of a thematized form OHG **mahan*, which during its development into MHG *mān* lost its intervocalic *h*. As a consequence, G *Mohn* is unable to substantiate a PGM. form **mēhan*-.

In fact, direct evidence against **mēhan*- is furnished by Schaffner (2001: 561), who adduces the form *maan* 'poppy' from the Dutch dialect of the Zaan area. As this dialect used to differentiate between PGM. lengthened **a* and **ē* as [ā] vs [ē], the word is more likely to represent **mahan*- than **mēhan*-.¹²¹⁶ Similarly, the evidence from other modern dialects precludes the reconstruction of OHG *mago* as **māgo* from PGM. **mēgan*-. In Upper German, there are Cimb. *mago* and Swab. *mag^e*, which point to **magan*-. E *maw-seed* presupposes short **a* as well, because **mēg*- would have resulted in ***mow* (cf. PGM. **lēga*- > ME *lāh* > E *low*).

Everything considered, the Germanic material carries evidence for only three forms, viz. **mōgan*-, **mahan*- and **magan*-. This type of variation is best explained by reconstructing an ablauting *n*-stem nom. **mōhō*, loc. **mageni*. Notably, this paradigm is in perfect agreement with Gr. μήκων and Dor. μάκων f. 'poppy' (cf. Tsakonian μάκος 'peony'¹²¹⁷), so that we are allowed to reconstruct a PIE paradigm **méh₂k-ōn*, **mh₂k-én-i*.

¹²¹⁶ However, the distinction has practically disappeared in this dialect.

¹²¹⁷ Cf. Heldreich 1910: 6.

Since the alleged variants with Proto-Germanic **ē* can be dropped, the polymorphism of the word is no longer problematic from the etymological perspective. This removes the necessity to analyze the lexeme as a loanword, as has been suggested by e.g. Boutkan (2003a: 15) and Kluge/Seebold (p. 627).

The ablauting *n*-stem was also reconstructed by Schaffner (p. 562). His analysis, however, contains two problems. First, Schaffner assumed that **mh₂k-* would yield **unk-* with vocalization of the *m*, and consequently that the Germanic forms with **a* were due to analogy. It was demonstrated by Beekes (1988), however, that in roots of the structure RHC- the laryngeal is vocalized, not the resonant. Compelling evidence for this vocalization is extant as PGM. **magra-*, which must be derived from PIE **mh₂k-ró-*. As a result, the stem **magan-* is likely to regularly continue **mh₂k-ón-*.

Schaffner further assumed that the paradigm **méh₂k-ôn*, **mh₂k-ón-* was replaced by ***mh₂k-ôn*, **mh₂k-ón-* before the operation of Verner's law, so as to explain the paradigm **mahō*, **magan-*. This early split-up, however, offers no explanation for the **g* of **mōgan-*, which seems to be adopted from the locative **mageni* after the operation of Verner's law. It is therefore more attractive to assume that the consonantal analogies took place at a late stage, and that the loss of the paradigmatic ablaut was posterior to these analogies: in West Germanic, the zero grade was generalized, so as to yield a paradigm **mahō*, **mageni*. The zero grade was lost in North Germanic, but not before the full-grade nominative **mōhō* could adopt the consonantism of the locative **mageni*. In other words, the apophonic paradigm remained intact until after the breaking up of the Proto-Northwest Germanic unity.

Beside the Germanic and Greek *n*-stems, a thematic formation must be reconstructed for Slavic, cf. Ru. *mak*, gen. *máka* m. 'poppy' < **meh₂k-o-*¹²¹⁸, and probably also for Alb. *mokth* m. 'pheasant's eye'¹²¹⁹ (= **meh₂k-* plus the diminutive suffix *-th* < *-ko-*¹²²⁰). The emerging linguistic distribution is not congruent with the historical spread of the poppy as a

¹²¹⁸ The Slavic word was borrowed into Old Prussian as *moke*. Lith. *māg(u)onė* and its enigmatic variant *aguonà* are generally assumed to have been adopted from Germanic.

¹²¹⁹ Taken from Newmark 1999: 536.

¹²²⁰ Cf. Alb. *kurpth* (beside *kurpën*) 'old-men's-beard', *elbth* 'barley' (Camaj 1966: 121-2).

cultivar. The plant was probably first cultivated for its seeds in Southern France and the surrounding area. Remains of poppy seeds are found in middle and late Neolithic sites in Central Europe, but carbonized specimens have also been recovered in West Germany from an Aldenhoven Linear Pottery (5500–4500 BC) find.¹²²¹ Since the archaeological distribution has no overlap with the Indo-European homeland to the North of the Black Sea, it must be assumed that the Indo-European term *méh₂k-on- originally referred to a species of wild poppy, and that its use was extended to the cultivated variety only later in the individual daughter languages.

***slōgō, *slakkaz ‘sludge’†**

- *slōga-: MLG *slōch* ‘swamp’¹²²², OE *slōh* mn. ‘miry place’¹²²³, E *slough*
- *slōk^ka-, -ō(n)-: Nw. dial. *slok* m. ‘pool on the floor’, MHG *sluoche* f. ‘ditch’¹²²⁴, G *Schluche* ‘waterfall’¹²²⁵, Du. dial. *sloek* ‘lump of dung’¹²²⁶
- *slaga(n)-: Icel. *slagi* m. ‘indoor puddle, moist’¹²²⁷, Far. *slag* n. ‘moisture’¹²²⁸, MLG *slage* ‘lump of butter’¹²²⁹
- *slakan-: Icel. *slaki* m. ‘moist’
⇒ *slaknan-: Icel. *slakna* ‘to become wet’¹²³⁰
- *slagga(n)-, -ōn-: Sw. *slagg(-väder)* ‘rainy weather’¹²³¹, G *Schlack* m. ‘mush’, *Schlacke* f. ‘slag’¹²³² (= Du. *slak* ‘slag’¹²³³), MLG *slagge* m. ‘slag, rainy weather’ (= ON *slaggi* ‘slag’)

¹²²¹ Cf. Zohary/Hopf 2000: 135-8.

¹²²² Lübben 355.

¹²²³ Bosworth/Toller 886; Holthausen 1934: 300.

¹²²⁴ Lexer 2, 992.

¹²²⁵ Neuestes Conversations-Lexicon VIII, 254.

¹²²⁶ WLD I/1, 16.

¹²²⁷ Böðvarsson 899.

¹²²⁸ Poulsen 1074.

¹²²⁹ Lübben 351.

¹²³⁰ Böðvarsson 899.

¹²³¹ Hellquist 782.

¹²³² Kluge/Seebold 805.

¹²³³ According to Franck/Van Wijk (p. 613), the word is borrowed from G *Schlacke*.

- ⇒ **slaggō(ja)n-*: MLG *slaggen* ‘to be rainy’¹²³⁴
- ⇒ **slaggjōn-*: MDu. *slegge* f. ‘drizzle, fine snow, damp fog’¹²³⁵, EDu. *slegghe* ‘cloud, moisture, continuous rain, hail’, Du. dial. *slegge* ‘swampy spot, puddle, wet snow’¹²³⁶
- **slakka(n)-, -ōn-*: G *Schlack* m. ‘mush, daub’¹²³⁷, MDu. *slac(ke)* f. ‘snail, slag’¹²³⁸, Du. dial. *sjlak* ‘puddle’¹²³⁹

⇒ **slakkjōn-*: MDu. *slec(ke)* f. ‘snail, slag’, EDu. *slecke* ‘slag’

The North and West Germanic dialects contain traces of an *n*-stem with a meaning ranging from ‘damp weather, drizzle’ to ‘mud’ and ‘slag’. The evidence points to the usual consonant variation **slag-*, **slakk-*, **slagg-*, which can be explained by a normal paradigm **slagō*, **slakkaz*. This *n*-stem is in ablaut correlation with OE *slōh*, gen. *slōges* ‘miry place’, Du. dial. *sloek* ‘lump of dung’, Nw. dial. *slok* ‘pool’ and probably also with the obscure G *Schluche* ‘waterfall’. The link between **slagō*, **slakkaz* and **slōga-/slōk^ka-* seems to be confirmed by the spread of gemination to the full-grade forms. It is not entirely certain, however, whether both of the ablaut grades once formed one single paradigm, i.e. **slōgo*, **slakkaz*, or that the full grades arose in independent thematic derivations.¹²⁴⁰ It seems quite likely, however, that the different formations stem from a verbal complex, e.g. a strong verb **slahan-* in combination with an iterative **slakkōpi*, **slagunanpi*.

The vocalic alternation of MDu. *slacke* and *slecke*, MLG *slagge* and MDu. *slegge* is not entirely clear. The most direct way to explain these forms with *e*-vocalism is to assume that they reflect the derivations **slakkjō(n)-* and **slaggjō(n)-* (cf. MHG *krebe* < **kreban-* vs *kribbe* < **krebjō(n)-*, see p. 179).

¹²³⁴ Lübben 351.

¹²³⁵ Verdam 546.

¹²³⁶ Kocks/Vording 1109; Weijnen 182.

¹²³⁷ Grimm 15, 254; Kluge/Seebold 805.

¹²³⁸ Verdam 545.

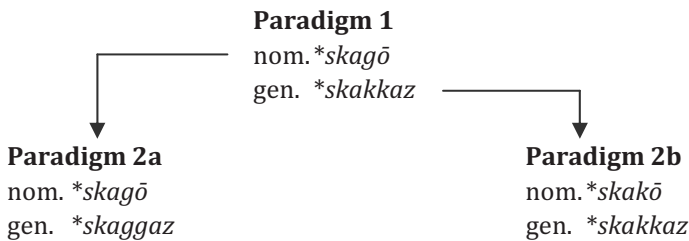
¹²³⁹ Weijnen 179.

¹²⁴⁰ Boutkan (2003: 248) took the alternation of **a* with **ō* to be an indication of a substrate origin. This is unlikely given the systematic functioning of both vowels in Proto-Germanic morphology.

***skōgō, *skakkaz ‘tip, brush’†**

- *skagan-: ON *skagi* m. ‘low cape, ness’¹²⁴¹, Icel. *skagi* m. ‘peninsula’¹²⁴², OE *sceaga* m. ‘brush’¹²⁴³, E *shaw*
 - *skaggan-: OE *sceagga* m. ‘hair’¹²⁴⁴ (⇒ *sceaggede* ‘comosus’¹²⁴⁵), E *shag*
⇒ *skagja-: ON *skegg* n. ‘beard’
 - *skakan-: OHG *scahho* ‘promuntorium’, MHG *schache* m. ‘isolated grove’¹²⁴⁶
-
- *skōga-: ON *skógr* m. ‘forest’¹²⁴⁷

In view of the material collected here, the reconstruction of an *n*-stem **skagō*, **skakkaz* seems beyond serious doubt; the consonant alternations can all be explained by Kluge’s law and the usual paradigmatic analogies. ON *skagi* and OE *sceaga* both continue a stem **skagan*-. A second root **skakk*- is presupposed by the obsolete English adjective *shack* ‘shaggy’.¹²⁴⁸ OHG *scahho* reflects a third, analogical stem **skakan*-, which must have arisen through contamination of **skag*- with **skakk*-. Finally, OE *sceagga* presupposes a fourth variant **skaggan*- with an analogically voiced long stop. Together, the four different variants point to the usual paradigmatic split:



¹²⁴¹ De Vries 1962: 480.

¹²⁴² Böðvarson 845.

¹²⁴³ Holthausen 1934: 272.

¹²⁴⁴ Cf. Holthausen 1934: 272; CIGI 1, 1500: *coma* feax, *sceacga*.

¹²⁴⁵ CIGI 1, 1514.

¹²⁴⁶ Lexer 2, 662.

¹²⁴⁷ De Vries 1962: 497.

¹²⁴⁸ OED, s.v. *shack*.

In addition to the roots with **a*-vocalism, a root **skōg-* occurs in the North Germanic word for ‘forest’, viz. ON *skógr* < **skōga-*. The question is whether this root originally functioned as the nominative allomorph of an apophonic *n*-stem **skōgō*, **skakkaz*, as Kauffmann (1887: 521), in fact, assumed, or that it represents a separate formation. Since the word is not inflected as an *n*-stem, but rather as an *a*-stem, the latter option seems preferable.

Both **skagō*, **skakkaz* and **skōga-* were probably derived from a verbal root, cf. ON *skaga*, *-ði* ‘to jut out’ < **skagējan-*. Etymologically, this verb has been connected with OIr. *der-scaigim* ‘to protrude’.¹²⁴⁹ The link with OCS *skočiti*, Lith. *šókti* ‘to jump’¹²⁵⁰ is semantically less attractive.

****tōgō*, **takkaz* ‘twig’**

- **tōga(n)-*: OHG *zuogo* m. ‘brachium, palmes, surculus’¹²⁵¹, Tyr. *zueggn* m. ‘prong, jag’¹²⁵², OS *tōgo* m. ‘twig’, MLG *tōch*, pl. *tōge(re)* ‘twig’¹²⁵³, MDu. *tooch* ‘twig, shoot’¹²⁵⁴, Du. dial. *toeg(e)*, *toog* ‘branch’¹²⁵⁵
- **tōk^kan-*: Du. obs., dial. *toek(e)* m. ‘branch (with leaves)’¹²⁵⁶
- **takan-*: LG *take* ‘prickle’¹²⁵⁷, ?WFri. *toake* ‘branch’¹²⁵⁸
- **tagga(n)-*: OSw. *tagger* m. ‘spike’, Sw. *tagg*¹²⁵⁹, Da. *tagge* (= Far. *tagga* f. ‘edge’¹²⁶⁰), Nw. *tagg(e)* m. ‘edge, tip’, MLG *tagge* ‘twig’¹²⁶¹, E *tag*
 ⇒ **tagla-*: Go. *tagl* n. ‘hair’, ON *tagl* n. ‘tail’, OHG *zagal* ‘verber, syrma, cauda’

¹²⁴⁹ De Vries 1962: 480.

¹²⁵⁰ Pokorny 922-923.

¹²⁵¹ Starck/Wells 10, 772.

¹²⁵² Schatz/Finsterwalder 735.

¹²⁵³ Lübben 406.

¹²⁵⁴ Verdam 613.

¹²⁵⁵ Kocks/Vording 1239; Weijnen 1996: 206.

¹²⁵⁶ WNT, s.v. *toek*; Kocks/Vording 1239; Weijnen 1996: 206.

¹²⁵⁷ Doornkaat-Koolman 3, 386.

¹²⁵⁸ Buitenrust Hetteema 1891: 244.

¹²⁵⁹ Hellquist 948.

¹²⁶⁰ Poulsen 1199.

¹²⁶¹ Lübben 398.

- **takka(n)*:- MHG *zacken* pl. 'jags'¹²⁶², G *Zacke(n)* 'edge, jag, prong'¹²⁶³, MLG *tack(e)* m. 'branch'¹²⁶⁴, MDu. *tac(ke)* m. 'jag, branch'¹²⁶⁵, Du. *tak* 'branch', ME *takke* 'button, clasp', E *tack* 'small nail'

The Northwest Germanic languages offer overwhelming evidence for the ablaut of the *n*-stem **tōgō*, **takkaz*, **tageni*. The reconstruction of such a paradigm is necessary to account for the vocalic and consonantal variation in these languages.

A survey of the material shows that the zero grade is prevalent in both North and West Germanic in a variety of stems with different consonants. A variant **takkan-* must be reconstructed on the basis of e.g. MHG *zacke*, MDu. *tac(ke)*, ME *takke*.¹²⁶⁶ The root **tagg-* occurs as an *a*- and *n*-stem in forms such as OSw. *tagger*, MLG *tagge*, E *tag* etc.¹²⁶⁷ It is further possible that LG *take* and WFri. *toake* continue a third variant **takan-*. With these forms alone, the reconstruction of a Northwest Germanic *n*-stem with consonant gradation seems ascertained.

The full grade is less widespread and only occurs in West Germanic. Within this branch, its attestation is excellent, however. OHG *zuogo* is well-attested as a gloss, and OS *tōgo* is found in the Heliand phrase *mid bōmo tōgun* 'with tree branches'. This pushes the attestation of the word back to the oldest West Germanic languages. The word furthermore appears to have lived through the Middle Germanic stage into the present, as is demonstrated by e.g. Tyr. *zuegg*n and Du. dial. *toeg(e)*.

Importantly, the direct appurtenance of **tōgan-* to the zero-grade variants is backed up by the Dutch dialectal variant *toek(e)* with a conspicuous final **k*. In the literature, this *toeke* is usually explained as a contamination form of *toege* with *tak*¹²⁶⁸, so as to account for the consonantism. Such a contamination indeed adequately clarifies the

¹²⁶² Lexer 3, 1017.

¹²⁶³ Grimm 31, 11-3.

¹²⁶⁴ Lübben 398.

¹²⁶⁵ Verdam 959.

¹²⁶⁶ Icel. *takki* m. 'switch, knob' (Böðvarsson 1029) must given its meaning be a loanword from Middle English *takke* or its unattested Old English fore-runner.

¹²⁶⁷ The modern Scandinavian forms can be borrowed from Low German. At least Far. *tagga*, with its conspicuous *-a*, looks like a loanword from Da. or MLG *tagge*.

¹²⁶⁸ Cf. WNT, s.v. *toek*; De Vries 1972: 24.

morphology of *toeke*, but the contamination must have taken place at the paradigmatic level rather than the lexical level: the geminate of **takkaz* spread to the nominative **tōgō* at a time when the ablaut of the paradigm had not yet been leveled. Thus, the variant **tōk^kan-* represents the missing link between the full-grade and the zero-grade forms.

The reconstruction of the paradigm **tōgō*, **takkaz* has far-reaching implications for the etymology of the word. On the basis of the Germanic evidence, it must be reconstructed as PIE **déh_{2/3}g^h-ōn*, ***dh_{2/3}g^h-n-ós*, **dh_{2/3}g^h-én-i*. This paradigm obviously precludes the old connection with **twīgō*, **twikkaz* (see p. 256).¹²⁶⁹ In *Wortschatz der germanischen Spracheinheit* (p. 173), for instance, OHG *zuogo* is cited under **twīha-*, even though it is clear that the roots **twīh-* and **tōg-* are impossible to reconcile. Pokorny (p. 228-232), too, argues that *zuogo* belongs to **du(e)i-g^ho-*, assuming that it was remodeled after the cardinal number **twō* ‘2’ (< **duoh₁*).

Equally problematic is the common connection of Go. *tagl*, ON *tagl*, OHG *zagal* (etc.) with Skt. *daśá-* ‘fag end’ < **deḱ-eh₂-* and Ir. *dúal* ‘frill’¹²⁷⁰, because the Sanskrit form does not contain a laryngeal. Instead, PGM. **tagla-* can better be regarded as a diminutive formation to the zero-grade root **tag-* < **dh_{2/3}g^h-*. This analysis is particularly attractive in view of the semantic field of MHG *zagal* m. ‘tail, prick, prickle’ (also cf. *zagal-holz* ‘top branches’).¹²⁷¹

In the end, the only extra-Germanic connection that remains compatible with the paradigm **déh_{2/3}g^h-ōn*, **dh_{2/3}g^h-n-ós* is Alb. *degë* f. ‘branch’ (< **doig^h-* or **dōg^h-*). Note, however, that Demiraj (1997: 125) claimed this formation to be purely Albanian.

9.9 The **ō ~ *ǔ* type

An **ō ~ *ǔ* alternation is shown by at least three Germanic nouns. Of the instances that display this alternation, there are two heteroclitics: *fōr*, **funaz* ‘fire’ and **sōl*, **sunaz* ‘sun’. Only **krōhō*, **krukkaz* / **krūk^kaz* ‘jug’ unambiguously classifies as an original *n*-stem.

¹²⁶⁹ Fick/Falk/Torp 173; Holthausen 1921: 136; Pokorny 228-232.

¹²⁷⁰ Cf. Pokorny 191.

¹²⁷¹ Lexer 3, 1019.

The $\bar{o} \sim \bar{u}$ type looks like a mixture of the $\bar{o} \sim \bar{a}$ type and the $\bar{u} \sim \bar{u}$ type. It arose in ablauting nouns whose root contained a coloring laryngeal plus $\bar{u}$. In the full grade, the cluster $\bar{e}h_{2/3}u-$ resulted in a diphthong $\bar{o}u$, which by regular loss of the labial off-glide developed into PGm. $\bar{o}$. In the zero grade, the sequence $\bar{e}h_{2/3}u-$ seems to have given short $\bar{u}$ in the heteroclitics $\bar{f}ōr$, $\bar{f}unaz$ ‘fire’ and $\bar{s}ōl$, $\bar{s}unaz$ ‘sun’, but long $\bar{u}$ in the real n -stem $\bar{k}rōhō$, $\bar{k}rūk^k az$ ‘jug’. The cause of this differentiation probably is Dybo’s law, by which long vowels were shortened before a resonant when the stress was on the next syllable. Prior to Dybo’s law, the zero-grade cluster $\bar{u}h_{2/3}-$ must first have developed into long $\bar{u}$ by laryngeal metathesis.

$\bar{f}ōr$, $\bar{f}unaz$ ‘fire’

- $\bar{f}ōn$, $\bar{f}un(en)az$: Go. *fon*, *funins* n. ‘id.’, ON *funi* m. ‘id.’¹²⁷²
- $\bar{f}ū(i)r-$: ON poet. *fúrr*, *fýrr* m. ‘id.’¹²⁷³, OHG *fiur*, *fuir*, *vugir* n. ‘id.’, OE *fȳr* n. ‘fire, hearth’¹²⁷⁴

In Proto-Indo-European, the word for ‘fire’ is generally reconstructed as a heteroclititic, i.e. either $\bar{p}éh_2u-r$, $\bar{p}h_2u-n-ós$ or $\bar{p}éh_2u-r$, $\bar{p}h_2u-úén-s$, cf. Hitt. *paḥḥur*, *paḥḥwenas* n. ‘fire’. The ablaut of this paradigm is preserved by Go. *fon*, gen. *funins* < $\bar{f}ōn$, $\bar{f}unenaz$.¹²⁷⁵ The heteroclitism is no longer attested in Gothic, but must be reconstructed on the basis of the Northwest Germanic evidence, which reveals a clear alternation between stems in $-r$ and $-n$, cf. ON *fúrr* < $\bar{f}ūr$ < $\bar{p}h_2u-r$ (cf. Gr. $\pi\bar{\upsilon\rho}$), ON *funi* < $\bar{p}h_2u-n$.

The different forms contain at least two separate roots, $\bar{f}ō-$ and $\bar{f}u-$, which respectively point to PIE $\bar{p}eh_{2/3}u-$ and $\bar{p}h_{2/3}u-$. The vocalism of OHG *fiur*, *fuir* and ON *fýrr* is somewhat ambiguous. De Vries (p. 149) reconstructs $\bar{f}eura-$ as if from $\bar{p}eu(H)r-$, but this reconstruction would have produced ON $\bar{f}fórr$. More probably, ON *fýrr* as well as OE *fȳr* and OHG *fiur*, *fuir* (= [fy:r]) contain the stem of the original locative $\bar{f}ueri$,

¹²⁷² De Vries 1962: 147.

¹²⁷³ De Vries 1962: 147, 149.

¹²⁷⁴ Bosworth/Toller 351.

¹²⁷⁵ Cf. Beekes 1996: 5; Kluge/Seebold 289-9: “Ausgangspunkt ist ig. $\bar{p}ehwr/\bar{p}hwnos$ [...]”

which replaced PIE **puH-én-i*.¹²⁷⁶ Note that in this form, just like in the original genitive **ph₂u-n-ós*, any long **ū* would have been shortened by Dybo's law of pretonic shortening.¹²⁷⁷

As opposed to **sōl*, **sunnaz* 'sun' (see p. 323) the paradigm of **fōr*, **funaz* did not receive an analogical geminate. The motivation behind this difference is not entirely clear, but it seems to have had something to do with the fact that **sōl*, **sunnaz* transgressed to the common gender *n*-stems, where consonant gradation was regular, whereas **fōr*, **funaz* remained a neuter.

****krōhō*, **krūk^kaz* 'jug'?**

- **krōga-*: OHG *chruog* m. 'jug', G *Krug*¹²⁷⁸, MDu. *croegh* 'id.', OE *crōg* m. 'crock'¹²⁷⁹
- **krūk^kōn-*: MHG *krūche* f. 'jar', OS *krūka* f. '*cambuca*'¹²⁸⁰, MDu. *cruke* f. 'jar', Du. *kruik* 'id.'¹²⁸¹, OE *crūce* f. 'crock', E *crouke*
- **krukkan-*, *-ōn-*: ?ON *leir-krukka* f. 'leather jug'¹²⁸², Icel., Far. *krukka* [f.] 'pot, jar', OE *crocca* m. 'crock', *crocce* f. 'id.'¹²⁸³
- **kruhhan-*: OFri. *krocha* m. 'scuttle'¹²⁸⁴, NFri. Wdh. *krōge* m. 'pot'¹²⁸⁵, OE *crohha* '*luteum*'¹²⁸⁶

The word for 'jug' has four different root variants, i.e. **krūk^k-*, **kruk^k-*, **kruh^h-* and **krōg-*, the first three of which are inflected as *n*-stems. It is clear, as the OED observed, that †*crouke* is "in ablaut relation to the family of *crock*" and that the underlying root **krūk-* contains a shortened

¹²⁷⁶ Seebold's reconstruction **fewur* is impossible from the Proto-Indo-European point of view, since the nominative was **péh₂ur* (thus Beekes 1996: 6).

¹²⁷⁷ Beekes l.c.

¹²⁷⁸ Kluge/Seebold 542.

¹²⁷⁹ Bosworth/Toller l.c.

¹²⁸⁰ Gallée 185.

¹²⁸¹ Franck/Van Wijk 354.

¹²⁸² De Vries 1962: 332: "möglich < ae. *crocca* [...] oder aus mnd. *krukke* [...]."

¹²⁸³ Bosworth/Toller 171.

¹²⁸⁴ Holthausen 1925: 61; Hofmann/Popkema 283.

¹²⁸⁵ Jensen 296. With *-g-* < **-hh-* (Löfstedt 1, 241).

¹²⁸⁶ Bosworth/Toller 134-5.

geminate.¹²⁸⁷ The variation between **ū* and **ǔ*, on the one hand, and **kk* and **hh*, on the other, thus points to an original paradigm **krūhō*, **krukkaz*, which was split-up into 1. **krūkō*, **krukkaz* and 2. **krūhō*, **kruhhaz*. Given the irregularity of fricative geminates, the variant **kruhh-* must at any rate be secondary, cf. **kliþþōn-* ‘burdock’ (see p. 235) and **mupþan-* ‘moth’ (see p. 218).

The root **krōg-* is difficult to explain from the above paradigms. Since it is attested as an *a*-stem, it should probably be reconstructed as **groHuk-ō-*, i.e. a thematic *o*-grade. The problem is that this reconstruction implies a laryngealic root, and that, as a result, the *n*-stem should be reconstructed accordingly, viz. **gréHuk-ōn*, **grHuk-n-ós*. In Proto-Germanic, this paradigm would develop into **krōhō*, **krūk^kaz* with an ablaut pattern similar to the one exhibited by the heteroclitics **sōl*, **sunaz* ‘sun’ < **séh₂-ul*, **sh₂-un-ós* and **fōr*, **funaz* ‘fire’ < **péh₂-ur*, **ph₂-un-ós*. This is problematic, however, because the expected stem **krōhan-* is not extant. It is clear, on the other hand, that the root **krūk^k*, with its combination of a long **ū* and a shortened geminate, points to an original genitive **krūk^kaz* < **gruHk-n-ós*. The paradigm **krōhō*, **krūk^kaz* thus finds some support.

Etymologically, the word for ‘jug’ is usually connected with Gr. κρῶσσός ‘jug’¹²⁸⁸ < **krōkjo*-(?), OCS *krugla* ‘cup’ and Alb. *karroqe* f. ‘wooden bucket’¹²⁸⁹, but the reconstruction of the Greek word is ambiguous and Alb. *karroqe* looks like a loanword (from Lat. *cambuca*?). OCS *krugla* can, just as W *crochan* and OIr. *crogán*, be borrowed from Germanic.

It has also been suggested that the Germanic and Greek words were adopted from an unknown language, so as to explain the vowel alternation of **ō* and **ū* in Germanic.¹²⁹⁰ Plausible as this possibility may seem, the consonant alternations can by no means be labeled as “un-Germanic”. So, even if we are dealing with an old loanword, it must have

¹²⁸⁷ Vercoullie (p. 187): “met *k* na langen klank uit *kk* = *gn*”; Falk/Torp (p. 583): “Die germ. Formen sind also **krōg-*, **krūk-* und **krukk-*, wo *k* und *kk* aus *gn*- entstanden sein können.”

¹²⁸⁸ Frisk 2, 30: “Schon das σσ-Element, gewissermaßen auch die technische Bed., läßt auf mediterranen Ursprung schließen.”

¹²⁸⁹ Cf. Pokorny 385-390.

¹²⁹⁰ Kluge/Seebold 542.

been adopted and incorporated into the category of the *n*-stems before the major sound shifts.

***sōl, *sunnaz ‘sun’**

- *sō(e)l-: Go. *sauil* n. ‘id.’, ON *sól* f. ‘id.’
- *sunna/ōn-: Go. *sunno* f., dat. *sunnin* m. ‘id.’, ON *sunna* f. ‘id.’, OHG *sunno* m. ‘id.’, *sunna* f. ‘id.’, OE *sunna* m., *sunne* f. ‘id.’
- *suil-: ?Go. *sugil*, OE *sigel-hweorfa* m. ‘eliotropum’

In Proto-Indo-European, the word for ‘sun’ is commonly reconstructed as a heteroclitc. It is still attested as such in e.g. Avestan, where the paradigm is nom. *huuarē*, gen. *xʷān̥g* ‘sun’ < *suH-_ḡ, *sHu-en-s. In combination with e.g. Gr. Dor. *ἥλιος*, Lat. *sōl*, Lith. *saulė* and, of course, Go. *sauil* < *seh₂u(e)-, the PIE paradigm is usually reconstructed as either *séhu-l, *sh₂u-n-ós or *séhu-l, *sh₂u-én-s.

Germanic seems to have preserved the root ablaut as well as the heteroclisys quite well. Both alternations were still more or less intact in Gothic. In this language, the neuter *sauil* < *sōel (with lowering of *ō* to *ī* in open syllables) and the feminine *sunno* < *sunnōn- < *sh₂un- share a masculine dative *sunnin*. For Indo-European, Schindler (1975: 1) and Beekes (1984: 5 fn.) reconstructed nom. *séhu-l, gen. *sh₂uéns. Beekes (1984: 6) further argued that the proterokinetic genitive of this paradigm could have been replaced by *sh₂unós already in late PIE, so as to explain e.g. Skt. gen. *sūras* < *sh₂u-l-ós < *sh₂u-n-ós. This form could have given PGm. *sūnaz directly, or indirectly through a metathesized form *suh₂nós that was affected by Dybo’s law of pretonic shortening.¹²⁹¹

The derivation of the geminate root of *sunna/ōn- has always been problematic.¹²⁹² Ever since Brugmann (1906: 303), it has been assumed that it came about as the “weak-case stem with a zero-grade of the *n*-suffix”¹²⁹³, i.e. *sun- plus *-n-, after the generalization of this root in the paradigm. A problem concerning this solution is that it requires a nominative *sunō of which there is no trace.

Alternatively, there is Hilmarsson’s (1987) idea that *sunnōn- is a secondary *n*-stem *sunþō, *sunþnaz derived from the adjective *sunþa-

¹²⁹¹ Schrijver 1991: 351-6.

¹²⁹² Cf. Benediktsson 1968: 11, 13.

¹²⁹³ Hilmarsson 1987: 62.

‘south’ < **sh₂un-to-*. This solution does not convince either, because in view of the retained heteroclisy in Gothic, it is unattractive to draw the nominative *sauil* from **sé_h2uel*, while at the same time reconstructing a different formation for the dative *sunnin*.

Probably the best way to deal with the geminate of the root **sunn-* is to assume that after the model of other *n*-stems, gemination was grammaticalized in the genitive case. This is, after all, the origin of consonant gradation. According to this principle, the paradigm **sōl*, **sunaz* may have been remodeled into **sōl*, **sunnaz*. Apparently, the morphologization of gemination was prior to the dissolution of heteroclisy.

9.10 The **ē ~ *a* type

In his article on the Germanic consonantism, Kauffmann listed a small number of *n*-stems with a vocalism that shifts between what looks like PGm. **ē* and **a*. The present section consists of a collection of potential cases that can be isolated from the Germanic dialects: **dēbō*, **dappaz* ‘paw’ (p. 327); **hēhō*, **hakkaz* ‘hook’ (p. 327); **krēbō*, **krappaz* ‘crook, clasp’ (p. 329); **krēgō*, **krakkaz* ‘crook’ (p. 331) and **snēgō*, **snakkaz* ‘snake’ (p. 333). The case of Swab. *zāk(e)* ‘hook, jag’ < **tēggan-* will be discussed in passing.

In spite of the fact that most of the attested *n*-stems with an **ē ~ *a* alternation have no sound Indo-European etymology¹²⁹⁴, an obvious way to deal with the ablaut pattern is to assume that it came about in *n*-stems with **h₁* in the root: PIE **Cé_h1C-ōn*, **Ch₁C-n-ós* > PGm. **CēCō*, **CaCCaz*. The postulation of this sub-type is indeed unproblematic theoretically, but, as will be demonstrated, the individual cases nevertheless seem to resist a PIE reconstruction. The main problem consists of the fact that none of the *n*-stems with **ē : *a* ablaut can be given a solid Indo-European etymology that unambiguously points to a **h₁* in the root. This problem was also encountered by Lühr (1988: 286) while attempting to reconstruct to PIE form of **hēhō*, **hakkaz* ‘hook’. Lühr in the end rejected

¹²⁹⁴ Lühr (1988: 319): “Ein solcher Typ hätte ebenfalls keine außergermanische Entsprechung.”

this Proto-Indo-European reconstruction, because there seemed to be no evidence for a root **keh₁g^h*- outside Germanic.¹²⁹⁵

Another complication is that the zero grade in **a* can only be regular in roots consisting of stops, as the laryngeal would never be vocalized in roots with an additional resonant. An old paradigm **snéh₁g^h-ōn*, **snh₁g^h-n-ós*, for instance, would develop into **snēgō*, **sunk^kaz*, not into **snēgō*, **snakkaz* ‘snake’. In fact, since the same line of reasoning is also valid for **krēbō*, **krappaz* (not ***kurp^paz*) and **krēgō*, **krakkaz* (not ***kurk^kaz*), the only possibly regular example of the **eh₁ ~ *h₁* type would be **hēhō*, **hakkaz*. Yet, exactly for this *n*-stem no laryngeal can be demonstrated outside Germanic.

The solution that is proposed here, is that the **ē : *a* ablaut type is not an Indo-European phenomenon, but rather an internally Germanic innovation. If this is correct, the most likely origin for the type is that it arose analogically on the basis of the **ī ~ *i* alternation, just like the ablaut of **ū : *u*.

The obvious *terminus post quem* for this analogical innovation is the lowering of PGm. **ē* to **ā* in Proto-Northwest Germanic. Such a time depth is indeed implied by the dialectal distribution. The *n*-stems with **ā* vocalism are most frequent in Upper German, viz. OHG *chrācco*, *hācco*, *krāpfo*, *snācco*. The process that led to the analogical introduction of **ā* must be older than the PWGm. stage, however, because the vocalism of OHG *snācco* and *chrācco* is neatly mirrored by ON *snákr* ‘snake’ and *krákr* ‘crook’. Similarly, the long **ā* of OHG *hācco* re-emerges in the Low and Middle German dialects, cf. Du. Stw. *haoke* ‘hook’¹²⁹⁶, G Rhl. *hōk*, *hōx* ‘id.’¹²⁹⁷ < **hāk^kan*-.¹²⁹⁸ Most importantly, the North Frisian form Wdh. *krēk* m. ‘hook on clothes’¹²⁹⁹ < **krāk^k*- seems to indicate that Anglo-Frisian, too, partook in the processes that led to the rise of long **ā*. With

¹²⁹⁵ Lühr (1988: 286): “In diesem Fall hätte man einen starken Stamm **χēgan*- und einen schwachen Stamm **χakk*- (‘Gekrummtes’?) zu postulieren. Doch ist eine Wurzel vorurgerm. **keh₁g^h*- sonst nicht nachweisbar, weshalb dieser Ansatz unsicher bleibt.”

¹²⁹⁶ For the vocalism, cf. *haeze* ‘hare’ < **hasan*-.

¹²⁹⁷ Franck/Müller 3, 119.

¹²⁹⁸ But WPhal. *hāken* ‘id.’ (Woeste 90), on the other hand, seems to go back to **hākan*- (cf. *hår* ‘hair’ < **hār*-).

¹²⁹⁹ Jensen 294.

this final piece of evidence, the rise of the $*\bar{a} \sim *a$ type can confidently be given a Northwest Germanic date.¹³⁰⁰

Parenthetically, the Upper German dialects show clear signs of lexical congruence, as it can hardly be coincidental that the OHG *hācco*, *chrācco* and *chrāpfo* all mean 'hook'. This *lexical huddling* must have been driven by the centripetal forces exerted by the formal and the semantic similarities between these stems. The continued growth of the *huddle* in Upper German seems to be demonstrated by the modern Swabian *n*-stem *zāk(eⁿ)* m. 'hook, jag'¹³⁰¹ < **tēggan-*. Etymologically, this *zāk(eⁿ)* clearly belongs to the otherwise regular ablauting *n*-stem **tōgō*, **takkaz* 'twig, jag' (cf. OHG *zuogo*), but it can nevertheless not be fit into this paradigm in a regular way.

One possibility is that *zāk(eⁿ)* arose through a process by which $*\bar{o} : *a$ ablaut was replaced by the purely quantitative and therefore more simple $*\bar{a} \sim *a$ alternation. The motivation for this analogy may very well have been the regular development of Pre-Gm. $*\bar{a}$ to PGm. $*\bar{o}$, by which the quantitative character of the $*\bar{a} : *a$ ablaut from PIE $*eh_2 : *h_2$ was destroyed. For obvious reasons, this analogy cannot have taken place before the lowering of PGm. $*\bar{e}$ to PNWGM. $*\bar{a}$.

If the scenario suggested here is correct, the chronology may be as follows. First, the regular Pre-Gm. $*\bar{i} \sim *i$ and $*\bar{a} \sim *a$ alternations triggered the rise of the $*\bar{u} \sim *u$ type (see § 8.2). When, at a later stage, Pre-Gm. $*\bar{a}$ developed into $*\bar{o}$, and – subsequently – PGm. $*\bar{e}$ into PNWGM. $*\bar{a}$, the right morphological and phonological conditions had arisen for the analogical reintroduction of $*\bar{a} : *a$ ablaut. It therefore seems most probable that the secondary ablaut type started off somewhere in the Proto-Northwest Germanic period. It may have petered out much later, however, in the individual West Germanic dialects. If this is true, Swab. *zāk(eⁿ)* would be one of the last instances to be affected by this process.¹³⁰²

¹³⁰⁰ The rise of the $*\bar{a} \sim *a$ alternation thus has an important bearing on the old question whether Anglo-Frisian partook in the lowering of PGm. $*\bar{e}$ to $*\bar{a}$, or that the lowering of PGm. $*\bar{e}$ occurred in the other dialects at a time when the Anglo-Frisian had already left the proto-Northwest Germanic continuum. The development of NFri. *krēk* < **krāk^k* is in support of the former scenario.

¹³⁰¹ Fischer/Taigel 436.

¹³⁰² A similar proces may have operated in the verbal system. The de-iterative MHG *zāfen*, for instance, probably has secondary $*\bar{a}$ (see p. 344).

****dēbō*, **dappaz* ‘paw’?**

- **dēbban*:- MHG *tāpe* m. ‘paw’, G *Dape*, *Tape*, Rhdl. *tape* ‘id.’¹³⁰³, Swi. App. *tōppə* m. ‘id.’¹³⁰⁴, Visp. *daappo* ‘paw, hand’¹³⁰⁵
- **dabban*:- G *Dappe*, *Tappe* ‘paw, (foot)print’¹³⁰⁶
- **dappan*:- G *Tapfe* m. ‘paw’¹³⁰⁷

It is revealed by the Alemannic dialects, including e.g. App. *tōppə* and Visp. *daappo* < OHG **dāppo* ~ **tāppo*, that Modern German *Dape* and *Tape* must continue a form with both a long vowel and a long consonant. The quasi-Proto-Germanic form underlying this formation is **dēbban*-. Since, however, geminates were shortened after long vowels in Proto-Germanic, the voiced geminate must have been introduced from somewhere else after the West Germanic period, possibly at a time when *j*-gemination had given rise to new superheavy syllables. A possible source could be the expected genitive **dabbaz* with a short vowel, cf. G *Tappe* < **dabban*-. The voiced geminate of this form cannot be primary either, however, and itself seems to have replaced the regular, voiceless geminate that is still found in G *Tapfe* < **dappan*-.

On the basis of the material discussed here, it is theoretically possible to reconstruct a quasi-PGm. paradigm **dēbō*, **dappaz*, **dabeni*. Since, however, the **ā* ~ **a* alternation is likely to have arisen after the lowering of PGm. **ē* to **ā*, it is probably better to reconstruct a Proto-Northwest Germanic paradigm **dābō*, **dabbenaz*, **dabeni*.

What seems certain, at any rate, is that the *n*-stem was derived from the Proto-Northwest Germanic iterative **dappōpi*, **dabunanpi* as evidenced by e.g. SFri. *dafen* ‘to knock’, MDu. *dabben* ‘to toddle’, Nw. dial. *dabba* ‘id.’, G *tappen* ‘to pat’.

****hēhō*, **hakkaz* ‘hook’?**

- ?**hēhan*:- OHG *hāho* m. ‘id.’¹³⁰⁸

¹³⁰³ Franck/Müller 8, 1061.

¹³⁰⁴ Vetsch 1910: 143.

¹³⁰⁵ Zimmermann-Heinzmann, s.v. *daappo*.

¹³⁰⁶ Grimm 21, 139-40.

¹³⁰⁷ Grimm 21, 134.

¹³⁰⁸ Grimm 10, 177.

- **hēg(g)an-*: OHG *hāc(c)o* m. 'id.', MHG *hā(c)ke*, *hōcke* m. 'id.', G *Haken*, Als. *hōkə* m. 'id.', Swi. App. *hɔɔckə*, pl. *hæckə* m. 'id.'¹³⁰⁹, Visp. *haacko* m. 'id.'
 - **hēk^kan-*: OS *hácon* 'uncis', ?MDu. *hake*, *haek* m. 'id.', ?Du. *haak*, dial. *haoke* 'id.'
 - **hakan-*: Icel. *haki* m. 'pickaxe', Nw. *hake* m. 'crook', OFri. *haka* m. 'id.', OE *haca* m. 'id.'
-
- **hōk^ka-*: OE *hōc* m. 'hook', MLG *hōk* m. 'corner', Du. *hoek* 'corner'
 ⇒ (?) **hōkjōn-*: ON *hækja* f. 'crutch'

The different forms, especially the Upper German ones, point to an *n*-stem with **ā* ~ **a* ablaut. OHG *hācco*, MHG *hā(c)ke* and G *Haken* go back to an apparent *ē*-grade form **hēggan-*. The length of the vowel is ascertained by those dialects that shift long OHG *ā* to *ō*. This shift spread from the 12th century onwards (Moser 1975: 70), and is exhibited by MHG *hōcke*, Als. *hōkə* and App. *hɔɔckə* (but not by Visp. *haacko*). The Swiss forms are especially interesting, because they preserve both vowel and consonant length up to the present day. The *ē*-grade is further supported by the form *haoke* 'hook' < **hēkan-* in the Saxon dialect of Stellingwerven, where **hakan-* would have given ***haeke*.

Other ablaut variants that occur are the zero grade and the *o*-grade. The zero grade is ascertained by Icel. *haki*, OFri. *haka*, OE *haca*, which all seem to have an analogical singulate. An *o*-grade is present in Saxonian and Franconian: OE *hōc*, MLG *hōk*, MDu. *hoek* 'hook'. Possibly, ON *hækja* 'crutch' was derived from this formation.

All the different forms can theoretically be united by reconstructing a paradigm **hēhō*, **hakkaz*, **hageni* and an *o*-grade thematization **hōk^ka-*. At first sight, this paradigm seems to presuppose PIE **kéh₁k-ōn*, **kh₁k-n-ós*, **kh₁k-éni*, but there is no extra-Germanic evidence for a laryngeal in the root. Alternatively, the long **ā* can be explained as analogical to the *n*-stems with **ī* ~ **i* and **ū* ~ **u* alternations. The Upper German dialects may then have generalized the full grade and the geminate **g*, which resulted into a paradigm **hāggō*, gen. **hāggen*, dat.

¹³⁰⁹ Vetsch 73, 90.

**hāggin*. In Low German, the root **hāk^k* seems to dominate, although Westphalian *hāke* seems to point to **ǎ*.

The etymology of the word is unclear. It is possible that Go. *hoha* m. 'plow' < **hōhan-* and OHG *huohila* m. 'small plow' belong here. They are related to Skt. *śākhā* f. 'twig', Lith. *šākė* f. 'fork, pitchfork', Ru. *soxá* f. '(wooden) plow', SCr. *sòha* f. 'stick with a fork'.¹³¹⁰ The semantic variation between 'twig' and 'plow' implies that a curved stick was used as a plow. Could the PGM. **ō* have been replaced by PNWGM. **ǎ*?

The word for 'plow' may have become conflated with the root **k(o)nk-*, which is found in several Indo-European languages, cf. Skt. *śānkú-* 'peg, post', OCS *spkъ* < **konk-*, W. *cainc* 'branch', OIr. *cécht* 'plow' < **knk(-to)-*. In Germanic, the root **konk-* is present in ON *hár* 'rowlock'¹³¹¹ (= Fi. *hanka* 'oarlock, rowlock'), and in OHG *hāhala*, *hāhila*, Swi. Visp. *heeli* f. 'kettle hook' < **hanhilō-*. In view of these forms, the question arises whether the outcome of **hanh-*, viz. **hāh-*, can have formed the basis for the paradigm **hāhō*, **hakkaz*.

Regarding the etymology, it is difficult to separate OE *hēla*, MDu. *hiele*, Du. *hiel* 'heel'¹³¹², a word with a North Sea Germanic distribution that is derived from **hanhila-*.¹³¹³ The primary form is OE *hōh* m. 'heel, promontory'¹³¹⁴, which is identical to *há-* in ON *há-mót* 'ankle-joint' and *há-sin* f. 'Achilles tendon'. Presumably, the meaning 'hook' was used metaphorically to designate the 'heel'.

The formation MDu. *honc*, Du. *honk*, WFri. *honk*, SFri. *hunk* 'corner, base (in games)' probably does not belong here: these forms continue PGM. **hunk^ka-*, which may be based on the original genitive of the paradigm of **hnekkō*, **hnukkaz* 'neck' (see p. 167).

****krēbō*, **krappaz* 'hook'?**

- **krēppan-*: OHG *chrāpfo* 'fusicinula, uncinus', MHG *krāpfe* m. 'hook, bracket'¹³¹⁵, G Pal. *krāpfe* m. 'id.'¹³¹⁶, Swi. Visp. *xraapfo* m. 'crooked cane'

¹³¹⁰ Cf. Pokorny 523.

¹³¹¹ In the *First Grammatical Treatise*, this form has a nasal vowel.

¹³¹² De Vries/Tollenaere 256.

¹³¹³ Fick/Falk/Torp 67; De Vries/Tollenaere 256.

¹³¹⁴ Bosworth/Toller 557.

¹³¹⁵ Lexer 1, 1712.

- ?**krēpan*:- OHG *chrāf(f)o* ‘*ungula, uncus*’, *fuscinula*’
- **krēbban*:- OHG *chrāppo* ‘*aspidiscus, uncinus*’, MHG *krāpe* m. ‘hook, bracket’, G *Krape* m. ‘hook, claw’, Pal. *krāpe*, *krōpe* m. ‘muck shovel’¹³¹⁷
- **krappan*:- MDu. *crappe* m. ‘hook, clamp’¹³¹⁸, G *Krapfen* m. ‘doughnut’¹³¹⁹, Swi. Ja. *krapfə* ‘two-pronged hoe’¹³²⁰
- ?**krabbōn*:- Sw. dial. *krabba* f. ‘grappling iron’¹³²¹

The German dialects show a wild variety of forms for the word for ‘shovel’ or ‘prong’. Two different vowel grades must be reconstructed, viz. **ē* and **a*¹³²², and they occur in combination with several different consonantisms. Notably, much of the total dialectal variation can be retrieved from the Modern Palatinate dialects alone (see the map on the next page).

A full grade is found in the formation **krēbban*- as supported by OHG *chrāppo*. The length of the OHG vowel is ascertained by the “Verdumpfung” in the German dialects, e.g. Palatinate German *krōpe*. The same vowel grade occurs in the variant OHG *chrāpfo*, the vowel length being ascertained by the many attestations with marked length, e.g. *crápho*, *crāpho*, as was shown by Lühr (1988), and further by Visp. *xraapfo* and Pal. *krāpfe*.¹³²³ Lühr further reconstructed a proto-form **krēpan*- on the basis of OHG *chrāf(f)o*. This seems to be correct in view of Pal. *krōwe*, which displays the regular shift of intervocalic *f* to *w*.

Then, there are some forms with a short vowel. Pal. *krappe* (= [*grabə*]) seems to continue a proto-form **krabban*-, and can thus be compared to Sw. *krabba* < **krabbōn*-. The same short **a* occurs in MDu. *crappe* ‘hook, clamp’ < **krappan*-, a form that further seems to be paralleled by Pal. *krapfe* and standard Modern German *Krapfen* ‘culler, donut’.

¹³¹⁶ Christmann 4, 547-50.

¹³¹⁷ Ibid.

¹³¹⁸ Verdam 312.

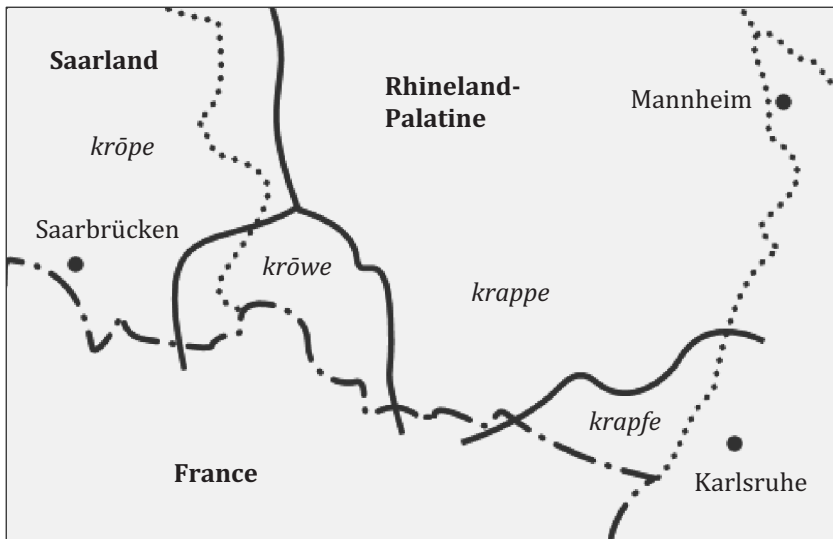
¹³¹⁹ Kluge/Seebold 535: “Ein etymologischer Zusammenhang mit *Krampf* legt sich nahe; es müßte eine frühe, unnasalierte Form vorliegen.”

¹³²⁰ Stucki 49.

¹³²¹ SAOB K2594.

¹³²² Fick/Falk/Torp 52; Lühr 1988: 288.

¹³²³ The form is cited as *grāpfə* by *Pfälzisches Wörterbuch*.



The dialectal distribution of G *Krapfen* in Palatinate German (adapted from *Pfälzisches Wörterbuch*, 1965-1998, p. 547).

On the basis of the different forms, it is theoretically possible to reconstruct a PNWGM. paradigm **krābō*, **krabbenaz* reflecting quasi-PGM. **krēbō*, **krappaz*. On the basis of the extensive variation in the modern Palatinate dialects, it seems attractive to assume that an ablauting paradigm indeed existed up to late stage. This does not necessarily mean, however, that all of the cited forms must have belonged to this paradigm. It remains possible that some of the formations belonging to the etymological cluster of the *n*-stem **krēb(b)an-* were derived from the iterative **krappōpi*, **krabunanpi*, cf. Du. *krabben*, *krappen*, dial. *kraven* 'to scratch' independently. Sw. *krabba*, for instance, seems especially peripheral to the reconstructed *n*-stem.

****krēgō*, **krakkaz* 'crook'?**

- **krēggan-*: OHG *chrācco* 'uncinus, fuscina', G Als. *krāge*ⁿ [krākə] f. 'crooked twig on a vine, vine with grapes'¹³²⁴, Pal. *krāke* [grägə], pl. *krāke* [grægə] m. 'old vine'¹³²⁵

¹³²⁴ Martin/Lienhart 1, 515a.

¹³²⁵ Christmann 4, 531.

- **krēk^a(n)-*: ON *krákr* m. 'crook to loosen frozen soil', NFri. Wdh. *krēk* m. 'hook on clothes'¹³²⁶, ?OHG *krāhho* 'uncinus'
 - **kragōn-*: MHG *krage* f. 'hoe'¹³²⁷
 - **kragga-*: Nw. *kragg* m. 'crooked tree'
 - **krakan-*: ON *kraki* m. 'crook', Nw. *krake* 'crooked tree, dial. curved stick', OHG *chracho* m. 'crook'
 - **krakka-*: G *Krack* m. 'crook'¹³²⁸
-
- **krōk^a-*: ON *krókr* m. 'corner, crook' (= OE *crōc* 'crook')

Just like the word for 'hook', the word for 'crook' shows two different vowel grades, i.e. PGm. **ē* and **a* or PNWgm. **ā* and **a*.

On the basis of the OHG gloss *crácco*, Lühr (1988: 286-7) assumed a form *chrācco* < **krēggan-* with what looks like an *ē*-grade of the root (cf. Fick/Falk/Torp 51: **krēkan-*). The reality of this vocalism is further confirmed by Pal. *krākā* and probably also by NFri. *krēk* < **krēka-*.

As opposed to the **ē*-grade forms, the attestation of short **a* is limited. It has been identified in OHG forms such as *kracko*, *krago* < **krag(g)an-* and *krahho* < **krakan-*, but the vowel length is uncertain in these forms. The short vowel seems to be unambiguous in Modern German *Krack* < **krakka-*, on the other hand. In addition, a short **a* is supported by ON *kraki* < **krakan*.

Additionally, an *o*-grade thematization is represented by ON *krókr* < **krōk^a-*. It is considered to be the source of OE *crōk* with roughly the same meaning.

In view of the material, it is theoretically possible to reconstruct a PNWgm. paradigm **krāgō*, **krakkenaz*, **krageni* or PGm. **krēgō*, **krakkaz*, **krageni*. A deeper reconstruction PIE **gréh₁k-ōn*, **grh₁k-n-ōs*, **grh₁k-én-i* makes no sense, because it would have yielded PGm. **krēhō*, **kurk^aaz*, **kurgeni*¹³²⁹, for which the material offers no support. It therefore seems preferable to assume that the ablaut pattern was introduced analogically in Proto-Northwest Germanic on the basis of the

¹³²⁶ Jensen 294.

¹³²⁷ Lexer 1, 1703; Benecke/Müller/Zarncke 1, 873.

¹³²⁸ Grimm 11, 1926.

¹³²⁹ Lühr (1988: 287): "mit analogischer Syllabifizierung urgerm. **kra^o-* < vorurgerm. **grā₁k/g^h-* anstelle von **gr₁k/g^h-*".

* $\bar{i}$ ~ * i and the * $\bar{u}$ ~ * u types. If this is correct, the NFri. form may be of great significance to the relative chronology of the Germanic sound changes, because it would prove that Anglo-Frisian * $\bar{e}$ is not a retention, but rather the continuation of analogically introduced * $\bar{a}$.

Etymologically, the n -stem can be related to the strong verb **kragan*- as evidenced by MHG *kragen* ‘to scratch, carve’.¹³³⁰

****snēgō*, **snakkaz* ‘snake’?**

- **snēk^a*:- ON *snákr* m. ‘snake’¹³³¹, Icel. *snákur* m. ‘snake, viper’¹³³², Far. *snákur* m. ‘snake, snout’ (⇒ *snáki* m. ‘snout’)¹³³³, Nw. dial. *snåk* m. ‘viper’
 - **snēggan*-, - $\bar{o}n$:- MHG *snācke*, *snōcke* m. ‘midge’, G *Schnake* m. ‘snake, midge’, Swi. App. *šnǝckə* ‘gnat’¹³³⁴, Visp. **šnaacko* (⇒ Visp. *šnaacku* ‘to crawl’)
 - **snagan*:- Icel. *snagi* m. ‘pin’¹³³⁵, Nw. *snage* m. ‘tip, pin, bud’
 - **snakan*-, - $\bar{o}n$:- OE *snaca* m. ‘snake’, MLG *snake* f. ‘id.’
-
- **snōk^a*:- Icel. *snókur* m. ‘trunk, snout, small shark, front part of a ship, snake’¹³³⁶, Nw. *snok* m. ‘snout, snail’¹³³⁷, Sw. *snok* ‘viper’, MDu. *snoek* m. ‘pike’¹³³⁸

The etymon that is at stake here is thoroughly discussed by Lühr (1988: 301), who noted several ablaut variants within this string of clearly related words. A form **snēggan*- can be reconstructed on the basis of MHG *schnācke*, G *Schnake* and the additional dialectal forms Als. *schnōke* and App. *šnǝckə*, which show the regular “Verdumpfung” of long $\bar{a}$. The evidence for the $\bar{e}$ -grade is further strengthened by Nordic forms such as ON *snákr*, Far. *snákur* and Nw. dial. *snåk* < **snēkan*-. By the addition of

¹³³⁰ Lexer 1, 1703.

¹³³¹ De Vries 1962: 522.

¹³³² Böðvarsson 915.

¹³³³ Poulsen 1094.

¹³³⁴ Vetsch 159.

¹³³⁵ Böðvarsson 915.

¹³³⁶ Böðvarsson 921.

¹³³⁷ Torp 873.

¹³³⁸ Franck/Van Wijk 634.

this evidence, the word acquires a clear Proto-Northwest Germanic horizon. It is further notable that ON *snákr* and cognates are completely parallel to the Old Norse formation *krákr* ‘crook’ < **krēka-*. Beside the *ē*-grade, a zero grade is present in OE *snaca* and MLG *snake* < **snakan-*.

Lühr was hesitant towards the possibility that the roots **snēgg-* and **snakk-* “ursprünglich in einem Paradigma gestanden haben.” She also correctly noted that the zero-grade vocalism must be analogical anyway, because **snh₁k-n-ós* would have yielded ***sunk^kaz* rather than **snakkaz*. The morphological unification of both vowel grades into a single paradigm nevertheless has the advantage of being able to account for the superheavy syllable of **snāggan-*, which may have arisen as a conflation of the nom. **snāgō* and an analogical genitive **snaggenaz*. The alternative, to explain **snēggan-*, **snēkan-* and **snakan-* as independent formations, seems more intricate.

Etymologically, the *n*-stems seem to be related to ON *snagi* ‘pin’. The original meaning of the word may have been “pointed one” (cf. “*Stechendes*” = Lühr 1988: 301), which reconciles ‘snake’ with ‘mosquito’. An alternative is offered by the dialectal Nw. has *snaka* ‘to snatch (said of animals)’, which just like OHG *snahhan* ‘to crouch’ is conjugated as a strong verb. Perhaps, the *n*-stem was somehow derived from this verbal base. This must at any rate be assumed for MDu. *snoek* m. ‘pike’ < **snōk^ka-*, which clearly represents a thematic *o*-grade. The position of Visp. *šnaacku* ‘to crawl’, on the other hand, is not entirely clear, but it may well be by a denominative verb OHG **snākkōn*.

10 Pseudo-ablaut

In both North and West Germanic, the process of umlaut had a far-reaching effect on the morphology of the different dialects. Umlaut gave rise to many new sorts of vowels, and they were phonologized and grammaticalized at many different stages. As a result, some paradigms received a more or less morphologically conditioned alternation between umlauted and non-umlauted vowels, cf. G *Baum* ‘tree’, pl. *Bäume*. Because of this secondary vowel alternation, it can often be difficult to determine whether a vocalic interchange reflects old ablaut or recent umlaut.

The present chapter consist of a discussion of *n*-stems that have been interpreted as displaying roots with different Proto-Germanic vowel grades, but which actually acquired a vocalic interchange due to the effects of umlaut. The majority of the material is abstracted from the Upper German dialects, in which front mutation has reached a considerable degree of productivity. The cases that are discussed are OHG *chretto* ~ *chratto* ‘basket’, OHG *zepfo* ~ *zapfo* ‘peg’, G *Schretz* ~ MHG *scräte* ‘demon’ and OHG *chreta* ~ *chrota* ‘toad’.

The chapter is concluded with a West Norse case. West Norse is the Nordic dialect that has been affected most extensively by several different vowel mutations such as *i*- and *u*-umlaut. Grammaticalization of these mutations seem to have given rise to clear pseudo-ablaut in the case of Icel. *hneta* ~ *hnót* ‘nut’.

10.1 Upper German

In Old High German, a small number of *n*-stems, such as the already mentioned *chretto*, *chretzo* ~ *chratto* ‘basket’ and *zepfo* ~ *zapfo* ‘wisp, peg, cone’, waver between *e*- and *a*-vocalism in the root. In the literature, these alternations have been explained in two different ways. First, it was suggested by Kauffmann (1887), that the doublets attest to old Proto-Germanic **e* : **a* ablaut such as described in § 9.2. Others have isolated the different formal variants from each other, and reconstructed

jan-stems for the variants with fronted vowels (e.g. Pokorny 385-90; Lühr 1988: 282). In this way, OHG *chretto*, *chretzo* and *zepfo* have been given the reconstructions **kraddjan-*, *krattjan-* and **tappjan-*.

The hesitation between the two solutions is chiefly the result of the ambiguity of the OHG grapheme <e>, which may stem from PGm. **e*, or from PGm. **a* with primary umlaut (*ä*₁), e.g. *felt* 'field' < **felpa-* and *gast*, pl. *gesti* 'guest' < **gasti-*. The grapheme <a> was actually ambiguous, too: it indicated both OHG /a/, which directly continues PGm. **a*, as well as /*ä*₂/ from PGm. **a* with secondary umlaut. Secondary umlaut typically occurred when there was a velar fricative between the root vowel and the mutation factor. The reflex of secondary umlaut was lower than the one of primary umlaut, and it was not expressed in writing until in the Middle High German period, cf. OHG *naht*, pl. *nahti*, MHG *nacht*, pl. *nächte* 'night', G *Nacht*, pl. *Nächte*.

The problem of the graphemic ambiguity of OHG <e> and <a> can be tackled by including the material from the modern Alemannic dialects. Most of these dialects, like Jaun Swiss, Visperterminen Swiss and Swabian, have a binary opposition between high *e* ([e]) from PGm. **a* with primary umlaut, and low *e* ([ɛ], [æ], [a]) from PGm. **e* and **a* with secondary umlaut. The Swiss Appenzell and Sankt Gallen dialects are known for their preservation of a threeway opposition between *e* from PGm. **a* with primary umlaut, *ɛ* from PGm. **e*, and *ä* [æ] from PGm. **a* with secondary umlaut. By using the data available from these dialects, it often becomes possible to reconstruct the phonemes underlying the OHG graphemes <e> and <a>.

A survey of the modern Alemannic evidence corresponding to OHG *chretto*, *chretzo* ~ *chratto* and *zepfo* ~ *zapfo* reveals that the forms with *e*-vocalism are likely to represent PGm. **a* with analogical umlaut, as App. *krää(n)tsə*, Swab. *(arm-)krätz^e*, App. *zäpfe*, Visp. *zäpfo*, Ja. *zäpfə* all point to OHG **ä*₂. This is perhaps not very remarkable, as it is *a priori* unlikely that a paradigm with **e* : **a* ablaut would have been preserved exclusively in the Upper German dialects. It is probably more surprising, then, that the correspondence of OHG <e> with dialectal Swiss **ä*₂ also excludes the alternative reconstruction of *chretto*, *chretzo* and *zepfo* as **kraddjan-*, *krattjan-* and **tappjan-*. The problem with these *jan*-formations is that, had they existed, they would have triggered *primary* umlaut, i.e. App., Ja., Swab. ***kretzə*, ***zəpfə*. Since this is not the case, the vocalism must be explained in another way.

No doubt, the answer to the problem is to be found in the widespread introduction of analogical umlaut in the plural of the *n*-stems. This “morphological” umlaut, which arose on the basis of regular umlaut in, for instance, the masculine and feminine *i*-stems, is largely limited to the old *a*-stems in the standard language. In Middle and Upper German, on the other hand, it to a large extent affected the *n*-stems as well. In this category, either primary or secondary umlaut was introduced in the plural to disambiguate or overambiguate it from the singular.

Interestingly, the exact distribution of the analogical umlaut in the *n*-stems, either primary or secondary, differs from dialect to dialect.¹³³⁹ This can be illustrated with the help of the Alemannic material. According to Stucki (p. 264), the Jaun dialect has *xrage*, pl. *xrägə* ‘collar’ < **kragan*-, *xrattə*, pl. *xrättə* ‘basket’ < **kraddan*-, *grabə*, pl. *gräbe* ‘ditch’ < **graban*-, *mage*, pl. *mägə* ‘stomach’ < **magan*-, while, for instance, *hasə* ‘hare’ < **hasan*- and *hanə* ‘cock’ < **hanan*- have plurals both with and without secondary umlaut. In the Vorderland dialects of Appenzell, more or less the same distribution is found, but here the plural forms appear with *primary* umlaut, cf. Vetsch (p. 57): *xragə*, pl. *xregə*, *magə*, pl. *megə*, *grabə*, pl. *grebə*, *xrattə*, pl. *xrettə*, *ladə*, pl. *ledə* ‘shop’ < **laþan*-, *zapfə*, pl. *zəpfə* < **tappan*-. Apparently, primary umlaut prevailed over secondary umlaut as pluralizing marker in this area.¹³⁴⁰

Umlaut appears in fewer cases in the south and the north: the Visperterminen dialect has generalized (secondary) umlaut in e.g. *palko*, pl. *pälku* ‘shutter’, *namo*, pl. *nemu* ‘name’, *gaarto*, pl. *gäärtu* ‘garden’ and *zapfo*, pl. *zäpfu* (Wipf 27, 129), and the other Valais dialects reveal a similar picture (cf. Bohnenberger 193). In Swabian, the same words are *grabe*ⁿ, pl. *gräbe*ⁿ (p. 207), *mage*ⁿ, pl. *mäge*ⁿ (p. 308), *lade*ⁿ, pl. *läde*ⁿ ‘schutter, bar, store’ (p. 293), but *zapfe*ⁿ, pl. *zapfe*ⁿ (p. 437).

In view of the modern dialectal situation, the spread of morphological umlaut to the *n*-stems can be regarded as a very likely source of the vocalic alternation of OHG *chretto*, *chretzo* ~ *chratto* and *zepfo* ~ *zapfo*. This seems to be supported by the fact that, in some *n*-stems, the analogical umlaut of the plural also became intrusive in the singular. Good examples of such intrusive umlaut are Visp. *güogo* ~

¹³³⁹ Cf. Hotzenköcherle (1956) on the South-Wallis dialects, esp. § 1 *Abneigung gegen analogischen Umlaut in der Pluralbildung der Maskulina*.

¹³⁴⁰ The Kurzenberg dialects, on the other hand, have *-ä-* < **ä*₂ in the same words.

gjogo, pl. *gjoge* ‘worm’, Ja. *guogə*, dpl. *güegne* ‘id.’ < OHG **guogo*, Visp. *blüoma* ~ *bljoma*, pl. *bljome* ‘flower’ and App. *magə*, pl. *megə* ‘stomach’ ~ *megə* sg. ‘rennin’.

Importantly, the Old High German singular doublet *chretto*, *chretzo* ~ *chratto* finds an exact parallel in modern Alemannic, cf. Swab. *krätte*ⁿ, (*arm-*)*krätz*^e ~ *kratte*ⁿ. The Old High German interchange of *zapfo* and *zepfo* similarly has a counterpart in the Visperterminen dialect, where the singulars *zapfo* and *zäpfə* occur beside each other, and, indeed, the latter variant is explained as analogical after the plural by Wipf.¹³⁴¹ The fact that the same analogy can be observed in Jaun Swiss *zäpfə*¹³⁴² and Bavarian *zēpfā* presupposes a time depth for this development that at least partly comprises the Old High German period. The conclusion therefore seems inescapable that the intrusion of morphological umlaut (either primary or secondary) from the plural into the singular dates back to the Old High German period at least in some cases. This solution harmonizes the Old High German vowel alternations with the modern Alemannic dialects, and at the same time removes the necessity to reconstruct *ad hoc* Proto-Germanic *e*-grades or *jan*-formations.

***kredō, *krattaz ‘basket’†**

- †**kredan-*: OHG *chretto* m. ‘basket’, Swab. *krätte*ⁿ m. ‘arm basket’¹³⁴³
- †**krettan-*: OHG *chretzo* m. ‘basket’, MHG *kretze* mf. ‘pannier’¹³⁴⁴, G *Krätze* m. ‘pack basket’¹³⁴⁵, Swab. (*arm-*)*krätz*^e f. ‘arm basket’¹³⁴⁶, Swi. App. *krää(n)tsə* f. ‘pannier’¹³⁴⁷
- **kraddan-*: OHG *chratto* m. ‘basket’, MHG *kratte* m. ‘id.’¹³⁴⁸, G *Kratte* m. ‘basket, cart’¹³⁴⁹, Car. *grätte* m. ‘cart’¹³⁵⁰, Cimb.

¹³⁴¹ P. 28: “*zapfo* oder analogisch nach dem Plur. *zæpfə*.”

¹³⁴² Stucki 264: “Die Form mit Umlaut hat auch für den Sing. Geltung gewonnen bei *tsæpfə* Tannzapfen (selten -a-[]).”

¹³⁴³ Fischer/Taigel 284.

¹³⁴⁴ Lexer 1, 1723.

¹³⁴⁵ Grimm 11, 2073-4.

¹³⁴⁶ Fischer/Taigel 40.

¹³⁴⁷ Vetsch 74, 172.

¹³⁴⁸ Lexer 1, 1712.

gratto m. 'cart with two wheels', Swab. *kratte*¹³⁵¹ m. 'arm basket', Swi. Ja. *xrattə* m. 'basket'¹³⁵², Rhtl. *kxrattə* m. 'basket'¹³⁵³

- **kradan-* ⇒ **krad(i)la-*: OE *cradol*, *credel* n. 'cradle'¹³⁵⁴
- **kratta(n)-*, *-ōn-*: ?ON *kartr* m. 'cart', OE *cræt* n. 'chariot'¹³⁵⁵, ME *cart(e)*, E *cart*, MDu. *cratte* m. 'wicker-work, hurdle, chariot'¹³⁵⁶, Du. *krat* 'crate'¹³⁵⁷, WFri. *kret* n. 'crate, dungcart'¹³⁵⁸ (= Du. dial. *kret* n. 'basket, wooden frame'¹³⁵⁹)

On the basis of the material presented here, the reconstruction of an *n*-stem with consonant gradation is beyond serious doubt (cf. Lühr 1988: 282ff). The forms with *a*-vocalism clearly point to a paradigm **kradō*, **krattaz*, **kradeni*. The root variant **kratt-* is primarily attested in the more Northern dialects, cf. OE *cræt*, WFri. *kret*, MDu. *cratte*, Du. *krat*.¹³⁶⁰ The variant **krad-* is somewhat isolated and only occurs in OE *cradol*, *credel*, two diminutives presupposing **krad(a)la-* and **kradila-*. The contamination of the roots **kratt-* and **krad-* led to the creation of the third variant **kradd-*, which is characteristic for the Upper German area, cf. OHG *chratto*, MHG *kratte*, G *Kratte*, Swi. *(k)xratte*. There is no necessity to assume expressivity here: "aufgrund der Wortbedeutung [ist] keine eindeutige lautsymbolische Funktion erkennbar" (Lühr 1988: 283).

¹³⁴⁹ Grimm 11, 2070.

¹³⁵⁰ Lexer 1862: 122.

¹³⁵¹ Fischer/Taigel 284.

¹³⁵² Stucki 264.

¹³⁵³ Berger 26.

¹³⁵⁴ Holthausen 1934: 59, 60.

¹³⁵⁵ Bosworth/Toller 169; Holthausen 1934: 59.

¹³⁵⁶ Verdam 312.

¹³⁵⁷ Franck/Van Wijk 345.

¹³⁵⁸ Zantema 1, 535.

¹³⁵⁹ Kocks/Vording 621; WNT, s.v. *kret*: "O.a. aan de Zaan en in Friesland."

¹³⁶⁰ The position of ON *kartr* is disputed. If directly related, it has unexpected metathesis. This metathesis has been ascribed to influence of ON *karmr* 'cart' (cf. De Vries 1962: 303). The word can also have been borrowed from Old English, which in view of W *cartwen* seems to have had a metathesized form **ceart-wæn* beside attested *cræt-wæn* 'chariot, waggon'. The OED (s.v. *cart*), on the other hand, assumes that ON *kartr* was adopted as ME *cart(e)*.

The reconstruction of the paradigm **kradō*, **krattaz* offers an elegant explanation for the consonant variation that is encountered in the West Germanic dialects. It does not, however, account for the different forms with *e*-vocalism in Upper German, such as OHG *chretto* and *chretzo*. In order to explain this interchange of *a* with *e*, Kauffmann (1887: 533, 544) proposed to reconstruct an ablauting *n*-stem, thus enriching the proto-language with such formal variants as **kreddan*- and **krettan*-. Fick/Falk/Torp (p. 51), Pokorny (p. 385-90) and Lühr (1988: 282), on the other hand, derived *chretzo* and *chretto* from **krattjan*- and **kraddjan*-, so as to explicate the *e*-vocalism by primary umlaut. In the end, however, neither of these solutions can be correct.

As pointed out, the main difficulty in deciding between **krettan*-/**kreddan*- and **krattjan*-/**kraddjan*- is the opacity of the grapheme <e> in OHG *chretzo*, *chretto*; it can represent two, perhaps three different vowels, i.e. OHG **e* < PGM. **e*, OHG **ä₁* and **ä₂* < PGM. **a*. The modern Upper German dialects, though, offer decisive information on which one of these three vowels is correct.

In the Swabian dialects, the distinction between the three vowels has partly been maintained, **e* and **ä₂* having merged into [ɛ], **ä₁* being continued by [e].¹³⁶¹ The Swabian form *krätze* – with low *e* – thus points to either OHG **chretzo* or **chrä₂tzo*, excluding **krä₁tto* with primary umlaut. Since any **j* in the second syllable would have caused primary umlaut, the reconstruction **krattjan*- (OHG **chrä₁tzo*) can be ruled out.

In order to decide between the two remaining possibilities, i.e. **chretzo* and **chrä₂tzo*, the Swiss Appenzell dialect can be consulted, this system preserving the distinction between OHG **e*, **ä₁* and **ä₂* as [ɛ], [e] and [æ]. Vetsch's 1910 description of the dialect gives the form *krätzə*, which appears to have developed out of a secondarily nasalized form *kräntzə* (cf. Vetsch § 96 *Vokalisierung des n*). The vocalism clearly points to OHG **chrä₂tzo* with secondary umlaut of **a*, and as such obliterates the PGM. reconstruction **kreddan*- that is encountered in the literature.

Everything considered, the *e*-vocalism of OHG *chretzo* and *chretto* appears to be due to the generalization of the analogical umlaut that characterized the plural forms. This scenario is confirmed by the

¹³⁶¹ Kauffmann (1890: 50): "In späteren Zeit ist hier ein jüngerer Umlaut aufgetreten, [...] und während die erste Umlautsperiode *e* ergeben hatte, war das Resultat des jüngeren Lautwandels *e*."

Swabian doublet *krätte*ⁿ ~ *kratte*ⁿ (beside *krätz*^e), which neatly mirrors the Old High German alternation of *chretto* with *chratto*. For late Old High German, the paradigm must therefore be reconstructed as sg. **chratto*, pl. **chrätton*. It continued PGm. **kradō*, **krattaz*, **kradeni* without ablaut.

The Proto-Germanic paradigm can be reconstructed as **grót-ōn*, **grot-n-ós*, **grot-én-i*. This etymon may be related to Skt. *grathnáti* ‘to fasten, tie or string together’ and OIr. *grinne* ‘bundle of twigs’ (< **grt(H)-nio-*).¹³⁶² Note that it is at any rate incorrect, as Lühr already pointed out, to reconstruct a root **gred-* (pace Pokorny IEW: 385-390) on the basis of OE *cræt* and cognates.

****tebō*, **tappaz* ‘tuft, knot, peg’†**

- †**teppan-*: OHG *zepfo* m. ‘plug, peg, broom’, MHG *zepfe* m. ‘bud, panicle, ear’, G Bav. *zēpfē* ‘lump, ear, grape’¹³⁶³, Tyr. *zepfe* m. ‘lappet, stub, fir-cone’¹³⁶⁴, Swi. Visp. *zäpfo* ‘pine nut’¹³⁶⁵
- **tappa(n)-*: OHG *zapfo* m. ‘plug, peg, broom’, G *Zapfen*, Als. *zapfe*ⁿ, pl. *zapfe/zäpfē* m. ‘tap, mais cone, vine stub’¹³⁶⁶, Bav. *zapfen* [zàpfē], pl. *zäpfen* [zápfē] m. ‘tap, fir cone’¹³⁶⁷, Swab. *zapfe*ⁿ m. ‘lump, uvula, fir cone’¹³⁶⁸, Tyr. *zapfn* m. ‘bell’¹³⁶⁹, Swi. App. *zapfē* m. ‘tap, cone’¹³⁷⁰, Rhntl. *zapfē* m. ‘id.’¹³⁷¹, Val. *zaffo* m. ‘id.’¹³⁷², Visp. *zapfo* m. ‘pine nut’¹³⁷³, OE *tæppa* m.

¹³⁶² Pokorny 385-90.

¹³⁶³ Grimm 31, 643; Schmeller/Frommann 2, 1148: *zēpfē*.

¹³⁶⁴ Schatz/Finsterwalder 725.

¹³⁶⁵ Vetsch 1910: 53.

¹³⁶⁶ Martin/Lienhart 2, 910b-911a.

¹³⁶⁷ Schmeller/Frommann 2, 1142.

¹³⁶⁸ Fischer/Taigel 431.

¹³⁶⁹ Schatz/Finsterwalder 720.

¹³⁷⁰ Vetsch 57.

¹³⁷¹ Berger 31.

¹³⁷² Bohnenberger 169.

¹³⁷³ Wipf 33.

- ‘tap, cone, strip of cloth’, ME *tappe* ‘ribbon’, MLG *tappe* m.
‘peg, tap’¹³⁷⁴, MDu. *tap(pe)* m. ‘id.’, SFri. *tappe* m. ‘plug’
⇒ **tappjan-*: ON *teppa* ‘to confine, close’, G *zepfen* ‘to
milch’, Bav. *zepfen* ‘to reap ears’
- **tapan-*: OE *tæpan* mf. pl. ‘strip of cloth’, ME *tape* ‘tape,
ribbon’, E *tape*
 - **taban-*: Nw. *tave* m. ‘piece of cloth, shred, tangle’, Sw. dial.
tave ‘piece of cloth’¹³⁷⁵, Da. *tave* ‘fiber, shred, tuft’
⇒ **tabnan-*: Far. *tavna* ‘to fray’¹³⁷⁶
 - **tabban-*: E *tab* ‘latchet, strap’, SFri. *tabbe* m. ‘plug’, G Als.
*zappe*ⁿ m. ‘tap, mais cone, vine stub’, Bav. *zappen* [zàppə] m.
‘tap, fir cone, lump’, Pal. *zappe* m. ‘plug, tap, fir cone, vine
stub’¹³⁷⁷

The cited forms clearly show that the *n*-stem meaning ‘tuft’ or ‘peg’ must have had a paradigm with consonant gradation in the proto-language. The material can be accounted for by reconstructing a paradigm **tabō*, **tappaz* that was split up into **tabō*, **tabbaz*, on the one hand, and **tapō*, **tappaz*, on the other. There is no need to attribute the consonant alternations to extra-linguistic factors such as “emphaticness”¹³⁷⁸ or “intensiver Konsonantverschärfung”.¹³⁷⁹

The root variant **tab-* is continued by Nw., Sw., Da. *tave* ‘fiber, shred’ and possibly by the Old Norse nickname *Tafi*.¹³⁸⁰ ME *tavele* ‘narrow lace’ is a diminutive with the same root. The second variant, PGM. **tapp*-¹³⁸¹ is found throughout the Germanic dialects, e.g. OE *tæppa* m. ‘tap, cone, strip of cloth’, ME *tappe* ‘ribbon’, *tappe* ‘plug’, OHG *zapfo* m. ‘plug, peg’, MDu. *tap(pe)* m. ‘peg, tap’. Together, these two roots gave rise to the contamination forms **tabb-*, as in E *tab* ‘latchet’, SFri. *tabbe* ‘plug’, and **tap-* as in ME *tape* ‘ribbon, tap’, E *tape*.

¹³⁷⁴ Vries (1962: 582) argues that ON *tappr* m. ‘tap’ (cf. Icel. *tappi* m. ‘cork, stopper’) was borrowed from MLG *tappe*, “weil das Wort erst spät auftritt”.

¹³⁷⁵ SAOB T554.

¹³⁷⁶ Poulsen 1215.

¹³⁷⁷ Christmann 6, 1533.

¹³⁷⁸ De Vries/Tollenaere 1991: 370.

¹³⁷⁹ Pokorny 227.

¹³⁸⁰ De Vries 1962: 579; Heggstad 689.

¹³⁸¹ Fick/Falk/Torp 155: **dap-n-*; Grimm 31, 258: **tabn-*.

The presence of OHG *zepfo* 'broom', MHG *zepfe* 'bud, panicle, ear', Bav. *zepfen* 'panicle, lump, ear', *tannen-zepfen* 'fir-cone' again seems to ascertain a Proto-Germanic variant form **teppan-*. This stem would then be in ablaut correlation with OHG *zapfo*, Bav. *zàpfā*¹³⁸², Swi. *zapfā* etc. Alternatively, it has been suggested that the forms with *e*-vocalism represent a *jan*-derivation, i.e. **tappjan-*.¹³⁸³ The Modern Upper German dialects, however, again provide evidence that precludes both of these reconstructions.

The dialect of Visperterminen, for instance, has both *zapfo* and *zäpfo* 'pine nut' (= Jaun Swiss *zäpfā*). Since this dialect differentiates between high *e* <*e*> from PGm. **a* with primary umlaut, and low *e* <*ä*> from both PGm. **e* and **a* with secondary umlaut, the second form *zäpfo* must go back to either OHG **zepfo* or **zäzpfō*, i.e. PGm. **teppan-* or **tappjan-* with secondary umlaut. This means that the reconstruction **tappjan-*, which would have resulted in OHG **zäzpfō*, Visp. ***zepfo*, can be canceled out.

The choice between the two remaining options can again be made with the help of the Appenzell dialect, which preserves the three-way distinction of OHG **e*, **ä₁* and **ä₂*. The form given by Vetsch is *zäpfe* with <*ä*>. Since PGm. **teppan-* should have developed into ***zepfā* in this dialect, and **tappjan-* into ***zepfā*, the actual form *zäpfe* can only be derived from **tappjan-* with secondary umlaut. This probably implies that OHG *zepfo*, too, must have had analogical umlaut, most probably **ä₂*. It is possible, however, that the umlaut vowel was **ä₁* in some dialects.

The consequence of this discussion is that the Proto-Germanic paradigm can be reconstructed as **tabō*, **tappaz*, **tabeni* without ablaut. Just as in the case of OHG *chratzo* ~ *chretzo*, the umlaut was presumably introduced in the plural in late Old High German, so as to result in a paradigm sg. **zapfo*, pl. **zäpfon*. Later, this analogical umlaut became intrusive in the singular, a process that Wipf and Stücker, too, consider for the Visperterminen and Jaun forms *zäpfo* and *zäpfā*.¹³⁸⁴ The fact that umlauted forms occur in a large area stretching from Jaun (*zäpfā*) in the

¹³⁸² Note that the alternant *zàppā* may go back to PGm. **tabban-*.

¹³⁸³ Grimm 31, 258; 31, 643; Kluge/Mitzka 874.

¹³⁸⁴ Wipf (p. 28): "*zapfo* oder analogisch nach dem Plur. *zæpfō*."; Stücker (p. 264): "Die Form mit Umlaut hat auch für den Sing. Geltung gewonnen bei *tsæpfā* Tannzapfen (selten -a-[])."

West to Bavaria (*zēpfē*) in the East, implies a relatively great time depth for this development.

Etymologically, the *n*-stem **tabō*, **tappaz* belongs to the ablauting iteratives **tappōhi*, **tabunanhi* (cf. G *zapfen* ‘to pull’, OHG *zabalōn*, G *zappeln* ‘to fidget’¹³⁸⁵) and **tuppōhi*, **tubunanhi* (cf. G *zupfen* ‘to reap’, G dial. *zobeln* ‘pull someone’s hair, tousle’¹³⁸⁶). The variant *zupfen* has given rise to the strong verb G *zaufen* ‘to pull’ < **tūp^{an}*- (see p. 108) as well as some nominal formations, e.g. **tuppa(n)*:- ON *toppr* m. ‘top, tuft of hair’, Nw. *topp(e)* m. ‘tap, tuft of hair, little peg’ (also *toppe* f. ‘cork, tuft’), OHG *zopf* ‘tip, tail’, G *Zopf* ‘tuft’¹³⁸⁷, Tyr. *zopfe* m. ‘braid’¹³⁸⁸, MDu. *top* ‘tip, (peg)top’, OE *toppa* m. ‘thread’, *top* m. ‘tip, tuft, pegtop’; **tubban-* MLG *tobbe*, *tubbe* ‘plug’.¹³⁸⁹ G *Zapfen*, on the other hand, seems to have served as the basis for the de-iterative verb MHG *zāfen* ‘to pull’ from Proto-Northwest Germanic **tāp^p*- with long **ā*.

As to the semantics, Grimm (31, 258) reconstructs a primary meaning “plucker” or “the plucked one” for the *n*-stem. This seems to be a profitable suggestion. It is conceivable that a tuft of wool or textile would have been used as a stopper, for example, to plug a vat. From here, it is just a small step to ‘peg’ and the relatively modern meaning ‘tap’. The semantic shift from ‘pluck’ to ‘tuft’ and ‘summit’ is trivial.

***skredō, *skrattaz ‘demon’†**

- †**skretta(n)*:- OHG *scretz(o)* ‘larva’, G *Schretz* m. ‘demon’¹³⁹⁰
 ⇒ ?**skrettjan*:- OE *scritta* m. ‘hermaphrodite’¹³⁹¹
- **skrada(n)*:- OHG *scrato* ‘faunus, larva, daemon’, MHG *schrat(e)* m. ‘(forest) goblin’¹³⁹² (⇒ MHG *schretel*, *schretzel* m. ‘small goblin’¹³⁹³), G *Schrat* m. ‘id.’¹³⁹⁴

¹³⁸⁵ Grimm 31, 276.

¹³⁸⁶ The link with Ru. *dybat* ‘to tiptoe’ (Holthausen 1934: 351; Vasmer 1, 557; De Vries 1962: 595) must be rejected.

¹³⁸⁷ Grimm 32, 76-84.

¹³⁸⁸ Schatz/Finsterwalder 733.

¹³⁸⁹ Schiller/Lübben 553.

¹³⁹⁰ Grimm 15, 1736; Kluge/Mitzka 678.

¹³⁹¹ Bosworth/Toller 65, 849.

¹³⁹² Lexer 2, 788.

¹³⁹³ Lexer 2, 792.

¹³⁹⁴ Kluge/Seebold 825.

- **skraddan-*: OHG *scratto* ‘larva, lar malus’
- **skrata(n)-*: ON *skrati* m. ‘troll’, Sw. dial. *skrate* ‘ghost, demon’¹³⁹⁵, MHG *schraz* m. ‘faun’¹³⁹⁶
- **skratta(n)-, -ōn-*: ON *skratti* m. ‘wizard, troll’, Icel. *skratti* m. ‘devil’, Sw. *skratte* ‘fool, devil’¹³⁹⁷, OE *scrætte* f. ‘adulteress’¹³⁹⁸, ME *skrat(te)* ‘hermaphrodite, goblin’, OHG *scratz, scretz*, pl. *scratza, scretz(a)* ‘porca, incubus, pilosus’, MHG *schraz*, pl. *schretze* m. ‘ghost, demon’¹³⁹⁹
- **skrudōn-*: OHG *scrota* ‘larva’
- **skrutta-*: Sw. dial. *skrutt* ‘devil’¹⁴⁰⁰

The consonant alternations that are found in the given forms have been explained by Lühr (1988: 252-4) as the result of an *n*-stem **skradō*, **skrattaz*. They are certainly not due to “eufemistiska o. hypokoristiska inflytelser”, as Hellquist claimed in his dictionary (p. 747).

The root **skrad-* is found only in e.g. OHG *scrato* and MHG *schrat(e)*. The geminated variant, on the other hand, is attested in both North and West Germanic, cf. ON *skratti*, OE *scrætte*, OHG *scratz*. Contamination of **skrad-* and **skratt-* seems to have led to the formation of a third root **skrat-*, which occurs in e.g. ON *skrati* and MHG *schraz*. The creation of this third root implies that Northwest Germanic possessed an analogical paradigm **skratō*, **skrattaz*.¹⁴⁰¹

The presence of OHG *scretz(o)*, G *Schretz* is reason to consider the possibility that the Proto-Germanic paradigm once contained an *e*-grade. Lühr (p. 253) indeed postulated a root **skrett-*, because if the vowel of OHG *scretz* were due to umlaut, she argued, the required umlaut factor would have left a trace in Old High German, e.g. ***scretzi* < **skrattja-*. To

¹³⁹⁵ Rietz 596.

¹³⁹⁶ Lexer 2, 788.

¹³⁹⁷ SAOB S4779.

¹³⁹⁸ Bosworth/Toller 840.

¹³⁹⁹ Lexer, l.c.

¹⁴⁰⁰ Hellquist 746-7; Rietz 596, 601.

¹⁴⁰¹ Lühr further connected *skradd* ‘wretch’, which with its voiced geminate may point to an analogical paradigm **skradō*, **skraddaz*. The different meaning of the word nevertheless makes the appurtenance of the word unconvincing. The same goes for Nw. *skrede* ‘scrag’, *krede* f. ‘miserable animal, person’ and Icel. *kreða* f. ‘mother’s darling, scrag’.

further strengthen the reconstruction of a root **skrett-*, Lühr (l.c.) pointed to OE *scritta* ‘hermaphroditus’, which with its *i* looks like a formation **skrettjan-* (cf. Fick/Falk/Torp 472).

In the end, however, it seems better to reject the possibility of **e : *a* ablaut, because both of Lühr’s arguments in favor of a root **skrett-* can be countered. OE *scritta* occurs only once, and is outweighed by the expected outcome of *skrattōn-*, viz. OE *scrætte* and ME *skrat(te)* ‘hermaphrodite’. More importantly, the analysis of OHG *scretz* as continuing PGM. **skrett-* does not seem to be compelling. Of all the attested forms in Old High German, the *e*-vocalism is predominantly found in the plural, e.g. *screza*, *screzza*, *scre(i)z*. Since the umlaut was analogical in the plural, this means that the OHG paradigm should probably be reconstructed as **scratz*, pl. **scrätza*.¹⁴⁰² In the singulars OHG *scretz(o)* and G *Schretz*, the umlaut must then be intrusive.

There is some evidence for **a : *u* ablaut. A formation **skrudōn-* is supported by the Old High German hapax *scrota* ‘larva’. In addition, a zero-grade stem **skrutta-* can be reconstructed for the Finland Swedish form *skrutt* ‘devil’. In spite of the marginal attestation of the root variants with **u*, they can theoretically be accounted for by reconstructing a paradigm **skradō*, **skrut taz*. If this is correct, the *n*-stem can be added to section § 9.3.

****kredō*, **kruttaz* ‘toad’†**

- †**kredōn-*: OHG *chreta* ‘*bufo*, *rubeta*, *rana*’, MHG *krete* f. ‘toad’¹⁴⁰³, MRhnl. *crede* ‘id.’
- **krudōn-*: OHG *chrota* ‘*rubeta*, *bufo*, *rana*’, MHG *krot(e)*, *kröte* f. ‘id.’, G *Kröte*¹⁴⁰⁴, Als. *krot*, pl. *krot* f., *kret*, pl. *kret* m. ‘id.’¹⁴⁰⁵, Swab. *krot^e*, *kröt^e*, pl. *kroteⁿ*, *kröteⁿ* f. ‘id.’¹⁴⁰⁶, Lus. *kröt*, *kröter*

¹⁴⁰² OHG *screz* may have developed out of the plural **scrätza* by apocope and the spelling *screiz* presupposes a long vowel that resulted from compensatory lengthening after this apocope. It seems unlikely, at any rate, that the word continues an *i*-stem **skratti-*.

¹⁴⁰³ Only in Herbort von Fritzlar’s *Lied von Troye*: “Ginge ich als ein crete gat” (Fromman 1837: 69).

¹⁴⁰⁴ Grimm 11, 2414-19; Kluge/Seebold 542.

¹⁴⁰⁵ Martin/Lienhart 1, Spalten 527a-527b.

¹⁴⁰⁶ Fischer/Taigel 287.

- f. 'id.'¹⁴⁰⁷, Zarz *kxroute*, pl. *kxroute*, *kxröute* f. 'id.'¹⁴⁰⁸, Swi. App. *kxrōt* f. 'id.'¹⁴⁰⁹, Visp. *xrotta* f. 'id.', MLG *krode* f. 'id.'¹⁴¹⁰, MDu. *crode* f. 'id.'¹⁴¹¹
- **kruddan-*, *-ōn-*: OHG *chrotta* 'rubeta', MHG *krotte* f. 'id.', G Als. *krotte*ⁿ m. 'id.', *krott*, *krett* f. 'toad, small person'¹⁴¹², Rhnl. *krutte* f. 'toad, frog, stunted child'¹⁴¹³, Swi. App. *kxrōt*¹⁴¹⁴ 'toad', Visp. *xrotta* f. 'id.', EDu. *krodde* 'rubeta, bufo', Du. *krod(de)* 'toad, chick, small child'¹⁴¹⁵
 - **kruttōn-*: G *Krotz* f. 'toad, irritable child, wizened person'¹⁴¹⁶, Loth. *krotze-mann* 'water goblin', ?E *croot*, *crut* 'feeble child, dwarf'¹⁴¹⁷

The formal variation of forms such as OHG *chrota*, MLG *krode*, Zarz *kxroute* < **krudōn-*, OHG *chrotta*, MHG *krotte*, Als. *krotte*ⁿ, Visp. *xrotta*, EDu. *krodde* < **kruddōn-* and G *Krotz* < **kruttōn-* straightforwardly points to an *n*-stem **krudō*, **kruttaz* with consonant gradation. The origin of the vocalism of OHG *chreta*, on the other hand, is more problematic. Traditionally, *chreta* is reconstructed as PGm. **kredōn-* with an ablauting full grade.¹⁴¹⁸ If this were correct, the *n*-stem should probably be reconstructed as **kredō*, **kruttaz*. Still, given the fact that there are other *n*-stems in Upper German that were affected by analogical umlaut, it is much more probable that the interchange of OHG *chreta* and *chrota* was created by this process as well.

The hypothesis that *chreta* represents a fronted form has a number of advantages over reconstructing ablaut variant **kredōn-*. For instance, it can account for the lack of singular forms with *e*-vocalism in the

¹⁴⁰⁷ Zingerle 39.

¹⁴⁰⁸ Kranzmayer/Lessiak 99.

¹⁴⁰⁹ Vetsch 1560.

¹⁴¹⁰ Lübben 190.

¹⁴¹¹ Verdam 313.

¹⁴¹² Martin/Lienhart 1, 527a.

¹⁴¹³ Franck/Müller 4, 1621.

¹⁴¹⁴ Vetsch 1560.

¹⁴¹⁵ Vercoullie 186, 187; WNT, s.v. *krod*.

¹⁴¹⁶ Höfler 1899: 336; Franck/Müller 5, 1575.

¹⁴¹⁷ OED, s.v. *croot*.

¹⁴¹⁸ Grimm 11, 2414-30; Fick/Falk/Torp 51.

modern Upper German dialects, e.g. Hess. (Wetterau) *krott*, pl. *kräte*, Lus. *krôt*, pl. *kröter* etc. Presumably, the original Old High German paradigm, i.e. sg. **chrota*, pl. **chroton*, was supplanted by a secondary paradigm sg. **chrota*, pl. **chröton*¹⁴¹⁹ with analogical umlaut in the plural. The reality of this process is confirmed by the modern dialects, which typically have umlaut in the plural, or waver between fronted and unfronted plural forms, such as Zarz *kxroute*, pl. *kxroute*, *kxröute*.

The question must therefore be whether in this *n*-stem, too, the umlaut became intrusive in the singular. Again, this solution indeed seems to be corroborated by the material. The most salient indication for intrusive umlaut actually comes from the standard High German form *Kröte* itself. It has been suggested that it represents a “Mischung” of *krete* and *krote*¹⁴²⁰, but this solution fails to explain where *krete* and *krote* come from in the first place. Most probably, *Kröte* must be regarded as a Luther form based on a dialect with intrusive umlaut in the singular. Such forms are especially abundant in the Southern dialects.

In Swabian, forms with and without umlaut compete with each other in both the singular and the plural, cf. *krot^e*, *kröt^e*, pl. *kroteⁿ*, *kröteⁿ*. The same competition is, in fact, found in Alsatian German, where a feminine *krot*, *krotə* and a masculine *kret*, *kretə* occur side by side. On the basis of these facts, it can safely be assumed that the paradigm **chrota*, **chroton* was in the process of being replaced by **chrota*, **chröton* in late Old High German, and that the fronted root vowel became generalized in at least some dialects.

It further follows that the vacillation of OHG *chreta* and *chrota* is likely to be a reflection of the form **chröta* with <ö>.¹⁴²¹ After all, the scribes did not have a separate symbol for this phone. Later forms with *e*-vocalism (cf. MHG *krete*, Middle Rhinelandish *creda*, *credda*, *crede*¹⁴²²) can probably not be equated with OHG *chreta* directly, though, because they may be due to the wide-spread unrounding of front vowels.

¹⁴¹⁹ It is actually superfluous to differentiate between primary and secondary umlaut of OHG *o*. Umlaut of this vowel is always secondary, because it arose out of PGm. **u* that was first affected by *a*-umlaut.

¹⁴²⁰ Pace Kluge/Mitzka 408; Kluge/Seebold 542.

¹⁴²¹ Cf. Swab. *kratteⁿ*, *krätteⁿ*, *krätzeⁿ* ‘basket’.

¹⁴²² Grimm 11, 2415.

Unrounding probably also led to the rise of some forms with ostensible *a*-vocalism in the Middle German area, cf. MHG *krate* f. 'id.'¹⁴²³, MRhnl. *crade* (= MDu. *crade* f.¹⁴²⁴) 'id.', G Rhnl. *krade* f.¹⁴²⁵ 'id.', WPhal. *kradde* f. 'id.'¹⁴²⁶, Lux. *kratz* 'toad, small child'.¹⁴²⁷ The limitation of these forms to this particular area makes it unattractive to reconstruct an old ablauting variant **kradōn-*. So, if the forms with *a*-vocalism are not due to a dialectal change of *o* to *a* in this phonetic environment, they may have come into existence due to backformation from an unrounded plural: 1) **krode*, *krōde* > 2) *krode*, *krāde* >> 3) *krade*, *krāde*.

The etymology of *Kröte* has not yet been clarified.¹⁴²⁸ Fick, Falk and Torp (p. 51) give Gr. βάτραχος, Ion. βρόταχος, βάθρακος 'frog' as possible cognates, reconstructing a PIE root **g^wred^h-*. The Greek forms are highly irregular, however, and even if βάτραχος and βάθρακος continue a root **βράθρ-* < **g^wrd^h-* through dissimilation of the first or the second rho, the suffix still displays a conspicuously non-Indo-European alternation between -αχος and -ακος. It is therefore probably better to connect the *n*-stem **krudō*, **kruttaz* to the verb G *krotten*, which is attested in Paracelsus's *Chirurgische Schriften* (p. 401b): "wann der schenkel oder das glid geschwillt und krottet sich, da ist kein heilung zu thun".¹⁴²⁹ Flabbiness is a common *Benennungsmotiv* for the toad, cf. Du. *kwab* 'flab' and EDu. *quabbe* 'rubeta, bufo, rana', and it is possible that **krudōn-* is just another example of this semantic association. Additionally, cognates such as EDu. *krotte* 'lutum vestibus haerens' and E *crote* 'clod of earth' can be taken into consideration. Grimm (l.c.) further mentions the apparently Rhinelandish gloss *croz* for Lat. *tabes* 'corruption'. Outside Germanic, a possible link is encountered in the form of W *crwth* 'any body swelling out or bulging; paunch; box' < **krutto-*, but in view of the geminate of this formation, it runs the risk of being a Germanic loanword in Celtic.

¹⁴²³ Lexer 1, 1712.

¹⁴²⁴ Verdam 313.

¹⁴²⁵ Franck/Müller 4, 1328.

¹⁴²⁶ Woeste 1882: 141.

¹⁴²⁷ Grimm 11, 2418.

¹⁴²⁸ Kluge/Seebold 542.

¹⁴²⁹ Grimm 11, 2424.

10.2 West Norse

The formal problems that surround the Nordic word for ‘nut’ are comparable to the difficulties of the seemingly ablauting *n*-stems in the Upper German dialects. At first sight, the word seems to have three different, ablauting root variants. On closer inspection, however, this variation turns out to be the result of a combination of vowel mutation and analogy.

**hnetō, *hnuttaz* ‘nut’†

- †**hnetōn*:- Icel. *hnet* f. ‘id.’¹⁴³⁰
- **hnut*:- ON *hnot*, pl. *hnøtr*, *hnetr* f. ‘id.’¹⁴³¹, Icel. *hnot* f., pl. *hnetur*, *hnotir*, *hnotur* ‘nut, clew’¹⁴³², OE *hnutu*, pl. *hnyte* f. ‘id.’, OHG *nuz* f. ‘id.’¹⁴³³
- **hnutōn*:- Icel. *val-hnota* ‘walnut’¹⁴³⁴
- †**hnat*-, *-ōn*:- ON *hnata-skógr* ‘nut grove’, Far. *nøt*, *nøta* f. ‘nut’¹⁴³⁵, Nw. dial. *nate-kjerne* ‘stone of a nut’, *nate-hams* ‘nutshell’

The alternation of the roots *hnet*-, *hnat*- and *hnot*- in the West Norse dialects seems to be a clear case of ablaut. Since the Icelandic forms *hnet* and *val-hnota* are inflected as *n*-stems, it is theoretically possible to postulate a paradigm **hnetō, *hnuttaz*.

The reconstruction of an ablauting *n*-stem is unfeasible, however, in view of the absence of the consonant gradation that is usually coupled with primary *n*-stems. What makes it even less attractive, is that the word for ‘nut’ is inflected as a root noun in most of the older Germanic dialects, cf. ON *hnot*, pl. *hnøtr*, OE *hnutu*, pl. *hnyte*. The paradigm must accordingly be reconstructed as **hnut*-z, gen. **hnut*-iz < **knud*-s, **knud*-es.

¹⁴³⁰ Böðvarsson 390.

¹⁴³¹ Zoëga 206.

¹⁴³² Böðvarsson 393.

¹⁴³³ Braune 1891: § 219: “Eine Anzahl der hierher gehörigen Fem. folgte früher der consonantischen Declination[...]: *eih*, eiche, *gans*, *geiz*, *nuz*, [...]” etc.

¹⁴³⁴ Böðvarsson 392.

¹⁴³⁵ Poulsen 839.

An alternative way to deal with the seemingly ablauting root variants is to assume that they result from a number of backformations. The Old Norse plural *hnøtr* was already unrounded to *hnetr* in late West Norse (cf. ON *kømr* 'comes' > Icel. *kemur*, Nn. *kjem*). When, in Middle Icelandic, the endings *-r* and *-ur* merged into *-ur*, the plural *hnetur* may well have been reanalyzed as belonging to a feminine *n*-stem *hnetā* (cf. *tunga*, pl. *tungur* 'tongue').

A similar explanation works for Far. *nøta*. Since the merger of *-r* and *-ur* occurred in Faroese just as much as in Icelandic, *nøta* is likely to be a backformation from the Old Faroese plural **nøtur* < ON *hnøtr*. Morphologically, *nøta* thus became similar to feminine *n*-stems of the *køka*-type, which generalized the mutated stems from the oblique, cf. ON nom. *kaka*, obl. *køku*, pl. *køkur* f. 'cake'.

No doubt, *u*-mutation played a role, as well, at the creation of the root *hnat-* in ON *hnata-skógr* and Nw. *nate-kjerne*. Morphologically, the first element of these compounds represents a genitive plural *hnata*. It is conceivable, that this *hnata* was created on the basis of the unrounded plural *hnetr* according to the umlaut proportions that are found in other consonant stems, e.g. ON npl. *merkr*, gpl. *marka*, dpl. *mørkum* 'forest' < **mark-iz*, **markōan*, **markumiz*. Apparently, *hnetr* was reanalyzed as reflecting **hnatiz*, and accordingly given a gpl. *hnata*.

The analogical explanation of the vocalism of *hnetā* and *hnata-* is supported by the etymology of the word. PGm. **hnut-* is clearly related to OIr. *cnú* 'nut' < **knū-*, obl. **knuw-* (Schrijver 1995: 329-30.) and Lat. *nux* < **knu-k-* which point to a primitive root **knu-*. With its **u*, this root could never have developed into PGm. **hnet-* in a phonetically regular way.

11 Summary and outlook

11.1 Summary

In this study, it has been argued that the peculiar typology of the Germanic *n*-stems is fully understandable from the Proto-Indo-European situation. The consonant and vowel alternations displayed by this morphological category are the direct result of the interaction of the sound laws and analogies that took place during the evolution of a West Indo-European dialect into what nowadays is known as Proto-Germanic.

The consonantal alternations exhibited by the *n*-stems can be understood by assuming that assimilation of the suffix gave rise to geminate consonants in the genitive singular and plural and in the accusative plural of the original paradigm. By this assimilation, which is known as Kluge's law, nasals were absorbed by a preceding stop in pretonic position. The resulting voiced geminates were later devoiced by Grimm's law, so as to yield long voiceless stops.

Since not all cases were affected by Kluge's law, the Proto-Germanic *n*-stems became highly allomorphic: they developed a paradigmatic interchange of singulate and geminate roots. This allomorphy was subsequently reduced by the generalization of the articulation of either the singulate or the geminate. Presumably, this happened through the split of the original paradigm into two new paradigms. The reduction of the allomorphy thus led to a paradoxical increase of the total amount of formal variation. This variation can be observed in all Germanic dialects, including Gothic.

A similar combination of sound law and analogy must be assumed for the directional adverbs and – most notably – the iterative verbs, which appear to have evolved out of those Proto-Indo-European *neh*₂-presents that were derived from primary aorists. The iterative verbs formed a derivationally crucial category, because they closely interacted with the strong verbs. As a result of this interaction, they frequently transmitted their (geminated) consonantism to the originally thematic

verbs. In fact, it is plausible that quite a few strong verbs were actually derived from iteratives.

The vowel alternations displayed by the *n*-stems are likely to be a continuation of the **e* : *zero* ablaut that is traditionally reconstructed in the root of the Proto-Indo-European paradigm. The evidence is in corroboration with the standard reconstruction of a full grade in the nominative and accusative and a zero grade in the other cases.

The **e* : *zero* ablaut gave rise to several different sub-types by a combination of regular sound law and analogy. The **e* : **u*, **a* : **u*, **ī* : **i* and **ō* : **a* types appear to have arisen in a more or less phonetically regular way. The **ū* : **u* and **ē* : *a* types, on the other hand, rather seem to have come about analogically after the model of the other types.

In addition to the **e* : *zero* ablaut, the Germanic *n*-stems give proof of an **o* ~ *zero* alternation comparable to the ablaut of the PIE word for 'path', viz. **pónt-eh₁*, gen. **pnt-h₁-ós*. This follows, for instance, from the words for 'moth' and 'rat', which exhibit a clear **a* ~ **u* alternation.

Finally, there is marginal evidence for *n*-stems with a triple **e* : **o* : *zero* alternation. Such ablaut is implied by the word for 'beam', which in Proto-Germanic can be reconstructed as **belk^kō*, gsg. **bulk^kaz*, apl. **balk^kuns*. In this way, the Germanic material supports the hypothesis by Beekes (1985) that the *o*-grade arose in the unstressed accusative root by the analogical spread of **e* from the nominative.

The main conclusions concerning the consonant and vowel alternations in the *n*-stems can be summarized with the following reconstruction of the amphikinetic paradigm in Proto-Germanic and Proto-Indo-European:

	Proto-Germanic	Proto-Indo-European
nsg.	CeC-ō	CéC-ōn
gsg.	CøCC-az	CøC-n-ós
dsg.	CøG-eni	CøC-én-i
asg.	CéC-anun	CéC-on-m
npl.	CeC-aniz	CéC-on-es
gsg.	CøCC-an	CøC-n-óm
dpl.	CøCC-ummiz	CøC-n-mis
apl.	CaCC-uns	CóCC-n-ńs

11.2 Outlook

In the present study, it has been argued that the Expressivity Theory and the Leiden Substrate Theory – two alternative explanations to the consonant and/or vowel alternations in the Germanic lexicon – cannot be maintained. They are both based on an imperfect understanding of the processes that gave rise to Proto-Germanic morphophonology in general, and the impact of Kluge's law in particular. Due to the broad acceptance of the Expressivity Theory and its dominance in handbooks and dictionaries, a considerable amount of received etymologies are in need of reconsideration.

The inclusion of Kluge's law into the canon of Indo-European Studies will expectedly have a considerable impact not just on etymology, but also on the discussion on Winter's law in Balto-Slavic, i.e. the debated lengthening of vowels before original PIE **b*, *d*, *g* (as opposed to **b^h*, *d^h*, *g^h*). Much of the counter-evidence to this law is based on Germanic material only, and thus runs the risk of being polluted by (shortened) geminates (cf. Kroonen forthc.).

It is further to be expected that the acceptance of Kluge's law and its consequences will shed new light upon the prehistoric interaction between Germanic and Celtic. It is well-known that these two branches share a number of isoglosses. Yet, in absence of a Proto-Germano-Celtic subclade, these can hardly be explained as inherited, at least not in both languages. The majority of the isoglosses are therefore attributed to borrowing, typically from Celtic to Germanic, e.g. PGm. **rīkja*- 'powerful' < PCelt. **rīgīo*-. The direction of borrowing must be the reverse, however, in the case of PGm. **hrauk^ka*- 'haystack' and PCelt. **kroukā*- 'id.', because the geminate is perfectly understandable from internally Germanic processes.

Finally, it has been suggested by Simon Mulder in his MA thesis (2010) that the Germanic overlong syllables, which disappeared on the way to Proto-Germanic, can sometimes be retrieved from Germanic loanwords in Finnish. Within Germanic, geminates in overlong syllables were probably shortened before the dissolution of the proto-language. It is possible, however, that some of these geminates were preserved by Finnish, in which there are no constraints against overlong syllables.

Bibliography

Abbreviations

EWA	Lloyd/Lühr/Springer: <i>Etymologisches Wörterbuch des Althochdeutschen</i> .
LIV	Kümmel/Rix (et al.): <i>Lexikon der indogermanischen Verben</i> .
MED	McSparran (ed.): <i>Middle English Dictionary</i> .
ODS	Det Danske Sprog- og Litteraturselskab: <i>Ordbog over det danske sprog</i> .
OEC	DiPaolo (ed.): <i>Dictionary of Old English corpus</i> .
OED	Simpson/Weiner (eds.): <i>Oxford English dictionary</i> .
PLAND	Brok/Kruijsen (et al.): <i>Plantennamen in de Nederlandse dialecten</i> .
RLGA	Hoops (et al.): <i>Reallexikon der germanischen Altertumskunde</i> .
SAOB	<i>Svenska akademiens ordbok</i> .
WBD	Weijnen (et al.): <i>Woordenboek van de Brabantse dialecten</i> .
WLD	Weijnen/Goossens/Hagen: <i>Woordenboek van de Limburgse dialecten</i> .
WNT	Instituut voor Nederlandse Lexicologie: <i>Woordenboek der Nederlandsche taal</i> .
WTM	Schatz: <i>Wörterbuch der Tiroler Mundarten</i> .
WVD	Ryckeboer (et al.): <i>Woordenboek van de Vlaamse dialecten</i> .

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Index of cited forms

1) Germanic languages

East Germanic

Gothic

aba, 56
aihva-tunþus, 22
akran, 39
alls, 43
ana-trimpan, 110, 111
and-bundnan, 102
arms, 185
aþns, 46
atta, 80, 110
auga-dauro, 38
augo, 39
auhns, 46
auhsa, 30, 56, 62, 63
auso, 39
bairandei, 37
bandwo, 37
barn, 39, 47
barnilo, 33, 39
bi-mampjan, 107, 111
bindan, 102
brunna, 83
daura-wardo, 37
digan, 105, 115
dis-hniupan, 106
dis-skreitan, 109, 111
diups, 45
fairra, 42, 43

fauho, 23, 70
fi(j)an, 241
fiskja, 43
fon, 136
fra-hinþan, 45
fraihnan, 43, 129
fullnan, 101
fulls, 42
ga-geigan, 129
gaitein, 230
ga-smeitan, 109, 111
ga-waknan, 101
ga-wigan, 104
graban, 105
greipan, 109, 111
guma, 29, 35
hairto, 39
hauhs, 68
heþjo, 306
himins, 61, 163, 305
hiuhma, 68
hlahjan, 104
huhjan, 68
hvað, 85
hvar, 85
hvaþ, 85
hvaþro, 85
hveits, 45
ibns, 48
in, 86
inn, 86
in-sailjan, 244
iup, 88

iupana, 88
jainar, 85
jaind, 85
jainþro, 85
*kaur*n, 174
kaurno, 39, 174, 175
lais, 102
lats, 310
lauhmuni, 182
liudan, 217
liugan, 103
lofa, 129, 133, 134, 136, 309
lukan, 115
malma, 184, 185
manna, 41
maþa, 135, 139, 218
mizdo, 218
mukamodei, 280
namo, 31, 38
nota, 45
qairrus, 42
qino, 36, 37
rahnjan, 46
razn, 48
rign, 48, 122
sakkus, 110
salba, 93
salbon, 93
sauil, 83, 136, 323
siuns, 49
skaban, 45
skatts, 41

skeima, 143, 246
skiuban, 104
skuft, 285
slepan, 107
smakka, 42, 111
sneiþan, 103
speiwan, 259
stairno, 225
stairo, 190
staua, 29
steigan, 115
sunno, 83
swamm, 223
tagl, 317, 319
tekan, 104
þairh, 91
þairko, 39
þrut-fill, 43
timrja, 43
tiuhan, 104
trudan, 104, 115
tuggo, 36
uf, 88
us, 87
us-keinan, 129
us-kijanata, 129
ut, 87
wato, 38, 48
winnan, 108
witubni, 183
wulan, 115
wulla, 43

North Germanic

Old Norse

akarn, 39
almr, 143, 156
alnbogi, 311
ari, 61
auga, 39
ax-helma, 162
barn, 47
barð, 150
barr, 209
**bí-fluga*, 229
birna, 214
bjalki, 134, 148
bjúga, 32, 38
björn, 61
bækr, 69
borð, 149
borði, 149
botn, 65, 122
broddr, 149, 150
brosma, 210
bý, 230
bølkr, 61, 148
dafla, 212
dapi, 211
dálkr, 192
depill, 212
des, 228
doki, 23
dokka, 23
drjúpa, 113
dropi, 106
dumba, 151, 152
dúfa, 128
dvena, 97
døf, 212
eikinn, 264
einir, 32
eista, 39
Elmi-kjarr, 156
enni, 39
eyra, 39
fara, 82
farri, 82
fjarri, 42, 43
fjá, 241
ffjúka, 109
ffögur, 158

flag-brjóska, 39
flík, 133
fljóta, 103
flóki, 133
flota, 103
fluga, 230
fraukr, 133
galti, 212, 213
gana, 97, 302
gaupn, 47
geimi, 231, 232
gelda, 213
geldr, 213
gíma, 143, 231
gíma, 231
gjá, 231
glotta, 93, 96
gómr, 302
gróp, 45
gryffa, 162
gumi, 35
gylta, -r, 213, 214
gyltr, 213
goltr, 61, 212, 213
haðna, 306
halmr, 162, 163
hamarr, 165
há-mót, 329
hár, 329
há-sin, 329
heðan, 201
heðinn, 61, 304, 305
hegri, 266, 267
heri, 200
hetta, 304
himinn, 36, 61, 163, 305
hjalmr, 162
hjarn(i), 166, 174
hjarri, 23, 79
hjarsi, *hjassi*, 81, 165, 166

- hjarta*, 39
hjú, 230
hnakki, 167, 168
hnata-skógr, 350, 351
hnoða, 33, 38
hnokki, 168
hnot, 350
hækja, 328
hoka, 110
hrafñ, 47, 60, 73, 171
hraukr, 143, 269, 271
hrip, 181
hríð, 234
hrím(i), 58, 60, 66, 67
hrjóta, 106, 113
hroði, 106
hrogn, 47, 60, 122, 171
hroki, -r, 24, 59, 269
hrúga, 24, 268, 269, 271
hvann-jóli, 197
**hofða*, 39
hofuð, 39
høm, 81
høss, 202
høttr, 58, 61, 149, 224, 304
inn, 86
iup, 88
í, 86
íkorni, 263
jafñ, 48
jú(g)r, 39, 157, 158
kaka, 306, 351
karmr, 339
kartr, 60, 339
kárr, 277
kerf, 179, 182
kippa, 235
kirna, 174
kjalki, 172, 173
kjarf, 179, 182
kjarni, 174, 198
klappa, 125
kleppr, 178
klé, 176, 177, 259
klífa, 129
kljúfa, 113
klót, 273
klumba, 178
klumbu-fótr, 178
klútr, 272
kløpp, 121, 178
knapi, 73
knappr, 298
knauss, 300, 301
knífr, 141
knoða, 115
knútr, 45, 267, 299
knýfill, 298
knøttr, 61, 144, 149, 224, 299, 300
koddi, 59, 175, 176
kók, 203
koma, 115
kona, 83
korf, 181
korn, 174
korn-kippa, 234
krabbi, 73, 80
kraki, 332
krákr, 325, 332, 334
krjúpa, 113
krókr, 144, 332
leika, 32, 38
leir-depill, 211
leir-krukka, 321
lim, 62
limr, 62
lík-síma, 38
lími, 62
ljá, 241
ljómi, 182
ljúka, 114
lófi, 309
logn, 122
lokka, 44, 98
lokkr, 42, 121
lófi, 129, 309
luma, 156
lúka, 114
maðkr, 218
malmr, 184, 185
motti, 135, 139, 219, 220
múgi, -r, 278
mý, 230
nefi, 260
nyra, 39
noðkur, 264
noðkviðr, 263
ngs, 208
of, 88
ofan, 88
okka, 125
opinn, 89
orri, 214
oxi, 63
ór, 87
padda, 281
poki, 23
posi, 291
púss, 291
rauðr, 102
reigjask, 243
rífa, 107
riga, 95, 96, 124
rippa, 107
rist, 262
rífa, 107, 108
rígr, 243
ríta, 109
rjómi, 186
rjúfa, 104, 283

- rjúka*, 113, 294
rjúma-rauðr, 186
roðna, 102
rugga, 94, 98
seil, 243
seli, sili, 143, 243
sími, 38
sjón, 49
sjúga, 114, 294
skaga, 317
skagi, 316
skaka, 105
skarn, 48
skauf, 284, 285
skegg, 316
skimi, 246
skíð, 246
skoft, 285
skógr, 316, 317
skopa, 95, 96
skrapa, 125
skrat(t)i, 345
skupla, 285
skúfr, 283
skvakka, 125
slaggi, 314
snagi, 334
snapa, 96, 249
snákr, 137, 144, 249, 325
snákr, 333
snípa, 248
sól, 83
sópa, 107
soppr, 224
sparri, 62
spjör-, 62
sproti, 293
steka, 104, 105
stinga, 104
stjarna, 174, 225
stjöl, 62
stofn, 286
strjúka, 106
strjúpa, -i, 39, 294
strúpi, 294
strútr, 295
stubbi, stubbr, 286
stúfr, 286, 287
sunna, 83
súga, 114, 294
svefn, 46
sveim(r), 143, 251
svimma, 129
svíri, 252
svoppr, 61
symja, 129
sopp, 224
tafn, 46
tagl, 317, 319
taka, 104
tappr, 342
teðja, 227
teikn, 47
telgja, 192
teppa, 342
tjalga, 191
toddi, 73, 80, 226, 227
toga, 95
toppr, 344
trani, 307
troða, 115
tveim(r), 36
tyggva, 206
Týr, 230
tönn, 22
töturr, 226, 227
ull, 43
upp, 88
uppi, 42
uxi, 63
úr, 87
út, 87
vagn, 48
vaka, 103
vakna, 103
vatn, 38, 48
vega, 105
**vé-höfða*, 39
vekja, 103
viðbeina, 39
vífandi, 108
vottr, 45
ylgr, 213
**yrna*, 214
þjóta, 113
þó, 269
þrjú, 230
þrúttinn, 296
þrútr, 295
þumall, 288
ögr, 209
ökla, 39
ökli, 39
önn, 48, 301
örn, 61
økk, 32
Icelandic
bjalli, 39
bý(-fluga), 229, 230
demba, 151
depill, 211
des, 228
dumba, 151
geimur, 143, 231, 232
gíma, 143, 231
gómur, 302
gubba, 104
gufa, 104
gylta, 213
haki, 328
héðan, 201
hegri, 266, 267

- hér*, 201
héri, 200
hjar(r)i, 23, 69, 79
hn(j)úkur, 294
hneta, 350
hnjóta, 299
hnjóti, 299
hnoða, -i, 299
hnot, 350
hnota, 299
hnotti, 144, 299
hnúði, 136, 299
hnúður, 299
hnúfa, 297, 298
hnullóttur, 170
hnúta, 299
hnútur, 299
hrái, 171, 172
hraukur, 269
hró, 24, 136, 143, 268, 269
hroki, -ur, 269
hrúga, 268
hrúka, 24, 268, 269
hvann-jóli, 197
íkorni, 263
júfur, *júgur*, 157
káلكur, 143, 172, 173
karfa, 179, 181
ker, 201
kjálki, *kjálkur*, 143, 172, 173
klé, 176
kljá, 176
koddi, 175
koðri, 176
kok, 203
kók, 203
kóka upp, 203
kot, 276
kraumur, 274, 275
kreða, 345
krubba, 179
krukka, 321
krumur, 274, 275
kvok, 203
labba, 309
lim, 62
ljúka, 114
lög-regla, 81
lögga, 81
löpp, 129, 309
loppa, 310
lúka, 114
Mogga, 81
Morgunnblaðið, 81
múgi, 278
mý-fluga, 230
njóli, 197
posi, 291
púsi, 291
riga, 243, 262
rjómi, 143, 186
rjúp-karri, 70, 80
rjúp-keri, 70, 80
rotta, 222
rubba, 104, 283
síla, 243, 244
sjúga, 114, 116, 294
skagi, 316
skratti, 345
skúfur, 283
skupla, 285
slabba, 96, 107, 111
slafa-st, 96, 107
slagi, 314
slaki, 314
slakna, 314
slapa, 96, 107, 111
slappa, 111
snagi, 333
snákur, 333
sníff(a), 249
snípa, 247
snípur, 247
snókur, 333
soppa, -i, 224
soppur, 224
strjúpi, 294
sveppur, 224
svía, 251
svími, 143, 251
takki, 318
tappi, 342
tenna, 22
puma, 288
pumall, 288
toddi, 226, 227
tönn, 22
val-hnota, 350
valmúi, 311
ví, 260
vía, 260
Faroese
breddi, 134, 143, 149
bý-fluga, 229, 230
des, 228
dokki, 23
dumba, 151
eta, 52
finger-gómur, 302
gøltur, 212
gómi, 302
hara, 200
hegri, 266
hvann-jóli, -ur, 197
hvökka, 44
íkorni, 263
jólur, 197
kjálki, 172, 173
klavi, 177
klíggja-steinur, 176
knabbi, 298
knappur, 298
knasi, 80, 300, 301

- kneysur*, 300
knobbi, 298
knota, 299
knubbi, -ur, 297
knúki, 58
knúkur, 58
knúta, 299
knútur, 299
koddi, 175
koka, 203
køka, 351
kot, 276
krukka, 321
kulka, 172
labbi, 59, 129, 309
leika, 39
múgva, -i, 278
nakki, -ur, 167
nokki, 168
nøt(a), 350, 351
posi, 291
ranga-strúpi, 294
reykur, 269
rógv, 268, 269
roki, 269
rómi, 186
rúka, 294
seil, 243
skankur, 188
skú(g)vur, 283
slag, 314
snáki, 333
snákur, 333
snípa, 247
snípi, 58, 247
snippur, 58, 247
soppur, 224
strútur, 295
sveim, 251
svíri, 252
tagga, 317, 318
tonn, 22
- tummi*, 288
tummil, 288
úr, 87
út, 87
uttan, 87
valmua, 311
- Norwegian**
abbor, *åbor*, 209
auga, 39
au(g)ur (dial.), 209
auka (dial.), 39
aul (dial.), 197
bard(e), 150
bie, 135, 228, 229
bi-fluga (Nn.), 229
borre, 209
bradd (dial.), 150
bredd(e), 149
brodd(e), 149
brosme, 210
dabba (dial.), 327
dabbe (dial.), 211, 212
damb, 151
dape, 211
dave (dial.), 211
dembe (dial.), 151
dep (dial.), 211
depel (Nn.), 211, 212
dope(l) (dial.), 211
dov (Nn.), 212
dove (dial.), 211
drekka, 32
drubba, 94, 106
dubba (dial.), 104, 127, 128
dubbe, 94, 96
dumbe (Nn.), 151
duppa (dial.), 128, 212
duppe, 94, 96, 102
- eista* (dial.), 39
ekorn, 263
enna (dial.), 39
fere (Nn.), 137
galt(e), 212, 213
geit-aul(e) (dial.), 197
gime (dial.), 231
gjeme (dial.), 231
gom(me), 302
grove (dial.), 162
guva (dial.), 104
gylt (dial.), 213
hake, 328
hare, 200
hegre, 266
heigre (dial.), 266
helme (dial.), 162
hette, 304, 306
higre, 266
hjar(r)e (dial.), 81, 165, 166
hjarta, 39
hjasse (Nn.), 165
hjerne, 166
hork, 172
hovda (dial.), 39
hruk(e) (dial.), 268
hyrna (dial.), 39
ikorn (Nn.), *ikorna* (dial.), 39, 263
jase (Nn.), 200, 201
jol (dial.), 197
jura (dial.), 39, 159
kall-hovda (dial.), 39
kaur(e), 277
kjake, 202, 203
kjelk(e), 172, 173
kjem (Nn.), 351
kjok (dial.), 202, 203
kjuke (dial.), 204
klamp, 178

- klepp*, 178
kljå(-stein), 176
klump, 178
knabb(e), 298
knape (dial.), 298
knapp, 298
knarre (dial.), 80, 300
knaus, 300
knubb (Nn.), 297
knupp, 297
knuv, 297, 298
kodd(e), 73, 80, 175
kok (dial.), 203
koke (dial.), 306
koppe, 285
korna (dial.), 39, 174
kott (dial.), 276
kott, 277
kragg, 332
krake, 332
krede (dial.), 345
krjuka (dial.), 113
krubbe (dial.), 179
kukla (dial.), 39
kvann-aule (dial.), 197
kvann-jol, 197
kvit-mo(ge) (dial.), 311
labb (dial.), 59, 309
lafta (dial.), 39
lette-drekka (dial.), 39
ljon (dial.), 182
luma (dial.), 156
lyn, 182
moke (dial.), 278
moro-leika (dial.), 39
mott, 219
mugge (dial.), 278
mukke (dial.), 278
mære (dial.), 219
mår(e) (dial.), 219
nakk(e), 167
nate-hams (dial.), 350
nate-kjerne (dial.), 350, 351
nipla (dial.), 39
nokk(e), 168
nos (Nn.), 208
null(e) (dial.), 170
nykla (dial.), 39
nysta (dial.), 39
oka (dial.), 39
oke (dial.), 195
okla (dial.), 39
pus, 291
reig, 240, 242
rig(g)e (dial.), 103
riga (dial.), 96
rige (dial.), 96
rigge (dial.), 96
ripa (dial.), 94, 107, 108
rjome (Nn.), 186
rjå (dial.), 239, 241, 242, 243
roke (dial.), 269
rotte, 222
ruk (dial.), 268
rømme, 186
rå (dial.), 240, 242
sele, 243
silje (dial.), 243
skank, 188
skine, skjene (dial.), 245
skrede, 345
skuv(e) (dial.), 283
slok (dial.), 314, 315
snage, 333
snaka (dial.), 334
snipe, 247, 248
snipp, 247
snive, 249
snok, 333
snåk (dial.), 333
sopp, 224
stabba (dial.), 110
strop(e) (dial.), 294
strup(e), 294
strut, 295
strype, 295
strøype, 295
stubb(e), 286
stuv(e) (dial.), 286
tagg(e), 317
tamp, 193
tave, 342
tikk(e) (dial.), 254
tikka, 125
tomme(l), 288
toppe, 344
trut, 295
vall-mo(g) (dial.), 311
valmue, 311
ve-hovda (dial.), 39
yrkne (dial.), 214
øyra, 39
åkorn(a) (dial.), 39, 263

Old Swedish

- agh-borre*, 209
balker, 148
bi(-fluga), 229, 230
biæлке, 148
brædder, 149
by, 229
damb, 151, 152
dimba, 152
dimba, dimma, 151
ekorne, 263

galder, 213
gōme, 302
hæggher, 266
hare, hære, 200, 201
heri (OGutn.), 200, 201
ikorne, 263
kærve, 179, 182
kiæke, 202, 203
klimper, 178
kratta, 125
malmber, 184
moghe, 278
niura, 39
ri (OGutn.), 239
sele, sile, 243
skoppa, 95, 96
swamper, 224
tagger, 59, 317, 318
thum, 288
pume, 268, 288
val-mōgha, -e, 29, 134, 136, 311
vrist, 262
ylva, 213
ypin, 89

Swedish

abborre, 209
bia, -e (dial.), 228, 229
bäiä (Gutn.), 228
dabba (dial.), 212
dimba (dial.), 109
dimba, dimma, 151, 152
ekorre, 263
fjas (dial.), 199
fot-bjälle, 39
fös (dial.), 199
gjäim (dial.), 231
gomme, 302

grjopa (dial.), 162
hare, 200, 201
hjelm (dial.), 162
hjärna, 166
hjassa, 165
häger, 266
hätta, 304
kippa (dial.), 234, 235
kjåk (dial.), 202
klamp, 178
klimp, 178
klund, klunn, 177
knagg(e), 301
knap(p)e (dial.), 298
knave (dial.), 298
knös (dial.), 300
kok, 306
koka, 306
kolk (dial.), 172
kotte, 276
krabba (dial.), 330
käke, 202
kälke, 172, 173
labb, 309
malm, 184
mäckå (Gutn.), 278
måuä (Gutn.), 278
nasist, 82
nasse, 82
rie (dial.), 239
rugge, 268, 269
rugä (Gutn.), 268
ruka, rukå (Gutn.), 268, 269
råga (dial.), 70, 268, 269
rätta, 222
räckå (Gutn.), 240
räj (Gutn.), 239, 241
römme, 186
sele, 243

silja, silla, 243
skank, 188
skena, 245
skrat(t)e, 345
skrutt (dial.), 345, 346
slagg(-väder), 314
snok, 144, 333
socialist, 82
sopp, 224
sosse, 82
strupe, 294
stråpe (dial.), 294
svamp, 224
svire, 252
tagg, 317
tamp, 193
tumme, 39, 288
vall-mo(ge), 311
veke, 194
ynn (dial.), 214
älm (dial.), 156
älme (dial.), 156
älmä (Gutn.), 156

Old Danish

ag-borræ, 209
albuæ, 311
f(j)øs, 199
jessæ, 165
kiæge, 202, 203
thumæ, 288
val-mu(gh)æ, 311

Early Danish

drene, 152, 153
egerne, 263
kippe, 234
ljun, 182
tvege, tvige, 256

Danish

aborre, 209
agern, 263
bi, 229
bi-flue (obs.), 229, 230
borre, *burre*, 209
egern, 263
gumme (dial.), 302
hare, 200
hejre, 266
hjerne, 166
hætte, 204
isse, 165
kaje (dial.), 202
kippe, 235
klamp(e), 178
klimp(e), 178
klump(e), 178
klyne, 176
knag, 301
knøs, 300
kulk (dial.), 143, 172
lab(be), 309
ri(e), 239
rotte, 222
rømme, 186
sele, 243
skank, 188
skinne, 245
snibe, 249
snippe, 247
snive, 249
sopp, 224
strube, 294
svamp, 224
tagge, 317, 318
tomme(l), 288
tveg(g)e, 256
valmue, 311
væge, 194

West Germanic**Old English**

ācurna, *ācwe(o)rn(a)*, 263
ǣl-pūte, 280, 281
bærs, 209
bannan, 107
bealca, 148
bearn, 47
beard, 150
bears, 209
bēaw, 230
bera, 61
bīa, 228
bīecþ, *bīehþ*, 220
bīo, 228, 229
blæge, 214
bodan, 65
bolca, 134, 148
bord, 149
borda, 149
borde, 149
botm, 43, 65, 68, 123
brerd, *breard*, *breord*, 150
brord, 149, 150
brunna, 83
būgan, 113
cēace, 202, 204
ceart, 339, 341
cēoce, 202
cēod(a), 175
ceole, 174
cēowan, 206
cian, *ciun*, 207
cīpa, 234, 235
cippian, 235
clæg, 239
clāte, 236, 238
clēat, 133
clēofan, 113

clēowen, 176, 177
clīðan, 238, 239
clīðe, 135, 235
clīfe, 249
climban, 129
clīte, 235, 236, 249
clod, 273
clot, 272
cluccian, 125
clūd, 272
clut, 272
clyne, 177
cnæp, 298
cnafa, 70, 128
cnapa, 70, 73, 128
cnoppa, 136, 144, 297, 298
cnotta, 56, 60, 136, 144, 267, 299
cnyttan, 299
cod, 175
corn, 174
cornuc, 307
crabba, 73, 80
cradol, 339
cræt, 59, 60, 339, 341
crammian, 275
cran(oc), 307
credel, 339
crēopan, 113
crimman, 275
crincan, 108
cringan, 108
crōc, 332
crocca, -e, 321
crōg, 321
crohha, 321
croma, 274
crūce, 321
cruma, 274
cryb, 180
crymmian, 275

- cūðe*, 102
cyren, 174
cyrnel, 174
dīc, 45
dora, 153, 154
dran(e), *dræn*, 153, 154
drēopan, 113
dryge, 47
dūfan, 104
elm, 155
fæs, 199
flēotan, 103
flotian, 103
fogge, 23, 70
friccea, 43
frocca, *frogga*, 133
furh, 137
gealt-bearg, -*borg*, 212
geofen, *geofon*, 231
gēopan, 47
gielde, 213
gifen, 231
gilte, 213
gīw, 245
goma, 302
grēofa, 135, 161
gripu, 161
gropa, 161
guma, 35
haca, 133, 137, 144, 328
hæt, 304
ham, 36
hamar, 165
hand-brede, 151
hara, 200
he(o)fen, *hiofen*, 61, 163, 183
healm, 162
hēap, 271
hearra, 23
hēdan, 306
heden, 61, 304, 305
hēla, 329
helma, 162
hēopa, 203
hleonian, 97
hnecca, 167, 168
hnēopan, 106
hnoc, 168
hnoll, 170
hnoppian, 106
hnutu, 350
hōc, 133, 328
hōd, 304
hofer, 271
hōh, 329
hoppe, 271
hræfn, 73
hræmn, 60
hrāgra, 266
hrēac, 269
hrehhe, 185
hrið, 232
hrið, 234
hridian, 232
hrīm, 60, 66
huntian, 45
hyl, 43
in, 86
læccan, 310
læt, 215
lapian, 107
lēoma, 182
leornian, 101
liccian, 44, 98
lim, 62
lirnian, 48
loc, 42
lōcian, 94
lūkan, 122
maða, 218
mealm-stān, 184
mohþa, *mohþe*, 139, 218, 220
molda, 30
mohþa, *mohþe*, 73, 74, 139, 216, 218
mūha, *mūwa*, 134, 278
nefa, 260
nosu, 208
or-, 87
oxa, 62, 63
padde, 281
piða, 23, 67, 68, 70
poc, 176
pocca, 23, 176
pohha, 23, 74
posa, 291
puduc, 280, 281
ræge, 214
ræt, 222
ræw, 242
rāh(a), 213
rēam, 186
rēama, 186
regn, 73
rēocan, 113
rēodan, 104
rēofan, 104, 283
rēoma, 186
ribb, 78
rīpa, 107, 108
rīpan, 107
roccian, 44, 94, 98
ruhha, 185
sælan, 244
sāl, 244
sc(e)anca, 188
scæp, 45
sceacan, 105, 107
scēaf, 284, 285
sceag(g)a, 316

- scearn*, 48
scēofan, 113
sceoppa, 284
scēotan, 110
scīa, 244, 245
scīd, 246
scīma, 246
scinu, 244, 245
sconca, 188
scrætte, 345, 346
screpan, 109
scrimman, 108
scritta, 344, 346
scūdan, 104
scūfan, 113
scyfele, 285
scypen, 284, 285
sēoc, 254
slincan, 109
slingan, 109
slōh, 314, 315
slūpan, 113
smēocan, 108
smittian, 109, 111
smūgan, 108, 113
snaca, 137, 144, 249, 324, 333, 334
snīcan, 106
snīte, 249
snofl, 292
socian, 98, 108
soppian, 106, 117
spīwan, 245
sprēot, 293
sprēotan, 113
sprot(a), 293
sprūtan, 113
stān-clūd, 272
stefn, 183
ste(o)la, 62
steorra, 23, 56
stician, 105
stingan, 104
stofn, 286
stroccian, 95, 106
stub, 286
styb, 286
sūcan, 108
sūgan, 108
sunna, -e, 83
sūpan, 106, 117
swe(o)r(a), 58, 252, 253
swef(e)n, 46, 122
sweord, 253
sweoster, 253
swīma, 251
swīor, 253
swom, 223
swura, 252
swurd, 253
swuster, 253
tācor, 206, 266
tæcan, 45, 47
tæp(p)a, 121, 341, 342
tættec, 226, 227
telga, *telge*, 190, 191
telgor, *telgra*, 190
þaccian, 44, 98, 123
þēotan, 113
þrindan, 108
þrintan, 108
þrote, 296
þrūtian, 296
þūma, 287
þūtan, 113
þȳmel, 287
þȳwan, 289
ticia, 254
tæðan, 22
tōþ, 22
tungan tulg, 191, 192
tusc, *tux* 22
twæm, 36
twig(a), *twig(g)e*, 58, 60, 256
ūðe, 102
ūder, 157, 158
ufan, 88
ulm-trēow, 143, 156, 185
up, 88
uppe, 42
ūt, 87
wæcnan, 129
wæcnian, 101
wægn, 73
wecca, 194, 195
wēoce, 194, 195, 196
weornan, 48
wiccian, 42
wisnan, 48
wōcig, 196
wrēon, *wrīon*, 262
wrigian, 262
wrist, *wyrst*, 262
Middle English
aquerne, 263
brain-wōd, 167
cart(e), 339
ch(e)oke, 202
cheke, *chieke*, *chik*, 202
clēte, 235
cod, 175
crouchen, 113
dorre, 153, 154
drane, 153, 154
fonke, *funke*, 159
heven, 183
knarre, 80, 300
knorre, *knurre*, 300
lāh, 312
latthe, 74

- levene*, 182, 183
lōve, 309
maðek, 218
mohthe, 218
nol, 170
raie, *raize*, 185
reihe, *rezge*, *righe*, 185
rowe, 171
ruggen, 94, 98
shoggen, 107
skrat(te), 345, 346
snipe, 247, 248
stev(e)ne, 183
tabbe, *tap(p)e*, 342
takke, 318
tavele, 342
teke, 254
tike, 254
toggen, 95, 103, 104
wrāh, 262
wricken, 95, 124
wriggen, 96
English
acre, 73
balk, 148
bass, 209
cart, 339
cheek, 202
choke, 202
cleat, 273
clew, 177
clite, 235, 236
clithe (obs.), 235
clod, 274
clot, 272, 274
clote, 238
cloud, 272
clout, 272, 273
cookie, 82
cot, 276
cot-gare, 276
cotted, 276
cotty, 276
cradle, 339
crook, 321
crook, 332
crote, 349
crouke, 321
crut, 347
dab, 212
dabble, 212
doggy, 82
dor, 153
drane (dial.), 153, 154
drone, 153
duke, 115
elder (dial.), 157, 158
elm, 155
feaze, 199
fozy, 199
funk, 159
fuzz, 199
gilt, 213
gloat, 96
grub, 98, 105
gum, 302
harns, 166
helm, 162
junkie, 82
kipe (dial.), 234
knar, 300
knit, 299
knob, 297
knop, 297
knur, 300
lat (dial.), 215
lath, 215
levin, 182
low, 312
maddock, 218
maggot, 218
malm, 184
maw-seed, 311, 312
mogthe (Scot.), 219
moth, 218
mow, 278
neck, 167
news, 115
picht (Scot.), 220
pith, 67
rat, 222
ream (obs.), 186
ruck (dial.), 268
shack, 316
shag, 316
shank, 188
shaw, 316
shippon, 284
shop, 284
shy (dial.), 244
sick, 254
slat, 218
slough, 314
snob, 292
sole, 244
sump, 225
swab, 107
swamp, 224, 225
swap, 107
tab, 342
tack, 318
tag, 317, 318
tape, 342
tatter, 226
tellow, 191
thimble, 287
throat, 296
throttle, 296
tick, 254
tow, 125
tug, 125
tump (dial.), 192
tyke, 254, 255

udder, 157
wick, 195

Old Frisian

balka, 148
berd, 150
ber-skinze, 188
bobbe, 88
bodem, 65
boppa, 88
bova, 88
buppa, 88
ciāke, 202, 203, 204
driāpa, 113
epen, 89
ers-knop, 297
fan, 92
haka, 328
hap, 271
hasa, 200
himul, *himel*, 163
hlakkia, 42, 93, 104
hnekka, 167
hod, 304
hrūta, 113
hwit, 121
jāder, 157, 158
klat, 273
kloppa, 125
knap, 298
knap(p)a, 70, 128
kriāpa, 113
krocha, 136, 321
lerna, 101
lirna, 101
lūka, 122
nos(e), 208
omma, 36
opa, 88
ova, 88
riāka, 113
rīva, 107, 108

sēla, 244
sil-rāp, 243
skēp(p)ena, 64
skern, 48
skidel, 246
skūfa, 113
skunka, 188
sliāpa, 113
slūta, 113
smakia, 111
smūga, 113
snabba, 249
snavel, 249
sprūta, 110, 113, 293
stapa, 106, 110
stera, 23, 56
strot-bolla, 295
swīma, 251
throt-bol(l)a, 79, 296
thūma, 287
tosk, 22
tōth, 22
tusk, 22
up(pa), 88
wind-sēl, 244
wrigia, 262
wrist, 262

West (Lauwers) Frisian

honk, 169, 329
iik, 265
iik-hoarntsje, 263
jaar, 157
kladde, 238
kret, 339
lange leat, 215, 217
leane, 217
leat, 215, 216, 217
loat, 215
lod(de), 215

pich, *piid*, *piik*, 67, 220
rjemme, 186
robbe, 283
rûp, 282
sile, 243
skonk, 188
skyl, 246
sobje, 106
toake, 317, 318
tosk, 22
tsjeak, 202
tyk, 254
wjuk(ke), 194, 195

Saterlandic Frisian

äil, 154
budde, 280
dafen, 327
droane, 153, 154
dumpen, 127, 128
hunk, 169, 329
juuke, 194
knuufe, 297, 298
krääf, *krääwe*, 24, 179
krouns-bäie, 307
näkke, 167
siele, 243
smugen, 108
sompe, 225
sooke, 202, 203
tabbe, 342
tappe, 342
tieke, 254, 255
todde, 226, 227
twiech, 256

North Frisian

ik, 265
ik-hörn, 263
jüp, 203

kēk (Wdh.), 204
krēk (Wdh.), 325,
 326, 332
krōge (Wdh.), 321
nope, 23
selle, 243
sīk (Wdh.), 202, 203,
 204
skidjel, 246
teg, 254

Old Saxon

ambo, 167
balko, 148
barn, 47
bord, 149
dag-skimo, 246
dran, 154
drano, 153
drukian, 47
drukno, 47
fan(a), 92
geban, 231
hāko, 328
heban, 61, 163
hegiro, 266
himil, 163
hlinon, 97
hod, 304
hōp, 271
hrīpo, 66
in, 86
kleuwin, 176, 177
kot(t), 276
kribbia, 179
likkon, 82
liomo, 182
magon-hōbud, 311
maho, 311
matho, 218
melm, 184

oċan(a), 88
ratta, 222
ribbi, 78
rido, 232
sēl, 243
skatt, 82
skimo, 246
sneppa, 247
strota, 295
stroton, 296
sumar-lada, 215, 216
sumar-loda, 215
swīmo, 251
tōgo, 24, 138, 317
up(p), 82, 88
uppa, 82
uppan, 82, 88
ur-, 87
ūt, 87
wokko, 194, 195, 196,
 316

Middle Low

German

balke, 148
bars, 209
bēne, 229
bleie, 214
borde, 149
brāgen, 47
brassem, 210
brēgen, 47
bres(s)em, 210
būgen, 113
damp(e), 151, 152
dempen, 151
dobbe, 211
drane, 153
**drēne*, 152
drēpen, 113, 114
drone, 153
drūpen, 113, 114

dūken, 110
dumpe, 152
dumpen, 109
ēkern(e), *ēk-horn(e)*,
 263, 264
elm, 155, 156
gedumpen, 152
gelte, 213
grabben, 96
grape(n), *grope(n)*,
 161
gropen, 98, 105
groppe, 135
hame, 81
harne, 165
harte, 181
hase, 200
helm, 162
herne, 165
hocke, 68
hōk, 328
hūpe, 270
jēder, 157, 158
kake, 202
karke, 181
karpe, 181
karve, 179, 181
keke, 202
keken, 202
kerne, 174
kerve, 139, 180
kewe, 207
kip, 234
kīpe, 234, 235
kirne, 174
kladderer, 238
klampe, 178
klatte, 238
klave, 177
kletze, 236, 237
klōt, 273
klove, 177

- klūs*, 177
klūt(e), 272
knagge, 301
knape, 70
knīpen, 106
knobbe, 297
knōp, 298
knorre, 300
knubbe, 297
knutte, 299
kodde, 175
kōke, 306
korf, 181
kran, 307
kranekes-snavel, 307
krēpen, 113, 114
krōn, 307, 308
krume, 274
krūpen, 114
kuddeken, 276
kūdel, 175
kuse, 207
lade, 214
lak, 310
late, 70, 215, 216
lode, 215
med(d)ek(e), 218
mucken, 279
mugge, 279
mūke, 278, 279
mutte, 218
necke, 167
nobbe, 23, 106
nocke, 168
noppe, 23
olm, 156
ped(d)ik, 67
pedde, 281
pit(te), 67, 70
ratte, 222
rede, 232
rege(l), 239
rēger, 266
rēken, 113, 114
repen, 94, 107
rīge, 239
rīm, 66
roche, 185
rogen, 171
rōm(e), 186, 187
rotte, 135, 222, 223
rubbe, 283
ruche, 185
rūken, 114
rūpe, 282
schēdel, 246
scheme, 143, 246
schēne, 244
schenke, schinke, 187, 189
schepen, 64
schobbe, 284
sele, 243
sēlen, 244
skūven, 113
slag(g)e, 314, 315
slaggen, 315
slūpen, 113
slūten, 113
smūgen, 113
smūken, 108
snake, 333, 334
snappen, 249
snaven, 249
snebbe, 248, 249
snep(p)el, 248
sneppe, 247
snibbe, 248, 249
snigge, 106
snippe, 247
snoppe, 292
snoppen, 104
snove, 292
snūf, snūve, 292
snūven, 292
som(m)er-lade, 216
sprēt, 293
sprēten, 113, 114
sprote, 293
sprūte, 293
sprūten, 113, 114
sticken, 105
stoppe, 286
stoppel, 286
streme(l), 250
strīme, 250
strot(t)e, 69, 295, 296
stubbe, 286
stūpe, 286, 287
stūve, 286
stūven, 113
sump, 225
swāien, 252
tack(e), 24, 318
tagge, 24, 59, 138, 317, 318
tant, 22
tappe, 342
tas, 227
teke, 254
telch, 190
telgere, 190
timpe, 24, 192, 193
tobbe, 344
tōch, 317
toddelen, 226
tumpe, 192
twīch, 256
wark, 181
webbe, 195
wecke, 194, 195
wēke, 194
wepse, 195
wobbe, 195
wocke, 194, 195

wocken-blat, 194
wopse, 195
wricken, 262
wriggen, 95, 96, 124,
 262

Old Low Franconian

skepeno, 64

Middle Dutch

ba(e)rse, 208, 209
balk(e), 148
barne, 153
bedompen, 152
bleie, 214
bocken, 102
borne, 153
brasem, 210
bressem, 210
bügen, 113
cladden, 239
clamp(e), 178
classe, 238
clatten, 239
classe, clisse, clitte,
 74, 76, 236, 237
clompe, 178
clos(se), clot(te), 272,
 273
clouwen, clu(w)en,
 176
cnoop, 298
cnop, 297
cnor(re), 300
cnovel, 297
codde, 73, 80
coeke, 306
corf, 180
crade, 349
craen, cran(e), 307
crappe, 330
cratte, 339

crode, 347
croegh, 321
crome, 274
cruke, 321
crume, 274
cudel(e), cuil, 175,
 176
dabben, 212, 327
damp, 151, 152
darne, dorne, 153
dieten, 113
dobben, 94, 96, 99
docken, 110
domp, 151, 152
drüpen, 113
druppen, 94
dubben, 128, 212
ducken, 110
duken, 110
dume, 287
duwen, 289
ee(n)coren, 263
gelte, 213, 214
grapen, 96
grobben, 98, 105
groop, grop(p)e 161
hake, haek, 328
harsen, 166
hase, 200
hēger, 266
heiger, 266
helm, 162
herne, 165
hersene, 166
hiele, 329
honc, 169, 329
hūken, 110
kake, 202
keke, 204
kieuwe, 207
kijp, 234, 235
klieven, 113

kloot, 273
klūven, 113
knape, 70
knoppe, 59, 297
knutte, 299
kratte, 59
krūpen, 113
lade, 214
latte, 215
lode, 215
lote, 215, 216
matte, 218
melm(e), 184
mol(e)m, molle(n),
 184, 185
molen, 185
molsem, 185
mot(te), mutte, 218
mūke, 278
necke, 167
nol(le), 170
noppe, 23
olme, 156
ont-fenken, 159, 160
ont-vengen, 160
pit(te), 67, 70
podde, pudde, 280,
 281
puut, 280, 281
ratte, 222
rede, 232, 233
rēger, 266
regghe, 239, 240
reghel, 239
reiger, 266
repen, 94, 107, 108
ridde, 232, 233
rie, 239
rige, 239
rigghe, 239, 240
rijen, 241
rīm, 66

- rīp(e)*, 66
robbe, 283
roc, 24, 70, 136, 143, 269, 271
roche, *rogghe*, 185
rog(g)e, 23
roge, 171
room, *rome*, 186, 187
roppen, 44, 98
rot(te), 222, 223
rūten, 113
scheme, 246
schene, 245
schepen(e), 64
schime, 246
schinke, 189
schove, 284
schrabben,
 schrapp(en),
 schraven 109
seel, 243
skūven, 113
slac(ke), 315
slec(ke), 315
slegge, 315
slūpen, 113
slūten, 113
smūgen, 113
smūken, 108
snappen, 249
sneppe, *snippe*, 248
snoek, 333, 334
snof, 292
snop, 292
snoppen, 292
snūven, 113, 292
soken, 98
somer-lade, *-late*, 216
somer-lode, 215, 216
somp, 225
sporte, 293
spouwen, 258
spriet, 293
sprote, 293
sprute, 293
sprūten, 113
starte, *sterte*, *storte*,
 153, 296
stobbe, 286
stoof, 286
stoppe, 286
stoppel(e), 286
stove, 286
stricken, 44, 98
strieme, 250
strop(p)en, 110
strot(t)e, 296
stubbe, 286
stūpe, 286, 287
stūven, 113
tac(ke), 24, 136, 318
taken, 98, 104
tant, 22
tap(pe), 342
tas, 228
tel(e)ch, *telgh(e)*, 190
timp(e), 192
to(c)ken, 95, 96, 104,
 125
token, 103, 125
tooch, 317
top, 344
trecken, 109
treken, 109
trocken, 109
twijch, 256
uder, 157, 158
vas(e), 199
vese, 199
vliegen, 108
vlieken, 108
vlot(t)en, 103
vocken, 109
vonke, 159
vonken, 160
wieke, 194
wocke, 195
wouwe(r), 258
wouwere, 258
wrijch, *wrijf*, *wrijghe*,
 261, 262
Early Dutch
ael-puyt, 280
be-mullen, 185
cnorre, 300
dabbelen, 212
dabben, 212
dampen, 151
dom, 288
domp, 151
dompen, 152
doppen, 104
fijcken, 113
fuycken, 109, 113
grape, *grope* (Sax.),
 161
gumme (Sax.), 302
helm, 162
herssen, 166
herssen-woedig, 167
hobbe, 270
hobbel, 270
kauwe, 207
kieme, 250
kijme, 250
kladde, 238
klijt(e), 236
klodde, 273
kloet, 272
kloot, 272
klos, 273
klot(te), 272
kluyksen, 177
kluyte, 272
knodde, 299

- kodde*, 175, 176
kossem, 176
kouwe, 207
krodde, 347
kronken, 108
krotte, 349
kuyse, 207
labben, 98, 99, 107
laede, 214
lappen, 44, 98, 107, 123
latte, 215
loef, 309
loeve, 309
loote, 215
maen, 311
molm, 184
nocke, 168
pee, *peën*, 67, 68
pette, 67
pit(te), 23, 67
puyt-ael, 280
quabbe, 281, 349
red(d)e, 232
reghe, 239
reppen, 94, 107
ridde, 232
rijchel, 239
rijde, 232, 233
rijghe, 239, 240
ritse (NRhnl.), 232, 233
ritsigh, 232
robbe(ken), 283
roch, 185
rock, 269
ruype, 282
scheene, 245
schemel, 246
schie(de)r, 246
schobben, 95, 96
schoppen, 95, 96
sleckke, 315
slegghe, 315
snabben, 96
snebbe, 248
sneppe, 248
snof, *snuf*, 292
snuyfelen, 292
sproten, 110, 293
streme, 250
stroocken, 95, 106, 125
swamme, 223
tacken, 104
teecke, 254
telghe(r), 190, 191
wiecke, 194
wikkelen, 196
woack, 195
wocke, 195
wreken, 95
wrijf, 261, 262
Dutch
adem, 68
asem (dial.), 68
baars, 209
balk, 148
bij, 228, 229
blijken, 129
blinken, 129
blom, 251
brasem, 210
censuur, 114
dabbe (dial.), 211
damp, 151, 152
dar, 153, 154
dempen, 151
dobbe (dial.), 211
dol, 166
dop (dial.), 211
drene (dial.), 152, 153
dri'jen (Stw.), 252
droog, 47
drubben (dial.), 94
druipen, 114
duim, 217, 287
duimelot, 215, 217
duipen, 128
eek-hoorn(tje), 263, 264
elder (dial.), 157, 158
gewricht, 262
grabben, 96
gribbe(le)n, 109, 111
haak, 328
haas, 200
haoke (Stw.), 325, 328
helm, 162
hersen, -s, 166
hiel, 329
hobbel, 270
hoek, 328
honk, 169, 329
hoorn, 166
hoorn-dol, 166, 167
hoorn-woedig, 166, 167
jaar, *jadder* (dial.), 157
jagen, 96
jakken (obs.), 93, 96
kaak, 202, 204
kaakje, 306
kib(be), 234, 235
kiem, 250
kieuw, 207
kladden, 239
klei, 239
klijt, 236
klijte (Flem.), 236
klis, *klit*, 68, 236, 237
klits (dial.), 237

- klodder*, 273
kloede (dial.), 272
kloen (dial.), 176
klomp, 178
kloot, 273
klos, 273
kluwen, klouwen (dial.), 176, 177
knaag (dial.), 70, 301
knaak, 70, 301
knag(ge) (dial.), 70, 301
knar, 300
knippen, 106
knor, 300
kobbe (Flem.), 285
kodde, 175, 276
koek, 306
kok-halzen, 203
korf, 180
kossem, 176
kraan-vogel, 307
krabben, krappen, 331
krat, 339
kraven (dial.), 331
kret (dial.), 339
krib(be), 179
krod(de), 347
kroene-krane (dial.), 308
*krui*k, 321
kruim(el), 274
kruipen, 114
kuif, 285
kuil, 175
kwab, 281, 349
land-rot, 222
langelot, 215, 217
lat, 215
loef-zijde, 309
loot, 215, 216
lotten, 217
maan (dial.), 312
maan-zaad, 311
made, 218
mi'jen (Stw.), 252
mikken, 44, 98
mok (obs., dial.), 278, 279
molm, 184
mot, 218
mugge (dial.), 279
muik, 279, 280
nek, 167
neus, 208
nok, 168
olm, 184
pad, 281
padden (dial.), 282
pedde (dial.), 281
peem, 67
peen, 67, 68
pessem (dial.), 67, 176
pettem (dial.), 67
pit, 67
podde (dial.), 280
poede (dial.), 280
poo(i) (dial.), 280, 281
poon, 280
*pudd*e (dial.), 280
puid (dial.), 280, 281
puut-aal, 280, 281
putte-kol (dial.), 280, 281
reeg (dial.), 239
reiger, 266
richel, 239
rieken, 114
riet, 217
rij, 239, 240
rijm, 66
rijp, 66
ritsig, 232
rob, 283
rog, 185
rook, 269
room, 186
ruiken, 114
ruip (dial.), 282
schape-tijk (dial.), 254, 255
scheen, 245
schenk (dial.), 187
schenkel, 188
schepen, 64
schier (Flem.), 246
schijmel (dial.), 246
schink(e) (dial.), 187
schoggen (Limb.), 107
schonk, 188
schoot, 217
schop (dial.), 284
slak, 314
slegge (dial.), 315
sloek (dial.), 314, 315
slokken, 110
smokken (obs.), 108
snappen, 96
sneb, 248, 249
sneep, 248, 249
snip, 248
snobben (dial.), 292
snoepen, 249
snuiven, 292
soppen, 106, 117
sport, 293
spriet, 293
spruit, 293
spruiten, 114
storre (dial.), 190
striem, 250
stroot (dial.), 296

- strot*, 296
stuiten, 104
tak, 318
tamp, 143, 193
tand, 22
teek, 254
teen, 281
telg, 190
tepel, 193
tijg (dial.), 135, 254, 256
tijk (dial.), 255
timp, 192
tod(de), 73, 80, 226, 227
todden (dial.), 228
toege (dial.), 138, 317, 318
toeke (dial.), 24, 138, 317, 318
tolk, 191
tomp (dial.), 192
toog (dial.), 317
tooien (dial.), 228
tump(e) (dial.), 24, 192
twijg, 256
ui, 154
vezel, 199
vonk, 159
wiek(e), 194
wit, 44
wouw, 258
wree (dial.), 261
wreef, 261, 262
wreeg, *wrege* (obs., dial.), 135, 261, 262
wrijg, 261
wrijven, 262
wrikken, 96, 103, 262
zeel, 243
zijl(e) (dial.), 243
zomer-lat, 216
zomer-lot, 216
zwaaien (Stw.), 252
zwam, 223
zwijm, 251
- Afrikaans**
klits-gras, 237
- Old High German**
ab(a), 92
aftirnel, 169
ahta, 93
ahtōn, 93
ancho, 32
bachan, 109
bachōn, 109
bahhan, 109
balcho, 148
barn, 47
bars, 209
bart, 150
bero, 61
bīa, 228, 229
bian, 229
bina, 229
bini, 229
biogan, 113
bodem, 65, 66
bolcho, 148
bort, 149
borto, 134, 143, 149
brahsa, 210
brahsma, 210
brart, 143, 150
brehsa, 210
breta, 151
brit, 151
brort, 149
brunno, 83
channa, 224
chanta, 224
chela, 174, 206
chelah, 172, 174, 206
chern, 198
chīmo, 250
chiot, 175
chiuwa, 207
chiuwan, 206
chizzi(n), 230
chledda, 236, 237
chleddo, 236
chlenan, 239
chleta, 235, 238
chletha, 74, 235
chletta, 76, 135, 236
chletto, 77, 236
chlimban, 109
chliuwa, 176
chliuwi, 176
chlōz, 144, 273
chnabo, 70
chnappo, 128
chnodo, 45, 56, 60, 267, 299
chnopf, 59, 297
chnotho, 267
chnoto, 56, 267, 299
chorb, 180
chorn, 174
chotzo, 276
chowe, 207
chrācco, 137, 140, 144, 326, 331, 332
chracho, 332
chrāff(o), 330
chranih, -oh, -uh, 307
chrano, 307
chrāpfo, 140, 326, 329, 330
chrāppo, 140, 330
chratto, 70, 133, 335, 338

- chresan*, 82
chresso, 82
chreta, 133, 346
chretto, 335-340
chretzo, 70, 134-135, 335-338, 340, 343
chripfa, 179, 181
chrippa, 179, 181
chrot(t)a, 133, 346, 347
chruog, 136, 321
chuman, 115
chuohho, 306
chupfa, 285
chuppa, 285
churb, 180
dampf, 151, 152
dampfo, 152
dempfo, 152
deni-chleta, 235
diozzan, 113
dona, 217
drozza, 296
dūhen, 289
dūmo, 268, 287
duruh, 91
eban, 48
eihhorn(o), *eihhurno*, 263
elm(o), 143, 155, 156
elm-boum, 155
fan(a), 92
fasa, 199
faso, 199
felt, 336
fesa, 199
flecho, 133
flocho, 133
fon(a), 92
fuir, 158
funcho, 159
galt, 213
galza, 212, 214
gast, 336
gelza, 213, 214
gingēn, 129
gomo, 35
goumo, 302
**guogo*, 338
guomo, 302
hācco, 133, 137, 140, 141, 142, 144, 325, 326, 328
hadara, 306
hāhala, *hāhila*, 329
hāho, 327
halm, 162
ham(m)a, 36, 81, 163
hamar, 165
hano, 59, 309
hasan, 202
haso, 200, 202
heigar, 266
heigiro, 266
herza, 39
himil, 163, 164
hirni, 165
hiufa, 158
hiufo, 203
hnach, 167
houf, 271
hovar, 271
hreigir, 266
hūfo, 270
humel, 163, 164
hunno, 224
huohila, 329
huon, 309
huot, 304
huoten, 306
ilma, 155
in, 86
īn, 86
int-rīhan, 262
jagōn, 96
klioban, 113
krāhho, 332
lad(d)a, 135, 214, 216
lado, 135, 214
laffa, 129, 133, 134, 309, 310
laffan, 107
lappo, 129, 136, 309, 310
lat(t)a, 214, 215, 216
letz, 70, 215
lechōn, 120, 122
lēhan, 46
lernōn, 101
liohhan, 122
lochōn, *lohhōn*, 98, 103
luogēn, 94
mado, 218
mago, 134, 136, 311
maho, 134, 136, 311
melm, 184
nachot, *nachut*, 263
naht, 336
nasa, 208
nel, 169
nevo, 260
nilla, 169
nuz, 350
ob(a), 88
obana, 88
offan, 89
olmoht, 184
ōra, 39
ouga, 39
ovan, 46
pfoso, 291
raban, 47, 60, 171
rabo, 47, 60, 171
radda, 221, 222

- ram*, 60
rato, 135, 221
ratta, 221, 222
ratza, 222
redan, 105
regan, 48, 73
rēh(o), 213
rēia, 214
reigar, 266
retzōn, 109
rīdo, 74, 134, 135, 232, 233
rīdōn, 232, 234
rīffo, 66, 67
rīga, 239, 240
rigil, 239
rīho, 261
riohhan, 113
riozan, 106
rit(t)o, 77, 135, 232, 233
ritzōn, 93, 109
riuhhan, 109
rogan, *rogo*, 47, 60, 171
ruowa, 48
rūzzan, 113
scahho, 316
sceffin(o), 64
scelo, 123
scena, 244, 245
scepfō, 64
scīmo, 246
scina, 244
scincha, -o, 187, 188
scīt, 246
sco(p)f, 284, 285
scorro, 58
scoup, 284, 285
scrat(t)o, 344, 345
scratz, 345
scretz(o), 344, 345
scrota, 345, 346
scubil, 285
scūfla, *scūvala*, 66
scuft, 285
silo, 243
skioban, 113
sliozzan, 113
sliufan, 113
slopfāri, 111
slucho, 120
smiogān, 113
snabul, 249
snācco, 325
snahhan, 334
snepfa, -o, 247, 248
snoffizen, *snopfizen*, 292
sparro, 62
spriozzan, 113
sprozzo, 293
stafel, 110
stapfōn, 106, 110
stechōn, 104
stehhan, 104
sterno, 174
stero, 189
sterro, 23
stioban, 113
stopfela, 286
stornēn, *storrēn*, 97
stotzōn, 104
strīmo, 250
stunchōn, 104
stupfula, 286
sumar-lata, 215, 216
sumar-lota, 135, 215, 216
sunno, 136
swam, 223
swamp, 223, 224
swirōn, 252, 253
tempfen, 151
tiof, 141
tocha, 23, 69
treno, 152
tretan, 104
trettōn, 104
triufan, 113
troffo, *tropfo*, 106
ūf, 88, 90
ūffana, 88
ur-, 87
ūter, 157, 158
ūtro, 157
ūz, 87
wagan, 48, 73
wanga, 39
weho, 258
wella, 43
weval, 66
wewo, 258
wī(w)o, 258
wioh, *wiohha*, 194, 195, 196
wipfōn, 108
wīz, 141
zagal, 317, 319
zan(t), 22
zapfo, 335, 341, 342
zata, 225, 227
zato, 225
zatta, 225, 227
zecho, *zehho*, 254
zeihhan, 47
zepfo, 335
zinna, 224
zochōn, 95, 120
zogōn, 95, 104
zopf, 344
zota, 226, 227
zumpo, 24, 192
zuogo, 24, 136, 258, 317, 319, 326
zwech(o), 256

zwīg, 256
zwirn, 48
zwirnōn, 48

Middle High

German

ag, 209
balke, 148
bars, *bers(e)*, 209
bīe, 228, 229
bīn(e), 229
bort, 149
brart, 150
brort, 149
crade (Rhnl.), 349
crede (Rhnl.), 346
dāhe, 154
dampf, 151
dempfe, 152
dempfen, 151
dimpfen, 109, 152
doum, 288
doume, 287
doumen, 288
drozze, 296
dumpfe, 152
dümpfen, 151
eich-horn, 263, 264
elm-boum, 155
fliegen, 103
flocken, 103, 108
fochen, 109
funke, 159
galze, *gelze*, 212, 213
goum(e), 302
gripfen, 109
grop(p)e, 161
gumme, 251
guome, 302
gupfen, 104
hā(c)ke, 328
hader, 306

hamme, 81
hangen, 45
hase, 200
hatele, 306
heiger, 266
hirn(e), 165
hirn-wüetec, 166
hō(c)ke, 328
hover, 271
hūfe, 270
ilme, 155
iuter, 157, 158
kanne, *kante*, 224
karb, *karp*, 180, 181
keibe, 234
kelch, 172
kerne, 174
kewe, 207
kieme, 250
kīme, 250
kiutel, 176
kiuwe, 207
klampe, 178
klete, 235
klimme, 178
klimpfen, 109, 178
kliuwe(l), 176, 177
klæzen, 273
klotze, 59, 133, 134, 144
kloz, 59, 272
klōz, 273
klūde, 144, 272
klumpe, 178
knapfe, *knappe*, 70, 75
knode, 267
knorre, 300, 301
knote, 267
knouf, 144, 298
knūr(e), 300
korb(e), 24, 180, 181

kotz(e), 276
kotzeht, 276
kouwe, 207
krage, 332
kragen, 333
kran(e)ch(e), 307
krāpe, 330
kräpfe, 324, 329
krate, 349
kratte, 338
krebbe, 24
krebe, 24, 134, 179, 180
kren(i)ch, *kreneche*, 307
kreppe, 179
krete, 346
kretze, 338
kripfe, 179
kroppe, 179
kröte, 346
krotte, 347
krüche, 321
krume, 274
krümmen, 275
kruon(e), 307, 308
krupfe, 24, 134, 139, 179
kruppe, 24, 179
küte, 275, 277
lade, 70, 214, 216
laffe, 309
lat(t)e, 70, 214
made, 70
magen, 311
mahen, *mān*, 311, 312
matte, 70, 139, 218
melm, 184
miete, 218
mocke, 134, 279
motte, 218, 220

- mū(l)-, molt-werf(e)*, 278
mūche, 279
mutte, 218
nacht, 336
nel(le), 169
nol, 170
pedde (MRhnl.), 281
pfose, 291
pfūsen, 291
radde, 221, 222
rat(t)e, 221, 222
ratze, 222
ric, 240, 243
riche, 261
rīden, 232
rīhe, 239, 261
riste, 262
rit(t)e, 74, 232, 233
rocken, 44, 98
rog(g)e, 23
ropfen, 104, 283
roum, 186
ruchen, rucken, 94
rūp(p)e, 282
rupfen, 44, 98
ruppe, 282
schache, 316
scheffene, 64, 65
scheim, 143, 246
schel(l)e, 123
scheme, 246
schepfe(ne), schepfen, 64, 65
schīden, 246
schīe, 244
schīm(e), 246, 247
schin(e), 244
schinke, 187, 188
schnäcke, 333
shocken, 105, 107
schopf(e), 284
schopfen, schoppen, 104
schor(re), 58
schrat(e), 344
schraz, 345
schretel, 344
schreven, 109
schrimpfen, 108
schrumpfen, 108
schuff, 284
seilen, 244
sil(le), 143, 243
slāt, slōt, 218
slāte, 218
slūchen, 110
sluoche, 314
smucken, 108
snaben, 96, 249
snäcke, 333
snepfe, 247
snitzen, 42, 93
snōcke, 333
snūfen, 104, 113, 292
snupfe, 292
spriezen, 110, 293
sproz(ze), spruz(ze), 293
ster(e), sterre, 189, 190
stocken, 105
streim(e), 250, 251
striefen, 110
strīme, 250
strozze, 296
strūben, 295
strupfen, 110
stupfe, 286
stutzen, 44, 123
sumer-lat(t)e, 214, 216
sumpf, 225
swamme, 223
sweim, 251
swir(re), 58, 252, 253
tampfe, 152
tāpe, 140, 327
trene, 152
trot(t)en, 104
trumpfen, 110, 111
ulm-boum, 156
ūter, 157
vanke, 159
vase, 199
venken, 159, 160
vesel, 199
vinc, 159, 160
**vinken*, 160
vude-nol, 170
vunken, 160
wacken, 104
wagen, 104
webel, 66
wehe, wewe, 258
wī(w)e, 258
wickeln, 196
wieche, 194
wīfen, 108
zacke, 318
zāfen, 344
zan(t), 22
zat(t)e, 225, 227
zeche, zecke, 121, 135, 254
zelch, zelge, 190, 191
zenden, 22
zepfe, 341
zettel, 226
zolcher, zolker, 191
zoll, 191
zot(t)e, 226, 227
zoten, 228
zump(e), 192, 193
zumpf(e), 135, 143, 192, 193

zūpe, 141

zwec, 256

zwīc, 256

German

aal-putte, 280, 281

Aal-raupe, 282

Acker, 73

auf, 90

aus-rotten, 104

auter (Bav.), 157,
158

*bach*ⁿ (Swab.), 109

Barsch, 209

Baum, 335

Baum-klette, 236

beie (dial.), 228, 229,
245

bein (Bav.), 229

bibo (Cimb.), 258

Biene, 135, 229, 245

bine (Swab.), 229

Boden, 65

Brett, 151

Dape, 327

Dappe, 327

darf (Bav.), 181

Daumen, 287

Dohne, 217

Drohne, 153, 154

Drossel, 296

dum (dial.), 288

Eichhörnchen, 264

ein, 86, 90

Euter, 157, 158

Faser, 199

fēse (Car.), 199

Feuer, 158

finkeln (obs.), 160

**finken*, 159

Funken, 159

gàlt (Car.), 213

galz (Bav.), 212

Galz(e), 212

Gaumen, 302

Gehirn, 165, 166

gelt (Fra.), 214

gelte (Fra.), 214

Gelze, 213

genagge, gnaggn
(Tyr.), 167

Genick, 167

gomme (obs.), 302

gratto (Cimbr.), 339

grappen (LG), 96, 99

grippen (Als.), 109,
111

Groppen, 161

gummen (Pal.), 302

Hader, 306

Haken, 141, 328

hāken (WPhal.), 325

hår (WPhal.), 325

Hase, 200

Haufen, 270

Himmel, 251

Hirn, 165, 166

hirn-toll, 167

hiuken (WPhal.), 110

hobbe (LG), 270

hock (Tyr.), 68

hocken, 42, 110

hocken (Cimb.), 110

hōk (Rhnl.), 325

hōkə (Als.), 328

hōx (Rhnl.), 325

Hunke, 169

hüpfen, 120

huppe (Tyr.), 270

kake (dial.), 202

karb (Bav.), 181

Käu, 207

Kauder, 277

Kaute, 275, 277

kauzen (Bav., Swab.),
141, 275, 277

Keim, 250

Keipe, 234

keke (LG), 202

Keutel, 175, 176

Kiepe, 234, 235

kīpe (WPhal.), 234,
235

kipfen (dial.), 235

klatteren (Swab.),
238

Kleise, 237

Klette, 236, 237

kletz, 238

kletze? (Tyr.), 236

Kließe, 235, 238

kliezen (Rhnl.), 274

klimpe (LG), 178

klitz (dial.), 236, 237

Kloß, 273

Klotz, 272

Klumpen, 178

klüte (Hess.), 133,
134

knabe (dial.), 298

knarre (LG), 300, 301

Knäuel, 176, 177

Knauer, 300

knaupe (Swab.), 136,
141, 142, 144, 297,
298

knaus (Swab.), 300

kneip(eⁿ) (Swab.),
141

Knopf, 297

Knorre(n), 300

Knoten, 299

knotto (Cimbr.), 299

Koder, 176

kommen, 251

Korb, 180

- korba* (Cimbr.), 180
Kotze, kotzen (dial.), 276
Krack, 137, 144, 332
kradde (WPhal.), 349
krāgeⁿ (Als.), 331
krāke (Pal.), 331
Kranich, 307
Krape, 330
krāpe (Pal.), 330
krāpfe (Pal.), 329, 330
Krapfen, 330, 331
krappe (Pal.), 330
krat (Rhnl.), 349
Kratte, 338
kratteⁿ (Swab.), 339
krätteⁿ (Swab.), 338
Kratz (Lux.), 349
Krätze, 338
krätz^e (Swab.), 338, 341
Krebe, 179
kreb^a (Swab.), 179
krett (Als.), 347
Krippe, 179, 181
kröpe (Pal.), 330
kröt (Lus.), 348
krot^e, kröt^e (Swab.), 346, 348
krott(eⁿ) (Als.), 347
krotten, 349
Krotz, 347
krotze-mann (Loth.), 347
kroune (LG), 307
krōwe (Pal.), 330
Krug, 321
krune-krane (LG), 308
Krupfe, Kröpfe, 179, 181
Krüppe, 180, 181
kruppen (Cimbr.), 106
krutte (Rhnl.), 347
kuddel (Rhnl.), 277
kudden-tol (Rhnl.), 276
kützche (Rhnl.), 275, 277
küz (Rhnl.), 275, 277
kxroute (Zarz), 347, 348
Laden, 214
lappeⁿ (Als.), 309
Latte, 214, 216, 217
latz(e) (dial.), 215
lock (Cimbr.), 122
locken, 102
Lote, 215, 217
mago (Cimbr.), 311, 312
Mauke, 279
Maul-wurf, 278
Miete, 218
Milch, 191
Mocke, 279
Mohn, 311, 312
Mond, 154, 312
mulm (dial.), 184
Nacht, 336
Nacken, 167
Näsling, 249
(n)élle (Car.), 170
nock (Tyr.), 168, 169
nocke (LG), 168
Nollen, 170
paia (Cimbr.), 228
pedde (WPhal.), 281
pfaupe (Swab.), 280, 281
pfitze (Fra.), 67
placken, 119
plagen, 119
puddek (LG), 280
Rahm, 186
raifo (Cimbr.), 66
raim (Cimbr.), 66
Ratte, 221, 222
Ratz, 222
ratze (Bav.), 222, 223
Raupe, 141, 282
Reck(e), 240
reffen (Pal.), 107, 108
Reif, 66
Reihe, 239, 240
Reihen, 135, 261
Reiher, 266
ribben (LG), 94, 108
Rick, 240, 243
Ricke, 213
ricke (dial.), 240
Riege, 239, 240
Riegel, 239
rih(əⁿ) (Als.), 261, 262
Ritte(n), 232, 233
**Ritze*, 233
ritze-rot (Swab.), 232
ritzig (dial.), 232
Robbe, 283
Rogen, 171
roppe (Thur.), 282
roppen (Als.), 283
rotzen, 106
rucken (Cimb.), 109
Ruppe, 282
schaffen, 64
schanke (LG), 188
Schaufel, 66
schaupe (Pal.), 282, 284
Scheie, 244, 245
Schemen, 246
Schiene, 244, 245

- schinke* (Car.), 187
Schinken, 187
schinko (Cimb.), 187
Schlack, 314, 315
Schlacke, 314
Schlot, 218
Schluche, 314, 315
schlucken, 110
Schnabel, 249
Schnake, 333
Schnäpel, *Schnepel*, 248
schnauben,
 schnaufen, 292
schnaupe (Pal.), 141
Schnauze, 141
schnauzen, 142
schnecken, 106
Schnepf, 247
Schnepfe, 247
Schneppe, *Schnibbe*, 248
schniefen, 113, 292
Schnippe, 248
schnöke (Als.), 333
schnupfen, 292
Schnupfen, 292
Schober, 285
Schöffe, 64
Schopf, 284
schöpfen, 64
schotten, 104
Schrat, 344
Schretz, 344
schritzen (Bav.), 109, 111
schunkchn (Bav.), 188
schunke (LG, Car., Swab.), 188
Schupfe, 284
Schuppen, 284
schutzen, 110
Schwier (dial.), 252
Schwir(re)n (dial.), 252
Seil, 143, 243
Seilen, 143, 243, 244
Siele, 243, 244
Sille, 243
snitzen, 103
Sommer-lat(t)e, 214, 216, 217
Sommer-lot(t)e, 216, 217
Spross(e), 293
Stär, 189
Stärke, 190
stochen, 105
Stoppel, 286
Storre, 190
Strieme, 250
strosse (Rhnl.), 296
strupfen, 295
struppig, 295
tachter (Bav.), 181
taddel (LG), 226
take (LG), 317, 318
Tape, 327
Tapfe, 327
Tappe, 327
tappen, 212, 327
telg (Rhnl.), 191
tief, 141
Ton, 154
Tran, 153
tricken (Bav.), 47
trocken, 47
Truhe, 23
twick (WPhal.), 257
twiak (WPhal.), 257
twig (LG), 256
Ulm, 184
Ulme, 156
wart (Bav.), 181
Weihe, 258
weiß, 141
Weizen, 142
wicke (dial.), 194
Wieche, *Wieke*, 194
Wocken, 194
Wolke, 191
Zacke(n), 318
Zahn, 22
zäk(eⁿ) (Swab.), 324, 326
zambel (Pal.), 193
zampf (Zarz), 193
Zapfen, 341
zapfeⁿ (Als., Swab.), 341
zapfen (Bav.), 341
zapfn (Tyr.), 341
zapfo (Visp.), 341
zappe(n) (Bav., Pal.), 342
Zatte, 225
Zattel, 226
zaufen, 108, 116
Zaupe, 141
zecho (Cimb.), 125, 255
Zecke, 254
zecko (Cimb.), 254, 255
Zelge, 190, 191
zepfä (Bav.), 338, 341, 344
zepfe (Tyr.), 341
zepfen, 342
Zettel, 226
zetten, 227
Zimpel, *zimpel* (Pal.), 192, 193
Zimp(en), 24, 135, 143, 192, 193

- zobeln* (dial.), 108, 344
Zolch, 191
Zopf, 344
zopfe (Tyr.), 344
Zot(t)e, 226
Zottel, 226
Zotter (Swab.), 226
zotze (Swab.), 226
zotzlen (Swab.), 226
zoute (Tyr.), 226
zucken, 103
zueggn (Tyr.), 317, 318
zulch (Hess.), 191
Zump, -e(n), 192
Zumpf, 192
Zungen-zolch, 192
zupfen, 108
zutzn (Tyr.), 226
Zweck(e), 256, 257
Zweig, 256
Zwick(e), 60, 256, 257
Zwickel, 257
- Swiss German**
bäix (Visp.), 189
bijji (Visp.), 228
biili (App.), 228
daappo (Visp.), 327
dəppə (App.), 140
düümo (Visp.), 287
fəəsə (Rhtl.), 199
gaarto (Visp.), 337
galz, 212
gniippə (App.), 141
haacko (Visp.), 140, 328
hällfa (Visp.), 158
heeli (Visp.), 329
heenkxə (App.), 45
həockə (App.), 328
hubol (Visp.), 270
hüüfo (Visp.), 270
k(x)uuder, 277
kipf, 234, 235
krää(n)tsə (App.), 338
krapfə (Ja.), 330
kuuz, 275, 277
kuuzig, 275
kxrət (App.), 347
magə (App.), 338
mauch, 280
namo (Visp.), 337
palko (Visp.), 337
reəhə (Rhtl.), 261
reijo (Visp.), 239
reicku (Visp.), 45
ruum(me), 143, 186, 187
šeixo (Visp.), 187
šeixu (Visp.), 189
šijja (Visp.), 244, 245
silo (Visp.), 243
**šnaacko* (Visp.), 333
šnaacku (Visp.), 333, 334
šnəockə (App.), 333
šprotza, 293
štriimo (Visp.), 250
suckə (App.), 44, 98, 108
sugə (App.), 98, 108
šuppo (Visp.), 284
šwiro (Visp.), 252, 253
təppə (App.), 327
toxxa (Visp.), 23, 69
tree (App.), 152, 153
trənə (Ja.), 152
triixu (Visp.), 189
trukxa (Ja., Val.), 23
uuttər (App.), *üütter* (Visp.), 157, 158
xlätta, 236
xlüüxji (Visp.), 176
xnədə (Ja.), *xnodo* (Visp.), 299
xnuupa (Bern), 297
xnuus, 300, 301
xraapfo (Visp.), 329, 330
xrep (App.), 179
xripfa (Visp.), 179, 181
xrotta (Visp.), 347
xüöxo (Visp.), 306
zaffo (Val.), 341
zäpfə (App., Rhntl.), 341
zäpfə (Ja.), *zäpfo* (Visp.), 338, 341, 343
zäxxo (Visp.), 254
zeixu (Visp.), 45
zettu (Visp.), 227
zolgge, 191
zöukx, 142
zwäck (Visp.), 257

2) Other Indo-European languages

Hittite

anda, 91
andan, 91
appa, 91
appan, 91
dāi, 30
ējan-, 32
ēyan-, 32
ḫanza, 86
ḫenkan-, 32
karza(n)-, 33
katta, 91
kattan, 91
kuššan-, 32
lagan-, 32
mūdan-, 32
neḫaḫḫi, 102
parā, 91
paḫḫur, 320
pēran, 91
šaḫḫan-, 32
šāk(a)n-, 32
šarā, 91
šēr, 91
takšan-, 32
ti(i)anzi, 30
tēkan, 29, 32
uātar, 31

Tocharian

kñuk (A), 169
šalype (B), 93
śuwaṃ (B), 207

Sanskrit

abhi-praśnín-, 43
áhar, 31
ákṣi, 32
ánakti, 32

ánti, 86
ápa, 92
ási, 41, 83
asmákam, 28, 59
ásman-, 61, 164, 187
aśmará-, 165
ástḥi, 32
aṣṭhīvā(nt)-, 245
ātmán-, 28, 142
badhnāti, 97, 100
bambhara-, 155
bhṛṣṭí-, 151
bhugná-, 103
budhná-, 43, 123, 55, 65
chāyá-, 247
chyáte, 245
daśá-, 319
dviká-, 257
ējati, 264
gláu-, 177
gr̥bhñāti, 95, 97, 98, 102
huḍu-, 213
īrmá-, 157, 185
jīrṇá-, 174
kruñcati, 270
lékhā-, 241
limpāti, 100
lunāti, 100
mathnāti, 100
matkuṇa-, 221
mṛṇāti, 100
mūrdhán-, 30
náman, 31
nāsá-, 208
nīdā-, 230
nīdī-, 229
pánthāḥ, 208
piṃśati, 97
plīhán-, 30
praśná-, 44
praśnín-, 43
pūrṇá-, 42
racáyati, 46
ráditi, 223
rájan-, 28
rátha-, 272
rekhá-, 241, 242
rékṇas-, 46
rikhāti, 242
riṇāti, 101, 102
rīyate, 101, 102
ṛṇóti, 99
śákhā-, 329
sákthi-, 189
śaṅkú-, 329
śása-, 201
sināti, 100
śíras-, 32, 167
skabhñāti, 97
starí-, 190
stṛṇāti, 100
sūpa-, 106, 117
śvā-, 29
svápna-, 49
sváru-, 254
śvetá-, 45
svítna-, 44, 45, 123
†svitná-, 121
syāti, 244
tákṣan-, 29
údhar, 32, 159
ukṣán-, 30, 62, 63
úpa, 88
úrṇā-, 43
utkuṇa-, 221
vanóti, 108

varšmán-, 253
vé-, 260
vínā, 85
yáva-, 32
yunákti, 97

Avestan

ānhan-, 32
ascūm, 245
asman-, 187
azan-, 31
dūma-, 194
gaona-, 277
haxti-, 189
huuarā, 83, 323
kaofa-, 272
karapan-, 30
kata-, 306
masan-, 32
mazan-, 32
mąθra-, 29
mąθran-, 29
pañtā, 135, 208
raoyna-, 187
raθa-, 272
sraiiian-, 32
uruuaēsa-, 262
uruuisiia-, 262
usan-, 32
vanhan-, 32
zaurura-, 31
zauuan-, 32
zruuan-, 31

Persian

varvarah, 264, 265

Armenian

lakem, 120
mat'il, 221
siwn, 246
sunk, 225

tiz, 256

Mycenaean

ki-wo, 245

Greek

ἄγνυμι, 99
ἀγοράζωσι, 100
ἀγοράσωσι, 100
ἀδελφός, 260
ἀδήν, 30, 32
ἄελιος (Dor.), 83, 323
ἀηδών, 155
ἄκμων, 29, 164
ἀλγηδών, 155
ἀμφήν (Hsch.), 30
ἀνεψιός, 260
ἀνθηδών, 155
ἄνθος, 155
ἀνθρηδών, 155
ἀνθρήνη, 155
ἀντί, 86
ἄξων, 29
ἄπο, ἀπό, 92
ἀποθανεῖν, 101
ἀποθνήσκειν, 101
ἀρήν, 30
ἄρσην, 30
αὐλός, 197
ἄχθηδών, 155
βάτραχος, 349
βεῦδος, 277, 278
βλήχων, 29
βραχίων, 29
βρεχμός, 47
γείτων, 29
γέρανος, 308
γέρην, 260, 308
γλία, 239
γλουτός, 274
γρίπος, γρίφος, 181
δαίμων, 29

δαπάνη, 46
δείκνυμι, 45
ἐδηδών, 155
εἷ, 41, 83
εἶρων, 29
εἰς (Att.), 87
έν, 86
ἐξ, 87
ἐρείκω, 242
ἐρωή, 48
ζεύγνυμι, 97
θρῶναξ, 155
καλάμη, 162
κάλαμος, 163
Κηληδόνες, 155
κίθων, 305
κίων, 29, 245
κνήμη, 81, 163
κρίνω, 105
κρωσσός, 322
κύων, 29
λαγάρος, 310
λάζομαι, 310
ληδεῖν, 310
λιχνεύω, 44, 123
λιχνός, 122
μάκων (Dor.), 312
μέμφομαι, 107, 111
μήκων, 29, 301, 312
μύσσομαι, 97
νεφροί, 39
νοσεῖν, 101
νοσῆσαι, 101
νῶτον, 45
ὄνομα, 31
ὄρνυμι, 99
οὔθαρ, 159
πεμφρηδών, 155
πίτνημι, 97
Πλάτων, 82
πλεύμων, πνεύμων,
29

ποιμήν, 30
 πυθμήν, 30, 43, 65
 ροικός, 262
 Σειρήν, 30
 σκάζω, 189
 σκιά, 247
 σπλήν, 30
 σπόγγος, 225
 στεῖρα, 190
 Στράβων, 29, 82
 στύπος, 287
 σφόγγος, 225
 τεῖχος, 45
 τέκμαρ, -ωρ, 165
 τέκτων, 29
 τενθρηδών, 155
 τενθρήνη, 155
 τενθρήνιον, 155
 τερηδών, 155
 τεταγών, 98
 ὕδωρ, 31, 38
 ὕμην, 30
 ὕπνος, 49
 ὑπό, 88
 φθίνω, 97
 φλέδων, 29
 φρέαρ, 83
 φώγω, 109
 χαίνω, 97, 302
 χήν, 30
 χιτών, 305

Modern Greek

μᾶκος (Tsak.), 312
 ουρά, 264
 σκιά, 264
 σκίουρος, 264

Albanian

degë, 257, 319
elbth, 313
grimë, 275

gjalpë, 93
gjumë, 49
karroqe, 322
kurpth, 313
mokth, 313
qipí, 272
shtjérrë, 190

Oscan

allo, 43

Latin

armus, 157, 185
avis, 260
barba, 150
calamus, 163
cānus, 201, 202
carō, 29
cassis, 306
Catō, 82
catulus, 306
collis, 43
corbis, 180
cornīx, 260
corvus, 260
crux, 270
culmus, 163
cumulus, 68
damnum, 46
femur, 32
fūcus, 230
fundus, 43, 55, 65
fungus, 225
glūs, 239
gluten, 32
grānum, 174, 198
grūmus, 275
grūs, 308
hiāre, 231
homō, 29, 167
inguen, 32
instīgō, 105

iūniperus, 32
lambō, 44, 98, 123
liēn, 30
lingō, 44, 55, 98, 123
lingua, 36
lippus, 120
mateola, 221
mungō, 97
mutilus, 221
Nāsō, 29
natis, 45
nefronēs, 39
nux, 351
pandō, 97
pellis, 43
pingō, 97
pollen, 32
pōne, 90
praeco, 44
pulmō, 29
rāia, 185
rāmus, 157, 185
rapidus, 222
rīma, 242
rumpō, 44, 98
runcō, 44, 98
sanguen, 32
sciūrus, 264
sōl, 323
spiriolus, squiriolus, 264
sterilis, 190
stria, 250
stringō, 44, 98
strūma, 296
surculus, 254
tabes, 349
tangō, 44, 98, 123
tundō, 44, 123
über, 159
ungen, 32
Varrō, 82

vincō, 105
virgō, 29
vīvārium, 258
vīverra, 265

Italian

latta, 218
nuca, 168
ratto, 222

Spanish

nuca, 168
rato, 222

Old French

ésclat, 218

French

bec, 248
bécasse, 248
éclat, 218
écureuil, 264
latte, 218
nuque, 168
oignon, 154
rat, 222

Old Irish

ainm, 31
animm, 36
bech, 230
ben, 36, 83
brot, 150
búas, 291
cécht, 329
cnáim, 81, 163
cnocc, 169
cnú, 351
crogán, 322
crúach, 228, 270
cruim, 157
daiss, 228, 291

delg, *delgae*, 192
der-scaigim, 317
dlongid, 192
domain, 43, 45, 103
fiach, 260
fo, 88
glenaid, 239
grán, 174
grinne, 341
imb, 32
macc, 127
mug, 127
net, 218
rondid, 101, 102
rúad, 101, 102
scendim, 189
slat, 217
sluccim, 120

Middle Irish

cadla, 306
crith, 234
cúan, 272
dega, 256
farr, 253
figid, 196
gúaire, 277
sруб, 295

Irish

dúal, 319
feoróg, 264
iora, 266
naoscach, *noasga*,
 250
naosga, 250
rata, 223

Scottish Gaelic

feòrag, 264

Middle Welsh

ceinach, 201

Welsh

begegryr, 230
cainc, 329
calaf, 163
cartwen, 339
cnwch, 169
crochan, 322
crug, 270
crwth, 349
crydd, 234
crynu, 234
chwyf, 251
dwfn, 43, 45
garan, 308
grawn, 174
gwar, 253
gwiwar, 265
llath, 217
llwyf, 157
nyth, 218
sugnaf, 44, 98
ystlath, 217

Breton

raz, 223

Old Church Slavonic

brada, 150
бѣѣла, 230
čeznŋti, 44
dlanь, 310
dręgati, 109
дѣнь, 175
goba, 225
grobъ, 45
imę, 31
koreъ, 30
krugla, 322
kupъ, 270

nosъ, 208
skočiti, 317
skotъ, 111
slama, 163
smoky, 111
smučati, 108
sъkъ, 329
stopa, 106
strъgati, 106
sъsati, 44, 98
věverica, 264
вѣнѣ, 90
вѣнѣ, 90
žely, 177, 178
žena, 36
зѣно, 174
žъvati, 207
žъly, 177, 178

Old Russian *věverica*, 265

Russian
bobr, 265, 267
bólozno, 148
cěvka, 245
čubъ, *čupъ*, 285
děrgat', 109
glýba, 274
glýda, 273
glýza, 274
ilem, 155
kórob, 180
korob'já, 180
krjak, 171
lápa, 310
lotók, 218
mak, 313
mignuť, 44, 98
motýl', 219
pásmo, 199
pčelá, 230

solóma, 163
soxá, 329
stópka, 287
trúten', 155
úlej, 198
volná, 43
xvéjat' sja, 252
žórav (dial.), 308

Byelorussian *kudel'*, 306

Polish
dłapa (dial.), 310
dłóń, 310
korb, 180
kuczeć, 110

Czech
céva, 245
čub, *čup*, 285
čupa, 285
dlaň, 310
dlapa (dial.), 310
tlapa, 310

Slovene
glúta, *glûta*, 273, 274
krâk, 171
krék, 171, 172
metúlj, 219
trôt, 155

Serbian & Croatian
čûpa, 285
glib, 239
kûpa, 271
mètîlj, 219
pčèla, 230
sòha, 329
trût, 155
žërāv, 260, 308

Old Prussian
bitte, 231
gegalis, 265
gerwe, 308
moke, 313
perrēist, 262
sasins, 201

Lithuanian
aguonà, 313
akmuō, 164
añtras, 101
añtrinti, 101
aũlas, 197
aulỹs, 198
bābras, 265
balžienas, 148
barzdà, 150
bēbras, 265
bitė, 231
blizgėti, 129
budėti, 101, 102
būdinti, 101
būsti, 102
daĩgis, 192
demblỹs, 194
dienà, 175
dilgė, 192
dilgùs, 192
dubùs, 128
dveigỹs, 257
gaĩgalas, 265
gaũras, 277
gėrvė, 260, 308
gliėti, 239
gomurė, 303
grėžti, 108
guogas, 307
kařbas, 180
káupas, 270, 271
kėlmas, 162
kėlti, 100

- krakulaĩ*, 171
kúgis, 68
kuodēlis, 306
lakù, 120
laz(d)à, 217
lìmti, 156
lópa, 310
lù(n)ginti, 44, 55, 98
lùgnas, 42, 102, 121, 122
māg(u)onē, 313
mākatas, 218
melmuō, 184
mìgti, 44, 98
nósis, 208
pérnai, 43, 90
pilkas, 202
pilkšis, 202
plīkas, 102
plīkti, 102
rāugas, 187
reivē, 242
riēkti, 242
rieša(s), 261
rievē, 242
rīkē, 243
rīšti, 262
sáulē, 323
seīlas, 244
siēti, 244
skrēbti, 108
slastaĩ, 217
snāpas, 249
snāpēlis, 249
strūbas, 295
šaivà, 245
šākē, 329
šeivà, 245
šiŗvas, 202
šiŗvis, 202
šókti, 317
trānas, 155
vaīveris, 264, 265
várna, *vaŗnas*, 361
vēveris, 265
voverē, *vóverē*, 264, 265
žiáunos, 207
žirnis, 198
žmuō, 29

Latvian

- budināt*, 101
cilināt, 100
dzērve, 308
gāmurs, 303
lazda, 217
puōsma, -s, 199
slasts, *slazds*, 217
stups, 287
tran(i)s, 155
vāvere, 264, 265

3) Non-Indo-European languages

Finnish

- hanka*, 329
karpio, 180
matikka, 218
nukun, 79
oppi, 79
tukki, 79

Mansi

- pōōŋk*, 225

Mari

- poŋgō*, 225

Mordvinic

- paŋgo*, 225

Khanty

- paŋk*, 225

Kartvelian

- ma-ṭl-*, 221